

```

{ 'heisenberg-ric2': { 'noiseoff': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': {'value': 'random'},
'noise_profile': { 'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': False,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': True, 'type': 'png'},
'file_name_prefix': 'heisenberg_noise_off'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'vqe',
'state': 'dmatrix',
'vqe': { 'iteration': 10,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': True}},
'zne': { 'data_points': None,
'degree': 1,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 '
'X 3 + 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + '
'1 Z 4 + 1 X 5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Y 0 '
'Y 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Y 1 '
'Y 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Y 2 '
'Y 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Y 3 '
'Y 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Y 4 '
'Y 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Y 5 '
'Y 6 + 0.5 Z 5 Z 6']},
'output': { 'exact_sol': -8.56677223350563,
'initial_cost_history': [ 6.121384846939922,
6.150953723335025,
6.28813394059272,
6.195576097324661,
5.996275986067059,
6.155907118612996,
6.197882180139617,
6.276491195599097,
6.126423666072617,
6.100624170709524],
'noise_details': { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],

```

```
'status': False,  
'value': 0},
```

```
'noise_level': [0, 0, 0, 0],  
'noise_profile': { 'noise_on_init_param': {  
  
    'noise_prob': [ 0.001,  
                    0.001,  
                    0.001,  
                    0.001],  
    'status': False,  
    'type': 'depolarizing'},  
    'odd_wires': 4},  
'optimized_minimum_cost': [ -7.113819716474412,  
                             -6.821323437950204,  
                             -7.2322812477454015,  
                             -7.168061781988564,  
                             -7.037022966690968,  
                             -6.764659140688094,  
                             -6.70772123506257,  
                             -6.951929490175616,  
                             -6.765271591343842,  
                             -6.643370531408768],  
'optimized_parameters': [ [ 0.010202312572582629,  
                             0.01020231257258252,  
                             0.010202312572582303,  
                             1.3908062838956354,  
                             1.3908062838956354,  
                             2.0164416760130464,  
                             2.592046705326477,  
                             2.592046705326477,  
                             2.592046705326477,  
                             3.724080378311745,  
                             3.724080378311745,  
                             4.739480012897176,  
                             4.739480012897176,  
                             5.945308900554525,  
                             5.945308900554525,  
                             5.945308900554525,  
                             5.945308900554525,  
                             6.51261715666153,  
                             6.516103684733277,  
                             7.99085874039394,  
                             7.99085874039394,  
                             8.16277333661163,  
                             8.99858956109671,  
                             9.406132081871055,  
                             9.406132081871055,  
                             9.406132081871055,  
                             10.1331640829029,  
                             10.184186623923145,  
                             11.15918741223016,  
                             11.16716557378114,  
                             2.8450840989270483,  
                             -0.611567672261254,  
                             -0.551095997013421,  
                             -1.0131107784954883,  
                             0.6825070788337171,  
                             -0.6162852792596847,  
                             0.053241771237976535,  
                             -0.12516614014524466,  
                             1.004304378969064,  
                             1.0874046621334863,  
                             -0.263039228744755,  
                             -0.10249623387196242,  
                             1.6330709573383448,  
                             0.7555807809408442,  
                             -1.304918472993802,  
                             -1.1870918453507433,  
                             -0.1330475605934714,  
                             -0.5047081392764299,  
                             0.7756728723196425,  
                             -0.49278801094288127,  
                             0.42847067786702214,  
                             0.5383171641852699,  
                             -0.1052976007723638,  
                             -1.2720333725347934,
```

0.1951924272458347,
-0.04277330847061178,
0.9574239041786494,
-0.23071352324401823,
0.594076188896681,
0.34935689199185954,
1.343124516909016,
0.09733282827416712,
0.30944030008743273,
0.4457207862171129,
-0.9683728985035202,
-1.120837258235465,
0.5707161794308084,
1.2158524027070072,
0.7958721804993727,
-0.9645697723054423,
-0.8960337800506504,
0.19569442936859724,
0.40224575611741115,
0.043062302517518566,
-1.1857970207577537,
0.1737802328972237,
-0.6433879386848066,
-0.09230835161436789,
-0.5865305292280132,
-0.4370825940325887,
-0.2558137517355887,
0.0011609522641742864,
0.6539099100207111,
0.5336009667754404,
-0.066252982145182,
-0.5359243057678726,
-1.3834098071980059,
1.0212180272076503,
0.5141425620577187,
-0.3002279094652444,
1.9431365651467594,
1.0552610834283143,
0.03764625526605462,
1.0189467943495765,
-1.1805772107399133,
0.2812463759993353,
1.4321423259624626,
0.004331180480864721,
-0.009722915333733264,
0.6451162813005181,
-1.3276490840851547,
-0.022791981388308465,
0.6904663535208042,
-0.08291995023229079,
-0.4292297662993819,
0.08451212085763125,
1.730745239172763,
-0.4572165915626394,
-0.5304437424739927,
0.6917595603044201,
-0.0834050618602206,
-0.18753214427540912,
0.03305271309241428,
-2.06126875311761,
1.5412249435528216,
0.6658854058101124,
-0.36436553183642006,
-0.9193216873412987,
-0.12900157901067746,
0.00542745656150127,
-2.171309732970475,
0.7554637248132821,
1.249093251376782,
-0.6445658098419218,
0.14255909437901,
1.9827929295292588,
1.7044112257377888,
1.3052476731654217,
-1.2421280078558083,

0.9803773552514545,
0.624656679395339,
1.5450717726138348,
1.3377635200462674,
1.6713162630334013,
1.1434539739578018,
0.7924845860796002,
0.7488178629415702,
-0.22907755928161938,
0.02068614393688229,
0.7608745118642098,
-0.9304086893990576,
0.32284859123973336,
1.1891889033470526,
-2.2276657670493667,
-0.3847051133881006,
0.34685342333136693,
0.549782335235608,
-0.741267714199706,
0.26835698412135817,
-0.3932040836252281],
[0.1668593898896292,
0.17040712573292452,
0.1805381301837931,
0.5412292500635668,
0.5489811947718172,
1.6527475146851556,
2.55163116002801,
2.5519072608250526,
3.5532683674980303,
3.6288880821260254,
3.628888082126026,
4.805536133422,
4.805536133422,
5.495812833389341,
7.061809444427505,
7.061809444427505,
7.091809082073759,
7.091809082073759,
7.177303629807857,
7.177303629807857,
8.014949756247965,
8.530162577551986,
8.530162577551986,
8.546061750342906,
9.04867098177819,
9.074898072060186,
9.632422305371618,
9.634373167303918,
10.38794675566215,
10.6266488059183,
1.098346025464491,
1.5064109656875124,
-1.345000798816457,
-1.0112785952191263,
1.6300712383138607,
0.8148371701480261,
-0.28350002839102345,
-0.33372360425933684,
1.6208682076493302,
0.8121027533840469,
1.424723017228629,
-2.424990019545176,
0.082574342220046,
1.5280227698708746,
-1.15137171364767,
0.8821561188139568,
-1.0160789387304932,
1.6746733218779515,
2.4704340348759835,
0.8258460042877822,
0.21603078517896931,
-0.05951632881852065,
-0.38389794767629976,
1.262782578763767,

0.2784323534326924,
-1.1033559163681899,
-0.6528860571486432,
0.2962709573582783,
0.7373896958863383,
-0.19469473272588958,
-0.5648694322067066,
-0.2865513283590244,
-0.1568370013695695,
0.5493402156787225,
-0.06660698090283595,
-0.03359282593361995,
2.386303225482439,
2.5818639178838447,
-0.2777126348162844,
0.9824372498806634,
-0.39422047955069633,
-0.2578521646674435,
-2.3189661125012333,
3.704222502127313,
0.0786550943979888,
-0.4292299615806753,
1.9657979836656132,
-2.0589782426096743,
0.5074656887310783,
0.529845836294102,
0.3333899546946687,
-2.293287544626138,
0.38495704078578347,
0.11228231507748758,
-0.5441382535456334,
-0.10489456654410802,
0.20011773016694326,
0.2071371298105447,
-2.3651747840346373,
-4.642753244408169,
0.31152670002610416,
0.5022276983536407,
0.868938175568576,
0.20348366213849262,
1.6475591691674232,
0.3723999247005541,
-0.6589367341470095,
-1.1662574597011768,
-0.6243580701563078,
0.7131540586709431,
0.1202680893039496,
-0.035429417928639004,
-0.764191314011073,
1.2086109032075552,
-1.051983148544319,
0.9246153479653275,
-1.1427863274072698,
1.332115914696049,
-0.27763903875773216,
-0.8366665559266224,
1.231317184155771,
1.0584874216549616,
-1.2625782973678872,
-0.13344263101573678,
-0.5438546353631553,
0.32212207070493337,
-1.303226737690094,
-0.01337174800709397,
1.5589050571537895,
0.3479397882311206,
-1.87970134892204,
2.977494205144941,
2.126179176803338,
0.873413305092612,
0.8500950046844986,
0.23032349681334924,
0.3909007718762141,
1.27926383991169,
0.4053229000380187,

0.18190897779649284,
0.3777935390994293,
0.16772342823060155,
0.7412973026981022,
0.639129360404442,
0.018341913161260836,
-1.4541884523192488,
0.28134413046256895,
2.4001077951469534,
1.6611157476326641,
0.7905510314185135,
0.3404203970178551,
2.183912805046456,
-0.3997518053650227,
0.053570896670028414,
-0.357205442227867,
2.465639862334342,
0.5537972989863772,
-0.39728466664110995,
-0.027817223268321305,
1.9807706991981826],
[0.5951590437264768,
0.5951597797898418,
0.596302078053195,
1.4089369768482796,
1.4331037442558932,
1.9334285070314567,
3.062681311500457,
3.0626813115004623,
3.701592986553387,
3.7015929865611663,
4.692693983743045,
4.692707272250184,
4.692707272250258,
4.692707272250313,
5.273956511551061,
5.2739565115534885,
5.274711944180805,
5.292162333185471,
6.388031186249758,
6.706092770308958,
6.706092773942781,
7.548703938317144,
7.681022457823495,
8.397974590110634,
8.397974590138377,
8.714530791128386,
8.714530791128386,
10.075092988071805,
10.075092988073155,
11.341131260513714,
1.2164966908304873,
0.5225196146354087,
-0.34542368185043265,
-1.252295733833459,
1.5735101799719247,
1.0464982053397573,
1.2394468649734893,
0.21595542567535536,
0.6527940529325085,
1.4145936861598034,
-0.5380512317672027,
-1.9014686324238337,
1.451754133219536,
2.1987565378949516,
0.938092448651857,
-1.6378088277288243,
0.38538945046186207,
1.1341127264381634,
0.4639756379522118,
0.6078279078278344,
0.6166736770850927,
-0.2105783577152824,
-0.35613934438782024,
0.019700965332415492,

-0.1607988588148763,
2.193665487035956,
-0.06423342915568941,
-1.9546338684001463,
-0.18633224680849803,
0.7857721800679403,
-0.6042593641712383,
0.7490426193215785,
-1.6440993970557098,
0.1929003499686422,
-0.13169470171686798,
0.1181804780373785,
0.15693401681144475,
-0.28658811756127545,
-0.20708135879200276,
0.09230795188954338,
-0.4114468277172402,
-0.8587939345722652,
-0.29108154529906144,
0.5427012646910324,
-0.25834402839896575,
0.08607676608574079,
-1.2954984273935897,
0.1813204379953717,
-0.34769281236878496,
-0.09683800499273743,
-0.9220596744797915,
-0.67003274699539,
0.4888752813745547,
1.2659094572448748,
0.11946751252820795,
-0.2586531113468492,
0.8859866115247531,
1.5010226653220677,
1.2253090413370418,
-1.41480051306012,
0.4438287250399343,
1.1348546909542183,
-0.26356600881424186,
-2.0062796648632446,
-1.5997185613370128,
0.19365670408843416,
1.9542746653083636,
-1.16753432767107,
-0.8341998310634118,
-0.8078884986396416,
1.057969787908329,
0.11874076530060441,
-0.37823536664235385,
-0.26520820180860055,
-0.05623142087332937,
-0.6351447036627758,
1.2477313546900863,
0.09475828495367859,
1.1570291529547316,
-0.3159140153065273,
-0.5798756886838644,
0.5049917679316769,
0.17933499272174766,
-0.6452919120449363,
0.3093583753904067,
0.6269418409971109,
0.536228963471926,
-0.39297460666924544,
-0.8585561386083093,
0.42310038622809626,
0.4625446959490829,
0.46551290354297115,
-0.006225430338097464,
-0.9498487290783498,
0.9283489563799108,
0.5288015885952413,
-0.6423745714782674,
0.055608107183735184,
-0.19141640271906557,

1.6106743132208,
-0.930502866063725,
0.13606233511650948,
0.7450273046384331,
-1.862998799714745,
0.8639045442938115,
2.2050766695222275,
2.1209593580448147,
1.5928713117159927,
-0.646261951983125,
0.18338232206466956,
1.9195145046471642,
0.9340015251080405,
0.22314701152636945,
0.22339358941062268,
-0.49545999196264406,
1.6497345768092624,
-0.37841357670502584,
0.19313501042389203,
0.07433308037827455,
1.831606295240235],
[0.013945989287166007,
0.014482887559355881,
0.7020792700957467,
1.2708090216179468,
1.8771473361142745,
1.898764043280349,
2.8886374873795146,
2.8887394286496386,
2.889554540070078,
2.8895545400700837,
2.889778536891296,
2.890175029071649,
2.8901764691110965,
3.881547037638128,
4.589967577258263,
4.916994542918821,
5.4213951120408375,
5.421396236977594,
5.937049320865825,
6.879796302074518,
6.931299553407329,
7.597215453538154,
7.941093252873733,
7.941093304737446,
9.23283235128487,
9.232832768748333,
9.232832769084892,
11.10587443204972,
11.182791630116155,
11.297769627933807,
0.7007490003921656,
0.8357798277265018,
-0.006144500286149594,
-2.002647677865005,
0.3078229546122665,
0.6497081019565148,
-0.5263987300225429,
-1.3591736982237872,
0.4978537208919288,
-1.0703987652029276,
1.4894572070841317,
0.10507602812754031,
1.2060409491957005,
0.37232709611155723,
-0.9426439100006672,
-0.36387040355717715,
0.9248005896093184,
1.0383864274136485,
-0.2845687665108692,
-1.3700022658840743,
0.197561542582957,
-0.32357133547759104,
-0.5540850172472152,
0.5749264875292693,

-0.8790799137852104,
1.0945413836746756,
0.3793043522465329,
-0.4056052817915047,
-0.03473792548266608,
0.06466238365347084,
0.06911551547308,
0.1689218888799635,
-0.24055379179177064,
1.0229644579244672,
-0.3032356624759471,
0.7939776889924924,
0.9000504980210274,
1.1922086788205504,
-0.09256238768786877,
0.5764640092401906,
0.20671868793560919,
1.5137094250639866,
-0.17335127063306519,
0.8566284172009878,
-0.14321647694006984,
1.3125145841881363,
1.3592268317355658,
0.1512217749645529,
0.40406540571588334,
0.048383985681794354,
0.19259007229061106,
0.09976438753283409,
-0.8546571821237718,
0.6621369033183476,
-0.1702351863826035,
0.36026352288591695,
-0.8538518608539489,
0.5392868667124087,
1.1279014251126462,
-0.36128436957709814,
0.388999106940785,
0.6046833425816769,
0.22322275018396126,
0.19171695999278726,
-0.909915166057891,
0.21776193961543494,
0.28996753900858246,
-0.5842323028487328,
-0.23864599488371552,
0.5243326463982184,
0.32056321937714066,
0.5356836550343521,
-0.3036011562711703,
0.28802877152454115,
-0.33781577627567244,
0.7960023934710929,
-0.34238272839391165,
-0.755152052016237,
-0.44352996180502197,
0.07213680417312635,
-0.10190373256064952,
0.6647215855084265,
-0.9303498099091333,
0.9785859773141862,
2.090431130104361,
-0.3615944398761197,
-0.05657593308707301,
-0.4893006812698228,
2.169089562284795,
-0.2512716878005756,
0.5324061192252376,
-0.35101147654456727,
0.809248070031391,
1.4509839161975397,
-0.14911304722907706,
0.009330045236156378,
0.16685413556340573,
-1.0472061077948698,
1.6539137014575498,

1.3611819463379937,
-0.1612641139265809,
-1.2962247052632323,
-1.3797066485510583,
0.7614302908946665,
0.5091243097447496,
-1.034844537010192,
1.0872752511102444,
-0.476692079414531,
0.49649227376774574,
-1.1491063122969838,
0.5155370403129137,
2.9855835289421178,
1.332632811562063,
-0.4511213152168112,
-0.18076672982111605,
-0.6499435359167832,
-0.1697346678000319,
1.4081723136905224,
0.442588820936982,
1.722678503402018],
[0.6039929180328771,
0.6039929180328772,
1.1870458270689,
1.1870458270688995,
1.412745515623535,
2.7932578893846336,
3.721244973412882,
4.900800320776727,
6.39210348373028,
6.39210348373028,
7.061133772685592,
7.061133772685592,
7.061133772685592,
7.064294030691084,
7.896781244010901,
7.896781244010901,
7.998168023342792,
8.09271241991372,
8.09271241991372,
9.328885229004163,
9.328906326502109,
9.56647323827175,
9.587868480377287,
9.606795901338938,
9.632809215929381,
9.925261433896223,
10.67745113859835,
11.226267254217802,
11.25813692634961,
11.259219336255413,
1.5276356326766196,
-1.2318130242699923,
0.07924764105056724,
-3.11015247863868,
1.71979592089301,
2.1636204056140302,
-0.44284897868332995,
-1.9599373218939633,
0.06248244737264575,
1.4749130816585614,
0.6808242800852113,
-2.8378068229153226,
1.7023278591356004,
1.50553848610602,
0.9365133390345636,
-2.1922613442274193,
2.8432021818280635,
0.1855585942894919,
-1.305416067894503,
0.1905478234118404,
0.7393995701939485,
0.19798052248890954,
-0.8075009715521676,
0.3948611135767452,

0.2802743807361451,
-1.4516873620037638,
1.5556285183865062,
-1.0754052466574715,
-0.26931261584286686,
1.2336702516634759,
-0.8384600779602994,
0.1263891070094113,
1.371252832802525,
0.215177045818575,
0.6381629880602765,
-0.10309421117565558,
0.4124383487994645,
-0.655499258712117,
0.4988310520659948,
0.9884276098863405,
1.4367219570983862,
2.175763151095631,
0.4657848221151968,
-0.43932071710829274,
0.9992343531072873,
2.543795366459747,
1.4484903663678823,
-1.0884458694866728,
1.5946427529603922,
1.6036026746991174,
0.5049418933876622,
-1.0871394525284908,
1.0140119509472394,
0.5109229333717124,
0.8743589388253838,
0.09626392588768701,
-1.263331036369507,
0.7924497462635596,
0.7602243892770352,
-0.02425703489356225,
-0.9488939939602019,
2.6693155936644737,
0.5100002820942202,
0.595619937330169,
-0.8296192082845288,
0.03973961449691636,
-1.7144893205098422,
1.1592271204386744,
0.04896218554826232,
0.6673584031026698,
-0.8454442747231521,
1.0304518931414424,
0.8556412792926265,
0.538780781704616,
-2.308898512383933,
1.5082121217292006,
0.9522091008412682,
-0.6026951427085929,
-0.561943926102551,
0.0993983558744849,
0.4791162049419018,
0.10320414203165484,
0.3730055252741846,
-0.47083232270591907,
1.8673303303286475,
0.11712430229385423,
-0.6044688726688213,
-0.9010581315425702,
0.8870944114042272,
-0.6585959519901412,
1.4737812112349316,
-0.08789784470103781,
2.094313922775301,
-0.33880630598079,
2.0864279335765668,
-0.3541510297313768,
-0.1447960879675017,
1.1237218765827641,
0.7472353955833465,

0.43980261574830926,
0.6534205210717992,
1.2606259200129655,
-0.17935209951086067,
1.6469967332630842,
-1.0541245665671384,
-1.9424469299865808,
-0.18975499615277758,
1.967627931161089,
-0.5610576072704585,
-1.4036970670846107,
2.6946925238411343,
-0.9860662529475024,
-0.6528571620700129,
-0.6551571328219318,
0.9777230807240078,
0.003267369094500521,
-1.3217863730621935,
0.3048209868145306,
2.1104722024376037,
-0.13744008360720306],
[0.25527267369507034,
0.25565792243175783,
0.2647851751522132,
0.5363821708593047,
0.5363873360294482,
0.7233633669048282,
1.469495303301716,
2.6782106232033085,
3.0738233934797257,
3.0740346152756928,
3.0740346154225127,
3.5401415295205747,
4.384558950933806,
4.384564402805264,
5.187694707345863,
5.431645815108105,
5.489249393218873,
5.489270645046236,
5.4892706450864015,
6.745887429448721,
6.751180124796475,
6.975059797459636,
7.4852253234058015,
7.485225323406553,
7.485225336458803,
8.65422617922401,
9.588402512165866,
9.885475791973308,
10.59890299271783,
10.69671178550786,
1.7193952291225598,
-2.699905058742405,
-1.223597261476082,
1.010311557825714,
2.412320728895392,
0.01205800479020808,
0.07148434641220561,
-0.1717056232391439,
0.7382566227367261,
1.1624421524382496,
-0.9704132592114074,
0.007350927807351759,
2.3582361698297074,
1.1487298862005386,
-0.5298367489795935,
-0.09976600187590766,
0.7543852753586129,
1.401553978295213,
0.3078871075376248,
-0.35588965288410024,
0.15554254602359127,
0.3425968302644435,
-1.202384247417669,
-0.987347192012539,

0.4529373843839761,
0.4903191821294036,
-0.3596344497958935,
-0.17493820576367525,
1.3619720378391473,
-0.2517632245160441,
0.6511229549349082,
-0.07544040749490964,
-0.04771613185909407,
-0.3755018325798389,
0.9938372515787193,
-2.617459009880976,
-1.1600518599196699,
1.385641845308893,
1.3078548392623544,
-1.3438342667259302,
0.06225569718179881,
2.197693302377489,
1.042516443745511,
0.03537145740203309,
-0.4361466628046714,
0.5235506730441506,
-0.526852610863259,
0.2683619694707144,
0.4190531675420984,
1.1917182306749494,
-0.22315973823529625,
0.3724063269005885,
-0.2016113706580773,
-0.6608684665627687,
-0.5566572791655187,
-0.6080637139710359,
-1.3764025932169088,
0.06381643974581075,
-0.12239416020448966,
0.8768915165514672,
0.9734732325585186,
0.43086234391719236,
-1.2868778636436715,
0.2387140460098626,
0.9650786815364466,
-0.8088454790489977,
0.3379873670823351,
-0.5777804082620341,
0.5336495835071627,
2.9920059240641073,
0.6079052241489462,
0.5309531439532562,
-0.3169545414432997,
-2.3335729587340226,
1.4959043429555212,
-0.5568960671660346,
1.653532167337357,
0.35120291709985135,
-1.2609025397237739,
-0.1931706005433399,
-0.7761167208985412,
-0.07661975220377984,
-0.5351965211836152,
-0.0031199845298622592,
-1.0423575145047397,
0.1404336078605159,
-0.2659485743030148,
-1.6615430856264528,
0.43884229259462737,
0.6812026744892391,
0.39460126400851775,
0.9189185317254136,
1.0545864345697524,
1.1559068061784519,
1.0720705313960006,
2.514687033544068,
2.0122769373294087,
1.2260219872848024,
-1.3734724775008675,

2.3904026332666466,
-2.404362121009983,
-0.8308162974608817,
-0.49116355387064226,
1.4544417316540983,
1.2021789553657038,
-0.045966888650276455,
-0.5647889892892385,
0.6643832758564343,
1.3569537910410745,
-0.9183847616510102,
0.945220285616687,
-0.2507148012652629,
0.7511336622169466,
-0.9416337312429676,
-1.4793772681330912,
0.4334847087508488,
-0.02256141084202589,
-0.8421263780740565,
0.7532984525691246,
-0.4880194119115941],
[0.008743077103622827,
0.009882832507877682,
0.7050944318623401,
0.710375902327237,
1.3735081730499452,
1.9102715342756955,
2.4433320146909057,
2.4516882946549723,
2.451688347950266,
2.451688347950339,
2.843659132454273,
2.8437290613274957,
4.1600447531010465,
4.306699526502985,
4.306699526503065,
5.241123177068926,
5.241123178389404,
5.24112317839451,
5.241123178416889,
5.24112317841696,
5.540975568888287,
5.540975570417886,
5.941112811664419,
8.170699047520403,
8.170699047520507,
8.170699047530466,
8.255093472901638,
10.272293989986627,
10.424665220800923,
11.465171522837265,
3.9395345563718758,
0.4691052202571919,
-1.862560190849311,
-0.5409749194015872,
0.6809788356765686,
1.8042992706436805,
-0.26820425656414737,
0.1583592390805078,
2.570139180175488,
1.9730281302693196,
-1.463840299189443,
-0.6685636516618708,
1.7639202171290544,
-0.25593154759594605,
-0.2985776439588054,
-2.380396900296592,
0.702730788267215,
0.1111013414139888,
0.8409172298864358,
-1.2895389501906573,
-0.6495780843509739,
-0.7851296452736322,
0.551111177160635,
-0.7726813911354989,

0.5538939215319534,
0.03671507307592432,
0.3611092546557634,
-1.1244512927039525,
0.9979933722488747,
2.1458567471834726,
0.6045511633065316,
-2.057454443497708,
1.252679002491011,
1.430968651076373,
0.1975347289978355,
-1.299395231227167,
1.3709528213003457,
0.1431677222477642,
-0.5468176167431326,
-1.363138690512715,
3.0723179570358097,
0.09197418207209145,
0.552028513182037,
-0.14805587495457423,
0.6542791830341044,
-0.6463721623376274,
0.594458988807259,
0.36313400085196706,
-0.22487670595801615,
-0.5600947295073319,
-0.13194082373612193,
-0.7877948735702447,
0.37453010501171596,
0.4466259555663157,
1.308811697533367,
0.18300565834449778,
0.28722746055568227,
0.11593248147227052,
-0.22092771987873516,
-0.3830228539100271,
-1.1848305159650747,
0.5197091232696622,
-0.20788874837533922,
-0.6085287569162429,
0.622048119692812,
0.36154402486320714,
-1.283336122338554,
0.06614563071195226,
-0.10338801392549199,
-0.6587068471684263,
1.34100892693596,
1.611824691291148,
0.6916042008757854,
1.6327439292692163,
-0.6648342512722216,
0.37654710496224697,
-0.24718958516166675,
-0.7065038375487233,
0.11746925267221191,
-0.5800210884330035,
-0.5819296114685866,
0.01696046668851879,
0.08164598903007107,
0.39930256073308995,
-0.22041497304878216,
-0.13569368036844073,
-0.734071731705797,
-0.10448732621745443,
0.4350514428273773,
-0.4345336137549378,
0.39154417527507906,
1.0175423593468684,
2.0822612023831226,
2.2518057581506135,
-0.4050083292569126,
-1.3225723626396935,
0.2066177197668252,
0.3518219705658924,
0.7692056892056433,

1.1078718838725619,
0.6955122293015593,
2.6391129283967607,
-0.5193467200233968,
0.16205999751797937,
1.0319480004914923,
0.0466744549888964,
0.8866401912062555,
0.6199118226017315,
0.9594432756697039,
0.30783399339708145,
2.007890427791846,
-0.42290702105473665,
-1.4981000967272742,
0.1246045734342663,
1.4175139007816862,
0.730422807509675,
-1.0203109099547352,
1.2475764531977165,
1.172103834016867,
2.0417923602468857],
[0.3006295792025693,
0.5011825648484141,
1.736075427951676,
1.7894431127664152,
2.4859557870150857,
3.6898895058559917,
3.73290444462633,
3.73290444462633,
3.741728259130309,
4.1342678631525835,
4.145528657970977,
4.615983426127272,
5.6679892124870985,
5.692197651053977,
5.773115344270512,
6.56178194421021,
6.56178194421021,
6.56178194421021,
6.56178194421021,
6.56178194421021,
6.561820835625042,
6.6476324960056985,
7.649631249906491,
8.361638748929934,
8.576867085540188,
8.77036061445696,
9.827286891841206,
10.5379658375992,
10.558826131428928,
11.83004188936984,
1.5987671786815294,
-1.5058826811902317,
2.0440908601375476,
-1.481172398738803,
1.4265067306792671,
1.3375234069756785,
-1.0089917144708827,
-1.8440331252413595,
-0.062239877224771915,
0.18483713323292328,
1.8795821846896128,
-2.058552713287731,
0.9708425302661001,
0.8840527876953188,
0.5964490041287303,
-1.4043070469033054,
0.14533141919970372,
0.022743431858071796,
-0.3923086391326145,
0.11023448615420071,
-0.4204112521530586,
0.5529792020785508,
-0.6330031478534407,
0.23236293228773558,

-0.4459800436834905,
2.5876988326738504,
0.6844024813933239,
0.1732372067049823,
0.7384853056565549,
2.8826092918303563,
-0.11204335589185775,
-0.2241007440171772,
-0.4845273525281558,
0.2744214554588425,
0.26177442491958286,
0.4512467910671607,
-0.6063187184164835,
-1.2386469417803163,
-1.1719897599242013,
1.9705273482015357,
1.945221133626245,
0.15309311442469994,
0.690693149150703,
0.22188036413222653,
0.47722534320651633,
0.12444414704260098,
-1.0057594761976716,
-0.3351811680967541,
1.0750707752291546,
-0.04597654655420585,
-1.4770936089010136,
0.1839119320523196,
1.0453831678930239,
1.6851715686961402,
2.1701829049155683,
1.692817047863365,
0.44152889324253053,
-0.3402272157802259,
-0.4600811596334244,
2.365566181908148,
1.166608300050211,
-0.12230572309949261,
-0.07192631995381693,
-0.48825404466402955,
1.202761995004197,
0.503733093641068,
-0.2613796073125891,
0.7323467617888468,
0.30547819787235264,
0.46207631030173263,
0.034964002783702264,
-0.30773087511026637,
2.2827399003210256,
0.773042400324002,
0.07466204753764587,
-2.812532590652055,
0.7925074856015164,
3.3306642432255886,
-0.8558810509247932,
-0.9283746444488663,
0.3079774400122357,
0.5580467297921388,
0.5588934104639726,
0.6444350594161403,
1.5119717259179621,
0.14573434197560214,
0.7631266139944589,
-0.43452714539522225,
0.38832086457140946,
0.36234082213113916,
-1.3222695581402877,
-0.6140190631377459,
2.2873566508452354,
1.0349767974312705,
-0.5734592440854204,
0.5234040439377924,
-0.8213429806368828,
-0.8621355796854082,
-0.5355268774606846,

1.8804920616270637,
0.022682713405259486,
1.3625464560687972,
-0.48449497738356756,
1.6018376883784673,
-1.3963097787223904,
-3.511711037561943,
1.3223211880663501,
2.1298031669093946,
0.24542774430512565,
0.7034284248706182,
-0.37346698797179273,
-0.43051209966139187,
-0.1034377630705127,
-0.07417862116176438,
0.2595758000914042,
0.9523286668350031,
0.2556898901330524,
0.22603519843353895,
0.9866656966967039,
2.5031147117731094],
[0.6608465720820748,
0.9671982645454077,
1.5458030244804597,
1.5458030244804597,
2.2589946156031733,
2.2589946156031733,
2.2589946156031733,
2.6595607382107107,
3.734699662125929,
4.07592175958283,
4.611631942641649,
4.831287600721991,
4.831287600721991,
5.914025831134952,
5.914025831134952,
6.072969066301945,
6.991999801396327,
6.991999801396326,
7.161687480905546,
7.673022292320241,
7.821921961804373,
7.912193421610441,
8.287160283389671,
8.956173402171897,
8.956173402171897,
9.75268721078065,
9.941478738047602,
9.941478738047602,
9.945621910117605,
10.438880525762482,
1.4707024445054038,
1.5353178524994073,
0.8263465992797864,
0.5104925709601137,
0.0752285958369072,
-0.1142202547489564,
-1.4526020976761953,
-1.3531650678039584,
-0.5250403673431315,
2.237334263415747,
0.2921144645165799,
-1.3764237769851546,
0.48788675650681135,
0.8685067508645514,
0.519063599878773,
1.1970126362568978,
0.35376457746361467,
2.1361591146989944,
1.704664258187609,
1.1956277779681033,
-0.8317273346515459,
1.1961203419780542,
0.6738815365004568,
2.0453196594697167,

-0.3197500197249862,
0.3940149430820266,
-0.03491413991246975,
1.9231424652296574,
-0.7553371587587144,
0.5536882090799309,
-1.1151355813488422,
0.06532487835880797,
-0.03610632383025197,
-0.721099071948806,
-0.7712693743746337,
0.2581737620009136,
0.6969815452179084,
-0.4519889309472695,
-0.6781990341468844,
0.3896669141200773,
0.746036455743935,
0.10534513312025831,
-0.7953867627526213,
-0.17196672908166516,
1.1080637640216904,
-0.36906579226317304,
0.3903518027422727,
0.7340380536410088,
0.16483545086429002,
0.12911772875241279,
-0.11978924907966522,
-0.5410781247103603,
0.27894738356501725,
-0.04603280575778243,
0.3301412555435043,
0.020942359039511207,
-0.535470327679227,
0.26379415288467284,
0.7549681986050077,
-0.774180168996079,
0.3042677541816752,
0.2314702730743271,
0.4347582057672544,
0.5269984906212896,
-0.5883800499089586,
0.4216788732962846,
-0.39154710450095487,
-0.20962806789618654,
0.008988716066278926,
0.42160932452985955,
-0.4362625059483254,
0.7149808650234762,
-0.9606565556866927,
0.02867923202735973,
0.6474453926631074,
0.3125034088891559,
0.24941070182093517,
-1.0996459114603814,
0.42918254303405223,
1.1958230549534523,
-0.23035300079654375,
0.43394127898307777,
-1.1958360530234091,
0.2490392548916114,
0.956337072693143,
-0.9098880983840639,
0.24421397010525564,
0.32228144879356985,
-0.04019115817841665,
0.601865593110944,
-1.0390070893344725,
-0.3870853595246112,
1.5254374970388465,
-0.06482825900051802,
1.508536156596221,
-0.4193438972827081,
-0.2124704995099638,
0.7982743594716017,
0.625924999124879,

-1.2144964531756246,
-0.7306315051046057,
-1.3775043670320717,
-0.5254300493631905,
0.24182169388976885,
0.05618119509583439,
-1.1649033986061088,
0.4414909596355956,
1.1243259379005652,
0.9623501696515389,
-1.0198159738874641,
0.6746700507964939,
0.9071271136722682,
-1.490119256375095,
-1.8915932031257203,
1.4861820000050194,
0.020835967248483293,
0.2672579153437498,
1.1487299094399863,
1.7394956603554965,
2.161406077992875],
[0.009700543149373924,
0.009700543149373951,
0.00970054314937392,
1.382070146567131,
1.7587904521787645,
2.560406827400793,
3.0924393369175114,
3.3853584719898073,
3.3853584719898073,
3.3855655528609647,
3.9151369396989866,
3.9151369396989866,
5.054043379727587,
5.058642146366078,
6.082229999853835,
6.348636465887238,
6.754028454322125,
6.777976106859175,
7.783524996765277,
7.783524996765276,
7.783524996765276,
7.995280883445485,
7.995280883445485,
8.645089634948334,
8.925040343314908,
9.356899631457969,
9.439062141544271,
9.717990345174004,
9.717990345174004,
10.595210012065888,
0.8074121230882934,
-0.11817978440691497,
2.3039735343549332,
-2.1294851108225332,
0.6714134591558454,
0.45516684502263977,
-0.8318799457604262,
-1.6937930072137053,
0.039287662842651484,
0.5223639661903685,
-1.2122069881806565,
0.27186944236208743,
0.22304871394760792,
-0.08855679301585977,
1.1988033553075006,
-0.9660581453773212,
1.393585655420428,
0.524497607950108,
-0.22549193465024678,
-0.20940428560116026,
-0.22546417020605838,
-0.06641646114084936,
0.3531991811801793,
-0.06811222553567531,

1.0762936524340476,
0.17762298521732672,
-0.24437612021854904,
-1.0622868736463833,
0.41092319903165814,
-0.5877006275418265,
1.2299752305066975,
-1.5519910987794148,
0.6657478217748831,
-0.027661876184918333,
0.011781258679136937,
0.9082970931589502,
0.006644189925206108,
0.3712893817061966,
-0.04659784477089089,
0.2991695519878787,
1.4480032246106211,
0.3765628272855095,
0.3810659512210775,
-0.4053198419433054,
-0.06957688452126994,
0.08768201200938007,
-0.16514507742174195,
0.05923670456485792,
-1.0025267882306543,
-0.041476053822733706,
0.4728058870419665,
-0.09799142504381375,
-0.642516889827151,
-0.521645479249954,
0.18062390558973918,
-0.5137105837376863,
0.315588961893638,
1.0102820620809385,
-1.412161058774387,
0.5136468128759858,
-0.5616754178138471,
0.11099281329082958,
-0.4065421947836274,
-0.8520422495977672,
1.5662481725135406,
-2.09542418518098,
1.2116955825236206,
0.8221816705323761,
0.041900809919046114,
-0.09736226691448804,
-0.6600925025551557,
-1.1656564284287168,
0.8214103261157102,
0.28477271669010623,
0.5992804506866763,
-0.21193327493741926,
0.6490050812244267,
0.8872065875645788,
0.2758198961026023,
-0.3259804283588707,
0.5954957550354935,
0.8064107897831347,
0.15673794452304002,
-0.5606863141571317,
1.1354810000650886,
0.29170605524887394,
-0.6471431342469229,
-0.29364595444579666,
0.05289373230476578,
0.03489286830366231,
-0.10405686076243598,
0.4547289014370767,
-0.1747513431047453,
-0.28439942445332933,
0.010516254927139741,
0.11813385237732216,
0.3966163987420368,
-1.37250350754582,
0.06601400397758364,

```

-0.3662241908629097,
-0.07601408247331909,
1.737776242565087,
2.1849230369400683,
-0.5785032754175823,
-0.5063968590594791,
-0.23335270192508997,
2.320012252653636,
0.6182382779358941,
0.7875200859675175,
-0.009873677663214134,
0.7482624361366613,
-0.35875384806276184,
-0.5357091554695692,
0.4397736267416906,
0.16706548411788932,
0.9251966251074774,
0.053252937203382235,
-0.23592171730002592,
1.2877035765438765,
2.1426518726936172]],
'run_time_sec': 43067.655497312546}}],
'redundant': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}}},
'init_param': { 'value': [ 0.38848003251483587,
0.3884800325148359,
1.2841735835398218,
1.284173583539822,
1.2841735835398218,
1.818138220668234,
1.818138220668234,
2.7575469841005327,
2.7575469841005327,
2.7575469841005327,
3.0053468968599275,
3.0053468968599275,
3.005346896859927,
4.184079661691832,
4.83635165164968,
4.83635165164968,
5.97926077140412,
6.458805266455043,
6.513102391504654,
6.513102391504654,
7.122105903846713,
7.2188422331450415,
7.988590448127284,
8.268095077127937,
8.268095077127937,
9.164456488648115,
9.164456488648115,
9.782953992889732,
9.782953992889732,
10.467486029517685,
1.5496959718296535,
-1.0300437645832343,
0.425343365508412,
-1.7136674729251273,
0.33645197441382346,

```

0.2288357961575219,
0.10470230685228102,
-0.6612661793744108,
0.11823315872480306,
0.48571962162230753,
-1.3748501535311368,
0.2563120795974374,
-0.38871520150353267,
-0.18786175076151226,
1.0017996241286973,
0.05598903629217323,
1.501328526357405,
-0.32299727271195017,
0.5715295079657807,
-0.12201961934486968,
2.2747823054657927,
0.16289086097520028,
0.027619518366160244,
-0.10165606616500937,
-0.06321979816820887,
-0.27543819802246033,
-0.0048370064103128475,
0.1330138167350915,
-0.06428151371988705,
0.21905201010909556,
0.25697175483980955,
-1.462905966720411,
0.8014511304369155,
3.456137165485204,
-0.3560002683832885,
-0.4886877133354182,
0.32039145066707037,
1.6153352350920127,
0.5940099487759896,
0.18140087044059483,
0.27123106498617,
0.7444341922951362,
0.170480557625751,
-0.4688462519246396,
0.5235839184929864,
-0.5470964469389679,
-0.011921500492307424,
1.4116230981753706,
0.49972453659747124,
-0.7800673920851233,
0.4234227404504273,
-0.2071635265929523,
0.14064328933496192,
-1.2007096247825992,
0.12047098237640114,
-0.4125103841815656,
1.0447620953563077,
0.0020814149928196773,
0.4988991062112368,
-0.34495773827123155,
-0.12849063482979717,
0.4280934542561602,
0.7138444629946918,
-0.6497712710787925,
-0.43652659674625793,
0.917432486791049,
-0.8523477361785018,
0.5648987974980545,
0.3183226113969618,
0.14295784675778261,
0.9589824340171335,
0.9194478619356266,
1.1001359353915339,
1.5417377267389083,
-0.5376003771957187,
-0.5404528215706331,
0.21624687748210675,
0.8814937279455825,
-0.371639532103812,
-0.2668100917972658,

```

1.3133163210428074,
0.44574090697479807,
-1.0940261852319737,
-0.313830643067749,
-0.06100275437545863,
-0.049560783686653447,
-0.3025056724120473,
-0.9891529613817422,
-0.35099332176432035,
0.5436069706516342,
-0.722828417704955,
1.6968661444846564,
-2.4402971962208975,
0.7865745542588627,
1.3437881169309889,
0.36108471326554326,
-0.1749159211919777,
-0.014334278530919132,
0.02978377629472008,
1.8533978555002608,
-1.8470426695738211,
0.6272356375919823,
0.44775499409400255,
1.3281153539688288,
0.4583217606753929,
0.6443670392327268,
0.5254233581973354,
2.3087652110195127,
-0.7761319091913014,
-0.7605421959233766,
-0.6461580464596497,
0.7075597766411752,
0.985778508379752,
-1.3130974173457381,
2.8711331564400684,
1.2993543284582694,
1.9493614761217355,
1.1647493000584068,
-0.42731613935458074,
2.7038346000669766]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_ric2_sample#6'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,

```


2.730288811725591,
4.476350841356259,
4.476350841356259,
4.476350841356259,
6.331658137341824,
6.331658137341824,
6.7621011201584285,
6.77851096220289,
7.490043179529316,
7.490043179529316,
7.490043179529316,
7.490043179529316,
7.511664240193972,
8.302259372586072,
8.36899822047307,
8.607405767895836,
9.631163052898822,
9.874876363637043,
10.225975825453373,
2.711573403856553,
0.20239905137011172,
-0.4220883549830381,
-0.8404830976891227,
1.794332340210785,
1.5095243517856047,
0.22326098118939897,
0.07733371735403745,
1.280548885023134,
1.4121496210624682,
0.9815935680907694,
-2.975348552150871,
1.6316643284140941,
0.8239502945369961,
0.0314505340316768,
-1.0892891749800817,
1.720694113273713,
0.6634696927679957,
-0.836512780686814,
-1.1286873737196692,
-0.09500137106776224,
1.0428135848317974,
-0.4828772957450712,
-1.129334172148368,
0.7253737318795237,
1.4108675361353635,
0.7588540034470418,
-0.5039485441412638,
0.10346857579652805,
1.2668095290766983,
0.7994757345637048,
-0.7393240316081237,
-0.7818304990031031,
0.25375980918776664,
2.228975199542585,
2.2487625387645873,
0.6384105819815235,
0.39434544396871446,
1.3679129840939837,
-0.27336699910726836,
1.3170761670781417,
1.1159622861844822,
1.176824183488981,
0.40583799697229034,
0.8786168217534174,
0.2914083315983707,
-0.8464697846173548,
-0.032455291822620566,
-2.0755386823554396,
-0.11439663923099666,
0.7312672791487281,
0.4263196833485598,
0.11802610051350722,
1.0445881812565523,
-1.9566462184988007,
0.9492303984073598,

```
-1.7129593951518396,  
-0.3230601534209437,  
0.3161713641922788,  
-0.4811213940350071,  
0.08327140785409295,  
0.37526634808329173,  
-1.9990814349080968,  
-0.15397700748904192,  
-0.9549084255901845,  
1.1120455683467105,  
-0.42952128451280097,  
0.17929609797141985,  
1.0446016069356951,  
0.846774755857218,  
-0.2784586201121851,  
0.23000407229986752,  
-0.9198047598183884,  
0.996529675185623,  
0.34608382599103005,  
0.5099375110708237,  
2.1948932092248885,  
1.3625694142279887,  
-0.9306056124011278,  
0.6012804875871011,  
-0.7788968212470156,  
-1.5884735532174328,  
1.0361309111992296,  
0.015937704739038126,  
1.4003797043998856,  
2.429422654005534,  
-0.27504014116509645,  
0.8256449732433587,  
1.2146087003914199,  
-0.6435553984602482,  
0.3046269503778821,  
1.0182549333954134,  
1.4202224852266074,  
1.3511302325707892,  
-0.7261672378416364,  
1.5992752660640235,  
-0.5764295578986515,  
-0.8318409785507642,  
1.1375736468462807,  
0.8529850561064739,  
0.9553972898831558,  
-0.09161160854677679,  
1.617622889574723,  
1.4551513261723394,  
-0.08817813881493969,  
0.8045852859105778,  
1.9892741470924016,  
1.8347816167556374,  
-0.8050652823779443,  
0.008511392950551585,  
2.1460096989270747,  
0.49277581514942953,  
-2.4311684786278165,  
0.5134791015266132,  
0.9751974520662461,  
-0.38582649358518695,  
0.7746227771853464,  
1.7307944301597258,  
2.1320333699599043,  
1.1654652968447545]],  
'noise_profile': { 'noise_on_init_param': { 'status': False,  
                                              'value': 0},  
                    'noise_prob': [0.001, 0.001, 0, 0],  
                    'status': True,  
                    'type': 'depolarizing'},  
'nqubits': 7,  
'observable': { 'coefficients': { 'bn': [ 1.0,  
                                           1.0,  
                                           1.0,  
                                           1.0,  
                                           1.0,
```



```
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [ 0.6189603325174557,
0.7142372255934472,
0.818217611349809,
1.1433975540074377,
1.9121137997887092,
1.912113799788709,
1.9189666863881127,
2.194219841514053,
2.734492156740232,
4.250258815642276,
4.250258815642276,
5.774076859791501,
6.527612880112256,
6.532834731882501,
8.049613322901676,
8.049613322901676,
8.050730377555883,
8.561643142317116,
8.561643142317116,
8.758357012835646,
8.758357012835646,
8.78353884989931,
9.477076383056275,
9.536600626467342,
9.536600626467342,
9.536600626467342,
10.241295505019522,
10.318321333031077,
10.579129354925781,
10.82886994059623,
3.662494394083504,
0.08309129295248968,
1.5295376506525808,
-1.5964000102221196,
1.8988970610647178,
-0.07218621526676876,
0.7238532160702689,
-1.2917423920622229,
1.8532222971765195,
-0.40818119715800666,
0.469598065996274,
-1.2652675561295137,
0.41465882817859046,
0.48683539382624846,
-0.15336518315146752,
-1.6419783820068856,
1.2136622236743368,
0.5933123377118196,
0.9666225223219593,
0.9539776194708611,
-0.3793986531891993,
-0.2570964298405315,
1.5884302353807402,
0.19867970725527498,
1.0036205245072367,
-1.7266220384949087,
0.6519864384701275,
0.173373648086925,
1.8697046853700252,
0.5263944462203128,
2.9785129063723943,
-0.5937256494260224,
```

0.22753174799850245,
0.24918335157216837,
1.5438671949833176,
-0.2981477150404249,
-0.0503664733834821,
0.542442835193611,
1.8096859498837612,
-2.2260935596321993,
0.28693396030012636,
0.9986781622414097,
-0.01940263773226835,
-1.8805315707929653,
-0.1513202473661221,
0.1836836320118686,
-0.5655439481125303,
-0.30323205019871646,
-0.4186198812206128,
0.2357002564872809,
2.8296979465953833,
-1.0776781071762758,
-0.08378558447273823,
1.7446476514004516,
-0.4606832586075214,
-1.762173446350097,
0.10241780783627007,
1.6018254886002428,
0.2621004786967086,
1.349261138919249,
0.14987623247602613,
0.5068543254371847,
-1.3474781551208477,
1.0890436270679893,
1.6662048029626129,
0.3608718991529565,
-0.11137748070221386,
0.5641204379289552,
2.2546014997074266,
1.8888740160956543,
-0.23345107582518032,
-1.5772444286238798,
2.1408383357361265,
-0.4968079560788485,
0.1533667959779154,
-0.40238715350257875,
0.25591729793510365,
2.947382531044255,
-0.32162602471486745,
-0.7587223675916386,
-0.8983111566200868,
0.006114119281372181,
2.9372824954589514,
0.48288231335139287,
0.7939297064609614,
1.9454360656955985,
2.092519500518508,
0.8995449331251366,
0.10171517677156572,
0.45235144869603666,
1.02501950409702,
0.24331156538914245,
0.48973195386607005,
2.672871435464254,
1.313352678455905,
2.010046934709317,
-0.42153209835304317,
-0.13436700176919486,
-0.6950305532982513,
0.6026553555930264,
1.6681913778840949,
0.4125109526933123,
0.514533549588801,
1.0794873063708696,
-0.9571806916209341,
0.27630815458061647,
0.6001387806934945,

```

0.5059087494947612,
-0.5722342732268321,
0.3986919058890637,
1.0044202485382987,
0.3352295947729849,
1.3777059541173544,
-2.835684252944388,
0.6104690716589696,
1.838086538234438,
-0.3106134411398272,
-2.2167272073692903,
0.10706372359431235,
-0.17469850626418476]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_ric2_sample#3'},
'redundant': { 'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]],
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X '
                                '2 X 3 + 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X '
                                '5 + 1 Z 4 + 1 X 5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Y 0 '
                                                         'Y 1 + 0.5 Z 0 Z 1 + '
                                                         '0.5 X 1 X 2 + 0.5 Y 1 '
                                                         'Y 2 + 0.5 Z 1 Z 2 + '
                                                         '0.5 X 2 X 3 + 0.5 Y 2 '
                                                         'Y 3 + 0.5 Z 2 Z 3 + '
                                                         '0.5 X 3 X 4 + 0.5 Y 3 '
                                                         'Y 4 + 0.5 Z 3 Z 4 + '
                                                         '0.5 X 4 X 5 + 0.5 Y 4 '
                                                         'Y 5 + 0.5 Z 4 Z 5 + '
                                                         '0.5 X 5 X 6 + 0.5 Y 5 '
                                                         'Y 6 + 0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -6.218690041461777],
                             [12, 6, 0, 0, -4.805156509769757]],
            'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                'identity_factors': [0, 0, 0, 0],
                                'noise_level': [4, 2, 0, 0],
                                'noise_profile': { 'noise_on_init_param': {
'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
```

```

                                0,
                                0],
                                'status': True,
                                'type': 'depolarizing'},
                                'odd_wires': 4},
                                { 'gates_num': [12, 3, 1, 0],
                                'identity_factors': [1, 1, 0, 0],
                                'noise_level': [12, 6, 0, 0],
                                'noise_profile': { 'noise_on_init_param': {
'noise_prob': [ 0.001,
0.001,
0,
0],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4}],
'run_time_sec': 4.225522518157959}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [ 0.4022850038273885,
0.4222288083135141,
0.7644599371513524,
0.7900179023917507,
1.1026508993249646,
1.1630326759467198,
1.1630326759467198,
1.2119340062926192,
1.9859325565890973,
1.985932556589097,
2.096166401632718,
3.378647998605865,
3.378647998605865,
3.378647998605865,
4.354848115861854,
6.006218342726225,
6.006218342726225,
6.009974297352436,
8.225327865566033,
8.242093148909708,
9.222354355246733,
9.369468664000962,
9.432712219232048,
9.692289963799956,
9.692829942577658,
10.391360826688166,
10.391360826688166,
10.529867035412478,
10.529867035412478,
11.06290493838992,
2.9062307172979733,
-1.7788694936155562,
-0.1145266982224492,
-0.9311559923392093,
0.9691998209476543,
1.7463479137973288,
-0.7663862807329105,
-0.9694574896037749,

```


2.8979112386638413,
1.1400097878698292,
0.8096361549155084,
-1.5900494197653285,
0.09829667109053167,
0.31839493135320834,
0.3886777840539921,
0.12229184789371997,
1.7250225844898464,
1.0065795676456808,
1.4535641205083867,
-0.3486770669120958,
0.2556790893233589,
1.778531141883236,
0.269972269049403,
1.0155903957099945,
0.8041163886037216,
0.5381821926966456,
-0.30108604163317454,
0.6396533041678052,
2.1819991440037696,
1.4978086658278347,
2.5181040929106753,
0.4624415671193703,
-0.17701736987873673,
0.6910098122382194,
2.947172461882987,
-1.0727539032453073,
0.4023854197196741,
1.650490750838525,
0.9313998963798695,
-0.6769162055305248,
-0.5620785927454173,
-0.20885673940707247,
2.5019757090972443,
-0.37561416876024456,
-0.9329392032282731,
0.7818961506889994,
0.44350434198669075,
-0.10644929099379355,
-2.498928032568615,
-1.2046085786256906,
-0.06365416567297402,
0.17257236645054272,
1.2516209341031546,
-0.3591527663232026,
0.46789807326225935,
0.24374638092155315,
0.02110383811013195,
0.12714835746845662,
-0.5505736947218767,
-0.03497727012222505,
-2.2168334497478877,
-0.21485512988314215,
1.7552902914988067,
0.6189795903733311,
0.3191491062237487,
0.522632432692797,
-1.1121089236913335,
1.1371618962620422,
-0.2410043783817164,
0.24168543661014927,
0.39418565996934685,
1.2107667809264395,
2.398781467533486,
0.6948257569073976,
0.6170229142231725,
0.7050419245924102,
1.768546262611762,
1.5562956304120967,
-0.4269237214365518,
0.6302835616447369,
1.7710848704009405,
1.2281100264507194,
1.5036530549577243,

```

-0.7435268012971183,
0.7613541379199313,
0.8523298100708424,
-0.6655959646469325,
1.1543911681084875,
2.2675628634479494,
1.0299985170060308,
1.400963430022522,
0.27759304330136975,
0.6960561589746208,
-0.4210335016880339,
0.0336463669485127,
-2.099660094290342,
0.6674819238895001,
0.5049668435179431,
0.48001405045948287,
1.3574871122533543,
-2.2986993170736225,
9.601426007351498e-05,
1.3312470798272018,
0.7582238004326416,
-0.35033967277151296,
-0.4108552112493075,
0.1982910129710702,
0.6700017270602091,
0.26559360983179486,
-0.14563210230012705,
-0.39610135234418875,
1.1727729846793147,
0.6713573564879365,
0.8527070052929181,
0.8268283343075853,
2.2918575835381745,
0.5082103460913493,
1.8600493489550152,
0.49576811175353974,
2.232387985311215]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_ric2_sample#8'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X '

```

```

'2 X 3 + 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X '
'5 + 1 Z 4 + 1 X 5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Y 0 '
'Y 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Y 1 '
'Y 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Y 2 '
'Y 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Y 3 '
'Y 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Y 4 '
'Y 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Y 5 '
'Y 6 + 0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -5.743389283741309],
[12, 6, 0, 0, -4.3560016590478785]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 0, 0],
'noise_profile': { 'noise_on_init_param': {
'noise_prob': [ 0.001,
0.001,
0,
0],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': { 'noise_on_init_param': {
'noise_prob': [ 0.001,
0.001,
0,
0],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4}],
'run_time_sec': 4.8073859214782715}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [ 0.16477895719846114,
0.16477895719846108,
0.7183189156962347,
1.8697093861419474,
2.105082615571597,
3.05308344502201,
4.199393321043376,
4.199393321043376,
4.199393321043375,
6.490210948425593,
6.4902109484255925,
6.490210948425593,
6.4902109484255925,
6.4902109484255925,
6.4902109484255925,

```

```

'status': False,
'value': 0},

```

```

'status': False,
'value': 0},

```

6.490210948425593,
6.882554174728105,
6.882554174728105,
6.882554174728105,
6.882554174728104,
6.8877583035732775,
6.8877583035732775,
6.8877583035732775,
8.21014824639056,
8.21014824639056,
9.071469420716642,
9.120928605341893,
9.714944098401723,
11.474131735950095,
11.474131735950095,
11.794551582273902,
1.341188628911811,
1.6554894539990426,
-2.1068974621953425,
-1.4171475675653578,
0.33211669069513333,
0.11331792925356984,
-1.3610911099963097,
0.46439741779020816,
0.9313501387866817,
1.0938593094543279,
-0.3591727738608774,
-0.4732622270970327,
1.6321598874818157,
-0.2782091348514778,
-1.0526781349295347,
0.0708703831522549,
2.6982701224663113,
-0.038867313655444914,
-0.10508583152652556,
-0.40759365792756236,
0.6284849004330764,
0.5192751929971827,
-1.6282040637266806,
0.0014422433007245106,
-1.8299719264591412,
2.3182597876656823,
0.8875167319646391,
1.5664268415036142,
1.5052778791259855,
2.935955436800565,
1.533101581959507,
1.8091829332960188,
0.45080966818917967,
0.1506883627647002,
-0.6843548866459184,
-0.5544883854492364,
1.2466562002431278,
-1.1866110136199353,
1.4521358231995756,
-1.125608676420736,
-0.24054383618816966,
-0.058272145415006006,
0.410408128597049,
-1.6441753460137252,
0.4019816715685168,
0.4588365987304068,
1.354346078964423,
-1.7153251360741875,
-1.1039978646265602e-06,
0.7940670488104785,
0.17979772520560053,
-0.24385394325110826,
0.6367160741723578,
0.8500459957371875,
0.0559416907316065,
0.32418668587130467,
-1.4797636873635434,
0.11902177942387202,
0.5185165439317966,

```

-0.08399222686641401,
0.8650918037299421,
1.4911587670088011,
0.12792248933645675,
0.3372262149750348,
-0.6553154954729696,
1.1995792367396254,
0.5132294030998703,
0.44835115860783636,
-0.37116980512777586,
0.7805194563374558,
-0.854334789795377,
0.5008527672922033,
-0.2302685629030402,
0.6145182391921147,
-0.6575500079215796,
0.8657035637529482,
1.1874516653159721,
-0.16284802367068918,
-0.26462171656665856,
-0.9815102056659819,
0.23198397744948207,
-1.087097195512072,
0.6616385654966559,
-0.2929637257774683,
0.03471390505932934,
0.2048751570049573,
0.9170017118070749,
0.39617956766989404,
0.12006199846995962,
0.1922705913940127,
0.8784478594386566,
0.055479957240883566,
-0.6634105431569992,
1.0742951419164057,
-0.37818200958777604,
0.28548758229245597,
-0.6594827056613419,
1.0235832498557866,
-0.19428253177041227,
-0.5531991468121416,
-1.189626464718852,
0.2519110119226656,
-0.6708462316627377,
2.7994352657547057,
-3.8764436680175813,
0.5152327138619494,
-0.3749347648847944,
2.9042167722097862,
1.1265965037549552,
-1.850959186134046,
2.165374611027145,
0.8818414791702859,
1.5039524911129019,
0.06564767423285973,
0.6486215314149069,
-0.08854143781823305,
1.8396223528529987,
-0.00792246728000446,
2.8805879469479487,
3.6337353861266473]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,

```

```
0.5,  
0.5,  
0.5,  
0.5],  
    'r': 1},  
    'def': 'ising'},  
    'output': { 'draw': { 'fig_dpi': 100,  
        'status': False,  
        'type': 'png'},  
        'file_name_prefix':  
'AUTOMATE_heisenberg_noise_free_time_evo_ric2_sample#9'},  
    'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},  
    'run': 'redundant',  
    'state': 'dmatrix',  
    'vqe': { 'iteration': 1,  
        'optimization': { 'algorithm': 'SLSQP',  
            'constraint': True,  
            'status': False}},  
    'zne': { 'data_points': None,  
        'degree': 0,  
        'method': 'richardson',  
        'sampling': 'default'}},  
    'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X '  
        '2 X 3 + 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X '  
        '5 + 1 Z 4 + 1 X 5 X 6 + 1 Z 5 + 1 Z 6',  
        'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Y 0 '  
            'Y 1 + 0.5 Z 0 Z 1 + '  
            '0.5 X 1 X 2 + 0.5 Y 1 '  
            'Y 2 + 0.5 Z 1 Z 2 + '  
            '0.5 X 2 X 3 + 0.5 Y 2 '  
            'Y 3 + 0.5 Z 2 Z 3 + '  
            '0.5 X 3 X 4 + 0.5 Y 3 '  
            'Y 4 + 0.5 Z 3 Z 4 + '  
            '0.5 X 4 X 5 + 0.5 Y 4 '  
            'Y 5 + 0.5 Z 4 Z 5 + '  
            '0.5 X 5 X 6 + 0.5 Y 5 '  
            'Y 6 + 0.5 Z 5 Z 6']],  
    'output': { 'data_points': [ [4, 2, 0, 0, -5.966171999362981],  
        [12, 6, 0, 0, -4.491721403204036]],  
        'noise_details': [ { 'gates_num': [4, 1, 1, 0],  
            'identity_factors': [0, 0, 0, 0],  
            'noise_level': [4, 2, 0, 0],  
            'noise_profile': { 'noise_on_init_param': {  
  
                'noise_prob': [ 0.001,  
                    0.001,  
                    0,  
                    0],  
                'status': True,  
                'type': 'depolarizing'},  
            'odd_wires': 4},  
        { 'gates_num': [12, 3, 1, 0],  
            'identity_factors': [1, 1, 0, 0],  
            'noise_level': [12, 6, 0, 0],  
            'noise_profile': { 'noise_on_init_param': {  
  
                'noise_prob': [ 0.001,  
                    0.001,  
                    0,  
                    0],  
                'status': True,  
                'type': 'depolarizing'},  
            'odd_wires': 4}],  
        'run_time_sec': 4.566856145858765}},  
    { 'config': { 'ansatz': { 'gateset': 1,  
        'layer': 30,  
        'ugate': { 'coefficients': { 'bn': [ 0,  
            0,  
            0,  
            0,  
            0,
```

```
                                0],
                                'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                'r': 0},
                                'time': {'max': 10.0, 'min': 0.0},
                                'type': 'heisenberg'}},
'init_param': { 'value': [ 0.11853625011402234,
                             0.11853625011402236,
                             1.1173235551611367,
                             1.7342580765110573,
                             1.7342580765110573,
                             2.108144483620023,
                             3.054129947242328,
                             3.054129947242328,
                             3.054129947242328,
                             3.056306360633301,
                             4.638414750363569,
                             5.0621851887275495,
                             5.7503913724004905,
                             5.774998407238176,
                             6.917933224474368,
                             6.917933224474368,
                             8.061549209526286,
                             8.061549209526286,
                             8.061549209526286,
                             8.591251503132888,
                             8.591251503132888,
                             9.051235537435975,
                             9.081856762230144,
                             9.467876152039107,
                             9.545932016874406,
                             9.591419209848118,
                             10.465616352275608,
                             10.465616352275608,
                             10.465735173309618,
                             10.980942228105905,
                             2.0734674059611633,
                             0.32892201175743896,
                             -0.007356440825095051,
                             0.7856576253246221,
                             2.089457548986907,
                             0.7744884921126718,
                             -0.35049309306395837,
                             -0.6556831494452968,
                             2.0369525201874903,
                             0.4739652080570001,
                             -0.8564345639983049,
                             1.25716179603495,
                             1.7181260437331554,
                             2.3978160665502384,
                             1.1152024575020885,
                             -1.3335737674295711,
                             0.016650799076572978,
                             1.6607129762234538,
                             0.648059834763191,
                             0.2438725527060928,
                             -0.27307896000752346,
                             1.4369017198677916,
                             3.5672448343800007,
                             -0.5379702113035718,
                             -0.7420664562717484,
                             -0.529699891930992,
                             0.7301471483152846,
                             0.781913925757813,
                             0.6004078048346045,
                             1.7324865479818374,
                             0.4548236607944829,
                             -1.4419813608219112,
                             -0.684662976785234,
                             0.8455512814500156,
                             1.97751114408285,
```

-0.833202517811791,
1.110524460863052,
2.6403320691876733,
-0.3129427957408566,
-0.06674204626492235,
0.5842857778318762,
-0.21132485649138796,
-0.011842019288790376,
-0.25743970610802486,
0.7486616881233721,
-0.1302029141156864,
0.09497091477412227,
-0.3715265429351211,
0.39570077921274605,
0.28643868832003017,
0.7848413456471106,
-0.7157146817058562,
0.09411764456438232,
-0.10910076647953601,
1.1262761892344808,
-0.27334997823872975,
-0.44027652197942063,
0.6740415100156434,
-0.336256645457715,
0.4268157265313067,
-0.8611696251566416,
-0.34560928156726306,
0.5618074077714486,
0.5866360379174992,
1.8662876827821717,
2.21592310898005,
1.972223656383108,
1.6141187926643297,
0.856255815565121,
-0.34275929095614754,
-1.0642528745182613,
-1.0618686751734654,
0.2379582550153795,
-0.6501770361983872,
-0.19408645485522444,
1.501330829125914,
1.5726602727591479,
0.7066066097903853,
-1.4580936665368467,
0.46940440748349077,
-0.1287015677341077,
-0.9805925033056946,
-0.3754741669636888,
-0.026817919501009202,
1.1067346586651607,
0.17547923725022047,
0.8902143171896091,
-0.22777698610446503,
0.8082253583749343,
-1.0674083601412958,
-0.022706971436593337,
0.34227899244110566,
1.3527736523760083,
2.0767841716125894,
0.5778568145668401,
-0.6555502504392442,
1.2670274746873995,
0.507694022232518,
-0.2629991187543805,
1.7048253080726326,
1.4088065522309612,
2.823918413203317,
1.2763232744235296,
-1.4348344213353486,
-0.605496445996402,
0.09645158262695042,
0.024639736182597342,
0.23997550724405606,
0.9979895378690888,
-0.40612123795993227,


```

1.826790650755685,
0.029599395583551916,
-0.4592861834683686,
0.0328358806035505,
1.3612120706287358,
0.8210402794880411,
0.05688164146575431,
0.03152531580318861,
2.0316547976974904,
2.795039607579165]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_ric2_sample#10'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X '
                                '2 X 3 + 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X '
                                '5 + 1 Z 4 + 1 X 5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Y 0 '
                                                         'Y 1 + 0.5 Z 0 Z 1 + '
                                                         '0.5 X 1 X 2 + 0.5 Y 1 '
                                                         'Y 2 + 0.5 Z 1 Z 2 + '
                                                         '0.5 X 2 X 3 + 0.5 Y 2 '
                                                         'Y 3 + 0.5 Z 2 Z 3 + '
                                                         '0.5 X 3 X 4 + 0.5 Y 3 '
                                                         'Y 4 + 0.5 Z 3 Z 4 + '
                                                         '0.5 X 4 X 5 + 0.5 Y 4 '
                                                         'Y 5 + 0.5 Z 4 Z 5 + '
                                                         '0.5 X 5 X 6 + 0.5 Y 5 '
                                                         'Y 6 + 0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -6.2813019572269795],
                             [12, 6, 0, 0, -4.778373581659957]],
            'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                'identity_factors': [0, 0, 0, 0],
                                'noise_level': [4, 2, 0, 0],
                                'noise_profile': { 'noise_on_init_param': {
'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0,
0],
'status': True,

```

'status': False,
'value': 0},

```
        'type': 'depolarizing'},
        'odd_wires': 4},
    { 'gates_num': [12, 3, 1, 0],
      'identity_factors': [1, 1, 0, 0],
      'noise_level': [12, 6, 0, 0],
      'noise_profile': { 'noise_on_init_param': {
          'noise_prob': [ 0.001,
                          0.001,
                          0,
                          0],
          'status': True,
          'type': 'depolarizing'},
        'odd_wires': 4}},
    'run_time_sec': 4.533947706222534}},
  { 'config': { 'ansatz': { 'gateset': 1,
                           'layer': 30,
                           'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                           'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                           'r': 0},
                           'time': {'max': 10.0, 'min': 0.0},
                           'type': 'heisenberg'}}},
    'init_param': { 'value': [ 0.12420123662801165,
                                0.138246244841176,
                                0.34185010800102933,
                                0.8771280515547424,
                                0.8771280515547423,
                                1.9326762312100456,
                                1.9326762312100456,
                                2.879806034127527,
                                3.4467495190209787,
                                3.460588109395402,
                                3.4605881093954016,
                                3.4605881093954016,
                                3.911934955717517,
                                3.9564836804547703,
                                4.188240551814426,
                                5.098640553346409,
                                5.098640553346409,
                                5.098640553346408,
                                6.466332587579958,
                                7.457565848827182,
                                7.712724508543908,
                                7.890302861063261,
                                7.898577524812402,
                                8.020037295803547,
                                8.020037295803547,
                                9.289723192898348,
                                9.68251578824504,
                                9.68251578824504,
                                10.66694519433398,
                                11.464898013549973,
                                1.8095532965205083,
                                -1.6633278493215808,
                                -1.2281091611401596,
                                -0.40155726479865544,
                                1.7903243850816337,
                                0.18661159996017582,
                                1.2217850039169107,
                                -0.48128156078104645,
                                1.438546033666569,
                                -0.9466383819139143,
                                -0.21792270060606069,
```

-0.07921201994836616,
0.6692914930417907,
-0.16791294486623565,
-0.3278463005181308,
1.2883193159418245,
1.9001761172388045,
0.7646920019853412,
-0.6325477821570596,
-0.5987552945744136,
0.5669730115381587,
0.5983941343774113,
-0.06179171362000896,
-0.6867235859693906,
0.5211737756628548,
-0.19354588250953578,
0.6072816218147872,
0.3472677197742769,
-0.08031734075212016,
1.4024126217748347,
-0.5867181790012919,
0.809620813948269,
0.35997425529443994,
-0.8722967249300235,
-0.5874818244923429,
-0.5567926036837068,
-0.6652314874983639,
-0.6042054992483096,
-0.13918430829641074,
-0.08277488470584692,
-0.4972293169762719,
-0.4421668735352916,
0.6160601456858428,
-1.1972657609969646,
-1.1558488364748807,
-1.4501782622641872,
0.03989088007781949,
-0.8684923327267209,
-0.5569704341339503,
0.15823054121075,
0.6345578215095505,
0.3601853745371903,
-0.42594808878819745,
0.08979804115834313,
-0.153043281560756,
-0.603228229875778,
-1.1264410002736531,
0.6895580235946861,
-0.11045941959152497,
0.5174109375139814,
1.3269460586908306,
-0.3852298542423827,
-2.4120095755015893,
0.08807412161753436,
0.1228551365077638,
-0.6982591234727157,
-1.1618488040328845,
2.5586909942385017,
-0.24921574033555105,
-1.9870969070050777,
-1.0279498295884981,
-0.7796802357622239,
0.48428681889730063,
0.7162815161060118,
-0.15547297036103372,
1.0522591887013215,
1.772822944479854,
-0.2654037211186137,
-0.2720110866721671,
-0.03929478442656517,
0.34606077605631835,
0.5533331494282367,
1.0574846609095763,
-0.5747611578732846,
0.4916772893793279,
0.9875709691916216,

```

-0.17857479553023695,
-1.147281220275154,
0.37370382177065037,
2.456815155407461,
0.6567596591037266,
0.9507316292553318,
1.5113928877114242,
1.8380989093110256,
-0.1736919900840187,
-1.2666253947041328,
-0.554819304070435,
1.1931643666366285,
0.4614862178419797,
1.2040726557184918,
0.3083618234293392,
-0.4066692475369416,
0.3740368412278018,
-1.684611974494111,
-0.8650016725760176,
-1.4838330341878547,
-0.14344887350922086,
4.203596189627334,
1.1133202614474196,
1.391124117957397,
1.8627332105499828,
-0.28192336755973296,
-1.5416482647725873,
-1.9612892990030817,
0.8836928087418303,
0.300761821046295,
0.6785848024508535,
-0.5500277556546446,
0.5742776326882599,
-1.181756808209963]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_ric2_sample#5'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X '
                                '2 X 3 + 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X '
                                '5 + 1 Z 4 + 1 X 5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Y 0 '

```

```

'Y 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Y 1 '
'Y 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Y 2 '
'Y 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Y 3 '
'Y 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Y 4 '
'Y 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Y 5 '
'Y 6 + 0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -5.814676484708358],
                             [12, 6, 0, 0, -4.248447309496402]],
            'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                'identity_factors': [0, 0, 0, 0],
                                'noise_level': [4, 2, 0, 0],
                                'noise_profile': { 'noise_on_init_param': {
'noise_prob': [ 0.001,
                 0.001,
                 0,
                 0],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [1, 1, 0, 0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': { 'noise_on_init_param': {
'noise_prob': [ 0.001,
                 0.001,
                 0,
                 0],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4}],
'run_time_sec': 4.216556787490845}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [ 0,
                                                             0,
                                                             0,
                                                             0,
                                                             0,
                                                             0,
                                                             0],
                                                           'cn': [ 0.5,
                                                             0.5,
                                                             0.5,
                                                             0.5,
                                                             0.5,
                                                             0.5],
                                                           'r': 0},
                        'time': {'max': 10.0, 'min': 0.0},
                        'type': 'heisenberg'}},
  'init_param': { 'value': [ 0.43940602753254654,
                             0.4524980474487914,
                             0.6462847510297367,
                             0.6535145711164267,
                             0.7019965052930044,
                             1.0830061860275473,
                             1.4380021973026182,
                             2.1860837771467474,
                             2.1860837771467474,
                             2.1860837771467474,
                             2.4875804812430995,
                             2.911818356328843,
                             4.9134595358452735,
                             5.043089843653786,
                             5.994354774193697,
                             6.658321281734746,
                             6.666230857319696,

```

8.458454027735183,
8.458454027735183,
8.458454027735183,
8.938931772570164,
8.938931772570164,
9.876451730548837,
9.876451730548837,
10.140618079102744,
10.453495471248813,
10.878653713196195,
11.282429749250388,
11.726474998572407,
11.93229290545428,
2.9640535214497064,
-0.6206168106317247,
-0.4035747848788786,
-2.067222048924027,
2.0576259596946604,
1.364315765000901,
-0.1480698866383175,
-1.95043906046908,
0.7897528850889669,
1.350582507885341,
-0.3349044763190947,
-2.3429833820345465,
2.04351680973891,
0.8083219796841106,
-0.06520770630024467,
-0.8776318722364843,
2.4862506381076748,
0.8345121096737285,
-0.03071661620417745,
-0.6225183255138825,
0.7450777062518694,
0.12619297185303674,
0.15620862918286052,
-0.47540100870445906,
0.6345315438244892,
1.451076347224231,
0.10985833383721234,
-1.6475696409761669,
1.1918822544398533,
1.3235539868194213,
-0.24222060474147966,
0.2418944215339802,
1.0397651583041008,
1.6773252160624976,
0.16975786500059598,
-0.9211027317156172,
0.7590471244619662,
1.76540125315151,
2.43856734239768,
-0.19488067618654012,
0.8462629124439717,
0.9979674075340036,
2.5775834810620677,
0.5650956422366618,
-0.5475934153287456,
-0.0253449469292375,
1.2524069066350432,
-0.605696708865831,
0.7019502861337156,
0.2803153244338447,
-1.0831925759777479,
1.0082140101851143,
-0.3762046176346905,
-0.9535933533256993,
-1.0422804614435257,
0.13948594332192887,
0.5835070530053111,
0.16086524965091817,
0.06403444071667018,
-0.06194188553710435,
-0.4026223625881385,
0.2750028468648149,

```
0.6789909068622377,
0.05162849323802493,
-0.7735332926339995,
-0.14862918062202837,
0.7661757453229759,
-0.16119730427373616,
0.6482389226852195,
3.1459456036848104,
-0.4760920139337192,
1.6197254159790668,
-0.5506328506738539,
0.9968299487743895,
-0.3106735286083523,
-1.2388662013594425,
0.8898580979157581,
0.43256773745092003,
-0.7272220041247079,
0.8716071923554094,
1.5193589139179793,
0.9787540447681606,
-1.4044929205930872,
-0.1984876316951887,
-0.46744984056178585,
-0.7765225748074835,
0.04344696088727837,
0.32373965212791267,
0.9049555024172572,
0.7139278873651953,
-0.8122282267372439,
0.5323405298774778,
-0.08437719753251166,
-0.14277007957413837,
0.7912147308716768,
0.26055493673650715,
0.11357082745857092,
0.3153698930365579,
3.1981441448638614,
0.32109035951393206,
-0.23017443442945684,
0.0316864124540797,
0.20875812581978664,
0.5466872162452653,
1.2918521189433907,
-0.7312547980705957,
1.4958457475939604,
-0.053193686203739404,
-0.4437772137226961,
-0.5271126400646005,
2.5136810835026426,
0.6674244883024821,
-1.0140291771816496,
-1.177197116527103,
2.058315434744553,
0.050372102721390984,
-0.09022974351195023,
-2.296438941834702,
0.3261688389091777,
0.3072223897800251]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
```

[illegible]


```
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [ -2.397634216504411e-17,
1.429450133825379e-17,
0.9192297980258759,
0.9192297980258759,
2.8984599305110854,
3.1993874679766523,
3.199387467976652,
3.715279848679994,
3.715919998165357,
5.126008182976459,
5.749875525815113,
5.785995839176617,
5.785995839176617,
5.785995839176617,
5.785995839176617,
6.315437962379428,
6.678845841951143,
7.3578167224936415,
7.415694801781823,
7.531263738212295,
7.949087697575632,
7.949087697575632,
8.111169110068786,
9.407094598577077,
9.495286031274873,
9.495286031274873,
10.993257661103009,
11.063439993331317,
11.639888600691357,
11.75175692668691,
2.1152038588685538,
-0.7839988468841306,
-1.0579861702584952,
-0.5122509123294313,
1.0012369417099964,
0.9275037440003397,
0.5343037326701707,
-0.014459274586663672,
2.31692048955396,
2.1400689678453872,
-0.9468421716760329,
1.3225342455296385,
2.4229762260600483,
1.4561600175543838,
-0.4592924007287594,
0.6389204073366491,
0.5578650700802136,
1.458345504678432,
-0.11931303728638508,
0.04406428844113783,
-0.26144087074192934,
0.6459890964472403,
1.2243061606997268,
0.09770103868863204,
0.23645607828472368,
0.3280111145811677,
0.5390648705936556,
-0.6034481934191781,
-1.1942560416831098,
0.3295059316644399,
-0.0732996553139216,
0.5043071925129845,
0.7399437462813884,
0.3595686242698361,
1.4643835231854074,
-0.1697776944282527,
0.2588380732750666,
0.44311592273109757,
```

-0.6031987466514527,
-0.06131410155451245,
-0.3299395710828798,
-0.45492250592824107,
0.5555861976799495,
1.1612177689175078,
1.833550163228831,
1.5629483318007886,
-0.2455585474033218,
-0.4534768786824923,
-1.2225865843307935,
0.20468879325769473,
1.3398561861341207,
-0.58709602862573,
0.07181500566168061,
0.17010271339345,
0.36406988692952136,
-1.689809967880289,
-1.1677426672173294,
-0.7977758879892671,
0.531032719362766,
-1.7722607960867016,
0.6332033998224597,
0.6669814362912183,
0.218182728116887,
1.0415656581640258,
-1.4544893086294475,
0.5175174925258392,
-0.9742907090362619,
-0.30648595673224416,
1.2400541074634985,
-0.30003161306087445,
-0.14406503395842435,
-0.43061285539013316,
0.6914697219419143,
-0.48545379734819916,
-0.2166375004086177,
1.2014709434123243,
1.0734635496959384,
0.43666289070672293,
-0.9550440807047608,
1.2817732673505913,
0.6254203972471953,
0.8730111716749767,
1.1814458389257443,
-0.5855214911540807,
0.33647224367417283,
0.914891547403245,
-0.10924028322724566,
0.6731472714644756,
-0.8672547313943412,
-1.1628903969680444,
1.2968110405185924,
-0.7478342770444015,
1.0157432273471354,
0.05866260951598884,
0.1549884083832883,
2.402954000164725,
-0.8510405490988947,
-0.13394876321797194,
0.35128800256256376,
0.03985349181110424,
1.0891408951797659,
0.49683980443602116,
1.3618410332836905,
0.31089249400158553,
0.007823999182329802,
-1.0325531315281817,
-0.13635517978577905,
-0.4783006843261583,
-0.9648987725589863,
-0.4094326055257113,
0.0620151507481985,
-0.3631850012398663,
-0.9119123598789223,

```

-0.3458539889181142,
0.8501459643039054,
1.6461875633189593,
0.08799287143090098,
-0.82318299513447,
1.17289644669955,
-0.9455811832961388]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_ric2_sample#4'},
'redundant': { 'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]],
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X '
                                '2 X 3 + 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X '
                                '5 + 1 Z 4 + 1 X 5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Y 0 '
                                                         'Y 1 + 0.5 Z 0 Z 1 + '
                                                         '0.5 X 1 X 2 + 0.5 Y 1 '
                                                         'Y 2 + 0.5 Z 1 Z 2 + '
                                                         '0.5 X 2 X 3 + 0.5 Y 2 '
                                                         'Y 3 + 0.5 Z 2 Z 3 + '
                                                         '0.5 X 3 X 4 + 0.5 Y 3 '
                                                         'Y 4 + 0.5 Z 3 Z 4 + '
                                                         '0.5 X 4 X 5 + 0.5 Y 4 '
                                                         'Y 5 + 0.5 Z 4 Z 5 + '
                                                         '0.5 X 5 X 6 + 0.5 Y 5 '
                                                         'Y 6 + 0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -6.241465922765087],
                             [12, 6, 0, 0, -4.714681794898787]],
            'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                'identity_factors': [0, 0, 0, 0],
                                'noise_level': [4, 2, 0, 0],
                                'noise_profile': { 'noise_on_init_param': {
'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0,
0],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],

```

```
'status': False,  
'value': 0},
```

```
'identity_factors': [1, 1, 0, 0],  
'noise_level': [12, 6, 0, 0],  
'noise_profile': { 'noise_on_init_param': {  
  
    'noise_prob': [ 0.001,  
                    0.001,  
                    0,  
                    0],  
    'status': True,  
    'type': 'depolarizing'},  
    'odd_wires': 4}},  
    'run_time_sec': 5.401231288909912}},  
{ 'config': { 'ansatz': { 'gateset': 1,  
                           'layer': 30,  
                           'ugate': { 'coefficients': { 'bn': [ 0,  
                                                                0,  
                                                                0,  
                                                                0,  
                                                                0,  
                                                                0],  
                                                           'cn': [ 0.5,  
                                                                0.5,  
                                                                0.5,  
                                                                0.5,  
                                                                0.5],  
                                                           'r': 0},  
                           'time': {'max': 10.0, 'min': 0.0},  
                           'type': 'heisenberg'}},  
    'init_param': { 'value': [ 0.05238332857976487,  
                                0.05238332857976488,  
                                0.05238332857976461,  
                                0.5882470284735626,  
                                1.6186245942059996,  
                                2.2470777294271223,  
                                2.8315596913723913,  
                                2.8315596913723913,  
                                2.8333251330604914,  
                                3.7474015252366555,  
                                3.7474015252366555,  
                                3.7474015252366555,  
                                3.7474015252366555,  
                                4.452994588443841,  
                                4.625910007504636,  
                                4.625910007504636,  
                                5.314362007876739,  
                                7.185299991262795,  
                                7.185299991262795,  
                                8.165312898322489,  
                                8.165312898322489,  
                                8.165312898322489,  
                                8.684979613482101,  
                                9.661121823219345,  
                                9.674169964387936,  
                                10.525655511340814,  
                                10.525655511340814,  
                                10.525655511340814,  
                                11.154963299169733,  
                                11.154963299169733,  
                                2.50282655917642,  
                                0.9490947670548783,  
                                0.70056532212722,  
                                -0.9370006834105179,  
                                2.127395945212884,  
                                1.5529950009292526,  
                                -0.5538339242405286,  
                                -0.45109357069990613,  
                                3.2608593101153462,  
                                2.1691098922881027,  
                                0.924351518772188,  
                                0.7799372105219657,  
                                2.588492585292434,  
                                1.4914484649202893,
```

0.5111989894903198,
-1.6854828839595362,
0.709524102178921,
-0.3201993643863164,
0.37756499567776974,
0.11160463007405702,
1.4506751210554825,
-0.1158349928449886,
-0.4172082421780774,
-0.47420587439641587,
2.1318004152619214,
1.5026342770773617,
0.9629412747697549,
-1.017935070811068,
2.2274033566257083,
0.9541631607884562,
2.2865374842341715,
-0.6901760390806929,
1.4488496207691695,
-0.15529093800821045,
1.378510821815928,
-0.1238908488185843,
1.0420944500710088,
1.8938950235205334,
-0.40270933293735856,
-0.4838468076888962,
-0.7769068081513464,
2.0570355933263196,
0.33843722154840217,
-0.21296684820549192,
-0.3087457919151385,
1.6713665097762895,
0.7955513549984788,
-0.5031484152839188,
0.2783608989762586,
1.4322280472670237,
0.12038812465567636,
0.8648150791824706,
0.13552183280667268,
0.07182610915111512,
0.3477791234991623,
-0.1584859230597403,
0.5851519083687279,
0.578565105996425,
-1.7921034519484391,
0.9189192684840749,
-0.6139572052593133,
-0.023527142002935514,
-0.6495330854871428,
0.7767670181361933,
0.13616638059771954,
-0.22568119775616352,
0.4192866974084779,
0.1401009023541104,
1.419355971488301,
1.1103093112149183,
-0.8728413553259085,
1.014691351510677,
2.35757700845783,
1.131891326153365,
-1.2979835358031357,
0.42365777146015526,
1.0982786992551208,
1.698510255572029,
1.2090506157632,
-1.3597097700308411,
-0.9049577551064396,
1.0199036275847497,
2.9541844671388584,
-0.42829772740772204,
0.6542829736223691,
3.7916586259391143,
-0.1983670801457003,
0.7855289833525587,
-0.1534574040841844,

[illegible]

```

'0.5 X 2 X 3 + 0.5 Y 2 '
'Y 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Y 3 '
'Y 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Y 4 '
'Y 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Y 5 '
'Y 6 + 0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -6.220556999324399],
                             [12, 6, 0, 0, -4.682225394272194]],
  'noise_details': [ { 'gates_num': [4, 1, 1, 0],
    'identity_factors': [0, 0, 0, 0],
    'noise_level': [4, 2, 0, 0],
    'noise_profile': { 'noise_on_init_param': {
      'noise_prob': [ 0.001,
                     0.001,
                     0,
                     0],
      'status': True,
      'type': 'depolarizing'},
    'odd_wires': 4},
    { 'gates_num': [12, 3, 1, 0],
      'identity_factors': [1, 1, 0, 0],
      'noise_level': [12, 6, 0, 0],
      'noise_profile': { 'noise_on_init_param': {
        'noise_prob': [ 0.001,
                       0.001,
                       0,
                       0],
        'status': True,
        'type': 'depolarizing'},
        'odd_wires': 4}],
      'run_time_sec': 4.927646160125732}}],
'vqe': [ { 'config': { 'ansatz': { 'gateset': 1,
    'layer': 30,
    'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
      'cn': [ 0.5,
              0.5,
              0.5,
              0.5,
              0.5,
              0.5],
      'r': 0},
      'time': {'max': 10.0, 'min': 0.0},
      'type': 'heisenberg'}},
    'init_param': {'value': 'random'},
    'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
      'noise_prob': [0.001, 0.001, 0, 0],
      'status': True,
      'type': 'depolarizing'},
    'nqubits': 7,
    'observable': { 'coefficients': { 'bn': [ 1.0,
        1.0,
        1.0,
        1.0,
        1.0,
        1.0,
        1.0],
        'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
        'r': 1},
      'def': 'ising'},
    'output': { 'draw': {'fig_dpi': 100, 'status': True, 'type': 'png'},
      'file_name_prefix': 'heisenberg_noise_free_time_evo'},
    'redundant': { 'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
    'run': 'vqe',
    'state': 'dmatrix',
    'vqe': { 'iteration': 10,
      'optimization': { 'algorithm': 'SLSQP',
        'constraint': True,
        'status': True}},
    'zne': { 'data_points': None,

```

```

        'degree': 1,
        'method': 'richardson',
        'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 '
        '+ 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + '
        '1 X 5 X 6 + 1 Z 5 + 1 Z 6',
        'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Y 0 Y 1 + '
        '0.5 Z 0 Z 1 + 0.5 X 1 X 2 + '
        '0.5 Y 1 Y 2 + 0.5 Z 1 Z 2 + '
        '0.5 X 2 X 3 + 0.5 Y 2 Y 3 + '
        '0.5 Z 2 Z 3 + 0.5 X 3 X 4 + '
        '0.5 Y 3 Y 4 + 0.5 Z 3 Z 4 + '
        '0.5 X 4 X 5 + 0.5 Y 4 Y 5 + '
        '0.5 Z 4 Z 5 + 0.5 X 5 X 6 + '
        '0.5 Y 5 Y 6 + 0.5 Z 5 Z 6']],
'output': { 'exact_sol': -8.56677223350563,
        'initial_cost_history': [ 5.796866365657828,
        5.94197444368484,
        5.78867192910188,
        5.996879673940564,
        5.865062503342141,
        5.991390959337936,
        5.72582024636959,
        5.872963620288267,
        5.948869313039893,
        5.964755550858089],
        'noise_details': { 'gates_num': [4, 1, 1, 0],
        'identity_factors': [0, 0, 0, 0],
        'noise_level': [4, 2, 0, 0],
        'noise_profile': { 'noise_on_init_param': { 'status':
        'value':
        'noise_prob': [0.001, 0.001, 0, 0],
        'status': True,
        'type': 'depolarizing'},
        'odd_wires': 4},
        'optimized_minimum_cost': [ -5.966837375471974,
        -6.220556999324403,
        -6.218690041461848,
        -6.241465922765082,
        -5.8146764847083485,
        -6.144691518305702,
        -5.966791417158719,
        -5.743389283741366,
        -5.966171999363111,
        -6.281301957226982],
        'optimized_parameters': [ [ 0.43940602753254654,
        0.4524980474487914,
        0.6462847510297367,
        0.6535145711164267,
        0.7019965052930044,
        1.0830061860275473,
        1.4380021973026182,
        2.1860837771467474,
        2.1860837771467474,
        2.1860837771467474,
        2.4875804812430995,
        2.911818356328843,
        4.9134595358452735,
        5.043089843653786,
        5.994354774193697,
        6.658321281734746,
        6.666230857319696,
        8.458454027735183,
        8.458454027735183,
        8.458454027735183,
        8.938931772570164,
        8.938931772570164,
        9.876451730548837,
        9.876451730548837,
        10.140618079102744,
        10.453495471248813,
        10.878653713196195,
        11.282429749250388,

```

False,
0},

11.726474998572407,
11.93229290545428,
2.9640535214497064,
-0.6206168106317247,
-0.4035747848788786,
-2.067222048924027,
2.0576259596946604,
1.364315765000901,
-0.1480698866383175,
-1.95043906046908,
0.7897528850889669,
1.350582507885341,
-0.3349044763190947,
-2.3429833820345465,
2.04351680973891,
0.8083219796841106,
-0.06520770630024467,
-0.8776318722364843,
2.4862506381076748,
0.8345121096737285,
-0.03071661620417745,
-0.6225183255138825,
0.7450777062518694,
0.12619297185303674,
0.15620862918286052,
-0.47540100870445906,
0.6345315438244892,
1.451076347224231,
0.10985833383721234,
-1.6475696409761669,
1.1918822544398533,
1.3235539868194213,
-0.24222060474147966,
0.2418944215339802,
1.0397651583041008,
1.6773252160624976,
0.16975786500059598,
-0.9211027317156172,
0.7590471244619662,
1.76540125315151,
2.43856734239768,
-0.19488067618654012,
0.8462629124439717,
0.9979674075340036,
2.5775834810620677,
0.5650956422366618,
-0.5475934153287456,
-0.0253449469292375,
1.2524069066350432,
-0.605696708865831,
0.7019502861337156,
0.2803153244338447,
-1.0831925759777479,
1.0082140101851143,
-0.3762046176346905,
-0.9535933533256993,
-1.0422804614435257,
0.13948594332192887,
0.5835070530053111,
0.16086524965091817,
0.06403444071667018,
-0.06194188553710435,
-0.4026223625881385,
0.2750028468648149,
0.6789909068622377,
0.05162849323802493,
-0.7735332926339995,
-0.14862918062202837,
0.7661757453229759,
-0.16119730427373616,
0.6482389226852195,
3.1459456036848104,
-0.4760920139337192,
1.6197254159790668,
-0.5506328506738539,

0.9968299487743895,
-0.3106735286083523,
-1.2388662013594425,
0.8898580979157581,
0.43256773745092003,
-0.7272220041247079,
0.8716071923554094,
1.5193589139179793,
0.9787540447681606,
-1.4044929205930872,
-0.1984876316951887,
-0.46744984056178585,
-0.7765225748074835,
0.04344696088727837,
0.32373965212791267,
0.9049555024172572,
0.7139278873651953,
-0.8122282267372439,
0.5323405298774778,
-0.08437719753251166,
-0.14277007957413837,
0.7912147308716768,
0.26055493673650715,
0.11357082745857092,
0.3153698930365579,
3.1981441448638614,
0.32109035951393206,
-0.23017443442945684,
0.0316864124540797,
0.20875812581978664,
0.5466872162452653,
1.2918521189433907,
-0.7312547980705957,
1.4958457475939604,
-0.053193686203739404,
-0.4437772137226961,
-0.5271126400646005,
2.5136810835026426,
0.6674244883024821,
-1.0140291771816496,
-1.177197116527103,
2.058315434744553,
0.050372102721390984,
-0.09022974351195023,
-2.296438941834702,
0.3261688389091777,
0.3072223897800251],
[0.05238332857976487,
0.05238332857976488,
0.05238332857976461,
0.5882470284735626,
1.6186245942059996,
2.2470777294271223,
2.8315596913723913,
2.8315596913723913,
2.8333251330604914,
3.7474015252366555,
3.7474015252366555,
3.7474015252366555,
3.7474015252366555,
4.452994588443841,
4.625910007504636,
4.625910007504636,
5.314362007876739,
7.185299991262795,
7.185299991262795,
8.165312898322489,
8.165312898322489,
8.165312898322489,
8.684979613482101,
9.661121823219345,
9.674169964387936,
10.525655511340814,
10.525655511340814,
10.525655511340814,

11.154963299169733,
11.154963299169733,
2.50282655917642,
0.9490947670548783,
0.70056532212722,
-0.9370006834105179,
2.127395945212884,
1.5529950009292526,
-0.5538339242405286,
-0.45109357069990613,
3.2608593101153462,
2.1691098922881027,
0.924351518772188,
0.7799372105219657,
2.588492585292434,
1.4914484649202893,
0.5111989894903198,
-1.6854828839595362,
0.709524102178921,
-0.3201993643863164,
0.37756499567776974,
0.11160463007405702,
1.4506751210554825,
-0.1158349928449886,
-0.4172082421780774,
-0.47420587439641587,
2.1318004152619214,
1.5026342770773617,
0.9629412747697549,
-1.017935070811068,
2.2274033566257083,
0.9541631607884562,
2.2865374842341715,
-0.6901760390806929,
1.4488496207691695,
-0.15529093800821045,
1.378510821815928,
-0.1238908488185843,
1.0420944500710088,
1.8938950235205334,
-0.40270933293735856,
-0.4838468076888962,
-0.7769068081513464,
2.0570355933263196,
0.33843722154840217,
-0.21296684820549192,
-0.3087457919151385,
1.6713665097762895,
0.7955513549984788,
-0.5031484152839188,
0.2783608989762586,
1.4322280472670237,
0.12038812465567636,
0.8648150791824706,
0.13552183280667268,
0.07182610915111512,
0.3477791234991623,
-0.1584859230597403,
0.5851519083687279,
0.578565105996425,
-1.7921034519484391,
0.9189192684840749,
-0.6139572052593133,
-0.023527142002935514,
-0.6495330854871428,
0.7767670181361933,
0.13616638059771954,
-0.22568119775616352,
0.4192866974084779,
0.1401009023541104,
1.419355971488301,
1.1103093112149183,
-0.8728413553259085,
1.014691351510677,
2.35757700845783,

1.131891326153365,
-1.2979835358031357,
0.42365777146015526,
1.0982786992551208,
1.698510255572029,
1.2090506157632,
-1.3597097700308411,
-0.9049577551064396,
1.0199036275847497,
2.9541844671388584,
-0.42829772740772204,
0.6542829736223691,
3.7916586259391143,
-0.1983670801457003,
0.7855289833525587,
-0.1534574040841844,
1.809122493112309,
0.35293394466185096,
1.0573849517470935,
2.461665111905985,
1.0784030177718114,
-1.585709677044719,
0.8985620342798133,
2.2360691692951646,
0.24022620726925167,
2.6922129347449117,
0.15599094623602128,
2.799936411115923,
0.9657316063780467,
-0.2732772147086647,
-1.448572189075712,
-0.7280589250410919,
0.057973620624421296,
0.9321748335747359,
-1.4338621772181848,
1.3976338029473028,
-0.25504403617708243,
3.210257057781798,
0.03448775329128267,
0.05070056715723011,
-1.3727823101385919,
0.8243579770218447,
0.13142013414460954,
-0.4933237632531999,
1.1105137227416793,
0.7155364379215329,
-0.21063252058278284],
[0.6189603325174557,
0.7142372255934472,
0.818217611349809,
1.1433975540074377,
1.9121137997887092,
1.912113799788709,
1.9189666863881127,
2.194219841514053,
2.734492156740232,
4.250258815642276,
4.250258815642276,
5.774076859791501,
6.527612880112256,
6.532834731882501,
8.049613322901676,
8.049613322901676,
8.050730377555883,
8.561643142317116,
8.561643142317116,
8.758357012835646,
8.758357012835646,
8.78353884989931,
9.477076383056275,
9.536600626467342,
9.536600626467342,
9.536600626467342,
10.241295505019522,
10.318321333031077,

10.579129354925781,
10.82886994059623,
3.662494394083504,
0.08309129295248968,
1.5295376506525808,
-1.5964000102221196,
1.8988970610647178,
-0.07218621526676876,
0.7238532160702689,
-1.2917423920622229,
1.8532222971765195,
-0.40818119715800666,
0.469598065996274,
-1.2652675561295137,
0.41465882817859046,
0.48683539382624846,
-0.15336518315146752,
-1.6419783820068856,
1.2136622236743368,
0.5933123377118196,
0.9666225223219593,
0.9539776194708611,
-0.3793986531891993,
-0.2570964298405315,
1.5884302353807402,
0.19867970725527498,
1.0036205245072367,
-1.7266220384949087,
0.6519864384701275,
0.173373648086925,
1.8697046853700252,
0.5263944462203128,
2.9785129063723943,
-0.5937256494260224,
0.22753174799850245,
0.24918335157216837,
1.5438671949833176,
-0.2981477150404249,
-0.0503664733834821,
0.542442835193611,
1.8096859498837612,
-2.2260935596321993,
0.28693396030012636,
0.9986781622414097,
-0.01940263773226835,
-1.8805315707929653,
-0.1513202473661221,
0.1836836320118686,
-0.5655439481125303,
-0.30323205019871646,
-0.4186198812206128,
0.2357002564872809,
2.8296979465953833,
-1.0776781071762758,
-0.08378558447273823,
1.7446476514004516,
-0.4606832586075214,
-1.762173446350097,
0.10241780783627007,
1.6018254886002428,
0.2621004786967086,
1.349261138919249,
0.14987623247602613,
0.5068543254371847,
-1.3474781551208477,
1.0890436270679893,
1.6662048029626129,
0.3608718991529565,
-0.11137748070221386,
0.5641204379289552,
2.2546014997074266,
1.8888740160956543,
-0.23345107582518032,
-1.5772444286238798,
2.1408383357361265,

-0.4968079560788485,
0.1533667959779154,
-0.40238715350257875,
0.25591729793510365,
2.947382531044255,
-0.32162602471486745,
-0.7587223675916386,
-0.8983111566200868,
0.006114119281372181,
2.9372824954589514,
0.48288231335139287,
0.7939297064609614,
1.9454360656955985,
2.092519500518508,
0.8995449331251366,
0.10171517677156572,
0.45235144869603666,
1.02501950409702,
0.24331156538914245,
0.48973195386607005,
2.672871435464254,
1.313352678455905,
2.010046934709317,
-0.42153209835304317,
-0.13436700176919486,
-0.6950305532982513,
0.6026553555930264,
1.6681913778840949,
0.4125109526933123,
0.514533549588801,
1.0794873063708696,
-0.9571806916209341,
0.27630815458061647,
0.6001387806934945,
0.5059087494947612,
-0.5722342732268321,
0.3986919058890637,
1.0044202485382987,
0.3352295947729849,
1.3777059541173544,
-2.835684252944388,
0.6104690716589696,
1.838086538234438,
-0.3106134411398272,
-2.2167272073692903,
0.10706372359431235,
-0.17469850626418476],
[-2.397634216504411e-17,
1.429450133825379e-17,
0.9192297980258759,
0.9192297980258759,
2.8984599305110854,
3.1993874679766523,
3.199387467976652,
3.715279848679994,
3.715919998165357,
5.126008182976459,
5.749875525815113,
5.785995839176617,
5.785995839176617,
5.785995839176617,
5.785995839176617,
6.315437962379428,
6.678845841951143,
7.3578167224936415,
7.415694801781823,
7.531263738212295,
7.949087697575632,
7.949087697575632,
8.111169110068786,
9.407094598577077,
9.495286031274873,
9.495286031274873,
10.993257661103009,
11.063439993331317,

11.639888600691357,
11.75175692668691,
2.1152038588685538,
-0.7839988468841306,
-1.0579861702584952,
-0.5122509123294313,
1.0012369417099964,
0.9275037440003397,
0.5343037326701707,
-0.014459274586663672,
2.31692048955396,
2.1400689678453872,
-0.9468421716760329,
1.3225342455296385,
2.4229762260600483,
1.4561600175543838,
-0.4592924007287594,
0.6389204073366491,
0.5578650700802136,
1.458345504678432,
-0.11931303728638508,
0.04406428844113783,
-0.26144087074192934,
0.6459890964472403,
1.2243061606997268,
0.09770103868863204,
0.23645607828472368,
0.3280111145811677,
0.5390648705936556,
-0.6034481934191781,
-1.1942560416831098,
0.3295059316644399,
-0.0732996553139216,
0.5043071925129845,
0.7399437462813884,
0.3595686242698361,
1.4643835231854074,
-0.1697776944282527,
0.2588380732750666,
0.44311592273109757,
-0.6031987466514527,
-0.06131410155451245,
-0.3299395710828798,
-0.45492250592824107,
0.5555861976799495,
1.1612177689175078,
1.833550163228831,
1.5629483318007886,
-0.2455585474033218,
-0.4534768786824923,
-1.2225865843307935,
0.20468879325769473,
1.3398561861341207,
-0.58709602862573,
0.07181500566168061,
0.17010271339345,
0.36406988692952136,
-1.689809967880289,
-1.1677426672173294,
-0.7977758879892671,
0.531032719362766,
-1.7722607960867016,
0.6332033998224597,
0.6669814362912183,
0.218182728116887,
1.0415656581640258,
-1.4544893086294475,
0.5175174925258392,
-0.9742907090362619,
-0.30648595673224416,
1.2400541074634985,
-0.30003161306087445,
-0.14406503395842435,
-0.43061285539013316,
0.6914697219419143,

-0.48545379734819916,
-0.2166375004086177,
1.2014709434123243,
1.0734635496959384,
0.43666289070672293,
-0.9550440807047608,
1.2817732673505913,
0.6254203972471953,
0.8730111716749767,
1.1814458389257443,
-0.5855214911540807,
0.33647224367417283,
0.914891547403245,
-0.10924028322724566,
0.6731472714644756,
-0.8672547313943412,
-1.1628903969680444,
1.2968110405185924,
-0.7478342770444015,
1.0157432273471354,
0.05866260951598884,
0.1549884083832883,
2.402954000164725,
-0.8510405490988947,
-0.13394876321797194,
0.35128800256256376,
0.03985349181110424,
1.0891408951797659,
0.49683980443602116,
1.3618410332836905,
0.31089249400158553,
0.007823999182329802,
-1.0325531315281817,
-0.13635517978577905,
-0.4783006843261583,
-0.9648987725589863,
-0.4094326055257113,
0.0620151507481985,
-0.3631850012398663,
-0.9119123598789223,
-0.3458539889181142,
0.8501459643039054,
1.6461875633189593,
0.08799287143090098,
-0.82318299513447,
1.17289644669955,
-0.9455811832961388],
[0.12420123662801165,
0.138246244841176,
0.34185010800102933,
0.8771280515547424,
0.8771280515547423,
1.9326762312100456,
1.9326762312100456,
2.879806034127527,
3.4467495190209787,
3.460588109395402,
3.4605881093954016,
3.4605881093954016,
3.911934955717517,
3.9564836804547703,
4.188240551814426,
5.098640553346409,
5.098640553346409,
5.098640553346408,
6.466332587579958,
7.457565848827182,
7.712724508543908,
7.890302861063261,
7.898577524812402,
8.020037295803547,
8.020037295803547,
9.289723192898348,
9.68251578824504,
9.68251578824504,

10.66694519433398,
11.464898013549973,
1.8095532965205083,
-1.6633278493215808,
-1.2281091611401596,
-0.40155726479865544,
1.7903243850816337,
0.18661159996017582,
1.2217850039169107,
-0.48128156078104645,
1.438546033666569,
-0.9466383819139143,
-0.2179227006066069,
-0.07921201994836616,
0.6692914930417907,
-0.16791294486623565,
-0.3278463005181308,
1.2883193159418245,
1.9001761172388045,
0.7646920019853412,
-0.6325477821570596,
-0.5987552945744136,
0.5669730115381587,
0.5983941343774113,
-0.06179171362000896,
-0.6867235859693906,
0.5211737756628548,
-0.19354588250953578,
0.6072816218147872,
0.3472677197742769,
-0.08031734075212016,
1.4024126217748347,
-0.5867181790012919,
0.809620813948269,
0.35997425529443994,
-0.8722967249300235,
-0.5874818244923429,
-0.5567926036837068,
-0.6652314874983639,
-0.6042054992483096,
-0.13918430829641074,
-0.08277488470584692,
-0.4972293169762719,
-0.4421668735352916,
0.6160601456858428,
-1.1972657609969646,
-1.1558488364748807,
-1.4501782622641872,
0.03989088007781949,
-0.8684923327267209,
-0.5569704341339503,
0.15823054121075,
0.6345578215095505,
0.3601853745371903,
-0.42594808878819745,
0.08979804115834313,
-0.153043281560756,
-0.603228229875778,
-1.1264410002736531,
0.6895580235946861,
-0.11045941959152497,
0.5174109375139814,
1.3269460586908306,
-0.3852298542423827,
-2.4120095755015893,
0.08807412161753436,
0.1228551365077638,
-0.6982591234727157,
-1.1618488040328845,
2.5586909942385017,
-0.24921574033555105,
-1.9870969070050777,
-1.0279498295884981,
-0.7796802357622239,
0.48428681889730063,

0.7162815161060118,
-0.15547297036103372,
1.0522591887013215,
1.772822944479854,
-0.2654037211186137,
-0.2720110866721671,
-0.03929478442656517,
0.34606077605631835,
0.5533331494282367,
1.0574846609095763,
-0.5747611578732846,
0.4916772893793279,
0.9875709691916216,
-0.17857479553023695,
-1.147281220275154,
0.37370382177065037,
2.456815155407461,
0.6567596591037266,
0.9507316292553318,
1.5113928877114242,
1.8380989093110256,
-0.1736919900840187,
-1.2666253947041328,
-0.554819304070435,
1.1931643666366285,
0.4614862178419797,
1.2040726557184918,
0.3083618234293392,
-0.4066692475369416,
0.3740368412278018,
-1.684611974494111,
-0.8650016725760176,
-1.4838330341878547,
-0.14344887350922086,
4.203596189627334,
1.1133202614474196,
1.391124117957397,
1.8627332105499828,
-0.28192336755973296,
-1.5416482647725873,
-1.9612892990030817,
0.8836928087418303,
0.300761821046295,
0.6785848024508535,
-0.5500277556546446,
0.5742776326882599,
-1.181756808209963],
[0.38848003251483587,
0.3884800325148359,
1.2841735835398218,
1.284173583539822,
1.2841735835398218,
1.818138220668234,
1.818138220668234,
2.7575469841005327,
2.7575469841005327,
2.7575469841005327,
3.0053468968599275,
3.0053468968599275,
3.005346896859927,
4.184079661691832,
4.83635165164968,
4.83635165164968,
5.97926077140412,
6.458805266455043,
6.513102391504654,
6.513102391504654,
7.122105903846713,
7.2188422331450415,
7.988590448127284,
8.268095077127937,
8.268095077127937,
9.164456488648115,
9.164456488648115,
9.782953992889732,

9.782953992889732,
10.467486029517685,
1.5496959718296535,
-1.0300437645832343,
0.425343365508412,
-1.7136674729251273,
0.33645197441382346,
0.2288357961575219,
0.10470230685228102,
-0.6612661793744108,
0.11823315872480306,
0.48571962162230753,
-1.3748501535311368,
0.2563120795974374,
-0.38871520150353267,
-0.18786175076151226,
1.0017996241286973,
0.05598903629217323,
1.501328526357405,
-0.32299727271195017,
0.5715295079657807,
-0.12201961934486968,
2.2747823054657927,
0.16289086097520028,
0.027619518366160244,
-0.10165606616500937,
-0.06321979816820887,
-0.27543819802246033,
-0.0048370064103128475,
0.1330138167350915,
-0.06428151371988705,
0.21905201010909556,
0.25697175483980955,
-1.462905966720411,
0.8014511304369155,
3.456137165485204,
-0.3560002683832885,
-0.4886877133354182,
0.32039145066707037,
1.6153352350920127,
0.5940099487759896,
0.18140087044059483,
0.27123106498617,
0.7444341922951362,
0.170480557625751,
-0.4688462519246396,
0.5235839184929864,
-0.5470964469389679,
-0.011921500492307424,
1.4116230981753706,
0.49972453659747124,
-0.7800673920851233,
0.4234227404504273,
-0.2071635265929523,
0.14064328933496192,
-1.2007096247825992,
0.12047098237640114,
-0.4125103841815656,
1.0447620953563077,
0.0020814149928196773,
0.4988991062112368,
-0.34495773827123155,
-0.12849063482979717,
0.4280934542561602,
0.7138444629946918,
-0.6497712710787925,
-0.43652659674625793,
0.917432486791049,
-0.8523477361785018,
0.5648987974980545,
0.3183226113969618,
0.14295784675778261,
0.9589824340171335,
0.9194478619356266,
1.1001359353915339,

1.5417377267389083,
-0.5376003771957187,
-0.5404528215706331,
0.21624687748210675,
0.8814937279455825,
-0.371639532103812,
-0.2668100917972658,
1.3133163210428074,
0.44574090697479807,
-1.0940261852319737,
-0.313830643067749,
-0.06100275437545863,
-0.049560783686653447,
-0.3025056724120473,
-0.9891529613817422,
-0.35099332176432035,
0.5436069706516342,
-0.722828417704955,
1.6968661444846564,
-2.4402971962208975,
0.7865745542588627,
1.3437881169309889,
0.36108471326554326,
-0.1749159211919777,
-0.014334278530919132,
0.02978377629472008,
1.8533978555002608,
-1.8470426695738211,
0.6272356375919823,
0.44775499409400255,
1.3281153539688288,
0.4583217606753929,
0.6443670392327268,
0.5254233581973354,
2.3087652110195127,
-0.7761319091913014,
-0.7605421959233766,
-0.6461580464596497,
0.7075597766411752,
0.985778508379752,
-1.3130974173457381,
2.8711331564400684,
1.2993543284582694,
1.9493614761217355,
1.1647493000584068,
-0.42731613935458074,
2.7038346000669766],
[-1.8784742854218807e-17,
-7.763750530158576e-17,
0.3802827375258553,
0.44096449652630626,
0.8040692975576897,
1.9224694434519336,
1.9224694434519338,
1.9224694434519336,
1.9224694434519336,
1.9224694434519336,
1.9224694434519338,
2.730288811725591,
4.476350841356259,
4.476350841356259,
4.476350841356259,
6.331658137341824,
6.331658137341824,
6.7621011201584285,
6.77851096220289,
7.490043179529316,
7.490043179529316,
7.490043179529316,
7.490043179529316,
7.511664240193972,
8.302259372586072,
8.36899822047307,
8.607405767895836,
9.631163052898822,

9.874876363637043,
10.225975825453373,
2.711573403856553,
0.20239905137011172,
-0.4220883549830381,
-0.8404830976891227,
1.794332340210785,
1.5095243517856047,
0.22326098118939897,
0.07733371735403745,
1.280548885023134,
1.4121496210624682,
0.9815935680907694,
-2.975348552150871,
1.6316643284140941,
0.8239502945369961,
0.0314505340316768,
-1.0892891749800817,
1.720694113273713,
0.6634696927679957,
-0.836512780686814,
-1.1286873737196692,
-0.09500137106776224,
1.0428135848317974,
-0.4828772957450712,
-1.129334172148368,
0.7253737318795237,
1.4108675361353635,
0.7588540034470418,
-0.5039485441412638,
0.10346857579652805,
1.2668095290766983,
0.7994757345637048,
-0.7393240316081237,
-0.7818304990031031,
0.25375980918776664,
2.228975199542585,
2.2487625387645873,
0.6384105819815235,
0.39434544396871446,
1.3679129840939837,
-0.27336699910726836,
1.3170761670781417,
1.1159622861844822,
1.176824183488981,
0.40583799697229034,
0.8786168217534174,
0.2914083315983707,
-0.8464697846173548,
-0.032455291822620566,
-2.0755386823554396,
-0.11439663923099666,
0.7312672791487281,
0.4263196833485598,
0.11802610051350722,
1.0445881812565523,
-1.9566462184988007,
0.9492303984073598,
-1.7129593951518396,
-0.3230601534209437,
0.3161713641922788,
-0.4811213940350071,
0.08327140785409295,
0.37526634808329173,
-1.9990814349080968,
-0.15397700748904192,
-0.9549084255901845,
1.1120455683467105,
-0.42952128451280097,
0.17929609797141985,
1.0446016069356951,
0.846774755857218,
-0.2784586201121851,
0.23000407229986752,
-0.9198047598183884,

0.996529675185623,
0.34608382599103005,
0.5099375110708237,
2.1948932092248885,
1.3625694142279887,
-0.9306056124011278,
0.6012804875871011,
-0.7788968212470156,
-1.5884735532174328,
1.0361309111992296,
0.015937704739038126,
1.4003797043998856,
2.429422654005534,
-0.27504014116509645,
0.8256449732433587,
1.2146087003914199,
-0.6435553984602482,
0.3046269503778821,
1.0182549333954134,
1.4202224852266074,
1.3511302325707892,
-0.7261672378416364,
1.5992752660640235,
-0.5764295578986515,
-0.8318409785507642,
1.1375736468462807,
0.8529850561064739,
0.9553972898831558,
-0.09161160854677679,
1.617622889574723,
1.4551513261723394,
-0.08817813881493969,
0.8045852859105778,
1.9892741470924016,
1.8347816167556374,
-0.8050652823779443,
0.008511392950551585,
2.1460096989270747,
0.49277581514942953,
-2.4311684786278165,
0.5134791015266132,
0.9751974520662461,
-0.38582649358518695,
0.7746227771853464,
1.7307944301597258,
2.1320333699599043,
1.1654652968447545],
[0.4022850038273885,
0.422288083135141,
0.7644599371513524,
0.7900179023917507,
1.1026508993249646,
1.1630326759467198,
1.1630326759467198,
1.2119340062926192,
1.9859325565890973,
1.985932556589097,
2.096166401632718,
3.378647998605865,
3.378647998605865,
3.378647998605865,
4.354848115861854,
6.006218342726225,
6.006218342726225,
6.009974297352436,
8.225327865566033,
8.242093148909708,
9.222354355246733,
9.369468664000962,
9.432712219232048,
9.692289963799956,
9.692829942577658,
10.391360826688166,
10.391360826688166,
10.529867035412478,

10.529867035412478,
11.06290493838992,
2.9062307172979733,
-1.7788694936155562,
-0.1145266982224492,
-0.9311559923392093,
0.9691998209476543,
1.7463479137973288,
-0.7663862807329105,
-0.9694574896037749,
2.8979112386638413,
1.1400097878698292,
0.8096361549155084,
-1.5900494197653285,
0.09829667109053167,
0.31839493135320834,
0.3886777840539921,
0.12229184789371997,
1.7250225844898464,
1.0065795676456808,
1.4535641205083867,
-0.3486770669120958,
0.2556790893233589,
1.778531141883236,
0.269972269049403,
1.0155903957099945,
0.8041163886037216,
0.5381821926966456,
-0.30108604163317454,
0.6396533041678052,
2.1819991440037696,
1.4978086658278347,
2.5181040929106753,
0.4624415671193703,
-0.17701736987873673,
0.6910098122382194,
2.947172461882987,
-1.0727539032453073,
0.4023854197196741,
1.650490750838525,
0.9313998963798695,
-0.6769162055305248,
-0.5620785927454173,
-0.20885673940707247,
2.5019757090972443,
-0.37561416876024456,
-0.9329392032282731,
0.7818961506889994,
0.44350434198669075,
-0.10644929099379355,
-2.498928032568615,
-1.2046085786256906,
-0.06365416567297402,
0.17257236645054272,
1.2516209341031546,
-0.3591527663232026,
0.46789807326225935,
0.24374638092155315,
0.02110383811013195,
0.12714835746845662,
-0.5505736947218767,
-0.03497727012222505,
-2.2168334497478877,
-0.21485512988314215,
1.7552902914988067,
0.6189795903733311,
0.3191491062237487,
0.522632432692797,
-1.1121089236913335,
1.1371618962620422,
-0.2410043783817164,
0.24168543661014927,
0.39418565996934685,
1.2107667809264395,
2.398781467533486,

0.6948257569073976,
0.6170229142231725,
0.7050419245924102,
1.768546262611762,
1.5562956304120967,
-0.4269237214365518,
0.6302835616447369,
1.7710848704009405,
1.2281100264507194,
1.5036530549577243,
-0.7435268012971183,
0.7613541379199313,
0.8523298100708424,
-0.6655959646469325,
1.1543911681084875,
2.2675628634479494,
1.0299985170060308,
1.400963430022522,
0.27759304330136975,
0.6960561589746208,
-0.4210335016880339,
0.0336463669485127,
-2.099660094290342,
0.6674819238895001,
0.5049668435179431,
0.48001405045948287,
1.3574871122533543,
-2.2986993170736225,
9.601426007351498e-05,
1.3312470798272018,
0.7582238004326416,
-0.35033967277151296,
-0.4108552112493075,
0.1982910129710702,
0.6700017270602091,
0.26559360983179486,
-0.14563210230012705,
-0.39610135234418875,
1.1727729846793147,
0.6713573564879365,
0.8527070052929181,
0.8268283343075853,
2.2918575835381745,
0.5082103460913493,
1.8600493489550152,
0.49576811175353974,
2.232387985311215],
[0.16477895719846114,
0.16477895719846108,
0.7183189156962347,
1.8697093861419474,
2.105082615571597,
3.05308344502201,
4.199393321043376,
4.199393321043376,
4.199393321043375,
6.490210948425593,
6.4902109484255925,
6.490210948425593,
6.4902109484255925,
6.4902109484255925,
6.490210948425593,
6.882554174728105,
6.882554174728105,
6.882554174728105,
6.882554174728104,
6.8877583035732775,
6.8877583035732775,
6.8877583035732775,
8.21014824639056,
8.21014824639056,
9.071469420716642,
9.120928605341893,
9.714944098401723,
11.474131735950095,

11.474131735950095,
11.794551582273902,
1.341188628911811,
1.6554894539990426,
-2.1068974621953425,
-1.4171475675653578,
0.33211669069513333,
0.11331792925356984,
-1.3610911099963097,
0.46439741779020816,
0.9313501387866817,
1.0938593094543279,
-0.3591727738608774,
-0.4732622270970327,
1.6321598874818157,
-0.2782091348514778,
-1.0526781349295347,
0.0708703831522549,
2.6982701224663113,
-0.038867313655444914,
-0.10508583152652556,
-0.40759365792756236,
0.6284849004330764,
0.5192751929971827,
-1.6282040637266806,
0.0014422433007245106,
-1.8299719264591412,
2.3182597876656823,
0.8875167319646391,
1.5664268415036142,
1.5052778791259855,
2.935955436800565,
1.533101581959507,
1.8091829332960188,
0.45080966818917967,
0.1506883627647002,
-0.6843548866459184,
-0.5544883854492364,
1.2466562002431278,
-1.1866110136199353,
1.4521358231995756,
-1.125608676420736,
-0.24054383618816966,
-0.058272145415006006,
0.410408128597049,
-1.6441753460137252,
0.4019816715685168,
0.4588365987304068,
1.354346078964423,
-1.7153251360741875,
-1.1039978646265602e-06,
0.7940670488104785,
0.17979772520560053,
-0.24385394325110826,
0.6367160741723578,
0.8500459957371875,
0.0559416907316065,
0.32418668587130467,
-1.4797636873635434,
0.11902177942387202,
0.5185165439317966,
-0.08399222686641401,
0.8650918037299421,
1.4911587670088011,
0.12792248933645675,
0.3372262149750348,
-0.6553154954729696,
1.1995792367396254,
0.5132294030998703,
0.44835115860783636,
-0.37116980512777586,
0.7805194563374558,
-0.854334789795377,
0.5008527672922033,
-0.2302685629030402,

0.6145182391921147,
-0.6575500079215796,
0.8657035637529482,
1.1874516653159721,
-0.16284802367068918,
-0.26462171656665856,
-0.9815102056659819,
0.23198397744948207,
-1.087097195512072,
0.6616385654966559,
-0.2929637257774683,
0.03471390505932934,
0.2048751570049573,
0.9170017118070749,
0.39617956766989404,
0.12006199846995962,
0.1922705913940127,
0.8784478594386566,
0.055479957240883566,
-0.6634105431569992,
1.0742951419164057,
-0.37818200958777604,
0.28548758229245597,
-0.6594827056613419,
1.0235832498557866,
-0.19428253177041227,
-0.5531991468121416,
-1.189626464718852,
0.2519110119226656,
-0.6708462316627377,
2.7994352657547057,
-3.8764436680175813,
0.5152327138619494,
-0.3749347648847944,
2.9042167722097862,
1.1265965037549552,
-1.850959186134046,
2.165374611027145,
0.8818414791702859,
1.5039524911129019,
0.06564767423285973,
0.6486215314149069,
-0.08854143781823305,
1.8396223528529987,
-0.00792246728000446,
2.8805879469479487,
3.6337353861266473],
[0.11853625011402234,
0.11853625011402236,
1.1173235551611367,
1.7342580765110573,
1.7342580765110573,
2.108144483620023,
3.054129947242328,
3.054129947242328,
3.054129947242328,
3.056306360633301,
4.638414750363569,
5.0621851887275495,
5.7503913724004905,
5.774998407238176,
6.917933224474368,
6.917933224474368,
8.061549209526286,
8.061549209526286,
8.061549209526286,
8.591251503132888,
8.591251503132888,
9.051235537435975,
9.081856762230144,
9.467876152039107,
9.545932016874406,
9.591419209848118,
10.465616352275608,
10.465616352275608,

10.465735173309618,
10.980942228105905,
2.0734674059611633,
0.32892201175743896,
-0.007356440825095051,
0.7856576253246221,
2.089457548986907,
0.7744884921126718,
-0.35049309306395837,
-0.6556831494452968,
2.0369525201874903,
0.4739652080570001,
-0.8564345639983049,
1.25716179603495,
1.7181260437331554,
2.3978160665502384,
1.1152024575020885,
-1.3335737674295711,
0.016650799076572978,
1.6607129762234538,
0.648059834763191,
0.2438725527060928,
-0.27307896000752346,
1.4369017198677916,
3.5672448343800007,
-0.5379702113035718,
-0.7420664562717484,
-0.529699891930992,
0.7301471483152846,
0.781913925757813,
0.6004078048346045,
1.7324865479818374,
0.4548236607944829,
-1.4419813608219112,
-0.684662976785234,
0.8455512814500156,
1.97751114408285,
-0.833202517811791,
1.110524460863052,
2.6403320691876733,
-0.3129427957408566,
-0.06674204626492235,
0.5842857778318762,
-0.21132485649138796,
-0.011842019288790376,
-0.25743970610802486,
0.7486616881233721,
-0.1302029141156864,
0.09497091477412227,
-0.3715265429351211,
0.39570077921274605,
0.28643868832003017,
0.7848413456471106,
-0.7157146817058562,
0.09411764456438232,
-0.10910076647953601,
1.1262761892344808,
-0.27334997823872975,
-0.44027652197942063,
0.6740415100156434,
-0.336256645457715,
0.4268157265313067,
-0.8611696251566416,
-0.34560928156726306,
0.5618074077714486,
0.5866360379174992,
1.8662876827821717,
2.21592310898005,
1.972223656383108,
1.6141187926643297,
0.856255815565121,
-0.34275929095614754,
-1.0642528745182613,
-1.0618686751734654,
0.2379582550153795,

```

-0.6501770361983872,
-0.19408645485522444,
1.501330829125914,
1.5726602727591479,
0.7066066097903853,
-1.4580936665368467,
0.46940440748349077,
-0.1287015677341077,
-0.9805925033056946,
-0.3754741669636888,
-0.026817919501009202,
1.1067346586651607,
0.17547923725022047,
0.8902143171896091,
-0.22777698610446503,
0.8082253583749343,
-1.0674083601412958,
-0.022706971436593337,
0.34227899244110566,
1.3527736523760083,
2.0767841716125894,
0.5778568145668401,
-0.6555502504392442,
1.2670274746873995,
0.507694022232518,
-0.2629991187543805,
1.7048253080726326,
1.4088065522309612,
2.823918413203317,
1.2763232744235296,
-1.4348344213353486,
-0.605496445996402,
0.09645158262695042,
0.024639736182597342,
0.23997550724405606,
0.9979895378690888,
-0.40612123795993227,
1.826790650755685,
0.029599395583551916,
-0.4592861834683686,
0.0328358806035505,
1.3612120706287358,
0.8210402794880411,
0.05688164146575431,
0.03152531580318861,
2.0316547976974904,
2.795039607579165]],
'run_time_sec': 69142.54766130447}}],
'zne': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}}},
'init_param': { 'value': [ 0.16477895719846114,
0.16477895719846108,
0.7183189156962347,
1.8697093861419474,
2.105082615571597,
3.05308344502201,
4.199393321043376,
4.199393321043376,
4.199393321043375,
6.490210948425593,
6.4902109484255925,
6.490210948425593,
6.4902109484255925,
6.4902109484255925,
6.490210948425593,

```

6.882554174728105,
6.882554174728105,
6.882554174728105,
6.882554174728104,
6.8877583035732775,
6.8877583035732775,
6.8877583035732775,
8.21014824639056,
8.21014824639056,
9.071469420716642,
9.120928605341893,
9.714944098401723,
11.474131735950095,
11.474131735950095,
11.794551582273902,
1.341188628911811,
1.6554894539990426,
-2.1068974621953425,
-1.4171475675653578,
0.33211669069513333,
0.11331792925356984,
-1.3610911099963097,
0.46439741779020816,
0.9313501387866817,
1.0938593094543279,
-0.3591727738608774,
-0.4732622270970327,
1.6321598874818157,
-0.2782091348514778,
-1.0526781349295347,
0.0708703831522549,
2.6982701224663113,
-0.038867313655444914,
-0.10508583152652556,
-0.40759365792756236,
0.6284849004330764,
0.5192751929971827,
-1.6282040637266806,
0.0014422433007245106,
-1.8299719264591412,
2.3182597876656823,
0.8875167319646391,
1.5664268415036142,
1.5052778791259855,
2.935955436800565,
1.533101581959507,
1.8091829332960188,
0.45080966818917967,
0.1506883627647002,
-0.6843548866459184,
-0.5544883854492364,
1.2466562002431278,
-1.1866110136199353,
1.4521358231995756,
-1.125608676420736,
-0.24054383618816966,
-0.058272145415006006,
0.410408128597049,
-1.6441753460137252,
0.4019816715685168,
0.4588365987304068,
1.354346078964423,
-1.7153251360741875,
-1.1039978646265602e-06,
0.7940670488104785,
0.17979772520560053,
-0.24385394325110826,
0.6367160741723578,
0.8500459957371875,
0.0559416907316065,
0.32418668587130467,
-1.4797636873635434,
0.11902177942387202,
0.5185165439317966,
-0.08399222686641401,

```

0.8650918037299421,
1.4911587670088011,
0.12792248933645675,
0.3372262149750348,
-0.6553154954729696,
1.1995792367396254,
0.5132294030998703,
0.44835115860783636,
-0.37116980512777586,
0.7805194563374558,
-0.854334789795377,
0.5008527672922033,
-0.2302685629030402,
0.6145182391921147,
-0.6575500079215796,
0.8657035637529482,
1.1874516653159721,
-0.16284802367068918,
-0.26462171656665856,
-0.9815102056659819,
0.23198397744948207,
-1.087097195512072,
0.6616385654966559,
-0.2929637257774683,
0.03471390505932934,
0.2048751570049573,
0.9170017118070749,
0.39617956766989404,
0.12006199846995962,
0.1922705913940127,
0.8784478594386566,
0.055479957240883566,
-0.6634105431569992,
1.0742951419164057,
-0.37818200958777604,
0.28548758229245597,
-0.6594827056613419,
1.0235832498557866,
-0.19428253177041227,
-0.5531991468121416,
-1.189626464718852,
0.2519110119226656,
-0.6708462316627377,
2.7994352657547057,
-3.8764436680175813,
0.5152327138619494,
-0.3749347648847944,
2.9042167722097862,
1.1265965037549552,
-1.850959186134046,
2.165374611027145,
0.8818414791702859,
1.5039524911129019,
0.06564767423285973,
0.6486215314149069,
-0.08854143781823305,
1.8396223528529987,
-0.00792246728000446,
2.8805879469479487,
3.6337353861266473]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0, 0],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                'r': 1},
                'def': 'ising'},

```

```

'output': { 'draw': { 'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_ric2_sample#9'},
            'redundant': { 'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
            'run': 'zne',
            'state': 'dmatrix',
            'vqe': { 'iteration': 1,
                    'optimization': { 'algorithm': 'SLSQP',
                                      'constraint': True,
                                      'status': False}},
            'zne': { 'data_points': [ [4, 2, 0, 0, -5.966171999362981],
                                      [12, 6, 0, 0, -4.491721403204036]],
                    'degree': 0,
                    'method': 'richardson',
                    'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.9591064453125e-05,
            'zne_values': { 'degree': 0,
                            'extrapolated_value': -6.703397297442454,
                            'others': { 'beta_coeffieicients': [1.5, -0.5],
                                        'cost_richardson_zne': 2.5,
                                        'sorted_expectation_vals': [
-5.966171999362981,
-4.491721403204036],

                                'sorted_noise': [6, 18]],
                                'sampled data': [ [4, 2, 0, 0, -5.966171999362981],
                                                  [12, 6, 0, 0, -4.491721403204036]],
                                'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                         'r': 0},
                                      'time': {'max': 10.0, 'min': 0.0},
                                      'type': 'heisenberg'}}},
  'init_param': { 'value': [ 0.38848003251483587,
                             0.3884800325148359,
                             1.2841735835398218,
                             1.284173583539822,
                             1.2841735835398218,
                             1.818138220668234,
                             1.818138220668234,
                             2.7575469841005327,
                             2.7575469841005327,
                             2.7575469841005327,
                             3.0053468968599275,
                             3.0053468968599275,
                             3.005346896859927,
                             4.184079661691832,
                             4.83635165164968,
                             4.83635165164968,
                             5.97926077140412,
                             6.458805266455043,
                             6.513102391504654,
                             6.513102391504654,
                             7.122105903846713,
                             7.2188422331450415,
                             7.988590448127284,
                             8.268095077127937,
                             8.268095077127937,
                             9.164456488648115,
                             9.164456488648115,
                             9.782953992889732,
                             9.782953992889732,
                             10.467486029517685,
                             1.5496959718296535,
                             -1.0300437645832343,
                             0.425343365508412,
                             -1.7136674729251273,
                             0.33645197441382346,

```

0.2288357961575219,
0.10470230685228102,
-0.6612661793744108,
0.11823315872480306,
0.48571962162230753,
-1.3748501535311368,
0.2563120795974374,
-0.38871520150353267,
-0.18786175076151226,
1.0017996241286973,
0.05598903629217323,
1.501328526357405,
-0.32299727271195017,
0.5715295079657807,
-0.12201961934486968,
2.2747823054657927,
0.16289086097520028,
0.027619518366160244,
-0.10165606616500937,
-0.06321979816820887,
-0.27543819802246033,
-0.0048370064103128475,
0.1330138167350915,
-0.06428151371988705,
0.21905201010909556,
0.25697175483980955,
-1.462905966720411,
0.8014511304369155,
3.456137165485204,
-0.3560002683832885,
-0.4886877133354182,
0.32039145066707037,
1.6153352350920127,
0.5940099487759896,
0.18140087044059483,
0.27123106498617,
0.7444341922951362,
0.170480557625751,
-0.4688462519246396,
0.5235839184929864,
-0.5470964469389679,
-0.011921500492307424,
1.4116230981753706,
0.49972453659747124,
-0.7800673920851233,
0.4234227404504273,
-0.2071635265929523,
0.14064328933496192,
-1.2007096247825992,
0.12047098237640114,
-0.4125103841815656,
1.0447620953563077,
0.0020814149928196773,
0.4988991062112368,
-0.34495773827123155,
-0.12849063482979717,
0.4280934542561602,
0.7138444629946918,
-0.6497712710787925,
-0.43652659674625793,
0.917432486791049,
-0.8523477361785018,
0.5648987974980545,
0.3183226113969618,
0.14295784675778261,
0.9589824340171335,
0.9194478619356266,
1.1001359353915339,
1.5417377267389083,
-0.5376003771957187,
-0.5404528215706331,
0.21624687748210675,
0.8814937279455825,
-0.371639532103812,
-0.2668100917972658,


```

1.3133163210428074,
0.44574090697479807,
-1.0940261852319737,
-0.313830643067749,
-0.06100275437545863,
-0.049560783686653447,
-0.3025056724120473,
-0.9891529613817422,
-0.35099332176432035,
0.5436069706516342,
-0.722828417704955,
1.6968661444846564,
-2.4402971962208975,
0.7865745542588627,
1.3437881169309889,
0.36108471326554326,
-0.1749159211919777,
-0.014334278530919132,
0.02978377629472008,
1.8533978555002608,
-1.8470426695738211,
0.6272356375919823,
0.44775499409400255,
1.3281153539688288,
0.4583217606753929,
0.6443670392327268,
0.5254233581973354,
2.3087652110195127,
-0.7761319091913014,
-0.7605421959233766,
-0.6461580464596497,
0.7075597766411752,
0.985778508379752,
-1.3130974173457381,
2.8711331564400684,
1.2993543284582694,
1.9493614761217355,
1.1647493000584068,
-0.42731613935458074,
2.7038346000669766]],
'noise_profile': { 'noise_on_init_param': { 'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0, 0],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_ric2_sample#6'},
'redundant': { 'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.1446915183056525],
                           [12, 6, 0, 0, -4.667338994079639]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 5.4836273193359375e-05,
            'zne_values': { 'degree': 0,
                            'extrapolated_value': -6.883367780418659,
                            'others': { 'beta_coefficients': [1.5, -0.5],

```

-6.1446915183056525,
-4.667338994079639],

```
'cost_richardson_zne': 2.5,  
'sorted_expectation_vals': [  
  
    'sorted_noise': [6, 18]],  
    'sampled data': [ [4, 2, 0, 0, -6.1446915183056525],  
                      [12, 6, 0, 0, -4.667338994079639]],  
    'sampling': 'default']},  
{ 'config': { 'ansatz': { 'gateset': 1,  
                          'layer': 30,  
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],  
                                                         'cn': [ 0.5,  
                                                             0.5,  
                                                             0.5,  
                                                             0.5,  
                                                             0.5],  
                                                         'r': 0},  
                          'time': {'max': 10.0, 'min': 0.0},  
                          'type': 'heisenberg'}},  
  'init_param': { 'value': [ 0.4022850038273885,  
                             0.422288083135141,  
                             0.7644599371513524,  
                             0.7900179023917507,  
                             1.1026508993249646,  
                             1.1630326759467198,  
                             1.1630326759467198,  
                             1.2119340062926192,  
                             1.9859325565890973,  
                             1.985932556589097,  
                             2.096166401632718,  
                             3.378647998605865,  
                             3.378647998605865,  
                             3.378647998605865,  
                             4.354848115861854,  
                             6.006218342726225,  
                             6.006218342726225,  
                             6.009974297352436,  
                             8.225327865566033,  
                             8.242093148909708,  
                             9.222354355246733,  
                             9.369468664000962,  
                             9.432712219232048,  
                             9.692289963799956,  
                             9.692829942577658,  
                             10.391360826688166,  
                             10.391360826688166,  
                             10.529867035412478,  
                             10.529867035412478,  
                             11.06290493838992,  
                             2.9062307172979733,  
                             -1.7788694936155562,  
                             -0.1145266982224492,  
                             -0.9311559923392093,  
                             0.9691998209476543,  
                             1.7463479137973288,  
                             -0.7663862807329105,  
                             -0.9694574896037749,  
                             2.8979112386638413,  
                             1.1400097878698292,  
                             0.8096361549155084,  
                             -1.5900494197653285,  
                             0.09829667109053167,  
                             0.31839493135320834,  
                             0.3886777840539921,  
                             0.12229184789371997,  
                             1.7250225844898464,  
                             1.0065795676456808,  
                             1.4535641205083867,  
                             -0.3486770669120958,  
                             0.2556790893233589,  
                             1.778531141883236,  
                             0.269972269049403,  
                             1.0155903957099945,  
                             0.8041163886037216,
```

0.5381821926966456,
-0.30108604163317454,
0.6396533041678052,
2.1819991440037696,
1.4978086658278347,
2.5181040929106753,
0.4624415671193703,
-0.17701736987873673,
0.6910098122382194,
2.947172461882987,
-1.0727539032453073,
0.4023854197196741,
1.650490750838525,
0.9313998963798695,
-0.6769162055305248,
-0.5620785927454173,
-0.20885673940707247,
2.5019757090972443,
-0.37561416876024456,
-0.9329392032282731,
0.7818961506889994,
0.44350434198669075,
-0.10644929099379355,
-2.498928032568615,
-1.2046085786256906,
-0.06365416567297402,
0.17257236645054272,
1.2516209341031546,
-0.3591527663232026,
0.46789807326225935,
0.24374638092155315,
0.02110383811013195,
0.12714835746845662,
-0.5505736947218767,
-0.03497727012222505,
-2.2168334497478877,
-0.21485512988314215,
1.7552902914988067,
0.6189795903733311,
0.3191491062237487,
0.522632432692797,
-1.1121089236913335,
1.1371618962620422,
-0.2410043783817164,
0.24168543661014927,
0.39418565996934685,
1.2107667809264395,
2.398781467533486,
0.6948257569073976,
0.6170229142231725,
0.7050419245924102,
1.768546262611762,
1.5562956304120967,
-0.4269237214365518,
0.6302835616447369,
1.7710848704009405,
1.2281100264507194,
1.5036530549577243,
-0.7435268012971183,
0.7613541379199313,
0.8523298100708424,
-0.6655959646469325,
1.1543911681084875,
2.2675628634479494,
1.0299985170060308,
1.400963430022522,
0.27759304330136975,
0.6960561589746208,
-0.4210335016880339,
0.0336463669485127,
-2.099660094290342,
0.6674819238895001,
0.5049668435179431,
0.48001405045948287,
1.3574871122533543,

```

-2.2986993170736225,
9.601426007351498e-05,
1.3312470798272018,
0.7582238004326416,
-0.35033967277151296,
-0.4108552112493075,
0.1982910129710702,
0.6700017270602091,
0.26559360983179486,
-0.14563210230012705,
-0.39610135234418875,
1.1727729846793147,
0.6713573564879365,
0.8527070052929181,
0.8268283343075853,
2.2918575835381745,
0.5082103460913493,
1.8600493489550152,
0.49576811175353974,
2.232387985311215]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_ric2_sample#8'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -5.743389283741309],
[12, 6, 0, 0, -4.3560016590478785]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 3.266334533691406e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -6.437083096088025,
'others': { 'beta_coefficients': [1.5, -0.5],
'cost_richardson_zne': 2.5,
'sorted_expectation_vals': [
-5.743389283741309,
-4.3560016590478785],
'sorted_noise': [6, 18]],
'sampled data': [ [4, 2, 0, 0, -5.743389283741309],
[12, 6, 0, 0, -4.3560016590478785]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}}},

```

```
'init_param': { 'value': [ 0.6189603325174557,  
0.7142372255934472,  
0.818217611349809,  
1.1433975540074377,  
1.9121137997887092,  
1.912113799788709,  
1.9189666863881127,  
2.194219841514053,  
2.734492156740232,  
4.250258815642276,  
4.250258815642276,  
5.774076859791501,  
6.527612880112256,  
6.532834731882501,  
8.049613322901676,  
8.049613322901676,  
8.050730377555883,  
8.561643142317116,  
8.561643142317116,  
8.758357012835646,  
8.758357012835646,  
8.78353884989931,  
9.477076383056275,  
9.536600626467342,  
9.536600626467342,  
9.536600626467342,  
10.241295505019522,  
10.318321333031077,  
10.579129354925781,  
10.82886994059623,  
3.662494394083504,  
0.08309129295248968,  
1.5295376506525808,  
-1.5964000102221196,  
1.8988970610647178,  
-0.07218621526676876,  
0.7238532160702689,  
-1.2917423920622229,  
1.8532222971765195,  
-0.40818119715800666,  
0.469598065996274,  
-1.2652675561295137,  
0.41465882817859046,  
0.48683539382624846,  
-0.15336518315146752,  
-1.6419783820068856,  
1.2136622236743368,  
0.5933123377118196,  
0.9666225223219593,  
0.9539776194708611,  
-0.3793986531891993,  
-0.2570964298405315,  
1.5884302353807402,  
0.19867970725527498,  
1.0036205245072367,  
-1.7266220384949087,  
0.6519864384701275,  
0.173373648086925,  
1.8697046853700252,  
0.5263944462203128,  
2.9785129063723943,  
-0.5937256494260224,  
0.22753174799850245,  
0.24918335157216837,  
1.5438671949833176,  
-0.2981477150404249,  
-0.0503664733834821,  
0.542442835193611,  
1.8096859498837612,  
-2.2260935596321993,  
0.28693396030012636,  
0.9986781622414097,  
-0.01940263773226835,  
-1.8805315707929653,  
-0.1513202473661221,
```

0.1836836320118686,
-0.5655439481125303,
-0.30323205019871646,
-0.4186198812206128,
0.2357002564872809,
2.8296979465953833,
-1.0776781071762758,
-0.08378558447273823,
1.7446476514004516,
-0.4606832586075214,
-1.762173446350097,
0.10241780783627007,
1.6018254886002428,
0.2621004786967086,
1.349261138919249,
0.14987623247602613,
0.5068543254371847,
-1.3474781551208477,
1.0890436270679893,
1.6662048029626129,
0.3608718991529565,
-0.11137748070221386,
0.5641204379289552,
2.2546014997074266,
1.8888740160956543,
-0.23345107582518032,
-1.5772444286238798,
2.1408383357361265,
-0.4968079560788485,
0.1533667959779154,
-0.40238715350257875,
0.25591729793510365,
2.947382531044255,
-0.32162602471486745,
-0.7587223675916386,
-0.8983111566200868,
0.006114119281372181,
2.9372824954589514,
0.48288231335139287,
0.7939297064609614,
1.9454360656955985,
2.092519500518508,
0.8995449331251366,
0.10171517677156572,
0.45235144869603666,
1.02501950409702,
0.24331156538914245,
0.48973195386607005,
2.672871435464254,
1.313352678455905,
2.010046934709317,
-0.42153209835304317,
-0.13436700176919486,
-0.6950305532982513,
0.6026553555930264,
1.6681913778840949,
0.4125109526933123,
0.514533549588801,
1.0794873063708696,
-0.9571806916209341,
0.27630815458061647,
0.6001387806934945,
0.5059087494947612,
-0.5722342732268321,
0.3986919058890637,
1.0044202485382987,
0.3352295947729849,
1.3777059541173544,
-2.835684252944388,
0.6104690716589696,
1.838086538234438,
-0.3106134411398272,
-2.2167272073692903,
0.10706372359431235,
-0.17469850626418476]]},

8.165312898322489,
8.165312898322489,
8.684979613482101,
9.661121823219345,
9.674169964387936,
10.525655511340814,
10.525655511340814,
10.525655511340814,
11.154963299169733,
11.154963299169733,
2.50282655917642,
0.9490947670548783,
0.70056532212722,
-0.9370006834105179,
2.127395945212884,
1.5529950009292526,
-0.5538339242405286,
-0.45109357069990613,
3.2608593101153462,
2.1691098922881027,
0.924351518772188,
0.7799372105219657,
2.588492585292434,
1.4914484649202893,
0.5111989894903198,
-1.6854828839595362,
0.709524102178921,
-0.3201993643863164,
0.37756499567776974,
0.11160463007405702,
1.4506751210554825,
-0.1158349928449886,
-0.4172082421780774,
-0.47420587439641587,
2.1318004152619214,
1.5026342770773617,
0.9629412747697549,
-1.017935070811068,
2.2274033566257083,
0.9541631607884562,
2.2865374842341715,
-0.6901760390806929,
1.4488496207691695,
-0.15529093800821045,
1.378510821815928,
-0.1238908488185843,
1.0420944500710088,
1.8938950235205334,
-0.40270933293735856,
-0.4838468076888962,
-0.7769068081513464,
2.0570355933263196,
0.33843722154840217,
-0.21296684820549192,
-0.3087457919151385,
1.6713665097762895,
0.7955513549984788,
-0.5031484152839188,
0.2783608989762586,
1.4322280472670237,
0.12038812465567636,
0.8648150791824706,
0.13552183280667268,
0.07182610915111512,
0.3477791234991623,
-0.1584859230597403,
0.5851519083687279,
0.578565105996425,
-1.7921034519484391,
0.9189192684840749,
-0.6139572052593133,
-0.023527142002935514,
-0.6495330854871428,
0.7767670181361933,
0.13616638059771954,


```

-0.22568119775616352,
0.4192866974084779,
0.1401009023541104,
1.419355971488301,
1.1103093112149183,
-0.8728413553259085,
1.014691351510677,
2.35757700845783,
1.131891326153365,
-1.2979835358031357,
0.42365777146015526,
1.0982786992551208,
1.698510255572029,
1.2090506157632,
-1.3597097700308411,
-0.9049577551064396,
1.0199036275847497,
2.9541844671388584,
-0.42829772740772204,
0.6542829736223691,
3.7916586259391143,
-0.1983670801457003,
0.7855289833525587,
-0.1534574040841844,
1.809122493112309,
0.35293394466185096,
1.0573849517470935,
2.461665111905985,
1.0784030177718114,
-1.585709677044719,
0.8985620342798133,
2.2360691692951646,
0.24022620726925167,
2.6922129347449117,
0.15599094623602128,
2.799936411115923,
0.9657316063780467,
-0.2732772147086647,
-1.448572189075712,
-0.7280589250410919,
0.057973620624421296,
0.9321748335747359,
-1.4338621772181848,
1.3976338029473028,
-0.25504403617708243,
3.210257057781798,
0.03448775329128267,
0.05070056715723011,
-1.3727823101385919,
0.8243579770218447,
0.13142013414460954,
-0.4933237632531999,
1.1105137227416793,
0.7155364379215329,
-0.21063252058278284]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_ric2_sample#2'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',

```

```

'state': 'dmatrix',
'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.220556999324399],
                          [12, 6, 0, 0, -4.682225394272194]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 3.528594970703125e-05,
            'zne_values': { 'degree': 0,
                          'extrapolated_value': -6.989722801850501,
                          'others': { 'beta_coefficients': [1.5, -0.5],
                                      'cost_richardson_zne': 2.5,
                                      'sorted_expectation_vals': [
-6.220556999324399,
-4.682225394272194],

                          'sorted_noise': [6, 18]],
                          'sampled data': [ [4, 2, 0, 0, -6.220556999324399],
                                             [12, 6, 0, 0, -4.682225394272194]],
                          'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                    'cn': [ 0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5],
                                                    'r': 0},
                        'time': {'max': 10.0, 'min': 0.0},
                        'type': 'heisenberg'}},
  'init_param': { 'value': [ -1.8784742854218807e-17,
                              -7.763750530158576e-17,
                              0.3802827375258553,
                              0.44096449652630626,
                              0.8040692975576897,
                              1.9224694434519336,
                              1.9224694434519338,
                              1.9224694434519336,
                              1.9224694434519336,
                              1.9224694434519336,
                              1.9224694434519338,
                              2.730288811725591,
                              4.476350841356259,
                              4.476350841356259,
                              4.476350841356259,
                              6.331658137341824,
                              6.331658137341824,
                              6.7621011201584285,
                              6.77851096220289,
                              7.490043179529316,
                              7.490043179529316,
                              7.490043179529316,
                              7.490043179529316,
                              7.511664240193972,
                              8.302259372586072,
                              8.36899822047307,
                              8.607405767895836,
                              9.631163052898822,
                              9.874876363637043,
                              10.225975825453373,
                              2.711573403856553,
                              0.20239905137011172,
                              -0.4220883549830381,
                              -0.8404830976891227,
                              1.794332340210785,
                              1.5095243517856047,
                              0.22326098118939897,
                              0.07733371735403745,
                              1.280548885023134,
                              1.4121496210624682,

```

0.9815935680907694,
-2.975348552150871,
1.6316643284140941,
0.8239502945369961,
0.0314505340316768,
-1.0892891749800817,
1.720694113273713,
0.6634696927679957,
-0.836512780686814,
-1.1286873737196692,
-0.09500137106776224,
1.0428135848317974,
-0.4828772957450712,
-1.129334172148368,
0.7253737318795237,
1.4108675361353635,
0.7588540034470418,
-0.5039485441412638,
0.10346857579652805,
1.2668095290766983,
0.7994757345637048,
-0.7393240316081237,
-0.7818304990031031,
0.25375980918776664,
2.228975199542585,
2.2487625387645873,
0.6384105819815235,
0.39434544396871446,
1.3679129840939837,
-0.27336699910726836,
1.3170761670781417,
1.1159622861844822,
1.176824183488981,
0.40583799697229034,
0.8786168217534174,
0.2914083315983707,
-0.8464697846173548,
-0.032455291822620566,
-2.0755386823554396,
-0.11439663923099666,
0.7312672791487281,
0.4263196833485598,
0.11802610051350722,
1.0445881812565523,
-1.9566462184988007,
0.9492303984073598,
-1.7129593951518396,
-0.3230601534209437,
0.3161713641922788,
-0.4811213940350071,
0.08327140785409295,
0.37526634808329173,
-1.9990814349080968,
-0.15397700748904192,
-0.9549084255901845,
1.1120455683467105,
-0.42952128451280097,
0.17929609797141985,
1.0446016069356951,
0.846774755857218,
-0.2784586201121851,
0.23000407229986752,
-0.9198047598183884,
0.996529675185623,
0.34608382599103005,
0.5099375110708237,
2.1948932092248885,
1.3625694142279887,
-0.9306056124011278,
0.6012804875871011,
-0.7788968212470156,
-1.5884735532174328,
1.0361309111992296,
0.015937704739038126,
1.4003797043998856,

```

2.429422654005534,
-0.27504014116509645,
0.8256449732433587,
1.2146087003914199,
-0.6435553984602482,
0.3046269503778821,
1.0182549333954134,
1.4202224852266074,
1.3511302325707892,
-0.7261672378416364,
1.5992752660640235,
-0.5764295578986515,
-0.8318409785507642,
1.1375736468462807,
0.8529850561064739,
0.9553972898831558,
-0.09161160854677679,
1.617622889574723,
1.4551513261723394,
-0.08817813881493969,
0.8045852859105778,
1.9892741470924016,
1.8347816167556374,
-0.8050652823779443,
0.008511392950551585,
2.1460096989270747,
0.49277581514942953,
-2.4311684786278165,
0.5134791015266132,
0.9751974520662461,
-0.38582649358518695,
0.7746227771853464,
1.7307944301597258,
2.1320333699599043,
1.1654652968447545]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0, 0],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_ric2_sample#7'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -5.9667914171586585],
                          [12, 6, 0, 0, -4.541488857163632]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.76837158203125e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -6.679442697156171,
                           'others': { 'beta_coefficients': [1.5, -0.5],
                                       'cost_richardson_zne': 2.5,
                                       'sorted_expectation_vals': [
-5.9667914171586585,
-4.541488857163632],
'sorted_noise': [6, 18]},

```

```
        'sampled data': [ [4, 2, 0, 0, -5.9667914171586585],
                          [12, 6, 0, 0, -4.541488857163632]],
        'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                         'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'heisenberg'}}},
  'init_param': { 'value': [ 0.43940602753254654,
                              0.4524980474487914,
                              0.6462847510297367,
                              0.6535145711164267,
                              0.7019965052930044,
                              1.0830061860275473,
                              1.4380021973026182,
                              2.1860837771467474,
                              2.1860837771467474,
                              2.1860837771467474,
                              2.4875804812430995,
                              2.911818356328843,
                              4.9134595358452735,
                              5.043089843653786,
                              5.994354774193697,
                              6.658321281734746,
                              6.666230857319696,
                              8.458454027735183,
                              8.458454027735183,
                              8.458454027735183,
                              8.938931772570164,
                              8.938931772570164,
                              9.876451730548837,
                              9.876451730548837,
                              10.140618079102744,
                              10.453495471248813,
                              10.878653713196195,
                              11.282429749250388,
                              11.726474998572407,
                              11.93229290545428,
                              2.9640535214497064,
                              -0.6206168106317247,
                              -0.4035747848788786,
                              -2.067222048924027,
                              2.0576259596946604,
                              1.364315765000901,
                              -0.1480698866383175,
                              -1.95043906046908,
                              0.7897528850889669,
                              1.350582507885341,
                              -0.3349044763190947,
                              -2.3429833820345465,
                              2.04351680973891,
                              0.8083219796841106,
                              -0.06520770630024467,
                              -0.8776318722364843,
                              2.4862506381076748,
                              0.8345121096737285,
                              -0.03071661620417745,
                              -0.6225183255138825,
                              0.7450777062518694,
                              0.12619297185303674,
                              0.15620862918286052,
                              -0.47540100870445906,
                              0.6345315438244892,
                              1.451076347224231,
                              0.10985833383721234,
                              -1.6475696409761669,
                              1.1918822544398533,
                              1.3235539868194213,
```

-0.24222060474147966,
0.2418944215339802,
1.0397651583041008,
1.6773252160624976,
0.16975786500059598,
-0.9211027317156172,
0.7590471244619662,
1.76540125315151,
2.43856734239768,
-0.19488067618654012,
0.8462629124439717,
0.9979674075340036,
2.5775834810620677,
0.5650956422366618,
-0.5475934153287456,
-0.0253449469292375,
1.2524069066350432,
-0.605696708865831,
0.7019502861337156,
0.2803153244338447,
-1.0831925759777479,
1.0082140101851143,
-0.3762046176346905,
-0.9535933533256993,
-1.0422804614435257,
0.13948594332192887,
0.5835070530053111,
0.16086524965091817,
0.06403444071667018,
-0.06194188553710435,
-0.4026223625881385,
0.2750028468648149,
0.6789909068622377,
0.05162849323802493,
-0.7735332926339995,
-0.14862918062202837,
0.7661757453229759,
-0.16119730427373616,
0.6482389226852195,
3.1459456036848104,
-0.4760920139337192,
1.6197254159790668,
-0.5506328506738539,
0.9968299487743895,
-0.3106735286083523,
-1.2388662013594425,
0.8898580979157581,
0.43256773745092003,
-0.7272220041247079,
0.8716071923554094,
1.5193589139179793,
0.9787540447681606,
-1.4044929205930872,
-0.1984876316951887,
-0.46744984056178585,
-0.7765225748074835,
0.04344696088727837,
0.32373965212791267,
0.9049555024172572,
0.7139278873651953,
-0.8122282267372439,
0.5323405298774778,
-0.08437719753251166,
-0.14277007957413837,
0.7912147308716768,
0.26055493673650715,
0.11357082745857092,
0.3153698930365579,
3.1981441448638614,
0.32109035951393206,
-0.23017443442945684,
0.0316864124540797,
0.20875812581978664,
0.5466872162452653,
1.2918521189433907,

```

-0.7312547980705957,
1.4958457475939604,
-0.053193686203739404,
-0.4437772137226961,
-0.5271126400646005,
2.5136810835026426,
0.6674244883024821,
-1.0140291771816496,
-1.177197116527103,
2.058315434744553,
0.050372102721390984,
-0.09022974351195023,
-2.296438941834702,
0.3261688389091777,
0.3072223897800251]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_ric2_sample#1'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -5.966837375471956],
[12, 6, 0, 0, -4.6298501952375855]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 3.266334533691406e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -6.635330965589142,
'others': { 'beta_coefficients': [1.5, -0.5],
'cost_richardson_zne': 2.5,
'sorted_expectation_vals': [
-5.966837375471956,
-4.6298501952375855],
'sorted_noise': [6, 18]],
'sampled data': [ [4, 2, 0, 0, -5.966837375471956],
[12, 6, 0, 0, -4.6298501952375855]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [ -2.397634216504411e-17,
1.429450133825379e-17,
0.9192297980258759,
0.9192297980258759,
2.8984599305110854,

```

3.1993874679766523,
3.199387467976652,
3.715279848679994,
3.715919998165357,
5.126008182976459,
5.749875525815113,
5.785995839176617,
5.785995839176617,
5.785995839176617,
5.785995839176617,
6.315437962379428,
6.678845841951143,
7.3578167224936415,
7.415694801781823,
7.531263738212295,
7.949087697575632,
7.949087697575632,
8.111169110068786,
9.407094598577077,
9.495286031274873,
9.495286031274873,
10.993257661103009,
11.063439993331317,
11.639888600691357,
11.75175692668691,
2.1152038588685538,
-0.7839988468841306,
-1.0579861702584952,
-0.5122509123294313,
1.0012369417099964,
0.9275037440003397,
0.5343037326701707,
-0.014459274586663672,
2.31692048955396,
2.1400689678453872,
-0.9468421716760329,
1.3225342455296385,
2.4229762260600483,
1.4561600175543838,
-0.4592924007287594,
0.6389204073366491,
0.5578650700802136,
1.458345504678432,
-0.11931303728638508,
0.04406428844113783,
-0.26144087074192934,
0.6459890964472403,
1.2243061606997268,
0.09770103868863204,
0.23645607828472368,
0.3280111145811677,
0.5390648705936556,
-0.6034481934191781,
-1.1942560416831098,
0.3295059316644399,
-0.0732996553139216,
0.5043071925129845,
0.7399437462813884,
0.3595686242698361,
1.4643835231854074,
-0.1697776944282527,
0.2588380732750666,
0.44311592273109757,
-0.6031987466514527,
-0.06131410155451245,
-0.3299395710828798,
-0.45492250592824107,
0.5555861976799495,
1.1612177689175078,
1.833550163228831,
1.5629483318007886,
-0.2455585474033218,
-0.4534768786824923,
-1.2225865843307935,
0.20468879325769473,


```

'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_ric2_sample#4'},
'redundant': { 'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.241465922765087],
[12, 6, 0, 0, -4.714681794898787]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 4.267692565917969e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -7.004857986698237,
'others': { 'beta_coefficients': [1.5, -0.5],
'cost_richardson_zne': 2.5,
'sorted_expectation_vals': [
-6.241465922765087,
-4.714681794898787],
'sorted_noise': [6, 18]},
'sampled data': [ [4, 2, 0, 0, -6.241465922765087],
[12, 6, 0, 0, -4.714681794898787]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': { 'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [ 0.11853625011402234,
0.11853625011402236,
1.1173235551611367,
1.7342580765110573,
1.7342580765110573,
2.108144483620023,
3.054129947242328,
3.054129947242328,
3.054129947242328,
3.056306360633301,
4.638414750363569,
5.0621851887275495,
5.7503913724004905,
5.774998407238176,
6.917933224474368,
6.917933224474368,
8.061549209526286,
8.061549209526286,
8.061549209526286,
8.591251503132888,
8.591251503132888,
9.051235537435975,
9.081856762230144,
9.467876152039107,
9.545932016874406,

```

9.591419209848118,
10.465616352275608,
10.465616352275608,
10.465735173309618,
10.980942228105905,
2.0734674059611633,
0.32892201175743896,
-0.007356440825095051,
0.7856576253246221,
2.089457548986907,
0.7744884921126718,
-0.35049309306395837,
-0.6556831494452968,
2.0369525201874903,
0.4739652080570001,
-0.8564345639983049,
1.25716179603495,
1.7181260437331554,
2.3978160665502384,
1.1152024575020885,
-1.3335737674295711,
0.016650799076572978,
1.6607129762234538,
0.648059834763191,
0.2438725527060928,
-0.27307896000752346,
1.4369017198677916,
3.5672448343800007,
-0.5379702113035718,
-0.7420664562717484,
-0.529699891930992,
0.7301471483152846,
0.781913925757813,
0.6004078048346045,
1.7324865479818374,
0.4548236607944829,
-1.4419813608219112,
-0.684662976785234,
0.8455512814500156,
1.97751114408285,
-0.833202517811791,
1.110524460863052,
2.6403320691876733,
-0.3129427957408566,
-0.06674204626492235,
0.5842857778318762,
-0.21132485649138796,
-0.011842019288790376,
-0.25743970610802486,
0.7486616881233721,
-0.1302029141156864,
0.09497091477412227,
-0.3715265429351211,
0.39570077921274605,
0.28643868832003017,
0.7848413456471106,
-0.7157146817058562,
0.09411764456438232,
-0.10910076647953601,
1.1262761892344808,
-0.27334997823872975,
-0.44027652197942063,
0.6740415100156434,
-0.336256645457715,
0.4268157265313067,
-0.8611696251566416,
-0.34560928156726306,
0.5618074077714486,
0.5866360379174992,
1.8662876827821717,
2.21592310898005,
1.972223656383108,
1.6141187926643297,
0.856255815565121,
-0.34275929095614754,

```

-1.0642528745182613,
-1.0618686751734654,
0.2379582550153795,
-0.6501770361983872,
-0.19408645485522444,
1.501330829125914,
1.5726602727591479,
0.7066066097903853,
-1.4580936665368467,
0.46940440748349077,
-0.1287015677341077,
-0.9805925033056946,
-0.3754741669636888,
-0.026817919501009202,
1.1067346586651607,
0.17547923725022047,
0.8902143171896091,
-0.22777698610446503,
0.8082253583749343,
-1.0674083601412958,
-0.022706971436593337,
0.34227899244110566,
1.3527736523760083,
2.0767841716125894,
0.5778568145668401,
-0.6555502504392442,
1.2670274746873995,
0.507694022232518,
-0.2629991187543805,
1.7048253080726326,
1.4088065522309612,
2.823918413203317,
1.2763232744235296,
-1.4348344213353486,
-0.605496445996402,
0.09645158262695042,
0.024639736182597342,
0.23997550724405606,
0.9979895378690888,
-0.40612123795993227,
1.826790650755685,
0.029599395583551916,
-0.4592861834683686,
0.0328358806035505,
1.3612120706287358,
0.8210402794880411,
0.05688164146575431,
0.03152531580318861,
2.0316547976974904,
2.795039607579165]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0, 0],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_ric2_sample#10'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},

```

```

'zne': { 'data_points': [ [4, 2, 0, 0, -6.2813019572269795],
                           [12, 6, 0, 0, -4.778373581659957]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default' }},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 3.2901763916015625e-05,
            'zne_values': { 'degree': 0,
                            'extrapolated_value': -7.032766145010491,
                            'others': { 'beta_coeffieicients': [1.5, -0.5],
                                          'cost_richardson_zne': 2.5,
                                          'sorted_expectation_vals': [
-6.2813019572269795,
-4.778373581659957],

                            'sorted_noise': [6, 18]],
                            'sampled data': [ [4, 2, 0, 0, -6.2813019572269795],
                                                [12, 6, 0, 0, -4.778373581659957]],
                            'sampling': 'default' } }},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                         'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'heisenberg' } }},
  'init_param': { 'value': [ 0.12420123662801165,
                             0.138246244841176,
                             0.34185010800102933,
                             0.8771280515547424,
                             0.8771280515547423,
                             1.9326762312100456,
                             1.9326762312100456,
                             2.879806034127527,
                             3.4467495190209787,
                             3.460588109395402,
                             3.4605881093954016,
                             3.4605881093954016,
                             3.911934955717517,
                             3.9564836804547703,
                             4.188240551814426,
                             5.098640553346409,
                             5.098640553346409,
                             5.098640553346408,
                             6.466332587579958,
                             7.457565848827182,
                             7.712724508543908,
                             7.890302861063261,
                             7.898577524812402,
                             8.020037295803547,
                             8.020037295803547,
                             9.289723192898348,
                             9.68251578824504,
                             9.68251578824504,
                             10.66694519433398,
                             11.464898013549973,
                             1.8095532965205083,
                             -1.6633278493215808,
                             -1.2281091611401596,
                             -0.40155726479865544,
                             1.7903243850816337,
                             0.18661159996017582,
                             1.2217850039169107,
                             -0.48128156078104645,
                             1.438546033666569,
                             -0.9466383819139143,
                             -0.2179227006066069,
                             -0.07921201994836616,
                             0.6692914930417907,
                             -0.16791294486623565,
                             -0.3278463005181308,

```

1.2883193159418245,
1.9001761172388045,
0.7646920019853412,
-0.6325477821570596,
-0.5987552945744136,
0.5669730115381587,
0.5983941343774113,
-0.06179171362000896,
-0.6867235859693906,
0.5211737756628548,
-0.19354588250953578,
0.6072816218147872,
0.3472677197742769,
-0.08031734075212016,
1.4024126217748347,
-0.5867181790012919,
0.809620813948269,
0.35997425529443994,
-0.8722967249300235,
-0.5874818244923429,
-0.5567926036837068,
-0.6652314874983639,
-0.6042054992483096,
-0.13918430829641074,
-0.08277488470584692,
-0.4972293169762719,
-0.4421668735352916,
0.6160601456858428,
-1.1972657609969646,
-1.1558488364748807,
-1.4501782622641872,
0.03989088007781949,
-0.8684923327267209,
-0.5569704341339503,
0.15823054121075,
0.6345578215095505,
0.3601853745371903,
-0.42594808878819745,
0.08979804115834313,
-0.153043281560756,
-0.603228229875778,
-1.1264410002736531,
0.6895580235946861,
-0.11045941959152497,
0.5174109375139814,
1.3269460586908306,
-0.3852298542423827,
-2.4120095755015893,
0.08807412161753436,
0.1228551365077638,
-0.6982591234727157,
-1.1618488040328845,
2.5586909942385017,
-0.24921574033555105,
-1.9870969070050777,
-1.0279498295884981,
-0.7796802357622239,
0.48428681889730063,
0.7162815161060118,
-0.15547297036103372,
1.0522591887013215,
1.772822944479854,
-0.2654037211186137,
-0.2720110866721671,
-0.03929478442656517,
0.34606077605631835,
0.5533331494282367,
1.0574846609095763,
-0.5747611578732846,
0.4916772893793279,
0.9875709691916216,
-0.17857479553023695,
-1.147281220275154,
0.37370382177065037,
2.456815155407461,

```

0.6567596591037266,
0.9507316292553318,
1.5113928877114242,
1.8380989093110256,
-0.1736919900840187,
-1.2666253947041328,
-0.554819304070435,
1.1931643666366285,
0.4614862178419797,
1.2040726557184918,
0.3083618234293392,
-0.4066692475369416,
0.3740368412278018,
-1.684611974494111,
-0.8650016725760176,
-1.4838330341878547,
-0.14344887350922086,
4.203596189627334,
1.1133202614474196,
1.391124117957397,
1.8627332105499828,
-0.28192336755973296,
-1.5416482647725873,
-1.9612892990030817,
0.8836928087418303,
0.300761821046295,
0.6785848024508535,
-0.5500277556546446,
0.5742776326882599,
-1.181756808209963]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0, 0],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_ric2_sample#5'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -5.814676484708358],
                           [12, 6, 0, 0, -4.248447309496402]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.0531158447265625e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -6.597791072314337,
                           'others': { 'beta_coeffieicients': [1.5, -0.5],
                                         'cost_richardson_zne': 2.5,
                                         'sorted_expectation_vals': [
-5.814676484708358,
-4.248447309496402],
                                         'sorted_noise': [6, 18]],
                                         'sampled data': [ [4, 2, 0, 0, -5.814676484708358],
                                                             [12, 6, 0, 0, -4.248447309496402]],
                                         'sampling': 'default'}}}},
'heisenberg-ric3-uniform-scaling': { 'noiseoff': [ { 'config': { 'ansatz': { 'gateset': 1,
                                         'layer': 30,

```

```

0.5],
False,
0},
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': { 'max': 10.0,
'min': 0.0},
'type': 'heisenberg'}},
'init_param': {'value': 'random'},
'noise_profile': { 'noise_on_init_param': { 'status':
'value':
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': False,
'type': 'depolarizing'}},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': True,
'type': 'png'},
'file_name_prefix':
'heisenberg_noise_off'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0]]},
'run': 'vqe',
'state': 'dmatrix',
'vqe': { 'iteration': 10,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': True}},
'zne': { 'data_points': None,
'degree': 1,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 '
'+ 1 Z 1 + 1 X 2 X 3 + 1 Z 2 + '
'1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 '
'+ 1 Z 4 + 1 X 5 X 6 + 1 Z 5 + '
'1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X '
'0 X 1 '
'+ 0.5 '
'Y 0 Y '
'1 + '
'0.5 Z '
'0 Z 1 '
'+ 0.5 '
'X 1 X '

```



```
'2 + '
'0.5 Y '
'1 Y 2 '
'+ 0.5 '
'Z 1 Z '
'2 + '
'0.5 X '
'2 X 3 '
'+ 0.5 '
'Y 2 Y '
'3 + '
'0.5 Z '
'2 Z 3 '
'+ 0.5 '
'X 3 X '
'4 + '
'0.5 Y '
'3 Y 4 '
'+ 0.5 '
'Z 3 Z '
'4 + '
'0.5 X '
'4 X 5 '
'+ 0.5 '
'Y 4 Y '
'5 + '
'0.5 Z '
'4 Z 5 '
'+ 0.5 '
'X 5 X '
'6 + '
'0.5 Y '
'5 Y 6 '
'+ 0.5 '
'Z 5 Z '
'6']},

'output': { 'exact_sol': -8.56677223350563,
            'initial_cost_history': [ 6.121384846939922,
                                     6.150953723335025,
                                     6.28813394059272,
                                     6.195576097324661,
                                     5.996275986067059,
                                     6.155907118612996,
                                     6.197882180139617,
                                     6.276491195599097,
                                     6.126423666072617,
                                     6.100624170709524],
            'noise_details': { 'gates_num': [4, 1, 1, 0],
                              'identity_factors': [0, 0, 0, 0],
                              'noise_level': [0, 0, 0, 0],
                              'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

'noise_prob': [

0.001,
0.001,
0.001,
0.001],

'status':

False,

'type':

'depolarizing'},

'odd_wires': 4},
'optimized_minimum_cost': [ -7.113819716474412,
                             -6.821323437950204,
                             -7.2322812477454015,
                             -7.168061781988564,
                             -7.037022966690968,
                             -6.764659140688094,
                             -6.70772123506257,
                             -6.951929490175616,
                             -6.765271591343842,
                             -6.643370531408768],
'optimized_parameters': [ [ 0.010202312572582629,
                             0.01020231257258252,
```

0.010202312572582303,
1.3908062838956354,
1.3908062838956354,
2.0164416760130464,
2.592046705326477,
2.592046705326477,
2.592046705326477,
3.724080378311745,
3.724080378311745,
4.739480012897176,
4.739480012897176,
5.945308900554525,
5.945308900554525,
5.945308900554525,
5.945308900554525,
6.51261715666153,
6.516103684733277,
7.99085874039394,
7.99085874039394,
8.16277333661163,
8.99858956109671,
9.406132081871055,
9.406132081871055,
9.406132081871055,
10.1331640829029,
10.184186623923145,
11.15918741223016,
11.16716557378114,
2.8450840989270483,
-0.611567672261254,
-0.551095997013421,
-1.0131107784954883,
0.6825070788337171,
-0.6162852792596847,
0.053241771237976535,
-0.12516614014524466,
1.004304378969064,
1.0874046621334863,
-0.263039228744755,
-0.10249623387196242,
1.6330709573383448,
0.7555807809408442,
-1.304918472993802,
-1.1870918453507433,
-0.1330475605934714,
-0.5047081392764299,
0.7756728723196425,
-0.49278801094288127,
0.42847067786702214,
0.5383171641852699,
-0.1052976007723638,
-1.2720333725347934,
0.1951924272458347,
-0.04277330847061178,
0.9574239041786494,
-0.23071352324401823,
0.594076188896681,
0.34935689199185954,
1.343124516909016,
0.09733282827416712,
0.30944030008743273,
0.4457207862171129,
-0.9683728985035202,
-1.120837258235465,
0.5707161794308084,
1.2158524027070072,
0.7958721804993727,
-0.9645697723054423,
-0.8960337800506504,
0.19569442936859724,
0.40224575611741115,
0.043062302517518566,
-1.1857970207577537,
0.1737802328972237,
-0.6433879386848066,

-0.09230835161436789,
-0.5865305292280132,
-0.4370825940325887,
-0.2558137517355887,
0.0011609522641742864,
0.6539099100207111,
0.5336009667754404,
-0.066252982145182,
-0.5359243057678726,
-1.3834098071980059,
1.0212180272076503,
0.5141425620577187,
-0.3002279094652444,
1.9431365651467594,
1.0552610834283143,
0.03764625526605462,
1.0189467943495765,
-1.1805772107399133,
0.2812463759993353,
1.4321423259624626,
0.004331180480864721,
-0.009722915333733264,
0.6451162813005181,
-1.3276490840851547,
-0.022791981388308465,
0.6904663535208042,
-0.08291995023229079,
-0.4292297662993819,
0.08451212085763125,
1.730745239172763,
-0.4572165915626394,
-0.5304437424739927,
0.6917595603044201,
-0.0834050618602206,
-0.18753214427540912,
0.03305271309241428,
-2.06126875311761,
1.5412249435528216,
0.6658854058101124,
-0.36436553183642006,
-0.9193216873412987,
-0.12900157901067746,
0.00542745656150127,
-2.171309732970475,
0.7554637248132821,
1.249093251376782,
-0.6445658098419218,
0.14255909437901,
1.9827929295292588,
1.7044112257377888,
1.3052476731654217,
-1.2421280078558083,
0.9803773552514545,
0.624656679395339,
1.5450717726138348,
1.3377635200462674,
1.6713162630334013,
1.1434539739578018,
0.7924845860796002,
0.7488178629415702,
-0.22907755928161938,
0.02068614393688229,
0.7608745118642098,
-0.9304086893990576,
0.32284859123973336,
1.1891889033470526,
-2.2276657670493667,
-0.3847051133881006,
0.34685342333136693,
0.549782335235608,
-0.741267714199706,
0.26835698412135817,
-0.3932040836252281],
[0.1668593898896292,
0.17040712573292452,

0.1805381301837931,
0.5412292500635668,
0.5489811947718172,
1.6527475146851556,
2.55163116002801,
2.5519072608250526,
3.5532683674980303,
3.6288880821260254,
3.628888082126026,
4.805536133422,
4.805536133422,
5.495812833389341,
7.061809444427505,
7.061809444427505,
7.091809082073759,
7.091809082073759,
7.177303629807857,
7.177303629807857,
8.014949756247965,
8.530162577551986,
8.530162577551986,
8.546061750342906,
9.04867098177819,
9.074898072060186,
9.632422305371618,
9.634373167303918,
10.38794675566215,
10.6266488059183,
1.098346025464491,
1.5064109656875124,
-1.345000798816457,
-1.0112785952191263,
1.6300712383138607,
0.8148371701480261,
-0.28350002839102345,
-0.33372360425933684,
1.6208682076493302,
0.8121027533840469,
1.424723017228629,
-2.424990019545176,
0.082574342220046,
1.5280227698708746,
-1.15137171364767,
0.8821561188139568,
-1.0160789387304932,
1.6746733218779515,
2.4704340348759835,
0.8258460042877822,
0.21603078517896931,
-0.05951632881852065,
-0.38389794767629976,
1.262782578763767,
0.2784323534326924,
-1.1033559163681899,
-0.6528860571486432,
0.2962709573582783,
0.7373896958863383,
-0.19469473272588958,
-0.5648694322067066,
-0.2865513283590244,
-0.1568370013695695,
0.5493402156787225,
-0.06660698090283595,
-0.03359282593361995,
2.386303225482439,
2.5818639178838447,
-0.2777126348162844,
0.9824372498806634,
-0.39422047955069633,
-0.2578521646674435,
-2.3189661125012333,
3.704222502127313,
0.0786550943979888,
-0.4292299615806753,
1.9657979836656132,

-2.0589782426096743,
0.5074656887310783,
0.529845836294102,
0.3333899546946687,
-2.293287544626138,
0.38495704078578347,
0.11228231507748758,
-0.5441382535456334,
-0.10489456654410802,
0.20011773016694326,
0.2071371298105447,
-2.3651747840346373,
-4.642753244408169,
0.31152670002610416,
0.5022276983536407,
0.868938175568576,
0.20348366213849262,
1.6475591691674232,
0.3723999247005541,
-0.6589367341470095,
-1.1662574597011768,
-0.6243580701563078,
0.7131540586709431,
0.1202680893039496,
-0.035429417928639004,
-0.764191314011073,
1.2086109032075552,
-1.051983148544319,
0.9246153479653275,
-1.1427863274072698,
1.332115914696049,
-0.27763903875773216,
-0.8366665559266224,
1.231317184155771,
1.0584874216549616,
-1.2625782973678872,
-0.13344263101573678,
-0.5438546353631553,
0.32212207070493337,
-1.303226737690094,
-0.01337174800709397,
1.5589050571537895,
0.3479397882311206,
-1.87970134892204,
2.977494205144941,
2.126179176803338,
0.873413305092612,
0.8500950046844986,
0.23032349681334924,
0.3909007718762141,
1.27926383991169,
0.4053229000380187,
0.18190897779649284,
0.3777935390994293,
0.16772342823060155,
0.7412973026981022,
0.639129360404442,
0.018341913161260836,
-1.4541884523192488,
0.28134413046256895,
2.4001077951469534,
1.6611157476326641,
0.7905510314185135,
0.3404203970178551,
2.183912805046456,
-0.3997518053650227,
0.053570896670028414,
-0.357205442227867,
2.465639862334342,
0.5537972989863772,
-0.39728466664110995,
-0.027817223268321305,
1.9807706991981826],
[0.5951590437264768,
0.5951597797898418,

0.596302078053195,
1.4089369768482796,
1.4331037442558932,
1.9334285070314567,
3.062681311500457,
3.0626813115004623,
3.701592986553387,
3.7015929865611663,
4.692693983743045,
4.692707272250184,
4.692707272250258,
4.692707272250313,
5.273956511551061,
5.2739565115534885,
5.274711944180805,
5.292162333185471,
6.388031186249758,
6.706092770308958,
6.706092773942781,
7.548703938317144,
7.681022457823495,
8.397974590110634,
8.397974590138377,
8.714530791128386,
8.714530791128386,
10.075092988071805,
10.075092988073155,
11.341131260513714,
1.2164966908304873,
0.5225196146354087,
-0.34542368185043265,
-1.252295733833459,
1.5735101799719247,
1.0464982053397573,
1.2394468649734893,
0.21595542567535536,
0.6527940529325085,
1.4145936861598034,
-0.5380512317672027,
-1.9014686324238337,
1.451754133219536,
2.1987565378949516,
0.938092448651857,
-1.6378088277288243,
0.38538945046186207,
1.1341127264381634,
0.4639756379522118,
0.6078279078278344,
0.6166736770850927,
-0.2105783577152824,
-0.35613934438782024,
0.019700965332415492,
-0.1607988588148763,
2.193665487035956,
-0.06423342915568941,
-1.9546338684001463,
-0.18633224680849803,
0.7857721800679403,
-0.6042593641712383,
0.7490426193215785,
-1.6440993970557098,
0.1929003499686422,
-0.13169470171686798,
0.1181804780373785,
0.15693401681144475,
-0.28658811756127545,
-0.20708135879200276,
0.09230795188954338,
-0.4114468277172402,
-0.8587939345722652,
-0.29108154529906144,
0.5427012646910324,
-0.25834402839896575,
0.08607676608574079,
-1.2954984273935897,

0.1813204379953717,
-0.34769281236878496,
-0.09683800499273743,
-0.9220596744797915,
-0.67003274699539,
0.4888752813745547,
1.2659094572448748,
0.11946751252820795,
-0.2586531113468492,
0.8859866115247531,
1.5010226653220677,
1.2253090413370418,
-1.41480051306012,
0.4438287250399343,
1.1348546909542183,
-0.26356600881424186,
-2.0062796648632446,
-1.5997185613370128,
0.19365670408843416,
1.9542746653083636,
-1.16753432767107,
-0.8341998310634118,
-0.8078884986396416,
1.057969787908329,
0.11874076530060441,
-0.37823536664235385,
-0.26520820180860055,
-0.05623142087332937,
-0.6351447036627758,
1.2477313546900863,
0.09475828495367859,
1.1570291529547316,
-0.3159140153065273,
-0.5798756886838644,
0.5049917679316769,
0.17933499272174766,
-0.6452919120449363,
0.3093583753904067,
0.6269418409971109,
0.536228963471926,
-0.39297460666924544,
-0.8585561386083093,
0.42310038622809626,
0.4625446959490829,
0.46551290354297115,
-0.006225430338097464,
-0.9498487290783498,
0.9283489563799108,
0.5288015885952413,
-0.6423745714782674,
0.055608107183735184,
-0.19141640271906557,
1.6106743132208,
-0.930502866063725,
0.13606233511650948,
0.7450273046384331,
-1.862998799714745,
0.8639045442938115,
2.2050766695222275,
2.1209593580448147,
1.5928713117159927,
-0.646261951983125,
0.18338232206466956,
1.9195145046471642,
0.9340015251080405,
0.22314701152636945,
0.22339358941062268,
-0.49545999196264406,
1.6497345768092624,
-0.37841357670502584,
0.19313501042389203,
0.07433308037827455,
1.831606295240235],
[0.013945989287166007,
0.014482887559355881,

0.7020792700957467,
1.2708090216179468,
1.8771473361142745,
1.898764043280349,
2.8886374873795146,
2.8887394286496386,
2.889554540070087,
2.8895545400700837,
2.889778536891296,
2.890175029071649,
2.8901764691110965,
3.881547037638128,
4.589967577258263,
4.916994542918821,
5.4213951120408375,
5.421396236977594,
5.937049320865825,
6.879796302074518,
6.931299553407329,
7.597215453538154,
7.941093252873733,
7.941093304737446,
9.23283235128487,
9.232832768748333,
9.232832769084892,
11.10587443204972,
11.182791630116155,
11.297769627933807,
0.7007490003921656,
0.8357798277265018,
-0.006144500286149594,
-2.002647677865005,
0.3078229546122665,
0.6497081019565148,
-0.5263987300225429,
-1.3591736982237872,
0.4978537208919288,
-1.0703987652029276,
1.4894572070841317,
0.10507602812754031,
1.2060409491957005,
0.37232709611155723,
-0.9426439100006672,
-0.36387040355717715,
0.9248005896093184,
1.0383864274136485,
-0.2845687665108692,
-1.3700022658840743,
0.197561542582957,
-0.32357133547759104,
-0.5540850172472152,
0.5749264875292693,
-0.8790799137852104,
1.0945413836746756,
0.3793043522465329,
-0.4056052817915047,
-0.03473792548266608,
0.06466238365347084,
0.06911551547308,
0.1689218888799635,
-0.24055379179177064,
1.0229644579244672,
-0.3032356624759471,
0.7939776889924924,
0.9000504980210274,
1.1922086788205504,
-0.09256238768786877,
0.5764640092401906,
0.20671868793560919,
1.5137094250639866,
-0.17335127063306519,
0.8566284172009878,
-0.14321647694006984,
1.3125145841881363,
1.3592268317355658,

0.1512217749645529,
0.40406540571588334,
0.048383985681794354,
0.19259007229061106,
0.09976438753283409,
-0.8546571821237718,
0.6621369033183476,
-0.1702351863826035,
0.36026352288591695,
-0.8538518608539489,
0.5392868667124087,
1.1279014251126462,
-0.36128436957709814,
0.388999106940785,
0.6046833425816769,
0.22322275018396126,
0.1917169599278726,
-0.909915166057891,
0.21776193961543494,
0.28996753900858246,
-0.5842323028487328,
-0.23864599488371552,
0.5243326463982184,
0.32056321937714066,
0.5356836550343521,
-0.3036011562711703,
0.28802877152454115,
-0.33781577627567244,
0.7960023934710929,
-0.34238272839391165,
-0.755152052016237,
-0.44352996180502197,
0.07213680417312635,
-0.10190373256064952,
0.6647215855084265,
-0.9303498099091333,
0.9785859773141862,
2.090431130104361,
-0.3615944398761197,
-0.05657593308707301,
-0.4893006812698228,
2.169089562284795,
-0.2512716878005756,
0.5324061192252376,
-0.35101147654456727,
0.809248070031391,
1.4509839161975397,
-0.14911304722907706,
0.009330045236156378,
0.16685413556340573,
-1.0472061077948698,
1.6539137014575498,
1.3611819463379937,
-0.1612641139265809,
-1.2962247052632323,
-1.3797066485510583,
0.7614302908946665,
0.5091243097447496,
-1.034844537010192,
1.0872752511102444,
-0.476692079414531,
0.49649227376774574,
-1.1491063122969838,
0.5155370403129137,
2.9855835289421178,
1.332632811562063,
-0.4511213152168112,
-0.18076672982111605,
-0.6499435359167832,
-0.1697346678000319,
1.4081723136905224,
0.442588820936982,
1.722678503402018],
[0.6039929180328771,
0.6039929180328772,

1.1870458270689,
1.1870458270688995,
1.412745515623535,
2.7932578893846336,
3.721244973412882,
4.900800320776727,
6.39210348373028,
6.39210348373028,
7.061133772685592,
7.061133772685592,
7.061133772685592,
7.064294030691084,
7.896781244010901,
7.896781244010901,
7.998168023342792,
8.09271241991372,
8.09271241991372,
9.328885229004163,
9.328906326502109,
9.56647323827175,
9.587868480377287,
9.606795901338938,
9.632809215929381,
9.925261433896223,
10.67745113859835,
11.226267254217802,
11.25813692634961,
11.259219336255413,
1.5276356326766196,
-1.2318130242699923,
0.07924764105056724,
-3.11015247863868,
1.71979592089301,
2.1636204056140302,
-0.44284897868332995,
-1.9599373218939633,
0.06248244737264575,
1.4749130816585614,
0.6808242800852113,
-2.8378068229153226,
1.7023278591356004,
1.50553848610602,
0.9365133390345636,
-2.1922613442274193,
2.8432021818280635,
0.1855585942894919,
-1.305416067894503,
0.1905478234118404,
0.7393995701939485,
0.19798052248890954,
-0.8075009715521676,
0.3948611135767452,
0.2802743807361451,
-1.4516873620037638,
1.5556285183865062,
-1.0754052466574715,
-0.26931261584286686,
1.2336702516634759,
-0.8384600779602994,
0.1263891070094113,
1.371252832802525,
0.215177045818575,
0.6381629880602765,
-0.10309421117565558,
0.4124383487994645,
-0.655499258712117,
0.4988310520659948,
0.9884276098863405,
1.4367219570983862,
2.175763151095631,
0.4657848221151968,
-0.43932071710829274,
0.9992343531072873,
2.543795366459747,
1.4484903663678823,

-1.0884458694866728,
1.5946427529603922,
1.6036026746991174,
0.5049418933876622,
-1.0871394525284908,
1.0140119509472394,
0.5109229333717124,
0.8743589388253838,
0.09626392588768701,
-1.263331036369507,
0.7924497462635596,
0.7602243892770352,
-0.02425703489356225,
-0.9488939939602019,
2.6693155936644737,
0.5100002820942202,
0.595619937330169,
-0.8296192082845288,
0.03973961449691636,
-1.7144893205098422,
1.1592271204386744,
0.04896218554826232,
0.6673584031026698,
-0.8454442747231521,
1.0304518931414424,
0.8556412792926265,
0.538780781704616,
-2.308898512383933,
1.5082121217292006,
0.9522091008412682,
-0.6026951427085929,
-0.561943926102551,
0.0993983558744849,
0.4791162049419018,
0.10320414203165484,
0.3730055252741846,
-0.47083232270591907,
1.8673303303286475,
0.11712430229385423,
-0.6044688726688213,
-0.9010581315425702,
0.8870944114042272,
-0.6585959519901412,
1.4737812112349316,
-0.08789784470103781,
2.094313922775301,
-0.33880630598079,
2.0864279335765668,
-0.3541510297313768,
-0.1447960879675017,
1.1237218765827641,
0.7472353955833465,
0.43980261574830926,
0.6534205210717992,
1.2606259200129655,
-0.17935209951086067,
1.6469967332630842,
-1.0541245665671384,
-1.9424469299865808,
-0.18975499615277758,
1.967627931161089,
-0.5610576072704585,
-1.4036970670846107,
2.6946925238411343,
-0.9860662529475024,
-0.6528571620700129,
-0.6551571328219318,
0.9777230807240078,
0.003267369094500521,
-1.3217863730621935,
0.3048209868145306,
2.1104722024376037,
-0.13744008360720306],
[0.25527267369507034,
0.25565792243175783,

0.2647851751522132,
0.5363821708593047,
0.5363873360294482,
0.7233633669048282,
1.469495303301716,
2.6782106232033085,
3.0738233934797257,
3.0740346152756928,
3.0740346154225127,
3.5401415295205747,
4.384558950933806,
4.384564402805264,
5.187694707345863,
5.431645815108105,
5.489249393218873,
5.489270645046236,
5.4892706450864015,
6.745887429448721,
6.751180124796475,
6.975059797459636,
7.4852253234058015,
7.485225323406553,
7.485225336458803,
8.65422617922401,
9.588402512165866,
9.885475791973308,
10.59890299271783,
10.69671178550786,
1.7193952291225598,
-2.699905058742405,
-1.223597261476082,
1.010311557825714,
2.412320728895392,
0.01205800479020808,
0.07148434641220561,
-0.1717056232391439,
0.7382566227367261,
1.1624421524382496,
-0.9704132592114074,
0.007350927807351759,
2.3582361698297074,
1.1487298862005386,
-0.5298367489795935,
-0.09976600187590766,
0.7543852753586129,
1.401553978295213,
0.3078871075376248,
-0.35588965288410024,
0.15554254602359127,
0.3425968302644435,
-1.202384247417669,
-0.987347192012539,
0.4529373843839761,
0.4903191821294036,
-0.3596344497958935,
-0.17493820576367525,
1.3619720378391473,
-0.2517632245160441,
0.6511229549349082,
-0.07544040749490964,
-0.04771613185909407,
-0.3755018325798389,
0.9938372515787193,
-2.617459009880976,
-1.1600518599196699,
1.385641845308893,
1.3078548392623544,
-1.3438342667259302,
0.06225569718179881,
2.197693302377489,
1.042516443745511,
0.03537145740203309,
-0.4361466628046714,
0.5235506730441506,
-0.526852610863259,

0.2683619694707144,
0.4190531675420984,
1.1917182306749494,
-0.22315973823529625,
0.3724063269005885,
-0.2016113706580773,
-0.6608684665627687,
-0.5566572791655187,
-0.6080637139710359,
-1.3764025932169088,
0.06381643974581075,
-0.12239416020448966,
0.8768915165514672,
0.9734732325585186,
0.43086234391719236,
-1.2868778636436715,
0.2387140460098626,
0.9650786815364466,
-0.8088454790489977,
0.3379873670823351,
-0.5777804082620341,
0.5336495835071627,
2.9920059240641073,
0.6079052241489462,
0.5309531439532562,
-0.3169545414432997,
-2.3335729587340226,
1.4959043429555212,
-0.5568960671660346,
1.653532167337357,
0.35120291709985135,
-1.2609025397237739,
-0.1931706005433399,
-0.7761167208985412,
-0.07661975220377984,
-0.5351965211836152,
-0.0031199845298622592,
-1.0423575145047397,
0.1404336078605159,
-0.2659485743030148,
-1.6615430856264528,
0.43884229259462737,
0.6812026744892391,
0.39460126400851775,
0.9189185317254136,
1.0545864345697524,
1.1559068061784519,
1.0720705313960006,
2.514687033544068,
2.0122769373294087,
1.2260219872848024,
-1.3734724775008675,
2.3904026332666466,
-2.404362121009983,
-0.8308162974608817,
-0.49116355387064226,
1.4544417316540983,
1.2021789553657038,
-0.045966888650276455,
-0.5647889892892385,
0.6643832758564343,
1.3569537910410745,
-0.9183847616510102,
0.945220285616687,
-0.2507148012652629,
0.7511336622169466,
-0.9416337312429676,
-1.4793772681330912,
0.4334847087508488,
-0.02256141084202589,
-0.8421263780740565,
0.7532984525691246,
-0.4880194119115941],
[0.008743077103622827,
0.009882832507877682,

0.7050944318623401,
0.710375902327237,
1.3735081730499452,
1.9102715342756955,
2.4433320146909057,
2.4516882946549723,
2.451688347950266,
2.451688347950339,
2.843659132454273,
2.8437290613274957,
4.1600447531010465,
4.306699526502985,
4.306699526503065,
5.241123177068926,
5.241123178389404,
5.24112317839451,
5.241123178416889,
5.24112317841696,
5.540975568888287,
5.540975570417886,
5.941112811664419,
8.170699047520403,
8.170699047520507,
8.170699047530466,
8.255093472901638,
10.272293989986627,
10.424665220800923,
11.465171522837265,
3.9395345563718758,
0.4691052202571919,
-1.862560190849311,
-0.5409749194015872,
0.6809788356765686,
1.8042992706436805,
-0.26820425656414737,
0.1583592390805078,
2.570139180175488,
1.9730281302693196,
-1.463840299189443,
-0.6685636516618708,
1.7639202171290544,
-0.25593154759594605,
-0.2985776439588054,
-2.380396900296592,
0.702730788267215,
0.1111013414139888,
0.8409172298864358,
-1.2895389501906573,
-0.6495780843509739,
-0.7851296452736322,
0.551111177160635,
-0.7726813911354989,
0.5538939215319534,
0.03671507307592432,
0.3611092546557634,
-1.1244512927039525,
0.9979933722488747,
2.1458567471834726,
0.6045511633065316,
-2.057454443497708,
1.252679002491011,
1.430968651076373,
0.1975347289978355,
-1.299395231227167,
1.3709528213003457,
0.1431677222477642,
-0.5468176167431326,
-1.363138690512715,
3.0723179570358097,
0.09197418207209145,
0.552028513182037,
-0.14805587495457423,
0.6542791830341044,
-0.6463721623376274,
0.594458988807259,

0.36313400085196706,
-0.22487670595801615,
-0.5600947295073319,
-0.13194082373612193,
-0.7877948735702447,
0.37453010501171596,
0.4466259555663157,
1.308811697533367,
0.18300565834449778,
0.28722746055568227,
0.11593248147227052,
-0.22092771987873516,
-0.3830228539100271,
-1.1848305159650747,
0.5197091232696622,
-0.20788874837533922,
-0.6085287569162429,
0.622048119692812,
0.36154402486320714,
-1.283336122338554,
0.06614563071195226,
-0.10338801392549199,
-0.6587068471684263,
1.34100892693596,
1.611824691291148,
0.6916042008757854,
1.6327439292692163,
-0.6648342512722216,
0.37654710496224697,
-0.24718958516166675,
-0.7065038375487233,
0.11746925267221191,
-0.5800210884330035,
-0.5819296114685866,
0.01696046668851879,
0.08164598903007107,
0.39930256073308995,
-0.22041497304878216,
-0.13569368036844073,
-0.734071731705797,
-0.10448732621745443,
0.4350514428273773,
-0.4345336137549378,
0.39154417527507906,
1.0175423593468684,
2.0822612023831226,
2.2518057581506135,
-0.4050083292569126,
-1.3225723626396935,
0.2066177197668252,
0.3518219705658924,
0.7692056892056433,
1.1078718838725619,
0.6955122293015593,
2.6391129283967607,
-0.5193467200233968,
0.16205999751797937,
1.0319480004914923,
0.0466744549888964,
0.8866401912062555,
0.6199118226017315,
0.9594432756697039,
0.30783399339708145,
2.007890427791846,
-0.42290702105473665,
-1.4981000967272742,
0.1246045734342663,
1.4175139007816862,
0.730422807509675,
-1.0203109099547352,
1.2475764531977165,
1.172103834016867,
2.0417923602468857],
[0.3006295792025693,
0.5011825648484141,

1.736075427951676,
1.7894431127664152,
2.4859557870150857,
3.6898895058559917,
3.73290444462633,
3.73290444462633,
3.741728259130309,
4.1342678631525835,
4.145528657970977,
4.615983426127272,
5.6679892124870985,
5.692197651053977,
5.773115344270512,
6.56178194421021,
6.56178194421021,
6.56178194421021,
6.56178194421021,
6.56178194421021,
6.561820835625042,
6.6476324960056985,
7.649631249906491,
8.361638748929934,
8.576867085540188,
8.77036061445696,
9.827286891841206,
10.5379658375992,
10.558826131428928,
11.83004188936984,
1.5987671786815294,
-1.5058826811902317,
2.0440908601375476,
-1.481172398738803,
1.4265067306792671,
1.3375234069756785,
-1.0089917144708827,
-1.8440331252413595,
-0.062239877224771915,
0.18483713323292328,
1.8795821846896128,
-2.058552713287731,
0.9708425302661001,
0.8840527876953188,
0.5964490041287303,
-1.4043070469033054,
0.14533141919970372,
0.022743431858071796,
-0.3923086391326145,
0.11023448615420071,
-0.4204112521530586,
0.5529792020785508,
-0.6330031478534407,
0.23236293228773558,
-0.4459800436834905,
2.5876988326738504,
0.6844024813933239,
0.1732372067049823,
0.7384853056565549,
2.8826092918303563,
-0.11204335589185775,
-0.2241007440171772,
-0.4845273525281558,
0.2744214554588425,
0.26177442491958286,
0.4512467910671607,
-0.6063187184164835,
-1.2386469417803163,
-1.1719897599242013,
1.9705273482015357,
1.945221133626245,
0.15309311442469994,
0.690693149150703,
0.22188036413222653,
0.47722534320651633,
0.12444414704260098,
-1.0057594761976716,

-0.3351811680967541,
1.0750707752291546,
-0.04597654655420585,
-1.4770936089010136,
0.1839119320523196,
1.0453831678930239,
1.6851715686961402,
2.1701829049155683,
1.692817047863365,
0.44152889324253053,
-0.3402272157802259,
-0.4600811596334244,
2.365566181908148,
1.166608300050211,
-0.12230572309949261,
-0.07192631995381693,
-0.48825404466402955,
1.202761995004197,
0.503733093641068,
-0.2613796073125891,
0.7323467617888468,
0.30547819787235264,
0.46207631030173263,
0.034964002783702264,
-0.30773087511026637,
2.2827399003210256,
0.773042400324002,
0.07466204753764587,
-2.812532590652055,
0.7925074856015164,
3.3306642432255886,
-0.8558810509247932,
-0.9283746444488663,
0.3079774400122357,
0.5580467297921388,
0.5588934104639726,
0.6444350594161403,
1.5119717259179621,
0.14573434197560214,
0.7631266139944589,
-0.43452714539522225,
0.38832086457140946,
0.36234082213113916,
-1.3222695581402877,
-0.6140190631377459,
2.2873566508452354,
1.0349767974312705,
-0.5734592440854204,
0.5234040439377924,
-0.8213429806368828,
-0.8621355796854082,
-0.5355268774606846,
1.8804920616270637,
0.022682713405259486,
1.3625464560687972,
-0.48449497738356756,
1.6018376883784673,
-1.3963097787223904,
-3.511711037561943,
1.3223211880663501,
2.1298031669093946,
0.24542774430512565,
0.7034284248706182,
-0.37346698797179273,
-0.43051209966139187,
-0.1034377630705127,
-0.07417862116176438,
0.2595758000914042,
0.9523286668350031,
0.2556898901330524,
0.22603519843353895,
0.9866656966967039,
2.5031147117731094],
[0.6608465720820748,
0.9671982645454077,

1.5458030244804597,
1.5458030244804597,
2.2589946156031733,
2.2589946156031733,
2.2589946156031733,
2.6595607382107107,
3.734699662125929,
4.07592175958283,
4.611631942641649,
4.831287600721991,
4.831287600721991,
5.914025831134952,
5.914025831134952,
6.072969066301945,
6.991999801396327,
6.991999801396326,
7.161687480905546,
7.673022292320241,
7.821921961804373,
7.912193421610441,
8.287160283389671,
8.956173402171897,
8.956173402171897,
9.75268721078065,
9.941478738047602,
9.941478738047602,
9.945621910117605,
10.438880525762482,
1.4707024445054038,
1.5353178524994073,
0.8263465992797864,
0.5104925709601137,
0.0752285958369072,
-0.1142202547489564,
-1.4526020976761953,
-1.3531650678039584,
-0.5250403673431315,
2.237334263415747,
0.2921144645165799,
-1.3764237769851546,
0.48788675650681135,
0.8685067508645514,
0.519063599878773,
1.1970126362568978,
0.35376457746361467,
2.1361591146989944,
1.704664258187609,
1.1956277779681033,
-0.8317273346515459,
1.1961203419780542,
0.6738815365004568,
2.0453196594697167,
-0.3197500197249862,
0.3940149430820266,
-0.03491413991246975,
1.9231424652296574,
-0.7553371587587144,
0.5536882090799309,
-1.1151355813488422,
0.06532487835880797,
-0.03610632383025197,
-0.721099071948806,
-0.7712693743746337,
0.2581737620009136,
0.6969815452179084,
-0.4519889309472695,
-0.6781990341468844,
0.3896669141200773,
0.746036455743935,
0.10534513312025831,
-0.7953867627526213,
-0.17196672908166516,
1.1080637640216904,
-0.36906579226317304,
0.3903518027422727,

0.7340380536410088,
0.16483545086429002,
0.12911772875241279,
-0.11978924907966522,
-0.5410781247103603,
0.27894738356501725,
-0.04603280575778243,
0.3301412555435043,
0.020942359039511207,
-0.535470327679227,
0.26379415288467284,
0.7549681986050077,
-0.774180168996079,
0.3042677541816752,
0.2314702730743271,
0.4347582057672544,
0.5269984906212896,
-0.5883800499089586,
0.4216788732962846,
-0.39154710450095487,
-0.20962806789618654,
0.008988716066278926,
0.42160932452985955,
-0.4362625059483254,
0.7149808650234762,
-0.9606565556866927,
0.02867923202735973,
0.6474453926631074,
0.3125034088891559,
0.24941070182093517,
-1.0996459114603814,
0.42918254303405223,
1.1958230549534523,
-0.23035300079654375,
0.43394127898307777,
-1.1958360530234091,
0.2490392548916114,
0.956337072693143,
-0.9098880983840639,
0.24421397010525564,
0.32228144879356985,
-0.04019115817841665,
0.601865593110944,
-1.0390070893344725,
-0.3870853595246112,
1.5254374970388465,
-0.06482825900051802,
1.508536156596221,
-0.4193438972827081,
-0.2124704995099638,
0.7982743594716017,
0.625924999124879,
-1.2144964531756246,
-0.7306315051046057,
-1.3775043670320717,
-0.5254300493631905,
0.24182169388976885,
0.05618119509583439,
-1.1649033986061088,
0.4414909596355956,
1.1243259379005652,
0.9623501696515389,
-1.0198159738874641,
0.6746700507964939,
0.9071271136722682,
-1.490119256375095,
-1.8915932031257203,
1.4861820000050194,
0.020835967248483293,
0.2672579153437498,
1.1487299094399863,
1.7394956603554965,
2.161406077992875],
[0.009700543149373924,
0.009700543149373951,

0.00970054314937392,
1.382070146567131,
1.7587904521787645,
2.560406827400793,
3.0924393369175114,
3.3853584719898073,
3.3853584719898073,
3.3855655528609647,
3.9151369396989866,
3.9151369396989866,
5.054043379727587,
5.058642146366078,
6.082229999853835,
6.348636465887238,
6.754028454322125,
6.777976106859175,
7.783524996765277,
7.783524996765276,
7.783524996765276,
7.995280883445485,
7.995280883445485,
8.645089634948334,
8.925040343314908,
9.356899631457969,
9.439062141544271,
9.717990345174004,
9.717990345174004,
10.595210012065888,
0.8074121230882934,
-0.11817978440691497,
2.3039735343549332,
-2.1294851108225332,
0.6714134591558454,
0.45516684502263977,
-0.8318799457604262,
-1.6937930072137053,
0.039287662842651484,
0.5223639661903685,
-1.2122069881806565,
0.27186944236208743,
0.22304871394760792,
-0.08855679301585977,
1.1988033553075006,
-0.9660581453773212,
1.393585655420428,
0.524497607950108,
-0.22549193465024678,
-0.20940428560116026,
-0.22546417020605838,
-0.06641646114084936,
0.3531991811801793,
-0.06811222553567531,
1.0762936524340476,
0.17762298521732672,
-0.24437612021854904,
-1.0622868736463833,
0.41092319903165814,
-0.5877006275418265,
1.2299752305066975,
-1.5519910987794148,
0.6657478217748831,
-0.027661876184918333,
0.011781258679136937,
0.9082970931589502,
0.006644189925206108,
0.3712893817061966,
-0.04659784477089089,
0.2991695519878787,
1.4480032246106211,
0.3765628272855095,
0.3810659512210775,
-0.4053198419433054,
-0.06957688452126994,
0.08768201200938007,
-0.16514507742174195,

0.05923670456485792,
-1.0025267882306543,
-0.041476053822733706,
0.4728058870419665,
-0.09799142504381375,
-0.642516889827151,
-0.521645479249954,
0.18062390558973918,
-0.5137105837376863,
0.315588961893638,
1.0102820620809385,
-1.412161058774387,
0.5136468128759858,
-0.5616754178138471,
0.11099281329082958,
-0.4065421947836274,
-0.8520422495977672,
1.5662481725135406,
-2.09542418518098,
1.2116955825236206,
0.8221816705323761,
0.041900809919046114,
-0.09736226691448804,
-0.6600925025551557,
-1.1656564284287168,
0.8214103261157102,
0.28477271669010623,
0.5992804506866763,
-0.21193327493741926,
0.6490050812244267,
0.8872065875645788,
0.2758198961026023,
-0.3259804283588707,
0.5954957550354935,
0.8064107897831347,
0.15673794452304002,
-0.5606863141571317,
1.1354810000650886,
0.29170605524887394,
-0.6471431342469229,
-0.29364595444579666,
0.05289373230476578,
0.03489286830366231,
-0.10405686076243598,
0.4547289014370767,
-0.1747513431047453,
-0.28439942445332933,
0.010516254927139741,
0.11813385237732216,
0.3966163987420368,
-1.37250350754582,
0.06601400397758364,
-0.3662241908629097,
-0.07601408247331909,
1.737776242565087,
2.1849230369400683,
-0.5785032754175823,
-0.5063968590594791,
-0.23335270192508997,
2.320012252653636,
0.6182382779358941,
0.7875200859675175,
-0.009873677663214134,
0.7482624361366613,
-0.35875384806276184,
-0.5357091554695692,
0.4397736267416906,
0.16706548411788932,
0.9251966251074774,
0.053252937203382235,
-0.23592171730002592,
1.2877035765438765,
2.1426518726936172]],

'run_time_sec': 43067.655497312546}}],

'redundant': [{ 'config': { 'ansatz': { 'gateset': 1,

```
'layer': 30,  
'ugate': { 'coefficients': { 'bn': [ 0,  
0,  
0,  
0,  
0,  
0,  
0],  
'cn': [
```

```
0.5,  
0.5,  
0.5,  
0.5,  
0.5,  
0.5],
```

```
        'r': 0},  
        'time': { 'max': 10.0,  
                   'min': 0.0},  
        'type': 'heisenberg'}},  
'init_param': { 'value': [ 0.11853625011402234,  
0.11853625011402236,  
1.1173235551611367,  
1.7342580765110573,  
1.7342580765110573,  
2.108144483620023,  
3.054129947242328,  
3.054129947242328,  
3.054129947242328,  
3.056306360633301,  
4.638414750363569,  
5.0621851887275495,  
5.7503913724004905,  
5.774998407238176,  
6.917933224474368,  
6.917933224474368,  
8.061549209526286,  
8.061549209526286,  
8.061549209526286,  
8.591251503132888,  
8.591251503132888,  
9.051235537435975,  
9.081856762230144,  
9.467876152039107,  
9.545932016874406,  
9.591419209848118,  
10.465616352275608,  
10.465616352275608,  
10.465735173309618,  
10.980942228105905,  
2.0734674059611633,  
0.32892201175743896,  
-0.007356440825095051,  
0.7856576253246221,  
2.089457548986907,  
0.7744884921126718,  
-0.35049309306395837,  
-0.6556831494452968,  
2.0369525201874903,  
0.4739652080570001,  
-0.8564345639983049,  
1.25716179603495,  
1.7181260437331554,  
2.3978160665502384,  
1.1152024575020885,  
-1.3335737674295711,  
0.016650799076572978,  
1.6607129762234538,  
0.648059834763191,  
0.2438725527060928,  
-0.27307896000752346,  
1.4369017198677916,  
3.5672448343800007,  
-0.5379702113035718,  
-0.7420664562717484,  
-0.529699891930992,
```

0.7301471483152846,
0.781913925757813,
0.6004078048346045,
1.7324865479818374,
0.4548236607944829,
-1.4419813608219112,
-0.684662976785234,
0.8455512814500156,
1.97751114408285,
-0.833202517811791,
1.110524460863052,
2.6403320691876733,
-0.3129427957408566,
-0.06674204626492235,
0.5842857778318762,
-0.21132485649138796,
-0.011842019288790376,
-0.25743970610802486,
0.7486616881233721,
-0.1302029141156864,
0.09497091477412227,
-0.3715265429351211,
0.39570077921274605,
0.28643868832003017,
0.7848413456471106,
-0.7157146817058562,
0.09411764456438232,
-0.10910076647953601,
1.1262761892344808,
-0.27334997823872975,
-0.44027652197942063,
0.6740415100156434,
-0.336256645457715,
0.4268157265313067,
-0.8611696251566416,
-0.34560928156726306,
0.5618074077714486,
0.5866360379174992,
1.8662876827821717,
2.21592310898005,
1.972223656383108,
1.6141187926643297,
0.856255815565121,
-0.34275929095614754,
-1.0642528745182613,
-1.0618686751734654,
0.2379582550153795,
-0.6501770361983872,
-0.19408645485522444,
1.501330829125914,
1.5726602727591479,
0.7066066097903853,
-1.4580936665368467,
0.46940440748349077,
-0.1287015677341077,
-0.9805925033056946,
-0.3754741669636888,
-0.026817919501009202,
1.1067346586651607,
0.17547923725022047,
0.8902143171896091,
-0.22777698610446503,
0.8082253583749343,
-1.0674083601412958,
-0.022706971436593337,
0.34227899244110566,
1.3527736523760083,
2.0767841716125894,
0.5778568145668401,
-0.6555502504392442,
1.2670274746873995,
0.507694022232518,
-0.2629991187543805,
1.7048253080726326,
1.4088065522309612,

```

2.823918413203317,
1.2763232744235296,
-1.4348344213353486,
-0.605496445996402,
0.09645158262695042,
0.024639736182597342,
0.23997550724405606,
0.9979895378690888,
-0.40612123795993227,
1.826790650755685,
0.029599395583551916,
-0.4592861834683686,
0.0328358806035505,
1.3612120706287358,
0.8210402794880411,
0.05688164146575431,
0.03152531580318861,
2.0316547976974904,
2.795039607579165]],
'noise_profile': { 'noise_on_init_param': {
'value':
'noise_prob': [ 0.001,
0.001,
0,
0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_ric3_unifrom_scaling_sample#10'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '
'2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '
'2 + 1 X 3 X 4 + 1 Z 3 + 1 X '
'4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
'1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X '
'0 X 1 '
'+ 0.5 '
'Y 0 Y '
'1 + '
'0.5 Z '
'0 Z 1 '

```



```
'+ 0.5 '
'X 1 X '
'2 + '
'0.5 Y '
'1 Y 2 '
'+ 0.5 '
'Z 1 Z '
'2 + '
'0.5 X '
'2 X 3 '
'+ 0.5 '
'Y 2 Y '
'3 + '
'0.5 Z '
'2 Z 3 '
'+ 0.5 '
'X 3 X '
'4 + '
'0.5 Y '
'3 Y 4 '
'+ 0.5 '
'Z 3 Z '
'4 + '
'0.5 X '
'4 X 5 '
'+ 0.5 '
'Y 4 Y '
'5 + '
'0.5 Z '
'4 Z 5 '
'+ 0.5 '
'X 5 X '
'6 + '
'0.5 Y '
'5 Y 6 '
'+ 0.5 '
'Z 5 Z '
'6']],
'output': { 'data_points': [ [4, 2, 0, 0, -6.28130195722698],
[12, 6, 0, 0, -4.778373581659957],
[ 20,
10,
0,
0,
-3.6190179833186162]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [ 0,
0,
0,
0],
'noise_level': [4, 2, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0,
0],
'status':
True,
'type':
'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [ 1,
1,
0,
0],
'noise_level': [12, 6, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0,
```

```
0],
True,
'depolarizing'},

'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0,
                0],

True,
'depolarizing'},

'odd_wires': 4}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [
0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': { 'max': 10.0,
           'min': 0.0},
'type': 'heisenberg'}}},
'init_param': { 'value': [ -2.397634216504411e-17,
1.429450133825379e-17,
0.9192297980258759,
0.9192297980258759,
2.8984599305110854,
3.1993874679766523,
3.199387467976652,
3.715279848679994,
3.715919998165357,
5.126008182976459,
5.749875525815113,
5.785995839176617,
5.785995839176617,
5.785995839176617,
5.785995839176617,
6.315437962379428,
6.678845841951143,
7.3578167224936415,
7.415694801781823,
7.531263738212295,
7.949087697575632,
7.949087697575632,
8.111169110068786,
9.407094598577077,
9.495286031274873,
9.495286031274873,
10.993257661103009,
11.063439993331317,
11.639888600691357,
11.75175692668691,
```

2.1152038588685538,
-0.7839988468841306,
-1.0579861702584952,
-0.5122509123294313,
1.0012369417099964,
0.9275037440003397,
0.5343037326701707,
-0.014459274586663672,
2.31692048955396,
2.1400689678453872,
-0.9468421716760329,
1.3225342455296385,
2.4229762260600483,
1.4561600175543838,
-0.4592924007287594,
0.6389204073366491,
0.5578650700802136,
1.458345504678432,
-0.11931303728638508,
0.04406428844113783,
-0.26144087074192934,
0.6459890964472403,
1.2243061606997268,
0.09770103868863204,
0.23645607828472368,
0.3280111145811677,
0.5390648705936556,
-0.6034481934191781,
-1.1942560416831098,
0.3295059316644399,
-0.0732996553139216,
0.5043071925129845,
0.7399437462813884,
0.3595686242698361,
1.4643835231854074,
-0.1697776944282527,
0.2588380732750666,
0.44311592273109757,
-0.6031987466514527,
-0.06131410155451245,
-0.3299395710828798,
-0.45492250592824107,
0.5555861976799495,
1.1612177689175078,
1.833550163228831,
1.5629483318007886,
-0.2455585474033218,
-0.4534768786824923,
-1.2225865843307935,
0.20468879325769473,
1.3398561861341207,
-0.58709602862573,
0.07181500566168061,
0.17010271339345,
0.36406988692952136,
-1.689809967880289,
-1.1677426672173294,
-0.7977758879892671,
0.531032719362766,
-1.7722607960867016,
0.6332033998224597,
0.6669814362912183,
0.218182728116887,
1.0415656581640258,
-1.4544893086294475,
0.5175174925258392,
-0.9742907090362619,
-0.30648595673224416,
1.2400541074634985,
-0.30003161306087445,
-0.14406503395842435,
-0.43061285539013316,
0.6914697219419143,
-0.48545379734819916,
-0.2166375004086177,

```

1.2014709434123243,
1.0734635496959384,
0.43666289070672293,
-0.9550440807047608,
1.2817732673505913,
0.6254203972471953,
0.8730111716749767,
1.1814458389257443,
-0.5855214911540807,
0.33647224367417283,
0.914891547403245,
-0.10924028322724566,
0.6731472714644756,
-0.8672547313943412,
-1.1628903969680444,
1.2968110405185924,
-0.7478342770444015,
1.0157432273471354,
0.05866260951598884,
0.1549884083832883,
2.402954000164725,
-0.8510405490988947,
-0.13394876321797194,
0.35128800256256376,
0.03985349181110424,
1.0891408951797659,
0.49683980443602116,
1.3618410332836905,
0.31089249400158553,
0.007823999182329802,
-1.0325531315281817,
-0.13635517978577905,
-0.4783006843261583,
-0.9648987725589863,
-0.4094326055257113,
0.0620151507481985,
-0.3631850012398663,
-0.9119123598789223,
-0.3458539889181142,
0.8501459643039054,
1.6461875633189593,
0.08799287143090098,
-0.82318299513447,
1.17289644669955,
-0.9455811832961388]],
'noise_profile': { 'noise_on_init_param': {
'value':
'noise_prob': [ 0.001,
0.001,
0,
0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':

```



```

0],
'noise_level': [4, 2, 0, 0],
'noise_profile': {

'status':

'type':

'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [ 1,
1,
0,
0],
'noise_level': [12, 6, 0, 0],
'noise_profile': {

'status':

'type':

'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
'identity_factors': [ 2,
2,
0,
0],
'noise_level': [20, 10, 0, 0],
'noise_profile': {

'status':

'type':

'odd_wires': 4}},
'run_time_sec': 7.639997243881226}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [

'r': 0},
'time': { 'max': 10.0,
'min': 0.0},
'type': 'heisenberg'}}},
'init_param': { 'value': [ 0.12420123662801165,
0.138246244841176,
0.34185010800102933,
0.8771280515547424,
```

0.8771280515547423,
1.9326762312100456,
1.9326762312100456,
2.879806034127527,
3.4467495190209787,
3.460588109395402,
3.4605881093954016,
3.4605881093954016,
3.911934955717517,
3.9564836804547703,
4.188240551814426,
5.098640553346409,
5.098640553346409,
5.098640553346408,
6.466332587579958,
7.457565848827182,
7.712724508543908,
7.890302861063261,
7.898577524812402,
8.020037295803547,
8.020037295803547,
9.289723192898348,
9.68251578824504,
9.68251578824504,
10.66694519433398,
11.464898013549973,
1.8095532965205083,
-1.6633278493215808,
-1.2281091611401596,
-0.40155726479865544,
1.7903243850816337,
0.18661159996017582,
1.2217850039169107,
-0.48128156078104645,
1.438546033666569,
-0.9466383819139143,
-0.2179227006066069,
-0.07921201994836616,
0.6692914930417907,
-0.16791294486623565,
-0.3278463005181308,
1.2883193159418245,
1.9001761172388045,
0.7646920019853412,
-0.6325477821570596,
-0.5987552945744136,
0.5669730115381587,
0.5983941343774113,
-0.06179171362000896,
-0.6867235859693906,
0.5211737756628548,
-0.19354588250953578,
0.6072816218147872,
0.3472677197742769,
-0.08031734075212016,
1.4024126217748347,
-0.5867181790012919,
0.809620813948269,
0.35997425529443994,
-0.8722967249300235,
-0.5874818244923429,
-0.5567926036837068,
-0.6652314874983639,
-0.6042054992483096,
-0.13918430829641074,
-0.08277488470584692,
-0.4972293169762719,
-0.4421668735352916,
0.6160601456858428,
-1.1972657609969646,
-1.1558488364748807,
-1.4501782622641872,
0.03989088007781949,
-0.8684923327267209,
-0.5569704341339503,

0.15823054121075,
0.6345578215095505,
0.3601853745371903,
-0.42594808878819745,
0.08979804115834313,
-0.153043281560756,
-0.603228229875778,
-1.1264410002736531,
0.6895580235946861,
-0.11045941959152497,
0.5174109375139814,
1.3269460586908306,
-0.3852298542423827,
-2.4120095755015893,
0.08807412161753436,
0.1228551365077638,
-0.6982591234727157,
-1.1618488040328845,
2.5586909942385017,
-0.24921574033555105,
-1.9870969070050777,
-1.0279498295884981,
-0.7796802357622239,
0.48428681889730063,
0.7162815161060118,
-0.15547297036103372,
1.0522591887013215,
1.772822944479854,
-0.2654037211186137,
-0.2720110866721671,
-0.03929478442656517,
0.34606077605631835,
0.5533331494282367,
1.0574846609095763,
-0.5747611578732846,
0.4916772893793279,
0.9875709691916216,
-0.17857479553023695,
-1.147281220275154,
0.37370382177065037,
2.456815155407461,
0.6567596591037266,
0.9507316292553318,
1.5113928877114242,
1.8380989093110256,
-0.1736919900840187,
-1.2666253947041328,
-0.554819304070435,
1.1931643666366285,
0.4614862178419797,
1.2040726557184918,
0.3083618234293392,
-0.4066692475369416,
0.3740368412278018,
-1.684611974494111,
-0.8650016725760176,
-1.4838330341878547,
-0.14344887350922086,
4.203596189627334,
1.1133202614474196,
1.391124117957397,
1.8627332105499828,
-0.28192336755973296,
-1.5416482647725873,
-1.9612892990030817,
0.8836928087418303,
0.300761821046295,
0.6785848024508535,
-0.5500277556546446,
0.5742776326882599,
-1.181756808209963]],

'noise_profile': { 'noise_on_init_param': {

'status': False,

'value':

0},

[illegible]

[illegible]

```
        'odd_wires': 4}},
        'run_time_sec': 7.7689526081085205}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [
```

```
0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
```

```
        'r': 0},
        'time': { 'max': 10.0,
                  'min': 0.0},
        'type': 'heisenberg'}},
'init_param': { 'value': [ 0.16477895719846114,
                           0.16477895719846108,
                           0.7183189156962347,
                           1.8697093861419474,
                           2.105082615571597,
                           3.05308344502201,
                           4.199393321043376,
                           4.199393321043376,
                           4.199393321043375,
                           6.490210948425593,
                           6.4902109484255925,
                           6.490210948425593,
                           6.4902109484255925,
                           6.4902109484255925,
                           6.490210948425593,
                           6.882554174728105,
                           6.882554174728105,
                           6.882554174728105,
                           6.882554174728104,
                           6.8877583035732775,
                           6.8877583035732775,
                           6.8877583035732775,
                           8.21014824639056,
                           8.21014824639056,
                           9.071469420716642,
                           9.120928605341893,
                           9.714944098401723,
                           11.474131735950095,
                           11.474131735950095,
                           11.794551582273902,
                           1.341188628911811,
                           1.6554894539990426,
                           -2.1068974621953425,
                           -1.4171475675653578,
                           0.33211669069513333,
                           0.11331792925356984,
                           -1.3610911099963097,
                           0.46439741779020816,
                           0.9313501387866817,
                           1.0938593094543279,
                           -0.3591727738608774,
                           -0.4732622270970327,
                           1.6321598874818157,
                           -0.2782091348514778,
                           -1.0526781349295347,
                           0.0708703831522549,
                           2.6982701224663113,
                           -0.038867313655444914,
                           -0.10508583152652556,
                           -0.40759365792756236,
                           0.6284849004330764,
                           0.5192751929971827,
                           -1.6282040637266806,
```

0.0014422433007245106,
-1.8299719264591412,
2.3182597876656823,
0.8875167319646391,
1.5664268415036142,
1.5052778791259855,
2.935955436800565,
1.533101581959507,
1.8091829332960188,
0.45080966818917967,
0.1506883627647002,
-0.6843548866459184,
-0.5544883854492364,
1.2466562002431278,
-1.1866110136199353,
1.4521358231995756,
-1.125608676420736,
-0.24054383618816966,
-0.058272145415006006,
0.410408128597049,
-1.6441753460137252,
0.4019816715685168,
0.4588365987304068,
1.354346078964423,
-1.7153251360741875,
-1.1039978646265602e-06,
0.7940670488104785,
0.17979772520560053,
-0.24385394325110826,
0.6367160741723578,
0.8500459957371875,
0.0559416907316065,
0.32418668587130467,
-1.4797636873635434,
0.11902177942387202,
0.5185165439317966,
-0.08399222686641401,
0.8650918037299421,
1.4911587670088011,
0.12792248933645675,
0.3372262149750348,
-0.6553154954729696,
1.1995792367396254,
0.5132294030998703,
0.44835115860783636,
-0.37116980512777586,
0.7805194563374558,
-0.854334789795377,
0.5008527672922033,
-0.2302685629030402,
0.6145182391921147,
-0.6575500079215796,
0.8657035637529482,
1.1874516653159721,
-0.16284802367068918,
-0.26462171656665856,
-0.9815102056659819,
0.23198397744948207,
-1.087097195512072,
0.6616385654966559,
-0.2929637257774683,
0.03471390505932934,
0.2048751570049573,
0.9170017118070749,
0.39617956766989404,
0.12006199846995962,
0.1922705913940127,
0.8784478594386566,
0.055479957240883566,
-0.6634105431569992,
1.0742951419164057,
-0.37818200958777604,
0.28548758229245597,
-0.6594827056613419,
1.0235832498557866,

```

-0.19428253177041227,
-0.5531991468121416,
-1.189626464718852,
0.2519110119226656,
-0.6708462316627377,
2.7994352657547057,
-3.8764436680175813,
0.5152327138619494,
-0.3749347648847944,
2.9042167722097862,
1.1265965037549552,
-1.850959186134046,
2.165374611027145,
0.8818414791702859,
1.5039524911129019,
0.06564767423285973,
0.6486215314149069,
-0.08854143781823305,
1.8396223528529987,
-0.00792246728000446,
2.8805879469479487,
3.6337353861266473]],
'noise_profile': { 'noise_on_init_param': {
'value':

'noise_prob': [ 0.001,
0.001,
0,
0],
'status': True,
'type': 'depolarizing'}},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_ric3_unifrom_scaling_sample#9'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '
'2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '
'2 + 1 X 3 X 4 + 1 Z 3 + 1 X '
'4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
'1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X '
'0 X 1 '
'+ 0.5 '
'Y 0 Y '

```

```

'1 + '
'0.5 Z '
'0 Z 1 '
'+ 0.5 '
'X 1 X '
'2 + '
'0.5 Y '
'1 Y 2 '
'+ 0.5 '
'Z 1 Z '
'2 + '
'0.5 X '
'2 X 3 '
'+ 0.5 '
'Y 2 Y '
'3 + '
'0.5 Z '
'2 Z 3 '
'+ 0.5 '
'X 3 X '
'4 + '
'0.5 Y '
'3 Y 4 '
'+ 0.5 '
'Z 3 Z '
'4 + '
'0.5 X '
'4 X 5 '
'+ 0.5 '
'Y 4 Y '
'5 + '
'0.5 Z '
'4 Z 5 '
'+ 0.5 '
'X 5 X '
'6 + '
'0.5 Y '
'5 Y 6 '
'+ 0.5 '
'Z 5 Z '
'6']],
'output': { 'data_points': [ [4, 2, 0, 0, -5.966171999362981],
                              [12, 6, 0, 0, -4.4917214032040365],
                              [ 20,
                                10,
                                0,
                                0,
                                -3.364633070617193]],
  'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                       'identity_factors': [ 0,
                                              0,
                                              0,
                                              0],
                       'noise_level': [4, 2, 0, 0],
                       'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0,
                0],

True,

'depolarizing'},

'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [ 1,
                        1,
                        0,
                        0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},

```

```
'noise_prob': [ 0.001,
               0.001,
               0,
               0],

True,

'depolarizing'},

'noise_on_init_param': { 'status': False,
                        'value': 0},

'noise_prob': [ 0.001,
               0.001,
               0,
               0],

True,

'depolarizing'},

'odd_wires': 4}},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [ 2,
                       2,
                       0,
                       0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {

'status':

'type':

'odd_wires': 4}},
'run_time_sec': 7.615773439407349}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [ 0,
                                                             0,
                                                             0,
                                                             0,
                                                             0,
                                                             0,
                                                             0],
                                                           'cn': [

0.5,
0.5,
0.5,
0.5,
0.5,
0.5,
0.5],

'r': 0},
'time': { 'max': 10.0,
          'min': 0.0},
'type': 'heisenberg'}}},
'init_param': { 'value': [ 0.43940602753254654,
                           0.4524980474487914,
                           0.6462847510297367,
                           0.6535145711164267,
                           0.7019965052930044,
                           1.0830061860275473,
                           1.4380021973026182,
                           2.1860837771467474,
                           2.1860837771467474,
                           2.1860837771467474,
                           2.4875804812430995,
                           2.911818356328843,
                           4.9134595358452735,
                           5.043089843653786,
                           5.994354774193697,
                           6.658321281734746,
                           6.666230857319696,
                           8.458454027735183,
                           8.458454027735183,
                           8.458454027735183,
                           8.938931772570164,
                           8.938931772570164,
                           9.876451730548837,
                           9.876451730548837,
                           10.140618079102744,
                           10.453495471248813,
                           10.878653713196195,
```

11.282429749250388,
11.726474998572407,
11.93229290545428,
2.9640535214497064,
-0.6206168106317247,
-0.4035747848788786,
-2.067222048924027,
2.0576259596946604,
1.364315765000901,
-0.1480698866383175,
-1.95043906046908,
0.7897528850889669,
1.350582507885341,
-0.3349044763190947,
-2.3429833820345465,
2.04351680973891,
0.8083219796841106,
-0.06520770630024467,
-0.8776318722364843,
2.4862506381076748,
0.8345121096737285,
-0.03071661620417745,
-0.6225183255138825,
0.7450777062518694,
0.12619297185303674,
0.15620862918286052,
-0.47540100870445906,
0.6345315438244892,
1.451076347224231,
0.10985833383721234,
-1.6475696409761669,
1.1918822544398533,
1.3235539868194213,
-0.24222060474147966,
0.2418944215339802,
1.0397651583041008,
1.6773252160624976,
0.16975786500059598,
-0.9211027317156172,
0.7590471244619662,
1.76540125315151,
2.43856734239768,
-0.19488067618654012,
0.8462629124439717,
0.9979674075340036,
2.5775834810620677,
0.5650956422366618,
-0.5475934153287456,
-0.0253449469292375,
1.2524069066350432,
-0.605696708865831,
0.7019502861337156,
0.2803153244338447,
-1.0831925759777479,
1.0082140101851143,
-0.3762046176346905,
-0.9535933533256993,
-1.0422804614435257,
0.13948594332192887,
0.5835070530053111,
0.16086524965091817,
0.06403444071667018,
-0.06194188553710435,
-0.4026223625881385,
0.2750028468648149,
0.6789909068622377,
0.05162849323802493,
-0.7735332926339995,
-0.14862918062202837,
0.7661757453229759,
-0.16119730427373616,
0.6482389226852195,
3.1459456036848104,
-0.4760920139337192,
1.6197254159790668,


```

-0.5506328506738539,
0.9968299487743895,
-0.3106735286083523,
-1.2388662013594425,
0.8898580979157581,
0.43256773745092003,
-0.7272220041247079,
0.8716071923554094,
1.5193589139179793,
0.9787540447681606,
-1.4044929205930872,
-0.1984876316951887,
-0.46744984056178585,
-0.7765225748074835,
0.04344696088727837,
0.32373965212791267,
0.9049555024172572,
0.7139278873651953,
-0.8122282267372439,
0.5323405298774778,
-0.08437719753251166,
-0.14277007957413837,
0.7912147308716768,
0.26055493673650715,
0.11357082745857092,
0.3153698930365579,
3.1981441448638614,
0.32109035951393206,
-0.23017443442945684,
0.0316864124540797,
0.20875812581978664,
0.5466872162452653,
1.2918521189433907,
-0.7312547980705957,
1.4958457475939604,
-0.053193686203739404,
-0.4437772137226961,
-0.5271126400646005,
2.5136810835026426,
0.6674244883024821,
-1.0140291771816496,
-1.177197116527103,
2.058315434744553,
0.050372102721390984,
-0.09022974351195023,
-2.296438941834702,
0.3261688389091777,
0.3072223897800251]],
'noise_profile': { 'noise_on_init_param': {
'value':
'noise_prob': [ 0.001,
0.001,
0,
0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,

```

```

        'status': False,
        'type': 'png'},
        'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_ric3_uniform_scaling_sample#1'},
        'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                                [1, 1, 0, 0],
                                                [2, 2, 0, 0]]},
        'run': 'redundant',
        'state': 'dmatrix',
        'vqe': { 'iteration': 1,
                  'optimization': { 'algorithm': 'SLSQP',
                                    'constraint': True,
                                    'status': False}},
        'zne': { 'data_points': None,
                  'degree': 0,
                  'method': 'richardson',
                  'sampling': 'default'}},
        'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '
                                           '2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '
                                           '2 + 1 X 3 X 4 + 1 Z 3 + 1 X '
                                           '4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
                                           '1 Z 5 + 1 Z 6',
                    'time_evolution_gate_hamiltonian_string': [ '0.5 X '
                                                                '0 X 1 '
                                                                '+ 0.5 '
                                                                'Y 0 Y '
                                                                '1 + '
                                                                '0.5 Z '
                                                                '0 Z 1 '
                                                                '+ 0.5 '
                                                                'X 1 X '
                                                                '2 + '
                                                                '0.5 Y '
                                                                '1 Y 2 '
                                                                '+ 0.5 '
                                                                'Z 1 Z '
                                                                '2 + '
                                                                '0.5 X '
                                                                '2 X 3 '
                                                                '+ 0.5 '
                                                                'Y 2 Y '
                                                                '3 + '
                                                                '0.5 Z '
                                                                '2 Z 3 '
                                                                '+ 0.5 '
                                                                'X 3 X '
                                                                '4 + '
                                                                '0.5 Y '
                                                                '3 Y 4 '
                                                                '+ 0.5 '
                                                                'Z 3 Z '
                                                                '4 + '
                                                                '0.5 X '
                                                                '4 X 5 '
                                                                '+ 0.5 '
                                                                'Y 4 Y '
                                                                '5 + '
                                                                '0.5 Z '
                                                                '4 Z 5 '
                                                                '+ 0.5 '
                                                                'X 5 X '
                                                                '6 + '
                                                                '0.5 Y '
                                                                '5 Y 6 '
                                                                '+ 0.5 '
                                                                'Z 5 Z '
                                                                '6']},
        'output': { 'data_points': [ [4, 2, 0, 0, -5.966837375471956],
                                       [12, 6, 0, 0, -4.6298501952375855],
                                       [ 20,
                                        10,
                                        0,
                                        0,
                                        -3.5807206753108627]],
                    'noise_details': [ { 'gates_num': [4, 1, 1, 0],

```

```
'init_param': { 'value': [ 0.05238332857976487,
```

0.05238332857976488,
0.05238332857976461,
0.5882470284735626,
1.6186245942059996,
2.2470777294271223,
2.8315596913723913,
2.8315596913723913,
2.8333251330604914,
3.7474015252366555,
3.7474015252366555,
3.7474015252366555,
3.7474015252366555,
4.452994588443841,
4.625910007504636,
4.625910007504636,
5.314362007876739,
7.185299991262795,
7.185299991262795,
8.165312898322489,
8.165312898322489,
8.165312898322489,
8.684979613482101,
9.661121823219345,
9.674169964387936,
10.525655511340814,
10.525655511340814,
10.525655511340814,
11.154963299169733,
11.154963299169733,
2.50282655917642,
0.9490947670548783,
0.70056532212722,
-0.9370006834105179,
2.127395945212884,
1.5529950009292526,
-0.5538339242405286,
-0.45109357069990613,
3.2608593101153462,
2.1691098922881027,
0.924351518772188,
0.7799372105219657,
2.588492585292434,
1.4914484649202893,
0.5111989894903198,
-1.6854828839595362,
0.709524102178921,
-0.3201993643863164,
0.37756499567776974,
0.11160463007405702,
1.4506751210554825,
-0.1158349928449886,
-0.4172082421780774,
-0.47420587439641587,
2.1318004152619214,
1.5026342770773617,
0.9629412747697549,
-1.017935070811068,
2.2274033566257083,
0.9541631607884562,
2.2865374842341715,
-0.6901760390806929,
1.4488496207691695,
-0.15529093800821045,
1.378510821815928,
-0.1238908488185843,
1.0420944500710088,
1.8938950235205334,
-0.40270933293735856,
-0.4838468076888962,
-0.7769068081513464,
2.0570355933263196,
0.33843722154840217,
-0.21296684820549192,
-0.3087457919151385,
1.6713665097762895,

0.7955513549984788,
-0.5031484152839188,
0.2783608989762586,
1.4322280472670237,
0.12038812465567636,
0.8648150791824706,
0.13552183280667268,
0.07182610915111512,
0.3477791234991623,
-0.1584859230597403,
0.5851519083687279,
0.578565105996425,
-1.7921034519484391,
0.9189192684840749,
-0.6139572052593133,
-0.023527142002935514,
-0.6495330854871428,
0.7767670181361933,
0.13616638059771954,
-0.22568119775616352,
0.4192866974084779,
0.1401009023541104,
1.419355971488301,
1.1103093112149183,
-0.8728413553259085,
1.014691351510677,
2.35757700845783,
1.131891326153365,
-1.2979835358031357,
0.42365777146015526,
1.0982786992551208,
1.698510255572029,
1.2090506157632,
-1.3597097700308411,
-0.9049577551064396,
1.0199036275847497,
2.9541844671388584,
-0.42829772740772204,
0.6542829736223691,
3.7916586259391143,
-0.1983670801457003,
0.7855289833525587,
-0.1534574040841844,
1.809122493112309,
0.35293394466185096,
1.0573849517470935,
2.461665111905985,
1.0784030177718114,
-1.585709677044719,
0.8985620342798133,
2.2360691692951646,
0.24022620726925167,
2.6922129347449117,
0.15599094623602128,
2.799936411115923,
0.9657316063780467,
-0.2732772147086647,
-1.448572189075712,
-0.7280589250410919,
0.057973620624421296,
0.9321748335747359,
-1.4338621772181848,
1.3976338029473028,
-0.25504403617708243,
3.210257057781798,
0.03448775329128267,
0.05070056715723011,
-1.3727823101385919,
0.8243579770218447,
0.13142013414460954,
-0.4933237632531999,
1.1105137227416793,
0.7155364379215329,
-0.21063252058278284]],

'noise_profile': { 'noise_on_init_param': {

'status': False,

'value':

0},

'noise_prob': [0.001,
0.001,
0,
0],

'status': True,
'type': 'depolarizing'},

'nqubits': 7,

'observable': { 'coefficients': { 'bn': [1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},

'def': 'ising'},

'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':

'AUTOMATE_heisenberg_noisefree_time_evo_ric3_unifrom_scaling_sample#2'},

'redundant': { 'identity_factors': [[0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},

'run': 'redundant',

'state': 'dmatrix',

'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},

'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},

'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '
'2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '
'2 + 1 X 3 X 4 + 1 Z 3 + 1 X '
'4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
'1 Z 5 + 1 Z 6',

'time_evolution_gate_hamiltonian_string': ['0.5 X '
'0 X 1 '
'+ 0.5 '
'Y 0 Y '
'1 + '
'0.5 Z '
'0 Z 1 '
'+ 0.5 '
'X 1 X '
'2 + '
'0.5 Y '
'1 Y 2 '
'+ 0.5 '
'Z 1 Z '
'2 + '
'0.5 X '
'2 X 3 '
'+ 0.5 '
'Y 2 Y '
'3 + '
'0.5 Z '
'2 Z 3 '
'+ 0.5 '
'X 3 X '
'4 + '
'0.5 Y '
'3 Y 4 '

```

'+ 0.5 '
'Z 3 Z '
'4 + '
'0.5 X '
'4 X 5 '
'+ 0.5 '
'Y 4 Y '
'5 + '
'0.5 Z '
'4 Z 5 '
'+ 0.5 '
'X 5 X '
'6 + '
'0.5 Y '
'5 Y 6 '
'+ 0.5 '
'Z 5 Z '
'6']},
'output': { 'data_points': [ [4, 2, 0, 0, -6.220556999324399],
[12, 6, 0, 0, -4.682225394272194],
[ 20,
10,
0,
0,
-3.4953983793015833]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [ 0,
0,
0,
0],
'noise_level': [4, 2, 0, 0],
'noise_profile': {

'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0,
0],

'status':

True,

'type':

'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [ 1,
1,
0,
0],
'noise_level': [12, 6, 0, 0],
'noise_profile': {

'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0,
0],

'status':

True,

'type':

'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
'identity_factors': [ 2,
2,
0,
0],
'noise_level': [20, 10, 0, 0],
'noise_profile': {

'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0,
0],

'status':

```

```
True,
'depolarizing'},
                                'type':
                                'odd_wires': 4}},
                                'run_time_sec': 7.230367183685303}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [
0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
                                                                'r': 0},
                                                                'time': { 'max': 10.0,
                                                                'min': 0.0},
                                                                'type': 'heisenberg'}}},
'init_param': { 'value': [ 0.38848003251483587,
                           0.3884800325148359,
                           1.2841735835398218,
                           1.284173583539822,
                           1.2841735835398218,
                           1.818138220668234,
                           1.818138220668234,
                           2.7575469841005327,
                           2.7575469841005327,
                           2.7575469841005327,
                           3.0053468968599275,
                           3.0053468968599275,
                           3.005346896859927,
                           4.184079661691832,
                           4.83635165164968,
                           4.83635165164968,
                           5.97926077140412,
                           6.458805266455043,
                           6.513102391504654,
                           6.513102391504654,
                           7.122105903846713,
                           7.2188422331450415,
                           7.988590448127284,
                           8.268095077127937,
                           8.268095077127937,
                           9.164456488648115,
                           9.164456488648115,
                           9.782953992889732,
                           9.782953992889732,
                           10.467486029517685,
                           1.5496959718296535,
                           -1.0300437645832343,
                           0.425343365508412,
                           -1.7136674729251273,
                           0.33645197441382346,
                           0.2288357961575219,
                           0.10470230685228102,
                           -0.6612661793744108,
                           0.11823315872480306,
                           0.48571962162230753,
                           -1.3748501535311368,
                           0.2563120795974374,
                           -0.38871520150353267,
                           -0.18786175076151226,
                           1.0017996241286973,
                           0.05598903629217323,
                           1.501328526357405,
                           -0.32299727271195017,
                           0.5715295079657807,
                           -0.12201961934486968,
```


2.2747823054657927,
0.16289086097520028,
0.027619518366160244,
-0.10165606616500937,
-0.06321979816820887,
-0.27543819802246033,
-0.0048370064103128475,
0.1330138167350915,
-0.06428151371988705,
0.21905201010909556,
0.25697175483980955,
-1.462905966720411,
0.8014511304369155,
3.456137165485204,
-0.3560002683832885,
-0.4886877133354182,
0.32039145066707037,
1.6153352350920127,
0.5940099487759896,
0.18140087044059483,
0.27123106498617,
0.7444341922951362,
0.170480557625751,
-0.4688462519246396,
0.5235839184929864,
-0.5470964469389679,
-0.011921500492307424,
1.4116230981753706,
0.49972453659747124,
-0.7800673920851233,
0.4234227404504273,
-0.2071635265929523,
0.14064328933496192,
-1.2007096247825992,
0.12047098237640114,
-0.4125103841815656,
1.0447620953563077,
0.0020814149928196773,
0.4988991062112368,
-0.34495773827123155,
-0.12849063482979717,
0.4280934542561602,
0.7138444629946918,
-0.6497712710787925,
-0.43652659674625793,
0.917432486791049,
-0.8523477361785018,
0.5648987974980545,
0.3183226113969618,
0.14295784675778261,
0.9589824340171335,
0.9194478619356266,
1.1001359353915339,
1.5417377267389083,
-0.5376003771957187,
-0.5404528215706331,
0.21624687748210675,
0.8814937279455825,
-0.371639532103812,
-0.2668100917972658,
1.3133163210428074,
0.44574090697479807,
-1.0940261852319737,
-0.313830643067749,
-0.06100275437545863,
-0.049560783686653447,
-0.3025056724120473,
-0.9891529613817422,
-0.35099332176432035,
0.5436069706516342,
-0.722828417704955,
1.6968661444846564,
-2.4402971962208975,
0.7865745542588627,
1.3437881169309889,

```

0.36108471326554326,
-0.1749159211919777,
-0.014334278530919132,
0.02978377629472008,
1.8533978555002608,
-1.8470426695738211,
0.6272356375919823,
0.44775499409400255,
1.3281153539688288,
0.4583217606753929,
0.6443670392327268,
0.5254233581973354,
2.3087652110195127,
-0.7761319091913014,
-0.7605421959233766,
-0.6461580464596497,
0.7075597766411752,
0.985778508379752,
-1.3130974173457381,
2.8711331564400684,
1.2993543284582694,
1.9493614761217355,
1.1647493000584068,
-0.42731613935458074,
2.7038346000669766]],
'noise_profile': { 'noise_on_init_param': {
'value':
'noise_prob': [ 0.001,
0.001,
0,
0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_ric3_unifrom_scaling_sample#6'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '
'2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '
'2 + 1 X 3 X 4 + 1 Z 3 + 1 X '
'4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
'1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X '

```

```

'0 X 1 '
'+ 0.5 '
'Y 0 Y '
'1 + '
'0.5 Z '
'0 Z 1 '
'+ 0.5 '
'X 1 X '
'2 + '
'0.5 Y '
'1 Y 2 '
'+ 0.5 '
'Z 1 Z '
'2 + '
'0.5 X '
'2 X 3 '
'+ 0.5 '
'Y 2 Y '
'3 + '
'0.5 Z '
'2 Z 3 '
'+ 0.5 '
'X 3 X '
'4 + '
'0.5 Y '
'3 Y 4 '
'+ 0.5 '
'Z 3 Z '
'4 + '
'0.5 X '
'4 X 5 '
'+ 0.5 '
'Y 4 Y '
'5 + '
'0.5 Z '
'4 Z 5 '
'+ 0.5 '
'X 5 X '
'6 + '
'0.5 Y '
'5 Y 6 '
'+ 0.5 '
'Z 5 Z '
'6']],
'output': { 'data_points': [ [4, 2, 0, 0, -6.144691518305653],
[12, 6, 0, 0, -4.66733899407964],
[ 20,
10,
0,
0,
-3.5226008080029567]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [ 0,
0,
0,
0],
'noise_level': [4, 2, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0,
0],
'status':
True,
'type':
'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [ 1,
1,
0,
0],
'noise_level': [12, 6, 0, 0],

```

```

'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0,
                0],

True,

'depolarizing'},

'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0,
                0],

True,

'depolarizing'},

'noise_profile': {
    'status':
    'type':
    'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [ 2,
                       2,
                       0,
                       0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {
    'status':
    'type':
    'odd_wires': 4}},
'run_time_sec': 7.105733394622803}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [
0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': { 'max': 10.0,
          'min': 0.0},
'type': 'heisenberg'}}},
'init_param': { 'value': [ -1.8784742854218807e-17,
                           -7.763750530158576e-17,
                           0.3802827375258553,
                           0.44096449652630626,
                           0.8040692975576897,
                           1.9224694434519336,
                           1.9224694434519338,
                           1.9224694434519336,
                           1.9224694434519336,
                           1.9224694434519336,
                           1.9224694434519338,
                           2.730288811725591,
                           4.476350841356259,
                           4.476350841356259,
                           4.476350841356259,
                           6.331658137341824,
                           6.331658137341824,
                           6.7621011201584285,
                           6.77851096220289,
                           7.490043179529316,
                           7.490043179529316,
                           7.490043179529316,
                           7.490043179529316,
                           7.511664240193972,

```

8.302259372586072,
8.36899822047307,
8.607405767895836,
9.631163052898822,
9.874876363637043,
10.225975825453373,
2.711573403856553,
0.20239905137011172,
-0.4220883549830381,
-0.8404830976891227,
1.794332340210785,
1.5095243517856047,
0.22326098118939897,
0.07733371735403745,
1.280548885023134,
1.4121496210624682,
0.9815935680907694,
-2.975348552150871,
1.6316643284140941,
0.8239502945369961,
0.0314505340316768,
-1.0892891749800817,
1.720694113273713,
0.6634696927679957,
-0.836512780686814,
-1.1286873737196692,
-0.09500137106776224,
1.0428135848317974,
-0.4828772957450712,
-1.129334172148368,
0.7253737318795237,
1.4108675361353635,
0.7588540034470418,
-0.5039485441412638,
0.10346857579652805,
1.2668095290766983,
0.7994757345637048,
-0.7393240316081237,
-0.7818304990031031,
0.25375980918776664,
2.228975199542585,
2.2487625387645873,
0.6384105819815235,
0.39434544396871446,
1.3679129840939837,
-0.27336699910726836,
1.3170761670781417,
1.1159622861844822,
1.176824183488981,
0.40583799697229034,
0.8786168217534174,
0.2914083315983707,
-0.8464697846173548,
-0.032455291822620566,
-2.0755386823554396,
-0.11439663923099666,
0.7312672791487281,
0.4263196833485598,
0.11802610051350722,
1.0445881812565523,
-1.9566462184988007,
0.9492303984073598,
-1.7129593951518396,
-0.3230601534209437,
0.3161713641922788,
-0.4811213940350071,
0.08327140785409295,
0.37526634808329173,
-1.9990814349080968,
-0.15397700748904192,
-0.9549084255901845,
1.1120455683467105,
-0.42952128451280097,
0.17929609797141985,
1.0446016069356951,

```

0.846774755857218,
-0.2784586201121851,
0.23000407229986752,
-0.9198047598183884,
0.996529675185623,
0.34608382599103005,
0.5099375110708237,
2.1948932092248885,
1.3625694142279887,
-0.9306056124011278,
0.6012804875871011,
-0.7788968212470156,
-1.5884735532174328,
1.0361309111992296,
0.015937704739038126,
1.4003797043998856,
2.429422654005534,
-0.27504014116509645,
0.8256449732433587,
1.2146087003914199,
-0.6435553984602482,
0.3046269503778821,
1.0182549333954134,
1.4202224852266074,
1.3511302325707892,
-0.7261672378416364,
1.5992752660640235,
-0.5764295578986515,
-0.8318409785507642,
1.1375736468462807,
0.8529850561064739,
0.9553972898831558,
-0.09161160854677679,
1.617622889574723,
1.4551513261723394,
-0.08817813881493969,
0.8045852859105778,
1.9892741470924016,
1.8347816167556374,
-0.8050652823779443,
0.008511392950551585,
2.1460096989270747,
0.49277581514942953,
-2.4311684786278165,
0.5134791015266132,
0.9751974520662461,
-0.38582649358518695,
0.7746227771853464,
1.7307944301597258,
2.1320333699599043,
1.1654652968447545]],
'noise_profile': { 'noise_on_init_param': {
'value':
'noise_prob': [ 0.001,
0.001,
0,
0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5],

```

[illegible]

```

0,
-3.433794388677785]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                    'identity_factors': [ 0,
                                         0,
                                         0,
                                         0],
                    'noise_level': [4, 2, 0, 0],
                    'noise_profile': {

'noise_on_init_param': { 'status': False,
                        'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0,
                0],

True,

'depolarizing'},

                    'status':

                    'type':

                    'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [ 1,
                      1,
                      0,
                      0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {

'noise_on_init_param': { 'status': False,
                        'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0,
                0],

True,

'depolarizing'},

                    'status':

                    'type':

                    'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [ 2,
                      2,
                      0,
                      0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {

'noise_on_init_param': { 'status': False,
                        'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0,
                0],

True,

'depolarizing'},

                    'status':

                    'type':

                    'odd_wires': 4}},
'run_time_sec': 7.748016595840454}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [ 0,
                                                            0,
                                                            0,
                                                            0,
                                                            0,
                                                            0],
                                                            'cn': [

0.5,
0.5,
0.5,
0.5,
0.5,
0.5],

'r': 0},
'time': { 'max': 10.0,

```



```
        'min': 0.0},
    'type': 'heisenberg'}},
'init_param': { 'value': [ 0.6189603325174557,
    0.7142372255934472,
    0.818217611349809,
    1.1433975540074377,
    1.9121137997887092,
    1.912113799788709,
    1.9189666863881127,
    2.194219841514053,
    2.734492156740232,
    4.250258815642276,
    4.250258815642276,
    5.774076859791501,
    6.527612880112256,
    6.532834731882501,
    8.049613322901676,
    8.049613322901676,
    8.050730377555883,
    8.561643142317116,
    8.561643142317116,
    8.758357012835646,
    8.758357012835646,
    8.78353884989931,
    9.477076383056275,
    9.536600626467342,
    9.536600626467342,
    9.536600626467342,
    10.241295505019522,
    10.318321333031077,
    10.579129354925781,
    10.82886994059623,
    3.662494394083504,
    0.08309129295248968,
    1.5295376506525808,
    -1.5964000102221196,
    1.8988970610647178,
    -0.07218621526676876,
    0.7238532160702689,
    -1.2917423920622229,
    1.8532222971765195,
    -0.40818119715800666,
    0.469598065996274,
    -1.2652675561295137,
    0.41465882817859046,
    0.48683539382624846,
    -0.15336518315146752,
    -1.6419783820068856,
    1.2136622236743368,
    0.5933123377118196,
    0.9666225223219593,
    0.9539776194708611,
    -0.3793986531891993,
    -0.2570964298405315,
    1.5884302353807402,
    0.19867970725527498,
    1.0036205245072367,
    -1.7266220384949087,
    0.6519864384701275,
    0.173373648086925,
    1.8697046853700252,
    0.5263944462203128,
    2.9785129063723943,
    -0.5937256494260224,
    0.22753174799850245,
    0.24918335157216837,
    1.5438671949833176,
    -0.2981477150404249,
    -0.0503664733834821,
    0.542442835193611,
    1.8096859498837612,
    -2.2260935596321993,
    0.28693396030012636,
    0.9986781622414097,
    -0.01940263773226835,
```

-1.8805315707929653,
-0.1513202473661221,
0.1836836320118686,
-0.5655439481125303,
-0.30323205019871646,
-0.4186198812206128,
0.2357002564872809,
2.8296979465953833,
-1.0776781071762758,
-0.08378558447273823,
1.7446476514004516,
-0.4606832586075214,
-1.762173446350097,
0.10241780783627007,
1.6018254886002428,
0.2621004786967086,
1.349261138919249,
0.14987623247602613,
0.5068543254371847,
-1.3474781551208477,
1.0890436270679893,
1.6662048029626129,
0.3608718991529565,
-0.11137748070221386,
0.5641204379289552,
2.2546014997074266,
1.8888740160956543,
-0.23345107582518032,
-1.5772444286238798,
2.1408383357361265,
-0.4968079560788485,
0.1533667959779154,
-0.40238715350257875,
0.25591729793510365,
2.947382531044255,
-0.32162602471486745,
-0.7587223675916386,
-0.8983111566200868,
0.006114119281372181,
2.9372824954589514,
0.48288231335139287,
0.7939297064609614,
1.9454360656955985,
2.092519500518508,
0.8995449331251366,
0.10171517677156572,
0.45235144869603666,
1.02501950409702,
0.24331156538914245,
0.48973195386607005,
2.672871435464254,
1.313352678455905,
2.010046934709317,
-0.42153209835304317,
-0.13436700176919486,
-0.6950305532982513,
0.6026553555930264,
1.6681913778840949,
0.4125109526933123,
0.514533549588801,
1.0794873063708696,
-0.9571806916209341,
0.27630815458061647,
0.6001387806934945,
0.5059087494947612,
-0.5722342732268321,
0.3986919058890637,
1.0044202485382987,
0.3352295947729849,
1.3777059541173544,
-2.835684252944388,
0.6104690716589696,
1.838086538234438,
-0.3106134411398272,
-2.2167272073692903,

```

0.10706372359431235,
-0.17469850626418476]],
'noise_profile': { 'noise_on_init_param': {
'value':
'noise_prob': [ 0.001,
0.001,
0,
0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_ric3_unifrom_scaling_sample#3'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '
'2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '
'2 + 1 X 3 X 4 + 1 Z 3 + 1 X '
'4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
'1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X '
'0 X 1 '
'+ 0.5 '
'Y 0 Y '
'1 + '
'0.5 Z '
'0 Z 1 '
'+ 0.5 '
'X 1 X '
'2 + '
'0.5 Y '
'1 Y 2 '
'+ 0.5 '
'Z 1 Z '
'2 + '
'0.5 X '
'2 X 3 '
'+ 0.5 '
'Y 2 Y '
'3 + '
'0.5 Z '
'2 Z 3 '
'+ 0.5 '
'X 3 X '

```

```

'4 + '
'0.5 Y '
'3 Y 4 '
'+ 0.5 '
'Z 3 Z '
'4 + '
'0.5 X '
'4 X 5 '
'+ 0.5 '
'Y 4 Y '
'5 + '
'0.5 Z '
'4 Z 5 '
'+ 0.5 '
'X 5 X '
'6 + '
'0.5 Y '
'5 Y 6 '
'+ 0.5 '
'Z 5 Z '
'6']],
'output': { 'data_points': [ [4, 2, 0, 0, -6.218690041461777],
[12, 6, 0, 0, -4.805156509769758],
[ 20,
10,
0,
0,
-3.7079207759854302]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [ 0,
0,
0,
0],
'noise_level': [4, 2, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0,
0],
'status':
True,
'type':
'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [ 1,
1,
0,
0],
'noise_level': [12, 6, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0,
0],
'status':
True,
'type':
'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
'identity_factors': [ 2,
2,
0,
0],
'noise_level': [20, 10, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0,
0],
'status':
True,
'type':
'depolarizing'},
'odd_wires': 4}
}
}

```

```
0,
0],

True,

'depolarizing'},

                                'status':

                                'type':

                                'odd_wires': 4}},
                                'run_time_sec': 6.773735761642456}},
{ 'config': { 'ansatz': { 'gateset': 1,
                            'layer': 30,
                            'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [

0.5,
0.5,
0.5,
0.5,
0.5,
0.5],

                                                                'r': 0},
                                                                'time': { 'max': 10.0,
                                                                'min': 0.0},
                                                                'type': 'heisenberg'}}},
'init_param': { 'value': [ 0.4022850038273885,
                            0.4222288083135141,
                            0.7644599371513524,
                            0.7900179023917507,
                            1.1026508993249646,
                            1.1630326759467198,
                            1.1630326759467198,
                            1.2119340062926192,
                            1.9859325565890973,
                            1.985932556589097,
                            2.096166401632718,
                            3.378647998605865,
                            3.378647998605865,
                            3.378647998605865,
                            4.354848115861854,
                            6.006218342726225,
                            6.006218342726225,
                            6.009974297352436,
                            8.225327865566033,
                            8.242093148909708,
                            9.222354355246733,
                            9.369468664000962,
                            9.432712219232048,
                            9.692289963799956,
                            9.692829942577658,
                            10.391360826688166,
                            10.391360826688166,
                            10.529867035412478,
                            10.529867035412478,
                            11.06290493838992,
                            2.9062307172979733,
                            -1.7788694936155562,
                            -0.1145266982224492,
                            -0.9311559923392093,
                            0.9691998209476543,
                            1.7463479137973288,
                            -0.7663862807329105,
                            -0.9694574896037749,
                            2.8979112386638413,
                            1.1400097878698292,
                            0.8096361549155084,
                            -1.5900494197653285,
                            0.09829667109053167,
                            0.31839493135320834,
                            0.3886777840539921,
                            0.12229184789371997,
                            1.7250225844898464,
```

1.0065795676456808,
1.4535641205083867,
-0.3486770669120958,
0.2556790893233589,
1.778531141883236,
0.269972269049403,
1.0155903957099945,
0.8041163886037216,
0.5381821926966456,
-0.30108604163317454,
0.6396533041678052,
2.1819991440037696,
1.4978086658278347,
2.5181040929106753,
0.4624415671193703,
-0.17701736987873673,
0.6910098122382194,
2.947172461882987,
-1.0727539032453073,
0.4023854197196741,
1.650490750838525,
0.9313998963798695,
-0.6769162055305248,
-0.5620785927454173,
-0.20885673940707247,
2.5019757090972443,
-0.37561416876024456,
-0.9329392032282731,
0.7818961506889994,
0.44350434198669075,
-0.10644929099379355,
-2.498928032568615,
-1.2046085786256906,
-0.06365416567297402,
0.17257236645054272,
1.2516209341031546,
-0.3591527663232026,
0.46789807326225935,
0.24374638092155315,
0.02110383811013195,
0.12714835746845662,
-0.5505736947218767,
-0.03497727012222505,
-2.2168334497478877,
-0.21485512988314215,
1.7552902914988067,
0.6189795903733311,
0.3191491062237487,
0.522632432692797,
-1.1121089236913335,
1.1371618962620422,
-0.2410043783817164,
0.24168543661014927,
0.39418565996934685,
1.2107667809264395,
2.398781467533486,
0.6948257569073976,
0.6170229142231725,
0.7050419245924102,
1.768546262611762,
1.5562956304120967,
-0.4269237214365518,
0.6302835616447369,
1.7710848704009405,
1.2281100264507194,
1.5036530549577243,
-0.7435268012971183,
0.7613541379199313,
0.8523298100708424,
-0.6655959646469325,
1.1543911681084875,
2.2675628634479494,
1.0299985170060308,
1.400963430022522,
0.27759304330136975,

```

0.6960561589746208,
-0.4210335016880339,
0.0336463669485127,
-2.099660094290342,
0.6674819238895001,
0.5049668435179431,
0.48001405045948287,
1.3574871122533543,
-2.2986993170736225,
9.601426007351498e-05,
1.3312470798272018,
0.7582238004326416,
-0.35033967277151296,
-0.4108552112493075,
0.1982910129710702,
0.6700017270602091,
0.26559360983179486,
-0.14563210230012705,
-0.39610135234418875,
1.1727729846793147,
0.6713573564879365,
0.8527070052929181,
0.8268283343075853,
2.2918575835381745,
0.5082103460913493,
1.8600493489550152,
0.49576811175353974,
2.232387985311215]],
'noise_profile': { 'noise_on_init_param': {
'value':
'noise_prob': [ 0.001,
0.001,
0,
0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_ric3_unifrom_scaling_sample#8'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '
'2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '
'2 + 1 X 3 X 4 + 1 Z 3 + 1 X '

```

```

'4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
'1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X '
'0 X 1 '
'+ 0.5 '
'Y 0 Y '
'1 + '
'0.5 Z '
'0 Z 1 '
'+ 0.5 '
'X 1 X '
'2 + '
'0.5 Y '
'1 Y 2 '
'+ 0.5 '
'Z 1 Z '
'2 + '
'0.5 X '
'2 X 3 '
'+ 0.5 '
'Y 2 Y '
'3 + '
'0.5 Z '
'2 Z 3 '
'+ 0.5 '
'X 3 X '
'4 + '
'0.5 Y '
'3 Y 4 '
'+ 0.5 '
'Z 3 Z '
'4 + '
'0.5 X '
'4 X 5 '
'+ 0.5 '
'Y 4 Y '
'5 + '
'0.5 Z '
'4 Z 5 '
'+ 0.5 '
'X 5 X '
'6 + '
'0.5 Y '
'5 Y 6 '
'+ 0.5 '
'Z 5 Z '
'6']],
'output': { 'data_points': [ [4, 2, 0, 0, -5.743389283741309],
[12, 6, 0, 0, -4.3560016590478785],
[ 20,
10,
0,
0,
-3.277640180193779]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [ 0,
0,
0,
0],
'noise_level': [4, 2, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0,
0],
'status':
True,
'type':
'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [ 1,
1,
1,
1,
1,
1,
1,
1,
1,
1,
1,
1]
}
]
}

```



```

'heisenberg_noisefree_time_evo'},

'output': { 'draw': { 'fig_dpi': 100,
                      'status': True,
                      'type': 'png'},
            'file_name_prefix':

'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0]]},

'run': 'vqe',
'state': 'dmatrix',
'vqe': { 'iteration': 10,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': True}},

'zne': { 'data_points': None,
         'degree': 1,
         'method': 'richardson',
         'sampling': 'default'}},

'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
                                'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
                                '4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
                                'X 5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 '
                                                         '+ 0.5 Y 0 Y '
                                                         '1 + 0.5 Z 0 '
                                                         'Z 1 + 0.5 X '
                                                         '1 X 2 + 0.5 '
                                                         'Y 1 Y 2 + '
                                                         '0.5 Z 1 Z 2 '
                                                         '+ 0.5 X 2 X '
                                                         '3 + 0.5 Y 2 '
                                                         'Y 3 + 0.5 Z '
                                                         '2 Z 3 + 0.5 '
                                                         'X 3 X 4 + '
                                                         '0.5 Y 3 Y 4 '
                                                         '+ 0.5 Z 3 Z '
                                                         '4 + 0.5 X 4 '
                                                         'X 5 + 0.5 Y '
                                                         '4 Y 5 + 0.5 '
                                                         'Z 4 Z 5 + '
                                                         '0.5 X 5 X 6 '
                                                         '+ 0.5 Y 5 Y '
                                                         '6 + 0.5 Z 5 '
                                                         'Z 6']},

'output': { 'exact_sol': -8.56677223350563,
            'initial_cost_history': [ 5.796866365657828,
                                     5.94197444368484,
                                     5.78867192910188,
                                     5.996879673940564,
                                     5.865062503342141,
                                     5.991390959337936,
                                     5.72582024636959,
                                     5.872963620288267,
                                     5.948869313039893,
                                     5.964755550858089],
            'noise_details': { 'gates_num': [4, 1, 1, 0],
                              'identity_factors': [0, 0, 0, 0],
                              'noise_level': [4, 2, 0, 0],
                              'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

'noise_prob': [

0.001,
0.001,

0,
0],

'status': True,
'type':

'depolarizing'},

'odd_wires': 4},
'optimized_minimum_cost': [ -5.966837375471974,
                             -6.220556999324403,
                             -6.218690041461848,
                             -6.241465922765082,
                             -5.8146764847083485,
                             -6.144691518305702,

```

```
-5.966791417158719,  
-5.743389283741366,  
-5.966171999363111,  
-6.281301957226982],  
'optimized_parameters': [ [ 0.43940602753254654,  
0.4524980474487914,  
0.6462847510297367,  
0.6535145711164267,  
0.7019965052930044,  
1.0830061860275473,  
1.4380021973026182,  
2.1860837771467474,  
2.1860837771467474,  
2.1860837771467474,  
2.4875804812430995,  
2.911818356328843,  
4.9134595358452735,  
5.043089843653786,  
5.994354774193697,  
6.658321281734746,  
6.666230857319696,  
8.458454027735183,  
8.458454027735183,  
8.458454027735183,  
8.938931772570164,  
8.938931772570164,  
9.876451730548837,  
9.876451730548837,  
10.140618079102744,  
10.453495471248813,  
10.878653713196195,  
11.282429749250388,  
11.726474998572407,  
11.93229290545428,  
2.9640535214497064,  
-0.6206168106317247,  
-0.4035747848788786,  
-2.067222048924027,  
2.0576259596946604,  
1.364315765000901,  
-0.1480698866383175,  
-1.95043906046908,  
0.7897528850889669,  
1.350582507885341,  
-0.3349044763190947,  
-2.3429833820345465,  
2.04351680973891,  
0.8083219796841106,  
-0.06520770630024467,  
-0.8776318722364843,  
2.4862506381076748,  
0.8345121096737285,  
-0.03071661620417745,  
-0.6225183255138825,  
0.7450777062518694,  
0.12619297185303674,  
0.15620862918286052,  
-0.47540100870445906,  
0.6345315438244892,  
1.451076347224231,  
0.10985833383721234,  
-1.6475696409761669,  
1.1918822544398533,  
1.3235539868194213,  
-0.24222060474147966,  
0.2418944215339802,  
1.0397651583041008,  
1.6773252160624976,  
0.16975786500059598,  
-0.9211027317156172,  
0.7590471244619662,  
1.76540125315151,  
2.43856734239768,  
-0.19488067618654012,  
0.8462629124439717,
```

0.9979674075340036,
2.5775834810620677,
0.5650956422366618,
-0.5475934153287456,
-0.0253449469292375,
1.2524069066350432,
-0.605696708865831,
0.7019502861337156,
0.2803153244338447,
-1.0831925759777479,
1.0082140101851143,
-0.3762046176346905,
-0.9535933533256993,
-1.0422804614435257,
0.13948594332192887,
0.5835070530053111,
0.16086524965091817,
0.06403444071667018,
-0.06194188553710435,
-0.4026223625881385,
0.2750028468648149,
0.6789909068622377,
0.05162849323802493,
-0.7735332926339995,
-0.14862918062202837,
0.7661757453229759,
-0.16119730427373616,
0.6482389226852195,
3.1459456036848104,
-0.4760920139337192,
1.6197254159790668,
-0.5506328506738539,
0.9968299487743895,
-0.3106735286083523,
-1.2388662013594425,
0.8898580979157581,
0.43256773745092003,
-0.7272220041247079,
0.8716071923554094,
1.5193589139179793,
0.9787540447681606,
-1.4044929205930872,
-0.1984876316951887,
-0.46744984056178585,
-0.7765225748074835,
0.04344696088727837,
0.32373965212791267,
0.9049555024172572,
0.7139278873651953,
-0.8122282267372439,
0.5323405298774778,
-0.08437719753251166,
-0.14277007957413837,
0.7912147308716768,
0.26055493673650715,
0.11357082745857092,
0.3153698930365579,
3.1981441448638614,
0.32109035951393206,
-0.23017443442945684,
0.0316864124540797,
0.20875812581978664,
0.5466872162452653,
1.2918521189433907,
-0.7312547980705957,
1.4958457475939604,
-0.053193686203739404,
-0.4437772137226961,
-0.5271126400646005,
2.5136810835026426,
0.6674244883024821,
-1.0140291771816496,
-1.177197116527103,
2.058315434744553,
0.050372102721390984,

-0.09022974351195023,
-2.296438941834702,
0.3261688389091777,
0.3072223897800251],
[0.05238332857976487,
0.05238332857976488,
0.05238332857976461,
0.5882470284735626,
1.6186245942059996,
2.2470777294271223,
2.8315596913723913,
2.8315596913723913,
2.8333251330604914,
3.7474015252366555,
3.7474015252366555,
3.7474015252366555,
3.7474015252366555,
4.452994588443841,
4.625910007504636,
4.625910007504636,
5.314362007876739,
7.185299991262795,
7.185299991262795,
8.165312898322489,
8.165312898322489,
8.165312898322489,
8.684979613482101,
9.661121823219345,
9.674169964387936,
10.525655511340814,
10.525655511340814,
10.525655511340814,
11.154963299169733,
11.154963299169733,
2.50282655917642,
0.9490947670548783,
0.70056532212722,
-0.9370006834105179,
2.127395945212884,
1.5529950009292526,
-0.5538339242405286,
-0.45109357069990613,
3.2608593101153462,
2.1691098922881027,
0.924351518772188,
0.7799372105219657,
2.588492585292434,
1.4914484649202893,
0.5111989894903198,
-1.6854828839595362,
0.709524102178921,
-0.3201993643863164,
0.37756499567776974,
0.11160463007405702,
1.4506751210554825,
-0.1158349928449886,
-0.4172082421780774,
-0.47420587439641587,
2.1318004152619214,
1.5026342770773617,
0.9629412747697549,
-1.017935070811068,
2.2274033566257083,
0.9541631607884562,
2.2865374842341715,
-0.6901760390806929,
1.4488496207691695,
-0.15529093800821045,
1.378510821815928,
-0.1238908488185843,
1.0420944500710088,
1.8938950235205334,
-0.40270933293735856,
-0.4838468076888962,
-0.7769068081513464,

2.0570355933263196,
0.33843722154840217,
-0.21296684820549192,
-0.3087457919151385,
1.6713665097762895,
0.7955513549984788,
-0.5031484152839188,
0.2783608989762586,
1.4322280472670237,
0.12038812465567636,
0.8648150791824706,
0.13552183280667268,
0.07182610915111512,
0.3477791234991623,
-0.1584859230597403,
0.5851519083687279,
0.578565105996425,
-1.7921034519484391,
0.9189192684840749,
-0.6139572052593133,
-0.023527142002935514,
-0.6495330854871428,
0.7767670181361933,
0.13616638059771954,
-0.22568119775616352,
0.4192866974084779,
0.1401009023541104,
1.419355971488301,
1.1103093112149183,
-0.8728413553259085,
1.014691351510677,
2.35757700845783,
1.131891326153365,
-1.2979835358031357,
0.42365777146015526,
1.0982786992551208,
1.698510255572029,
1.2090506157632,
-1.3597097700308411,
-0.9049577551064396,
1.0199036275847497,
2.9541844671388584,
-0.42829772740772204,
0.6542829736223691,
3.7916586259391143,
-0.1983670801457003,
0.7855289833525587,
-0.1534574040841844,
1.809122493112309,
0.35293394466185096,
1.0573849517470935,
2.461665111905985,
1.0784030177718114,
-1.585709677044719,
0.8985620342798133,
2.2360691692951646,
0.24022620726925167,
2.6922129347449117,
0.15599094623602128,
2.799936411115923,
0.9657316063780467,
-0.2732772147086647,
-1.448572189075712,
-0.7280589250410919,
0.057973620624421296,
0.9321748335747359,
-1.4338621772181848,
1.3976338029473028,
-0.25504403617708243,
3.210257057781798,
0.03448775329128267,
0.05070056715723011,
-1.3727823101385919,
0.8243579770218447,
0.13142013414460954,

-0.4933237632531999,
1.1105137227416793,
0.7155364379215329,
-0.21063252058278284],
[0.6189603325174557,
0.7142372255934472,
0.818217611349809,
1.1433975540074377,
1.9121137997887092,
1.912113799788709,
1.9189666863881127,
2.194219841514053,
2.734492156740232,
4.250258815642276,
4.250258815642276,
5.774076859791501,
6.527612880112256,
6.532834731882501,
8.049613322901676,
8.049613322901676,
8.050730377555883,
8.561643142317116,
8.561643142317116,
8.758357012835646,
8.758357012835646,
8.78353884989931,
9.477076383056275,
9.536600626467342,
9.536600626467342,
9.536600626467342,
10.241295505019522,
10.318321333031077,
10.579129354925781,
10.82886994059623,
3.662494394083504,
0.08309129295248968,
1.5295376506525808,
-1.5964000102221196,
1.8988970610647178,
-0.07218621526676876,
0.7238532160702689,
-1.2917423920622229,
1.8532222971765195,
-0.40818119715800666,
0.469598065996274,
-1.2652675561295137,
0.41465882817859046,
0.48683539382624846,
-0.15336518315146752,
-1.6419783820068856,
1.2136622236743368,
0.5933123377118196,
0.9666225223219593,
0.9539776194708611,
-0.3793986531891993,
-0.2570964298405315,
1.5884302353807402,
0.19867970725527498,
1.0036205245072367,
-1.7266220384949087,
0.6519864384701275,
0.173373648086925,
1.8697046853700252,
0.5263944462203128,
2.9785129063723943,
-0.5937256494260224,
0.22753174799850245,
0.24918335157216837,
1.5438671949833176,
-0.2981477150404249,
-0.0503664733834821,
0.542442835193611,
1.8096859498837612,
-2.2260935596321993,
0.28693396030012636,

0.9986781622414097,
-0.01940263773226835,
-1.8805315707929653,
-0.1513202473661221,
0.1836836320118686,
-0.5655439481125303,
-0.30323205019871646,
-0.4186198812206128,
0.2357002564872809,
2.8296979465953833,
-1.0776781071762758,
-0.08378558447273823,
1.7446476514004516,
-0.4606832586075214,
-1.762173446350097,
0.10241780783627007,
1.6018254886002428,
0.2621004786967086,
1.349261138919249,
0.14987623247602613,
0.5068543254371847,
-1.3474781551208477,
1.0890436270679893,
1.6662048029626129,
0.3608718991529565,
-0.11137748070221386,
0.5641204379289552,
2.2546014997074266,
1.8888740160956543,
-0.23345107582518032,
-1.5772444286238798,
2.1408383357361265,
-0.4968079560788485,
0.1533667959779154,
-0.40238715350257875,
0.25591729793510365,
2.947382531044255,
-0.32162602471486745,
-0.7587223675916386,
-0.8983111566200868,
0.006114119281372181,
2.9372824954589514,
0.48288231335139287,
0.7939297064609614,
1.9454360656955985,
2.092519500518508,
0.8995449331251366,
0.10171517677156572,
0.45235144869603666,
1.02501950409702,
0.24331156538914245,
0.48973195386607005,
2.672871435464254,
1.313352678455905,
2.010046934709317,
-0.42153209835304317,
-0.13436700176919486,
-0.6950305532982513,
0.6026553555930264,
1.6681913778840949,
0.4125109526933123,
0.514533549588801,
1.0794873063708696,
-0.9571806916209341,
0.27630815458061647,
0.6001387806934945,
0.5059087494947612,
-0.5722342732268321,
0.3986919058890637,
1.0044202485382987,
0.3352295947729849,
1.3777059541173544,
-2.835684252944388,
0.6104690716589696,
1.838086538234438,

-0.3106134411398272,
-2.2167272073692903,
0.10706372359431235,
-0.17469850626418476],
[-2.397634216504411e-17,
1.429450133825379e-17,
0.9192297980258759,
0.9192297980258759,
2.8984599305110854,
3.1993874679766523,
3.199387467976652,
3.715279848679994,
3.715919998165357,
5.126008182976459,
5.749875525815113,
5.785995839176617,
5.785995839176617,
5.785995839176617,
5.785995839176617,
6.315437962379428,
6.678845841951143,
7.3578167224936415,
7.415694801781823,
7.531263738212295,
7.949087697575632,
7.949087697575632,
8.111169110068786,
9.407094598577077,
9.495286031274873,
9.495286031274873,
10.993257661103009,
11.063439993331317,
11.639888600691357,
11.75175692668691,
2.1152038588685538,
-0.7839988468841306,
-1.0579861702584952,
-0.5122509123294313,
1.0012369417099964,
0.9275037440003397,
0.5343037326701707,
-0.014459274586663672,
2.31692048955396,
2.1400689678453872,
-0.9468421716760329,
1.3225342455296385,
2.4229762260600483,
1.4561600175543838,
-0.4592924007287594,
0.6389204073366491,
0.5578650700802136,
1.458345504678432,
-0.11931303728638508,
0.04406428844113783,
-0.26144087074192934,
0.6459890964472403,
1.2243061606997268,
0.09770103868863204,
0.23645607828472368,
0.3280111145811677,
0.5390648705936556,
-0.6034481934191781,
-1.1942560416831098,
0.3295059316644399,
-0.0732996553139216,
0.5043071925129845,
0.7399437462813884,
0.3595686242698361,
1.4643835231854074,
-0.1697776944282527,
0.2588380732750666,
0.44311592273109757,
-0.6031987466514527,
-0.06131410155451245,
-0.3299395710828798,

-0.45492250592824107,
0.5555861976799495,
1.1612177689175078,
1.833550163228831,
1.5629483318007886,
-0.2455585474033218,
-0.4534768786824923,
-1.2225865843307935,
0.20468879325769473,
1.3398561861341207,
-0.58709602862573,
0.07181500566168061,
0.17010271339345,
0.36406988692952136,
-1.689809967880289,
-1.1677426672173294,
-0.7977758879892671,
0.531032719362766,
-1.7722607960867016,
0.6332033998224597,
0.6669814362912183,
0.218182728116887,
1.0415656581640258,
-1.4544893086294475,
0.5175174925258392,
-0.9742907090362619,
-0.30648595673224416,
1.2400541074634985,
-0.30003161306087445,
-0.14406503395842435,
-0.43061285539013316,
0.6914697219419143,
-0.48545379734819916,
-0.2166375004086177,
1.2014709434123243,
1.0734635496959384,
0.43666289070672293,
-0.9550440807047608,
1.2817732673505913,
0.6254203972471953,
0.8730111716749767,
1.1814458389257443,
-0.5855214911540807,
0.33647224367417283,
0.914891547403245,
-0.10924028322724566,
0.6731472714644756,
-0.8672547313943412,
-1.1628903969680444,
1.2968110405185924,
-0.7478342770444015,
1.0157432273471354,
0.05866260951598884,
0.1549884083832883,
2.402954000164725,
-0.8510405490988947,
-0.13394876321797194,
0.35128800256256376,
0.03985349181110424,
1.0891408951797659,
0.49683980443602116,
1.3618410332836905,
0.31089249400158553,
0.007823999182329802,
-1.0325531315281817,
-0.13635517978577905,
-0.4783006843261583,
-0.9648987725589863,
-0.4094326055257113,
0.0620151507481985,
-0.3631850012398663,
-0.9119123598789223,
-0.3458539889181142,
0.8501459643039054,
1.6461875633189593,

0.08799287143090098,
-0.82318299513447,
1.17289644669955,
-0.9455811832961388],
[0.12420123662801165,
0.138246244841176,
0.34185010800102933,
0.8771280515547424,
0.8771280515547423,
1.9326762312100456,
1.9326762312100456,
2.879806034127527,
3.4467495190209787,
3.460588109395402,
3.4605881093954016,
3.4605881093954016,
3.911934955717517,
3.9564836804547703,
4.188240551814426,
5.098640553346409,
5.098640553346409,
5.098640553346408,
6.466332587579958,
7.457565848827182,
7.712724508543908,
7.890302861063261,
7.898577524812402,
8.020037295803547,
8.020037295803547,
9.289723192898348,
9.68251578824504,
9.68251578824504,
10.66694519433398,
11.464898013549973,
1.8095532965205083,
-1.6633278493215808,
-1.2281091611401596,
-0.40155726479865544,
1.7903243850816337,
0.18661159996017582,
1.2217850039169107,
-0.48128156078104645,
1.438546033666569,
-0.9466383819139143,
-0.2179227006066069,
-0.07921201994836616,
0.6692914930417907,
-0.16791294486623565,
-0.3278463005181308,
1.2883193159418245,
1.9001761172388045,
0.7646920019853412,
-0.6325477821570596,
-0.5987552945744136,
0.5669730115381587,
0.5983941343774113,
-0.06179171362000896,
-0.6867235859693906,
0.5211737756628548,
-0.19354588250953578,
0.6072816218147872,
0.3472677197742769,
-0.08031734075212016,
1.4024126217748347,
-0.5867181790012919,
0.809620813948269,
0.35997425529443994,
-0.8722967249300235,
-0.5874818244923429,
-0.5567926036837068,
-0.6652314874983639,
-0.6042054992483096,
-0.13918430829641074,
-0.08277488470584692,
-0.4972293169762719,

-0.4421668735352916,
0.6160601456858428,
-1.1972657609969646,
-1.1558488364748807,
-1.4501782622641872,
0.03989088007781949,
-0.8684923327267209,
-0.5569704341339503,
0.15823054121075,
0.6345578215095505,
0.3601853745371903,
-0.42594808878819745,
0.08979804115834313,
-0.153043281560756,
-0.603228229875778,
-1.1264410002736531,
0.6895580235946861,
-0.11045941959152497,
0.5174109375139814,
1.3269460586908306,
-0.3852298542423827,
-2.4120095755015893,
0.08807412161753436,
0.1228551365077638,
-0.6982591234727157,
-1.1618488040328845,
2.5586909942385017,
-0.24921574033555105,
-1.9870969070050777,
-1.0279498295884981,
-0.7796802357622239,
0.48428681889730063,
0.7162815161060118,
-0.15547297036103372,
1.0522591887013215,
1.772822944479854,
-0.2654037211186137,
-0.2720110866721671,
-0.03929478442656517,
0.34606077605631835,
0.5533331494282367,
1.0574846609095763,
-0.5747611578732846,
0.4916772893793279,
0.9875709691916216,
-0.17857479553023695,
-1.147281220275154,
0.37370382177065037,
2.456815155407461,
0.6567596591037266,
0.9507316292553318,
1.5113928877114242,
1.8380989093110256,
-0.1736919900840187,
-1.2666253947041328,
-0.554819304070435,
1.1931643666366285,
0.4614862178419797,
1.2040726557184918,
0.3083618234293392,
-0.4066692475369416,
0.3740368412278018,
-1.684611974494111,
-0.8650016725760176,
-1.4838330341878547,
-0.14344887350922086,
4.203596189627334,
1.1133202614474196,
1.391124117957397,
1.8627332105499828,
-0.28192336755973296,
-1.5416482647725873,
-1.9612892990030817,
0.8836928087418303,
0.300761821046295,

0.6785848024508535,
-0.5500277556546446,
0.5742776326882599,
-1.181756808209963],
[0.38848003251483587,
0.3884800325148359,
1.2841735835398218,
1.284173583539822,
1.2841735835398218,
1.818138220668234,
1.818138220668234,
2.7575469841005327,
2.7575469841005327,
2.7575469841005327,
3.0053468968599275,
3.0053468968599275,
3.005346896859927,
4.184079661691832,
4.83635165164968,
4.83635165164968,
5.97926077140412,
6.458805266455043,
6.513102391504654,
6.513102391504654,
7.122105903846713,
7.2188422331450415,
7.988590448127284,
8.268095077127937,
8.268095077127937,
9.164456488648115,
9.164456488648115,
9.782953992889732,
9.782953992889732,
10.467486029517685,
1.5496959718296535,
-1.0300437645832343,
0.425343365508412,
-1.7136674729251273,
0.33645197441382346,
0.2288357961575219,
0.10470230685228102,
-0.6612661793744108,
0.11823315872480306,
0.48571962162230753,
-1.3748501535311368,
0.2563120795974374,
-0.38871520150353267,
-0.18786175076151226,
1.0017996241286973,
0.05598903629217323,
1.501328526357405,
-0.32299727271195017,
0.5715295079657807,
-0.12201961934486968,
2.2747823054657927,
0.16289086097520028,
0.027619518366160244,
-0.10165606616500937,
-0.06321979816820887,
-0.27543819802246033,
-0.0048370064103128475,
0.1330138167350915,
-0.06428151371988705,
0.21905201010909556,
0.25697175483980955,
-1.462905966720411,
0.8014511304369155,
3.456137165485204,
-0.3560002683832885,
-0.4886877133354182,
0.32039145066707037,
1.6153352350920127,
0.5940099487759896,
0.18140087044059483,
0.27123106498617,

0.7444341922951362,
0.170480557625751,
-0.4688462519246396,
0.5235839184929864,
-0.5470964469389679,
-0.011921500492307424,
1.4116230981753706,
0.49972453659747124,
-0.7800673920851233,
0.4234227404504273,
-0.2071635265929523,
0.14064328933496192,
-1.2007096247825992,
0.12047098237640114,
-0.4125103841815656,
1.0447620953563077,
0.0020814149928196773,
0.4988991062112368,
-0.34495773827123155,
-0.12849063482979717,
0.4280934542561602,
0.7138444629946918,
-0.6497712710787925,
-0.43652659674625793,
0.917432486791049,
-0.8523477361785018,
0.5648987974980545,
0.3183226113969618,
0.14295784675778261,
0.9589824340171335,
0.9194478619356266,
1.1001359353915339,
1.5417377267389083,
-0.5376003771957187,
-0.5404528215706331,
0.21624687748210675,
0.8814937279455825,
-0.371639532103812,
-0.2668100917972658,
1.3133163210428074,
0.44574090697479807,
-1.0940261852319737,
-0.313830643067749,
-0.06100275437545863,
-0.049560783686653447,
-0.3025056724120473,
-0.9891529613817422,
-0.35099332176432035,
0.5436069706516342,
-0.722828417704955,
1.6968661444846564,
-2.4402971962208975,
0.7865745542588627,
1.3437881169309889,
0.36108471326554326,
-0.1749159211919777,
-0.014334278530919132,
0.02978377629472008,
1.8533978555002608,
-1.8470426695738211,
0.6272356375919823,
0.44775499409400255,
1.3281153539688288,
0.4583217606753929,
0.6443670392327268,
0.5254233581973354,
2.3087652110195127,
-0.7761319091913014,
-0.7605421959233766,
-0.6461580464596497,
0.7075597766411752,
0.985778508379752,
-1.3130974173457381,
2.8711331564400684,
1.2993543284582694,

1.9493614761217355,
1.1647493000584068,
-0.42731613935458074,
2.7038346000669766],
[-1.8784742854218807e-17,
-7.763750530158576e-17,
0.3802827375258553,
0.44096449652630626,
0.8040692975576897,
1.9224694434519336,
1.9224694434519338,
1.9224694434519336,
1.9224694434519336,
1.9224694434519336,
1.9224694434519338,
2.730288811725591,
4.476350841356259,
4.476350841356259,
4.476350841356259,
6.331658137341824,
6.331658137341824,
6.7621011201584285,
6.77851096220289,
7.490043179529316,
7.490043179529316,
7.490043179529316,
7.490043179529316,
7.511664240193972,
8.302259372586072,
8.36899822047307,
8.607405767895836,
9.631163052898822,
9.874876363637043,
10.225975825453373,
2.711573403856553,
0.20239905137011172,
-0.4220883549830381,
-0.8404830976891227,
1.794332340210785,
1.5095243517856047,
0.22326098118939897,
0.07733371735403745,
1.280548885023134,
1.4121496210624682,
0.9815935680907694,
-2.975348552150871,
1.6316643284140941,
0.8239502945369961,
0.0314505340316768,
-1.0892891749800817,
1.720694113273713,
0.6634696927679957,
-0.836512780686814,
-1.1286873737196692,
-0.09500137106776224,
1.0428135848317974,
-0.4828772957450712,
-1.129334172148368,
0.7253737318795237,
1.4108675361353635,
0.7588540034470418,
-0.5039485441412638,
0.10346857579652805,
1.2668095290766983,
0.7994757345637048,
-0.7393240316081237,
-0.7818304990031031,
0.25375980918776664,
2.228975199542585,
2.2487625387645873,
0.6384105819815235,
0.39434544396871446,
1.3679129840939837,
-0.27336699910726836,
1.3170761670781417,

1.1159622861844822,
1.176824183488981,
0.40583799697229034,
0.8786168217534174,
0.2914083315983707,
-0.8464697846173548,
-0.032455291822620566,
-2.0755386823554396,
-0.11439663923099666,
0.7312672791487281,
0.4263196833485598,
0.11802610051350722,
1.0445881812565523,
-1.9566462184988007,
0.9492303984073598,
-1.7129593951518396,
-0.3230601534209437,
0.3161713641922788,
-0.4811213940350071,
0.08327140785409295,
0.37526634808329173,
-1.9990814349080968,
-0.15397700748904192,
-0.9549084255901845,
1.1120455683467105,
-0.42952128451280097,
0.17929609797141985,
1.0446016069356951,
0.846774755857218,
-0.2784586201121851,
0.23000407229986752,
-0.9198047598183884,
0.996529675185623,
0.34608382599103005,
0.5099375110708237,
2.1948932092248885,
1.3625694142279887,
-0.9306056124011278,
0.6012804875871011,
-0.7788968212470156,
-1.5884735532174328,
1.0361309111992296,
0.015937704739038126,
1.4003797043998856,
2.429422654005534,
-0.27504014116509645,
0.8256449732433587,
1.2146087003914199,
-0.6435553984602482,
0.3046269503778821,
1.0182549333954134,
1.4202224852266074,
1.3511302325707892,
-0.7261672378416364,
1.5992752660640235,
-0.5764295578986515,
-0.8318409785507642,
1.1375736468462807,
0.8529850561064739,
0.9553972898831558,
-0.09161160854677679,
1.617622889574723,
1.4551513261723394,
-0.08817813881493969,
0.8045852859105778,
1.9892741470924016,
1.8347816167556374,
-0.8050652823779443,
0.008511392950551585,
2.1460096989270747,
0.49277581514942953,
-2.4311684786278165,
0.5134791015266132,
0.9751974520662461,
-0.38582649358518695,

0.7746227771853464,
1.7307944301597258,
2.1320333699599043,
1.1654652968447545],
[0.4022850038273885,
0.4222288083135141,
0.7644599371513524,
0.7900179023917507,
1.1026508993249646,
1.1630326759467198,
1.1630326759467198,
1.2119340062926192,
1.9859325565890973,
1.985932556589097,
2.096166401632718,
3.378647998605865,
3.378647998605865,
3.378647998605865,
4.354848115861854,
6.006218342726225,
6.006218342726225,
6.009974297352436,
8.225327865566033,
8.242093148909708,
9.222354355246733,
9.369468664000962,
9.432712219232048,
9.692289963799956,
9.692829942577658,
10.391360826688166,
10.391360826688166,
10.529867035412478,
10.529867035412478,
11.06290493838992,
2.9062307172979733,
-1.7788694936155562,
-0.1145266982224492,
-0.9311559923392093,
0.9691998209476543,
1.7463479137973288,
-0.7663862807329105,
-0.9694574896037749,
2.8979112386638413,
1.1400097878698292,
0.8096361549155084,
-1.5900494197653285,
0.09829667109053167,
0.31839493135320834,
0.3886777840539921,
0.12229184789371997,
1.7250225844898464,
1.0065795676456808,
1.4535641205083867,
-0.3486770669120958,
0.2556790893233589,
1.778531141883236,
0.269972269049403,
1.0155903957099945,
0.8041163886037216,
0.5381821926966456,
-0.30108604163317454,
0.6396533041678052,
2.1819991440037696,
1.4978086658278347,
2.5181040929106753,
0.4624415671193703,
-0.17701736987873673,
0.6910098122382194,
2.947172461882987,
-1.0727539032453073,
0.4023854197196741,
1.650490750838525,
0.9313998963798695,
-0.6769162055305248,
-0.5620785927454173,

-0.20885673940707247,
2.5019757090972443,
-0.37561416876024456,
-0.9329392032282731,
0.7818961506889994,
0.44350434198669075,
-0.10644929099379355,
-2.498928032568615,
-1.2046085786256906,
-0.06365416567297402,
0.17257236645054272,
1.2516209341031546,
-0.3591527663232026,
0.46789807326225935,
0.24374638092155315,
0.02110383811013195,
0.12714835746845662,
-0.5505736947218767,
-0.03497727012222505,
-2.2168334497478877,
-0.21485512988314215,
1.7552902914988067,
0.6189795903733311,
0.3191491062237487,
0.522632432692797,
-1.1121089236913335,
1.1371618962620422,
-0.2410043783817164,
0.24168543661014927,
0.39418565996934685,
1.2107667809264395,
2.398781467533486,
0.6948257569073976,
0.6170229142231725,
0.7050419245924102,
1.768546262611762,
1.5562956304120967,
-0.4269237214365518,
0.6302835616447369,
1.7710848704009405,
1.2281100264507194,
1.5036530549577243,
-0.7435268012971183,
0.7613541379199313,
0.8523298100708424,
-0.6655959646469325,
1.1543911681084875,
2.2675628634479494,
1.0299985170060308,
1.400963430022522,
0.27759304330136975,
0.6960561589746208,
-0.4210335016880339,
0.0336463669485127,
-2.099660094290342,
0.6674819238895001,
0.5049668435179431,
0.48001405045948287,
1.3574871122533543,
-2.2986993170736225,
9.601426007351498e-05,
1.3312470798272018,
0.7582238004326416,
-0.35033967277151296,
-0.4108552112493075,
0.1982910129710702,
0.6700017270602091,
0.26559360983179486,
-0.14563210230012705,
-0.39610135234418875,
1.1727729846793147,
0.6713573564879365,
0.8527070052929181,
0.8268283343075853,
2.2918575835381745,

0.5082103460913493,
1.8600493489550152,
0.49576811175353974,
2.232387985311215],
[0.16477895719846114,
0.16477895719846108,
0.7183189156962347,
1.8697093861419474,
2.105082615571597,
3.05308344502201,
4.199393321043376,
4.199393321043376,
4.199393321043375,
6.490210948425593,
6.4902109484255925,
6.490210948425593,
6.4902109484255925,
6.4902109484255925,
6.490210948425593,
6.882554174728105,
6.882554174728105,
6.882554174728105,
6.882554174728104,
6.8877583035732775,
6.8877583035732775,
6.8877583035732775,
8.21014824639056,
8.21014824639056,
9.071469420716642,
9.120928605341893,
9.714944098401723,
11.474131735950095,
11.474131735950095,
11.794551582273902,
1.341188628911811,
1.6554894539990426,
-2.1068974621953425,
-1.4171475675653578,
0.33211669069513333,
0.11331792925356984,
-1.3610911099963097,
0.46439741779020816,
0.9313501387866817,
1.0938593094543279,
-0.3591727738608774,
-0.4732622270970327,
1.6321598874818157,
-0.2782091348514778,
-1.0526781349295347,
0.0708703831522549,
2.6982701224663113,
-0.038867313655444914,
-0.10508583152652556,
-0.40759365792756236,
0.6284849004330764,
0.5192751929971827,
-1.6282040637266806,
0.0014422433007245106,
-1.8299719264591412,
2.3182597876656823,
0.8875167319646391,
1.5664268415036142,
1.5052778791259855,
2.935955436800565,
1.533101581959507,
1.8091829332960188,
0.45080966818917967,
0.1506883627647002,
-0.6843548866459184,
-0.5544883854492364,
1.2466562002431278,
-1.1866110136199353,
1.4521358231995756,
-1.125608676420736,
-0.24054383618816966,

-0.058272145415006006,
0.410408128597049,
-1.6441753460137252,
0.4019816715685168,
0.4588365987304068,
1.354346078964423,
-1.7153251360741875,
-1.1039978646265602e-06,
0.7940670488104785,
0.17979772520560053,
-0.24385394325110826,
0.6367160741723578,
0.8500459957371875,
0.0559416907316065,
0.32418668587130467,
-1.4797636873635434,
0.11902177942387202,
0.5185165439317966,
-0.08399222686641401,
0.8650918037299421,
1.4911587670088011,
0.12792248933645675,
0.3372262149750348,
-0.6553154954729696,
1.1995792367396254,
0.5132294030998703,
0.44835115860783636,
-0.37116980512777586,
0.7805194563374558,
-0.854334789795377,
0.5008527672922033,
-0.2302685629030402,
0.6145182391921147,
-0.6575500079215796,
0.8657035637529482,
1.1874516653159721,
-0.16284802367068918,
-0.26462171656665856,
-0.9815102056659819,
0.23198397744948207,
-1.087097195512072,
0.6616385654966559,
-0.2929637257774683,
0.03471390505932934,
0.2048751570049573,
0.9170017118070749,
0.39617956766989404,
0.12006199846995962,
0.1922705913940127,
0.8784478594386566,
0.055479957240883566,
-0.6634105431569992,
1.0742951419164057,
-0.37818200958777604,
0.28548758229245597,
-0.6594827056613419,
1.0235832498557866,
-0.19428253177041227,
-0.5531991468121416,
-1.189626464718852,
0.2519110119226656,
-0.6708462316627377,
2.7994352657547057,
-3.8764436680175813,
0.5152327138619494,
-0.3749347648847944,
2.9042167722097862,
1.1265965037549552,
-1.850959186134046,
2.165374611027145,
0.8818414791702859,
1.5039524911129019,
0.06564767423285973,
0.6486215314149069,
-0.08854143781823305,

1.8396223528529987,
-0.00792246728000446,
2.8805879469479487,
3.6337353861266473],
[0.11853625011402234,
0.11853625011402236,
1.1173235551611367,
1.7342580765110573,
1.7342580765110573,
2.108144483620023,
3.054129947242328,
3.054129947242328,
3.054129947242328,
3.056306360633301,
4.638414750363569,
5.0621851887275495,
5.7503913724004905,
5.774998407238176,
6.917933224474368,
6.917933224474368,
8.061549209526286,
8.061549209526286,
8.061549209526286,
8.591251503132888,
8.591251503132888,
9.051235537435975,
9.081856762230144,
9.467876152039107,
9.545932016874406,
9.591419209848118,
10.465616352275608,
10.465616352275608,
10.465735173309618,
10.980942228105905,
2.0734674059611633,
0.32892201175743896,
-0.007356440825095051,
0.7856576253246221,
2.089457548986907,
0.7744884921126718,
-0.35049309306395837,
-0.6556831494452968,
2.0369525201874903,
0.4739652080570001,
-0.8564345639983049,
1.25716179603495,
1.7181260437331554,
2.3978160665502384,
1.1152024575020885,
-1.3335737674295711,
0.016650799076572978,
1.6607129762234538,
0.648059834763191,
0.2438725527060928,
-0.27307896000752346,
1.4369017198677916,
3.5672448343800007,
-0.5379702113035718,
-0.7420664562717484,
-0.529699891930992,
0.7301471483152846,
0.781913925757813,
0.6004078048346045,
1.7324865479818374,
0.4548236607944829,
-1.4419813608219112,
-0.684662976785234,
0.8455512814500156,
1.97751114408285,
-0.833202517811791,
1.110524460863052,
2.6403320691876733,
-0.3129427957408566,
-0.06674204626492235,
0.5842857778318762,

-0.21132485649138796,
-0.011842019288790376,
-0.25743970610802486,
0.7486616881233721,
-0.1302029141156864,
0.09497091477412227,
-0.3715265429351211,
0.39570077921274605,
0.28643868832003017,
0.7848413456471106,
-0.7157146817058562,
0.09411764456438232,
-0.10910076647953601,
1.1262761892344808,
-0.27334997823872975,
-0.44027652197942063,
0.6740415100156434,
-0.336256645457715,
0.4268157265313067,
-0.8611696251566416,
-0.34560928156726306,
0.5618074077714486,
0.5866360379174992,
1.8662876827821717,
2.21592310898005,
1.972223656383108,
1.6141187926643297,
0.856255815565121,
-0.34275929095614754,
-1.0642528745182613,
-1.0618686751734654,
0.2379582550153795,
-0.6501770361983872,
-0.19408645485522444,
1.501330829125914,
1.5726602727591479,
0.7066066097903853,
-1.4580936665368467,
0.46940440748349077,
-0.1287015677341077,
-0.9805925033056946,
-0.3754741669636888,
-0.026817919501009202,
1.1067346586651607,
0.17547923725022047,
0.8902143171896091,
-0.22777698610446503,
0.8082253583749343,
-1.0674083601412958,
-0.022706971436593337,
0.34227899244110566,
1.3527736523760083,
2.0767841716125894,
0.5778568145668401,
-0.6555502504392442,
1.2670274746873995,
0.507694022232518,
-0.2629991187543805,
1.7048253080726326,
1.4088065522309612,
2.823918413203317,
1.2763232744235296,
-1.4348344213353486,
-0.605496445996402,
0.09645158262695042,
0.024639736182597342,
0.23997550724405606,
0.9979895378690888,
-0.40612123795993227,
1.826790650755685,
0.029599395583551916,
-0.4592861834683686,
0.0328358806035505,
1.3612120706287358,
0.8210402794880411,

3.5672448343800007,
-0.5379702113035718,
-0.7420664562717484,
-0.529699891930992,
0.7301471483152846,
0.781913925757813,
0.6004078048346045,
1.7324865479818374,
0.4548236607944829,
-1.4419813608219112,
-0.684662976785234,
0.8455512814500156,
1.97751114408285,
-0.833202517811791,
1.110524460863052,
2.6403320691876733,
-0.3129427957408566,
-0.06674204626492235,
0.5842857778318762,
-0.21132485649138796,
-0.011842019288790376,
-0.25743970610802486,
0.7486616881233721,
-0.1302029141156864,
0.09497091477412227,
-0.3715265429351211,
0.39570077921274605,
0.28643868832003017,
0.7848413456471106,
-0.7157146817058562,
0.09411764456438232,
-0.10910076647953601,
1.1262761892344808,
-0.27334997823872975,
-0.44027652197942063,
0.6740415100156434,
-0.336256645457715,
0.4268157265313067,
-0.8611696251566416,
-0.34560928156726306,
0.5618074077714486,
0.5866360379174992,
1.8662876827821717,
2.21592310898005,
1.972223656383108,
1.6141187926643297,
0.856255815565121,
-0.34275929095614754,
-1.0642528745182613,
-1.0618686751734654,
0.2379582550153795,
-0.6501770361983872,
-0.19408645485522444,
1.501330829125914,
1.5726602727591479,
0.7066066097903853,
-1.4580936665368467,
0.46940440748349077,
-0.1287015677341077,
-0.9805925033056946,
-0.3754741669636888,
-0.026817919501009202,
1.1067346586651607,
0.17547923725022047,
0.8902143171896091,
-0.22777698610446503,
0.8082253583749343,
-1.0674083601412958,
-0.022706971436593337,
0.34227899244110566,
1.3527736523760083,
2.0767841716125894,
0.5778568145668401,
-0.6555502504392442,
1.2670274746873995,

False,

```
0.507694022232518,
-0.2629991187543805,
1.7048253080726326,
1.4088065522309612,
2.823918413203317,
1.2763232744235296,
-1.4348344213353486,
-0.605496445996402,
0.09645158262695042,
0.024639736182597342,
0.23997550724405606,
0.9979895378690888,
-0.40612123795993227,
1.826790650755685,
0.029599395583551916,
-0.4592861834683686,
0.0328358806035505,
1.3612120706287358,
0.8210402794880411,
0.05688164146575431,
0.03152531580318861,
2.0316547976974904,
2.795039607579165]],
'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value': 0},
                  'noise_prob': [0.001, 0.001, 0, 0],
                  'status': True,
                  'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_ric3_unifrom_scaling_sample#10'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.28130195722698],
                          [ 12,
                           6,
                           0,
                           0,
                           -4.778373581659957],
                          [ 20,
                           10,
                           0,
                           0,
                           -3.6190179833186162]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 7.677078247070312e-05,
```

```

'zne_values': { 'degree': 0,
                'extrapolated_value': -7.161605936470123,
                'others': { 'beta_coefficients': [ 1.875,
                                                    -1.25,
                                                    0.375],
                            'cost_richardson_zne':
                                                    5.21875,
                            'sorted_expectation_vals': [
                                                    -6.28130195722698,
                                                    -4.778373581659957,
                                                    -3.6190179833186162],
                            'sorted_noise': [6, 18, 30]},
                'sampled_data': [ [ 4,
                                     2,
                                     0,
                                     0,
                                     -6.28130195722698],
                                   [ 12,
                                     6,
                                     0,
                                     0,
                                     -4.778373581659957],
                                   [ 20,
                                     10,
                                     0,
                                     0,
                                     -3.6190179833186162]],
                'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                            'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                            'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'heisenberg'}}},
  'init_param': { 'value': [ 0.43940602753254654,
                              0.4524980474487914,
                              0.6462847510297367,
                              0.6535145711164267,
                              0.7019965052930044,
                              1.0830061860275473,
                              1.4380021973026182,
                              2.1860837771467474,
                              2.1860837771467474,
                              2.1860837771467474,
                              2.4875804812430995,
                              2.911818356328843,
                              4.9134595358452735,
                              5.043089843653786,
                              5.994354774193697,
                              6.658321281734746,
                              6.666230857319696,
                              8.458454027735183,
                              8.458454027735183,
                              8.458454027735183,
                              8.938931772570164,
                              8.938931772570164,
                              9.876451730548837,
                              9.876451730548837,
                              10.140618079102744,
                              10.453495471248813,
                              10.878653713196195,
                              11.282429749250388,
                              11.726474998572407,

```

11.93229290545428,
2.9640535214497064,
-0.6206168106317247,
-0.4035747848788786,
-2.067222048924027,
2.0576259596946604,
1.364315765000901,
-0.1480698866383175,
-1.95043906046908,
0.7897528850889669,
1.350582507885341,
-0.3349044763190947,
-2.3429833820345465,
2.04351680973891,
0.8083219796841106,
-0.06520770630024467,
-0.8776318722364843,
2.4862506381076748,
0.8345121096737285,
-0.03071661620417745,
-0.6225183255138825,
0.7450777062518694,
0.12619297185303674,
0.15620862918286052,
-0.47540100870445906,
0.6345315438244892,
1.451076347224231,
0.10985833383721234,
-1.6475696409761669,
1.1918822544398533,
1.3235539868194213,
-0.24222060474147966,
0.2418944215339802,
1.0397651583041008,
1.6773252160624976,
0.16975786500059598,
-0.9211027317156172,
0.7590471244619662,
1.76540125315151,
2.43856734239768,
-0.19488067618654012,
0.8462629124439717,
0.9979674075340036,
2.5775834810620677,
0.5650956422366618,
-0.5475934153287456,
-0.0253449469292375,
1.2524069066350432,
-0.605696708865831,
0.7019502861337156,
0.2803153244338447,
-1.0831925759777479,
1.0082140101851143,
-0.3762046176346905,
-0.9535933533256993,
-1.0422804614435257,
0.13948594332192887,
0.5835070530053111,
0.16086524965091817,
0.06403444071667018,
-0.06194188553710435,
-0.4026223625881385,
0.2750028468648149,
0.6789909068622377,
0.05162849323802493,
-0.7735332926339995,
-0.14862918062202837,
0.7661757453229759,
-0.16119730427373616,
0.6482389226852195,
3.1459456036848104,
-0.4760920139337192,
1.6197254159790668,
-0.5506328506738539,
0.9968299487743895,

```

-0.3106735286083523,
-1.2388662013594425,
0.8898580979157581,
0.43256773745092003,
-0.7272220041247079,
0.8716071923554094,
1.5193589139179793,
0.9787540447681606,
-1.4044929205930872,
-0.1984876316951887,
-0.46744984056178585,
-0.7765225748074835,
0.04344696088727837,
0.32373965212791267,
0.9049555024172572,
0.7139278873651953,
-0.8122282267372439,
0.5323405298774778,
-0.08437719753251166,
-0.14277007957413837,
0.7912147308716768,
0.26055493673650715,
0.11357082745857092,
0.3153698930365579,
3.1981441448638614,
0.32109035951393206,
-0.23017443442945684,
0.0316864124540797,
0.20875812581978664,
0.5466872162452653,
1.2918521189433907,
-0.7312547980705957,
1.4958457475939604,
-0.053193686203739404,
-0.4437772137226961,
-0.5271126400646005,
2.5136810835026426,
0.6674244883024821,
-1.0140291771816496,
-1.177197116527103,
2.058315434744553,
0.050372102721390984,
-0.09022974351195023,
-2.296438941834702,
0.3261688389091777,
0.3072223897800251]],
'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value': 0},
                  'noise_prob': [0.001, 0.001, 0, 0],
                  'status': True,
                  'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_ric3_unifrom_scaling_sample#1'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                      [1, 1, 0, 0],

```

```

[2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [ 4,
                           2,
                           0,
                           0,
                           -5.966837375471956],
                          [ 12,
                           6,
                           0,
                           0,
                           -4.6298501952375855],
                          [ 20,
                           10,
                           0,
                           0,
                           -3.5807206753108627]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 0.00010275840759277344,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -6.743277588204508,
                           'others': { 'beta_coeffieicients': [ 1.875,
                                                                -1.25,
                                                                0.375],
                                         'cost_richardson_zne':
                                         'sorted_expectation_vals': [
                                         'sorted_noise': [6, 18, 30]],
            'sampled data': [ [ 4,
                              2,
                              0,
                              0,
                              -5.966837375471956],
                             [ 12,
                              6,
                              0,
                              0,
                              -4.6298501952375855],
                             [ 20,
                              10,
                              0,
                              0,
                              -3.5807206753108627]],
            'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                           'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                           'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'heisenberg'}},
  'init_param': { 'value': [ -2.397634216504411e-17,
                             1.429450133825379e-17,

```

0.9192297980258759,
0.9192297980258759,
2.8984599305110854,
3.1993874679766523,
3.199387467976652,
3.715279848679994,
3.715919998165357,
5.126008182976459,
5.749875525815113,
5.785995839176617,
5.785995839176617,
5.785995839176617,
5.785995839176617,
6.315437962379428,
6.678845841951143,
7.3578167224936415,
7.415694801781823,
7.531263738212295,
7.949087697575632,
7.949087697575632,
8.111169110068786,
9.407094598577077,
9.495286031274873,
9.495286031274873,
10.993257661103009,
11.063439993331317,
11.639888600691357,
11.75175692668691,
2.1152038588685538,
-0.7839988468841306,
-1.0579861702584952,
-0.5122509123294313,
1.0012369417099964,
0.9275037440003397,
0.5343037326701707,
-0.014459274586663672,
2.31692048955396,
2.1400689678453872,
-0.9468421716760329,
1.3225342455296385,
2.4229762260600483,
1.4561600175543838,
-0.4592924007287594,
0.6389204073366491,
0.5578650700802136,
1.458345504678432,
-0.11931303728638508,
0.04406428844113783,
-0.26144087074192934,
0.6459890964472403,
1.2243061606997268,
0.09770103868863204,
0.23645607828472368,
0.3280111145811677,
0.5390648705936556,
-0.6034481934191781,
-1.1942560416831098,
0.3295059316644399,
-0.0732996553139216,
0.5043071925129845,
0.7399437462813884,
0.3595686242698361,
1.4643835231854074,
-0.1697776944282527,
0.2588380732750666,
0.44311592273109757,
-0.6031987466514527,
-0.06131410155451245,
-0.3299395710828798,
-0.45492250592824107,
0.5555861976799495,
1.1612177689175078,
1.833550163228831,
1.5629483318007886,
-0.2455585474033218,

-0.4534768786824923,
-1.2225865843307935,
0.20468879325769473,
1.3398561861341207,
-0.58709602862573,
0.07181500566168061,
0.17010271339345,
0.36406988692952136,
-1.689809967880289,
-1.1677426672173294,
-0.7977758879892671,
0.531032719362766,
-1.7722607960867016,
0.6332033998224597,
0.6669814362912183,
0.218182728116887,
1.0415656581640258,
-1.4544893086294475,
0.5175174925258392,
-0.9742907090362619,
-0.30648595673224416,
1.2400541074634985,
-0.30003161306087445,
-0.14406503395842435,
-0.43061285539013316,
0.6914697219419143,
-0.48545379734819916,
-0.2166375004086177,
1.2014709434123243,
1.0734635496959384,
0.43666289070672293,
-0.9550440807047608,
1.2817732673505913,
0.6254203972471953,
0.8730111716749767,
1.1814458389257443,
-0.5855214911540807,
0.33647224367417283,
0.914891547403245,
-0.10924028322724566,
0.6731472714644756,
-0.8672547313943412,
-1.1628903969680444,
1.2968110405185924,
-0.7478342770444015,
1.0157432273471354,
0.05866260951598884,
0.1549884083832883,
2.402954000164725,
-0.8510405490988947,
-0.13394876321797194,
0.35128800256256376,
0.03985349181110424,
1.0891408951797659,
0.49683980443602116,
1.3618410332836905,
0.31089249400158553,
0.007823999182329802,
-1.0325531315281817,
-0.13635517978577905,
-0.4783006843261583,
-0.9648987725589863,
-0.4094326055257113,
0.0620151507481985,
-0.3631850012398663,
-0.9119123598789223,
-0.3458539889181142,
0.8501459643039054,
1.6461875633189593,
0.08799287143090098,
-0.82318299513447,
1.17289644669955,
-0.9455811832961388]],

'noise_profile': { 'noise_on_init_param': { 'status':

False,

```

'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1}},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_ric3_unifrom_scaling_sample#4'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [ 4,
2,
0,
0,
-6.241465922765086],
[ 12,
6,
0,
0,
-4.714681794898787],
[ 20,
10,
0,
0,
-3.5395259571477817]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 3.910064697265625e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -7.136718595491471,
'others': { 'beta_coefficients': [ 1.875,
-1.25,
0.375],
'cost_richardson_zne':
5.21875,
'sorted_expectation_vals': [
-6.241465922765086,
-4.714681794898787,
-3.5395259571477817],
'sorted_noise': [6, 18, 30]},
'sampled data': [ [ 4,
2,
0,
0,
-6.241465922765086],
[ 12,
6,
0,
0,

```



```
-4.714681794898787],
[ 20,
 10,
 0,
 0,
-3.5395259571477817]],
'sampling': 'default']}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [ 0.16477895719846114,
0.16477895719846108,
0.7183189156962347,
1.8697093861419474,
2.105082615571597,
3.05308344502201,
4.199393321043376,
4.199393321043376,
4.199393321043375,
6.490210948425593,
6.4902109484255925,
6.490210948425593,
6.4902109484255925,
6.4902109484255925,
6.490210948425593,
6.882554174728105,
6.882554174728105,
6.882554174728105,
6.882554174728104,
6.8877583035732775,
6.8877583035732775,
6.8877583035732775,
8.21014824639056,
8.21014824639056,
9.071469420716642,
9.120928605341893,
9.714944098401723,
11.474131735950095,
11.474131735950095,
11.794551582273902,
1.341188628911811,
1.6554894539990426,
-2.1068974621953425,
-1.4171475675653578,
0.33211669069513333,
0.11331792925356984,
-1.3610911099963097,
0.46439741779020816,
0.9313501387866817,
1.0938593094543279,
-0.3591727738608774,
-0.4732622270970327,
1.6321598874818157,
-0.2782091348514778,
-1.0526781349295347,
0.0708703831522549,
2.6982701224663113,
-0.038867313655444914,
-0.10508583152652556,
-0.40759365792756236,
```

0.6284849004330764,
0.5192751929971827,
-1.6282040637266806,
0.0014422433007245106,
-1.8299719264591412,
2.3182597876656823,
0.8875167319646391,
1.5664268415036142,
1.5052778791259855,
2.935955436800565,
1.533101581959507,
1.8091829332960188,
0.45080966818917967,
0.1506883627647002,
-0.6843548866459184,
-0.5544883854492364,
1.2466562002431278,
-1.1866110136199353,
1.4521358231995756,
-1.125608676420736,
-0.24054383618816966,
-0.058272145415006006,
0.410408128597049,
-1.6441753460137252,
0.4019816715685168,
0.4588365987304068,
1.354346078964423,
-1.7153251360741875,
-1.1039978646265602e-06,
0.7940670488104785,
0.17979772520560053,
-0.24385394325110826,
0.6367160741723578,
0.8500459957371875,
0.0559416907316065,
0.32418668587130467,
-1.4797636873635434,
0.11902177942387202,
0.5185165439317966,
-0.08399222686641401,
0.8650918037299421,
1.4911587670088011,
0.12792248933645675,
0.3372262149750348,
-0.6553154954729696,
1.1995792367396254,
0.5132294030998703,
0.44835115860783636,
-0.37116980512777586,
0.7805194563374558,
-0.854334789795377,
0.5008527672922033,
-0.2302685629030402,
0.6145182391921147,
-0.6575500079215796,
0.8657035637529482,
1.1874516653159721,
-0.16284802367068918,
-0.26462171656665856,
-0.9815102056659819,
0.23198397744948207,
-1.087097195512072,
0.6616385654966559,
-0.2929637257774683,
0.03471390505932934,
0.2048751570049573,
0.9170017118070749,
0.39617956766989404,
0.12006199846995962,
0.1922705913940127,
0.8784478594386566,
0.055479957240883566,
-0.6634105431569992,
1.0742951419164057,
-0.37818200958777604,

```
'AUTOMATE_heisenberg_noisefree_time_evo_ric3_unifrom_scaling_sample#9'},
```

```
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]
```

[illegible]

```

-3.364633070617193]],
    'degree': 0,
    'method': 'richardson',
    'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 4.1484832763671875e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -6.833658146281991,
'others': { 'beta_coefficients': [ 1.875,
-1.25,
0.375],
'cost_richardson_zne':
5.21875,
'sorted_expectation_vals': [
-5.966171999362981,
-4.4917214032040365,
-3.364633070617193],
'sorted_noise': [6, 18, 30]},
'sampled data': [ [ 4,
2,
0,
0,
-5.966171999362981],
[ 12,
6,
0,
0,
-4.4917214032040365],
[ 20,
10,
0,
0,
-3.364633070617193]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [ 0.12420123662801165,
0.138246244841176,
0.34185010800102933,
0.8771280515547424,
0.8771280515547423,
1.9326762312100456,
1.9326762312100456,
2.879806034127527,
3.4467495190209787,
3.460588109395402,
3.4605881093954016,
3.4605881093954016,
3.911934955717517,
3.9564836804547703,
4.188240551814426,
5.098640553346409,
5.098640553346409,
5.098640553346408,
6.466332587579958,
7.457565848827182,
7.712724508543908,
7.890302861063261,
7.898577524812402,

```

8.020037295803547,
8.020037295803547,
9.289723192898348,
9.68251578824504,
9.68251578824504,
10.66694519433398,
11.464898013549973,
1.8095532965205083,
-1.6633278493215808,
-1.2281091611401596,
-0.40155726479865544,
1.7903243850816337,
0.18661159996017582,
1.2217850039169107,
-0.48128156078104645,
1.438546033666569,
-0.9466383819139143,
-0.21792270060606069,
-0.07921201994836616,
0.6692914930417907,
-0.16791294486623565,
-0.3278463005181308,
1.2883193159418245,
1.9001761172388045,
0.7646920019853412,
-0.6325477821570596,
-0.5987552945744136,
0.5669730115381587,
0.5983941343774113,
-0.06179171362000896,
-0.6867235859693906,
0.5211737756628548,
-0.19354588250953578,
0.6072816218147872,
0.3472677197742769,
-0.08031734075212016,
1.4024126217748347,
-0.5867181790012919,
0.809620813948269,
0.35997425529443994,
-0.8722967249300235,
-0.5874818244923429,
-0.5567926036837068,
-0.6652314874983639,
-0.6042054992483096,
-0.13918430829641074,
-0.08277488470584692,
-0.4972293169762719,
-0.4421668735352916,
0.6160601456858428,
-1.1972657609969646,
-1.1558488364748807,
-1.4501782622641872,
0.03989088007781949,
-0.8684923327267209,
-0.5569704341339503,
0.15823054121075,
0.6345578215095505,
0.3601853745371903,
-0.42594808878819745,
0.08979804115834313,
-0.153043281560756,
-0.603228229875778,
-1.1264410002736531,
0.6895580235946861,
-0.11045941959152497,
0.5174109375139814,
1.3269460586908306,
-0.3852298542423827,
-2.4120095755015893,
0.08807412161753436,
0.1228551365077638,
-0.6982591234727157,
-1.1618488040328845,
2.5586909942385017,

```

-0.24921574033555105,
-1.9870969070050777,
-1.0279498295884981,
-0.7796802357622239,
0.48428681889730063,
0.7162815161060118,
-0.15547297036103372,
1.0522591887013215,
1.772822944479854,
-0.2654037211186137,
-0.2720110866721671,
-0.03929478442656517,
0.34606077605631835,
0.5533331494282367,
1.0574846609095763,
-0.5747611578732846,
0.4916772893793279,
0.9875709691916216,
-0.17857479553023695,
-1.147281220275154,
0.37370382177065037,
2.456815155407461,
0.6567596591037266,
0.9507316292553318,
1.5113928877114242,
1.8380989093110256,
-0.1736919900840187,
-1.2666253947041328,
-0.554819304070435,
1.1931643666366285,
0.4614862178419797,
1.2040726557184918,
0.3083618234293392,
-0.4066692475369416,
0.3740368412278018,
-1.684611974494111,
-0.8650016725760176,
-1.4838330341878547,
-0.14344887350922086,
4.203596189627334,
1.1133202614474196,
1.391124117957397,
1.8627332105499828,
-0.28192336755973296,
-1.5416482647725873,
-1.9612892990030817,
0.8836928087418303,
0.300761821046295,
0.6785848024508535,
-0.5500277556546446,
0.5742776326882599,
-1.181756808209963]],
'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value': 0},
                    'noise_prob': [0.001, 0.001, 0, 0],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,

```

False,

[illegible]

```
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [ 0.05238332857976487,
0.05238332857976488,
0.05238332857976461,
0.5882470284735626,
1.6186245942059996,
2.2470777294271223,
2.8315596913723913,
2.8315596913723913,
2.8333251330604914,
3.7474015252366555,
3.7474015252366555,
3.7474015252366555,
3.7474015252366555,
4.452994588443841,
4.625910007504636,
4.625910007504636,
5.314362007876739,
7.185299991262795,
7.185299991262795,
8.165312898322489,
8.165312898322489,
8.165312898322489,
8.684979613482101,
9.661121823219345,
9.674169964387936,
10.525655511340814,
10.525655511340814,
10.525655511340814,
11.154963299169733,
11.154963299169733,
2.50282655917642,
0.9490947670548783,
0.70056532212722,
-0.9370006834105179,
2.127395945212884,
1.5529950009292526,
-0.5538339242405286,
-0.45109357069990613,
3.2608593101153462,
2.1691098922881027,
0.924351518772188,
0.7799372105219657,
2.588492585292434,
1.4914484649202893,
0.5111989894903198,
-1.6854828839595362,
0.709524102178921,
-0.3201993643863164,
0.37756499567776974,
0.11160463007405702,
1.4506751210554825,
-0.1158349928449886,
-0.4172082421780774,
-0.47420587439641587,
2.1318004152619214,
1.5026342770773617,
0.9629412747697549,
-1.017935070811068,
2.2274033566257083,
0.9541631607884562,
2.2865374842341715,
-0.6901760390806929,
1.4488496207691695,
-0.15529093800821045,
1.378510821815928,
-0.1238908488185843,
1.0420944500710088,
1.8938950235205334,
-0.40270933293735856,
-0.4838468076888962,
-0.7769068081513464,
```


2.0570355933263196,
0.33843722154840217,
-0.21296684820549192,
-0.3087457919151385,
1.6713665097762895,
0.7955513549984788,
-0.5031484152839188,
0.2783608989762586,
1.4322280472670237,
0.12038812465567636,
0.8648150791824706,
0.13552183280667268,
0.07182610915111512,
0.3477791234991623,
-0.1584859230597403,
0.5851519083687279,
0.578565105996425,
-1.7921034519484391,
0.9189192684840749,
-0.6139572052593133,
-0.023527142002935514,
-0.6495330854871428,
0.7767670181361933,
0.13616638059771954,
-0.22568119775616352,
0.4192866974084779,
0.1401009023541104,
1.419355971488301,
1.1103093112149183,
-0.8728413553259085,
1.014691351510677,
2.35757700845783,
1.131891326153365,
-1.2979835358031357,
0.42365777146015526,
1.0982786992551208,
1.698510255572029,
1.2090506157632,
-1.3597097700308411,
-0.9049577551064396,
1.0199036275847497,
2.9541844671388584,
-0.42829772740772204,
0.6542829736223691,
3.7916586259391143,
-0.1983670801457003,
0.7855289833525587,
-0.1534574040841844,
1.809122493112309,
0.35293394466185096,
1.0573849517470935,
2.461665111905985,
1.0784030177718114,
-1.585709677044719,
0.8985620342798133,
2.2360691692951646,
0.24022620726925167,
2.6922129347449117,
0.15599094623602128,
2.799936411115923,
0.9657316063780467,
-0.2732772147086647,
-1.448572189075712,
-0.7280589250410919,
0.057973620624421296,
0.9321748335747359,
-1.4338621772181848,
1.3976338029473028,
-0.25504403617708243,
3.210257057781798,
0.03448775329128267,
0.05070056715723011,
-1.3727823101385919,
0.8243579770218447,
0.13142013414460954,

```

-0.4933237632531999,
1.1105137227416793,
0.7155364379215329,
-0.21063252058278284]],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1}},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_ric3_unifrom_scaling_sample#2'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [ 4,
2,
0,
0,
-6.220556999324399],
[ 12,
6,
0,
0,
-4.682225394272194],
[ 20,
10,
0,
0,
-3.4953983793015833]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 0.00011301040649414062,
'zne_values': { 'degree': 0,
'extrapolated_value': -7.1215370231311,
'others': { 'beta_coeffieicients': [ 1.875,
-1.25,
0.375],
'cost_richardson_zne':
'sorted_expectation_vals': [
-6.220556999324399,
-4.682225394272194,
-3.4953983793015833],
'sorted_noise': [6, 18, 30]],
'sampled data': [ [ 4,
2,
0,

```

```
0,
-6.220556999324399],
[ 12,
 6,
 0,
 0,
-4.682225394272194],
[ 20,
 10,
 0,
 0,
-3.4953983793015833]],
'sampling': 'default']}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                                'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'heisenberg'}},
  'init_param': { 'value': [ -1.8784742854218807e-17,
                              -7.763750530158576e-17,
                              0.3802827375258553,
                              0.44096449652630626,
                              0.8040692975576897,
                              1.9224694434519336,
                              1.9224694434519338,
                              1.9224694434519336,
                              1.9224694434519336,
                              1.9224694434519336,
                              1.9224694434519338,
                              2.730288811725591,
                              4.476350841356259,
                              4.476350841356259,
                              4.476350841356259,
                              6.331658137341824,
                              6.331658137341824,
                              6.7621011201584285,
                              6.77851096220289,
                              7.490043179529316,
                              7.490043179529316,
                              7.490043179529316,
                              7.490043179529316,
                              7.511664240193972,
                              8.302259372586072,
                              8.36899822047307,
                              8.607405767895836,
                              9.631163052898822,
                              9.874876363637043,
                              10.225975825453373,
                              2.711573403856553,
                              0.20239905137011172,
                              -0.4220883549830381,
                              -0.8404830976891227,
                              1.794332340210785,
                              1.5095243517856047,
                              0.22326098118939897,
                              0.07733371735403745,
                              1.280548885023134,
                              1.4121496210624682,
                              0.9815935680907694,
                              -2.975348552150871,
                              1.6316643284140941,
                              0.8239502945369961,
```

0.0314505340316768,
-1.0892891749800817,
1.720694113273713,
0.6634696927679957,
-0.836512780686814,
-1.1286873737196692,
-0.09500137106776224,
1.0428135848317974,
-0.4828772957450712,
-1.129334172148368,
0.7253737318795237,
1.4108675361353635,
0.7588540034470418,
-0.5039485441412638,
0.10346857579652805,
1.2668095290766983,
0.7994757345637048,
-0.7393240316081237,
-0.7818304990031031,
0.25375980918776664,
2.228975199542585,
2.2487625387645873,
0.6384105819815235,
0.39434544396871446,
1.3679129840939837,
-0.27336699910726836,
1.3170761670781417,
1.1159622861844822,
1.176824183488981,
0.40583799697229034,
0.8786168217534174,
0.2914083315983707,
-0.8464697846173548,
-0.032455291822620566,
-2.0755386823554396,
-0.11439663923099666,
0.7312672791487281,
0.4263196833485598,
0.11802610051350722,
1.0445881812565523,
-1.9566462184988007,
0.9492303984073598,
-1.7129593951518396,
-0.3230601534209437,
0.3161713641922788,
-0.4811213940350071,
0.08327140785409295,
0.37526634808329173,
-1.9990814349080968,
-0.15397700748904192,
-0.9549084255901845,
1.1120455683467105,
-0.42952128451280097,
0.17929609797141985,
1.0446016069356951,
0.846774755857218,
-0.2784586201121851,
0.23000407229986752,
-0.9198047598183884,
0.996529675185623,
0.34608382599103005,
0.5099375110708237,
2.1948932092248885,
1.3625694142279887,
-0.9306056124011278,
0.6012804875871011,
-0.7788968212470156,
-1.5884735532174328,
1.0361309111992296,
0.015937704739038126,
1.4003797043998856,
2.429422654005534,
-0.27504014116509645,
0.8256449732433587,
1.2146087003914199,


```

0,
-4.541488857163632],
[ 20,
10,
0,
0,
-3.433794388677785]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 3.457069396972656e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -6.798545731472112,
'others': { 'beta_coefficients': [ 1.875,
-1.25,
0.375],
'cost_richardson_zne':
'sorted_expectation_vals': [
'sorted_noise': [6, 18, 30]],
'sampled data': [ [ 4,
2,
0,
0,
-5.966791417158658],
[ 12,
6,
0,
0,
-4.541488857163632],
[ 20,
10,
0,
0,
-3.433794388677785]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [ 0.38848003251483587,
0.3884800325148359,
1.2841735835398218,
1.284173583539822,
1.2841735835398218,
1.818138220668234,
1.818138220668234,
2.7575469841005327,
2.7575469841005327,
2.7575469841005327,
3.0053468968599275,
3.0053468968599275,
3.005346896859927,
4.184079661691832,
4.83635165164968,
4.83635165164968,
5.97926077140412,

```

6.458805266455043,
6.513102391504654,
6.513102391504654,
7.122105903846713,
7.2188422331450415,
7.988590448127284,
8.268095077127937,
8.268095077127937,
9.164456488648115,
9.164456488648115,
9.782953992889732,
9.782953992889732,
10.467486029517685,
1.5496959718296535,
-1.0300437645832343,
0.425343365508412,
-1.7136674729251273,
0.33645197441382346,
0.2288357961575219,
0.10470230685228102,
-0.6612661793744108,
0.11823315872480306,
0.48571962162230753,
-1.3748501535311368,
0.2563120795974374,
-0.38871520150353267,
-0.18786175076151226,
1.0017996241286973,
0.05598903629217323,
1.501328526357405,
-0.32299727271195017,
0.5715295079657807,
-0.12201961934486968,
2.2747823054657927,
0.16289086097520028,
0.027619518366160244,
-0.10165606616500937,
-0.06321979816820887,
-0.27543819802246033,
-0.0048370064103128475,
0.1330138167350915,
-0.06428151371988705,
0.21905201010909556,
0.25697175483980955,
-1.462905966720411,
0.8014511304369155,
3.456137165485204,
-0.3560002683832885,
-0.4886877133354182,
0.32039145066707037,
1.6153352350920127,
0.5940099487759896,
0.18140087044059483,
0.27123106498617,
0.7444341922951362,
0.170480557625751,
-0.4688462519246396,
0.5235839184929864,
-0.5470964469389679,
-0.011921500492307424,
1.4116230981753706,
0.49972453659747124,
-0.7800673920851233,
0.4234227404504273,
-0.2071635265929523,
0.14064328933496192,
-1.2007096247825992,
0.12047098237640114,
-0.4125103841815656,
1.0447620953563077,
0.0020814149928196773,
0.4988991062112368,
-0.34495773827123155,
-0.12849063482979717,
0.4280934542561602,

```

0.7138444629946918,
-0.6497712710787925,
-0.43652659674625793,
0.917432486791049,
-0.8523477361785018,
0.5648987974980545,
0.3183226113969618,
0.14295784675778261,
0.9589824340171335,
0.9194478619356266,
1.1001359353915339,
1.5417377267389083,
-0.5376003771957187,
-0.5404528215706331,
0.21624687748210675,
0.8814937279455825,
-0.371639532103812,
-0.2668100917972658,
1.3133163210428074,
0.44574090697479807,
-1.0940261852319737,
-0.313830643067749,
-0.06100275437545863,
-0.049560783686653447,
-0.3025056724120473,
-0.9891529613817422,
-0.35099332176432035,
0.5436069706516342,
-0.722828417704955,
1.6968661444846564,
-2.4402971962208975,
0.7865745542588627,
1.3437881169309889,
0.36108471326554326,
-0.1749159211919777,
-0.014334278530919132,
0.02978377629472008,
1.8533978555002608,
-1.8470426695738211,
0.6272356375919823,
0.44775499409400255,
1.3281153539688288,
0.4583217606753929,
0.6443670392327268,
0.5254233581973354,
2.3087652110195127,
-0.7761319091913014,
-0.7605421959233766,
-0.6461580464596497,
0.7075597766411752,
0.985778508379752,
-1.3130974173457381,
2.8711331564400684,
1.2993543284582694,
1.9493614761217355,
1.1647493000584068,
-0.42731613935458074,
2.7038346000669766]],
'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value': 0},
                    'noise_prob': [0.001, 0.001, 0, 0],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5]

```

False,

[illegible]

```
                                0],
                                'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                'r': 0},
                                'time': {'max': 10.0, 'min': 0.0},
                                'type': 'heisenberg'}},
'init_param': { 'value': [ 0.4022850038273885,
                             0.4222288083135141,
                             0.7644599371513524,
                             0.7900179023917507,
                             1.1026508993249646,
                             1.1630326759467198,
                             1.1630326759467198,
                             1.2119340062926192,
                             1.9859325565890973,
                             1.985932556589097,
                             2.096166401632718,
                             3.378647998605865,
                             3.378647998605865,
                             3.378647998605865,
                             4.354848115861854,
                             6.006218342726225,
                             6.006218342726225,
                             6.009974297352436,
                             8.225327865566033,
                             8.242093148909708,
                             9.222354355246733,
                             9.369468664000962,
                             9.432712219232048,
                             9.692289963799956,
                             9.692829942577658,
                             10.391360826688166,
                             10.391360826688166,
                             10.529867035412478,
                             10.529867035412478,
                             11.06290493838992,
                             2.9062307172979733,
                             -1.7788694936155562,
                             -0.1145266982224492,
                             -0.9311559923392093,
                             0.9691998209476543,
                             1.7463479137973288,
                             -0.7663862807329105,
                             -0.9694574896037749,
                             2.8979112386638413,
                             1.1400097878698292,
                             0.8096361549155084,
                             -1.5900494197653285,
                             0.09829667109053167,
                             0.31839493135320834,
                             0.3886777840539921,
                             0.12229184789371997,
                             1.7250225844898464,
                             1.0065795676456808,
                             1.4535641205083867,
                             -0.3486770669120958,
                             0.2556790893233589,
                             1.778531141883236,
                             0.269972269049403,
                             1.0155903957099945,
                             0.8041163886037216,
                             0.5381821926966456,
                             -0.30108604163317454,
                             0.6396533041678052,
                             2.1819991440037696,
                             1.4978086658278347,
                             2.5181040929106753,
                             0.4624415671193703,
                             -0.17701736987873673,
                             0.6910098122382194,
                             2.947172461882987,
```

-1.0727539032453073,
0.4023854197196741,
1.650490750838525,
0.9313998963798695,
-0.6769162055305248,
-0.5620785927454173,
-0.20885673940707247,
2.5019757090972443,
-0.37561416876024456,
-0.9329392032282731,
0.7818961506889994,
0.44350434198669075,
-0.10644929099379355,
-2.498928032568615,
-1.2046085786256906,
-0.06365416567297402,
0.17257236645054272,
1.2516209341031546,
-0.3591527663232026,
0.46789807326225935,
0.24374638092155315,
0.02110383811013195,
0.12714835746845662,
-0.5505736947218767,
-0.03497727012222505,
-2.2168334497478877,
-0.21485512988314215,
1.7552902914988067,
0.6189795903733311,
0.3191491062237487,
0.522632432692797,
-1.1121089236913335,
1.1371618962620422,
-0.2410043783817164,
0.24168543661014927,
0.39418565996934685,
1.2107667809264395,
2.398781467533486,
0.6948257569073976,
0.6170229142231725,
0.7050419245924102,
1.768546262611762,
1.5562956304120967,
-0.4269237214365518,
0.6302835616447369,
1.7710848704009405,
1.2281100264507194,
1.5036530549577243,
-0.7435268012971183,
0.7613541379199313,
0.8523298100708424,
-0.6655959646469325,
1.1543911681084875,
2.2675628634479494,
1.0299985170060308,
1.400963430022522,
0.27759304330136975,
0.6960561589746208,
-0.4210335016880339,
0.0336463669485127,
-2.099660094290342,
0.6674819238895001,
0.5049668435179431,
0.48001405045948287,
1.3574871122533543,
-2.2986993170736225,
9.601426007351498e-05,
1.3312470798272018,
0.7582238004326416,
-0.35033967277151296,
-0.4108552112493075,
0.1982910129710702,
0.6700017270602091,
0.26559360983179486,
-0.14563210230012705,

```

-0.39610135234418875,
1.1727729846793147,
0.6713573564879365,
0.8527070052929181,
0.8268283343075853,
2.2918575835381745,
0.5082103460913493,
1.8600493489550152,
0.49576811175353974,
2.232387985311215]],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
'value': 0},
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_ric3_unifrom_scaling_sample#8'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [ 4,
2,
0,
0,
-5.743389283741309],
[ 12,
6,
0,
0,
-4.3560016590478785],
[ 20,
10,
0,
0,
-3.277640180193779]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 3.838539123535156e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -6.552967900777775,
'others': { 'beta_coeffiecents': [ 1.875,
-1.25,
0.375],
'cost_richardson_zne':
5.21875,
'sorted_expectation_vals': [
-5.743389283741309,

```

```
'sorted_noise': [6, 18, 30]],
'sampled_data': [ [ 4,
                    2,
                    0,
                    0,
                    -5.743389283741309],
                  [ 12,
                    6,
                    0,
                    0,
                    -4.3560016590478785],
                  [ 20,
                    10,
                    0,
                    0,
                    -3.277640180193779]],
'sampling': 'default'}}},
'ansatz': { 'gateset': 1,
            'layer': 30,
            'ugate': { 'coefficients': { 'bn': [ 0,
                                                  0,
                                                  0,
                                                  0,
                                                  0,
                                                  0],
                                          'cn': [ 0.5,
                                                  0.5,
                                                  0.5,
                                                  0.5,
                                                  0.5,
                                                  0.5],
                                          'r': 0},
                    'time': {'max': 10.0, 'min': 0.0},
                    'type': 'heisenberg'}}},
'init_param': { 'value': [ 0.6189603325174557,
                            0.7142372255934472,
                            0.818217611349809,
                            1.1433975540074377,
                            1.9121137997887092,
                            1.912113799788709,
                            1.9189666863881127,
                            2.194219841514053,
                            2.734492156740232,
                            4.250258815642276,
                            4.250258815642276,
                            5.774076859791501,
                            6.527612880112256,
                            6.532834731882501,
                            8.049613322901676,
                            8.049613322901676,
                            8.050730377555883,
                            8.561643142317116,
                            8.561643142317116,
                            8.758357012835646,
                            8.758357012835646,
                            8.78353884989931,
                            9.477076383056275,
                            9.536600626467342,
                            9.536600626467342,
                            9.536600626467342,
                            10.241295505019522,
                            10.318321333031077,
                            10.579129354925781,
                            10.82886994059623,
                            3.662494394083504,
                            0.08309129295248968,
                            1.5295376506525808,
                            -1.5964000102221196,
                            1.8988970610647178,
                            -0.07218621526676876,
                            0.7238532160702689,
                            -1.2917423920622229,
```

1.8532222971765195,
-0.40818119715800666,
0.469598065996274,
-1.2652675561295137,
0.41465882817859046,
0.48683539382624846,
-0.15336518315146752,
-1.6419783820068856,
1.2136622236743368,
0.5933123377118196,
0.9666225223219593,
0.9539776194708611,
-0.3793986531891993,
-0.2570964298405315,
1.5884302353807402,
0.19867970725527498,
1.0036205245072367,
-1.7266220384949087,
0.6519864384701275,
0.173373648086925,
1.8697046853700252,
0.5263944462203128,
2.9785129063723943,
-0.5937256494260224,
0.22753174799850245,
0.24918335157216837,
1.5438671949833176,
-0.2981477150404249,
-0.0503664733834821,
0.542442835193611,
1.8096859498837612,
-2.2260935596321993,
0.28693396030012636,
0.9986781622414097,
-0.01940263773226835,
-1.8805315707929653,
-0.1513202473661221,
0.1836836320118686,
-0.5655439481125303,
-0.30323205019871646,
-0.4186198812206128,
0.2357002564872809,
2.8296979465953833,
-1.0776781071762758,
-0.08378558447273823,
1.7446476514004516,
-0.4606832586075214,
-1.762173446350097,
0.10241780783627007,
1.6018254886002428,
0.2621004786967086,
1.349261138919249,
0.14987623247602613,
0.5068543254371847,
-1.3474781551208477,
1.0890436270679893,
1.6662048029626129,
0.3608718991529565,
-0.11137748070221386,
0.5641204379289552,
2.2546014997074266,
1.8888740160956543,
-0.23345107582518032,
-1.5772444286238798,
2.1408383357361265,
-0.4968079560788485,
0.1533667959779154,
-0.40238715350257875,
0.25591729793510365,
2.947382531044255,
-0.32162602471486745,
-0.7587223675916386,
-0.8983111566200868,
0.006114119281372181,
2.9372824954589514,

False,

```
0.48288231335139287,
0.7939297064609614,
1.9454360656955985,
2.092519500518508,
0.8995449331251366,
0.10171517677156572,
0.45235144869603666,
1.02501950409702,
0.24331156538914245,
0.48973195386607005,
2.672871435464254,
1.313352678455905,
2.010046934709317,
-0.42153209835304317,
-0.13436700176919486,
-0.6950305532982513,
0.6026553555930264,
1.6681913778840949,
0.4125109526933123,
0.514533549588801,
1.0794873063708696,
-0.9571806916209341,
0.27630815458061647,
0.6001387806934945,
0.5059087494947612,
-0.5722342732268321,
0.3986919058890637,
1.0044202485382987,
0.3352295947729849,
1.3777059541173544,
-2.835684252944388,
0.6104690716589696,
1.838086538234438,
-0.3106134411398272,
-2.2167272073692903,
0.10706372359431235,
-0.17469850626418476]],
'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value': 0},
                    'noise_prob': [0.001, 0.001, 0, 0],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_ric3_unifrom_scaling_sample#3'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [ 4,
                           2,
```

```

0,
0,
-6.218690041461777],
[ 12,
6,
0,
0,
-4.805156509769758],
[ 20,
10,
0,
0,
-3.7079207759854302]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 7.319450378417969e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -7.044068481523171,
'others': { 'beta_coefficients': [ 1.875,
-1.25,
0.375],
'cost_richardson_zne':
5.21875,
'sorted_expectation_vals': [
-6.218690041461777,
-4.805156509769758,
-3.7079207759854302],
'sorted_noise': [6, 18, 30]},
'sampled data': [ [ 4,
2,
0,
0,
-6.218690041461777],
[ 12,
6,
0,
0,
-4.805156509769758],
[ 20,
10,
0,
0,
-3.7079207759854302]],
'sampling': 'default'}}}},
'heisenberg-ric4-uniform-scaling': { 'noiseoff': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': { 'max': 10.0,
'min': 0.0},
'type': 'heisenberg'}},
'init_param': {'value': 'random'},
'noise_profile': { 'noise_on_init_param': { 'status':
'value':
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': False,

```


[illegible]

'4 Z 5 '
'+ 0.5 '
'X 5 X '
'6 + '
'0.5 Y '
'5 Y 6 '
'+ 0.5 '
'Z 5 Z '
'6']},

'output': { 'exact_sol': -8.56677223350563,
 'initial_cost_history': [6.121384846939922,
 6.150953723335025,
 6.28813394059272,
 6.195576097324661,
 5.996275986067059,
 6.155907118612996,
 6.197882180139617,
 6.276491195599097,
 6.126423666072617,
 6.100624170709524],
 'noise_details': { 'gates_num': [4, 1, 1, 0],
 'identity_factors': [0, 0, 0, 0],
 'noise_level': [0, 0, 0, 0],
 'noise_profile': {

'noise_on_init_param': { 'status': False,
 'value': 0},

'noise_prob': [

0.001,
0.001,
0.001,
0.001],

'status':

False,

'type':

'depolarizing'},

 'odd_wires': 4},
'optimized_minimum_cost': [-7.113819716474412,
 -6.821323437950204,
 -7.2322812477454015,
 -7.168061781988564,
 -7.037022966690968,
 -6.764659140688094,
 -6.70772123506257,
 -6.951929490175616,
 -6.765271591343842,
 -6.643370531408768],
'optimized_parameters': [[0.010202312572582629,
 0.01020231257258252,
 0.010202312572582303,
 1.3908062838956354,
 1.3908062838956354,
 2.0164416760130464,
 2.592046705326477,
 2.592046705326477,
 2.592046705326477,
 3.724080378311745,
 3.724080378311745,
 4.739480012897176,
 4.739480012897176,
 5.945308900554525,
 5.945308900554525,
 5.945308900554525,
 5.945308900554525,
 6.51261715666153,
 6.516103684733277,
 7.99085874039394,
 7.99085874039394,
 8.16277333661163,
 8.99858956109671,
 9.406132081871055,
 9.406132081871055,
 9.406132081871055,
 10.1331640829029,
 10.184186623923145,
 11.15918741223016,

11.16716557378114,
2.8450840989270483,
-0.611567672261254,
-0.551095997013421,
-1.0131107784954883,
0.6825070788337171,
-0.6162852792596847,
0.053241771237976535,
-0.12516614014524466,
1.004304378969064,
1.0874046621334863,
-0.263039228744755,
-0.10249623387196242,
1.6330709573383448,
0.7555807809408442,
-1.304918472993802,
-1.1870918453507433,
-0.1330475605934714,
-0.5047081392764299,
0.7756728723196425,
-0.49278801094288127,
0.42847067786702214,
0.5383171641852699,
-0.1052976007723638,
-1.2720333725347934,
0.1951924272458347,
-0.04277330847061178,
0.9574239041786494,
-0.23071352324401823,
0.594076188896681,
0.34935689199185954,
1.343124516909016,
0.09733282827416712,
0.30944030008743273,
0.4457207862171129,
-0.9683728985035202,
-1.120837258235465,
0.5707161794308084,
1.2158524027070072,
0.7958721804993727,
-0.9645697723054423,
-0.8960337800506504,
0.19569442936859724,
0.40224575611741115,
0.043062302517518566,
-1.1857970207577537,
0.1737802328972237,
-0.6433879386848066,
-0.09230835161436789,
-0.5865305292280132,
-0.4370825940325887,
-0.2558137517355887,
0.0011609522641742864,
0.6539099100207111,
0.5336009667754404,
-0.066252982145182,
-0.5359243057678726,
-1.3834098071980059,
1.0212180272076503,
0.5141425620577187,
-0.3002279094652444,
1.9431365651467594,
1.0552610834283143,
0.03764625526605462,
1.0189467943495765,
-1.1805772107399133,
0.2812463759993353,
1.4321423259624626,
0.004331180480864721,
-0.009722915333733264,
0.6451162813005181,
-1.3276490840851547,
-0.022791981388308465,
0.6904663535208042,
-0.08291995023229079,

-0.4292297662993819,
0.08451212085763125,
1.730745239172763,
-0.4572165915626394,
-0.5304437424739927,
0.6917595603044201,
-0.0834050618602206,
-0.18753214427540912,
0.03305271309241428,
-2.06126875311761,
1.5412249435528216,
0.6658854058101124,
-0.36436553183642006,
-0.9193216873412987,
-0.12900157901067746,
0.00542745656150127,
-2.171309732970475,
0.7554637248132821,
1.249093251376782,
-0.6445658098419218,
0.14255909437901,
1.9827929295292588,
1.7044112257377888,
1.3052476731654217,
-1.2421280078558083,
0.9803773552514545,
0.624656679395339,
1.5450717726138348,
1.3377635200462674,
1.6713162630334013,
1.1434539739578018,
0.7924845860796002,
0.7488178629415702,
-0.22907755928161938,
0.02068614393688229,
0.7608745118642098,
-0.9304086893990576,
0.32284859123973336,
1.1891889033470526,
-2.2276657670493667,
-0.3847051133881006,
0.34685342333136693,
0.549782335235608,
-0.741267714199706,
0.26835698412135817,
-0.3932040836252281],
[0.1668593898896292,
0.17040712573292452,
0.1805381301837931,
0.5412292500635668,
0.5489811947718172,
1.6527475146851556,
2.55163116002801,
2.5519072608250526,
3.5532683674980303,
3.6288880821260254,
3.628888082126026,
4.805536133422,
4.805536133422,
5.495812833389341,
7.061809444427505,
7.061809444427505,
7.091809082073759,
7.091809082073759,
7.177303629807857,
7.177303629807857,
8.014949756247965,
8.530162577551986,
8.530162577551986,
8.546061750342906,
9.04867098177819,
9.074898072060186,
9.632422305371618,
9.634373167303918,
10.38794675566215,

10.6266488059183,
1.098346025464491,
1.5064109656875124,
-1.345000798816457,
-1.0112785952191263,
1.6300712383138607,
0.8148371701480261,
-0.28350002839102345,
-0.33372360425933684,
1.6208682076493302,
0.8121027533840469,
1.424723017228629,
-2.424990019545176,
0.082574342220046,
1.5280227698708746,
-1.15137171364767,
0.8821561188139568,
-1.0160789387304932,
1.6746733218779515,
2.4704340348759835,
0.8258460042877822,
0.21603078517896931,
-0.05951632881852065,
-0.38389794767629976,
1.262782578763767,
0.2784323534326924,
-1.1033559163681899,
-0.6528860571486432,
0.2962709573582783,
0.7373896958863383,
-0.19469473272588958,
-0.5648694322067066,
-0.2865513283590244,
-0.1568370013695695,
0.5493402156787225,
-0.06660698090283595,
-0.03359282593361995,
2.386303225482439,
2.5818639178838447,
-0.2777126348162844,
0.9824372498806634,
-0.39422047955069633,
-0.2578521646674435,
-2.3189661125012333,
3.704222502127313,
0.0786550943979888,
-0.4292299615806753,
1.9657979836656132,
-2.0589782426096743,
0.5074656887310783,
0.529845836294102,
0.3333899546946687,
-2.293287544626138,
0.38495704078578347,
0.11228231507748758,
-0.5441382535456334,
-0.10489456654410802,
0.20011773016694326,
0.2071371298105447,
-2.3651747840346373,
-4.642753244408169,
0.31152670002610416,
0.5022276983536407,
0.868938175568576,
0.20348366213849262,
1.6475591691674232,
0.3723999247005541,
-0.6589367341470095,
-1.1662574597011768,
-0.6243580701563078,
0.7131540586709431,
0.1202680893039496,
-0.035429417928639004,
-0.764191314011073,
1.2086109032075552,

-1.051983148544319,
0.9246153479653275,
-1.1427863274072698,
1.332115914696049,
-0.27763903875773216,
-0.8366665559266224,
1.231317184155771,
1.0584874216549616,
-1.2625782973678872,
-0.13344263101573678,
-0.5438546353631553,
0.32212207070493337,
-1.303226737690094,
-0.01337174800709397,
1.5589050571537895,
0.3479397882311206,
-1.87970134892204,
2.977494205144941,
2.126179176803338,
0.873413305092612,
0.8500950046844986,
0.23032349681334924,
0.3909007718762141,
1.27926383991169,
0.4053229000380187,
0.18190897779649284,
0.3777935390994293,
0.16772342823060155,
0.7412973026981022,
0.639129360404442,
0.018341913161260836,
-1.4541884523192488,
0.28134413046256895,
2.4001077951469534,
1.6611157476326641,
0.7905510314185135,
0.3404203970178551,
2.183912805046456,
-0.3997518053650227,
0.053570896670028414,
-0.357205442227867,
2.465639862334342,
0.5537972989863772,
-0.39728466664110995,
-0.027817223268321305,
1.9807706991981826],
[0.5951590437264768,
0.5951597797898418,
0.596302078053195,
1.4089369768482796,
1.4331037442558932,
1.9334285070314567,
3.062681311500457,
3.0626813115004623,
3.701592986553387,
3.7015929865611663,
4.692693983743045,
4.692707272250184,
4.692707272250258,
4.692707272250313,
5.273956511551061,
5.2739565115534885,
5.274711944180805,
5.292162333185471,
6.388031186249758,
6.706092770308958,
6.706092773942781,
7.548703938317144,
7.681022457823495,
8.397974590110634,
8.397974590138377,
8.714530791128386,
8.714530791128386,
10.075092988071805,
10.075092988073155,

11.341131260513714,
1.2164966908304873,
0.5225196146354087,
-0.34542368185043265,
-1.252295733833459,
1.5735101799719247,
1.0464982053397573,
1.2394468649734893,
0.21595542567535536,
0.6527940529325085,
1.4145936861598034,
-0.5380512317672027,
-1.9014686324238337,
1.451754133219536,
2.1987565378949516,
0.938092448651857,
-1.6378088277288243,
0.38538945046186207,
1.1341127264381634,
0.4639756379522118,
0.6078279078278344,
0.6166736770850927,
-0.2105783577152824,
-0.35613934438782024,
0.019700965332415492,
-0.1607988588148763,
2.193665487035956,
-0.06423342915568941,
-1.9546338684001463,
-0.18633224680849803,
0.7857721800679403,
-0.6042593641712383,
0.7490426193215785,
-1.6440993970557098,
0.1929003499686422,
-0.13169470171686798,
0.1181804780373785,
0.15693401681144475,
-0.28658811756127545,
-0.20708135879200276,
0.09230795188954338,
-0.4114468277172402,
-0.8587939345722652,
-0.29108154529906144,
0.5427012646910324,
-0.25834402839896575,
0.08607676608574079,
-1.2954984273935897,
0.1813204379953717,
-0.34769281236878496,
-0.09683800499273743,
-0.9220596744797915,
-0.67003274699539,
0.4888752813745547,
1.2659094572448748,
0.11946751252820795,
-0.2586531113468492,
0.8859866115247531,
1.5010226653220677,
1.2253090413370418,
-1.41480051306012,
0.4438287250399343,
1.1348546909542183,
-0.26356600881424186,
-2.0062796648632446,
-1.5997185613370128,
0.19365670408843416,
1.9542746653083636,
-1.16753432767107,
-0.8341998310634118,
-0.8078884986396416,
1.057969787908329,
0.11874076530060441,
-0.37823536664235385,
-0.26520820180860055,

-0.05623142087332937,
-0.6351447036627758,
1.2477313546900863,
0.09475828495367859,
1.1570291529547316,
-0.3159140153065273,
-0.5798756886838644,
0.5049917679316769,
0.17933499272174766,
-0.6452919120449363,
0.3093583753904067,
0.6269418409971109,
0.536228963471926,
-0.39297460666924544,
-0.8585561386083093,
0.42310038622809626,
0.4625446959490829,
0.46551290354297115,
-0.006225430338097464,
-0.9498487290783498,
0.9283489563799108,
0.5288015885952413,
-0.6423745714782674,
0.055608107183735184,
-0.19141640271906557,
1.6106743132208,
-0.930502866063725,
0.13606233511650948,
0.7450273046384331,
-1.862998799714745,
0.8639045442938115,
2.2050766695222275,
2.1209593580448147,
1.5928713117159927,
-0.646261951983125,
0.18338232206466956,
1.9195145046471642,
0.9340015251080405,
0.22314701152636945,
0.22339358941062268,
-0.49545999196264406,
1.6497345768092624,
-0.37841357670502584,
0.19313501042389203,
0.07433308037827455,
1.831606295240235],
[0.013945989287166007,
0.014482887559355881,
0.7020792700957467,
1.2708090216179468,
1.8771473361142745,
1.898764043280349,
2.8886374873795146,
2.8887394286496386,
2.889554540070078,
2.8895545400700837,
2.889778536891296,
2.890175029071649,
2.8901764691110965,
3.881547037638128,
4.589967577258263,
4.916994542918821,
5.4213951120408375,
5.421396236977594,
5.937049320865825,
6.879796302074518,
6.931299553407329,
7.597215453538154,
7.941093252873733,
7.941093304737446,
9.23283235128487,
9.232832768748333,
9.232832769084892,
11.10587443204972,
11.182791630116155,

11.297769627933807,
0.7007490003921656,
0.8357798277265018,
-0.006144500286149594,
-2.002647677865005,
0.3078229546122665,
0.6497081019565148,
-0.5263987300225429,
-1.3591736982237872,
0.4978537208919288,
-1.0703987652029276,
1.4894572070841317,
0.10507602812754031,
1.2060409491957005,
0.37232709611155723,
-0.9426439100006672,
-0.36387040355717715,
0.9248005896093184,
1.0383864274136485,
-0.2845687665108692,
-1.3700022658840743,
0.197561542582957,
-0.32357133547759104,
-0.5540850172472152,
0.5749264875292693,
-0.8790799137852104,
1.0945413836746756,
0.3793043522465329,
-0.4056052817915047,
-0.03473792548266608,
0.06466238365347084,
0.06911551547308,
0.1689218888799635,
-0.24055379179177064,
1.0229644579244672,
-0.3032356624759471,
0.7939776889924924,
0.9000504980210274,
1.1922086788205504,
-0.09256238768786877,
0.5764640092401906,
0.20671868793560919,
1.5137094250639866,
-0.17335127063306519,
0.8566284172009878,
-0.14321647694006984,
1.3125145841881363,
1.3592268317355658,
0.1512217749645529,
0.40406540571588334,
0.048383985681794354,
0.19259007229061106,
0.09976438753283409,
-0.8546571821237718,
0.6621369033183476,
-0.1702351863826035,
0.36026352288591695,
-0.8538518608539489,
0.5392868667124087,
1.1279014251126462,
-0.36128436957709814,
0.388999106940785,
0.6046833425816769,
0.22322275018396126,
0.19171695999278726,
-0.909915166057891,
0.21776193961543494,
0.28996753900858246,
-0.5842323028487328,
-0.23864599488371552,
0.5243326463982184,
0.32056321937714066,
0.5356836550343521,
-0.3036011562711703,
0.28802877152454115,

-0.33781577627567244,
0.7960023934710929,
-0.34238272839391165,
-0.755152052016237,
-0.44352996180502197,
0.07213680417312635,
-0.10190373256064952,
0.6647215855084265,
-0.9303498099091333,
0.9785859773141862,
2.090431130104361,
-0.3615944398761197,
-0.05657593308707301,
-0.4893006812698228,
2.169089562284795,
-0.2512716878005756,
0.5324061192252376,
-0.35101147654456727,
0.809248070031391,
1.4509839161975397,
-0.14911304722907706,
0.009330045236156378,
0.16685413556340573,
-1.0472061077948698,
1.6539137014575498,
1.3611819463379937,
-0.1612641139265809,
-1.2962247052632323,
-1.3797066485510583,
0.7614302908946665,
0.5091243097447496,
-1.034844537010192,
1.0872752511102444,
-0.476692079414531,
0.49649227376774574,
-1.1491063122969838,
0.5155370403129137,
2.9855835289421178,
1.332632811562063,
-0.4511213152168112,
-0.18076672982111605,
-0.6499435359167832,
-0.1697346678000319,
1.4081723136905224,
0.442588820936982,
1.722678503402018],
[0.6039929180328771,
0.6039929180328772,
1.1870458270689,
1.1870458270688995,
1.412745515623535,
2.7932578893846336,
3.721244973412882,
4.900800320776727,
6.39210348373028,
6.39210348373028,
7.061133772685592,
7.061133772685592,
7.061133772685592,
7.064294030691084,
7.896781244010901,
7.896781244010901,
7.998168023342792,
8.09271241991372,
8.09271241991372,
9.328885229004163,
9.328906326502109,
9.56647323827175,
9.587868480377287,
9.606795901338938,
9.632809215929381,
9.925261433896223,
10.67745113859835,
11.226267254217802,
11.25813692634961,

11.259219336255413,
1.5276356326766196,
-1.2318130242699923,
0.07924764105056724,
-3.11015247863868,
1.71979592089301,
2.1636204056140302,
-0.44284897868332995,
-1.9599373218939633,
0.06248244737264575,
1.4749130816585614,
0.6808242800852113,
-2.8378068229153226,
1.7023278591356004,
1.50553848610602,
0.9365133390345636,
-2.1922613442274193,
2.8432021818280635,
0.1855585942894919,
-1.305416067894503,
0.1905478234118404,
0.7393995701939485,
0.19798052248890954,
-0.8075009715521676,
0.3948611135767452,
0.2802743807361451,
-1.4516873620037638,
1.5556285183865062,
-1.0754052466574715,
-0.26931261584286686,
1.2336702516634759,
-0.8384600779602994,
0.1263891070094113,
1.371252832802525,
0.215177045818575,
0.6381629880602765,
-0.10309421117565558,
0.4124383487994645,
-0.655499258712117,
0.4988310520659948,
0.9884276098863405,
1.4367219570983862,
2.175763151095631,
0.4657848221151968,
-0.43932071710829274,
0.9992343531072873,
2.543795366459747,
1.4484903663678823,
-1.0884458694866728,
1.5946427529603922,
1.6036026746991174,
0.5049418933876622,
-1.0871394525284908,
1.0140119509472394,
0.5109229333717124,
0.8743589388253838,
0.09626392588768701,
-1.263331036369507,
0.7924497462635596,
0.7602243892770352,
-0.02425703489356225,
-0.9488939939602019,
2.6693155936644737,
0.5100002820942202,
0.595619937330169,
-0.8296192082845288,
0.03973961449691636,
-1.7144893205098422,
1.1592271204386744,
0.04896218554826232,
0.6673584031026698,
-0.8454442747231521,
1.0304518931414424,
0.8556412792926265,
0.538780781704616,

-2.308898512383933,
1.5082121217292006,
0.9522091008412682,
-0.6026951427085929,
-0.561943926102551,
0.0993983558744849,
0.4791162049419018,
0.10320414203165484,
0.3730055252741846,
-0.47083232270591907,
1.8673303303286475,
0.11712430229385423,
-0.6044688726688213,
-0.9010581315425702,
0.8870944114042272,
-0.6585959519901412,
1.4737812112349316,
-0.08789784470103781,
2.094313922775301,
-0.33880630598079,
2.0864279335765668,
-0.3541510297313768,
-0.1447960879675017,
1.1237218765827641,
0.7472353955833465,
0.43980261574830926,
0.6534205210717992,
1.2606259200129655,
-0.17935209951086067,
1.6469967332630842,
-1.0541245665671384,
-1.9424469299865808,
-0.18975499615277758,
1.967627931161089,
-0.5610576072704585,
-1.4036970670846107,
2.6946925238411343,
-0.9860662529475024,
-0.6528571620700129,
-0.6551571328219318,
0.9777230807240078,
0.003267369094500521,
-1.3217863730621935,
0.3048209868145306,
2.1104722024376037,
-0.13744008360720306],
[0.25527267369507034,
0.25565792243175783,
0.2647851751522132,
0.5363821708593047,
0.5363873360294482,
0.7233633669048282,
1.469495303301716,
2.6782106232033085,
3.0738233934797257,
3.0740346152756928,
3.0740346154225127,
3.5401415295205747,
4.384558950933806,
4.384564402805264,
5.187694707345863,
5.431645815108105,
5.489249393218873,
5.489270645046236,
5.4892706450864015,
6.745887429448721,
6.751180124796475,
6.975059797459636,
7.4852253234058015,
7.485225323406553,
7.485225336458803,
8.65422617922401,
9.588402512165866,
9.885475791973308,
10.59890299271783,

10.69671178550786,
1.7193952291225598,
-2.699905058742405,
-1.223597261476082,
1.010311557825714,
2.412320728895392,
0.01205800479020808,
0.07148434641220561,
-0.1717056232391439,
0.7382566227367261,
1.1624421524382496,
-0.9704132592114074,
0.007350927807351759,
2.3582361698297074,
1.1487298862005386,
-0.5298367489795935,
-0.09976600187590766,
0.7543852753586129,
1.401553978295213,
0.3078871075376248,
-0.35588965288410024,
0.15554254602359127,
0.3425968302644435,
-1.202384247417669,
-0.987347192012539,
0.4529373843839761,
0.4903191821294036,
-0.3596344497958935,
-0.17493820576367525,
1.3619720378391473,
-0.2517632245160441,
0.6511229549349082,
-0.07544040749490964,
-0.04771613185909407,
-0.3755018325798389,
0.9938372515787193,
-2.617459009880976,
-1.1600518599196699,
1.385641845308893,
1.3078548392623544,
-1.3438342667259302,
0.06225569718179881,
2.197693302377489,
1.042516443745511,
0.03537145740203309,
-0.4361466628046714,
0.5235506730441506,
-0.526852610863259,
0.2683619694707144,
0.4190531675420984,
1.1917182306749494,
-0.22315973823529625,
0.3724063269005885,
-0.2016113706580773,
-0.6608684665627687,
-0.5566572791655187,
-0.6080637139710359,
-1.3764025932169088,
0.06381643974581075,
-0.12239416020448966,
0.8768915165514672,
0.9734732325585186,
0.43086234391719236,
-1.2868778636436715,
0.2387140460098626,
0.9650786815364466,
-0.8088454790489977,
0.3379873670823351,
-0.5777804082620341,
0.5336495835071627,
2.9920059240641073,
0.6079052241489462,
0.5309531439532562,
-0.3169545414432997,
-2.3335729587340226,

1.4959043429555212,
-0.5568960671660346,
1.653532167337357,
0.35120291709985135,
-1.2609025397237739,
-0.1931706005433399,
-0.7761167208985412,
-0.07661975220377984,
-0.5351965211836152,
-0.0031199845298622592,
-1.0423575145047397,
0.1404336078605159,
-0.2659485743030148,
-1.6615430856264528,
0.43884229259462737,
0.6812026744892391,
0.39460126400851775,
0.9189185317254136,
1.0545864345697524,
1.1559068061784519,
1.0720705313960006,
2.514687033544068,
2.0122769373294087,
1.2260219872848024,
-1.3734724775008675,
2.3904026332666466,
-2.404362121009983,
-0.8308162974608817,
-0.49116355387064226,
1.4544417316540983,
1.2021789553657038,
-0.045966888650276455,
-0.5647889892892385,
0.6643832758564343,
1.3569537910410745,
-0.9183847616510102,
0.945220285616687,
-0.2507148012652629,
0.7511336622169466,
-0.9416337312429676,
-1.4793772681330912,
0.4334847087508488,
-0.02256141084202589,
-0.8421263780740565,
0.7532984525691246,
-0.4880194119115941],
[0.008743077103622827,
0.009882832507877682,
0.7050944318623401,
0.710375902327237,
1.3735081730499452,
1.9102715342756955,
2.4433320146909057,
2.4516882946549723,
2.451688347950266,
2.451688347950339,
2.843659132454273,
2.8437290613274957,
4.1600447531010465,
4.306699526502985,
4.306699526503065,
5.241123177068926,
5.241123178389404,
5.24112317839451,
5.241123178416889,
5.24112317841696,
5.540975568888287,
5.540975570417886,
5.941112811664419,
8.170699047520403,
8.170699047520507,
8.170699047530466,
8.255093472901638,
10.272293989986627,
10.424665220800923,

11.465171522837265,
3.9395345563718758,
0.4691052202571919,
-1.862560190849311,
-0.5409749194015872,
0.6809788356765686,
1.8042992706436805,
-0.26820425656414737,
0.1583592390805078,
2.570139180175488,
1.9730281302693196,
-1.463840299189443,
-0.6685636516618708,
1.7639202171290544,
-0.25593154759594605,
-0.2985776439588054,
-2.380396900296592,
0.702730788267215,
0.1111013414139888,
0.8409172298864358,
-1.2895389501906573,
-0.6495780843509739,
-0.7851296452736322,
0.551111177160635,
-0.7726813911354989,
0.5538939215319534,
0.03671507307592432,
0.3611092546557634,
-1.1244512927039525,
0.9979933722488747,
2.1458567471834726,
0.6045511633065316,
-2.057454443497708,
1.252679002491011,
1.430968651076373,
0.1975347289978355,
-1.299395231227167,
1.3709528213003457,
0.1431677222477642,
-0.5468176167431326,
-1.363138690512715,
3.0723179570358097,
0.09197418207209145,
0.552028513182037,
-0.14805587495457423,
0.6542791830341044,
-0.6463721623376274,
0.594458988807259,
0.36313400085196706,
-0.22487670595801615,
-0.5600947295073319,
-0.13194082373612193,
-0.7877948735702447,
0.37453010501171596,
0.4466259555663157,
1.308811697533367,
0.18300565834449778,
0.28722746055568227,
0.11593248147227052,
-0.22092771987873516,
-0.3830228539100271,
-1.1848305159650747,
0.5197091232696622,
-0.20788874837533922,
-0.6085287569162429,
0.622048119692812,
0.36154402486320714,
-1.283336122338554,
0.06614563071195226,
-0.10338801392549199,
-0.6587068471684263,
1.34100892693596,
1.611824691291148,
0.6916042008757854,
1.6327439292692163,

-0.6648342512722216,
0.37654710496224697,
-0.24718958516166675,
-0.7065038375487233,
0.11746925267221191,
-0.5800210884330035,
-0.5819296114685866,
0.01696046668851879,
0.08164598903007107,
0.39930256073308995,
-0.22041497304878216,
-0.13569368036844073,
-0.734071731705797,
-0.10448732621745443,
0.4350514428273773,
-0.4345336137549378,
0.39154417527507906,
1.0175423593468684,
2.0822612023831226,
2.2518057581506135,
-0.4050083292569126,
-1.3225723626396935,
0.2066177197668252,
0.3518219705658924,
0.7692056892056433,
1.1078718838725619,
0.6955122293015593,
2.6391129283967607,
-0.5193467200233968,
0.16205999751797937,
1.0319480004914923,
0.0466744549888964,
0.8866401912062555,
0.6199118226017315,
0.9594432756697039,
0.30783399339708145,
2.007890427791846,
-0.42290702105473665,
-1.4981000967272742,
0.1246045734342663,
1.4175139007816862,
0.730422807509675,
-1.0203109099547352,
1.2475764531977165,
1.172103834016867,
2.0417923602468857],
[0.3006295792025693,
0.5011825648484141,
1.736075427951676,
1.7894431127664152,
2.4859557870150857,
3.6898895058559917,
3.73290444462633,
3.73290444462633,
3.741728259130309,
4.1342678631525835,
4.145528657970977,
4.615983426127272,
5.6679892124870985,
5.692197651053977,
5.773115344270512,
6.56178194421021,
6.56178194421021,
6.56178194421021,
6.56178194421021,
6.56178194421021,
6.561820835625042,
6.6476324960056985,
7.649631249906491,
8.361638748929934,
8.576867085540188,
8.77036061445696,
9.827286891841206,
10.5379658375992,
10.558826131428928,

11.83004188936984,
1.5987671786815294,
-1.5058826811902317,
2.0440908601375476,
-1.481172398738803,
1.4265067306792671,
1.3375234069756785,
-1.0089917144708827,
-1.8440331252413595,
-0.062239877224771915,
0.18483713323292328,
1.8795821846896128,
-2.058552713287731,
0.9708425302661001,
0.8840527876953188,
0.5964490041287303,
-1.4043070469033054,
0.14533141919970372,
0.022743431858071796,
-0.3923086391326145,
0.11023448615420071,
-0.4204112521530586,
0.5529792020785508,
-0.6330031478534407,
0.23236293228773558,
-0.4459800436834905,
2.5876988326738504,
0.6844024813933239,
0.1732372067049823,
0.7384853056565549,
2.8826092918303563,
-0.11204335589185775,
-0.2241007440171772,
-0.4845273525281558,
0.2744214554588425,
0.26177442491958286,
0.4512467910671607,
-0.6063187184164835,
-1.2386469417803163,
-1.1719897599242013,
1.9705273482015357,
1.945221133626245,
0.15309311442469994,
0.690693149150703,
0.22188036413222653,
0.47722534320651633,
0.12444414704260098,
-1.0057594761976716,
-0.3351811680967541,
1.0750707752291546,
-0.04597654655420585,
-1.4770936089010136,
0.1839119320523196,
1.0453831678930239,
1.6851715686961402,
2.1701829049155683,
1.692817047863365,
0.44152889324253053,
-0.3402272157802259,
-0.4600811596334244,
2.365566181908148,
1.166608300050211,
-0.12230572309949261,
-0.07192631995381693,
-0.48825404466402955,
1.202761995004197,
0.503733093641068,
-0.2613796073125891,
0.7323467617888468,
0.30547819787235264,
0.46207631030173263,
0.034964002783702264,
-0.30773087511026637,
2.2827399003210256,
0.773042400324002,

0.07466204753764587,
-2.812532590652055,
0.7925074856015164,
3.3306642432255886,
-0.8558810509247932,
-0.9283746444488663,
0.3079774400122357,
0.5580467297921388,
0.5588934104639726,
0.6444350594161403,
1.5119717259179621,
0.14573434197560214,
0.7631266139944589,
-0.43452714539522225,
0.38832086457140946,
0.36234082213113916,
-1.3222695581402877,
-0.6140190631377459,
2.2873566508452354,
1.0349767974312705,
-0.5734592440854204,
0.5234040439377924,
-0.8213429806368828,
-0.8621355796854082,
-0.5355268774606846,
1.8804920616270637,
0.022682713405259486,
1.3625464560687972,
-0.48449497738356756,
1.6018376883784673,
-1.3963097787223904,
-3.511711037561943,
1.3223211880663501,
2.1298031669093946,
0.24542774430512565,
0.7034284248706182,
-0.37346698797179273,
-0.43051209966139187,
-0.1034377630705127,
-0.07417862116176438,
0.2595758000914042,
0.9523286668350031,
0.2556898901330524,
0.22603519843353895,
0.9866656966967039,
2.5031147117731094],
[0.6608465720820748,
0.9671982645454077,
1.5458030244804597,
1.5458030244804597,
2.2589946156031733,
2.2589946156031733,
2.2589946156031733,
2.6595607382107107,
3.734699662125929,
4.07592175958283,
4.611631942641649,
4.831287600721991,
4.831287600721991,
5.914025831134952,
5.914025831134952,
6.072969066301945,
6.991999801396327,
6.991999801396326,
7.161687480905546,
7.673022292320241,
7.821921961804373,
7.912193421610441,
8.287160283389671,
8.956173402171897,
8.956173402171897,
9.75268721078065,
9.941478738047602,
9.941478738047602,
9.945621910117605,

10.438880525762482,
1.4707024445054038,
1.5353178524994073,
0.8263465992797864,
0.5104925709601137,
0.0752285958369072,
-0.1142202547489564,
-1.4526020976761953,
-1.3531650678039584,
-0.5250403673431315,
2.237334263415747,
0.2921144645165799,
-1.3764237769851546,
0.48788675650681135,
0.8685067508645514,
0.519063599878773,
1.1970126362568978,
0.35376457746361467,
2.1361591146989944,
1.704664258187609,
1.1956277779681033,
-0.8317273346515459,
1.1961203419780542,
0.6738815365004568,
2.0453196594697167,
-0.3197500197249862,
0.3940149430820266,
-0.03491413991246975,
1.9231424652296574,
-0.7553371587587144,
0.5536882090799309,
-1.1151355813488422,
0.06532487835880797,
-0.03610632383025197,
-0.721099071948806,
-0.7712693743746337,
0.2581737620009136,
0.6969815452179084,
-0.4519889309472695,
-0.6781990341468844,
0.3896669141200773,
0.746036455743935,
0.10534513312025831,
-0.7953867627526213,
-0.17196672908166516,
1.1080637640216904,
-0.36906579226317304,
0.3903518027422727,
0.7340380536410088,
0.16483545086429002,
0.12911772875241279,
-0.11978924907966522,
-0.5410781247103603,
0.27894738356501725,
-0.04603280575778243,
0.3301412555435043,
0.020942359039511207,
-0.535470327679227,
0.26379415288467284,
0.7549681986050077,
-0.774180168996079,
0.3042677541816752,
0.2314702730743271,
0.4347582057672544,
0.5269984906212896,
-0.5883800499089586,
0.4216788732962846,
-0.39154710450095487,
-0.20962806789618654,
0.008988716066278926,
0.42160932452985955,
-0.4362625059483254,
0.7149808650234762,
-0.9606565556866927,
0.02867923202735973,

0.6474453926631074,
0.3125034088891559,
0.24941070182093517,
-1.0996459114603814,
0.42918254303405223,
1.1958230549534523,
-0.23035300079654375,
0.43394127898307777,
-1.1958360530234091,
0.2490392548916114,
0.956337072693143,
-0.9098880983840639,
0.24421397010525564,
0.32228144879356985,
-0.04019115817841665,
0.601865593110944,
-1.0390070893344725,
-0.3870853595246112,
1.5254374970388465,
-0.06482825900051802,
1.508536156596221,
-0.4193438972827081,
-0.2124704995099638,
0.7982743594716017,
0.625924999124879,
-1.2144964531756246,
-0.7306315051046057,
-1.3775043670320717,
-0.5254300493631905,
0.24182169388976885,
0.05618119509583439,
-1.1649033986061088,
0.4414909596355956,
1.1243259379005652,
0.9623501696515389,
-1.0198159738874641,
0.6746700507964939,
0.9071271136722682,
-1.490119256375095,
-1.8915932031257203,
1.4861820000050194,
0.020835967248483293,
0.2672579153437498,
1.1487299094399863,
1.7394956603554965,
2.161406077992875],
[0.009700543149373924,
0.009700543149373951,
0.00970054314937392,
1.382070146567131,
1.7587904521787645,
2.560406827400793,
3.0924393369175114,
3.3853584719898073,
3.3853584719898073,
3.3855655528609647,
3.9151369396989866,
3.9151369396989866,
5.054043379727587,
5.058642146366078,
6.082229999853835,
6.348636465887238,
6.754028454322125,
6.777976106859175,
7.783524996765277,
7.783524996765276,
7.783524996765276,
7.995280883445485,
7.995280883445485,
8.645089634948334,
8.925040343314908,
9.356899631457969,
9.439062141544271,
9.717990345174004,
9.717990345174004,

10.595210012065888,
0.8074121230882934,
-0.11817978440691497,
2.3039735343549332,
-2.1294851108225332,
0.6714134591558454,
0.45516684502263977,
-0.8318799457604262,
-1.6937930072137053,
0.039287662842651484,
0.5223639661903685,
-1.2122069881806565,
0.27186944236208743,
0.22304871394760792,
-0.08855679301585977,
1.1988033553075006,
-0.9660581453773212,
1.393585655420428,
0.524497607950108,
-0.22549193465024678,
-0.20940428560116026,
-0.22546417020605838,
-0.06641646114084936,
0.3531991811801793,
-0.06811222553567531,
1.0762936524340476,
0.17762298521732672,
-0.24437612021854904,
-1.0622868736463833,
0.41092319903165814,
-0.5877006275418265,
1.2299752305066975,
-1.5519910987794148,
0.6657478217748831,
-0.027661876184918333,
0.011781258679136937,
0.9082970931589502,
0.006644189925206108,
0.3712893817061966,
-0.04659784477089089,
0.2991695519878787,
1.4480032246106211,
0.3765628272855095,
0.3810659512210775,
-0.4053198419433054,
-0.06957688452126994,
0.08768201200938007,
-0.16514507742174195,
0.05923670456485792,
-1.0025267882306543,
-0.041476053822733706,
0.4728058870419665,
-0.09799142504381375,
-0.642516889827151,
-0.521645479249954,
0.18062390558973918,
-0.5137105837376863,
0.315588961893638,
1.0102820620809385,
-1.412161058774387,
0.5136468128759858,
-0.5616754178138471,
0.11099281329082958,
-0.4065421947836274,
-0.8520422495977672,
1.5662481725135406,
-2.09542418518098,
1.2116955825236206,
0.8221816705323761,
0.041900809919046114,
-0.09736226691448804,
-0.6600925025551557,
-1.1656564284287168,
0.8214103261157102,
0.28477271669010623,

```
0.5992804506866763,
-0.21193327493741926,
0.6490050812244267,
0.8872065875645788,
0.2758198961026023,
-0.3259804283588707,
0.5954957550354935,
0.8064107897831347,
0.15673794452304002,
-0.5606863141571317,
1.1354810000650886,
0.29170605524887394,
-0.6471431342469229,
-0.29364595444579666,
0.05289373230476578,
0.03489286830366231,
-0.10405686076243598,
0.4547289014370767,
-0.1747513431047453,
-0.28439942445332933,
0.010516254927139741,
0.11813385237732216,
0.3966163987420368,
-1.37250350754582,
0.06601400397758364,
-0.3662241908629097,
-0.07601408247331909,
1.737776242565087,
2.1849230369400683,
-0.5785032754175823,
-0.5063968590594791,
-0.23335270192508997,
2.320012252653636,
0.6182382779358941,
0.7875200859675175,
-0.009873677663214134,
0.7482624361366613,
-0.35875384806276184,
-0.5357091554695692,
0.4397736267416906,
0.16706548411788932,
0.9251966251074774,
0.053252937203382235,
-0.23592171730002592,
1.2877035765438765,
2.1426518726936172]],
'run_time_sec': 43067.655497312546}}],
'redundant': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [
0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': { 'max': 10.0,
'min': 0.0},
'type': 'heisenberg'}}},
'init_param': { 'value': [ 0.6189603325174557,
0.7142372255934472,
0.818217611349809,
1.1433975540074377,
1.9121137997887092,
1.912113799788709,
1.9189666863881127,
2.194219841514053,
```

2.734492156740232,
4.250258815642276,
4.250258815642276,
5.774076859791501,
6.527612880112256,
6.532834731882501,
8.049613322901676,
8.049613322901676,
8.050730377555883,
8.561643142317116,
8.561643142317116,
8.758357012835646,
8.758357012835646,
8.78353884989931,
9.477076383056275,
9.536600626467342,
9.536600626467342,
9.536600626467342,
10.241295505019522,
10.318321333031077,
10.579129354925781,
10.82886994059623,
3.662494394083504,
0.08309129295248968,
1.5295376506525808,
-1.5964000102221196,
1.8988970610647178,
-0.07218621526676876,
0.7238532160702689,
-1.2917423920622229,
1.8532222971765195,
-0.40818119715800666,
0.469598065996274,
-1.2652675561295137,
0.41465882817859046,
0.48683539382624846,
-0.15336518315146752,
-1.6419783820068856,
1.2136622236743368,
0.5933123377118196,
0.9666225223219593,
0.9539776194708611,
-0.3793986531891993,
-0.2570964298405315,
1.5884302353807402,
0.19867970725527498,
1.0036205245072367,
-1.7266220384949087,
0.6519864384701275,
0.173373648086925,
1.8697046853700252,
0.5263944462203128,
2.9785129063723943,
-0.5937256494260224,
0.22753174799850245,
0.24918335157216837,
1.5438671949833176,
-0.2981477150404249,
-0.0503664733834821,
0.542442835193611,
1.8096859498837612,
-2.2260935596321993,
0.28693396030012636,
0.9986781622414097,
-0.01940263773226835,
-1.8805315707929653,
-0.1513202473661221,
0.1836836320118686,
-0.5655439481125303,
-0.30323205019871646,
-0.4186198812206128,
0.2357002564872809,
2.8296979465953833,
-1.0776781071762758,
-0.08378558447273823,

1.7446476514004516,
-0.4606832586075214,
-1.762173446350097,
0.10241780783627007,
1.6018254886002428,
0.2621004786967086,
1.349261138919249,
0.14987623247602613,
0.5068543254371847,
-1.3474781551208477,
1.0890436270679893,
1.6662048029626129,
0.3608718991529565,
-0.11137748070221386,
0.5641204379289552,
2.2546014997074266,
1.8888740160956543,
-0.23345107582518032,
-1.5772444286238798,
2.1408383357361265,
-0.4968079560788485,
0.1533667959779154,
-0.40238715350257875,
0.25591729793510365,
2.947382531044255,
-0.32162602471486745,
-0.7587223675916386,
-0.8983111566200868,
0.006114119281372181,
2.9372824954589514,
0.48288231335139287,
0.7939297064609614,
1.9454360656955985,
2.092519500518508,
0.8995449331251366,
0.10171517677156572,
0.45235144869603666,
1.02501950409702,
0.24331156538914245,
0.48973195386607005,
2.672871435464254,
1.313352678455905,
2.010046934709317,
-0.42153209835304317,
-0.13436700176919486,
-0.6950305532982513,
0.6026553555930264,
1.6681913778840949,
0.4125109526933123,
0.514533549588801,
1.0794873063708696,
-0.9571806916209341,
0.27630815458061647,
0.6001387806934945,
0.5059087494947612,
-0.5722342732268321,
0.3986919058890637,
1.0044202485382987,
0.3352295947729849,
1.3777059541173544,
-2.835684252944388,
0.6104690716589696,
1.838086538234438,
-0.3106134411398272,
-2.2167272073692903,
0.10706372359431235,
-0.17469850626418476]],

'noise_profile': { 'noise_on_init_param': {

'value':

'noise_prob': [0.001,
0.001,
0,
0],

'status': False,

0},


```

        'status': True,
        'type': 'depolarizing'},
    'nqubits': 7,
    'observable': { 'coefficients': { 'bn': [ 1.0,
        1.0,
        1.0,
        1.0,
        1.0,
        1.0,
        1.0],
        'cn': [ 0.5,
        0.5,
        0.5,
        0.5,
        0.5,
        0.5],
        'r': 1},
        'def': 'ising'},
    'output': { 'draw': { 'fig_dpi': 100,
        'status': False,
        'type': 'png'},
        'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_ric4_unifrom_scaling_sample#3'},
    'redundant': { 'identity_factors': [ [0, 0, 0, 0],
        [1, 1, 0, 0],
        [2, 2, 0, 0],
        [3, 3, 0, 0]]},
    'run': 'redundant',
    'state': 'dmatrix',
    'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
        'constraint': True,
        'status': False}},
    'zne': { 'data_points': None,
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
    'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '
        '2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '
        '2 + 1 X 3 X 4 + 1 Z 3 + 1 X '
        '4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
        '1 Z 5 + 1 Z 6',
        'time_evolution_gate_hamiltonian_string': [ '0.5 X '
        '0 X 1 '
        '+ 0.5 '
        'Y 0 Y '
        '1 + '
        '0.5 Z '
        '0 Z 1 '
        '+ 0.5 '
        'X 1 X '
        '2 + '
        '0.5 Y '
        '1 Y 2 '
        '+ 0.5 '
        'Z 1 Z '
        '2 + '
        '0.5 X '
        '2 X 3 '
        '+ 0.5 '
        'Y 2 Y '
        '3 + '
        '0.5 Z '
        '2 Z 3 '
        '+ 0.5 '
        'X 3 X '
        '4 + '
        '0.5 Y '
        '3 Y 4 '
        '+ 0.5 '
        'Z 3 Z '
        '4 + '
        '0.5 X '
        '4 X 5 '
        '+ 0.5 '
        'X 5 X 6 '
        '+ 0.5 '
        'Y 6 Y '
        '+ 0.5 '
        'Z 6 Z '
        '7 + '
        '0.5 X '
        '7 X 8 '
        '+ 0.5 '
        'Y 8 Y '
        '+ 0.5 '
        'Z 8 Z '
        '9 + '
        '0.5 X '
        '9 X 10 '
        '+ 0.5 '
        'Y 10 Y '
        '+ 0.5 '
        'Z 10 Z '
        '11 + '
        '0.5 X '
        '11 X 12 '
        '+ 0.5 '
        'Y 12 Y '
        '+ 0.5 '
        'Z 12 Z '
        '13 + '
        '0.5 X '
        '13 X 14 '
        '+ 0.5 '
        'Y 14 Y '
        '+ 0.5 '
        'Z 14 Z '
        '15 + '
        '0.5 X '
        '15 X 16 '
        '+ 0.5 '
        'Y 16 Y '
        '+ 0.5 '
        'Z 16 Z '
        '17 + '
        '0.5 X '
        '17 X 18 '
        '+ 0.5 '
        'Y 18 Y '
        '+ 0.5 '
        'Z 18 Z '
        '19 + '
        '0.5 X '
        '19 X 20 '
        '+ 0.5 '
        'Y 20 Y '
        '+ 0.5 '
        'Z 20 Z '
        '21 + '
        '0.5 X '
        '21 X 22 '
        '+ 0.5 '
        'Y 22 Y '
        '+ 0.5 '
        'Z 22 Z '
        '23 + '
        '0.5 X '
        '23 X 24 '
        '+ 0.5 '
        'Y 24 Y '
        '+ 0.5 '
        'Z 24 Z '
        '25 + '
        '0.5 X '
        '25 X 26 '
        '+ 0.5 '
        'Y 26 Y '
        '+ 0.5 '
        'Z 26 Z '
        '27 + '
        '0.5 X '
        '27 X 28 '
        '+ 0.5 '
        'Y 28 Y '
        '+ 0.5 '
        'Z 28 Z '
        '29 + '
        '0.5 X '
        '29 X 30 '
        '+ 0.5 '
        'Y 30 Y '
        '+ 0.5 '
        'Z 30 Z '
        '31 + '
        '0.5 X '
        '31 X 32 '
        '+ 0.5 '
        'Y 32 Y '
        '+ 0.5 '
        'Z 32 Z '
        '33 + '
        '0.5 X '
        '33 X 34 '
        '+ 0.5 '
        'Y 34 Y '
        '+ 0.5 '
        'Z 34 Z '
        '35 + '
        '0.5 X '
        '35 X 36 '
        '+ 0.5 '
        'Y 36 Y '
        '+ 0.5 '
        'Z 36 Z '
        '37 + '
        '0.5 X '
        '37 X 38 '
        '+ 0.5 '
        'Y 38 Y '
        '+ 0.5 '
        'Z 38 Z '
        '39 + '
        '0.5 X '
        '39 X 40 '
        '+ 0.5 '
        'Y 40 Y '
        '+ 0.5 '
        'Z 40 Z '
        '41 + '
        '0.5 X '
        '41 X 42 '
        '+ 0.5 '
        'Y 42 Y '
        '+ 0.5 '
        'Z 42 Z '
        '43 + '
        '0.5 X '
        '43 X 44 '
        '+ 0.5 '
        'Y 44 Y '
        '+ 0.5 '
        'Z 44 Z '
        '45 + '
        '0.5 X '
        '45 X 46 '
        '+ 0.5 '
        'Y 46 Y '
        '+ 0.5 '
        'Z 46 Z '
        '47 + '
        '0.5 X '
        '47 X 48 '
        '+ 0.5 '
        'Y 48 Y '
        '+ 0.5 '
        'Z 48 Z '
        '49 + '
        '0.5 X '
        '49 X 50 '
        '+ 0.5 '
        'Y 50 Y '
        '+ 0.5 '
        'Z 50 Z '
        '51 + '
        '0.5 X '
        '51 X 52 '
        '+ 0.5 '
        'Y 52 Y '
        '+ 0.5 '
        'Z 52 Z '
        '53 + '
        '0.5 X '
        '53 X 54 '
        '+ 0.5 '
        'Y 54 Y '
        '+ 0.5 '
        'Z 54 Z '
        '55 + '
        '0.5 X '
        '55 X 56 '
        '+ 0.5 '
        'Y 56 Y '
        '+ 0.5 '
        'Z 56 Z '
        '57 + '
        '0.5 X '
        '57 X 58 '
        '+ 0.5 '
        'Y 58 Y '
        '+ 0.5 '
        'Z 58 Z '
        '59 + '
        '0.5 X '
        '59 X 60 '
        '+ 0.5 '
        'Y 60 Y '
        '+ 0.5 '
        'Z 60 Z '
        '61 + '
        '0.5 X '
        '61 X 62 '
        '+ 0.5 '
        'Y 62 Y '
        '+ 0.5 '
        'Z 62 Z '
        '63 + '
        '0.5 X '
        '63 X 64 '
        '+ 0.5 '
        'Y 64 Y '
        '+ 0.5 '
        'Z 64 Z '
        '65 + '
        '0.5 X '
        '65 X 66 '
        '+ 0.5 '
        'Y 66 Y '
        '+ 0.5 '
        'Z 66 Z '
        '67 + '
        '0.5 X '
        '67 X 68 '
        '+ 0.5 '
        'Y 68 Y '
        '+ 0.5 '
        'Z 68 Z '
        '69 + '
        '0.5 X '
        '69 X 70 '
        '+ 0.5 '
        'Y 70 Y '
        '+ 0.5 '
        'Z 70 Z '
        '71 + '
        '0.5 X '
        '71 X 72 '
        '+ 0.5 '
        'Y 72 Y '
        '+ 0.5 '
        'Z 72 Z '
        '73 + '
        '0.5 X '
        '73 X 74 '
        '+ 0.5 '
        'Y 74 Y '
        '+ 0.5 '
        'Z 74 Z '
        '75 + '
        '0.5 X '
        '75 X 76 '
        '+ 0.5 '
        'Y 76 Y '
        '+ 0.5 '
        'Z 76 Z '
        '77 + '
        '0.5 X '
        '77 X 78 '
        '+ 0.5 '
        'Y 78 Y '
        '+ 0.5 '
        'Z 78 Z '
        '79 + '
        '0.5 X '
        '79 X 80 '
        '+ 0.5 '
        'Y 80 Y '
        '+ 0.5 '
        'Z 80 Z '
        '81 + '
        '0.5 X '
        '81 X 82 '
        '+ 0.5 '
        'Y 82 Y '
        '+ 0.5 '
        'Z 82 Z '
        '83 + '
        '0.5 X '
        '83 X 84 '
        '+ 0.5 '
        'Y 84 Y '
        '+ 0.5 '
        'Z 84 Z '
        '85 + '
        '0.5 X '
        '85 X 86 '
        '+ 0.5 '
        'Y 86 Y '
        '+ 0.5 '
        'Z 86 Z '
        '87 + '
        '0.5 X '
        '87 X 88 '
        '+ 0.5 '
        'Y 88 Y '
        '+ 0.5 '
        'Z 88 Z '
        '89 + '
        '0.5 X '
        '89 X 90 '
        '+ 0.5 '
        'Y 90 Y '
        '+ 0.5 '
        'Z 90 Z '
        '91 + '
        '0.5 X '
        '91 X 92 '
        '+ 0.5 '
        'Y 92 Y '
        '+ 0.5 '
        'Z 92 Z '
        '93 + '
        '0.5 X '
        '93 X 94 '
        '+ 0.5 '
        'Y 94 Y '
        '+ 0.5 '
        'Z 94 Z '
        '95 + '
        '0.5 X '
        '95 X 96 '
        '+ 0.5 '
        'Y 96 Y '
        '+ 0.5 '
        'Z 96 Z '
        '97 + '
        '0.5 X '
        '97 X 98 '
        '+ 0.5 '
        'Y 98 Y '
        '+ 0.5 '
        'Z 98 Z '
        '99 + '
        '0.5 X '
        '99 X 100 '
        '+ 0.5 '
        'Y 100 Y '
        '+ 0.5 '
        'Z 100 Z '
        '101 + '
        '0.5 X '
        '101 X 102 '
        '+ 0.5 '
        'Y 102 Y '
        '+ 0.5 '
        'Z 102 Z '
        '103 + '
        '0.5 X '
        '103 X 104 '
        '+ 0.5 '
        'Y 104 Y '
        '+ 0.5 '
        'Z 104 Z '
        '105 + '
        '0.5 X '
        '105 X 106 '
        '+ 0.5 '
        'Y 106 Y '
        '+ 0.5 '
        'Z 106 Z '
        '107 + '
        '0.5 X '
        '107 X 108 '
        '+ 0.5 '
        'Y 108 Y '
        '+ 0.5 '
        'Z 108 Z '
        '109 + '
        '0.5 X '
        '109 X 110 '
        '+ 0.5 '
        'Y 110 Y '
        '+ 0.5 '
        'Z 110 Z '
        '111 + '
        '0.5 X '
        '111 X 112 '
        '+ 0.5 '
        'Y 112 Y '
        '+ 0.5 '
        'Z 112 Z '
        '113 + '
        '0.5 X '
        '113 X 114 '
        '+ 0.5 '
        'Y 114 Y '
        '+ 0.5 '
        'Z 114 Z '
        '115 + '
        '0.5 X '
        '115 X 116 '
        '+ 0.5 '
        'Y 116 Y '
        '+ 0.5 '
        'Z 116 Z '
       
```

```

'Y 4 Y '
'5 + '
'0.5 Z '
'4 Z 5 '
'+ 0.5 '
'X 5 X '
'6 + '
'0.5 Y '
'5 Y 6 '
'+ 0.5 '
'Z 5 Z '
'6']],
'output': { 'data_points': [ [4, 2, 0, 0, -6.218690041461777],
[12, 6, 0, 0, -4.805156509769758],
[ 20,
10,
0,
0,
-3.7079207759854302],
[ 28,
14,
0,
0,
-2.8469331605050203]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [ 0,
0,
0,
0],
'noise_level': [4, 2, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0,
0],
True,
'depolarizing'},
'status':
'type':
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [ 1,
1,
0,
0],
'noise_level': [12, 6, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0,
0],
True,
'depolarizing'},
'status':
'type':
'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
'identity_factors': [ 2,
2,
0,
0],
'noise_level': [20, 10, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0,
0],
True,
'depolarizing'},
'status':

```

```

'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0,
                0],

True,

'depolarizing'},

'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
  'identity_factors': [ 3,
                       3,
                       0,
                       0],
  'noise_level': [28, 14, 0, 0],
  'noise_profile': {

'status':

'type':

'odd_wires': 4}},
'run_time_sec': 10.127862930297852}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [

0.5,
0.5,
0.5,
0.5,
0.5,
0.5],

'r': 0},
'time': { 'max': 10.0,
          'min': 0.0},
'type': 'heisenberg'}}},
'init_param': { 'value': [ 0.05238332857976487,
                           0.05238332857976488,
                           0.05238332857976461,
                           0.5882470284735626,
                           1.6186245942059996,
                           2.2470777294271223,
                           2.8315596913723913,
                           2.8315596913723913,
                           2.8333251330604914,
                           3.7474015252366555,
                           3.7474015252366555,
                           3.7474015252366555,
                           3.7474015252366555,
                           4.452994588443841,
                           4.625910007504636,
                           4.625910007504636,
                           5.314362007876739,
                           7.185299991262795,
                           7.185299991262795,
                           8.165312898322489,
                           8.165312898322489,
                           8.165312898322489,
                           8.684979613482101,
                           9.661121823219345,
                           9.674169964387936,
                           10.525655511340814,
                           10.525655511340814,
                           10.525655511340814,
                           11.154963299169733,
                           11.154963299169733,
                           2.50282655917642,
                           0.9490947670548783,
                           0.70056532212722,

```

-0.9370006834105179,
2.127395945212884,
1.5529950009292526,
-0.5538339242405286,
-0.45109357069990613,
3.2608593101153462,
2.1691098922881027,
0.924351518772188,
0.7799372105219657,
2.588492585292434,
1.4914484649202893,
0.5111989894903198,
-1.6854828839595362,
0.709524102178921,
-0.3201993643863164,
0.37756499567776974,
0.11160463007405702,
1.4506751210554825,
-0.1158349928449886,
-0.4172082421780774,
-0.47420587439641587,
2.1318004152619214,
1.5026342770773617,
0.9629412747697549,
-1.017935070811068,
2.2274033566257083,
0.9541631607884562,
2.2865374842341715,
-0.6901760390806929,
1.4488496207691695,
-0.15529093800821045,
1.378510821815928,
-0.1238908488185843,
1.0420944500710088,
1.8938950235205334,
-0.40270933293735856,
-0.4838468076888962,
-0.7769068081513464,
2.0570355933263196,
0.33843722154840217,
-0.21296684820549192,
-0.3087457919151385,
1.6713665097762895,
0.7955513549984788,
-0.5031484152839188,
0.2783608989762586,
1.4322280472670237,
0.12038812465567636,
0.8648150791824706,
0.13552183280667268,
0.07182610915111512,
0.3477791234991623,
-0.1584859230597403,
0.5851519083687279,
0.578565105996425,
-1.7921034519484391,
0.9189192684840749,
-0.6139572052593133,
-0.023527142002935514,
-0.6495330854871428,
0.7767670181361933,
0.13616638059771954,
-0.22568119775616352,
0.4192866974084779,
0.1401009023541104,
1.419355971488301,
1.1103093112149183,
-0.8728413553259085,
1.014691351510677,
2.35757700845783,
1.131891326153365,
-1.2979835358031357,
0.42365777146015526,
1.0982786992551208,
1.698510255572029,

```

1.2090506157632,
-1.3597097700308411,
-0.9049577551064396,
1.0199036275847497,
2.9541844671388584,
-0.42829772740772204,
0.6542829736223691,
3.7916586259391143,
-0.1983670801457003,
0.7855289833525587,
-0.1534574040841844,
1.809122493112309,
0.35293394466185096,
1.0573849517470935,
2.461665111905985,
1.0784030177718114,
-1.585709677044719,
0.8985620342798133,
2.2360691692951646,
0.24022620726925167,
2.6922129347449117,
0.15599094623602128,
2.799936411115923,
0.9657316063780467,
-0.2732772147086647,
-1.448572189075712,
-0.7280589250410919,
0.057973620624421296,
0.9321748335747359,
-1.4338621772181848,
1.3976338029473028,
-0.25504403617708243,
3.210257057781798,
0.03448775329128267,
0.05070056715723011,
-1.3727823101385919,
0.8243579770218447,
0.13142013414460954,
-0.4933237632531999,
1.1105137227416793,
0.7155364379215329,
-0.21063252058278284]],
'noise_profile': { 'noise_on_init_param': {
'value':
'noise_prob': [ 0.001,
0.001,
0,
0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_ric4_unifrom_scaling_sample#2'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],

```

```
[2, 2, 0, 0],  
[3, 3, 0, 0]]},  
  
'run': 'redundant',  
'state': 'dmatrix',  
'vqe': { 'iteration': 1,  
          'optimization': { 'algorithm': 'SLSQP',  
                             'constraint': True,  
                             'status': False}},  
  
'zne': { 'data_points': None,  
          'degree': 0,  
          'method': 'richardson',  
          'sampling': 'default'}},  
  
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '  
      '2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '  
      '2 + 1 X 3 X 4 + 1 Z 3 + 1 X '  
      '4 X 5 + 1 Z 4 + 1 X 5 X 6 + '  
      '1 Z 5 + 1 Z 6',  
  
    'time_evolution_gate_hamiltonian_string': [ '0.5 X '  
      '0 X 1 '  
      '+ 0.5 '  
      'Y 0 Y '  
      '1 + '  
      '0.5 Z '  
      '0 Z 1 '  
      '+ 0.5 '  
      'X 1 X '  
      '2 + '  
      '0.5 Y '  
      '1 Y 2 '  
      '+ 0.5 '  
      'Z 1 Z '  
      '2 + '  
      '0.5 X '  
      '2 X 3 '  
      '+ 0.5 '  
      'Y 2 Y '  
      '3 + '  
      '0.5 Z '  
      '2 Z 3 '  
      '+ 0.5 '  
      'X 3 X '  
      '4 + '  
      '0.5 Y '  
      '3 Y 4 '  
      '+ 0.5 '  
      'Z 3 Z '  
      '4 + '  
      '0.5 X '  
      '4 X 5 '  
      '+ 0.5 '  
      'Y 4 Y '  
      '5 + '  
      '0.5 Z '  
      '4 Z 5 '  
      '+ 0.5 '  
      'X 5 X '  
      '6 + '  
      '0.5 Y '  
      '5 Y 6 '  
      '+ 0.5 '  
      'Z 5 Z '  
      '6']},  
  
'output': { 'data_points': [ [4, 2, 0, 0, -6.220556999324399],  
                              [12, 6, 0, 0, -4.682225394272194],  
                              [20, 10, 0, 0, -3.495398379301583],  
                              [ 28,  
                                14,  
                                0,  
                                0,  
                                -2.571924591189056]],  
  
    'noise_details': [ { 'gates_num': [4, 1, 1, 0],  
                        'identity_factors': [ 0,  
                                              0,  
                                              0,  
                                              0],
```

```
'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0,
                0],

True,

'depolarizing'},
```

```
'noise_level': [4, 2, 0, 0],
'noise_profile': {
```

[illegible]

0,
0],

'cn': [

0.5,
0.5,
0.5,
0.5,
0.5,
0.5],

'r': 0},
'time': { 'max': 10.0,
 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [0.12420123662801165,
 0.138246244841176,
 0.34185010800102933,
 0.8771280515547424,
 0.8771280515547423,
 1.9326762312100456,
 1.9326762312100456,
 2.879806034127527,
 3.4467495190209787,
 3.460588109395402,
 3.4605881093954016,
 3.4605881093954016,
 3.911934955717517,
 3.9564836804547703,
 4.188240551814426,
 5.098640553346409,
 5.098640553346409,
 5.098640553346408,
 6.466332587579958,
 7.457565848827182,
 7.712724508543908,
 7.890302861063261,
 7.898577524812402,
 8.020037295803547,
 8.020037295803547,
 9.289723192898348,
 9.68251578824504,
 9.68251578824504,
 10.66694519433398,
 11.464898013549973,
 1.8095532965205083,
 -1.6633278493215808,
 -1.2281091611401596,
 -0.40155726479865544,
 1.7903243850816337,
 0.18661159996017582,
 1.2217850039169107,
 -0.48128156078104645,
 1.438546033666569,
 -0.9466383819139143,
 -0.2179227006066069,
 -0.07921201994836616,
 0.6692914930417907,
 -0.16791294486623565,
 -0.3278463005181308,
 1.2883193159418245,
 1.9001761172388045,
 0.7646920019853412,
 -0.6325477821570596,
 -0.5987552945744136,
 0.5669730115381587,
 0.5983941343774113,
 -0.06179171362000896,
 -0.6867235859693906,
 0.5211737756628548,
 -0.19354588250953578,
 0.6072816218147872,
 0.3472677197742769,
 -0.08031734075212016,
 1.4024126217748347,
 -0.5867181790012919,
 0.809620813948269,

0.35997425529443994,
-0.8722967249300235,
-0.5874818244923429,
-0.5567926036837068,
-0.6652314874983639,
-0.6042054992483096,
-0.13918430829641074,
-0.08277488470584692,
-0.4972293169762719,
-0.4421668735352916,
0.6160601456858428,
-1.1972657609969646,
-1.1558488364748807,
-1.4501782622641872,
0.03989088007781949,
-0.8684923327267209,
-0.5569704341339503,
0.15823054121075,
0.6345578215095505,
0.3601853745371903,
-0.42594808878819745,
0.08979804115834313,
-0.153043281560756,
-0.603228229875778,
-1.1264410002736531,
0.6895580235946861,
-0.11045941959152497,
0.5174109375139814,
1.3269460586908306,
-0.3852298542423827,
-2.4120095755015893,
0.08807412161753436,
0.1228551365077638,
-0.6982591234727157,
-1.1618488040328845,
2.5586909942385017,
-0.24921574033555105,
-1.9870969070050777,
-1.0279498295884981,
-0.7796802357622239,
0.48428681889730063,
0.7162815161060118,
-0.15547297036103372,
1.0522591887013215,
1.772822944479854,
-0.2654037211186137,
-0.2720110866721671,
-0.03929478442656517,
0.34606077605631835,
0.5533331494282367,
1.0574846609095763,
-0.5747611578732846,
0.4916772893793279,
0.9875709691916216,
-0.17857479553023695,
-1.147281220275154,
0.37370382177065037,
2.456815155407461,
0.6567596591037266,
0.9507316292553318,
1.5113928877114242,
1.8380989093110256,
-0.1736919900840187,
-1.2666253947041328,
-0.554819304070435,
1.1931643666366285,
0.4614862178419797,
1.2040726557184918,
0.3083618234293392,
-0.4066692475369416,
0.3740368412278018,
-1.684611974494111,
-0.8650016725760176,
-1.4838330341878547,
-0.14344887350922086,

```
'status': False,  
0},
```

```
4.203596189627334,  
1.1133202614474196,  
1.391124117957397,  
1.8627332105499828,  
-0.28192336755973296,  
-1.5416482647725873,  
-1.9612892990030817,  
0.8836928087418303,  
0.300761821046295,  
0.6785848024508535,  
-0.5500277556546446,  
0.5742776326882599,  
-1.181756808209963]],  
'noise_profile': { 'noise_on_init_param': {  
  
    'value':  
  
        'noise_prob': [ 0.001,  
                        0.001,  
                        0,  
                        0],  
        'status': True,  
        'type': 'depolarizing'},  
    'nqubits': 7,  
    'observable': { 'coefficients': { 'bn': [ 1.0,  
                                              1.0,  
                                              1.0,  
                                              1.0,  
                                              1.0,  
                                              1.0,  
                                              1.0],  
                                          'cn': [ 0.5,  
                                                  0.5,  
                                                  0.5,  
                                                  0.5,  
                                                  0.5,  
                                                  0.5],  
                                          'r': 1},  
                    'def': 'ising'},  
    'output': { 'draw': { 'fig_dpi': 100,  
                          'status': False,  
                          'type': 'png'},  
                'file_name_prefix':  
    'AUTOMATE_heisenberg_noise_free_time_evo_ric4_unifrom_scaling_sample#5'},  
    'redundant': { 'identity_factors': [ [0, 0, 0, 0],  
                                         [1, 1, 0, 0],  
                                         [2, 2, 0, 0],  
                                         [3, 3, 0, 0]]},  
    'run': 'redundant',  
    'state': 'dmatrix',  
    'vqe': { 'iteration': 1,  
             'optimization': { 'algorithm': 'SLSQP',  
                               'constraint': True,  
                               'status': False}},  
    'zne': { 'data_points': None,  
            'degree': 0,  
            'method': 'richardson',  
            'sampling': 'default'}},  
    'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '  
              '2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '  
              '2 + 1 X 3 X 4 + 1 Z 3 + 1 X '  
              '4 X 5 + 1 Z 4 + 1 X 5 X 6 + '  
              '1 Z 5 + 1 Z 6',  
    'time_evolution_gate_hamiltonian_string': [ '0.5 X '  
                                              '0 X 1 '  
                                              '+ 0.5 '  
                                              'Y 0 Y '  
                                              '1 + '  
                                              '0.5 Z '  
                                              '0 Z 1 '  
                                              '+ 0.5 '  
                                              'X 1 X '  
                                              '2 + '  
                                              '0.5 Y '  
                                              '1 Y 2 '
```

```
'+ 0.5 '
'Z 1 Z '
'2 + '
'0.5 X '
'2 X 3 '
'+ 0.5 '
'Y 2 Y '
'3 + '
'0.5 Z '
'2 Z 3 '
'+ 0.5 '
'X 3 X '
'4 + '
'0.5 Y '
'3 Y 4 '
'+ 0.5 '
'Z 3 Z '
'4 + '
'0.5 X '
'4 X 5 '
'+ 0.5 '
'Y 4 Y '
'5 + '
'0.5 Z '
'4 Z 5 '
'+ 0.5 '
'X 5 X '
'0.5 Y '
'5 Y 6 '
'+ 0.5 '
'Z 5 Z '
'6']],
'output': { 'data_points': [ [4, 2, 0, 0, -5.814676484708359],
[12, 6, 0, 0, -4.248447309496402],
[ 20,
10,
0,
0,
-3.0694027528903187],
[ 28,
14,
0,
0,
-2.174125329701547]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [ 0,
0,
0,
0],
'noise_level': [4, 2, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0,
0],
'status':
True,
'type':
'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [ 1,
1,
0,
0],
'noise_level': [12, 6, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0,
```

```
0],
True,
'depolarizing'},

'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0,
                0],

True,
'depolarizing'},

'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0,
                0],

True,
'depolarizing'},

'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0,
                0],

True,
'depolarizing'},

'odd_wires': 4}},
'run_time_sec': 10.139310121536255}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [
0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': { 'max': 10.0,
          'min': 0.0},
'type': 'heisenberg'}}},
'init_param': { 'value': [ 0.43940602753254654,
                           0.4524980474487914,
                           0.6462847510297367,
                           0.6535145711164267,
                           0.7019965052930044,
                           1.0830061860275473,
                           1.4380021973026182,
                           2.1860837771467474,
                           2.1860837771467474,
                           2.1860837771467474,
                           2.4875804812430995,
                           2.911818356328843,
```

4.9134595358452735,
5.043089843653786,
5.994354774193697,
6.658321281734746,
6.666230857319696,
8.458454027735183,
8.458454027735183,
8.458454027735183,
8.938931772570164,
8.938931772570164,
9.876451730548837,
9.876451730548837,
10.140618079102744,
10.453495471248813,
10.878653713196195,
11.282429749250388,
11.726474998572407,
11.93229290545428,
2.9640535214497064,
-0.6206168106317247,
-0.4035747848788786,
-2.067222048924027,
2.0576259596946604,
1.364315765000901,
-0.1480698866383175,
-1.95043906046908,
0.7897528850889669,
1.350582507885341,
-0.3349044763190947,
-2.3429833820345465,
2.04351680973891,
0.8083219796841106,
-0.06520770630024467,
-0.8776318722364843,
2.4862506381076748,
0.8345121096737285,
-0.03071661620417745,
-0.6225183255138825,
0.7450777062518694,
0.12619297185303674,
0.15620862918286052,
-0.47540100870445906,
0.6345315438244892,
1.451076347224231,
0.10985833383721234,
-1.6475696409761669,
1.1918822544398533,
1.3235539868194213,
-0.24222060474147966,
0.2418944215339802,
1.0397651583041008,
1.6773252160624976,
0.16975786500059598,
-0.9211027317156172,
0.7590471244619662,
1.76540125315151,
2.43856734239768,
-0.19488067618654012,
0.8462629124439717,
0.9979674075340036,
2.5775834810620677,
0.5650956422366618,
-0.5475934153287456,
-0.0253449469292375,
1.2524069066350432,
-0.605696708865831,
0.7019502861337156,
0.2803153244338447,
-1.0831925759777479,
1.0082140101851143,
-0.3762046176346905,
-0.9535933533256993,
-1.0422804614435257,
0.13948594332192887,
0.5835070530053111,

```

0.16086524965091817,
0.06403444071667018,
-0.06194188553710435,
-0.4026223625881385,
0.2750028468648149,
0.6789909068622377,
0.05162849323802493,
-0.7735332926339995,
-0.14862918062202837,
0.7661757453229759,
-0.16119730427373616,
0.6482389226852195,
3.1459456036848104,
-0.4760920139337192,
1.6197254159790668,
-0.5506328506738539,
0.9968299487743895,
-0.3106735286083523,
-1.2388662013594425,
0.8898580979157581,
0.43256773745092003,
-0.7272220041247079,
0.8716071923554094,
1.5193589139179793,
0.9787540447681606,
-1.4044929205930872,
-0.1984876316951887,
-0.46744984056178585,
-0.7765225748074835,
0.04344696088727837,
0.32373965212791267,
0.9049555024172572,
0.7139278873651953,
-0.8122282267372439,
0.5323405298774778,
-0.08437719753251166,
-0.14277007957413837,
0.7912147308716768,
0.26055493673650715,
0.11357082745857092,
0.3153698930365579,
3.1981441448638614,
0.32109035951393206,
-0.23017443442945684,
0.0316864124540797,
0.20875812581978664,
0.5466872162452653,
1.2918521189433907,
-0.7312547980705957,
1.4958457475939604,
-0.053193686203739404,
-0.4437772137226961,
-0.5271126400646005,
2.5136810835026426,
0.6674244883024821,
-1.0140291771816496,
-1.177197116527103,
2.058315434744553,
0.050372102721390984,
-0.09022974351195023,
-2.296438941834702,
0.3261688389091777,
0.3072223897800251]],
'noise_profile': { 'noise_on_init_param': {
'value':
'noise_prob': [ 0.001,
0.001,
0,
0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,

```

```

1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_ric4_unifrom_scaling_sample#1'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0],
[3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '
'2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '
'2 + 1 X 3 X 4 + 1 Z 3 + 1 X '
'4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
'1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X '
'0 X 1 '
'+ 0.5 '
'Y 0 Y '
'1 + '
'0.5 Z '
'0 Z 1 '
'+ 0.5 '
'X 1 X '
'2 + '
'0.5 Y '
'1 Y 2 '
'+ 0.5 '
'Z 1 Z '
'2 + '
'0.5 X '
'2 X 3 '
'+ 0.5 '
'Y 2 Y '
'3 + '
'0.5 Z '
'2 Z 3 '
'+ 0.5 '
'X 3 X '
'4 + '
'0.5 Y '
'3 Y 4 '
'+ 0.5 '
'Z 3 Z '
'4 + '
'0.5 X '
'4 X 5 '
'+ 0.5 '
'Y 4 Y '
'5 + '
'0.5 Z '
'4 Z 5 '

```

```

        '+ 0.5 '
        'X 5 X '
        '6 + '
        '0.5 Y '
        '5 Y 6 '
        '+ 0.5 '
        'Z 5 Z '
        '6']],
'output': { 'data_points': [ [4, 2, 0, 0, -5.966837375471956],
                             [12, 6, 0, 0, -4.6298501952375855],
                             [ 20,
                              10,
                              0,
                              0,
                              -3.5807206753108622],
                             [ 28,
                              14,
                              0,
                              0,
                              -2.750174370397261]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                     'identity_factors': [ 0,
                                           0,
                                           0,
                                           0],
                     'noise_level': [4, 2, 0, 0],
                     'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0,
                0],
True,
'depolarizing'},

'status':

'type':

'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [ 1,
                       1,
                       0,
                       0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {
'status':

'type':

'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [ 2,
                       2,
                       0,
                       0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0,
                0],
True,
'depolarizing'},

'status':

'type':

'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],

```



```
'identity_factors': [ 3,
                      3,
                      0,
                      0],
'noise_level': [28, 14, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0,
                0],

'status':
'type':

'depolarizing'},

'odd_wires': 4}},
'run_time_sec': 10.158948183059692}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [
0.5,
0.5,
0.5,
0.5,
0.5,
0.5],

'r': 0},
'time': { 'max': 10.0,
          'min': 0.0},
'type': 'heisenberg'}}},
'init_param': { 'value': [ -2.397634216504411e-17,
                            1.429450133825379e-17,
                            0.9192297980258759,
                            0.9192297980258759,
                            2.8984599305110854,
                            3.1993874679766523,
                            3.199387467976652,
                            3.715279848679994,
                            3.715919998165357,
                            5.126008182976459,
                            5.749875525815113,
                            5.785995839176617,
                            5.785995839176617,
                            5.785995839176617,
                            5.785995839176617,
                            6.315437962379428,
                            6.678845841951143,
                            7.3578167224936415,
                            7.415694801781823,
                            7.531263738212295,
                            7.949087697575632,
                            7.949087697575632,
                            8.111169110068786,
                            9.407094598577077,
                            9.495286031274873,
                            9.495286031274873,
                            10.993257661103009,
                            11.063439993331317,
                            11.639888600691357,
                            11.75175692668691,
                            2.1152038588685538,
                            -0.7839988468841306,
                            -1.0579861702584952,
                            -0.5122509123294313,
                            1.0012369417099964,
                            0.9275037440003397,
                            0.5343037326701707,
```

-0.014459274586663672,
2.31692048955396,
2.1400689678453872,
-0.9468421716760329,
1.3225342455296385,
2.4229762260600483,
1.4561600175543838,
-0.4592924007287594,
0.6389204073366491,
0.5578650700802136,
1.458345504678432,
-0.11931303728638508,
0.04406428844113783,
-0.26144087074192934,
0.6459890964472403,
1.2243061606997268,
0.09770103868863204,
0.23645607828472368,
0.3280111145811677,
0.5390648705936556,
-0.6034481934191781,
-1.1942560416831098,
0.3295059316644399,
-0.0732996553139216,
0.5043071925129845,
0.7399437462813884,
0.3595686242698361,
1.4643835231854074,
-0.1697776944282527,
0.2588380732750666,
0.44311592273109757,
-0.6031987466514527,
-0.06131410155451245,
-0.3299395710828798,
-0.45492250592824107,
0.5555861976799495,
1.1612177689175078,
1.833550163228831,
1.5629483318007886,
-0.2455585474033218,
-0.4534768786824923,
-1.2225865843307935,
0.20468879325769473,
1.3398561861341207,
-0.58709602862573,
0.07181500566168061,
0.17010271339345,
0.36406988692952136,
-1.689809967880289,
-1.1677426672173294,
-0.7977758879892671,
0.531032719362766,
-1.7722607960867016,
0.6332033998224597,
0.6669814362912183,
0.218182728116887,
1.0415656581640258,
-1.4544893086294475,
0.5175174925258392,
-0.9742907090362619,
-0.30648595673224416,
1.2400541074634985,
-0.30003161306087445,
-0.14406503395842435,
-0.43061285539013316,
0.6914697219419143,
-0.48545379734819916,
-0.2166375004086177,
1.2014709434123243,
1.0734635496959384,
0.43666289070672293,
-0.9550440807047608,
1.2817732673505913,
0.6254203972471953,
0.8730111716749767,

```

1.1814458389257443,
-0.5855214911540807,
0.33647224367417283,
0.914891547403245,
-0.10924028322724566,
0.6731472714644756,
-0.8672547313943412,
-1.1628903969680444,
1.2968110405185924,
-0.7478342770444015,
1.0157432273471354,
0.05866260951598884,
0.1549884083832883,
2.402954000164725,
-0.8510405490988947,
-0.13394876321797194,
0.35128800256256376,
0.03985349181110424,
1.0891408951797659,
0.49683980443602116,
1.3618410332836905,
0.31089249400158553,
0.007823999182329802,
-1.0325531315281817,
-0.13635517978577905,
-0.4783006843261583,
-0.9648987725589863,
-0.4094326055257113,
0.0620151507481985,
-0.3631850012398663,
-0.9119123598789223,
-0.3458539889181142,
0.8501459643039054,
1.6461875633189593,
0.08799287143090098,
-0.82318299513447,
1.17289644669955,
-0.9455811832961388]],
'noise_profile': { 'noise_on_init_param': {
'value':
'noise_prob': [ 0.001,
0.001,
0,
0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_ric4_unifrom_scaling_sample#4'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0],
[3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'status': False,
0},

```

```

'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '
                                '2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '
                                '2 + 1 X 3 X 4 + 1 Z 3 + 1 X '
                                '4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
                                '1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X '
                                                         '0 X 1 '
                                                         '+ 0.5 '
                                                         'Y 0 Y '
                                                         '1 + '
                                                         '0.5 Z '
                                                         '0 Z 1 '
                                                         '+ 0.5 '
                                                         'X 1 X '
                                                         '2 + '
                                                         '0.5 Y '
                                                         '1 Y 2 '
                                                         '+ 0.5 '
                                                         'Z 1 Z '
                                                         '2 + '
                                                         '0.5 X '
                                                         '2 X 3 '
                                                         '+ 0.5 '
                                                         'Y 2 Y '
                                                         '3 + '
                                                         '0.5 Z '
                                                         '2 Z 3 '
                                                         '+ 0.5 '
                                                         'X 3 X '
                                                         '4 + '
                                                         '0.5 Y '
                                                         '3 Y 4 '
                                                         '+ 0.5 '
                                                         'Z 3 Z '
                                                         '4 + '
                                                         '0.5 X '
                                                         '4 X 5 '
                                                         '+ 0.5 '
                                                         'Y 4 Y '
                                                         '5 + '
                                                         '0.5 Z '
                                                         '4 Z 5 '
                                                         '+ 0.5 '
                                                         'X 5 X '
                                                         '6 + '
                                                         '0.5 Y '
                                                         '5 Y 6 '
                                                         '+ 0.5 '
                                                         'Z 5 Z '
                                                         '6']],
'output': { 'data_points': [ [4, 2, 0, 0, -6.241465922765086],
                              [12, 6, 0, 0, -4.714681794898787],
                              [ 20,
                               10,
                               0,
                               0,
                               -3.5395259571477817],
                              [ 28,
                               14,
                               0,
                               0,
                               -2.627473806451929]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                    'identity_factors': [ 0,
                                           0,
                                           0,
                                           0],

```

```
'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0,
                0],

True,

'depolarizing'},
```

```
'type':
```

[illegible]

0,
0],

'cn': [

0.5,
0.5,
0.5,
0.5,
0.5,
0.5],

'r': 0},
'time': { 'max': 10.0,
 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [0.16477895719846114,
 0.16477895719846108,
 0.7183189156962347,
 1.8697093861419474,
 2.105082615571597,
 3.05308344502201,
 4.199393321043376,
 4.199393321043376,
 4.199393321043375,
 6.490210948425593,
 6.4902109484255925,
 6.490210948425593,
 6.4902109484255925,
 6.4902109484255925,
 6.490210948425593,
 6.882554174728105,
 6.882554174728105,
 6.882554174728105,
 6.882554174728104,
 6.8877583035732775,
 6.8877583035732775,
 6.8877583035732775,
 6.8877583035732775,
 8.21014824639056,
 8.21014824639056,
 9.071469420716642,
 9.120928605341893,
 9.714944098401723,
 11.474131735950095,
 11.474131735950095,
 11.794551582273902,
 1.341188628911811,
 1.6554894539990426,
 -2.1068974621953425,
 -1.4171475675653578,
 0.33211669069513333,
 0.11331792925356984,
 -1.3610911099963097,
 0.46439741779020816,
 0.9313501387866817,
 1.0938593094543279,
 -0.3591727738608774,
 -0.4732622270970327,
 1.6321598874818157,
 -0.2782091348514778,
 -1.0526781349295347,
 0.0708703831522549,
 2.6982701224663113,
 -0.038867313655444914,
 -0.10508583152652556,
 -0.40759365792756236,
 0.6284849004330764,
 0.5192751929971827,
 -1.6282040637266806,
 0.0014422433007245106,
 -1.8299719264591412,
 2.3182597876656823,
 0.8875167319646391,
 1.5664268415036142,
 1.5052778791259855,
 2.935955436800565,
 1.533101581959507,
 1.8091829332960188,

0.45080966818917967,
0.1506883627647002,
-0.6843548866459184,
-0.5544883854492364,
1.2466562002431278,
-1.1866110136199353,
1.4521358231995756,
-1.125608676420736,
-0.24054383618816966,
-0.058272145415006006,
0.410408128597049,
-1.6441753460137252,
0.4019816715685168,
0.4588365987304068,
1.354346078964423,
-1.7153251360741875,
-1.1039978646265602e-06,
0.7940670488104785,
0.17979772520560053,
-0.24385394325110826,
0.6367160741723578,
0.8500459957371875,
0.0559416907316065,
0.32418668587130467,
-1.4797636873635434,
0.11902177942387202,
0.5185165439317966,
-0.08399222686641401,
0.8650918037299421,
1.4911587670088011,
0.12792248933645675,
0.3372262149750348,
-0.6553154954729696,
1.1995792367396254,
0.5132294030998703,
0.44835115860783636,
-0.37116980512777586,
0.7805194563374558,
-0.854334789795377,
0.5008527672922033,
-0.2302685629030402,
0.6145182391921147,
-0.6575500079215796,
0.8657035637529482,
1.1874516653159721,
-0.16284802367068918,
-0.26462171656665856,
-0.9815102056659819,
0.23198397744948207,
-1.087097195512072,
0.6616385654966559,
-0.2929637257774683,
0.03471390505932934,
0.2048751570049573,
0.9170017118070749,
0.39617956766989404,
0.12006199846995962,
0.1922705913940127,
0.8784478594386566,
0.055479957240883566,
-0.6634105431569992,
1.0742951419164057,
-0.37818200958777604,
0.28548758229245597,
-0.6594827056613419,
1.0235832498557866,
-0.19428253177041227,
-0.5531991468121416,
-1.189626464718852,
0.2519110119226656,
-0.6708462316627377,
2.7994352657547057,
-3.8764436680175813,
0.5152327138619494,
-0.3749347648847944,

```

2.9042167722097862,
1.1265965037549552,
-1.850959186134046,
2.165374611027145,
0.8818414791702859,
1.5039524911129019,
0.06564767423285973,
0.6486215314149069,
-0.08854143781823305,
1.8396223528529987,
-0.00792246728000446,
2.8805879469479487,
3.6337353861266473]],
'noise_profile': { 'noise_on_init_param': {
'value':
'noise_prob': [ 0.001,
0.001,
0,
0],
'status': True,
'type': 'depolarizing'}},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_ric4_unifrom_scaling_sample#9'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0],
[3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '
'2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '
'2 + 1 X 3 X 4 + 1 Z 3 + 1 X '
'4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
'1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X '
'0 X 1 '
'+ 0.5 '
'Y 0 Y '
'1 + '
'0.5 Z '
'0 Z 1 '
'+ 0.5 '
'X 1 X '
'2 + '
'0.5 Y '
'1 Y 2 '

```



```
'+ 0.5 '
'Z 1 Z '
'2 + '
'0.5 X '
'2 X 3 '
'+ 0.5 '
'Y 2 Y '
'3 + '
'0.5 Z '
'2 Z 3 '
'+ 0.5 '
'X 3 X '
'4 + '
'0.5 Y '
'3 Y 4 '
'+ 0.5 '
'Z 3 Z '
'4 + '
'0.5 X '
'4 X 5 '
'+ 0.5 '
'Y 4 Y '
'5 + '
'0.5 Z '
'4 Z 5 '
'+ 0.5 '
'X 5 X '
'0.5 Y '
'5 Y 6 '
'+ 0.5 '
'Z 5 Z '
'6']],
'output': { 'data_points': [ [4, 2, 0, 0, -5.966171999362981],
[12, 6, 0, 0, -4.491721403204036],
[ 20,
10,
0,
0,
-3.3646330706171934],
[ 28,
14,
0,
0,
-2.4951156781776946]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [ 0,
0,
0,
0],
'noise_level': [4, 2, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0,
0],
'status':
True,
'type':
'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [ 1,
1,
0,
0],
'noise_level': [12, 6, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0,
```

```

0],
True,
'depolarizing'},

'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0,
                0],

True,
'depolarizing'},

'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0,
                0],

True,
'depolarizing'},

'odd_wires': 4}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [
0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': { 'max': 10.0,
           'min': 0.0},
'type': 'heisenberg'}}},
'init_param': { 'value': [ 0.4022850038273885,
                           0.4222288083135141,
                           0.7644599371513524,
                           0.7900179023917507,
                           1.1026508993249646,
                           1.1630326759467198,
                           1.1630326759467198,
                           1.2119340062926192,
                           1.9859325565890973,
                           1.985932556589097,
                           2.096166401632718,
                           3.378647998605865,

```

3.378647998605865,
3.378647998605865,
4.354848115861854,
6.006218342726225,
6.006218342726225,
6.009974297352436,
8.225327865566033,
8.242093148909708,
9.222354355246733,
9.369468664000962,
9.432712219232048,
9.692289963799956,
9.692829942577658,
10.391360826688166,
10.391360826688166,
10.529867035412478,
10.529867035412478,
11.06290493838992,
2.9062307172979733,
-1.7788694936155562,
-0.1145266982224492,
-0.9311559923392093,
0.9691998209476543,
1.7463479137973288,
-0.7663862807329105,
-0.9694574896037749,
2.8979112386638413,
1.1400097878698292,
0.8096361549155084,
-1.5900494197653285,
0.09829667109053167,
0.31839493135320834,
0.3886777840539921,
0.12229184789371997,
1.7250225844898464,
1.0065795676456808,
1.4535641205083867,
-0.3486770669120958,
0.2556790893233589,
1.778531141883236,
0.269972269049403,
1.0155903957099945,
0.8041163886037216,
0.5381821926966456,
-0.30108604163317454,
0.6396533041678052,
2.1819991440037696,
1.4978086658278347,
2.5181040929106753,
0.4624415671193703,
-0.17701736987873673,
0.6910098122382194,
2.947172461882987,
-1.0727539032453073,
0.4023854197196741,
1.650490750838525,
0.9313998963798695,
-0.6769162055305248,
-0.5620785927454173,
-0.20885673940707247,
2.5019757090972443,
-0.37561416876024456,
-0.9329392032282731,
0.7818961506889994,
0.44350434198669075,
-0.10644929099379355,
-2.498928032568615,
-1.2046085786256906,
-0.06365416567297402,
0.17257236645054272,
1.2516209341031546,
-0.3591527663232026,
0.46789807326225935,
0.24374638092155315,
0.02110383811013195,

```

0.12714835746845662,
-0.5505736947218767,
-0.03497727012222505,
-2.2168334497478877,
-0.21485512988314215,
1.7552902914988067,
0.6189795903733311,
0.3191491062237487,
0.522632432692797,
-1.1121089236913335,
1.1371618962620422,
-0.2410043783817164,
0.24168543661014927,
0.39418565996934685,
1.2107667809264395,
2.398781467533486,
0.6948257569073976,
0.6170229142231725,
0.7050419245924102,
1.768546262611762,
1.5562956304120967,
-0.4269237214365518,
0.6302835616447369,
1.7710848704009405,
1.2281100264507194,
1.5036530549577243,
-0.7435268012971183,
0.7613541379199313,
0.8523298100708424,
-0.6655959646469325,
1.1543911681084875,
2.2675628634479494,
1.0299985170060308,
1.400963430022522,
0.27759304330136975,
0.6960561589746208,
-0.4210335016880339,
0.0336463669485127,
-2.099660094290342,
0.6674819238895001,
0.5049668435179431,
0.48001405045948287,
1.3574871122533543,
-2.2986993170736225,
9.601426007351498e-05,
1.3312470798272018,
0.7582238004326416,
-0.35033967277151296,
-0.4108552112493075,
0.1982910129710702,
0.6700017270602091,
0.26559360983179486,
-0.14563210230012705,
-0.39610135234418875,
1.1727729846793147,
0.6713573564879365,
0.8527070052929181,
0.8268283343075853,
2.2918575835381745,
0.5082103460913493,
1.8600493489550152,
0.49576811175353974,
2.232387985311215]],
'noise_profile': { 'noise_on_init_param': {
'value':
'noise_prob': [ 0.001,
0.001,
0,
0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,

```

```

1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_ric4_unifrom_scaling_sample#8'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0],
[3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '
'2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '
'2 + 1 X 3 X 4 + 1 Z 3 + 1 X '
'4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
'1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X '
'0 X 1 '
'+ 0.5 '
'Y 0 Y '
'1 + '
'0.5 Z '
'0 Z 1 '
'+ 0.5 '
'X 1 X '
'2 + '
'0.5 Y '
'1 Y 2 '
'+ 0.5 '
'Z 1 Z '
'2 + '
'0.5 X '
'2 X 3 '
'+ 0.5 '
'Y 2 Y '
'3 + '
'0.5 Z '
'2 Z 3 '
'+ 0.5 '
'X 3 X '
'4 + '
'0.5 Y '
'3 Y 4 '
'+ 0.5 '
'Z 3 Z '
'4 + '
'0.5 X '
'4 X 5 '
'+ 0.5 '
'Y 4 Y '
'5 + '
'0.5 Z '
'4 Z 5 '

```

```

'+ 0.5 '
'X 5 X '
'6 + '
'0.5 Y '
'5 Y 6 '
'+ 0.5 '
'Z 5 Z '
'6']],
'output': { 'data_points': [ [4, 2, 0, 0, -5.743389283741309],
                              [12, 6, 0, 0, -4.3560016590478785],
                              [ 20,
                                10,
                                0,
                                0,
                                -3.2776401801937793],
                              [28, 14, 0, 0, -2.43192951997479]],
  'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                       'identity_factors': [ 0,
                                              0,
                                              0,
                                              0],
                       'noise_level': [4, 2, 0, 0],
                       'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0,
                0],

                                'status':
True,
                                'type':
'depolarizing'},

                                'odd_wires': 4},
                              { 'gates_num': [12, 3, 1, 0],
                                'identity_factors': [ 1,
                                                       1,
                                                       0,
                                                       0],
                                'noise_level': [12, 6, 0, 0],
                                'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0,
                0],

                                'status':
True,
                                'type':
'depolarizing'},

                                'odd_wires': 4},
                              { 'gates_num': [20, 5, 1, 0],
                                'identity_factors': [ 2,
                                                       2,
                                                       0,
                                                       0],
                                'noise_level': [20, 10, 0, 0],
                                'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0,
                0],

                                'status':
True,
                                'type':
'depolarizing'},

                                'odd_wires': 4},
                              { 'gates_num': [28, 7, 1, 0],
                                'identity_factors': [ 3,
                                                       3,
                                                       0,
                                                       0],
                                'noise_level': [28, 14, 0, 0],
                                'noise_profile': {

```


1.25716179603495,
1.7181260437331554,
2.3978160665502384,
1.1152024575020885,
-1.3335737674295711,
0.016650799076572978,
1.6607129762234538,
0.648059834763191,
0.2438725527060928,
-0.27307896000752346,
1.4369017198677916,
3.5672448343800007,
-0.5379702113035718,
-0.7420664562717484,
-0.529699891930992,
0.7301471483152846,
0.781913925757813,
0.6004078048346045,
1.7324865479818374,
0.4548236607944829,
-1.4419813608219112,
-0.684662976785234,
0.8455512814500156,
1.97751114408285,
-0.833202517811791,
1.110524460863052,
2.6403320691876733,
-0.3129427957408566,
-0.06674204626492235,
0.5842857778318762,
-0.21132485649138796,
-0.011842019288790376,
-0.25743970610802486,
0.7486616881233721,
-0.1302029141156864,
0.09497091477412227,
-0.3715265429351211,
0.39570077921274605,
0.28643868832003017,
0.7848413456471106,
-0.7157146817058562,
0.09411764456438232,
-0.10910076647953601,
1.1262761892344808,
-0.27334997823872975,
-0.44027652197942063,
0.6740415100156434,
-0.336256645457715,
0.4268157265313067,
-0.8611696251566416,
-0.34560928156726306,
0.5618074077714486,
0.5866360379174992,
1.8662876827821717,
2.21592310898005,
1.972223656383108,
1.6141187926643297,
0.856255815565121,
-0.34275929095614754,
-1.0642528745182613,
-1.0618686751734654,
0.2379582550153795,
-0.6501770361983872,
-0.19408645485522444,
1.501330829125914,
1.5726602727591479,
0.7066066097903853,
-1.4580936665368467,
0.46940440748349077,
-0.1287015677341077,
-0.9805925033056946,
-0.3754741669636888,
-0.026817919501009202,
1.1067346586651607,
0.17547923725022047,


```

0.8902143171896091,
-0.22777698610446503,
0.8082253583749343,
-1.0674083601412958,
-0.022706971436593337,
0.34227899244110566,
1.3527736523760083,
2.0767841716125894,
0.5778568145668401,
-0.6555502504392442,
1.2670274746873995,
0.507694022232518,
-0.2629991187543805,
1.7048253080726326,
1.4088065522309612,
2.823918413203317,
1.2763232744235296,
-1.4348344213353486,
-0.605496445996402,
0.09645158262695042,
0.024639736182597342,
0.23997550724405606,
0.9979895378690888,
-0.40612123795993227,
1.826790650755685,
0.029599395583551916,
-0.4592861834683686,
0.0328358806035505,
1.3612120706287358,
0.8210402794880411,
0.05688164146575431,
0.03152531580318861,
2.0316547976974904,
2.795039607579165]],
'noise_profile': { 'noise_on_init_param': {
'value':
'noise_prob': [ 0.001,
0.001,
0,
0],
'status': True,
'type': 'depolarizing'}},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_ric4_unifrom_scaling_sample#10'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0],
[3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'status': False,
0},

```

```

'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '
                                '2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '
                                '2 + 1 X 3 X 4 + 1 Z 3 + 1 X '
                                '4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
                                '1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X '
                                                         '0 X 1 '
                                                         '+ 0.5 '
                                                         'Y 0 Y '
                                                         '1 + '
                                                         '0.5 Z '
                                                         '0 Z 1 '
                                                         '+ 0.5 '
                                                         'X 1 X '
                                                         '2 + '
                                                         '0.5 Y '
                                                         '1 Y 2 '
                                                         '+ 0.5 '
                                                         'Z 1 Z '
                                                         '2 + '
                                                         '0.5 X '
                                                         '2 X 3 '
                                                         '+ 0.5 '
                                                         'Y 2 Y '
                                                         '3 + '
                                                         '0.5 Z '
                                                         '2 Z 3 '
                                                         '+ 0.5 '
                                                         'X 3 X '
                                                         '4 + '
                                                         '0.5 Y '
                                                         '3 Y 4 '
                                                         '+ 0.5 '
                                                         'Z 3 Z '
                                                         '4 + '
                                                         '0.5 X '
                                                         '4 X 5 '
                                                         '+ 0.5 '
                                                         'Y 4 Y '
                                                         '5 + '
                                                         '0.5 Z '
                                                         '4 Z 5 '
                                                         '+ 0.5 '
                                                         'X 5 X '
                                                         '6 + '
                                                         '0.5 Y '
                                                         '5 Y 6 '
                                                         '+ 0.5 '
                                                         'Z 5 Z '
                                                         '6']},
'output': { 'data_points': [ [4, 2, 0, 0, -6.2813019572269795],
                              [12, 6, 0, 0, -4.778373581659957],
                              [ 20,
                                10,
                                0,
                                0,
                                -3.6190179833186162],
                              [ 28,
                                0,
                                0,
                                -2.7162489715378166]],
            'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                'identity_factors': [ 0,
                                                        0,
                                                        0,
                                                        0],
                                'noise_level': [4, 2, 0, 0],
                                'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},

```


0.5,
0.5,
0.5,
0.5,
0.5],

```
        'r': 0},  
    'time': { 'max': 10.0,  
              'min': 0.0},  
    'type': 'heisenberg'}},  
    'init_param': { 'value': [ -1.8784742854218807e-17,  
                                -7.763750530158576e-17,  
                                0.3802827375258553,  
                                0.44096449652630626,  
                                0.8040692975576897,  
                                1.9224694434519336,  
                                1.9224694434519338,  
                                1.9224694434519336,  
                                1.9224694434519336,  
                                1.9224694434519336,  
                                1.9224694434519338,  
                                2.730288811725591,  
                                4.476350841356259,  
                                4.476350841356259,  
                                4.476350841356259,  
                                6.331658137341824,  
                                6.331658137341824,  
                                6.7621011201584285,  
                                6.77851096220289,  
                                7.490043179529316,  
                                7.490043179529316,  
                                7.490043179529316,  
                                7.490043179529316,  
                                7.511664240193972,  
                                8.302259372586072,  
                                8.36899822047307,  
                                8.607405767895836,  
                                9.631163052898822,  
                                9.874876363637043,  
                                10.225975825453373,  
                                2.711573403856553,  
                                0.20239905137011172,  
                                -0.4220883549830381,  
                                -0.8404830976891227,  
                                1.794332340210785,  
                                1.5095243517856047,  
                                0.22326098118939897,  
                                0.07733371735403745,  
                                1.280548885023134,  
                                1.4121496210624682,  
                                0.9815935680907694,  
                                -2.975348552150871,  
                                1.6316643284140941,  
                                0.8239502945369961,  
                                0.0314505340316768,  
                                -1.0892891749800817,  
                                1.720694113273713,  
                                0.6634696927679957,  
                                -0.836512780686814,  
                                -1.1286873737196692,  
                                -0.09500137106776224,  
                                1.0428135848317974,  
                                -0.4828772957450712,  
                                -1.129334172148368,  
                                0.7253737318795237,  
                                1.4108675361353635,  
                                0.7588540034470418,  
                                -0.5039485441412638,  
                                0.10346857579652805,  
                                1.2668095290766983,  
                                0.7994757345637048,  
                                -0.7393240316081237,  
                                -0.7818304990031031,  
                                0.25375980918776664,  
                                2.228975199542585,  
                                2.2487625387645873,
```

0.6384105819815235,
0.39434544396871446,
1.3679129840939837,
-0.27336699910726836,
1.3170761670781417,
1.1159622861844822,
1.176824183488981,
0.40583799697229034,
0.8786168217534174,
0.2914083315983707,
-0.8464697846173548,
-0.032455291822620566,
-2.0755386823554396,
-0.11439663923099666,
0.7312672791487281,
0.4263196833485598,
0.11802610051350722,
1.0445881812565523,
-1.9566462184988007,
0.9492303984073598,
-1.7129593951518396,
-0.3230601534209437,
0.3161713641922788,
-0.4811213940350071,
0.08327140785409295,
0.37526634808329173,
-1.9990814349080968,
-0.15397700748904192,
-0.9549084255901845,
1.1120455683467105,
-0.42952128451280097,
0.17929609797141985,
1.0446016069356951,
0.846774755857218,
-0.2784586201121851,
0.23000407229986752,
-0.9198047598183884,
0.996529675185623,
0.34608382599103005,
0.5099375110708237,
2.1948932092248885,
1.3625694142279887,
-0.9306056124011278,
0.6012804875871011,
-0.7788968212470156,
-1.5884735532174328,
1.0361309111992296,
0.015937704739038126,
1.4003797043998856,
2.429422654005534,
-0.27504014116509645,
0.8256449732433587,
1.2146087003914199,
-0.6435553984602482,
0.3046269503778821,
1.0182549333954134,
1.4202224852266074,
1.3511302325707892,
-0.7261672378416364,
1.5992752660640235,
-0.5764295578986515,
-0.8318409785507642,
1.1375736468462807,
0.8529850561064739,
0.9553972898831558,
-0.09161160854677679,
1.617622889574723,
1.4551513261723394,
-0.08817813881493969,
0.8045852859105778,
1.9892741470924016,
1.8347816167556374,
-0.8050652823779443,
0.008511392950551585,
2.1460096989270747,

```

0.49277581514942953,
-2.4311684786278165,
0.5134791015266132,
0.9751974520662461,
-0.38582649358518695,
0.7746227771853464,
1.7307944301597258,
2.1320333699599043,
1.1654652968447545]],
'noise_profile': { 'noise_on_init_param': {
'value':
'noise_prob': [ 0.001,
0.001,
0,
0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_ric4_uniform_scaling_sample#7'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0],
[3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '
'2 + 1 X Z 1 + 1 X 2 X 3 + 1 Z '
'2 + 1 X 3 X 4 + 1 Z 3 + 1 X '
'4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
'1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X '
'0 X 1 '
'+ 0.5 '
'Y 0 Y '
'1 + '
'0.5 Z '
'0 Z 1 '
'+ 0.5 '
'X 1 X '
'2 + '
'0.5 Y '
'1 Y 2 '
'+ 0.5 '
'Z 1 Z '
'2 + '
'0.5 X '

```

```
'2 X 3 '
'+ 0.5 '
'Y 2 Y '
'3 + '
'0.5 Z '
'2 Z 3 '
'+ 0.5 '
'X 3 X '
'4 + '
'0.5 Y '
'3 Y 4 '
'+ 0.5 '
'Z 3 Z '
'4 + '
'0.5 X '
'4 X 5 '
'+ 0.5 '
'Y 4 Y '
'5 + '
'0.5 Z '
'4 Z 5 '
'+ 0.5 '
'X 5 X '
'6 + '
'0.5 Y '
'5 Y 6 '
'+ 0.5 '
'Z 5 Z '
'6']},
```

```
'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0,
                0],
```

```

'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [ 1,
                        1,
                        0,
                        0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {

```

```
'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0,
                0],
```

'depolarizing'},

'noise_on_init_param': { 'status': False,
 'value': 0},
'noise_prob': [0.001,
 0.001,
 0,
 0],

True,

'depolarizing'},

'noise_on_init_param': { 'status': False,
 'value': 0},
'noise_prob': [0.001,
 0.001,
 0,
 0],

True,

'depolarizing'},

0.5,
0.5,
0.5,
0.5,
0.5,
0.5],

'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
 'identity_factors': [2,
 2,
 0,
 0],
 'noise_level': [20, 10, 0, 0],
 'noise_profile': {

'status':

'type':

'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
 'identity_factors': [3,
 3,
 0,
 0],
 'noise_level': [28, 14, 0, 0],
 'noise_profile': {

'status':

'type':

'odd_wires': 4}},
 'run_time_sec': 10.504075527191162}},
{ 'config': { 'ansatz': { 'gateset': 1,
 'layer': 30,
 'ugate': { 'coefficients': { 'bn': [0,
 0,
 0,
 0,
 0,
 0,
 0],
 'cn': [

'r': 0},

'time': { 'max': 10.0,
 'min': 0.0},

'type': 'heisenberg'}},

'init_param': { 'value': [0.38848003251483587,
 0.3884800325148359,
 1.2841735835398218,
 1.284173583539822,
 1.2841735835398218,
 1.818138220668234,
 1.818138220668234,
 2.7575469841005327,
 2.7575469841005327,
 2.7575469841005327,
 3.0053468968599275,
 3.0053468968599275,
 3.005346896859927,
 4.184079661691832,
 4.83635165164968,
 4.83635165164968,

5.97926077140412,
6.458805266455043,
6.513102391504654,
6.513102391504654,
7.122105903846713,
7.2188422331450415,
7.988590448127284,
8.268095077127937,
8.268095077127937,
9.164456488648115,
9.164456488648115,
9.782953992889732,
9.782953992889732,
10.467486029517685,
1.5496959718296535,
-1.0300437645832343,
0.425343365508412,
-1.7136674729251273,
0.33645197441382346,
0.2288357961575219,
0.10470230685228102,
-0.6612661793744108,
0.11823315872480306,
0.48571962162230753,
-1.3748501535311368,
0.2563120795974374,
-0.38871520150353267,
-0.18786175076151226,
1.0017996241286973,
0.05598903629217323,
1.501328526357405,
-0.32299727271195017,
0.5715295079657807,
-0.12201961934486968,
2.2747823054657927,
0.16289086097520028,
0.027619518366160244,
-0.10165606616500937,
-0.06321979816820887,
-0.27543819802246033,
-0.0048370064103128475,
0.1330138167350915,
-0.06428151371988705,
0.21905201010909556,
0.25697175483980955,
-1.462905966720411,
0.8014511304369155,
3.456137165485204,
-0.3560002683832885,
-0.4886877133354182,
0.32039145066707037,
1.6153352350920127,
0.5940099487759896,
0.18140087044059483,
0.27123106498617,
0.7444341922951362,
0.170480557625751,
-0.4688462519246396,
0.5235839184929864,
-0.5470964469389679,
-0.011921500492307424,
1.4116230981753706,
0.49972453659747124,
-0.7800673920851233,
0.4234227404504273,
-0.2071635265929523,
0.14064328933496192,
-1.2007096247825992,
0.12047098237640114,
-0.4125103841815656,
1.0447620953563077,
0.0020814149928196773,
0.4988991062112368,
-0.34495773827123155,
-0.12849063482979717,


```

1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_ric4_unifrom_scaling_sample#6'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0],
[3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '
'2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '
'2 + 1 X 3 X 4 + 1 Z 3 + 1 X '
'4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
'1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X '
'0 X 1 '
'+ 0.5 '
'Y 0 Y '
'1 + '
'0.5 Z '
'0 Z 1 '
'+ 0.5 '
'X 1 X '
'2 + '
'0.5 Y '
'1 Y 2 '
'+ 0.5 '
'Z 1 Z '
'2 + '
'0.5 X '
'2 X 3 '
'+ 0.5 '
'Y 2 Y '
'3 + '
'0.5 Z '
'2 Z 3 '
'+ 0.5 '
'X 3 X '
'4 + '
'0.5 Y '
'3 Y 4 '
'+ 0.5 '
'Z 3 Z '
'4 + '
'0.5 X '
'4 X 5 '
'+ 0.5 '
'Y 4 Y '
'5 + '
'0.5 Z '
'4 Z 5 '
'+ 0.5 '
'X 5 X '
'6 + '
'0.5 Y '

```

```

'5 Y 6 '
'+ 0.5 '
'Z 5 Z '
'6']],
'output': { 'data_points': [ [4, 2, 0, 0, -6.144691518305653],
[12, 6, 0, 0, -4.667338994079639],
[20, 10, 0, 0, -3.522600808002957],
[ 28,
14,
0,
0,
-2.628229437680454]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [ 0,
0,
0,
0],
'noise_level': [4, 2, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0,
0],
True,
'type':
'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [ 1,
1,
0,
0],
'noise_level': [12, 6, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0,
0],
True,
'type':
'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
'identity_factors': [ 2,
2,
0,
0],
'noise_level': [20, 10, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0,
0],
True,
'type':
'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
'identity_factors': [ 3,
3,
0,
0],
'noise_level': [28, 14, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},

```

```

'noise_prob': [ 0.001,
0.001,
0,
0],

True,

'depolarizing'},

'odd_wires': 4}},
'run_time_sec': 10.210347890853882}}],
'vqe': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}}},
'init_param': {'value': 'random'},
'noise_profile': { 'noise_on_init_param': { 'status':
'value': 0},
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': True,
'type': 'png'},
'file_name_prefix':
'heisenberg_noise_free_time_evo'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0]]},
'run': 'vqe',
'state': 'dmatrix',
'vqe': { 'iteration': 10,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': True}},
'zne': { 'data_points': None,
'degree': 1,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
'4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
'X 5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 '
'+ 0.5 Y 0 Y '
'1 + 0.5 Z 0

```

```
'optimized_parameters': [ [ 0.43940602753254654,  
                             0.4524980474487914,  
                             0.6462847510297367,  
                             0.6535145711164267,  
                             0.7019965052930044,  
                             1.0830061860275473,  
                             1.4380021973026182,  
                             2.1860837771467474,  
                             2.1860837771467474,  
                             2.1860837771467474,  
                             2.4875804812430995,  
                             2.911818356328843,  
                             4.9134595358452735,  
                             5.043089843653786,  
                             5.994354774193697,  
                             6.658321281734746,  
                             6.666230857319696,  
                             8.458454027735183,  
                             8.458454027735183,  
                             8.458454027735183,
```

8.938931772570164,
8.938931772570164,
9.876451730548837,
9.876451730548837,
10.140618079102744,
10.453495471248813,
10.878653713196195,
11.282429749250388,
11.726474998572407,
11.93229290545428,
2.9640535214497064,
-0.6206168106317247,
-0.4035747848788786,
-2.067222048924027,
2.0576259596946604,
1.364315765000901,
-0.1480698866383175,
-1.95043906046908,
0.7897528850889669,
1.350582507885341,
-0.3349044763190947,
-2.3429833820345465,
2.04351680973891,
0.8083219796841106,
-0.06520770630024467,
-0.8776318722364843,
2.4862506381076748,
0.8345121096737285,
-0.03071661620417745,
-0.6225183255138825,
0.7450777062518694,
0.12619297185303674,
0.15620862918286052,
-0.47540100870445906,
0.6345315438244892,
1.451076347224231,
0.10985833383721234,
-1.6475696409761669,
1.1918822544398533,
1.3235539868194213,
-0.24222060474147966,
0.2418944215339802,
1.0397651583041008,
1.6773252160624976,
0.16975786500059598,
-0.9211027317156172,
0.7590471244619662,
1.76540125315151,
2.43856734239768,
-0.19488067618654012,
0.8462629124439717,
0.9979674075340036,
2.5775834810620677,
0.5650956422366618,
-0.5475934153287456,
-0.0253449469292375,
1.2524069066350432,
-0.605696708865831,
0.7019502861337156,
0.2803153244338447,
-1.0831925759777479,
1.0082140101851143,
-0.3762046176346905,
-0.9535933533256993,
-1.0422804614435257,
0.13948594332192887,
0.5835070530053111,
0.16086524965091817,
0.06403444071667018,
-0.06194188553710435,
-0.4026223625881385,
0.2750028468648149,
0.6789909068622377,
0.05162849323802493,
-0.7735332926339995,

-0.14862918062202837,
0.7661757453229759,
-0.16119730427373616,
0.6482389226852195,
3.1459456036848104,
-0.4760920139337192,
1.6197254159790668,
-0.5506328506738539,
0.9968299487743895,
-0.3106735286083523,
-1.2388662013594425,
0.8898580979157581,
0.43256773745092003,
-0.7272220041247079,
0.8716071923554094,
1.5193589139179793,
0.9787540447681606,
-1.4044929205930872,
-0.1984876316951887,
-0.46744984056178585,
-0.7765225748074835,
0.04344696088727837,
0.32373965212791267,
0.9049555024172572,
0.7139278873651953,
-0.8122282267372439,
0.5323405298774778,
-0.08437719753251166,
-0.14277007957413837,
0.7912147308716768,
0.26055493673650715,
0.11357082745857092,
0.3153698930365579,
3.1981441448638614,
0.32109035951393206,
-0.23017443442945684,
0.0316864124540797,
0.20875812581978664,
0.5466872162452653,
1.2918521189433907,
-0.7312547980705957,
1.4958457475939604,
-0.053193686203739404,
-0.4437772137226961,
-0.5271126400646005,
2.5136810835026426,
0.6674244883024821,
-1.0140291771816496,
-1.177197116527103,
2.058315434744553,
0.050372102721390984,
-0.09022974351195023,
-2.296438941834702,
0.3261688389091777,
0.3072223897800251],
[0.05238332857976487,
0.05238332857976488,
0.05238332857976461,
0.5882470284735626,
1.6186245942059996,
2.2470777294271223,
2.8315596913723913,
2.8315596913723913,
2.8333251330604914,
3.7474015252366555,
3.7474015252366555,
3.7474015252366555,
3.7474015252366555,
4.452994588443841,
4.625910007504636,
4.625910007504636,
5.314362007876739,
7.185299991262795,
7.185299991262795,
8.165312898322489,

8.165312898322489,
8.165312898322489,
8.684979613482101,
9.661121823219345,
9.674169964387936,
10.525655511340814,
10.525655511340814,
10.525655511340814,
11.154963299169733,
11.154963299169733,
2.50282655917642,
0.9490947670548783,
0.70056532212722,
-0.9370006834105179,
2.127395945212884,
1.5529950009292526,
-0.5538339242405286,
-0.45109357069990613,
3.2608593101153462,
2.1691098922881027,
0.924351518772188,
0.7799372105219657,
2.588492585292434,
1.4914484649202893,
0.5111989894903198,
-1.6854828839595362,
0.709524102178921,
-0.3201993643863164,
0.37756499567776974,
0.11160463007405702,
1.4506751210554825,
-0.1158349928449886,
-0.4172082421780774,
-0.47420587439641587,
2.1318004152619214,
1.5026342770773617,
0.9629412747697549,
-1.017935070811068,
2.2274033566257083,
0.9541631607884562,
2.2865374842341715,
-0.6901760390806929,
1.4488496207691695,
-0.15529093800821045,
1.378510821815928,
-0.1238908488185843,
1.0420944500710088,
1.8938950235205334,
-0.40270933293735856,
-0.4838468076888962,
-0.7769068081513464,
2.0570355933263196,
0.33843722154840217,
-0.21296684820549192,
-0.3087457919151385,
1.6713665097762895,
0.7955513549984788,
-0.5031484152839188,
0.2783608989762586,
1.4322280472670237,
0.12038812465567636,
0.8648150791824706,
0.13552183280667268,
0.07182610915111512,
0.3477791234991623,
-0.1584859230597403,
0.5851519083687279,
0.578565105996425,
-1.7921034519484391,
0.9189192684840749,
-0.6139572052593133,
-0.023527142002935514,
-0.6495330854871428,
0.7767670181361933,
0.13616638059771954,

-0.22568119775616352,
0.4192866974084779,
0.1401009023541104,
1.419355971488301,
1.1103093112149183,
-0.8728413553259085,
1.014691351510677,
2.35757700845783,
1.131891326153365,
-1.2979835358031357,
0.42365777146015526,
1.0982786992551208,
1.698510255572029,
1.2090506157632,
-1.3597097700308411,
-0.9049577551064396,
1.0199036275847497,
2.9541844671388584,
-0.42829772740772204,
0.6542829736223691,
3.7916586259391143,
-0.1983670801457003,
0.7855289833525587,
-0.1534574040841844,
1.809122493112309,
0.35293394466185096,
1.0573849517470935,
2.461665111905985,
1.0784030177718114,
-1.585709677044719,
0.8985620342798133,
2.2360691692951646,
0.24022620726925167,
2.6922129347449117,
0.15599094623602128,
2.799936411115923,
0.9657316063780467,
-0.2732772147086647,
-1.448572189075712,
-0.7280589250410919,
0.057973620624421296,
0.9321748335747359,
-1.4338621772181848,
1.3976338029473028,
-0.25504403617708243,
3.210257057781798,
0.03448775329128267,
0.05070056715723011,
-1.3727823101385919,
0.8243579770218447,
0.13142013414460954,
-0.4933237632531999,
1.1105137227416793,
0.7155364379215329,
-0.21063252058278284],
[0.6189603325174557,
0.7142372255934472,
0.818217611349809,
1.1433975540074377,
1.9121137997887092,
1.912113799788709,
1.9189666863881127,
2.194219841514053,
2.734492156740232,
4.250258815642276,
4.250258815642276,
5.774076859791501,
6.527612880112256,
6.532834731882501,
8.049613322901676,
8.049613322901676,
8.050730377555883,
8.561643142317116,
8.561643142317116,
8.758357012835646,

8.758357012835646,
8.78353884989931,
9.477076383056275,
9.536600626467342,
9.536600626467342,
9.536600626467342,
10.241295505019522,
10.318321333031077,
10.579129354925781,
10.82886994059623,
3.662494394083504,
0.08309129295248968,
1.5295376506525808,
-1.5964000102221196,
1.8988970610647178,
-0.07218621526676876,
0.7238532160702689,
-1.2917423920622229,
1.8532222971765195,
-0.40818119715800666,
0.469598065996274,
-1.2652675561295137,
0.41465882817859046,
0.48683539382624846,
-0.15336518315146752,
-1.6419783820068856,
1.2136622236743368,
0.5933123377118196,
0.9666225223219593,
0.9539776194708611,
-0.3793986531891993,
-0.2570964298405315,
1.5884302353807402,
0.19867970725527498,
1.0036205245072367,
-1.7266220384949087,
0.6519864384701275,
0.173373648086925,
1.8697046853700252,
0.5263944462203128,
2.9785129063723943,
-0.5937256494260224,
0.22753174799850245,
0.24918335157216837,
1.5438671949833176,
-0.2981477150404249,
-0.0503664733834821,
0.542442835193611,
1.8096859498837612,
-2.2260935596321993,
0.28693396030012636,
0.9986781622414097,
-0.01940263773226835,
-1.8805315707929653,
-0.1513202473661221,
0.1836836320118686,
-0.5655439481125303,
-0.30323205019871646,
-0.4186198812206128,
0.2357002564872809,
2.8296979465953833,
-1.0776781071762758,
-0.08378558447273823,
1.7446476514004516,
-0.4606832586075214,
-1.762173446350097,
0.10241780783627007,
1.6018254886002428,
0.2621004786967086,
1.349261138919249,
0.14987623247602613,
0.5068543254371847,
-1.3474781551208477,
1.0890436270679893,
1.6662048029626129,

0.3608718991529565,
-0.11137748070221386,
0.5641204379289552,
2.2546014997074266,
1.8888740160956543,
-0.23345107582518032,
-1.5772444286238798,
2.1408383357361265,
-0.4968079560788485,
0.1533667959779154,
-0.40238715350257875,
0.25591729793510365,
2.947382531044255,
-0.32162602471486745,
-0.7587223675916386,
-0.8983111566200868,
0.006114119281372181,
2.9372824954589514,
0.48288231335139287,
0.7939297064609614,
1.9454360656955985,
2.092519500518508,
0.8995449331251366,
0.10171517677156572,
0.45235144869603666,
1.02501950409702,
0.24331156538914245,
0.48973195386607005,
2.672871435464254,
1.313352678455905,
2.010046934709317,
-0.42153209835304317,
-0.13436700176919486,
-0.6950305532982513,
0.6026553555930264,
1.6681913778840949,
0.4125109526933123,
0.514533549588801,
1.0794873063708696,
-0.9571806916209341,
0.27630815458061647,
0.6001387806934945,
0.5059087494947612,
-0.5722342732268321,
0.3986919058890637,
1.0044202485382987,
0.3352295947729849,
1.3777059541173544,
-2.835684252944388,
0.6104690716589696,
1.838086538234438,
-0.3106134411398272,
-2.2167272073692903,
0.10706372359431235,
-0.17469850626418476],
[-2.397634216504411e-17,
1.429450133825379e-17,
0.9192297980258759,
0.9192297980258759,
2.8984599305110854,
3.1993874679766523,
3.199387467976652,
3.715279848679994,
3.715919998165357,
5.126008182976459,
5.749875525815113,
5.785995839176617,
5.785995839176617,
5.785995839176617,
5.785995839176617,
6.315437962379428,
6.678845841951143,
7.3578167224936415,
7.415694801781823,
7.531263738212295,

7.949087697575632,
7.949087697575632,
8.111169110068786,
9.407094598577077,
9.495286031274873,
9.495286031274873,
10.993257661103009,
11.063439993331317,
11.639888600691357,
11.75175692668691,
2.1152038588685538,
-0.7839988468841306,
-1.0579861702584952,
-0.5122509123294313,
1.0012369417099964,
0.9275037440003397,
0.5343037326701707,
-0.014459274586663672,
2.31692048955396,
2.1400689678453872,
-0.9468421716760329,
1.3225342455296385,
2.4229762260600483,
1.4561600175543838,
-0.4592924007287594,
0.6389204073366491,
0.5578650700802136,
1.458345504678432,
-0.11931303728638508,
0.04406428844113783,
-0.26144087074192934,
0.6459890964472403,
1.2243061606997268,
0.09770103868863204,
0.23645607828472368,
0.3280111145811677,
0.5390648705936556,
-0.6034481934191781,
-1.1942560416831098,
0.3295059316644399,
-0.0732996553139216,
0.5043071925129845,
0.7399437462813884,
0.3595686242698361,
1.4643835231854074,
-0.1697776944282527,
0.2588380732750666,
0.44311592273109757,
-0.6031987466514527,
-0.06131410155451245,
-0.3299395710828798,
-0.45492250592824107,
0.5555861976799495,
1.1612177689175078,
1.833550163228831,
1.5629483318007886,
-0.2455585474033218,
-0.4534768786824923,
-1.2225865843307935,
0.20468879325769473,
1.3398561861341207,
-0.58709602862573,
0.07181500566168061,
0.17010271339345,
0.36406988692952136,
-1.689809967880289,
-1.1677426672173294,
-0.7977758879892671,
0.531032719362766,
-1.7722607960867016,
0.6332033998224597,
0.6669814362912183,
0.218182728116887,
1.0415656581640258,
-1.4544893086294475,

0.5175174925258392,
-0.9742907090362619,
-0.30648595673224416,
1.2400541074634985,
-0.30003161306087445,
-0.14406503395842435,
-0.43061285539013316,
0.6914697219419143,
-0.48545379734819916,
-0.2166375004086177,
1.2014709434123243,
1.0734635496959384,
0.43666289070672293,
-0.9550440807047608,
1.2817732673505913,
0.6254203972471953,
0.8730111716749767,
1.1814458389257443,
-0.5855214911540807,
0.33647224367417283,
0.914891547403245,
-0.10924028322724566,
0.6731472714644756,
-0.8672547313943412,
-1.1628903969680444,
1.2968110405185924,
-0.7478342770444015,
1.0157432273471354,
0.05866260951598884,
0.1549884083832883,
2.402954000164725,
-0.8510405490988947,
-0.13394876321797194,
0.35128800256256376,
0.03985349181110424,
1.0891408951797659,
0.49683980443602116,
1.3618410332836905,
0.31089249400158553,
0.007823999182329802,
-1.0325531315281817,
-0.13635517978577905,
-0.4783006843261583,
-0.9648987725589863,
-0.4094326055257113,
0.0620151507481985,
-0.3631850012398663,
-0.9119123598789223,
-0.3458539889181142,
0.8501459643039054,
1.6461875633189593,
0.08799287143090098,
-0.82318299513447,
1.17289644669955,
-0.9455811832961388],
[0.12420123662801165,
0.138246244841176,
0.34185010800102933,
0.8771280515547424,
0.8771280515547423,
1.9326762312100456,
1.9326762312100456,
2.879806034127527,
3.4467495190209787,
3.460588109395402,
3.4605881093954016,
3.4605881093954016,
3.911934955717517,
3.9564836804547703,
4.188240551814426,
5.098640553346409,
5.098640553346409,
5.098640553346408,
6.466332587579958,
7.457565848827182,

7.712724508543908,
7.890302861063261,
7.898577524812402,
8.020037295803547,
8.020037295803547,
9.289723192898348,
9.68251578824504,
9.68251578824504,
10.66694519433398,
11.464898013549973,
1.8095532965205083,
-1.6633278493215808,
-1.2281091611401596,
-0.40155726479865544,
1.7903243850816337,
0.18661159996017582,
1.2217850039169107,
-0.48128156078104645,
1.438546033666569,
-0.9466383819139143,
-0.21792270060606069,
-0.07921201994836616,
0.6692914930417907,
-0.16791294486623565,
-0.3278463005181308,
1.2883193159418245,
1.9001761172388045,
0.7646920019853412,
-0.6325477821570596,
-0.5987552945744136,
0.5669730115381587,
0.5983941343774113,
-0.06179171362000896,
-0.6867235859693906,
0.5211737756628548,
-0.19354588250953578,
0.6072816218147872,
0.3472677197742769,
-0.08031734075212016,
1.4024126217748347,
-0.5867181790012919,
0.809620813948269,
0.35997425529443994,
-0.8722967249300235,
-0.5874818244923429,
-0.5567926036837068,
-0.6652314874983639,
-0.6042054992483096,
-0.13918430829641074,
-0.08277488470584692,
-0.4972293169762719,
-0.4421668735352916,
0.6160601456858428,
-1.1972657609969646,
-1.1558488364748807,
-1.4501782622641872,
0.03989088007781949,
-0.8684923327267209,
-0.5569704341339503,
0.15823054121075,
0.6345578215095505,
0.3601853745371903,
-0.42594808878819745,
0.08979804115834313,
-0.153043281560756,
-0.603228229875778,
-1.1264410002736531,
0.6895580235946861,
-0.11045941959152497,
0.5174109375139814,
1.3269460586908306,
-0.3852298542423827,
-2.4120095755015893,
0.08807412161753436,
0.1228551365077638,

-0.6982591234727157,
-1.1618488040328845,
2.5586909942385017,
-0.24921574033555105,
-1.9870969070050777,
-1.0279498295884981,
-0.7796802357622239,
0.48428681889730063,
0.7162815161060118,
-0.15547297036103372,
1.0522591887013215,
1.772822944479854,
-0.2654037211186137,
-0.2720110866721671,
-0.03929478442656517,
0.34606077605631835,
0.5533331494282367,
1.0574846609095763,
-0.5747611578732846,
0.4916772893793279,
0.9875709691916216,
-0.17857479553023695,
-1.147281220275154,
0.37370382177065037,
2.456815155407461,
0.6567596591037266,
0.9507316292553318,
1.5113928877114242,
1.8380989093110256,
-0.1736919900840187,
-1.2666253947041328,
-0.554819304070435,
1.1931643666366285,
0.4614862178419797,
1.2040726557184918,
0.3083618234293392,
-0.4066692475369416,
0.3740368412278018,
-1.684611974494111,
-0.8650016725760176,
-1.4838330341878547,
-0.14344887350922086,
4.203596189627334,
1.1133202614474196,
1.391124117957397,
1.8627332105499828,
-0.28192336755973296,
-1.5416482647725873,
-1.9612892990030817,
0.8836928087418303,
0.300761821046295,
0.6785848024508535,
-0.5500277556546446,
0.5742776326882599,
-1.181756808209963],
[0.38848003251483587,
0.3884800325148359,
1.2841735835398218,
1.284173583539822,
1.2841735835398218,
1.818138220668234,
1.818138220668234,
2.7575469841005327,
2.7575469841005327,
2.7575469841005327,
3.0053468968599275,
3.0053468968599275,
3.005346896859927,
4.184079661691832,
4.83635165164968,
4.83635165164968,
5.97926077140412,
6.458805266455043,
6.513102391504654,
6.513102391504654,

7.122105903846713,
7.2188422331450415,
7.988590448127284,
8.268095077127937,
8.268095077127937,
9.164456488648115,
9.164456488648115,
9.782953992889732,
9.782953992889732,
10.467486029517685,
1.5496959718296535,
-1.0300437645832343,
0.425343365508412,
-1.7136674729251273,
0.33645197441382346,
0.2288357961575219,
0.10470230685228102,
-0.6612661793744108,
0.11823315872480306,
0.48571962162230753,
-1.3748501535311368,
0.2563120795974374,
-0.38871520150353267,
-0.18786175076151226,
1.0017996241286973,
0.05598903629217323,
1.501328526357405,
-0.32299727271195017,
0.5715295079657807,
-0.12201961934486968,
2.2747823054657927,
0.16289086097520028,
0.027619518366160244,
-0.10165606616500937,
-0.06321979816820887,
-0.27543819802246033,
-0.0048370064103128475,
0.1330138167350915,
-0.06428151371988705,
0.21905201010909556,
0.25697175483980955,
-1.462905966720411,
0.8014511304369155,
3.456137165485204,
-0.3560002683832885,
-0.4886877133354182,
0.32039145066707037,
1.6153352350920127,
0.5940099487759896,
0.18140087044059483,
0.27123106498617,
0.7444341922951362,
0.170480557625751,
-0.4688462519246396,
0.5235839184929864,
-0.5470964469389679,
-0.011921500492307424,
1.4116230981753706,
0.49972453659747124,
-0.7800673920851233,
0.4234227404504273,
-0.2071635265929523,
0.14064328933496192,
-1.2007096247825992,
0.12047098237640114,
-0.4125103841815656,
1.0447620953563077,
0.0020814149928196773,
0.4988991062112368,
-0.34495773827123155,
-0.12849063482979717,
0.4280934542561602,
0.7138444629946918,
-0.6497712710787925,
-0.43652659674625793,

0.917432486791049,
-0.8523477361785018,
0.5648987974980545,
0.3183226113969618,
0.14295784675778261,
0.9589824340171335,
0.9194478619356266,
1.1001359353915339,
1.5417377267389083,
-0.5376003771957187,
-0.5404528215706331,
0.21624687748210675,
0.8814937279455825,
-0.371639532103812,
-0.2668100917972658,
1.3133163210428074,
0.44574090697479807,
-1.0940261852319737,
-0.313830643067749,
-0.06100275437545863,
-0.049560783686653447,
-0.3025056724120473,
-0.9891529613817422,
-0.35099332176432035,
0.5436069706516342,
-0.722828417704955,
1.6968661444846564,
-2.4402971962208975,
0.7865745542588627,
1.3437881169309889,
0.36108471326554326,
-0.1749159211919777,
-0.014334278530919132,
0.02978377629472008,
1.8533978555002608,
-1.8470426695738211,
0.6272356375919823,
0.44775499409400255,
1.3281153539688288,
0.4583217606753929,
0.6443670392327268,
0.5254233581973354,
2.3087652110195127,
-0.7761319091913014,
-0.7605421959233766,
-0.6461580464596497,
0.7075597766411752,
0.985778508379752,
-1.3130974173457381,
2.8711331564400684,
1.2993543284582694,
1.9493614761217355,
1.1647493000584068,
-0.42731613935458074,
2.7038346000669766],
[-1.8784742854218807e-17,
-7.763750530158576e-17,
0.3802827375258553,
0.44096449652630626,
0.8040692975576897,
1.9224694434519336,
1.9224694434519338,
1.9224694434519336,
1.9224694434519336,
1.9224694434519336,
1.9224694434519338,
2.730288811725591,
4.476350841356259,
4.476350841356259,
4.476350841356259,
6.331658137341824,
6.331658137341824,
6.7621011201584285,
6.77851096220289,
7.490043179529316,

7.490043179529316,
7.490043179529316,
7.490043179529316,
7.511664240193972,
8.302259372586072,
8.36899822047307,
8.607405767895836,
9.631163052898822,
9.874876363637043,
10.225975825453373,
2.711573403856553,
0.20239905137011172,
-0.4220883549830381,
-0.8404830976891227,
1.794332340210785,
1.5095243517856047,
0.22326098118939897,
0.07733371735403745,
1.280548885023134,
1.4121496210624682,
0.9815935680907694,
-2.975348552150871,
1.6316643284140941,
0.8239502945369961,
0.0314505340316768,
-1.0892891749800817,
1.720694113273713,
0.6634696927679957,
-0.836512780686814,
-1.1286873737196692,
-0.09500137106776224,
1.0428135848317974,
-0.4828772957450712,
-1.129334172148368,
0.7253737318795237,
1.4108675361353635,
0.7588540034470418,
-0.5039485441412638,
0.10346857579652805,
1.2668095290766983,
0.7994757345637048,
-0.7393240316081237,
-0.7818304990031031,
0.25375980918776664,
2.228975199542585,
2.2487625387645873,
0.6384105819815235,
0.39434544396871446,
1.3679129840939837,
-0.27336699910726836,
1.3170761670781417,
1.1159622861844822,
1.176824183488981,
0.40583799697229034,
0.8786168217534174,
0.2914083315983707,
-0.8464697846173548,
-0.032455291822620566,
-2.0755386823554396,
-0.11439663923099666,
0.7312672791487281,
0.4263196833485598,
0.11802610051350722,
1.0445881812565523,
-1.9566462184988007,
0.9492303984073598,
-1.7129593951518396,
-0.3230601534209437,
0.3161713641922788,
-0.4811213940350071,
0.08327140785409295,
0.37526634808329173,
-1.9990814349080968,
-0.15397700748904192,
-0.9549084255901845,

1.1120455683467105,
-0.42952128451280097,
0.17929609797141985,
1.0446016069356951,
0.846774755857218,
-0.2784586201121851,
0.23000407229986752,
-0.9198047598183884,
0.996529675185623,
0.34608382599103005,
0.5099375110708237,
2.1948932092248885,
1.3625694142279887,
-0.9306056124011278,
0.6012804875871011,
-0.7788968212470156,
-1.5884735532174328,
1.0361309111992296,
0.015937704739038126,
1.4003797043998856,
2.429422654005534,
-0.27504014116509645,
0.8256449732433587,
1.2146087003914199,
-0.6435553984602482,
0.3046269503778821,
1.0182549333954134,
1.4202224852266074,
1.3511302325707892,
-0.7261672378416364,
1.5992752660640235,
-0.5764295578986515,
-0.8318409785507642,
1.1375736468462807,
0.8529850561064739,
0.9553972898831558,
-0.09161160854677679,
1.617622889574723,
1.4551513261723394,
-0.08817813881493969,
0.8045852859105778,
1.9892741470924016,
1.8347816167556374,
-0.8050652823779443,
0.008511392950551585,
2.1460096989270747,
0.49277581514942953,
-2.4311684786278165,
0.5134791015266132,
0.9751974520662461,
-0.38582649358518695,
0.7746227771853464,
1.7307944301597258,
2.1320333699599043,
1.1654652968447545],
[0.4022850038273885,
0.4222288083135141,
0.7644599371513524,
0.7900179023917507,
1.1026508993249646,
1.1630326759467198,
1.1630326759467198,
1.2119340062926192,
1.9859325565890973,
1.985932556589097,
2.096166401632718,
3.378647998605865,
3.378647998605865,
3.378647998605865,
4.354848115861854,
6.006218342726225,
6.006218342726225,
6.009974297352436,
8.225327865566033,
8.242093148909708,

9.222354355246733,
9.369468664000962,
9.432712219232048,
9.692289963799956,
9.692829942577658,
10.391360826688166,
10.391360826688166,
10.529867035412478,
10.529867035412478,
11.06290493838992,
2.9062307172979733,
-1.7788694936155562,
-0.1145266982224492,
-0.9311559923392093,
0.9691998209476543,
1.7463479137973288,
-0.7663862807329105,
-0.9694574896037749,
2.8979112386638413,
1.1400097878698292,
0.8096361549155084,
-1.5900494197653285,
0.09829667109053167,
0.31839493135320834,
0.3886777840539921,
0.12229184789371997,
1.7250225844898464,
1.0065795676456808,
1.4535641205083867,
-0.3486770669120958,
0.2556790893233589,
1.778531141883236,
0.269972269049403,
1.0155903957099945,
0.8041163886037216,
0.5381821926966456,
-0.30108604163317454,
0.6396533041678052,
2.1819991440037696,
1.4978086658278347,
2.5181040929106753,
0.4624415671193703,
-0.17701736987873673,
0.6910098122382194,
2.947172461882987,
-1.0727539032453073,
0.4023854197196741,
1.650490750838525,
0.9313998963798695,
-0.6769162055305248,
-0.5620785927454173,
-0.20885673940707247,
2.5019757090972443,
-0.37561416876024456,
-0.9329392032282731,
0.7818961506889994,
0.44350434198669075,
-0.10644929099379355,
-2.498928032568615,
-1.2046085786256906,
-0.06365416567297402,
0.17257236645054272,
1.2516209341031546,
-0.3591527663232026,
0.46789807326225935,
0.24374638092155315,
0.02110383811013195,
0.12714835746845662,
-0.5505736947218767,
-0.03497727012222505,
-2.2168334497478877,
-0.21485512988314215,
1.7552902914988067,
0.6189795903733311,
0.3191491062237487,

0.522632432692797,
-1.1121089236913335,
1.1371618962620422,
-0.2410043783817164,
0.24168543661014927,
0.39418565996934685,
1.2107667809264395,
2.398781467533486,
0.6948257569073976,
0.6170229142231725,
0.7050419245924102,
1.768546262611762,
1.5562956304120967,
-0.4269237214365518,
0.6302835616447369,
1.7710848704009405,
1.2281100264507194,
1.5036530549577243,
-0.7435268012971183,
0.7613541379199313,
0.8523298100708424,
-0.6655959646469325,
1.1543911681084875,
2.2675628634479494,
1.0299985170060308,
1.400963430022522,
0.27759304330136975,
0.6960561589746208,
-0.4210335016880339,
0.0336463669485127,
-2.099660094290342,
0.6674819238895001,
0.5049668435179431,
0.48001405045948287,
1.3574871122533543,
-2.2986993170736225,
9.601426007351498e-05,
1.3312470798272018,
0.7582238004326416,
-0.35033967277151296,
-0.4108552112493075,
0.1982910129710702,
0.6700017270602091,
0.26559360983179486,
-0.14563210230012705,
-0.39610135234418875,
1.1727729846793147,
0.6713573564879365,
0.8527070052929181,
0.8268283343075853,
2.2918575835381745,
0.5082103460913493,
1.8600493489550152,
0.49576811175353974,
2.232387985311215],
[0.16477895719846114,
0.16477895719846108,
0.7183189156962347,
1.8697093861419474,
2.105082615571597,
3.05308344502201,
4.199393321043376,
4.199393321043376,
4.199393321043375,
6.490210948425593,
6.4902109484255925,
6.490210948425593,
6.4902109484255925,
6.4902109484255925,
6.490210948425593,
6.882554174728105,
6.882554174728105,
6.882554174728105,
6.882554174728104,
6.8877583035732775,

6.8877583035732775,
6.8877583035732775,
8.21014824639056,
8.21014824639056,
9.071469420716642,
9.120928605341893,
9.714944098401723,
11.474131735950095,
11.474131735950095,
11.794551582273902,
1.341188628911811,
1.6554894539990426,
-2.1068974621953425,
-1.4171475675653578,
0.332116690069513333,
0.11331792925356984,
-1.3610911099963097,
0.46439741779020816,
0.9313501387866817,
1.0938593094543279,
-0.3591727738608774,
-0.4732622270970327,
1.6321598874818157,
-0.2782091348514778,
-1.0526781349295347,
0.0708703831522549,
2.6982701224663113,
-0.038867313655444914,
-0.10508583152652556,
-0.40759365792756236,
0.6284849004330764,
0.5192751929971827,
-1.6282040637266806,
0.0014422433007245106,
-1.8299719264591412,
2.3182597876656823,
0.8875167319646391,
1.5664268415036142,
1.5052778791259855,
2.935955436800565,
1.533101581959507,
1.8091829332960188,
0.45080966818917967,
0.1506883627647002,
-0.6843548866459184,
-0.5544883854492364,
1.2466562002431278,
-1.1866110136199353,
1.4521358231995756,
-1.125608676420736,
-0.24054383618816966,
-0.058272145415006006,
0.410408128597049,
-1.6441753460137252,
0.4019816715685168,
0.4588365987304068,
1.354346078964423,
-1.7153251360741875,
-1.1039978646265602e-06,
0.7940670488104785,
0.17979772520560053,
-0.24385394325110826,
0.6367160741723578,
0.8500459957371875,
0.0559416907316065,
0.32418668587130467,
-1.4797636873635434,
0.11902177942387202,
0.5185165439317966,
-0.08399222686641401,
0.8650918037299421,
1.4911587670088011,
0.12792248933645675,
0.3372262149750348,
-0.6553154954729696,

1.1995792367396254,
0.5132294030998703,
0.44835115860783636,
-0.37116980512777586,
0.7805194563374558,
-0.854334789795377,
0.5008527672922033,
-0.2302685629030402,
0.6145182391921147,
-0.6575500079215796,
0.8657035637529482,
1.1874516653159721,
-0.16284802367068918,
-0.26462171656665856,
-0.9815102056659819,
0.23198397744948207,
-1.087097195512072,
0.6616385654966559,
-0.2929637257774683,
0.03471390505932934,
0.2048751570049573,
0.9170017118070749,
0.39617956766989404,
0.12006199846995962,
0.1922705913940127,
0.8784478594386566,
0.055479957240883566,
-0.6634105431569992,
1.0742951419164057,
-0.37818200958777604,
0.28548758229245597,
-0.6594827056613419,
1.0235832498557866,
-0.19428253177041227,
-0.5531991468121416,
-1.189626464718852,
0.2519110119226656,
-0.6708462316627377,
2.7994352657547057,
-3.8764436680175813,
0.5152327138619494,
-0.3749347648847944,
2.9042167722097862,
1.1265965037549552,
-1.850959186134046,
2.165374611027145,
0.8818414791702859,
1.5039524911129019,
0.06564767423285973,
0.6486215314149069,
-0.08854143781823305,
1.8396223528529987,
-0.00792246728000446,
2.8805879469479487,
3.6337353861266473],
[0.11853625011402234,
0.11853625011402236,
1.1173235551611367,
1.7342580765110573,
1.7342580765110573,
2.108144483620023,
3.054129947242328,
3.054129947242328,
3.054129947242328,
3.056306360633301,
4.638414750363569,
5.0621851887275495,
5.7503913724004905,
5.774998407238176,
6.917933224474368,
6.917933224474368,
8.061549209526286,
8.061549209526286,
8.061549209526286,
8.591251503132888,

8.591251503132888,
9.051235537435975,
9.081856762230144,
9.467876152039107,
9.545932016874406,
9.591419209848118,
10.465616352275608,
10.465616352275608,
10.465735173309618,
10.980942228105905,
2.0734674059611633,
0.32892201175743896,
-0.007356440825095051,
0.7856576253246221,
2.089457548986907,
0.7744884921126718,
-0.35049309306395837,
-0.6556831494452968,
2.0369525201874903,
0.4739652080570001,
-0.8564345639983049,
1.25716179603495,
1.7181260437331554,
2.3978160665502384,
1.1152024575020885,
-1.3335737674295711,
0.016650799076572978,
1.6607129762234538,
0.648059834763191,
0.2438725527060928,
-0.27307896000752346,
1.4369017198677916,
3.5672448343800007,
-0.5379702113035718,
-0.7420664562717484,
-0.529699891930992,
0.7301471483152846,
0.781913925757813,
0.6004078048346045,
1.7324865479818374,
0.4548236607944829,
-1.4419813608219112,
-0.684662976785234,
0.8455512814500156,
1.97751114408285,
-0.833202517811791,
1.110524460863052,
2.6403320691876733,
-0.3129427957408566,
-0.06674204626492235,
0.5842857778318762,
-0.21132485649138796,
-0.011842019288790376,
-0.25743970610802486,
0.7486616881233721,
-0.1302029141156864,
0.09497091477412227,
-0.3715265429351211,
0.39570077921274605,
0.28643868832003017,
0.7848413456471106,
-0.7157146817058562,
0.09411764456438232,
-0.10910076647953601,
1.1262761892344808,
-0.27334997823872975,
-0.44027652197942063,
0.6740415100156434,
-0.336256645457715,
0.4268157265313067,
-0.8611696251566416,
-0.34560928156726306,
0.5618074077714486,
0.5866360379174992,
1.8662876827821717,

```

2.21592310898005,
1.972223656383108,
1.6141187926643297,
0.856255815565121,
-0.34275929095614754,
-1.0642528745182613,
-1.0618686751734654,
0.2379582550153795,
-0.6501770361983872,
-0.19408645485522444,
1.501330829125914,
1.5726602727591479,
0.7066066097903853,
-1.4580936665368467,
0.46940440748349077,
-0.1287015677341077,
-0.9805925033056946,
-0.3754741669636888,
-0.026817919501009202,
1.1067346586651607,
0.17547923725022047,
0.8902143171896091,
-0.22777698610446503,
0.8082253583749343,
-1.0674083601412958,
-0.022706971436593337,
0.34227899244110566,
1.3527736523760083,
2.0767841716125894,
0.5778568145668401,
-0.6555502504392442,
1.2670274746873995,
0.507694022232518,
-0.2629991187543805,
1.7048253080726326,
1.4088065522309612,
2.823918413203317,
1.2763232744235296,
-1.4348344213353486,
-0.605496445996402,
0.09645158262695042,
0.024639736182597342,
0.23997550724405606,
0.9979895378690888,
-0.40612123795993227,
1.826790650755685,
0.029599395583551916,
-0.4592861834683686,
0.0328358806035505,
1.3612120706287358,
0.8210402794880411,
0.05688164146575431,
0.03152531580318861,
2.0316547976974904,
2.795039607579165]],
'run_time_sec': 69142.54766130447}}],
'zne': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}}},
'init_param': { 'value': [ 0.6189603325174557,

```

0.7142372255934472,
0.818217611349809,
1.1433975540074377,
1.9121137997887092,
1.912113799788709,
1.9189666863881127,
2.194219841514053,
2.734492156740232,
4.250258815642276,
4.250258815642276,
5.774076859791501,
6.527612880112256,
6.532834731882501,
8.049613322901676,
8.049613322901676,
8.050730377555883,
8.561643142317116,
8.561643142317116,
8.758357012835646,
8.758357012835646,
8.78353884989931,
9.477076383056275,
9.536600626467342,
9.536600626467342,
9.536600626467342,
10.241295505019522,
10.318321333031077,
10.579129354925781,
10.82886994059623,
3.662494394083504,
0.08309129295248968,
1.5295376506525808,
-1.5964000102221196,
1.8988970610647178,
-0.07218621526676876,
0.7238532160702689,
-1.2917423920622229,
1.8532222971765195,
-0.40818119715800666,
0.469598065996274,
-1.2652675561295137,
0.41465882817859046,
0.48683539382624846,
-0.15336518315146752,
-1.6419783820068856,
1.2136622236743368,
0.5933123377118196,
0.9666225223219593,
0.9539776194708611,
-0.3793986531891993,
-0.2570964298405315,
1.5884302353807402,
0.19867970725527498,
1.0036205245072367,
-1.7266220384949087,
0.6519864384701275,
0.173373648086925,
1.8697046853700252,
0.5263944462203128,
2.9785129063723943,
-0.5937256494260224,
0.22753174799850245,
0.24918335157216837,
1.5438671949833176,
-0.2981477150404249,
-0.0503664733834821,
0.542442835193611,
1.8096859498837612,
-2.2260935596321993,
0.28693396030012636,
0.9986781622414097,
-0.01940263773226835,
-1.8805315707929653,
-0.1513202473661221,
0.1836836320118686,

-0.5655439481125303,
-0.30323205019871646,
-0.4186198812206128,
0.2357002564872809,
2.8296979465953833,
-1.0776781071762758,
-0.08378558447273823,
1.7446476514004516,
-0.4606832586075214,
-1.762173446350097,
0.10241780783627007,
1.6018254886002428,
0.2621004786967086,
1.349261138919249,
0.14987623247602613,
0.5068543254371847,
-1.3474781551208477,
1.0890436270679893,
1.6662048029626129,
0.3608718991529565,
-0.11137748070221386,
0.5641204379289552,
2.2546014997074266,
1.8888740160956543,
-0.23345107582518032,
-1.5772444286238798,
2.1408383357361265,
-0.4968079560788485,
0.1533667959779154,
-0.40238715350257875,
0.25591729793510365,
2.947382531044255,
-0.32162602471486745,
-0.7587223675916386,
-0.8983111566200868,
0.006114119281372181,
2.9372824954589514,
0.48288231335139287,
0.7939297064609614,
1.9454360656955985,
2.092519500518508,
0.8995449331251366,
0.10171517677156572,
0.45235144869603666,
1.02501950409702,
0.24331156538914245,
0.48973195386607005,
2.672871435464254,
1.313352678455905,
2.010046934709317,
-0.42153209835304317,
-0.13436700176919486,
-0.6950305532982513,
0.6026553555930264,
1.6681913778840949,
0.4125109526933123,
0.514533549588801,
1.0794873063708696,
-0.9571806916209341,
0.27630815458061647,
0.6001387806934945,
0.5059087494947612,
-0.5722342732268321,
0.3986919058890637,
1.0044202485382987,
0.3352295947729849,
1.3777059541173544,
-2.835684252944388,
0.6104690716589696,
1.838086538234438,
-0.3106134411398272,
-2.2167272073692903,
0.10706372359431235,
-0.17469850626418476]],

'noise_profile': { 'noise_on_init_param': { 'status':

```
False,

'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_ric4_unifrom_scaling_sample#3'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0],
[3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [ 4,
2,
0,
0,
-6.218690041461777],
[ 12,
6,
0,
0,
-4.805156509769758],
[ 20,
10,
0,
0,
-3.7079207759854302],
[ 28,
14,
0,
0,
-2.8469331605050203]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 3.838539123535156e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -7.0690840063993505,
'others': { 'beta_coefficients': [
2.1875,
-2.1875,
1.3125,
-0.3125],
'cost_richardson_zne':
'sorted_expectation_vals': [
-6.218690041461777,
-4.805156509769758,
-3.7079207759854302,
-2.8469331605050203],
```

```

'sorted_noise': [ 6,
                  18,
                  30,
                  42]],
'sampled data': [ [ 4,
                    2,
                    0,
                    0,
                    -6.218690041461777],
                  [ 12,
                    6,
                    0,
                    0,
                    -4.805156509769758],
                  [ 20,
                    10,
                    0,
                    0,
                    -3.7079207759854302],
                  [ 28,
                    14,
                    0,
                    0,
                    -2.8469331605050203]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                                'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'heisenberg'}}},
  'init_param': { 'value': [ 0.11853625011402234,
                              0.11853625011402236,
                              1.1173235551611367,
                              1.7342580765110573,
                              1.7342580765110573,
                              2.108144483620023,
                              3.054129947242328,
                              3.054129947242328,
                              3.054129947242328,
                              3.056306360633301,
                              4.638414750363569,
                              5.0621851887275495,
                              5.7503913724004905,
                              5.774998407238176,
                              6.917933224474368,
                              6.917933224474368,
                              8.061549209526286,
                              8.061549209526286,
                              8.061549209526286,
                              8.591251503132888,
                              8.591251503132888,
                              9.051235537435975,
                              9.081856762230144,
                              9.467876152039107,
                              9.545932016874406,
                              9.591419209848118,
                              10.465616352275608,
                              10.465616352275608,
                              10.465735173309618,
                              10.980942228105905,
                              2.0734674059611633,
                              0.32892201175743896,

```

-0.007356440825095051,
0.7856576253246221,
2.089457548986907,
0.7744884921126718,
-0.35049309306395837,
-0.6556831494452968,
2.0369525201874903,
0.4739652080570001,
-0.8564345639983049,
1.25716179603495,
1.7181260437331554,
2.3978160665502384,
1.1152024575020885,
-1.3335737674295711,
0.016650799076572978,
1.6607129762234538,
0.648059834763191,
0.2438725527060928,
-0.27307896000752346,
1.4369017198677916,
3.5672448343800007,
-0.5379702113035718,
-0.7420664562717484,
-0.529699891930992,
0.7301471483152846,
0.781913925757813,
0.6004078048346045,
1.7324865479818374,
0.4548236607944829,
-1.4419813608219112,
-0.684662976785234,
0.8455512814500156,
1.97751114408285,
-0.833202517811791,
1.110524460863052,
2.6403320691876733,
-0.3129427957408566,
-0.06674204626492235,
0.5842857778318762,
-0.21132485649138796,
-0.011842019288790376,
-0.25743970610802486,
0.7486616881233721,
-0.1302029141156864,
0.09497091477412227,
-0.3715265429351211,
0.39570077921274605,
0.28643868832003017,
0.7848413456471106,
-0.7157146817058562,
0.09411764456438232,
-0.10910076647953601,
1.1262761892344808,
-0.27334997823872975,
-0.44027652197942063,
0.6740415100156434,
-0.336256645457715,
0.4268157265313067,
-0.8611696251566416,
-0.34560928156726306,
0.5618074077714486,
0.5866360379174992,
1.8662876827821717,
2.21592310898005,
1.972223656383108,
1.6141187926643297,
0.856255815565121,
-0.34275929095614754,
-1.0642528745182613,
-1.0618686751734654,
0.2379582550153795,
-0.6501770361983872,
-0.19408645485522444,
1.501330829125914,
1.5726602727591479,

```

0.7066066097903853,
-1.4580936665368467,
0.46940440748349077,
-0.1287015677341077,
-0.9805925033056946,
-0.3754741669636888,
-0.026817919501009202,
1.1067346586651607,
0.17547923725022047,
0.8902143171896091,
-0.22777698610446503,
0.8082253583749343,
-1.0674083601412958,
-0.022706971436593337,
0.34227899244110566,
1.3527736523760083,
2.0767841716125894,
0.5778568145668401,
-0.6555502504392442,
1.2670274746873995,
0.507694022232518,
-0.2629991187543805,
1.7048253080726326,
1.4088065522309612,
2.823918413203317,
1.2763232744235296,
-1.4348344213353486,
-0.605496445996402,
0.09645158262695042,
0.024639736182597342,
0.23997550724405606,
0.9979895378690888,
-0.40612123795993227,
1.826790650755685,
0.029599395583551916,
-0.4592861834683686,
0.0328358806035505,
1.3612120706287358,
0.8210402794880411,
0.05688164146575431,
0.03152531580318861,
2.0316547976974904,
2.795039607579165]],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_ric4_unifrom_scaling_sample#10'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0],
[3, 3, 0, 0]]},
'run': 'zne',

```



```
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [ -1.8784742854218807e-17,
-7.763750530158576e-17,
0.3802827375258553,
0.44096449652630626,
0.8040692975576897,
1.9224694434519336,
1.9224694434519338,
1.9224694434519336,
1.9224694434519336,
1.9224694434519336,
1.9224694434519338,
2.730288811725591,
4.476350841356259,
4.476350841356259,
4.476350841356259,
6.331658137341824,
6.331658137341824,
6.7621011201584285,
6.77851096220289,
7.490043179529316,
7.490043179529316,
7.490043179529316,
7.490043179529316,
7.511664240193972,
8.302259372586072,
8.36899822047307,
8.607405767895836,
9.631163052898822,
9.874876363637043,
10.225975825453373,
2.711573403856553,
0.20239905137011172,
-0.4220883549830381,
-0.8404830976891227,
1.794332340210785,
1.5095243517856047,
0.22326098118939897,
0.07733371735403745,
1.280548885023134,
1.4121496210624682,
0.9815935680907694,
-2.975348552150871,
1.6316643284140941,
0.8239502945369961,
0.0314505340316768,
-1.0892891749800817,
1.720694113273713,
0.6634696927679957,
-0.836512780686814,
-1.1286873737196692,
-0.09500137106776224,
1.0428135848317974,
-0.4828772957450712,
-1.129334172148368,
0.7253737318795237,
1.4108675361353635,
0.7588540034470418,
-0.5039485441412638,
0.10346857579652805,
1.2668095290766983,
0.7994757345637048,
-0.7393240316081237,
-0.7818304990031031,
```

0.25375980918776664,
2.228975199542585,
2.2487625387645873,
0.6384105819815235,
0.39434544396871446,
1.3679129840939837,
-0.27336699910726836,
1.3170761670781417,
1.1159622861844822,
1.176824183488981,
0.40583799697229034,
0.8786168217534174,
0.2914083315983707,
-0.8464697846173548,
-0.032455291822620566,
-2.0755386823554396,
-0.11439663923099666,
0.7312672791487281,
0.4263196833485598,
0.11802610051350722,
1.0445881812565523,
-1.9566462184988007,
0.9492303984073598,
-1.7129593951518396,
-0.3230601534209437,
0.3161713641922788,
-0.4811213940350071,
0.08327140785409295,
0.37526634808329173,
-1.9990814349080968,
-0.15397700748904192,
-0.9549084255901845,
1.1120455683467105,
-0.42952128451280097,
0.17929609797141985,
1.0446016069356951,
0.846774755857218,
-0.2784586201121851,
0.23000407229986752,
-0.9198047598183884,
0.996529675185623,
0.34608382599103005,
0.5099375110708237,
2.1948932092248885,
1.3625694142279887,
-0.9306056124011278,
0.6012804875871011,
-0.7788968212470156,
-1.5884735532174328,
1.0361309111992296,
0.015937704739038126,
1.4003797043998856,
2.429422654005534,
-0.27504014116509645,
0.8256449732433587,
1.2146087003914199,
-0.6435553984602482,
0.3046269503778821,
1.0182549333954134,
1.4202224852266074,
1.3511302325707892,
-0.7261672378416364,
1.5992752660640235,
-0.5764295578986515,
-0.8318409785507642,
1.1375736468462807,
0.8529850561064739,
0.9553972898831558,
-0.09161160854677679,
1.617622889574723,
1.4551513261723394,
-0.08817813881493969,
0.8045852859105778,
1.9892741470924016,
1.8347816167556374,

False,

```
-0.8050652823779443,
0.008511392950551585,
2.1460096989270747,
0.49277581514942953,
-2.4311684786278165,
0.5134791015266132,
0.9751974520662461,
-0.38582649358518695,
0.7746227771853464,
1.7307944301597258,
2.1320333699599043,
1.1654652968447545]],
'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value': 0},
                  'noise_prob': [0.001, 0.001, 0, 0],
                  'status': True,
                  'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_ric4_unifrom_scaling_sample#7'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [ 4,
                           2,
                           0,
                           0,
                           -5.9667914171586585],
                          [ 12,
                           6,
                           0,
                           0,
                           -4.541488857163632],
                          [ 20,
                           10,
                           0,
                           0,
                           -3.4337943886777844],
                          [ 28,
                           14,
                           0,
                           0,
                           -2.564526423455772]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 5.555152893066406e-05,
            'zne_values': { 'degree': 0,
```

2.1875,
-2.1875,
1.3125,
-0.3125],

11.390625,

-5.9667914171586585,
-4.541488857163632,
-3.4337943886777844,
-2.564526423455772],

'extrapolated_value': -6.823289977798783,
'others': { 'beta_coefficients': [

'cost_richardson_zne':

'sorted_expectation_vals': [

'sorted_noise': [6,
18,
30,
42]],

'sampled data': [[4,
2,
0,
0,
-5.9667914171586585],
[12,
6,
0,
0,
-4.541488857163632],
[20,
10,
0,
0,
-3.4337943886777844],
[28,
14,
0,
0,
-2.564526423455772]],

'sampling': 'default']]],
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0,
0,
0,
0,
0,
0],
'cn': [0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg']]],
'init_param': { 'value': [0.12420123662801165,
0.138246244841176,
0.34185010800102933,
0.8771280515547424,
0.8771280515547423,
1.9326762312100456,
1.9326762312100456,
2.879806034127527,
3.4467495190209787,
3.460588109395402,
3.4605881093954016,
3.4605881093954016,
3.911934955717517,
3.9564836804547703,
4.188240551814426,
5.098640553346409,
5.098640553346409,
5.098640553346408,
6.466332587579958,

7.457565848827182,
7.712724508543908,
7.890302861063261,
7.898577524812402,
8.020037295803547,
8.020037295803547,
9.289723192898348,
9.68251578824504,
9.68251578824504,
10.66694519433398,
11.464898013549973,
1.8095532965205083,
-1.6633278493215808,
-1.2281091611401596,
-0.40155726479865544,
1.7903243850816337,
0.18661159996017582,
1.2217850039169107,
-0.48128156078104645,
1.438546033666569,
-0.9466383819139143,
-0.2179227006066069,
-0.07921201994836616,
0.6692914930417907,
-0.16791294486623565,
-0.3278463005181308,
1.2883193159418245,
1.9001761172388045,
0.7646920019853412,
-0.6325477821570596,
-0.5987552945744136,
0.5669730115381587,
0.5983941343774113,
-0.06179171362000896,
-0.6867235859693906,
0.5211737756628548,
-0.19354588250953578,
0.6072816218147872,
0.3472677197742769,
-0.08031734075212016,
1.4024126217748347,
-0.5867181790012919,
0.809620813948269,
0.35997425529443994,
-0.8722967249300235,
-0.5874818244923429,
-0.5567926036837068,
-0.6652314874983639,
-0.6042054992483096,
-0.13918430829641074,
-0.08277488470584692,
-0.4972293169762719,
-0.4421668735352916,
0.6160601456858428,
-1.1972657609969646,
-1.1558488364748807,
-1.4501782622641872,
0.03989088007781949,
-0.8684923327267209,
-0.5569704341339503,
0.15823054121075,
0.6345578215095505,
0.3601853745371903,
-0.42594808878819745,
0.08979804115834313,
-0.153043281560756,
-0.603228229875778,
-1.1264410002736531,
0.6895580235946861,
-0.11045941959152497,
0.5174109375139814,
1.3269460586908306,
-0.3852298542423827,
-2.4120095755015893,
0.08807412161753436,

```

0.1228551365077638,
-0.6982591234727157,
-1.1618488040328845,
2.5586909942385017,
-0.24921574033555105,
-1.9870969070050777,
-1.0279498295884981,
-0.7796802357622239,
0.48428681889730063,
0.7162815161060118,
-0.15547297036103372,
1.0522591887013215,
1.772822944479854,
-0.2654037211186137,
-0.2720110866721671,
-0.03929478442656517,
0.34606077605631835,
0.5533331494282367,
1.0574846609095763,
-0.5747611578732846,
0.4916772893793279,
0.9875709691916216,
-0.17857479553023695,
-1.147281220275154,
0.37370382177065037,
2.456815155407461,
0.6567596591037266,
0.9507316292553318,
1.5113928877114242,
1.8380989093110256,
-0.1736919900840187,
-1.2666253947041328,
-0.554819304070435,
1.1931643666366285,
0.4614862178419797,
1.2040726557184918,
0.3083618234293392,
-0.4066692475369416,
0.3740368412278018,
-1.684611974494111,
-0.8650016725760176,
-1.4838330341878547,
-0.14344887350922086,
4.203596189627334,
1.1133202614474196,
1.391124117957397,
1.8627332105499828,
-0.28192336755973296,
-1.5416482647725873,
-1.9612892990030817,
0.8836928087418303,
0.300761821046295,
0.6785848024508535,
-0.5500277556546446,
0.5742776326882599,
-1.181756808209963]],
'noise_profile': { 'noise_on_init_param': { 'status':
'value': 0},
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5,

```

False,

```

0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_ric4_unifrom_scaling_sample#5'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [ 4,
                           2,
                           0,
                           0,
                           -5.814676484708359],
                          [ 12,
                           6,
                           0,
                           0,
                           -4.248447309496402],
                          [ 20,
                           10,
                           0,
                           0,
                           -3.0694027528903187],
                          [ 28,
                           14,
                           0,
                           0,
                           -2.174125329701547]],
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 5.316734313964844e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -6.775303268412966,
                           'others': { 'beta_coefficients': [
2.1875,
-2.1875,
1.3125,
-0.3125],
'cost_richardson_zne':
'sorted_expectation_vals': [
-5.814676484708359,
-4.248447309496402,
-3.0694027528903187,
-2.174125329701547],
'sorted_noise': [ 6,
                  18,
                  30,
                  42]},
'sampled data': [ [ 4,
                    2,
                    0,
                    0,
                    -5.814676484708359],
                   [ 12,
                    6,
                    0,
                    0,
                    -4.248447309496402],
                   [ 20,
                    10,
                    0,
                    0,

```



```
-3.0694027528903187],
[ 28,
 14,
 0,
 0,
-2.174125329701547]],
'sampling': 'default']}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                                'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'heisenberg'}},
  'init_param': { 'value': [ 0.05238332857976487,
                              0.05238332857976488,
                              0.05238332857976461,
                              0.5882470284735626,
                              1.6186245942059996,
                              2.2470777294271223,
                              2.8315596913723913,
                              2.8315596913723913,
                              2.8333251330604914,
                              3.7474015252366555,
                              3.7474015252366555,
                              3.7474015252366555,
                              3.7474015252366555,
                              4.452994588443841,
                              4.625910007504636,
                              4.625910007504636,
                              5.314362007876739,
                              7.185299991262795,
                              7.185299991262795,
                              8.165312898322489,
                              8.165312898322489,
                              8.165312898322489,
                              8.684979613482101,
                              9.661121823219345,
                              9.674169964387936,
                              10.525655511340814,
                              10.525655511340814,
                              10.525655511340814,
                              11.154963299169733,
                              11.154963299169733,
                              2.50282655917642,
                              0.9490947670548783,
                              0.70056532212722,
                              -0.9370006834105179,
                              2.127395945212884,
                              1.5529950009292526,
                              -0.5538339242405286,
                              -0.45109357069990613,
                              3.2608593101153462,
                              2.1691098922881027,
                              0.924351518772188,
                              0.7799372105219657,
                              2.588492585292434,
                              1.4914484649202893,
                              0.5111989894903198,
                              -1.6854828839595362,
                              0.709524102178921,
                              -0.3201993643863164,
                              0.37756499567776974,
                              0.11160463007405702,
```

1.4506751210554825,
-0.1158349928449886,
-0.4172082421780774,
-0.47420587439641587,
2.1318004152619214,
1.5026342770773617,
0.9629412747697549,
-1.017935070811068,
2.2274033566257083,
0.9541631607884562,
2.2865374842341715,
-0.6901760390806929,
1.4488496207691695,
-0.15529093800821045,
1.378510821815928,
-0.1238908488185843,
1.0420944500710088,
1.8938950235205334,
-0.40270933293735856,
-0.4838468076888962,
-0.7769068081513464,
2.0570355933263196,
0.33843722154840217,
-0.21296684820549192,
-0.3087457919151385,
1.6713665097762895,
0.7955513549984788,
-0.5031484152839188,
0.2783608989762586,
1.4322280472670237,
0.12038812465567636,
0.8648150791824706,
0.13552183280667268,
0.07182610915111512,
0.3477791234991623,
-0.1584859230597403,
0.5851519083687279,
0.578565105996425,
-1.7921034519484391,
0.9189192684840749,
-0.6139572052593133,
-0.023527142002935514,
-0.6495330854871428,
0.7767670181361933,
0.13616638059771954,
-0.22568119775616352,
0.4192866974084779,
0.1401009023541104,
1.419355971488301,
1.1103093112149183,
-0.8728413553259085,
1.014691351510677,
2.35757700845783,
1.131891326153365,
-1.2979835358031357,
0.42365777146015526,
1.0982786992551208,
1.698510255572029,
1.2090506157632,
-1.3597097700308411,
-0.9049577551064396,
1.0199036275847497,
2.9541844671388584,
-0.42829772740772204,
0.6542829736223691,
3.7916586259391143,
-0.1983670801457003,
0.7855289833525587,
-0.1534574040841844,
1.809122493112309,
0.35293394466185096,
1.0573849517470935,
2.461665111905985,
1.0784030177718114,
-1.585709677044719,

```

0.8985620342798133,
2.2360691692951646,
0.24022620726925167,
2.6922129347449117,
0.15599094623602128,
2.799936411115923,
0.9657316063780467,
-0.2732772147086647,
-1.448572189075712,
-0.7280589250410919,
0.057973620624421296,
0.9321748335747359,
-1.4338621772181848,
1.3976338029473028,
-0.25504403617708243,
3.210257057781798,
0.03448775329128267,
0.05070056715723011,
-1.3727823101385919,
0.8243579770218447,
0.13142013414460954,
-0.4933237632531999,
1.1105137227416793,
0.7155364379215329,
-0.21063252058278284]],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_ric4_unifrom_scaling_sample#2'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0],
[3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [ 4,
2,
0,
0,
-6.220556999324399],
[ 12,
6,
0,
0,
-4.682225394272194],
[ 20,
10,
0,
0,
-4.682225394272194],
[ 28,
14,
0,
0,
-4.682225394272194],
[ 36,
18,
0,
0,
-4.682225394272194],
[ 44,
22,
0,
0,
-4.682225394272194],
[ 52,
26,
0,
0,
-4.682225394272194],
[ 60,
30,
0,
0,
-4.682225394272194],
[ 68,
34,
0,
0,
-4.682225394272194],
[ 76,
38,
0,
0,
-4.682225394272194],
[ 84,
42,
0,
0,
-4.682225394272194],
[ 92,
46,
0,
0,
-4.682225394272194],
[ 100,
50,
0,
0,
-4.682225394272194],
[ 108,
54,
0,
0,
-4.682225394272194],
[ 116,
58,
0,
0,
-4.682225394272194],
[ 124,
62,
0,
0,
-4.682225394272194],
[ 132,
66,
0,
0,
-4.682225394272194],
[ 140,
70,
0,
0,
-4.682225394272194],
[ 148,
74,
0,
0,
-4.682225394272194],
[ 156,
78,
0,
0,
-4.682225394272194],
[ 164,
82,
0,
0,
-4.682225394272194],
[ 172,
86,
0,
0,
-4.682225394272194],
[ 180,
90,
0,
0,
-4.682225394272194],
[ 188,
94,
0,
0,
-4.682225394272194],
[ 196,
98,
0,
0,
-4.682225394272194],
[ 204,
102,
0,
0,
-4.682225394272194],
[ 212,
106,
0,
0,
-4.682225394272194],
[ 220,
110,
0,
0,
-4.682225394272194],
[ 228,
114,
0,
0,
-4.682225394272194],
[ 236,
118,
0,
0,
-4.682225394272194],
[ 244,
122,
0,
0,
-4.682225394272194],
[ 252,
126,
0,
0,
-4.682225394272194],
[ 260,
130,
0,
0,
-4.682225394272194],
[ 268,
134,
0,
0,
-4.682225394272194],
[ 276,
138,
0,
0,
-4.682225394272194],
[ 284,
142,
0,
0,
-4.682225394272194],
[ 292,
146,
0,
0,
-4.682225394272194],
[ 300,
150,
0,
0,
-4.682225394272194],
[ 308,
154,
0,
0,
-4.682225394272194],
[ 316,
158,
0,
0,
-4.682225394272194],
[ 324,
162,
0,
0,
-4.682225394272194],
[ 332,
166,
0,
0,
-4.682225394272194],
[ 340,
170,
0,
0,
-4.682225394272194],
[ 348,
174,
0,
0,
-4.682225394272194],
[ 356,
178,
0,
0,
-4.682225394272194],
[ 364,
182,
0,
0,
-4.682225394272194],
[ 372,
186,
0,
0,
-4.682225394272194],
[ 380,
190,
0,
0,
-4.682225394272194],
[ 388,
194,
0,
0,
-4.682225394272194],
[ 396,
198,
0,
0,
-4.682225394272194],
[ 404,
202,
0,
0,
-4.682225394272194],
[ 412,
206,
0,
0,
-4.682225394272194],
[ 420,
210,
0,
0,
-4.682225394272194],
[ 428,
214,
0,
0,
-4.682225394272194],
[ 436,
218,
0,
0,
-4.682225394272194],
[ 444,
222,
0,
0,
-4.682225394272194],
[ 452,
226,
0,
0,
-4.682225394272194],
[ 460,
230,
0,
0,
-4.682225394272194],
[ 468,
234,
0,
0,
-4.682225394272194],
[ 476,
238,
0,
0,
-4.682225394272194],
[ 484,
242,
0,
0,
-4.682225394272194],
[ 492,
246,
0,
0,
-4.682225394272194],
[ 500,
250,
0,
0,
-4.682225394272194],
[ 508,
254,
0,
0,
-4.682225394272194],
[ 516,
258,
0,
0,
-4.682225394272194],
[ 524,
262,
0,
0,
-4.682225394272194],
[ 532,
266,
0,
0,
-4.682225394272194],
[ 540,
270,
0,
0,
-4.682225394272194],
[ 548,
274,
0,
0,
-4.682225394272194],
[ 556,
278,
0,
0,
-4.682225394272194],
[ 564,
282,
0,
0,
-4.682225394272194],
[ 572,
286,
0,
0,
-4.682225394272194],
[ 580,
290,
0,
0,
-4.682225394272194],
[ 588,
294,
0,
0,
-4.682225394272194],
[ 596,
298,
0,
0,
-4.682225394272194],
[ 604,
302,
0,
0,
-4.682225394272194],
[ 612,
306,
0,
0,
-4.682225394272194],
[ 620,
310,
0,
0,
-4.682225394272194],
[ 628,
314,
0,
0,
-4.682225394272194],
[ 636,
318,
0,
0,
-4.682225394272194],
[ 644,
322,
0,
0,
-4.682225394272194],
[ 652,
326,
0,
0,
-4.682225394272194],
[ 660,
330,
0,
0,
-4.682225394272194],
[ 668,
334,
0,
0,
-4.682225394272194],
[ 676,
338,
0,
0,
-4.682225394272194],
[ 684,
342,
0,
0,
-4.682225394272194],
[ 6
```

```
2.1875,  
-2.1875,  
1.3125,  
-0.3125],  
  
11.390625,  
  
-6.220556999324399,  
-4.682225394272194,  
-3.495398379301583,  
-2.571924591189056],
```

```
0,  
-3.495398379301583],  
[ 28,  
14,  
0,  
0,  
-2.571924591189056]],  
    'degree': 0,  
    'method': 'richardson',  
    'sampling': 'default'}},  
'output': { 'exact_sol': -8.566772233505624,  
    'run_time_sec': 4.124641418457031e-05,  
    'zne_values': { 'degree': 0,  
        'extrapolated_value': -7.149084324138446,  
        'others': { 'beta_coefficients': [  
  
            'cost_richardson_zne':  
  
            'sorted_expectation_vals': [  
  
                'sorted_noise': [ 6,  
                                18,  
                                30,  
                                42]}},  
        'sampled data': [ [ 4,  
                            2,  
                            0,  
                            0,  
                            -6.220556999324399],  
                        [ 12,  
                          6,  
                          0,  
                          0,  
                          -4.682225394272194],  
                        [ 20,  
                          10,  
                          0,  
                          0,  
                          -3.495398379301583],  
                        [ 28,  
                          14,  
                          0,  
                          0,  
                          -2.571924591189056]],  
        'sampling': 'default'}},  
{ 'config': { 'ansatz': { 'gateset': 1,  
    'layer': 30,  
    'ugate': { 'coefficients': { 'bn': [ 0,  
                                        0,  
                                        0,  
                                        0,  
                                        0,  
                                        0],  
    'cn': [ 0.5,  
            0.5,  
            0.5,  
            0.5,  
            0.5,  
            0.5],  
    'r': 0},  
    'time': {'max': 10.0, 'min': 0.0},  
    'type': 'heisenberg'}},  
    'init_param': { 'value': [ 0.43940602753254654,  
                                0.4524980474487914,  
                                0.6462847510297367,  
                                0.6535145711164267,  
                                0.7019965052930044,  
                                1.0830061860275473,
```

1.4380021973026182,
2.1860837771467474,
2.1860837771467474,
2.1860837771467474,
2.4875804812430995,
2.911818356328843,
4.9134595358452735,
5.043089843653786,
5.994354774193697,
6.658321281734746,
6.666230857319696,
8.458454027735183,
8.458454027735183,
8.458454027735183,
8.938931772570164,
8.938931772570164,
9.876451730548837,
9.876451730548837,
10.140618079102744,
10.453495471248813,
10.878653713196195,
11.282429749250388,
11.726474998572407,
11.93229290545428,
2.9640535214497064,
-0.6206168106317247,
-0.4035747848788786,
-2.067222048924027,
2.0576259596946604,
1.364315765000901,
-0.1480698866383175,
-1.95043906046908,
0.7897528850889669,
1.350582507885341,
-0.3349044763190947,
-2.3429833820345465,
2.04351680973891,
0.8083219796841106,
-0.06520770630024467,
-0.8776318722364843,
2.4862506381076748,
0.8345121096737285,
-0.03071661620417745,
-0.6225183255138825,
0.7450777062518694,
0.12619297185303674,
0.15620862918286052,
-0.47540100870445906,
0.6345315438244892,
1.451076347224231,
0.10985833383721234,
-1.6475696409761669,
1.1918822544398533,
1.3235539868194213,
-0.24222060474147966,
0.2418944215339802,
1.0397651583041008,
1.6773252160624976,
0.16975786500059598,
-0.9211027317156172,
0.7590471244619662,
1.76540125315151,
2.43856734239768,
-0.19488067618654012,
0.8462629124439717,
0.9979674075340036,
2.5775834810620677,
0.5650956422366618,
-0.5475934153287456,
-0.0253449469292375,
1.2524069066350432,
-0.605696708865831,
0.7019502861337156,
0.2803153244338447,
-1.0831925759777479,

1.0082140101851143,
-0.3762046176346905,
-0.9535933533256993,
-1.0422804614435257,
0.13948594332192887,
0.5835070530053111,
0.16086524965091817,
0.06403444071667018,
-0.06194188553710435,
-0.4026223625881385,
0.2750028468648149,
0.6789909068622377,
0.05162849323802493,
-0.7735332926339995,
-0.14862918062202837,
0.7661757453229759,
-0.16119730427373616,
0.6482389226852195,
3.1459456036848104,
-0.4760920139337192,
1.6197254159790668,
-0.5506328506738539,
0.9968299487743895,
-0.3106735286083523,
-1.2388662013594425,
0.8898580979157581,
0.43256773745092003,
-0.7272220041247079,
0.8716071923554094,
1.5193589139179793,
0.9787540447681606,
-1.4044929205930872,
-0.1984876316951887,
-0.46744984056178585,
-0.7765225748074835,
0.04344696088727837,
0.32373965212791267,
0.9049555024172572,
0.7139278873651953,
-0.8122282267372439,
0.5323405298774778,
-0.08437719753251166,
-0.14277007957413837,
0.7912147308716768,
0.26055493673650715,
0.11357082745857092,
0.3153698930365579,
3.1981441448638614,
0.32109035951393206,
-0.23017443442945684,
0.0316864124540797,
0.20875812581978664,
0.5466872162452653,
1.2918521189433907,
-0.7312547980705957,
1.4958457475939604,
-0.053193686203739404,
-0.4437772137226961,
-0.5271126400646005,
2.5136810835026426,
0.6674244883024821,
-1.0140291771816496,
-1.177197116527103,
2.058315434744553,
0.050372102721390984,
-0.09022974351195023,
-2.296438941834702,
0.3261688389091777,
0.3072223897800251]],

'noise_profile': { 'noise_on_init_param': { 'status':

'value': 0},

'noise_prob': [0.001, 0.001, 0, 0],

'status': True,

'type': 'depolarizing'},

False,

```

'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_ric4_unifrom_scaling_sample#1'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0],
[3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [ 4,
2,
0,
0,
-5.966837375471956],
[ 12,
6,
0,
0,
-4.6298501952375855],
[ 20,
10,
0,
0,
-3.5807206753108622],
[ 28,
14,
0,
0,
-2.750174370397261]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 3.838539123535156e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -6.764925852359047,
'others': { 'beta_coefficients': [
2.1875,
-2.1875,
1.3125,
-0.3125],
'cost_richardson_zne':
11.390625,
'sorted_expectation_vals': [
-5.966837375471956,
-4.6298501952375855,
-3.5807206753108622,
-2.750174370397261],
'sorted_noise': [ 6,
18,
30,
42]},
'sampled data': [ [ 4,

```

```

2,
0,
0,
-5.966837375471956],
[ 12,
6,
0,
0,
-4.6298501952375855],
[ 20,
10,
0,
0,
-3.5807206753108622],
[ 28,
14,
0,
0,
-2.750174370397261]],
'sampling': 'default']]],
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [ -2.397634216504411e-17,
1.429450133825379e-17,
0.9192297980258759,
0.9192297980258759,
2.8984599305110854,
3.1993874679766523,
3.199387467976652,
3.715279848679994,
3.715919998165357,
5.126008182976459,
5.749875525815113,
5.785995839176617,
5.785995839176617,
5.785995839176617,
5.785995839176617,
6.315437962379428,
6.678845841951143,
7.3578167224936415,
7.415694801781823,
7.531263738212295,
7.949087697575632,
7.949087697575632,
8.111169110068786,
9.407094598577077,
9.495286031274873,
9.495286031274873,
10.993257661103009,
11.063439993331317,
11.639888600691357,
11.75175692668691,
2.1152038588685538,
-0.7839988468841306,
-1.0579861702584952,
-0.5122509123294313,
1.0012369417099964,
0.9275037440003397,
0.5343037326701707,

```


-0.014459274586663672,
2.31692048955396,
2.1400689678453872,
-0.9468421716760329,
1.3225342455296385,
2.4229762260600483,
1.4561600175543838,
-0.4592924007287594,
0.6389204073366491,
0.5578650700802136,
1.458345504678432,
-0.11931303728638508,
0.04406428844113783,
-0.26144087074192934,
0.6459890964472403,
1.2243061606997268,
0.09770103868863204,
0.23645607828472368,
0.3280111145811677,
0.5390648705936556,
-0.6034481934191781,
-1.1942560416831098,
0.3295059316644399,
-0.0732996553139216,
0.5043071925129845,
0.7399437462813884,
0.3595686242698361,
1.4643835231854074,
-0.1697776944282527,
0.2588380732750666,
0.44311592273109757,
-0.6031987466514527,
-0.06131410155451245,
-0.3299395710828798,
-0.45492250592824107,
0.5555861976799495,
1.1612177689175078,
1.833550163228831,
1.5629483318007886,
-0.2455585474033218,
-0.4534768786824923,
-1.2225865843307935,
0.20468879325769473,
1.3398561861341207,
-0.58709602862573,
0.07181500566168061,
0.17010271339345,
0.36406988692952136,
-1.689809967880289,
-1.1677426672173294,
-0.7977758879892671,
0.531032719362766,
-1.7722607960867016,
0.6332033998224597,
0.6669814362912183,
0.218182728116887,
1.0415656581640258,
-1.4544893086294475,
0.5175174925258392,
-0.9742907090362619,
-0.30648595673224416,
1.2400541074634985,
-0.30003161306087445,
-0.14406503395842435,
-0.43061285539013316,
0.6914697219419143,
-0.48545379734819916,
-0.2166375004086177,
1.2014709434123243,
1.0734635496959384,
0.4366289070672293,
-0.9550440807047608,
1.2817732673505913,
0.6254203972471953,
0.8730111716749767,

```
1.1814458389257443,
-0.5855214911540807,
0.33647224367417283,
0.914891547403245,
-0.10924028322724566,
0.6731472714644756,
-0.8672547313943412,
-1.1628903969680444,
1.2968110405185924,
-0.7478342770444015,
1.0157432273471354,
0.05866260951598884,
0.1549884083832883,
2.402954000164725,
-0.8510405490988947,
-0.13394876321797194,
0.35128800256256376,
0.03985349181110424,
1.0891408951797659,
0.49683980443602116,
1.3618410332836905,
0.31089249400158553,
0.007823999182329802,
-1.0325531315281817,
-0.13635517978577905,
-0.4783006843261583,
-0.9648987725589863,
-0.4094326055257113,
0.0620151507481985,
-0.3631850012398663,
-0.9119123598789223,
-0.3458539889181142,
0.8501459643039054,
1.6461875633189593,
0.08799287143090098,
-0.82318299513447,
1.17289644669955,
-0.9455811832961388]],
'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value': 0},
                    'noise_prob': [0.001, 0.001, 0, 0],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_ric4_unifrom_scaling_sample#4'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
```

[illegible]

```
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [ 0.16477895719846114,
0.16477895719846108,
0.7183189156962347,
1.8697093861419474,
2.105082615571597,
3.05308344502201,
4.199393321043376,
4.199393321043376,
4.199393321043375,
6.490210948425593,
6.4902109484255925,
6.490210948425593,
6.4902109484255925,
6.490210948425593,
6.882554174728105,
6.882554174728105,
6.882554174728104,
6.8877583035732775,
6.8877583035732775,
6.8877583035732775,
8.21014824639056,
8.21014824639056,
9.071469420716642,
9.120928605341893,
9.714944098401723,
11.474131735950095,
11.474131735950095,
11.794551582273902,
1.341188628911811,
1.6554894539990426,
-2.1068974621953425,
-1.4171475675653578,
0.33211669069513333,
0.11331792925356984,
-1.3610911099963097,
0.46439741779020816,
0.9313501387866817,
1.0938593094543279,
-0.3591727738608774,
-0.4732622270970327,
1.6321598874818157,
-0.2782091348514778,
-1.0526781349295347,
0.0708703831522549,
2.6982701224663113,
-0.038867313655444914,
-0.10508583152652556,
-0.40759365792756236,
0.6284849004330764,
0.5192751929971827,
-1.6282040637266806,
0.0014422433007245106,
-1.8299719264591412,
2.3182597876656823,
0.8875167319646391,
1.5664268415036142,
1.5052778791259855,
2.935955436800565,
1.533101581959507,
1.8091829332960188,
0.45080966818917967,
0.1506883627647002,
-0.6843548866459184,
-0.5544883854492364,
1.2466562002431278,
-1.1866110136199353,
```

1.4521358231995756,
-1.125608676420736,
-0.24054383618816966,
-0.058272145415006006,
0.410408128597049,
-1.6441753460137252,
0.4019816715685168,
0.4588365987304068,
1.354346078964423,
-1.7153251360741875,
-1.1039978646265602e-06,
0.7940670488104785,
0.17979772520560053,
-0.24385394325110826,
0.6367160741723578,
0.8500459957371875,
0.0559416907316065,
0.32418668587130467,
-1.4797636873635434,
0.11902177942387202,
0.5185165439317966,
-0.08399222686641401,
0.8650918037299421,
1.4911587670088011,
0.12792248933645675,
0.3372262149750348,
-0.6553154954729696,
1.1995792367396254,
0.5132294030998703,
0.44835115860783636,
-0.3711698051277586,
0.7805194563374558,
-0.854334789795377,
0.5008527672922033,
-0.2302685629030402,
0.6145182391921147,
-0.6575500079215796,
0.8657035637529482,
1.1874516653159721,
-0.16284802367068918,
-0.26462171656665856,
-0.9815102056659819,
0.23198397744948207,
-1.087097195512072,
0.6616385654966559,
-0.2929637257774683,
0.03471390505932934,
0.2048751570049573,
0.9170017118070749,
0.39617956766989404,
0.12006199846995962,
0.1922705913940127,
0.8784478594386566,
0.055479957240883566,
-0.6634105431569992,
1.0742951419164057,
-0.37818200958777604,
0.28548758229245597,
-0.6594827056613419,
1.0235832498557866,
-0.19428253177041227,
-0.5531991468121416,
-1.189626464718852,
0.2519110119226656,
-0.6708462316627377,
2.7994352657547057,
-3.8764436680175813,
0.5152327138619494,
-0.3749347648847944,
2.9042167722097862,
1.1265965037549552,
-1.850959186134046,
2.165374611027145,
0.8818414791702859,
1.5039524911129019,

False,

```
0.06564767423285973,
0.6486215314149069,
-0.08854143781823305,
1.8396223528529987,
-0.00792246728000446,
2.8805879469479487,
3.6337353861266473]],
'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value': 0},
                  'noise_prob': [0.001, 0.001, 0, 0],
                  'status': True,
                  'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [ 0.5,
                                         0.5,
                                         0.5,
                                         0.5,
                                         0.5,
                                         0.5],
                                'r': 1},
               'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_ric4_unifrom_scaling_sample#9'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [ 4,
                           2,
                           0,
                           0,
                           -5.966171999362981],
                        [ 12,
                           6,
                           0,
                           0,
                           -4.491721403204036],
                        [ 20,
                           10,
                           0,
                           0,
                           -3.3646330706171934],
                        [ 28,
                           14,
                           0,
                           0,
                           -2.4951156781776946]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.50611145019531e-05,
            'zne_values': { 'degree': 0,
                            'extrapolated_value': -6.861717934852231,
                            'others': { 'beta_coefficients': [
2.1875,
-2.1875,
1.3125,
```

-0.3125],

11.390625,

-5.966171999362981,

-4.491721403204036,

-3.3646330706171934,

-2.4951156781776946],

'cost_richardson_zne':

'sorted_expectation_vals': [

'sorted_noise': [6,
18,
30,
42]],

'sampled data': [[4,
2,
0,
0,
-5.966171999362981],
[12,
6,
0,
0,
-4.491721403204036],
[20,
10,
0,
0,
-3.3646330706171934],
[28,
14,
0,
0,
-2.4951156781776946]],

'sampling': 'default']]],

{ 'config': { 'ansatz': { 'gateset': 1,

'layer': 30,

'ugate': { 'coefficients': { 'bn': [0,

0,

0,

0,

0,

0],

'cn': [0.5,

0.5,

0.5,

0.5,

0.5],

'r': 0},

'time': {'max': 10.0, 'min': 0.0},

'type': 'heisenberg']]],

'init_param': { 'value': [0.38848003251483587,

0.3884800325148359,

1.2841735835398218,

1.284173583539822,

1.2841735835398218,

1.818138220668234,

1.818138220668234,

2.7575469841005327,

2.7575469841005327,

2.7575469841005327,

3.0053468968599275,

3.0053468968599275,

3.005346896859927,

4.184079661691832,

4.83635165164968,

4.83635165164968,

5.97926077140412,

6.458805266455043,

6.513102391504654,

6.513102391504654,

7.122105903846713,

7.2188422331450415,

7.988590448127284,

8.268095077127937,

8.268095077127937,
9.164456488648115,
9.164456488648115,
9.782953992889732,
9.782953992889732,
10.467486029517685,
1.5496959718296535,
-1.0300437645832343,
0.425343365508412,
-1.7136674729251273,
0.33645197441382346,
0.2288357961575219,
0.10470230685228102,
-0.6612661793744108,
0.11823315872480306,
0.48571962162230753,
-1.3748501535311368,
0.2563120795974374,
-0.38871520150353267,
-0.18786175076151226,
1.0017996241286973,
0.05598903629217323,
1.501328526357405,
-0.32299727271195017,
0.5715295079657807,
-0.12201961934486968,
2.2747823054657927,
0.16289086097520028,
0.027619518366160244,
-0.10165606616500937,
-0.06321979816820887,
-0.27543819802246033,
-0.0048370064103128475,
0.1330138167350915,
-0.06428151371988705,
0.21905201010909556,
0.25697175483980955,
-1.462905966720411,
0.8014511304369155,
3.456137165485204,
-0.3560002683832885,
-0.4886877133354182,
0.32039145066707037,
1.6153352350920127,
0.5940099487759896,
0.18140087044059483,
0.27123106498617,
0.7444341922951362,
0.170480557625751,
-0.4688462519246396,
0.5235839184929864,
-0.5470964469389679,
-0.011921500492307424,
1.4116230981753706,
0.49972453659747124,
-0.7800673920851233,
0.4234227404504273,
-0.2071635265929523,
0.14064328933496192,
-1.2007096247825992,
0.12047098237640114,
-0.4125103841815656,
1.0447620953563077,
0.0020814149928196773,
0.4988991062112368,
-0.34495773827123155,
-0.12849063482979717,
0.4280934542561602,
0.7138444629946918,
-0.6497712710787925,
-0.43652659674625793,
0.917432486791049,
-0.8523477361785018,
0.5648987974980545,
0.3183226113969618,


```

0.14295784675778261,
0.9589824340171335,
0.9194478619356266,
1.1001359353915339,
1.5417377267389083,
-0.5376003771957187,
-0.5404528215706331,
0.21624687748210675,
0.8814937279455825,
-0.371639532103812,
-0.2668100917972658,
1.3133163210428074,
0.44574090697479807,
-1.0940261852319737,
-0.313830643067749,
-0.06100275437545863,
-0.049560783686653447,
-0.3025056724120473,
-0.9891529613817422,
-0.35099332176432035,
0.5436069706516342,
-0.722828417704955,
1.6968661444846564,
-2.4402971962208975,
0.7865745542588627,
1.3437881169309889,
0.36108471326554326,
-0.1749159211919777,
-0.014334278530919132,
0.02978377629472008,
1.8533978555002608,
-1.8470426695738211,
0.6272356375919823,
0.44775499409400255,
1.3281153539688288,
0.4583217606753929,
0.6443670392327268,
0.5254233581973354,
2.3087652110195127,
-0.7761319091913014,
-0.7605421959233766,
-0.6461580464596497,
0.7075597766411752,
0.985778508379752,
-1.3130974173457381,
2.8711331564400684,
1.2993543284582694,
1.9493614761217355,
1.1647493000584068,
-0.42731613935458074,
2.7038346000669766]],
'noise_profile': { 'noise_on_init_param': { 'status':
'value': 0},
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,

```

False,

```

'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_ric4_unifrom_scaling_sample#6'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0],
[3, 3, 0, 0]]},

'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},

'zne': { 'data_points': [ [ 4,
2,
0,
0,
-6.144691518305653],
[ 12,
6,
0,
0,
-4.667338994079639],
[ 20,
10,
0,
0,
-3.522600808002957],
[ 28,
14,
0,
0,
-2.628229437680454]],

'degree': 0,
'method': 'richardson',
'sampling': 'default'}}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 4.38690185546875e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -7.033800507973144,
'others': { 'beta_coeffieicients': [
2.1875,
-2.1875,
1.3125,
-0.3125],

'cost_richardson_zne':
'sorted_expectation_vals': [
-6.144691518305653,
-4.667338994079639,
-3.522600808002957,
-2.628229437680454],

'sorted_noise': [ 6,
18,
30,
42]],

'sampled data': [ [ 4,
2,
0,
0,
-6.144691518305653],
[ 12,
6,
0,
0,
-4.667338994079639],
[ 20,
10,
0,
0,
-3.522600808002957],
[ 28,
14,
0,
0,

```

```
-2.628229437680454]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                                'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'heisenberg'}}},
  'init_param': { 'value': [ 0.4022850038273885,
                              0.4222288083135141,
                              0.7644599371513524,
                              0.7900179023917507,
                              1.1026508993249646,
                              1.1630326759467198,
                              1.1630326759467198,
                              1.2119340062926192,
                              1.9859325565890973,
                              1.985932556589097,
                              2.096166401632718,
                              3.378647998605865,
                              3.378647998605865,
                              3.378647998605865,
                              4.354848115861854,
                              6.006218342726225,
                              6.006218342726225,
                              6.009974297352436,
                              8.225327865566033,
                              8.242093148909708,
                              9.222354355246733,
                              9.369468664000962,
                              9.432712219232048,
                              9.692289963799956,
                              9.692829942577658,
                              10.391360826688166,
                              10.391360826688166,
                              10.529867035412478,
                              10.529867035412478,
                              11.06290493838992,
                              2.9062307172979733,
                              -1.7788694936155562,
                              -0.1145266982224492,
                              -0.9311559923392093,
                              0.9691998209476543,
                              1.7463479137973288,
                              -0.7663862807329105,
                              -0.9694574896037749,
                              2.8979112386638413,
                              1.1400097878698292,
                              0.8096361549155084,
                              -1.5900494197653285,
                              0.09829667109053167,
                              0.31839493135320834,
                              0.3886777840539921,
                              0.12229184789371997,
                              1.7250225844898464,
                              1.0065795676456808,
                              1.4535641205083867,
                              -0.3486770669120958,
                              0.2556790893233589,
                              1.778531141883236,
                              0.269972269049403,
                              1.0155903957099945,
                              0.8041163886037216,
```

0.5381821926966456,
-0.30108604163317454,
0.6396533041678052,
2.1819991440037696,
1.4978086658278347,
2.5181040929106753,
0.4624415671193703,
-0.17701736987873673,
0.6910098122382194,
2.947172461882987,
-1.0727539032453073,
0.4023854197196741,
1.650490750838525,
0.9313998963798695,
-0.6769162055305248,
-0.5620785927454173,
-0.20885673940707247,
2.5019757090972443,
-0.37561416876024456,
-0.9329392032282731,
0.7818961506889994,
0.44350434198669075,
-0.10644929099379355,
-2.498928032568615,
-1.2046085786256906,
-0.06365416567297402,
0.17257236645054272,
1.2516209341031546,
-0.3591527663232026,
0.46789807326225935,
0.24374638092155315,
0.02110383811013195,
0.12714835746845662,
-0.5505736947218767,
-0.03497727012222505,
-2.2168334497478877,
-0.21485512988314215,
1.7552902914988067,
0.6189795903733311,
0.3191491062237487,
0.522632432692797,
-1.1121089236913335,
1.1371618962620422,
-0.2410043783817164,
0.24168543661014927,
0.39418565996934685,
1.2107667809264395,
2.398781467533486,
0.6948257569073976,
0.6170229142231725,
0.7050419245924102,
1.768546262611762,
1.5562956304120967,
-0.4269237214365518,
0.6302835616447369,
1.7710848704009405,
1.2281100264507194,
1.5036530549577243,
-0.7435268012971183,
0.7613541379199313,
0.8523298100708424,
-0.6655959646469325,
1.1543911681084875,
2.2675628634479494,
1.0299985170060308,
1.400963430022522,
0.27759304330136975,
0.6960561589746208,
-0.4210335016880339,
0.0336463669485127,
-2.099660094290342,
0.6674819238895001,
0.5049668435179431,
0.48001405045948287,
1.3574871122533543,

False,

```

-2.2986993170736225,
9.601426007351498e-05,
1.3312470798272018,
0.7582238004326416,
-0.35033967277151296,
-0.4108552112493075,
0.1982910129710702,
0.6700017270602091,
0.26559360983179486,
-0.14563210230012705,
-0.39610135234418875,
1.1727729846793147,
0.6713573564879365,
0.8527070052929181,
0.8268283343075853,
2.2918575835381745,
0.5082103460913493,
1.8600493489550152,
0.49576811175353974,
2.232387985311215}},
'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value': 0},
                  'noise_prob': [0.001, 0.001, 0, 0],
                  'status': True,
                  'type': 'depolarizing'}},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
sample#8}},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                      [1, 1, 0, 0],
                                      [2, 2, 0, 0],
                                      [3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': False}},
'zne': { 'data_points': [ [ 4,
                           2,
                           0,
                           0,
                           -5.743389283741309],
                          [ 12,
                           6,
                           0,
                           0,
                           -4.3560016590478785],
                          [ 20,
                           10,
                           0,
                           0,
                           -3.2776401801937793],
                          [ 28,
                           14,
                           0,
                           0,
                           -2.2986993170736225],
                          [ 36,
                           18,
                           0,
                           0,
                           -1.601426007351498e-05],
                          [ 44,
                           22,
                           0,
                           0,
                           1.3312470798272018],
                          [ 52,
                           26,
                           0,
                           0,
                           0.7582238004326416],
                          [ 60,
                           30,
                           0,
                           0,
                           -0.35033967277151296],
                          [ 68,
                           34,
                           0,
                           0,
                           -0.4108552112493075],
                          [ 76,
                           38,
                           0,
                           0,
                           0.1982910129710702],
                          [ 84,
                           42,
                           0,
                           0,
                           0.6700017270602091],
                          [ 92,
                           46,
                           0,
                           0,
                           0.26559360983179486],
                          [ 100,
                           50,
                           0,
                           0,
                           -0.14563210230012705],
                          [ 108,
                           54,
                           0,
                           0,
                           -0.39610135234418875],
                          [ 116,
                           58,
                           0,
                           0,
                           1.1727729846793147],
                          [ 124,
                           62,
                           0,
                           0,
                           0.6713573564879365],
                          [ 132,
                           66,
                           0,
                           0,
                           0.8527070052929181],
                          [ 140,
                           70,
                           0,
                           0,
                           0.8268283343075853],
                          [ 148,
                           74,
                           0,
                           0,
                           2.2918575835381745],
                          [ 156,
                           78,
                           0,
                           0,
                           0.5082103460913493],
                          [ 164,
                           82,
                           0,
                           0,
                           1.8600493489550152],
                          [ 172,
                           86,
                           0,
                           0,
                           0.49576811175353974],
                          [ 180,
                           90,
                           0,
                           0,
                           2.232387985311215]
                        ]
    }
}

```

```

0,
-2.43192951997479]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 3.528594970703125e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -6.576835190529094,
'others': { 'beta_coeffieicients': [
2.1875,
-2.1875,
1.3125,
-0.3125],
'cost_richardson_zne':
'sorted_expectation_vals': [
-5.743389283741309,
-4.3560016590478785,
-3.2776401801937793,
-2.43192951997479],
'sorted_noise': [ 6,
18,
30,
42]],
'sampled data': [ [ 4,
2,
0,
0,
-5.743389283741309],
[ 12,
6,
0,
0,
-4.3560016590478785],
[ 20,
10,
0,
0,
-3.2776401801937793],
[ 28,
14,
0,
0,
-2.43192951997479]],
'sampling': 'default'}}}],
'ising-ric2': { 'noiseoff': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': {'value': 'random'},
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': False,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': True, 'type': 'png'},

```

```

        'file_name_prefix': 'ising_noise_off'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'vqe',
'state': 'dmatrix',
'vqe': { 'iteration': 10,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': True}},
'zne': { 'data_points': None,
        'degree': 1,
        'method': 'richardson',
        'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 '
        '+ 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + '
        '1 X 5 X 6 + 1 Z 5 + 1 Z 6',
        'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '
        '2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '
        '2 + 1 X 3 X 4 + 1 Z 3 + 1 X '
        '4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
        '1 Z 5 + 1 Z 6']},
'output': { 'exact_sol': -8.56677223350563,
        'initial_cost_history': [ 3.7901552515724086,
        3.396779462430669,
        3.6416626483897634,
        5.191980344851134,
        3.300383519902229,
        4.2302694452267025,
        5.5832854157035126,
        3.424340408479732,
        4.575860085354091,
        3.843028199719871],
        'noise_details': { 'gates_num': [4, 1, 1, 0],
        'identity_factors': [0, 0, 0, 0],
        'noise_level': [0, 0, 0, 0],
        'noise_profile': { 'noise_on_init_param': { 'status':
        'value':
        'noise_prob': [ 0.001,
        0.001,
        0.001,
        0.001],
        'status': False,
        'type': 'depolarizing'},
        'odd_wires': 4},
'optimized_minimum_cost': [ -7.3566708947935595,
        -7.585318923493526,
        -7.222021821934782,
        -7.637732542535524,
        -6.9717347656162385,
        -7.183899566070664,
        -7.312238847468353,
        -7.55546648538883,
        -6.871387778407254,
        -7.103750879845002],
'optimized_parameters': [ [ -8.981486547411966e-18,
        0.2355269100006643,
        0.6734351680867104,
        0.7281872519006076,
        0.8287057436356319,
        0.9968962697957876,
        0.9968962697957877,
        0.9968962697957875,
        0.9968962697957875,
        1.6732799868313266,
        1.9939389913716405,
        1.9939389913716405,
        1.9939389913716405,
        2.7786795690350794,
        2.7786795690350794,
        4.834741825646661,
        4.843194997300159,
        5.442664851556481,
        6.006041306704694,
        6.006041306704694,

```

False,

0},

6.42159854018659,
6.42159854018659,
7.172079495244572,
7.835006027334925,
7.835006027334925,
7.835006027334926,
9.27390458565848,
10.142272229921797,
10.547148172870333,
11.945646083493756,
0.30631338507375855,
0.950240081652421,
0.3197394260348401,
-0.8989315334324052,
-0.19972246213391193,
1.1515465399564957,
0.5794269610995502,
0.029234461946744308,
-0.3033609133811924,
-0.5068802243159589,
0.41891242044729254,
1.2272962791271989,
0.3217023759538914,
-0.0389595358377685,
-0.2818159573379153,
0.40344301100056285,
0.012601476380185236,
-0.7249227034698249,
-0.28859136632269766,
0.9177183017829931,
0.3192674751581855,
-0.6960323093564068,
0.9954617266198262,
-0.0960673339612117,
0.30357952531427623,
0.9148253966896076,
-0.41157308092318357,
-0.2244725515886058,
0.900815469140735,
-0.8704821270800919,
-0.2201886465573435,
-0.058451834208670006,
-0.5564504264806815,
1.0028418241392123,
0.6862825176439941,
-0.07515432929364027,
-0.132716959754057,
0.3156193912732197,
-0.5571752224261073,
-1.1266280252389607,
-0.2496012410462972,
0.4568813812573679,
0.5040642955504137,
-0.27382094178370653,
0.738802930675346,
-2.1017467172452315,
1.7134969759675782,
0.19045289169065632,
0.4959480509883087,
0.2725041426894662,
-0.020113534287112555,
-0.27730132318570966,
0.9121222342974794,
0.7771919661145833,
1.2410984938270941,
0.698373852916785,
0.049079898076347495,
0.4715079446630061,
-0.0012225173391828686,
1.2380103358912495,
0.18126256323161893,
0.02822806167971892,
0.9624172506978782,
-0.1017646649934255,
0.10712069411267353,

0.11059700546555196,
0.1764728469149672,
1.0536625920020957,
-0.43614672067179516,
1.1290800077402758,
-0.7544321351970764,
0.20178203991689547,
0.545960403958859,
-1.271331788368515,
0.3827893670869708,
0.514944994226915,
0.9987612988518954,
0.9130737051848579,
-0.33354580533033523,
0.4110572637290035,
-1.0065548546569214,
0.05504586363332423,
-0.8123464439408392,
0.9622209015425294,
-1.7981345279534349,
0.8489846299822246,
1.9191971402543575,
-2.019266402915899,
-1.6151954323496909,
1.693158829823371,
0.8368996699819713,
-0.7925788637806501,
-1.416264150106769,
-0.7484853165625206,
-1.026066230288144,
0.2591646216069995,
0.9485984853376032,
1.624644962520762,
0.1389264160191751,
1.7901079295930897,
0.28173803525757823,
-0.06727915670909099,
-0.39998196135024877,
-0.911581975644718,
-0.8802717995009537,
-0.6496644186002655,
0.9474433579476823,
0.5810351318438188,
-0.5928504020318089,
0.40429180463166087,
0.19955746243713188,
0.14940534343654552,
1.148493268597809,
-0.6171972430901704,
2.358530954611547,
1.6132804954395525,
1.5230819806307347,
0.07100911239249047,
-0.9821849785093056,
-1.2186503325370892],
[0.43242608326371745,
0.47036005705701023,
0.6035896599876003,
0.6406575109863006,
1.589410377648682,
1.589410377648682,
1.589410377648682,
1.9902163938334159,
1.9902163938334156,
2.6096176446013035,
2.6096176446013035,
3.039961339509649,
3.039961339509649,
3.0408788822093955,
3.0408788822093955,
4.1325863285768,
4.1325863285768,
4.801738885893826,
5.558430388585627,
5.558430388585628,

6.48703427557745,
6.599293033589678,
7.119130597501129,
7.119130597501129,
8.096879981584715,
8.096879981584715,
11.989689272176165,
12.289334556064002,
12.289334556064002,
12.695171119049377,
-2.1320303145314323,
1.164574641976517,
0.2281426394568994,
0.5098216286601104,
-0.15959745874570574,
-0.06246168319786304,
-1.335183444898994,
1.1253611654826523,
-2.453373641849758,
1.542670605477509,
-0.23459513075014657,
0.7629240608803527,
0.7066862387679962,
0.023354672976759586,
-0.26545157517882206,
0.8070755399487665,
0.621829996842247,
-0.8206657495436546,
-0.6456938685697818,
-0.8432536462697657,
1.5481421634749726,
0.030458967225015475,
-0.6362611752642527,
-0.5779049484598997,
0.15136987691256515,
0.37940285865801,
-0.13025289482332936,
-0.6088352960517447,
1.0137338691570092,
0.9312563664781934,
-0.06260332826127091,
0.16422266217724538,
-0.44919022316429513,
0.7120810093352475,
-0.045811140982536094,
1.4693096124875311,
1.3068674578470139,
0.3099696428239748,
0.05726716998373588,
1.4138231816748696,
1.3251150202531472,
2.4661338582136705,
-1.9221360663147242,
1.6309773912120944,
1.012572768899168,
1.1304323717308846,
-0.10014704960087417,
1.2138621646932324,
0.9150507542177275,
-0.1931179153053463,
-0.19983886142442753,
-0.37113288193216537,
1.0783926703777422,
0.7626802083907843,
-0.5396621733108105,
-1.0558614970998554,
-0.5031372556139323,
-0.1733591918410679,
-0.05810492443974475,
0.15252588011535415,
-1.3613790038949474,
0.14268023162896296,
2.1247307661901087,
-0.2996908036461345,
0.2682896527620792,

-0.6487468384363545,
0.0174900962257196,
0.7865320417875226,
-0.4365770411878935,
0.12280152026315115,
0.2668485276831852,
1.0055493052017264,
-0.13783144983938653,
0.16428177903351826,
-0.4107244428852575,
-0.5006989958309392,
0.4728442136126457,
-0.5756228808846314,
-0.2555605201147205,
-0.6655897533222209,
0.27354627255859315,
1.056546145946755,
-1.0108463744632321,
0.8570822622523233,
1.1003650202540056,
-0.13364446072464353,
0.19686136380997882,
-2.015333107317771,
-0.3474605408709381,
0.5127772990877489,
0.8873132714146144,
-0.1538651896386833,
-0.49691154598038334,
2.4337764506921564,
0.11687121421819974,
0.2637773919076603,
1.1119719051108947,
-1.1407171454187541,
0.293218856784229,
0.9089141681578352,
1.0587297353590328,
0.41354403585027527,
0.26331197174432663,
0.04763876798960274,
2.8376965303594455,
0.038400591301352385,
0.732378529151854,
1.259237861002353,
-0.09371603030550141,
-2.6524545055620123,
0.166545686266708,
0.8383948517861464,
0.5625425863873136,
0.6352389159281269,
-0.3793199251622615,
1.4353166704457385,
-0.06088962240819872,
0.5575840323453971,
-0.4370866693930651,
-1.3477496560670819],
[0.5485274765250757,
0.6924497584354614,
0.77926902254116,
1.4968846113254817,
1.4968846113254817,
1.617988193716464,
1.617988193716464,
2.3157329555892914,
2.3157329555892914,
2.3157329555892914,
2.6599448622122046,
2.6599448622122046,
3.198693955827293,
3.198693955827293,
4.93646112072991,
5.0183990860356085,
5.166530398168251,
5.732062846348599,
6.452081956191476,
6.950653712152026,

6.950653712152026,
7.527182443059662,
8.141470761550243,
8.141470761550243,
8.23161566953153,
8.23161566953153,
8.263835107953264,
9.693553204732913,
9.696078071259171,
10.987802073160276,
0.31236348113638823,
-1.370738548013981,
-0.19640846640845944,
-1.409184053738275,
-0.00366718677866519,
0.6361615892126344,
-0.6796219270984741,
-1.7803162192131878,
-1.2406782697025376,
-0.17521451360574944,
-0.2890470362486656,
-0.7870591534070972,
0.6214527622340972,
-0.5055171901576345,
0.3099897895999387,
0.6495941239175653,
0.4137491249725721,
0.292781258130194,
-0.18467758047440086,
-0.2764099099100952,
-0.12795113266111025,
0.14328500307003156,
0.1623766240720685,
0.09884205682735484,
0.07093196196575538,
0.17766773916349385,
0.1810420204025737,
-0.037056767883675096,
-0.38325428434024456,
-1.808138895166064,
-1.2174086587283277,
0.2723541238647832,
0.0911451526301191,
0.829454165789877,
-0.7076888645983359,
-1.1582847892183987,
-0.2789235155040628,
-0.1868914176693055,
-0.03618003850886602,
1.0797789643516258,
0.010710633626313776,
0.3041103423483441,
0.027606917069623423,
0.09715752300035431,
-0.025081467189021912,
0.7307804473426243,
-0.028312324696826655,
-0.5439567135695698,
1.0353579775860826,
0.03274794094556986,
0.338229999884877,
0.003923288912346083,
0.011059712392702849,
-0.24419721784886275,
0.0538500196620971,
-0.29704802197132957,
-0.9714959737951582,
-1.3776571886055275,
-0.4387141762272093,
-0.9013896619503009,
-0.2890811474761574,
-0.5664350037938545,
-1.265940570830118,
-0.36500389888052437,
-0.11634811830012508,

-1.0092353619257808,
-0.6849489189933503,
0.9856942680495377,
0.8277230147066852,
0.2903619348614201,
-0.3836733140597432,
-0.530059482018533,
0.8169168177018395,
0.5323656053240163,
0.649672850550592,
-0.3032256260984974,
-0.8829469814645045,
-0.00635052379210004,
0.35535568318583244,
-0.06185634463022188,
0.10533559478429602,
1.2902435643868964,
0.8825397316379378,
0.3070411861797082,
-0.07934132673084492,
0.09844318572655962,
0.7695889603351249,
0.14399374014816757,
1.8827194490661865,
0.013305728022345127,
1.2201617371808122,
0.13458781619262966,
-0.42760236848753097,
1.3132997483594182,
0.29586095969137777,
0.14782133458354627,
1.1539512755729704,
0.8255769419022573,
2.1284509830181357,
1.5067425074661367,
1.2286171121451426,
0.8625402575232176,
0.5254367589291086,
0.8683165111711829,
0.35804094605055825,
-1.3678290318224675,
-0.3952095755717416,
2.8373763512718595,
-1.2381307915304474,
0.21836664096244152,
-1.0373469190584237,
0.35616351546904684,
-0.48865262743063154,
0.5395297676370602,
-0.0068350558769426075,
-1.6240423184716037,
-1.005087252746991,
-0.324331708942255,
1.56936745287968,
1.3955283874562228],
[0.3026146462879133,
0.3869510743764029,
0.3869510743764029,
0.5382374643481795,
0.7459322218289088,
0.7459322218289088,
0.8083924716224152,
0.8083924716224152,
3.219627607910341,
3.2253302931763046,
3.9244184219369176,
4.114742178023509,
4.115220647252454,
4.115220647252454,
4.115220647252454,
4.65196408394345,
5.196968879903224,
5.736027504566688,
5.736027504566688,
6.321573696596394,

6.7009287553652,
6.7009287553652,
6.7009287553652,
8.383675014305652,
8.383675014305652,
8.383675014305652,
8.383675014305652,
9.235844817182018,
9.235844817182018,
12.415877726713473,
1.4636821166030158,
0.8436356038076634,
-0.6467230542935267,
0.2472607630354758,
0.028721536942984512,
0.5720112007049558,
-0.179388461505878,
0.6605644899444229,
-0.2997484495809327,
-0.5848737649454969,
0.7567373384637915,
-0.3113371232415861,
1.5925608798994126,
0.15551153490893768,
1.144256524486279,
-1.0086304067563407,
0.4692085043130235,
-0.5289416707772752,
0.6895537558918402,
-0.9023920029674053,
0.6687255575647186,
1.3626463195064957,
0.18435277462186048,
0.04101567185284027,
1.2334404589145347,
-0.8099334611041354,
0.47217953762833137,
-0.19021445340255347,
-0.47126223762254077,
0.21535783789089086,
-0.13160146544027274,
-0.14745399432080403,
-0.2131109368071074,
0.6116235033000932,
0.7187755412423116,
-0.7690585507561546,
0.333613273647484,
0.29688326543860977,
0.022443242058892903,
-0.12904022561026712,
1.1137507228270036,
-0.1754869613868649,
0.6791619259450926,
0.2688019369707634,
-0.33561997076117317,
-0.4910450017308718,
-0.2921449581476723,
0.07017675650466139,
-0.5628878746231101,
0.23633646017547374,
-0.4408658192483561,
0.6337746650828207,
-0.5275874765915898,
-0.11397090424872018,
0.7713694979315344,
-0.24077591679338844,
0.7998056369050096,
0.5449521212196076,
0.6440774810386566,
0.0012540722532346088,
0.4406697804838639,
-0.4697708168210915,
-0.38183296272166767,
0.4926783027265995,
0.012102728676941394,

0.9995268367701109,
-0.489600116502774,
-0.9378926015671718,
0.7538025600192173,
-0.9270581698564858,
0.23395173940139535,
-1.5047568644302813,
-0.6335501933328295,
-0.1851947866597326,
-0.18416301345971506,
-0.15010118329037675,
-0.30085163208099847,
-0.030658108221894752,
0.48141025496380413,
1.2807653372993688,
-0.27688172784497056,
-0.9404124429777033,
0.07772158149530461,
1.2206520244647234,
0.07167829739929571,
1.5514296753518346,
-0.22739832257997444,
-0.5457307695824859,
-0.05085143495318925,
1.4086504148457157,
-0.02595768387677813,
1.2057536466860386,
-0.5009834455732999,
1.0234750156328314,
-0.34572346527257924,
1.4975050133157242,
0.6424858734227697,
-0.5412025600486546,
-0.965195784498948,
0.8078930733387817,
-0.8747920106317195,
1.6444412493640082,
-1.3120143788008718,
0.7469227938848356,
0.7428309989793587,
0.25043937013438866,
-0.7176977899841391,
1.6961228018637329,
0.8286720852002186,
-0.08923208660690117,
-0.4564959840968765,
-0.10389957398275913,
-0.04652285532782134,
0.6308845069083873,
-0.025008046054888575,
-1.0109619026173717,
0.31698532529812806,
-0.07816069204265776,
2.517427548138029,
1.300984357739734],
[0.48696254731677435,
0.9433356258260287,
1.8515151481415904,
3.298386332934612,
3.298386332934612,
3.298386332934612,
3.298386332934612,
3.298386332934612,
3.374346159341203,
3.374346159341203,
5.043620376468771,
5.043620376468771,
5.218159502227583,
5.683079438253876,
5.683079438253876,
5.902572186502105,
5.902572186502105,
5.902572186502105,
6.17408053545979,
6.17408053545979,

6.17408053545979,
6.17408053545979,
8.600010985810668,
8.600010985810668,
9.424738936566825,
9.424738936566825,
9.43568443936613,
10.707654072137746,
10.707654072137748,
11.243950101093004,
0.30560526927017156,
-0.6023681078531374,
-0.5052271452160477,
-1.0944465846035194,
0.24881605409968105,
1.5229622975948298,
1.730509331816328,
-1.3834281548913137,
-0.8373914729231547,
-1.186133470726964,
-0.983750614650609,
1.3432121337567746,
1.6569377678026054,
0.14131844528439758,
2.9464329846729154,
1.3448953312948555,
0.7896925151532842,
-0.22765810177877452,
-1.6238177025568765,
1.1679097532601286,
1.5429785141845107,
0.44641536771583223,
-0.4990561654458825,
0.2890079097559007,
0.07664205286689205,
1.0463374337388847,
1.0582154531020835,
-0.7032713068941808,
-0.5914102053027229,
0.4481794192989112,
-0.597054072340342,
0.6503285491619032,
-1.3548677382206493,
0.3073247134537545,
-0.42900928099090013,
0.8290869769130189,
0.25661934620723303,
-0.41883276922189405,
-0.25409113110415216,
-0.2750324278847333,
1.8535327289298686,
0.7375047938415388,
0.21982546706442144,
0.3355829615333414,
-0.18885187042138774,
-0.09263125664317409,
0.8768150768205386,
-0.31105067773622297,
0.11882768143806702,
0.007542537022557154,
0.6689687664434811,
0.7431113887615772,
0.9447406593038509,
1.0052341467989638,
1.4439497926806117,
0.4202121214316228,
0.3604666875525164,
-1.6888556500060994,
0.1396130584965534,
0.10094819946612499,
0.5021137054548022,
0.3961649599525077,
0.4932381662816783,
1.3418440833635417,
0.09854079710503719,

-0.9989986964474209,
1.2672439763832217,
-0.07601477770236317,
0.46767000685442095,
0.3227070865546575,
-0.7175096141500144,
0.2758047915288214,
-0.24095833900456404,
0.058134638501602325,
0.6051082814423787,
0.5442716652871094,
0.3025208071236723,
-0.6679934596266708,
0.8499265434495678,
0.1803663678751026,
-0.3707988225819567,
0.6812577092957876,
1.2284831501706344,
0.20883808756632122,
0.03358446867125293,
0.026609877934757803,
-0.40919242519291843,
-0.07640273708110212,
-1.1643751408115468,
-0.16769301556045021,
-1.2911947285322207,
0.037223344859154565,
-0.05786091979914895,
-1.8047938425602224,
0.014158596153457605,
0.534864509903787,
0.006329420481561668,
1.239578876475735,
0.2639865904342132,
1.2160651224901187,
-0.2035728984920286,
0.6511657116444571,
0.3064364057804899,
2.45196454382623,
0.8340823063941794,
1.113505687918428,
0.049528306503774544,
0.12947314602533921,
-0.5399313265794791,
-1.2647403981540062,
-0.3482737106911781,
0.09697429466359891,
2.20804968215061,
-0.15489516431237316,
0.1250815357650574,
-0.13157603031624435,
-1.2324441762654297,
-0.4179909399666764,
0.3779753490492196,
1.6472121643813018],
[-1.1990978117137884e-16,
0.0939706539344674,
0.583570497583306,
0.583570497583306,
1.7474308983833136,
1.7474308983833136,
2.435631059267761,
2.4356310592677604,
2.435631059267761,
2.4356310592677612,
2.9757043839921162,
2.975704383992116,
4.1141402268542535,
4.1141402268542535,
4.889622687428616,
5.08553065152385,
5.14502423979873,
6.38608549233117,
6.405094454380333,
6.807050165246778,

6.807050165246778,
6.997225211250308,
7.82493752524197,
8.419597935808826,
8.419597935808826,
8.756502393683652,
9.514795580339912,
10.006329694888725,
10.008671195739254,
10.582332645019955,
-0.18325409339400697,
-1.8356335907661927,
-0.6014425812689169,
1.272854521157504,
0.38561099010389943,
-1.3354103698578268,
1.2089446007576305,
-0.16684037696101606,
1.5272415494549387,
-0.21803857489548387,
0.659293576456549,
-0.7629850988015884,
-2.0880787600158244,
-0.2200967935369742,
-0.21257917133409554,
0.05335328813343252,
0.740881120097457,
0.002552459203101907,
2.0381278489789163,
0.07009602349011958,
-0.027486349535234322,
-0.5162309983105858,
-0.018107734929441965,
0.3459554204227654,
1.1337361200247116,
0.6158218713410994,
-0.5251391279578093,
0.2758552924607382,
1.227089028823027,
-0.16405595633516176,
0.7587337472599636,
-1.1046949665522132,
0.7157795020690486,
-0.9666557350608269,
0.5941166908158014,
-0.3035426485157296,
-1.1713314559961463,
-0.3285214016177699,
0.15201600725182188,
1.158582769628082,
-0.2335509332665805,
0.021624313631101504,
-0.7981749327134954,
0.12400359670343174,
-0.00041124409742626503,
0.10665347356443583,
-0.030437347790731137,
-0.22360998336164295,
0.21313944649575595,
0.08788738041696632,
-1.2878790640853577,
-0.028446563531964782,
-0.12898316604526575,
-0.29469285393680966,
-0.07114900510733681,
-0.8104482788171727,
0.6394923422746198,
0.6811407094584414,
0.15127654478155184,
0.5799864823630293,
-0.55116978803857,
-0.5805840448115108,
0.9630930231089359,
0.9222833564508055,
-0.28733988453131243,

2.645322235680672,
0.02138581028992107,
-0.12421624460779133,
-1.4608772192659922,
-1.289711141721096,
-1.433469567857053,
-0.01041716118532159,
0.3377052351749707,
-1.74263853738129,
-0.6445874089376945,
0.6790378089998735,
0.0019732068694448615,
0.6260241949545162,
-1.3812938119718974,
0.06110738334511842,
0.919936405723882,
-2.761538990972181,
0.01773377957551518,
0.6041489324828,
-0.281847147959406,
0.36931104439326196,
0.14008962906529285,
1.0402771661903438,
0.40062198057764514,
-0.03775531840468896,
-0.4974133017905023,
0.5298490799692606,
-1.7184458207495306,
-0.7561470763062167,
-0.5888363413103725,
-0.9669502040570058,
1.3911426640670486,
-1.922807929753264,
-0.7347523142785206,
-0.6976214468215837,
-0.9022103009000106,
-0.17554401086360705,
-0.7004215075729389,
-1.1648825394224225,
-1.1285064327861984,
1.3794410327712907,
0.7524041737227146,
-0.600926776988818,
0.724704963452131,
0.24574404734520428,
1.3856850730269805,
0.505698953761027,
-0.030535393994814766,
-0.4836980977841504,
0.6166564120021005,
2.4235804988343936,
0.6827837636480912,
0.3673788389460804,
0.11748797515103059,
1.4924513209248702],
[1.112287766890805e-16,
0.11579095179987547,
1.7588846075757274,
1.7588846075757274,
1.998030831219005,
3.2454648629288743,
3.248106797240174,
3.248106797240174,
4.259596531772135,
4.958175754196259,
5.7210454174100525,
5.861260496697951,
5.861260496697951,
6.7588095170100075,
6.7588095170100075,
6.758809517010007,
7.044198424034494,
7.044198424034494,
7.044198424034494,
7.044198424034494,

8.232813771743604,
8.791432218349115,
8.791432218349115,
8.791432218349115,
9.616347292478379,
9.616347292478379,
9.869492613451865,
9.869492613451865,
10.182429226961672,
13.25758469112181,
0.5705572906646224,
0.09014038635403764,
0.1263845657016112,
1.0203985977202104,
-1.095649826264738,
0.019398597884167954,
-1.3544747652103108,
0.8866931148929111,
-0.18940184666069662,
1.0130589658671383,
0.15424354259331335,
-0.8117851912876255,
0.28413415623416743,
1.542699745821886,
1.836477396464815,
0.7160099188075301,
-0.9589348876176362,
0.00854407549207699,
0.36750908199249616,
0.7288091138524155,
0.9748756553152056,
0.7257579234395075,
-1.9128288506616224,
0.515489770530508,
-0.8485176532957428,
0.21376115901922352,
1.0834633235702664,
0.404913552038143,
-0.12863840971929708,
0.8333794222571905,
2.496813054711946,
-0.7354302042700512,
0.27199375272502646,
-1.0107781002238316,
-1.201703662816619,
-0.3331724830549245,
1.3030593505928814,
-0.8138384777541797,
1.9586180892259586,
1.1713603302631173,
1.0819642334098143,
-1.1784039537689197,
-0.12146178641002908,
-0.1717950232658355,
0.13833462928205204,
-0.8697724612180706,
0.7515471411202548,
1.166125613326729,
0.37941686042382794,
0.17074885851048915,
-0.33600856179263133,
-0.13669943521352546,
0.548834657399268,
-0.2522233718783481,
0.009665778262181637,
-0.5020120590858954,
0.16617168999240312,
-0.46390045473242114,
-0.05573261329441784,
1.1182956755615927,
0.0937037366596967,
-0.4441962997721035,
-0.21050297984767188,
-1.0381458298882043,
0.830180513233935,

-0.8157309834659043,
-0.5833742339287631,
-1.5354704640903527,
0.44493118131447734,
0.13057382050598434,
-0.7034976447329113,
0.6218444459919024,
-0.2420386731361673,
-0.7867967091164098,
-0.168935029173583,
-1.6928247400206122,
0.22571922404505704,
0.1126763335131705,
-0.22160264498444046,
2.299575126838168,
2.5415010654218975,
-0.5449604787415219,
-1.005432959888493,
-1.722931012111005,
-0.6732620373630841,
-0.0004890899721480893,
1.492586488782192,
-0.9055437852306896,
1.1298576359465664,
-0.5182549439372126,
0.6586606791613716,
0.7929632788713356,
0.08462141435268096,
0.5066414628201175,
0.6503902510816232,
0.06456835392989443,
-1.687283870544211,
-0.2816102525113523,
0.9601910446201565,
-1.0364808528540121,
0.32603884641661884,
-0.37671408634672476,
-0.16068551007585027,
0.6157148912247653,
2.0105220675930333,
1.7580796652062145,
1.7229426093703843,
-0.5199627188331776,
-1.3760925920420213,
0.8290526991464299,
-0.08765267414283408,
-0.456659399307068,
1.4313668535261046,
0.635052489001378,
2.136142869643981,
-1.5755126698917075,
-0.40116032394931217,
-0.004321541181387126,
3.301925406547275,
0.7673504276272846],
[0.49826187507447023,
0.6110785201427625,
0.6558173388435118,
1.2433731466916595,
1.2468055416987345,
1.9464512542038723,
2.109950686395802,
3.7879039161756585,
3.7879039161756585,
3.7879039161756585,
3.7879039161756585,
3.7879039161756585,
3.7879039161756585,
5.054414830549875,
5.106961698737131,
5.488040579082762,
5.488040579082762,
6.832391146999917,
6.967864562238176,
6.967864562238176,

7.490508971727458,
8.20118277218113,
8.612503180124596,
9.093659709060372,
9.807153774862888,
10.372385504551621,
10.372385504551621,
11.272736556325881,
11.767303580595307,
11.767303580595307,
0.035679258194077725,
-0.8723307875716305,
-1.7560663493087938,
1.9629136553479551,
1.9120807210302333,
-0.39357388790453784,
0.036182026687196715,
-0.21816401975370772,
0.8090968256508778,
0.22775227657680963,
-0.9402388060640036,
1.9199330583190146,
-0.6293921442826633,
-0.37521158500092616,
-2.4122776248196542,
0.47614923103710605,
-0.2356719275662886,
0.5930119757607211,
-0.08067436637158562,
-0.6975308853470512,
-0.1533962892581602,
0.6838091266999453,
0.243129377869582,
0.5896632240559074,
0.12048284254834898,
1.186903891075025,
0.22948374273850122,
-0.9592798973456383,
1.431786126792839,
1.099938502501145,
-0.6421065764259075,
-0.4570128400176702,
-0.40361390122591434,
-0.030699975686000303,
0.5709596867047622,
0.8129177753256275,
0.0978347154858261,
-0.1791413540365364,
1.316347643480991,
1.0669242344323897,
0.9542637314281797,
0.2732710750226029,
-0.24925074976969727,
0.14351675204102743,
-2.058043419369877,
-0.5772495977232713,
0.4759355346590471,
-0.3040209890415461,
0.07360159687611288,
0.11769956200453384,
0.01709491725825118,
1.1325236656251834,
0.008658928128351846,
-0.21852721169551936,
-0.1826844911726131,
0.39535377862993026,
0.035729500082236576,
-1.0747644579938396,
-0.15992271631488378,
0.8234408989300128,
2.679077704708449,
0.2559692131167249,
1.5592174030452497,
0.3453465662341248,
-0.24046357809578378,

-0.6876498012434461,
-0.01750342347144341,
0.34704200676536884,
0.2033679229118891,
-1.5920113668255755,
-0.5283783493172445,
0.4351355192977228,
-0.8281455342425702,
0.2974457926897622,
-1.336557258227178,
0.2885982325597283,
-1.2111847441179915,
0.698629224522434,
-1.072563817644129,
-1.2092878769637225,
-0.33153669598628793,
-0.2558676939050336,
-0.039300685015777354,
0.5730206646846686,
0.6772024906875076,
-0.5431857921175166,
-0.32242263836861185,
-1.2435651530921474,
0.6447338068594848,
-1.1796614873773261,
-0.5517178991847818,
-0.9950127711245446,
-0.750107619545704,
1.546476758712725,
0.7798072495873917,
-1.3969546244577882,
-1.3869625997614359,
-0.5499269608373658,
-2.0698482195340895,
-0.31472418351029113,
1.3858906356684872,
-2.3234389152233144,
0.8580050890595776,
0.5574876161662472,
0.12663545081840322,
0.0024286165622580055,
1.4351555562364873,
-0.018100010671135307,
-1.4466701603237728,
1.0091335554802272,
-0.25739419705547106,
0.8584486110461004,
-1.6353613496727522,
1.3481381905015994,
1.664215570319441,
-1.4688501218242656,
0.6630999772073086,
-1.3598408860414706,
-0.9086993193406631,
-0.7720609621308895],
[0.03916643192220903,
0.09260151508853934,
0.5125042140553085,
0.5342494495137248,
1.4070055183169814,
1.4076098722130033,
1.4080234932826805,
1.9886432369288551,
1.988643236970097,
2.4501760969485846,
2.4501760969572235,
2.6141080657383133,
2.614108094671839,
2.6141080978874367,
2.614108199851818,
2.614108200075538,
2.6141082008692407,
4.121207830152765,
4.126135109264967,
4.126137501750553,

4.129417665319773,
4.731890053629371,
4.734221022017094,
4.783748744413943,
6.888546551790504,
7.435120796142169,
9.27338967383107,
9.884598996480708,
9.884598996488108,
10.57466535326793,
0.042050619766987055,
1.2200437115279086,
-0.1897539908832087,
0.3883222993215419,
-0.3953259161886337,
0.9433481261689909,
0.03976732857871137,
0.380475950476648,
-1.6480061036631655,
0.19200671798130073,
1.8964236385178486,
0.5825466621902844,
-0.7017933974726621,
0.46167134168781887,
1.3455780978346443,
1.000341055668334,
0.3220859397336833,
0.4901021103591929,
1.7507755726531005,
0.22815940378303187,
-0.27415827045663116,
-0.06669089502017891,
-1.2681856141960832,
-1.4186851222947996,
-0.7484212831710693,
-0.2624358161393499,
0.22137315279362701,
0.8266616228870926,
-0.28382763940078026,
0.4526868734709094,
-0.03876262869445461,
0.028933904781200354,
0.2124235174903725,
0.5275601102845284,
-0.035224229477880745,
-0.11459385773709553,
0.4537223625267385,
1.3135706165796082,
-0.4103066311241145,
-0.06740865621875465,
0.07865262902016133,
-0.8220672300227196,
0.05131914396597353,
-0.012779296277522511,
1.6650491211728184,
-0.5271258879235212,
-0.2286028431320702,
0.9810985313826684,
0.5796827949136435,
-1.4216581678953852,
0.908254540613963,
0.7809397666182989,
0.23060365236949015,
-1.629683736567043,
0.08687379015275534,
0.5302421766128769,
0.858319872120405,
-1.2816124200556078,
0.08255098737682609,
0.49190330494839835,
1.2241314669476433,
-1.6043734869556,
0.35480117328822147,
-1.3692690448046365,
0.08801115679730828,

-0.3416815622800032,
0.5349891528238044,
0.23599371094677032,
-0.2955967206364773,
0.7351110566185237,
-0.1358629473726594,
-0.2538927978152746,
-0.18822579896800806,
0.023625415865148318,
-0.47104955898400697,
0.3192033261234258,
1.3512870657881308,
-0.7816632422470443,
-1.6469850516046918,
0.27638573599623373,
-0.3090269715855155,
-0.4193062343229584,
-0.1377874030219536,
0.0004776579492072504,
0.08325659247733753,
1.4611032338266408,
0.021431340914286272,
0.20066912699708425,
0.29328305426482204,
2.42997753887945,
-0.5690256391604135,
-1.480840559800266,
-0.1363946771767577,
-1.376439435407693,
0.06387998538825271,
-1.110029094439266,
0.9868538484965683,
0.962243917255456,
-0.41126656322102295,
-0.657103189069447,
1.1434637843768656,
-0.9757973897576084,
1.448315222432508,
1.3110742046722172,
2.3333185683193554,
0.7520157773640208,
-0.20203562372486106,
0.34414613752805073,
0.5926366260275047,
-1.985044802604821,
2.6078832019778186,
-1.6883872397325923,
-1.6784323905215754,
-0.4297482144956373,
0.6572781395405978,
0.02277885553642172,
1.3256214503718338,
0.25673693454286806,
-0.6358308725577859,
-1.6133180674191239],
[0.4366123913610663,
0.4768198412180804,
0.9843286193666714,
1.2674678431689148,
1.5775702096396949,
1.5775702096396946,
2.2591837513222837,
2.2591837513222837,
3.168921311818864,
3.1689213118188637,
3.1689213118188637,
3.2207076420855234,
3.220707642085524,
4.00024951310582,
4.479169433031019,
5.572620903211533,
5.572620903211533,
7.169574827830268,
7.200887753316444,
7.200887753316444,

7.4320670533634186,
7.4320670533634186,
7.668889448658714,
7.668889448658714,
8.088052955990618,
8.127936129548912,
8.127936129548912,
8.129700499301142,
8.129700499301142,
13.234708916500516,
-0.8409114197195183,
0.13809956722013594,
-1.3698910408430074,
-0.1271619514190957,
-1.0028759735951214,
-0.7519751568518348,
-1.7027753599694198,
-0.5003000098057334,
1.7803062126367422,
-0.6956057702973542,
-0.5292841478808504,
1.3371604427134443,
-0.6271315116651272,
-0.9849660401714492,
-1.0245042646602094,
1.7746289077120558,
0.789646250683077,
2.3274439120303834,
1.0250145144124314,
0.236963759763461,
0.6449585291401012,
0.5302895792637682,
-0.17332551626046228,
0.13296008629005646,
1.3001365941190501,
-0.5491163203391332,
0.8939708861040915,
-0.1183728413911535,
-0.27672563627574315,
1.7953720191393387,
-0.00825054985875676,
-0.6170579146731886,
1.801545679753067,
-0.6786704865717594,
-0.5702663839537956,
-1.2235657195316856,
0.059502931568887765,
0.15944146767128192,
-0.5445588526685949,
1.0645821939979259,
-0.3246614360002631,
1.6342475668463907,
-0.6365693512803821,
-0.25644806680599874,
0.9153791587574333,
-1.5555971818784813,
1.7626038696684259,
0.29250830199243494,
0.4597874603079029,
-0.4983420660721544,
0.22206047990995867,
0.11089125969329257,
1.9712857917750568,
-0.8482820514299444,
1.7711968287204813,
-0.3356021753581158,
-2.239225908466879,
-0.021402454584719684,
1.6641712424719513,
0.5059308187330879,
1.9296669762106724,
0.7155610478862724,
1.024294278133781,
0.2881031759975837,
0.06318491357276658,

```

0.8308741396333285,
0.16800005476886462,
-0.26845892946467115,
-0.378371667085078,
-2.374474400887392,
-0.26943703763693594,
-0.698442208248677,
1.5029972614102114,
0.7682274655447995,
1.7790823677975025,
1.1233449399645796,
-0.1962677980677998,
-0.6013220679983903,
1.243366811759978,
0.23851493156650586,
-1.3516830763738037,
0.4546075088056473,
0.45461165179145885,
0.29299766155391044,
0.9768506591689141,
-1.3614782254940123,
-0.1864841683590143,
0.03417328589702677,
-0.12392810093414776,
1.4042342901901068,
0.4181387945528863,
0.6171624447156228,
0.5887687464980093,
0.07995045192475739,
0.12406497411608935,
-0.1304256749231809,
0.5172871404602656,
1.1235454838537668,
1.5976273357881063,
-0.7195312648953682,
-0.7697726746437508,
-0.1718403740134208,
0.48757574872507614,
0.7291603816887677,
1.3754811895843486,
0.06384938170591244,
0.9009847125572167,
1.0744001081352503,
1.7305303874736466,
0.491454536253661,
0.11643540469065636,
0.592545584033324,
-1.2877724397991297,
0.052924560657905566,
-0.8760423919600488,
0.6313576901954856,
2.736167065037608,
0.029295634649204143,
2.1044233481865446,
0.6660786050345296]],
'run_time_sec': 43335.76831364632}}],
'redundant': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}}},
'init_param': { 'value': [ 0.27654223573266234,
0.27654223573266246,
0.27654223573266246,
0.4643738922966213,
0.6582164862517121,
0.6582164862517121,
0.6582164862517121,

```

1.3424523496852623,
1.47635793598756,
1.47635793598756,
3.385550316280557,
3.385550316280557,
3.385550316280557,
4.494062371253785,
4.494062371253785,
5.0204758683966135,
5.023404188660082,
7.05704094573064,
7.087391629397132,
7.533802135057517,
8.245970909352765,
8.390887842007308,
8.390887842007308,
8.703391612007538,
8.703391612007538,
9.485545381639698,
9.485545381639698,
9.849437794456351,
9.849437794456351,
10.799155842585746,
-0.12708089143278917,
1.4168148977025445,
-0.13519536748631492,
-0.32308714516775877,
0.06981336221044306,
-0.034321895006426455,
-0.18209443759399158,
-0.024858958594746107,
0.664488174834645,
0.6897317665852637,
-0.008774731192519907,
0.625642551337688,
0.6862410821653743,
-0.7670419540542446,
-0.48055390132433673,
1.573964677292243,
-1.0555745677813198,
1.3159721165631335,
-0.9138502497993347,
-0.4641566084568084,
-0.06966564000847691,
0.039012746519005555,
1.0827641697579717,
1.2190093772283843,
1.5391150382377632,
-0.6675581491933664,
0.09104662559232415,
0.5283818781579022,
0.04478118244404365,
1.5836075485651113,
0.251932560991488,
-0.3949560381811925,
0.8854639388207172,
-0.8611267164776638,
1.566928747241295,
-0.15445151319573347,
-0.01396050433941564,
0.8172620115356718,
-0.014658132241289147,
-0.6623010385304499,
0.6299284572753584,
-0.15026947869759505,
-1.2293112897827836,
-0.5768656731979434,
1.231557922062332,
-1.447872926116422,
1.4918722706131171,
0.31945486783385635,
0.43870804207253483,
1.3582636256308331,
-0.8478398442626237,
0.799518029891469,

```

0.5539369896048497,
0.13453197415145846,
0.7167498104862234,
-0.06589109804263343,
0.041038724923082276,
1.151975756926885,
-0.052026838729968425,
0.3952978000808587,
-1.1754748970457833,
-0.01965607042871015,
-0.2456324144680547,
-0.01136184217607491,
0.022503168314325715,
1.1777923723104653,
0.14554117253938412,
-1.2287946167385297,
-0.9683737383161766,
1.073329622962756,
-0.7143583802799215,
-0.3545002682598469,
-0.31906110571926777,
-0.9597094503848127,
0.8825903987559673,
-0.08273415521500459,
0.2895039435279609,
-0.3388594197258312,
0.2406459434933063,
0.3824128671502841,
-0.5390555829405563,
0.3757684495816124,
-0.5015719636333678,
0.3958257688275076,
-0.6428354095046468,
-0.9178624992467067,
0.29327306512581747,
-0.09193285724698198,
0.1770091195095364,
0.6803523512851801,
1.05525770275555,
-0.4646437845139076,
2.2560908642344026,
-0.7500939262299234,
1.92959092052175,
-0.36895777993274226,
-0.04538283416771957,
1.2529586558275392,
-0.57805717638909,
0.3080585810325818,
0.031910210614104206,
0.18519950010488512,
1.3888566879174211,
1.087260914864616,
0.15259355623633358,
-0.08810167996863522,
0.07467874103808375,
1.4444893382870363,
-1.446220680213735,
0.8215180856176293,
2.0151723649306295,
0.13454991965456886,
0.21923381008732426,
0.31112145477192793,
-0.8977876743954992,
0.10239876571985529,
-1.0132922278363503,
-0.3794863035801566,
0.8437803973608401,
1.111671691788496]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0, 0],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,

```

```

1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric2_sample#2'},
'redundant': { 'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 '
'+ 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + '
'1 X 5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 + 1 Z 0 + 1 X 1 '
'X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 '
'+ 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -6.51493088468928],
[12, 6, 0, 0, -5.310706408751286]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 0, 0],
'noise_profile': { 'noise_on_init_param': {
'status': False,
'value':
0},
'noise_prob': [ 0.001,
0.001,
0,
0],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': { 'noise_on_init_param': {
'status': False,
'value':
0},
'noise_prob': [ 0.001,
0.001,
0,
0],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4}],
'run_time_sec': 4.555899143218994}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': { 'max': 10.0, 'min': 0.0},
'type': 'ising'},
'init_param': { 'value': [ 0.48023040681030116,
0.7029544315470944,

```

0.722671987246733,
0.9307929858945541,
0.9307929858945543,
2.546598088688794,
2.546598088688794,
2.546598088688794,
2.546598088688794,
5.184334435330857,
5.195111458618343,
5.47436411399969,
5.47436411399969,
5.47436411399969,
5.47436411399969,
5.47436411399969,
6.068336244524409,
6.068336244524409,
7.838605251840715,
7.912250316107233,
7.912250316107233,
8.562517202814183,
9.063385768597811,
9.216218862905647,
9.216218862905647,
10.84047371877396,
10.84047371877396,
11.044358714867027,
11.375406566932424,
11.375406566932424,
0.18886549713610382,
0.6040291443391601,
-0.9727809979878402,
3.008209947509642,
0.3239743149147096,
-0.643987619924633,
1.0739882269552894,
0.4935103262912805,
-1.1713673989643745,
0.32339055167680386,
-0.8254170399860232,
3.387204293956005,
0.5437302598099015,
-0.6585961721148387,
1.0903163929414719,
-1.5261906384818182,
1.102349928722959,
-0.7373436929054379,
0.1712635196093208,
1.2086697581514199,
-0.4977298668106944,
-0.8828019932256734,
-0.13051782426094366,
0.048744185631251574,
-1.0535030780910155,
0.5224555306604155,
-0.9502111367985572,
-0.2573420205933925,
1.5833039752620888,
1.3806480812363842,
0.3864833677006273,
-0.0006284730712113316,
0.6487740408134013,
1.697829570896293,
-0.20023867528920775,
-0.32438662287273723,
1.3674639807988553,
-1.5592347723565667,
-0.6294582959326964,
1.111157394071659,
1.4895912958038178,
-0.7203901402661329,
-0.027076706863032294,
0.009574786781832467,
1.1437695218822916,
0.11720143179507149,
1.0973549625545624,

```
-1.0756168303605034,  
-1.5785125289756954,  
-0.42999007233668873,  
0.7592603373612552,  
0.7357293999316703,  
0.03766014972468404,  
1.450035910180884,  
-0.08978031589044846,  
0.7176267391661779,  
-2.3377280976646966,  
0.3878354316208249,  
-0.8070070584177742,  
0.07084117333557359,  
-0.008998168433091903,  
0.6612796195456235,  
0.6287929626789847,  
0.8601333603228948,  
-0.7786354708155643,  
-0.144476679155352,  
0.39492950957300044,  
-0.20012968159779548,  
0.01326107241550192,  
-0.5649340596325207,  
-0.020398755490040426,  
0.6333499066300691,  
0.400844220606507,  
0.5954749034740784,  
0.8700413681534388,  
1.315868035063538,  
0.9914334712755634,  
0.10418309057983809,  
1.1095991500751576,  
-0.1712743534684691,  
-0.2730581386141925,  
0.1453142903679217,  
-0.3813955795194703,  
0.1280814941289577,  
-0.5627986213807415,  
-1.0576722394029623,  
-1.98721560251358,  
0.519862592148374,  
1.03695228961559,  
0.21101375663462807,  
-1.750252459751414,  
-1.1450993785481438,  
-2.2517541986316916,  
-1.5456964825715331,  
0.6026638989139771,  
0.47561448130168205,  
0.08201672609434518,  
0.47464262797792917,  
-0.14585857152212733,  
0.08636040211648545,  
0.7388192170680543,  
1.071366965639021,  
2.5115163876343742,  
-0.4286872636197954,  
-0.2977998181526895,  
0.46569070003689333,  
0.2115019986130443,  
-0.5123638542481742,  
0.09024611293676764,  
-0.3380729423866059,  
1.6314769884628497,  
1.3338027138849353,  
-0.7442166070564612,  
1.2186722356710948,  
1.584709484399307,  
-0.6554175978902271,  
0.36887213174672207,  
-1.365021965092438,  
-0.4623126041921459,  
0.4869420326570461]],  
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},  
  'noise_prob': [0.001, 0.001, 0, 0],
```



```

        'status': True,
        'type': 'depolarizing'},
    'nqubits': 7,
    'observable': { 'coefficients': { 'bn': [ 1.0,
        1.0,
        1.0,
        1.0,
        1.0,
        1.0,
        1.0],
        'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
        'r': 1},
        'def': 'ising'},
    'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
        'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric2_sample#9'},
    'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
    'run': 'redundant',
    'state': 'dmatrix',
    'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
            'constraint': True,
            'status': False}},
    'zne': { 'data_points': None,
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
    'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 +
        + 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 +
        1 X 5 X 6 + 1 Z 5 + 1 Z 6',
        'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 + 1 Z 0 + 1 X 1 X
            X 2 + 1 Z 1 + 1 X 2 X 3 +
            1 Z 2 + 1 X 3 X 4 + 1 Z 3 +
            + 1 X 4 X 5 + 1 Z 4 + 1 X
            5 X 6 + 1 Z 5 + 1 Z 6']],
    'output': { 'data_points': [ [4, 2, 0, 0, -6.751241519687581],
        [12, 6, 0, 0, -5.575778108730956]],
    'noise_details': [ { 'gates_num': [4, 1, 1, 0],
        'identity_factors': [0, 0, 0, 0],
        'noise_level': [4, 2, 0, 0],
        'noise_profile': { 'noise_on_init_param': {
            'value':
                'noise_prob': [ 0.001,
                    0.001,
                    0,
                    0],
            'status': True,
            'type': 'depolarizing'},
        'odd_wires': 4},
        { 'gates_num': [12, 3, 1, 0],
            'identity_factors': [1, 1, 0, 0],
            'noise_level': [12, 6, 0, 0],
            'noise_profile': { 'noise_on_init_param': {
                'value':
                    'noise_prob': [ 0.001,
                        0.001,
                        0,
                        0],
                    'status': True,
                    'type': 'depolarizing'},
                'odd_wires': 4}],
        'run_time_sec': 3.5373356342315674}},
    { 'config': { 'ansatz': { 'gateset': 1,
        'layer': 30,
        'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
            'cn': [ 0.5,
                0.5,
                0.5,
                0.5,
                0.5,
                0.5],

```

```
        'r': 0},
    'time': {'max': 10.0, 'min': 0.0},
    'type': 'ising'}},
'init_param': { 'value': [ 0.5118290521669792,
    0.9034573591840084,
    1.0514460879099101,
    1.1283814725964603,
    1.1293980629167455,
    1.6874068541059999,
    1.8362001710030256,
    2.594529848453243,
    2.5949023258749775,
    4.384420392533986,
    4.804388533064279,
    4.8044974726668155,
    4.925049921883437,
    4.925049921883437,
    6.055903002618935,
    6.0650337145794,
    6.06510024924717,
    6.06510024924717,
    7.719643893437192,
    7.719643893437192,
    8.110116534451196,
    8.374448879129652,
    8.374448879129652,
    9.29045698473019,
    9.2929923765699,
    9.977025927736127,
    9.977025927736127,
    9.977438303479875,
    11.277250876863835,
    11.588901828478656,
    -0.5970483257864091,
    1.048395642841534,
    0.3563299433581316,
    2.04182497417867,
    -1.0659189653240388,
    -1.5711224255679066,
    -0.6854026923334804,
    1.3002799990574343,
    0.334979527312172,
    -0.41771661057992626,
    -0.5255960701407614,
    0.9200835209290303,
    0.5840294212204432,
    0.4309612994747514,
    0.010788324288593443,
    0.8398650983618882,
    0.35555521736822654,
    0.33383672576306345,
    -0.03972798241435421,
    -0.5808926346218187,
    -0.6217506829607715,
    -0.022449055124049946,
    -0.4671428723737863,
    1.3840583406465645,
    3.321823997583478,
    1.9516123562000514,
    2.109430068020724,
    0.17138900458361278,
    1.0734765554988877,
    1.1591682463434823,
    -1.5085197494005436,
    0.3709514368460093,
    -0.08354812013321816,
    -0.6849414455501801,
    -1.1807064172516142,
    0.1889544406654655,
    0.45125852432694835,
    0.40946867023093825,
    0.9591749175330944,
    0.862406962512403,
    -0.47565451021668054,
    0.14791469386649075,
```

0.012323642821740255,
-0.3613715829994197,
-0.6520104899254531,
1.1500641632479522,
0.11480411675389499,
0.37929728325133344,
-0.12818017639377846,
1.5548403554359806,
-0.08803696907532751,
0.16058007713541472,
-0.2835300528775952,
1.4317946638041426,
0.04218231832431045,
0.6164913379423986,
-0.5546192578057587,
-0.0016176435191898148,
0.3455446109244303,
-0.7983090776945799,
0.06548494602750325,
0.09601011255154575,
-1.274882751616874,
-0.6639296860891427,
0.08468574728059233,
0.09178155492125434,
-1.9746588496307496,
1.363083624655819,
-0.11754172722968637,
-1.7454296260872237,
0.689437867548643,
0.09536601432398076,
2.1875690275532618,
1.9737367040865754,
0.19623670188051764,
1.3618525198389628,
-1.6551377680800423,
-1.2796187335858575,
-0.02979150349567955,
0.3655877448114123,
-1.339933806945279,
-1.6131280858420662,
1.1268021774842836,
0.7853899025443938,
-1.214954346638235,
1.881374469676006,
-2.2108375055617597,
0.38299461206746305,
-0.062378913993248956,
-1.0812916442950207,
0.19113842494911862,
0.43947065709017896,
1.4340329690981122,
-2.1914706359927933,
1.683520172253517,
-0.4151753501170658,
0.2652280013518245,
-0.09829123134965576,
-0.4033858255054015,
-0.09488290407421365,
1.1087820722198931,
1.3758250784849964,
0.3693308876448478,
-1.4774754586810708,
0.45790576106810227,
-1.318961759734687,
1.2898775251023542,
0.5291693857421421,
-0.7225860054677886,
-0.6024865845145562,
1.1258609582335735,
1.4545810101615688,
0.4988359655120556,
0.39968483637691726,
1.4778005154845164,
1.6358502222921723,
0.07081811420225437,

```

1.5336807499569445,
-0.7595143926860392,
0.07947795439670366]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric2_sample#10'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 '
'+ 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + '
'1 X 5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 + 1 Z 0 + 1 X 1 '
'X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 '
'+ 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -6.406033238196167],
[12, 6, 0, 0, -5.147683680770249]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 0, 0],
'noise_profile': { 'noise_on_init_param': {
'status': False,
'value':
0},
'noise_prob': [ 0.001,
0.001,
0,
0],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': { 'noise_on_init_param': {
'status': False,
'value':
0},
'noise_prob': [ 0.001,
0.001,
0,
0],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4}],
'run_time_sec': 4.950466632843018}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,

```

```
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [ 0.23818326834520384,
0.242015507589488,
0.5292100600091615,
0.5598196968290768,
0.5598196968290768,
2.3451854381830244,
2.3451854381830244,
3.8453025861748578,
3.8453025861748578,
4.395137112369032,
4.408833101975425,
4.829585475162733,
4.867115107997544,
5.058958687371625,
5.058958687371625,
5.058958687371625,
5.581243202757385,
6.011433944644334,
6.011433944644334,
8.358003932454157,
8.358003932454157,
8.358003932454157,
8.371420896509267,
8.783744346022278,
8.783744346022278,
9.34189587380724,
9.34189587380724,
9.39608576624051,
9.726727464435275,
10.95102910414735,
0.14721560345391302,
1.0205767628001141,
0.40564926087548847,
-1.9238321605823567,
-0.04729505851188324,
0.482541600628026,
0.11521556777775581,
0.8524320040134185,
0.42380767990264745,
-0.6219665889019169,
-0.05759638705357143,
0.8161328242190486,
0.7425402472775383,
0.7535392998290764,
1.8314745216576223,
1.3334023135030668,
0.05552193989453376,
0.9691374289368472,
-0.07338372949421543,
-0.6690664310785921,
1.168533104165511,
-0.01962836639606316,
0.6576122461884292,
-0.037637936325233226,
-0.0009647519828692927,
0.4524839512418326,
-0.013115244288784337,
1.6270372625341938,
1.182284078721198,
0.9533845712158122,
1.0956662941001023,
-0.9647918240366662,
0.1897778948773928,
-0.04047520420354409,
0.5790278460715794,
0.631719469583885,
-0.3335418357294817,
```

1.0603186443819097,
-0.7449471003423949,
0.1630231005006795,
-0.07563007210422526,
-1.1009042311153838,
-0.34587951137957407,
-0.22408475871037167,
-0.6364303490666092,
-0.5245531829065874,
-0.7807390048523051,
-0.7767575864285188,
-0.9679768325508518,
-1.8691843526791756,
0.4783530568164554,
0.6133546874260105,
-0.20020986269221935,
-0.41532071148562183,
-0.8557118152516873,
-0.16985308585729458,
0.8428338023785568,
0.6750667051632169,
1.2795031602880573,
-0.41384224674171627,
0.3213362409234008,
1.2340309605381181,
1.0506369357289735,
-0.6415500741723684,
-0.9628453188140567,
1.935214357876566,
1.4159154638103797,
-2.0864443113151867,
-0.07426020994129089,
-1.0538315394110076,
0.2589745455351295,
-0.2955978932252484,
-0.030644528099012087,
-0.8602288471584197,
-0.09490606960216369,
0.4190622694749842,
-1.5667390101922392,
-0.7804920738383083,
0.1255849265457167,
0.5884338319364165,
-0.46028089859616994,
0.2488277677353687,
-0.7513271615815256,
-1.0561258734551267,
0.3697237180501531,
-0.7108432439071602,
0.5137418203820399,
0.946767478727941,
1.619957961978311,
1.2944079622561346,
0.21625880379379211,
0.007168703139473973,
-0.334764182643075,
-0.9086121674985949,
-0.8247776105092918,
-0.5734460393428641,
-0.4381826298821608,
-0.44306191141826123,
-0.23860269421390362,
-0.03343496826073402,
-0.008972211649351379,
1.2196121588758198,
1.5985131140815898,
0.060325450296581484,
-0.7312434635304047,
0.9413542695101463,
-0.6701271691421761,
-1.0680925542186157,
-0.5192506372311365,
1.5594784335685845,
0.17928043514789443,
0.2455246384495121,

```

-0.053106843359071614,
0.7488220810664211,
1.3030207698426615,
-1.787387332504871,
-1.1184519607509558,
-0.2511275398302805,
1.445676213762956,
1.5991220174606833]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric2_sample#4'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 '
'+ 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + '
'1 X 5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 + 1 Z 0 + 1 X 1 '
'X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 '
'+ 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -6.630764461698827],
[12, 6, 0, 0, -5.461578440310041]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 0, 0],
'noise_profile': { 'noise_on_init_param': {
'status': False,
'value':
0},
'noise_prob': [ 0.001,
0.001,
0,
0],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': { 'noise_on_init_param': {
'status': False,
'value':
0},
'noise_prob': [ 0.001,
0.001,
0,
0],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4}],

```

```
'run_time_sec': 4.446979761123657}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                    'cn': [ 0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5],
                                                    'r': 0},
                        'time': {'max': 10.0, 'min': 0.0},
                        'type': 'ising'}},
  'init_param': { 'value': [ 0.4349098035102067,
                             0.5281397824211302,
                             0.6402833431740305,
                             0.701719039924861,
                             0.8953768954552646,
                             1.4118719346144908,
                             2.230113621742168,
                             2.6548053250922607,
                             3.10433397641996,
                             3.1043339764199596,
                             4.146783795497047,
                             4.146783795497047,
                             4.149760661232294,
                             4.149760661232294,
                             4.223539194779984,
                             4.223539194779984,
                             6.7727886487081435,
                             6.7727886487081435,
                             7.243551717080481,
                             7.282491880183138,
                             7.282491880183137,
                             7.282491880183138,
                             8.806989173945695,
                             8.80698917394569,
                             9.706285648296737,
                             9.809777023075187,
                             9.837118725166619,
                             10.701586770517197,
                             11.185931122995514,
                             11.185931122995514,
                             0.39919688039300266,
                             1.4533249104921768,
                             0.7852252158845567,
                             -0.4957764525834907,
                             -0.3354719566017423,
                             0.16097458435951068,
                             -0.21971891747261646,
                             1.5370438656668401,
                             0.26531517225969137,
                             -0.6386119707402834,
                             0.37748437574064975,
                             1.2333814560270282,
                             0.5934062959304571,
                             1.633059089666515,
                             0.31170790405459564,
                             0.3995146629536743,
                             -0.11155094016301184,
                             0.8138063699882787,
                             0.019467271717771087,
                             -0.38186317682975884,
                             -2.778803401027002,
                             0.7887082829537533,
                             -3.1348050840084074,
                             -1.5109224736536222,
                             -0.38589710406642097,
                             -1.105927053527023,
                             0.3451707237315739,
                             0.6290043291617069,
                             -1.834722402137916,
                             -1.0551446159380258,
                             -1.3581260193610523,
                             0.9666180881078981,
```


-0.05759676522292653,
1.1866392284178628,
-1.1308471740259582,
0.34327443183487805,
0.17662613760194443,
-0.043135596735193626,
-0.03667546733606037,
-0.4901895513664937,
-1.0399227541005371,
0.4880459765792483,
0.053089003166299456,
-0.6259966396796703,
0.03159579372412852,
-0.09798805529491862,
0.052872660107769434,
1.490194875693949,
0.22167423812356596,
-0.33497064714209324,
1.0144679614160885,
0.01608415027143934,
0.49167899744128424,
0.12491439548600353,
-0.1329609067962543,
0.8639852596635488,
-1.6607871811173882,
0.2801513494447602,
0.3908501333463579,
0.6044891435983025,
0.08637544621327581,
1.2270127042754213,
0.11860545946003472,
-0.13066468162372125,
-0.998222267933649,
-1.2643438207146234,
0.7835210648968657,
0.8370312354478452,
0.06527740704106412,
-1.0235950282739155,
-0.705070781232052,
0.037682760290170314,
1.4778132318704873,
1.7053977225841355,
-0.1291331271719845,
0.1648924115529835,
2.208642048449397,
1.1885953055185114,
-0.8138037079028133,
0.6715189891287302,
0.7819572861394978,
-0.4022064636590845,
-0.3927127310055937,
-0.4118784323717791,
1.3463239978578432,
-0.3323606385924751,
0.6664683657180488,
1.58852177854836,
-1.212203169826861,
-0.031436536468697034,
0.5094262403458202,
-0.0616688967239856,
-0.0666169691754888,
-0.7277012125404719,
-0.12684196683036533,
0.5325519345188151,
2.2778420441533447,
0.8735933622003313,
0.9907279467768721,
1.1448787224294625,
0.4291077507842401,
-0.21356701746617238,
1.7174041100732933,
0.1885112766205208,
0.3630143936992631,
0.6395640633025061,
-0.2937369198596852,

```
-0.18128670495631546,  
0.9504587478277511,  
-0.6241293783488098,  
1.8733903679237267,  
0.4875344831143031,  
-0.9182747577406853,  
0.850791268752179,  
-0.46183623376516114,  
-0.634145646812118,  
-0.9165900492445601,  
-0.8973746003077631,  
-0.879107286657044,  
0.15645801311276636]],  
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},  
                    'noise_prob': [0.001, 0.001, 0, 0],  
                    'status': True,  
                    'type': 'depolarizing'},  
'nqubits': 7,  
'observable': { 'coefficients': { 'bn': [ 1.0,  
                                           1.0,  
                                           1.0,  
                                           1.0,  
                                           1.0,  
                                           1.0],  
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],  
                                   'r': 1},  
                  'def': 'ising'},  
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},  
             'file_name_prefix':  
'AUTOMATE_ising_noisefree_time_evo_ric2_sample#8'},  
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},  
'run': 'redundant',  
'state': 'dmatrix',  
'vqe': { 'iteration': 1,  
          'optimization': { 'algorithm': 'SLSQP',  
                            'constraint': True,  
                            'status': False}},  
'zne': { 'data_points': None,  
          'degree': 0,  
          'method': 'richardson',  
          'sampling': 'default'}},  
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 '  
                                '+ 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + '  
                                '1 X 5 X 6 + 1 Z 5 + 1 Z 6',  
            'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 + 1 Z 0 + 1 X 1 '  
                                                        'X 2 + 1 Z 1 + 1 X 2 X 3 + '  
                                                        '1 Z 2 + 1 X 3 X 4 + 1 Z 3 '  
                                                        '+ 1 X 4 X 5 + 1 Z 4 + 1 X '  
                                                        '5 X 6 + 1 Z 5 + 1 Z 6']},  
'output': { 'data_points': [ [4, 2, 0, 0, -6.672780190342767],  
                              [12, 6, 0, 0, -5.522238922989663]],  
            'noise_details': [ { 'gates_num': [4, 1, 1, 0],  
                                 'identity_factors': [0, 0, 0, 0],  
                                 'noise_level': [4, 2, 0, 0],  
                                 'noise_profile': { 'noise_on_init_param': {  
  
'status': False,  
  
0},  
  
'noise_prob': [ 0.001,  
                 0.001,  
                 0,  
                 0],  
                'status': True,  
                'type': 'depolarizing'},  
                'odd_wires': 4},  
              { 'gates_num': [12, 3, 1, 0],  
                'identity_factors': [1, 1, 0, 0],  
                'noise_level': [12, 6, 0, 0],  
                'noise_profile': { 'noise_on_init_param': {  
  
'status': False,  
  
0},  
  
'noise_prob': [ 0.001,  
                 0.001,
```

```

0,
0],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4]],
'run_time_sec': 4.373898983001709}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}}},
'init_param': { 'value': [ 0.3463233138801294,
0.3463233138801294,
0.3463233138801294,
0.3463233138801293,
1.2403205490491955,
1.38033446895038,
3.6939218351946623,
3.6939218351946623,
3.693921835194662,
3.693921835194662,
3.9412089126587,
3.9412089126587,
3.9412089126587007,
3.9412089126587007,
4.8310999601027795,
4.870844593548848,
4.900228671797898,
4.9002286717978984,
4.9002286717978984,
4.900228671797898,
5.70167657474683,
6.21188441483336,
6.21188441483336,
6.686419232103557,
6.723504371727371,
7.618130890836856,
7.618130890836856,
9.420364260644014,
9.420364260644014,
10.775132222422771,
-0.3525750090684208,
1.2748976558430758,
-0.16146998454247521,
1.3224575871686013,
0.0349265192498484,
0.36138711032415904,
-0.5043488106619078,
0.7634801084306476,
-0.3536814593617201,
0.9606809354246906,
-1.1143894720984433,
-0.45289061390722213,
-0.1605333784989486,
-1.594210268458609,
-0.09465166736510336,
0.11132251138190599,
1.004235666711761,
-0.23115493220428254,
0.9451761882751959,
-0.3707284170975133,
-0.17782948231634454,
-0.18433417990028922,
-0.06421460341189031,
-1.155489001827845,
-0.2074908306187579,
-0.036360325422879626,
0.5283181819327893,

```

0.3603171208434518,
0.6819349944270952,
-0.2733083993242672,
0.7075879191324372,
-0.042779623335368125,
0.7032435595714176,
-0.7363942983542189,
0.9583235184546098,
-0.5892472427173001,
-0.7139877518112785,
-0.12446625268371646,
-0.439646171766664,
0.7129808957109575,
0.3373075268097138,
0.3345550414369601,
-0.38068761370203624,
0.07864964370676188,
-0.3383553862900091,
0.6633419895127639,
-0.578720703894673,
0.486162302530746,
-0.06112668401522289,
0.34417654454775265,
-0.09182266750822224,
0.012215000731405658,
0.5658291917160285,
-0.16728488191683846,
0.3754616489195623,
-0.30497315054621493,
-0.7559846133933803,
0.9474959005813702,
-0.5904076686561035,
-0.2620011537974264,
0.1328515248519479,
-0.44066408562446224,
-0.7849425610737308,
0.44059474279745625,
-0.04646983134951628,
0.27897770882356554,
0.36838349877106535,
1.1610009997739648,
0.9058411506824863,
0.3119955912029684,
-0.5265683545543336,
1.0661822357546584,
0.8560044406571757,
0.5004409227892788,
0.5192086876555008,
-0.0019665558968806392,
-0.23949800878404184,
0.32563956832297103,
0.795323960796817,
1.5213873243487468,
0.8637563634588281,
0.07945031145196896,
1.7583315097859118,
0.5335864884320034,
-0.1366270407456156,
-0.9111102828185913,
-0.41650979281126865,
0.20926691692934202,
0.2776061293997892,
1.4257231544028384,
-0.10754544247245298,
1.0035199546983695,
1.1162390986419686,
1.7211522778091677,
-1.4332245925979028,
1.4417595659646931,
-2.106784537892007,
-0.6878309432904478,
-0.20360639354575213,
0.6373705186717347,
0.051333236332203484,
-0.35868991955772395,

```

-0.24365907607080922,
1.619736185871098,
0.5895182658245096,
-0.6086170057348131,
0.39774539931448055,
1.409837948786624,
0.17438880158294706,
-0.0415548590378924,
0.1413623732119859,
-0.8192475896231625,
0.5823982141653503,
-0.8462302367353051,
-0.60454163385331,
1.8216629921858374,
1.2634551976945552,
-0.40200420270308274,
-1.5954759495917508,
-1.7732438129518737]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                  'noise_prob': [0.001, 0.001, 0, 0],
                  'status': True,
                  'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
               'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric2_sample#3'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 '
                                '+ 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + '
                                '1 X 5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 + 1 Z 0 + 1 X 1 '
                                                         'X 2 + 1 Z 1 + 1 X 2 X 3 + '
                                                         '1 Z 2 + 1 X 3 X 4 + 1 Z 3 '
                                                         '+ 1 X 4 X 5 + 1 Z 4 + 1 X '
                                                         '5 X 6 + 1 Z 5 + 1 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -6.1514222647937595],
                             [12, 6, 0, 0, -4.906882414400225]],
            'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                'identity_factors': [0, 0, 0, 0],
                                'noise_level': [4, 2, 0, 0],
                                'noise_profile': { 'noise_on_init_param': {
'status': False,
'noise_prob': [ 0.001,
0.001,
0,
0],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': { 'noise_on_init_param': {
'value':

```

'status': False,

'value':

0},

'noise_prob': [0.001,
0.001,
0,
0],

'status': True,
'type': 'depolarizing'},

'odd_wires': 4]],

'run_time_sec': 4.746487617492676]],

{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
'cn': [0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [0.32980400918867037,

0.7818596814437996,
0.9680306497310072,
1.0586432954274991,
1.7663116348413028,
1.8956491295658664,
2.0436116769319477,
2.0436116769319477,
2.0436116769319477,
2.1846619555479685,
2.1846619555479685,
4.516997836642403,
4.516997836642403,
5.3166564351730035,
5.320130741320412,
5.399447086770547,
5.400046100307029,
5.400046100307029,
5.400046100307029,
5.400046100307029,
7.097750404951498,
7.618222769006738,
7.956666567531361,
7.956666567531361,
8.961109688308255,
8.961109688308255,
11.028500695271937,
11.5809418583784,
11.5809418583784,
11.5809418583784,
0.24114173221382337,
1.3754512844042854,
0.9720111739434927,
-0.29877116871955545,
-0.029737313498699618,
0.3185153459444675,
-0.31840745367437817,
0.6801602139502809,
-0.015407845998522466,
-1.114484654283524,
0.22997676136047449,
1.5201821520192487,
0.3217486942799661,
0.43678760134757944,
0.17618749246228715,
0.7010359103792874,
0.3446701255696699,
1.4700166379981474,
0.3471333265905846,
-0.968444017764766,
-1.1485269521507444,
0.3282504398407059,

-0.7898554413193151,
-0.22157985789400653,
0.4570931682042821,
-0.22158635023930806,
-0.36549836384277606,
-0.2366283880333898,
0.4114765778384287,
-0.7694643480660831,
-0.40507000819737765,
-0.6576468499882752,
0.2023600853039859,
-1.2917280200640486,
0.32702771491506877,
-0.07546137553843744,
-0.7713328193108817,
-0.8163562225239717,
-0.07772489804090182,
-0.32841576637855896,
-0.05728139293162625,
0.8564947637293437,
0.15384434981345546,
0.41944080031994563,
0.16884394015055976,
-0.010490661309829376,
-1.3198249528292274,
0.07128101806419536,
0.03515395872296211,
-0.09826644736389979,
-0.06882299183557238,
-0.7062443088386591,
-0.5280915567870916,
0.14860690208212682,
0.338235073517239,
0.012130023587481156,
-0.037483081002571056,
0.8881056862099682,
0.06312673968269537,
-0.44738898868242327,
-0.2463284936792685,
1.1708672891281489,
0.5297819270424092,
-0.5463182826640249,
-1.667403353493104,
1.3713545814180228,
0.9413366263710182,
-0.021951472934173096,
1.0345046397029698,
-0.8002739446090432,
1.076242700462428,
0.04994674089563993,
-1.446235170371471,
-0.3268246709561185,
0.6264234635598642,
-1.2465048991503196,
0.1169092097763034,
1.2453937722962254,
0.7220287063888942,
0.8114212311039525,
0.18333674920471413,
0.7345776889004225,
0.9807678368065545,
0.8703572591074042,
0.28720991162605214,
-0.9085817320602367,
1.1216621938695932,
1.1327477502874412,
-1.8285395156530388,
1.4231155243966302,
0.08156271721069372,
0.18027327076782884,
-0.10302960773112253,
-0.5560482514096969,
-0.09484541903718098,
0.5886430341587933,
-1.1674014904638685,

```

-0.03940950630369794,
0.2792515765657198,
0.08214125135098017,
0.04344924016389119,
-0.7787970529711157,
0.07887586110139429,
0.7532574519080738,
0.17107625040009872,
-0.4128415687827808,
1.3723943620028143,
1.4278581324344195,
0.5879977529754344,
0.28910810409117477,
-1.4412046993697643,
1.624844367134446,
0.19366683662945536,
0.3179841250458877,
0.5357240606734743,
1.5680682524509713,
0.04418323830986127,
0.2311137213204739,
2.8741283660232546,
0.9612408402592636]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0, 0],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric2_sample#5'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 '
                                '+ 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + '
                                '1 X 5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 + 1 Z 0 + 1 X 1 '
                                                         'X 2 + 1 Z 1 + 1 X 2 X 3 + '
                                                         '1 Z 2 + 1 X 3 X 4 + 1 Z 3 '
                                                         '+ 1 X 4 X 5 + 1 Z 4 + 1 X '
                                                         '5 X 6 + 1 Z 5 + 1 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -6.720969040841223],
                             [12, 6, 0, 0, -5.577003279994283]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                    'identity_factors': [0, 0, 0, 0],
                    'noise_level': [4, 2, 0, 0],
                    'noise_profile': { 'noise_on_init_param': {
'status': False,
'noise_prob': [ 0.001,
0.001,
0,
0],
'status': True,
'type': 'depolarizing'},
'value':
0},
'noise_prob': [ 0.001,
0.001,
0,
0],
'status': True,
'type': 'depolarizing'},

```



```

'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [1, 1, 0, 0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': { 'noise_on_init_param': {
    'value':
      'noise_prob': [ 0.001,
                      0.001,
                      0,
                      0],
    'status': True,
    'type': 'depolarizing'},
    'odd_wires': 4}},
  'run_time_sec': 5.059067964553833}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                         'r': 0},
                                      'time': {'max': 10.0, 'min': 0.0},
                                      'type': 'ising'}}},
  'init_param': { 'value': [ -4.1290471568429383e-17,
                              0.028387941050870547,
                              0.028387941050870516,
                              0.028387941050870544,
                              0.4764187545019148,
                              0.485154879376271,
                              0.7094706976105217,
                              2.998376901501724,
                              3.486264965503922,
                              3.487775573446841,
                              4.058897205630428,
                              5.020552086463093,
                              5.02158492362958,
                              5.02158492362958,
                              5.0846878698400815,
                              5.0846878698400815,
                              6.794704827143902,
                              6.794704827143902,
                              6.794704827143902,
                              6.794704827143902,
                              8.013356083289752,
                              8.21076169573117,
                              8.21076169573117,
                              8.633075344303478,
                              8.633075344303478,
                              8.930018414781724,
                              8.930018414781724,
                              8.930018414781724,
                              9.990329186240457,
                              9.990329186240457,
                              12.915390927357043,
                              0.8289215528981712,
                              0.6465354442587259,
                              -0.06922368073773534,
                              -2.0853926237797484,
                              0.25605409461249207,
                              -0.765586831155604,
                              0.9589723041272441,
                              0.27455658215646966,
                              -0.07189982532198122,
                              -0.3449139525561357,
                              -0.46800027418386986,
                              1.016480689947782,
                              -0.9706976058410632,
                              -0.9556868223128826,
                              0.9323574793538898,
                              -0.4376613900889883,
                              0.44138701419965265,

```

1.4418520162063444,
0.8420619129497768,
-0.025530845682239494,
-2.3076344184440156,
0.37458463155346333,
-0.47031338116346383,
0.07338001085955462,
-0.869249007732584,
1.6824786550253577,
1.719346297136726,
-0.9622509832303179,
0.5779076253870294,
0.4760527439731688,
-0.2451715046346434,
1.5906513054768954,
0.35918458386075697,
-0.44817294000256097,
-0.37416377940751877,
0.37048554789855764,
0.693137718643464,
2.288219219235325,
-0.26501198620577204,
0.3689053264668303,
0.10166782579927121,
-0.3356073800920907,
-0.6413954053758932,
0.28636656044125136,
-0.8690978212322298,
1.166030172665772,
-0.46365263204645707,
-0.8505396438729699,
2.4328989003329693,
-1.4554443951257197,
2.09134311034601,
1.7695145471360614,
1.1170184462314887,
0.20325573406500014,
0.34206355374635444,
0.016804831103183593,
-0.19882507083844791,
1.4011478969931443,
-0.0572616468882171,
-1.3773340344964602,
1.1031086912886459,
0.47433623341773395,
-0.037679845343088075,
0.7450954028512253,
1.3114871459373192,
-0.18651966717154955,
1.0750199871192778,
0.48905379729033566,
-0.6281827821606347,
1.322074656325554,
-1.4761484868509396,
0.2859918816514792,
-1.0033541595176405,
0.8264053813863532,
0.622010811103528,
1.1636747207646185,
-0.31883271288389914,
-0.0013981850979601159,
-0.09999405171802342,
1.3085280853238759,
-1.3727969089464465,
-0.08341159264840771,
1.6249369276484606,
-0.28082331570516694,
1.863294400254821,
2.2932652495309953,
1.4849536675442596,
0.6811668802751195,
1.5380110980874968,
-0.6615225065895972,
0.4000828281024264,
0.33932121588009856,

```

0.48855764825038867,
2.5371111232777053,
-0.7260430040504564,
0.09005940379987407,
-0.8762980817992282,
-0.23800314057977784,
0.9728027685352436,
-0.20857834368009137,
0.6148333457084475,
-1.5171898336040224,
0.8658542069129427,
-1.6478287868490311,
0.5770049367106069,
0.134497700004433,
-0.11612046371881249,
1.5826352840237945,
1.6045820840261675,
-0.08805669980595342,
-1.507745125921201,
-0.11276651576435986,
0.07206005105970639,
-1.7788122679933258,
-0.002988765738482436,
0.7255804452314191,
2.7661804504965604,
-3.1943627011957507,
-0.07088434770936546,
0.10949711075632612]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0, 0],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric2_sample#1'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 '
                                '+ 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + '
                                '1 X 5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 + 1 Z 0 + 1 X 1 '
                                                         'X 2 + 1 Z 1 + 1 X 2 X 3 + '
                                                         '1 Z 2 + 1 X 3 X 4 + 1 Z 3 '
                                                         '+ 1 X 4 X 5 + 1 Z 4 + 1 X '
                                                         '5 X 6 + 1 Z 5 + 1 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -6.833109640228285],
                             [12, 6, 0, 0, -5.610198560557998]],
            'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                'identity_factors': [0, 0, 0, 0],
                                'noise_level': [4, 2, 0, 0],
                                'noise_profile': { 'noise_on_init_param': {
'status': False,
'value':
0},
'noise_prob': [ 0.001,

```

```

0.001,
0,
0],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': { 'noise_on_init_param': {
'status': False,
0},
'value':
'noise_prob': [ 0.001,
0.001,
0,
0],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4}],
'run_time_sec': 4.242884874343872}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [ 0.4319322451408806,
0.4441934852523559,
0.5878315575931067,
0.5878315575931067,
0.5912427958750637,
0.5947187380620651,
0.5947187380620652,
1.5628074958209335,
1.568718083448373,
2.013035808837258,
2.013035808837258,
2.013035808837258,
2.0277481086167364,
4.114573484042011,
4.1154343976538055,
5.358852938788934,
5.358852938788934,
5.549639110185835,
5.667644288942511,
5.6706149635077026,
7.063797010237488,
7.658675772413245,
8.123010743902626,
8.633070512181217,
8.7849794748515,
8.7849794748515,
10.749769122252356,
10.89899813672382,
11.413879450074099,
11.413879450074099,
-0.04151001581527148,
-2.751209241920508,
-0.5056705094410174,
-0.406548832581749,
0.8633340895103881,
0.17259893989954428,
-0.21235924407313156,
-0.5485334561985523,
0.2012951527025238,
-0.23858034563229433,
2.1364176057207223,
0.7817014982866842,

```

-0.1108899918027214,
0.6048531661014985,
1.809261328558185,
-1.1786415575821785,
-1.0827170099436865,
0.4109384498547832,
0.8090945308822337,
-0.9356585252219308,
0.28697429006677655,
-0.15332459566133277,
1.2998079093160566,
0.3870967504567973,
-0.2413465423953762,
1.001771196320206,
0.1138394509020578,
1.3220786271535379,
-1.1806565063083097,
2.2789659185869464,
0.11875741943281123,
-0.15631084464392242,
-2.14680662310284,
0.18035127914430246,
-0.260880073181003,
0.21031876899961,
0.7641152777599183,
1.203568210035907,
1.2767673297334832,
1.2585129786889195,
0.24752046596725819,
0.7203551788145919,
-0.17420447991641552,
-1.3486030918499698,
0.2145369555808404,
0.8896874485404291,
-1.6470089821954121,
-0.37928143051952584,
-0.29903232984244293,
0.6984471945997809,
0.14906235834494008,
0.054873949093549826,
1.3877615433857144,
0.04193983551007985,
1.2014028569521673,
-0.02338341588216991,
-0.05255422822816251,
-0.1905875093002492,
0.16425167437144692,
0.2542835721118732,
1.1822299841866795,
1.2032222480152739,
0.30142488752060526,
0.09125464470109583,
-0.45638140154025475,
0.050260497156630585,
-0.04000614529085093,
0.9096133936431033,
-0.03555378439095736,
0.4405325841695877,
-0.48129885373423525,
-0.3617221403442356,
-2.5216506086368518,
-0.5081934521569421,
0.041332600726076626,
0.7413771443671474,
0.2204029987501584,
-0.7517920276590782,
-0.21022249714251526,
1.5607830526955413,
0.16615947769508344,
-0.17023919527139658,
-0.4295619963603494,
-1.1838803442651602,
0.6319564057885301,
-0.3935277768061919,
0.005550337258741166,

```

-0.4226065324972882,
-0.3715846249960284,
-0.48395498614671884,
-0.4790957128536121,
1.4427293239187895,
0.5875094653073935,
1.404722056846091,
0.6973465258635152,
0.8623508528879442,
-0.5460563718875903,
0.8326491581513842,
-2.4015389244466907,
-0.04583861440999851,
-0.36265388938085896,
-0.5347868312224912,
0.49674658852292397,
-0.6488379607428911,
1.1568053403721041,
0.30662269781395407,
-2.013151608263867,
-0.17383275003878457,
0.5036662539248705,
1.9638910157847163,
3.118155075975211,
0.8117246453310248,
-1.4517859264576565,
-1.4946803709676548,
-2.7792982516572895,
1.1174424252005506,
2.333430423546859,
0.6903927508165344,
-0.2883899927957993,
-0.8542826896862586]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric2_sample#7'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 '
'+ 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + '
'1 X 5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 + 1 Z 0 + 1 X 1 '
'X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 '
'+ 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -6.662379321078875],
[12, 6, 0, 0, -5.394498644176584]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 0, 0],

```

```

'noise_profile': { 'noise_on_init_param': {
'noise_prob': [ 0.001,
0.001,
0,
0],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': { 'noise_on_init_param': {
'noise_prob': [ 0.001,
0.001,
0,
0],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4}},
'run_time_sec': 4.039986610412598}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}}},
'init_param': { 'value': [ 0.46756606893882113,
0.46983859896604685,
0.5133985927484581,
0.5133985927484581,
0.5133985927484581,
1.4016794842576101,
1.6914116785996751,
1.691411678599675,
1.6941756924801525,
1.6941756924801528,
3.749864451084885,
3.749864451084885,
3.749864451084885,
3.749864451084885,
4.969636107010246,
4.969636107010246,
4.969636107010246,
4.969636107010246,
5.983107385388112,
5.983107385388112,
8.378949246097575,
8.378949246097575,
8.635680633753974,
8.635680633753974,
9.108807770997622,
9.108807770997622,
9.108807770997622,
9.960036411132767,
10.436643041587223,
10.716533289695375,
0.5466515694267952,
3.318518267704871,
0.40678728382369284,
0.6290091030083763,
-0.4770983795300032,
-0.7455483390932742,
0.1630064768968011,

```

0.4889724943271746,
0.10596004146863507,
-1.4759536729524354,
0.957604974713893,
0.6004514101153323,
0.0725728265044025,
1.4994700617905987,
0.15508641282397265,
-0.15494303336158094,
-0.5663845144966579,
2.6663065398173496,
-0.27086336898318036,
-1.1828784814640632,
-0.295045685212881,
0.6236994649007899,
1.1769360493970402,
1.7159795997719172,
-0.3019600780315931,
0.651545330446025,
-2.0782382222089724,
0.3186344830290561,
-0.2815476257680432,
0.4067635126151388,
-0.8228051832090519,
-0.4154478828430832,
2.3723163321732827,
0.931436930203099,
0.9804374423714106,
0.058895326584719035,
0.3167253661454302,
0.3390503539155689,
-0.5024758650485814,
-0.2425872659725522,
-0.6703364135359525,
-0.147558399690948,
1.1644296968010859,
0.5475507579506065,
0.1736068872874216,
0.30547527216138975,
0.5300773267023443,
0.7722201656261689,
-0.6858549484151734,
-0.16073113081558088,
-1.197458252004488,
-0.507307082981415,
-0.16437738354723153,
0.2647407403580543,
-0.45930133674563567,
-0.22673992276266391,
1.8127715449849986,
-1.4334128748253496,
-0.759404859696163,
1.2431110601081345,
0.8953271645792334,
0.6963251389999259,
-0.23147129445607595,
0.3638861144885483,
2.2394622330281906,
1.543816783446588,
1.7535830084845025,
-0.3169637330795974,
0.2453194636587019,
2.0076391670843647,
0.286017506499548,
0.6600449842625801,
0.19782118041633945,
0.05463577684798935,
0.4755005320975566,
0.055207215781380096,
0.06969685609102444,
0.2645328829673777,
-0.01235845210357445,
1.01982876424508,
2.262492417600915,
1.3443784687440867,


```

0.10698144982242018,
0.428081928456963,
-1.5494921893757727,
1.0701294130786736,
0.39987519173641656,
0.06301738855484718,
-1.6711192593807125,
0.26920159401030147,
-0.08358711295948693,
0.4054786916979446,
-0.5013194648307361,
2.126550690771933,
0.10959959792145137,
0.4205360694942305,
-0.586878230803869,
-0.3381608637889511,
-0.5055401953468239,
-0.45995744817477163,
0.07999668782445096,
-0.4269018063306222,
0.9588276598807326,
1.042187521522261,
0.07821502449201799,
0.6777167152964713,
0.4177387822172794,
-0.18721284346239236,
-0.8064458441104513,
2.8712851588196218,
0.7141862154478698,
-0.9976723444987857,
-0.8529435737233043,
1.9602636639541733,
1.2634521849205176,
-0.7170870121313951,
0.31227949023655405,
-1.0451293221946831,
-0.6787852158143914,
1.8581833428208436]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric2_sample#6'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 '
'+ 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + '
'1 X 5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 + 1 Z 0 + 1 X 1 '
'X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 '
'+ 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6']},

```

```

'output': { 'data_points': [ [4, 2, 0, 0, -6.4878948983695235],
                             [12, 6, 0, 0, -5.297876077058435]],
  'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                       'identity_factors': [0, 0, 0, 0],
                       'noise_level': [4, 2, 0, 0],
                       'noise_profile': { 'noise_on_init_param': {
'value':
'noise_prob': [ 0.001,
                0.001,
                0,
                0],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [1, 1, 0, 0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': { 'noise_on_init_param': {
'value':
'noise_prob': [ 0.001,
                0.001,
                0,
                0],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4}],
  'run_time_sec': 5.047370910644531}],
'vqe': [ { 'config': { 'ansatz': { 'gateset': 1,
                                  'layer': 30,
                                  'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                                'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                                'r': 0},
                                  'time': {'max': 10.0, 'min': 0.0},
                                  'type': 'ising'}},
'init_param': {'value': 'random'},
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                  'noise_prob': [0.001, 0.001, 0, 0],
                  'status': True,
                  'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                  'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                  'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': True, 'type': 'png'},
'file_name_prefix': 'ising_noisefree_time_evo'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'vqe',
'state': 'dmatrix',
'vqe': { 'iteration': 10,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': True}},
'zne': { 'data_points': None,
'degree': 1,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '
'2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
'1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + '
'1 Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X '
'3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z '
'4 + 1 X 5 X 6 + 1 Z 5 + 1 Z 6']},
'output': { 'exact_sol': -8.56677223350563,
'initial_cost_history': [ 4.2809785500030175,
2.611097744705238,
3.419414059277099,
1.5773527652373136,
3.36522638437751,
3.1574660994918307,

```

```
4.256553926466165,
4.577016272245339,
4.50890344694736,
4.056304045999429],
'noise_details': { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 0, 0],
'noise_profile': { 'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
'optimized_minimum_cost': [ -6.833109640228162,
-6.51493088468928,
-6.151422264793805,
-6.630764461698797,
-6.72096904084131,
-6.487894898369486,
-6.662379321078827,
-6.6727801903427855,
-6.751241519687556,
-6.406033238196173],
'optimized_parameters': [ [ -4.1290471568429383e-17,
0.028387941050870547,
0.028387941050870516,
0.028387941050870544,
0.4764187545019148,
0.485154879376271,
0.7094706976105217,
2.998376901501724,
3.486264965503922,
3.487775573446841,
4.058897205630428,
5.020552086463093,
5.02158492362958,
5.02158492362958,
5.0846878698400815,
5.0846878698400815,
6.794704827143902,
6.794704827143902,
6.794704827143902,
8.013356083289752,
8.21076169573117,
8.21076169573117,
8.633075344303478,
8.633075344303478,
8.930018414781724,
8.930018414781724,
8.930018414781724,
9.990329186240457,
9.990329186240457,
12.915390927357043,
0.8289215528981712,
0.6465354442587259,
-0.06922368073773534,
-2.0853926237797484,
0.25605409461249207,
-0.765586831155604,
0.9589723041272441,
0.27455658215646966,
-0.07189982532198122,
-0.3449139525561357,
-0.46800027418386986,
1.016480689947782,
-0.9706976058410632,
-0.9556868223128826,
0.9323574793538898,
-0.4376613900889883,
0.44138701419965265,
1.4418520162063444,
0.8420619129497768,
-0.025530845682239494,
-2.3076344184440156,
0.37458463155346333,
```

-0.47031338116346383,
0.07338001085955462,
-0.869249007732584,
1.6824786550253577,
1.719346297136726,
-0.9622509832303179,
0.5779076253870294,
0.4760527439731688,
-0.2451715046346434,
1.5906513054768954,
0.35918458386075697,
-0.44817294000256097,
-0.37416377940751877,
0.37048554789855764,
0.693137718643464,
2.288219219235325,
-0.26501198620577204,
0.3689053264668303,
0.10166782579927121,
-0.3356073800920907,
-0.6413954053758932,
0.28636656044125136,
-0.8690978212322298,
1.166030172665772,
-0.46365263204645707,
-0.8505396438729699,
2.4328989003329693,
-1.4554443951257197,
2.09134311034601,
1.7695145471360614,
1.1170184462314887,
0.20325573406500014,
0.34206355374635444,
0.016804831103183593,
-0.19882507083844791,
1.4011478969931443,
-0.0572616468882171,
-1.3773340344964602,
1.1031086912886459,
0.47433623341773395,
-0.037679845343088075,
0.7450954028512253,
1.3114871459373192,
-0.18651966717154955,
1.0750199871192778,
0.48905379729033566,
-0.6281827821606347,
1.322074656325554,
-1.4761484868509396,
0.2859918816514792,
-1.0033541595176405,
0.8264053813863532,
0.622010811103528,
1.1636747207646185,
-0.31883271288389914,
-0.0013981850979601159,
-0.09999405171802342,
1.3085280853238759,
-1.3727969089464465,
-0.08341159264840771,
1.6249369276484606,
-0.28082331570516694,
1.863294400254821,
2.2932652495309953,
1.4849536675442596,
0.6811668802751195,
1.5380110980874968,
-0.6615225065895972,
0.4000828281024264,
0.33932121588009856,
0.48855764825038867,
2.5371111232777053,
-0.7260430040504564,
0.09005940379987407,
-0.8762980817992282,

-0.23800314057977784,
0.9728027685352436,
-0.20857834368009137,
0.6148333457084475,
-1.5171898336040224,
0.8658542069129427,
-1.6478287868490311,
0.5770049367106069,
0.134497700004433,
-0.11612046371881249,
1.5826352840237945,
1.6045820840261675,
-0.08805669980595342,
-1.507745125921201,
-0.11276651576435986,
0.07206005105970639,
-1.7788122679933258,
-0.002988765738482436,
0.7255804452314191,
2.7661804504965604,
-3.1943627011957507,
-0.07088434770936546,
0.10949711075632612],
[0.27654223573266234,
0.27654223573266246,
0.27654223573266246,
0.4643738922966213,
0.6582164862517121,
0.6582164862517121,
0.6582164862517121,
1.3424523496852623,
1.47635793598756,
1.47635793598756,
3.385550316280557,
3.385550316280557,
3.385550316280557,
4.494062371253785,
4.494062371253785,
5.0204758683966135,
5.023404188660082,
7.05704094573064,
7.087391629397132,
7.533802135057517,
8.245970909352765,
8.390887842007308,
8.390887842007308,
8.703391612007538,
8.703391612007538,
9.485545381639698,
9.485545381639698,
9.849437794456351,
9.849437794456351,
10.799155842585746,
-0.12708089143278917,
1.4168148977025445,
-0.13519536748631492,
-0.32308714516775877,
0.06981336221044306,
-0.034321895006426455,
-0.18209443759399158,
-0.024858958594746107,
0.664488174834645,
0.6897317665852637,
-0.008774731192519907,
0.625642551337688,
0.6862410821653743,
-0.7670419540542446,
-0.48055390132433673,
1.573964677292243,
-1.0555745677813198,
1.3159721165631335,
-0.9138502497993347,
-0.4641566084568084,
-0.06966564000847691,
0.039012746519005555,

1.0827641697579717,
1.2190093772283843,
1.5391150382377632,
-0.6675581491933664,
0.09104662559232415,
0.5283818781579022,
0.04478118244404365,
1.5836075485651113,
0.251932560991488,
-0.3949560381811925,
0.8854639388207172,
-0.8611267164776638,
1.566928747241295,
-0.15445151319573347,
-0.01396050433941564,
0.8172620115356718,
-0.014658132241289147,
-0.6623010385304499,
0.6299284572753584,
-0.15026947869759505,
-1.2293112897827836,
-0.5768656731979434,
1.231557922062332,
-1.447872926116422,
1.4918722706131171,
0.31945486783385635,
0.43870804207253483,
1.3582636256308331,
-0.8478398442626237,
0.799518029891469,
0.5539369896048497,
0.13453197415145846,
0.7167498104862234,
-0.06589109804263343,
0.041038724923082276,
1.151975756926885,
-0.052026838729968425,
0.3952978000808587,
-1.1754748970457833,
-0.01965607042871015,
-0.2456324144680547,
-0.01136184217607491,
0.022503168314325715,
1.1777923723104653,
0.14554117253938412,
-1.2287946167385297,
-0.9683737383161766,
1.073329622962756,
-0.7143583802799215,
-0.3545002682598469,
-0.31906110571926777,
-0.9597094503848127,
0.8825903987559673,
-0.08273415521500459,
0.2895039435279609,
-0.3388594197258312,
0.2406459434933063,
0.3824128671502841,
-0.5390555829405563,
0.3757684495816124,
-0.5015719636333678,
0.3958257688275076,
-0.6428354095046468,
-0.9178624992467067,
0.29327306512581747,
-0.09193285724698198,
0.1770091195095364,
0.6803523512851801,
1.05525770275555,
-0.4646437845139076,
2.2560908642344026,
-0.7500939262299234,
1.92959092052175,
-0.36895777993274226,
-0.04538283416771957,

1.2529586558275392,
-0.57805717638909,
0.3080585810325818,
0.031910210614104206,
0.18519950010488512,
1.3888566879174211,
1.087260914864616,
0.15259355623633358,
-0.08810167996863522,
0.07467874103808375,
1.4444893382870363,
-1.446220680213735,
0.8215180856176293,
2.0151723649306295,
0.13454991965456886,
0.21923381008732426,
0.31112145477192793,
-0.8977876743954992,
0.10239876571985529,
-1.0132922278363503,
-0.3794863035801566,
0.8437803973608401,
1.111671691788496],
[0.3463233138801294,
0.34632331388012944,
0.3463233138801294,
0.3463233138801293,
1.2403205490491955,
1.38033446895038,
3.6939218351946623,
3.6939218351946623,
3.693921835194662,
3.693921835194662,
3.9412089126587,
3.9412089126587,
3.9412089126587007,
3.9412089126587007,
4.8310999601027795,
4.870844593548848,
4.900228671797898,
4.9002286717978984,
4.9002286717978984,
4.900228671797898,
5.70167657474683,
6.21188441483336,
6.21188441483336,
6.686419232103557,
6.723504371727371,
7.618130890836856,
7.618130890836856,
9.420364260644014,
9.420364260644014,
10.775132222422771,
-0.3525750090684208,
1.2748976558430758,
-0.16146998454247521,
1.3224575871686013,
0.0349265192498484,
0.36138711032415904,
-0.5043488106619078,
0.7634801084306476,
-0.3536814593617201,
0.9606809354246906,
-1.1143894720984433,
-0.45289061390722213,
-0.1605333784989486,
-1.594210268458609,
-0.09465166736510336,
0.11132251138190599,
1.004235666711761,
-0.23115493220428254,
0.9451761882751959,
-0.3707284170975133,
-0.17782948231634454,
-0.18433417990028922,

-0.06421460341189031,
-1.155489001827845,
-0.2074908306187579,
-0.036360325422879626,
0.5283181819327893,
0.3603171208434518,
0.6819349944270952,
-0.2733083993242672,
0.7075879191324372,
-0.042779623335368125,
0.7032435595714176,
-0.7363942983542189,
0.9583235184546098,
-0.5892472427173001,
-0.7139877518112785,
-0.12446625268371646,
-0.439646171766664,
0.7129808957109575,
0.3373075268097138,
0.3345550414369601,
-0.38068761370203624,
0.07864964370676188,
-0.3383553862900091,
0.6633419895127639,
-0.578720703894673,
0.486162302530746,
-0.06112668401522289,
0.34417654454775265,
-0.09182266750822224,
0.012215000731405658,
0.5658291917160285,
-0.16728488191683846,
0.3754616489195623,
-0.30497315054621493,
-0.7559846133933803,
0.9474959005813702,
-0.5904076686561035,
-0.2620011537974264,
0.1328515248519479,
-0.44066408562446224,
-0.7849425610737308,
0.44059474279745625,
-0.04646983134951628,
0.27897770882356554,
0.36838349877106535,
1.1610009997739648,
0.9058411506824863,
0.3119955912029684,
-0.5265683545543336,
1.0661822357546584,
0.8560044406571757,
0.5004409227892788,
0.5192086876555008,
-0.0019665558968806392,
-0.23949800878404184,
0.32563956832297103,
0.795323960796817,
1.5213873243487468,
0.8637563634588281,
0.07945031145196896,
1.7583315097859118,
0.5335864884320034,
-0.1366270407456156,
-0.9111102828185913,
-0.41650979281126865,
0.20926691692934202,
0.2776061293997892,
1.4257231544028384,
-0.10754544247245298,
1.0035199546983695,
1.1162390986419686,
1.7211522778091677,
-1.4332245925979028,
1.4417595659646931,
-2.106784537892007,

-0.6878309432904478,
-0.20360639354575213,
0.6373705186717347,
0.051333236332203484,
-0.35868991955772395,
-0.24365907607080922,
1.619736185871098,
0.5895182658245096,
-0.6086170057348131,
0.39774539931448055,
1.409837948786624,
0.17438880158294706,
-0.0415548590378924,
0.1413623732119859,
-0.8192475896231625,
0.5823982141653503,
-0.8462302367353051,
-0.60454163385331,
1.8216629921858374,
1.2634551976945552,
-0.40200420270308274,
-1.5954759495917508,
-1.7732438129518737],
[0.23818326834520384,
0.242015507589488,
0.5292100600091615,
0.5598196968290768,
0.5598196968290768,
2.3451854381830244,
2.3451854381830244,
3.8453025861748578,
3.8453025861748578,
4.395137112369032,
4.408833101975425,
4.829585475162733,
4.867115107997544,
5.058958687371625,
5.058958687371625,
5.058958687371625,
5.581243202757385,
6.011433944644334,
6.011433944644334,
8.358003932454157,
8.358003932454157,
8.358003932454157,
8.371420896509267,
8.783744346022278,
8.783744346022278,
9.34189587380724,
9.34189587380724,
9.39608576624051,
9.726727464435275,
10.95102910414735,
0.14721560345391302,
1.0205767628001141,
0.40564926087548847,
-1.9238321605823567,
-0.04729505851188324,
0.482541600628026,
0.11521556777775581,
0.8524320040134185,
0.42380767990264745,
-0.6219665889019169,
-0.05759638705357143,
0.8161328242190486,
0.7425402472775383,
0.7535392998290764,
1.8314745216576223,
1.3334023135030668,
0.05552193989453376,
0.9691374289368472,
-0.07338372949421543,
-0.6690664310785921,
1.168533104165511,
-0.01962836639606316,

0.6576122461884292,
-0.037637936325233226,
-0.0009647519828692927,
0.4524839512418326,
-0.013115244288784337,
1.6270372625341938,
1.182284078721198,
0.9533845712158122,
1.0956662941001023,
-0.9647918240366662,
0.1897778948773928,
-0.04047520420354409,
0.5790278460715794,
0.631719469583885,
-0.3335418357294817,
1.0603186443819097,
-0.7449471003423949,
0.1630231005006795,
-0.07563007210422526,
-1.1009042311153838,
-0.34587951137957407,
-0.22408475871037167,
-0.6364303490666092,
-0.5245531829065874,
-0.7807390048523051,
-0.7767575864285188,
-0.9679768325508518,
-1.8691843526791756,
0.4783530568164554,
0.6133546874260105,
-0.20020986269221935,
-0.41532071148562183,
-0.8557118152516873,
-0.16985308585729458,
0.8428338023785568,
0.6750667051632169,
1.2795031602880573,
-0.41384224674171627,
0.3213362409234008,
1.2340309605381181,
1.0506369357289735,
-0.6415500741723684,
-0.9628453188140567,
1.935214357876566,
1.4159154638103797,
-2.0864443113151867,
-0.07426020994129089,
-1.0538315394110076,
0.2589745455351295,
-0.2955978932252484,
-0.030644528099012087,
-0.8602288471584197,
-0.09490606960216369,
0.4190622694749842,
-1.5667390101922392,
-0.7804920738383083,
0.1255849265457167,
0.5884338319364165,
-0.46028089859616994,
0.2488277677353687,
-0.7513271615815256,
-1.0561258734551267,
0.3697237180501531,
-0.7108432439071602,
0.5137418203820399,
0.946767478727941,
1.619957961978311,
1.2944079622561346,
0.21625880379379211,
0.007168703139473973,
-0.334764182643075,
-0.9086121674985949,
-0.8247776105092918,
-0.5734460393428641,
-0.4381826298821608,

-0.44306191141826123,
-0.23860269421390362,
-0.03343496826073402,
-0.008972211649351379,
1.2196121588758198,
1.5985131140815898,
0.060325450296581484,
-0.7312434635304047,
0.9413542695101463,
-0.6701271691421761,
-1.0680925542186157,
-0.5192506372311365,
1.5594784335685845,
0.17928043514789443,
0.2455246384495121,
-0.053106843359071614,
0.7488220810664211,
1.3030207698426615,
-1.787387332504871,
-1.1184519607509558,
-0.2511275398302805,
1.445676213762956,
1.5991220174606833],
[0.32980400918867037,
0.7818596814437996,
0.9680306497310072,
1.0586432954274991,
1.7663116348413028,
1.8956491295658664,
2.0436116769319477,
2.0436116769319477,
2.0436116769319477,
2.1846619555479685,
2.1846619555479685,
4.516997836642403,
4.516997836642403,
5.3166564351730035,
5.320130741320412,
5.399447086770547,
5.400046100307029,
5.400046100307029,
5.400046100307029,
5.400046100307029,
7.097750404951498,
7.618222769006738,
7.956666567531361,
7.956666567531361,
8.961109688308255,
8.961109688308255,
11.028500695271937,
11.5809418583784,
11.5809418583784,
11.5809418583784,
0.24114173221382337,
1.3754512844042854,
0.9720111739434927,
-0.29877116871955545,
-0.029737313498699618,
0.3185153459444675,
-0.31840745367437817,
0.6801602139502809,
-0.015407845998522466,
-1.114484654283524,
0.22997676136047449,
1.5201821520192487,
0.3217486942799661,
0.43678760134757944,
0.17618749246228715,
0.7010359103792874,
0.3446701255696699,
1.4700166379981474,
0.3471333265905846,
-0.968444017764766,
-1.1485269521507444,
0.3282504398407059,

-0.7898554413193151,
-0.22157985789400653,
0.4570931682042821,
-0.22158635023930806,
-0.36549836384277606,
-0.2366283880333898,
0.4114765778384287,
-0.7694643480660831,
-0.40507000819737765,
-0.6576468499882752,
0.2023600853039859,
-1.2917280200640486,
0.32702771491506877,
-0.07546137553843744,
-0.7713328193108817,
-0.8163562225239717,
-0.07772489804090182,
-0.32841576637855896,
-0.05728139293162625,
0.8564947637293437,
0.15384434981345546,
0.41944080031994563,
0.16884394015055976,
-0.010490661309829376,
-1.3198249528292274,
0.07128101806419536,
0.03515395872296211,
-0.09826644736389979,
-0.06882299183557238,
-0.7062443088386591,
-0.5280915567870916,
0.14860690208212682,
0.338235073517239,
0.012130023587481156,
-0.037483081002571056,
0.8881056862099682,
0.06312673968269537,
-0.44738898868242327,
-0.2463284936792685,
1.1708672891281489,
0.5297819270424092,
-0.5463182826640249,
-1.667403353493104,
1.3713545814180228,
0.9413366263710182,
-0.021951472934173096,
1.0345046397029698,
-0.8002739446090432,
1.076242700462428,
0.04994674089563993,
-1.446235170371471,
-0.3268246709561185,
0.6264234635598642,
-1.2465048991503196,
0.1169092097763034,
1.2453937722962254,
0.7220287063888942,
0.8114212311039525,
0.18333674920471413,
0.7345776889004225,
0.9807678368065545,
0.8703572591074042,
0.28720991162605214,
-0.9085817320602367,
1.1216621938695932,
1.1327477502874412,
-1.8285395156530388,
1.4231155243966302,
0.08156271721069372,
0.18027327076782884,
-0.10302960773112253,
-0.5560482514096969,
-0.09484541903718098,
0.5886430341587933,
-1.1674014904638685,

-0.03940950630369794,
0.2792515765657198,
0.08214125135098017,
0.04344924016389119,
-0.7787970529711157,
0.07887586110139429,
0.7532574519080738,
0.17107625040009872,
-0.4128415687827808,
1.3723943620028143,
1.4278581324344195,
0.5879977529754344,
0.28910810409117477,
-1.4412046993697643,
1.624844367134446,
0.19366683662945536,
0.3179841250458877,
0.5357240606734743,
1.5680682524509713,
0.04418323830986127,
0.2311137213204739,
2.8741283660232546,
0.9612408402592636],
[0.46756606893882113,
0.46983859896604685,
0.5133985927484581,
0.5133985927484581,
0.5133985927484581,
1.4016794842576101,
1.6914116785996751,
1.691411678599675,
1.6941756924801525,
1.6941756924801528,
3.749864451084885,
3.749864451084885,
3.7498644510848855,
3.7498644510848855,
4.969636107010246,
4.969636107010246,
4.969636107010246,
4.969636107010246,
5.983107385388112,
5.983107385388112,
8.378949246097575,
8.378949246097575,
8.635680633753974,
8.635680633753974,
9.108807770997622,
9.108807770997622,
9.108807770997622,
9.960036411132767,
10.436643041587223,
10.716533289695375,
0.5466515694267952,
3.318518267704871,
0.40678728382369284,
0.6290091030083763,
-0.4770983795300032,
-0.7455483390932742,
0.1630064768968011,
0.4889724943271746,
0.10596004146863507,
-1.4759536729524354,
0.957604974713893,
0.6004514101153323,
0.0725728265044025,
1.4994700617905987,
0.15508641282397265,
-0.15494303336158094,
-0.5663845144966579,
2.6663065398173496,
-0.27086336898318036,
-1.1828784814640632,
-0.295045685212881,
0.6236994649007899,

1.1769360493970402,
1.7159795997719172,
-0.3019600780315931,
0.651545330446025,
-2.0782382222089724,
0.3186344830290561,
-0.2815476257680432,
0.4067635126151388,
-0.8228051832090519,
-0.4154478828430832,
2.3723163321732827,
0.931436930203099,
0.9804374423714106,
0.058895326584719035,
0.3167253661454302,
0.3390503539155689,
-0.5024758650485814,
-0.2425872659725522,
-0.6703364135359525,
-0.147558399690948,
1.1644296968010859,
0.5475507579506065,
0.1736068872874216,
0.30547527216138975,
0.5300773267023443,
0.7722201656261689,
-0.6858549484151734,
-0.16073113081558088,
-1.197458252004488,
-0.507307082981415,
-0.16437738354723153,
0.2647407403580543,
-0.45930133674563567,
-0.22673992276266391,
1.8127715449849986,
-1.4334128748253496,
-0.759404859696163,
1.2431110601081345,
0.8953271645792334,
0.6963251389999259,
-0.23147129445607595,
0.3638861144885483,
2.2394622330281906,
1.543816783446588,
1.7535830084845025,
-0.3169637330795974,
0.2453194636587019,
2.0076391670843647,
0.286017506499548,
0.6600449842625801,
0.19782118041633945,
0.05463577684798935,
0.4755005320975566,
0.055207215781380096,
0.06969685609102444,
0.2645328829673777,
-0.01235845210357445,
1.01982876424508,
2.262492417600915,
1.3443784687440867,
0.10698144982242018,
0.428081928456963,
-1.5494921893757727,
1.0701294130786736,
0.39987519173641656,
0.06301738855484718,
-1.6711192593807125,
0.26920159401030147,
-0.08358711295948693,
0.4054786916979446,
-0.5013194648307361,
2.126550690771933,
0.10959959792145137,
0.4205360694942305,
-0.586878230803869,

-0.3381608637889511,
-0.5055401953468239,
-0.45995744817477163,
0.07999668782445096,
-0.4269018063306222,
0.9588276598807326,
1.042187521522261,
0.07821502449201799,
0.6777167152964713,
0.4177387822172794,
-0.18721284346239236,
-0.8064458441104513,
2.8712851588196218,
0.7141862154478698,
-0.9976723444987857,
-0.8529435737233043,
1.9602636639541733,
1.2634521849205176,
-0.7170870121313951,
0.31227949023655405,
-1.0451293221946831,
-0.6787852158143914,
1.8581833428208436],
[0.4319322451408806,
0.4441934852523559,
0.5878315575931067,
0.5878315575931067,
0.5912427958750637,
0.5947187380620651,
0.5947187380620652,
1.5628074958209335,
1.568718083448373,
2.013035808837258,
2.013035808837258,
2.013035808837258,
2.0277481086167364,
4.114573484042011,
4.1154343976538055,
5.358852938788934,
5.358852938788934,
5.549639110185835,
5.667644288942511,
5.6706149635077026,
7.063797010237488,
7.658675772413245,
8.123010743902626,
8.633070512181217,
8.7849794748515,
8.7849794748515,
10.749769122252356,
10.89899813672382,
11.413879450074099,
11.413879450074099,
-0.04151001581527148,
-2.751209241920508,
-0.5056705094410174,
-0.406548832581749,
0.8633340895103881,
0.17259893989954428,
-0.21235924407313156,
-0.5485334561985523,
0.2012951527025238,
-0.23858034563229433,
2.1364176057207223,
0.7817014982866842,
-0.1108899918027214,
0.6048531661014985,
1.809261328558185,
-1.1786415575821785,
-1.0827170099436865,
0.4109384498547832,
0.8090945308822337,
-0.9356585252219308,
0.28697429006677655,
-0.15332459566133277,

1.2998079093160566,
0.3870967504567973,
-0.2413465423953762,
1.001771196320206,
0.1138394509020578,
1.3220786271535379,
-1.1806565063083097,
2.2789659185869464,
0.11875741943281123,
-0.15631084464392242,
-2.14680662310284,
0.18035127914430246,
-0.260880073181003,
0.21031876899961,
0.7641152777599183,
1.203568210035907,
1.2767673297334832,
1.2585129786889195,
0.24752046596725819,
0.7203551788145919,
-0.17420447991641552,
-1.3486030918499698,
0.2145369555808404,
0.8896874485404291,
-1.6470089821954121,
-0.37928143051952584,
-0.29903232984244293,
0.6984471945997809,
0.14906235834494008,
0.054873949093549826,
1.3877615433857144,
0.04193983551007985,
1.2014028569521673,
-0.02338341588216991,
-0.05255422822816251,
-0.1905875093002492,
0.16425167437144692,
0.2542835721118732,
1.1822299841866795,
1.2032222480152739,
0.30142488752060526,
0.09125464470109583,
-0.45638140154025475,
0.050260497156630585,
-0.04000614529085093,
0.9096133936431033,
-0.03555378439095736,
0.4405325841695877,
-0.48129885373423525,
-0.3617221403442356,
-2.5216506086368518,
-0.5081934521569421,
0.041332600726076626,
0.7413771443671474,
0.2204029987501584,
-0.7517920276590782,
-0.21022249714251526,
1.5607830526955413,
0.16615947769508344,
-0.17023919527139658,
-0.4295619963603494,
-1.1838803442651602,
0.6319564057885301,
-0.3935277768061919,
0.005550337258741166,
-0.4226065324972882,
-0.3715846249960284,
-0.48395498614671884,
-0.4790957128536121,
1.4427293239187895,
0.5875094653073935,
1.404722056846091,
0.6973465258635152,
0.8623508528879442,
-0.5460563718875903,

0.8326491581513842,
-2.4015389244466907,
-0.04583861440999851,
-0.36265388938085896,
-0.5347868312224912,
0.49674658852292397,
-0.6488379607428911,
1.1568053403721041,
0.30662269781395407,
-2.013151608263867,
-0.17383275003878457,
0.5036662539248705,
1.9638910157847163,
3.118155075975211,
0.8117246453310248,
-1.4517859264576565,
-1.4946803709676548,
-2.7792982516572895,
1.1174424252005506,
2.333430423546859,
0.6903927508165344,
-0.2883899927957993,
-0.8542826896862586],
[0.4349098035102067,
0.5281397824211302,
0.6402833431740305,
0.701719039924861,
0.8953768954552646,
1.4118719346144908,
2.230113621742168,
2.6548053250922607,
3.1043339764199596,
3.1043339764199596,
4.146783795497047,
4.146783795497047,
4.149760661232294,
4.149760661232294,
4.223539194779984,
4.223539194779984,
6.7727886487081435,
6.7727886487081435,
7.243551717080481,
7.282491880183138,
7.282491880183137,
7.282491880183138,
8.806989173945695,
8.80698917394569,
9.706285648296737,
9.809777023075187,
9.837118725166619,
10.701586770517197,
11.185931122995514,
11.185931122995514,
0.39919688039300266,
1.4533249104921768,
0.7852252158845567,
-0.4957764525834907,
-0.3354719566017423,
0.16097458435951068,
-0.21971891747261646,
1.5370438656668401,
0.26531517225969137,
-0.6386119707402834,
0.37748437574064975,
1.2333814560270282,
0.5934062959304571,
1.633059089666515,
0.31170790405459564,
0.3995146629536743,
-0.11155094016301184,
0.8138063699882787,
0.019467271717771087,
-0.38186317682975884,
-2.778803401027002,
0.7887082829537533,

-3.1348050840084074,
-1.5109224736536222,
-0.38589710406642097,
-1.105927053527023,
0.3451707237315739,
0.6290043291617069,
-1.834722402137916,
-1.0551446159380258,
-1.3581260193610523,
0.9666180881078981,
-0.05759676522292653,
1.1866392284178628,
-1.1308471740259582,
0.34327443183487805,
0.17662613760194443,
-0.043135596735193626,
-0.03667546733606037,
-0.4901895513664937,
-1.0399227541005371,
0.4880459765792483,
0.053089003166299456,
-0.6259966396796703,
0.03159579372412852,
-0.09798805529491862,
0.052872660107769434,
1.490194875693949,
0.22167423812356596,
-0.33497064714209324,
1.0144679614160885,
0.01608415027143934,
0.49167899744128424,
0.12491439548600353,
-0.1329609067962543,
0.8639852596635488,
-1.6607871811173882,
0.2801513494447602,
0.3908501333463579,
0.6044891435983025,
0.08637544621327581,
1.2270127042754213,
0.11860545946003472,
-0.13066468162372125,
-0.998222267933649,
-1.2643438207146234,
0.7835210648968657,
0.8370312354478452,
0.06527740704106412,
-1.0235950282739155,
-0.705070781232052,
0.037682760290170314,
1.4778132318704873,
1.7053977225841355,
-0.1291331271719845,
0.1648924115529835,
2.208642048449397,
1.1885953055185114,
-0.8138037079028133,
0.6715189891287302,
0.7819572861394978,
-0.4022064636590845,
-0.3927127310055937,
-0.4118784323717791,
1.3463239978578432,
-0.3323606385924751,
0.6664683657180488,
1.58852177854836,
-1.212203169826861,
-0.031436536468697034,
0.5094262403458202,
-0.0616688967239856,
-0.0666169691754888,
-0.7277012125404719,
-0.12684196683036533,
0.5325519345188151,
2.2778420441533447,

0.8735933622003313,
0.9907279467768721,
1.1448787224294625,
0.4291077507842401,
-0.21356701746617238,
1.7174041100732933,
0.1885112766205208,
0.3630143936992631,
0.6395640633025061,
-0.2937369198596852,
-0.18128670495631546,
0.9504587478277511,
-0.6241293783488098,
1.8733903679237267,
0.4875344831143031,
-0.9182747577406853,
0.850791268752179,
-0.46183623376516114,
-0.634145646812118,
-0.9165900492445601,
-0.8973746003077631,
-0.879107286657044,
0.15645801311276636],
[0.48023040681030116,
0.7029544315470944,
0.722671987246733,
0.9307929858945541,
0.9307929858945543,
2.546598088688794,
2.546598088688794,
2.546598088688794,
2.546598088688794,
5.184334435330857,
5.195111458618343,
5.47436411399969,
5.47436411399969,
5.47436411399969,
5.47436411399969,
5.47436411399969,
6.068336244524409,
6.068336244524409,
7.838605251840715,
7.912250316107233,
7.912250316107233,
8.562517202814183,
9.063385768597811,
9.216218862905647,
9.216218862905647,
10.84047371877396,
10.84047371877396,
11.044358714867027,
11.375406566932424,
11.375406566932424,
0.18886549713610382,
0.6040291443391601,
-0.9727809979878402,
3.008209947509642,
0.3239743149147096,
-0.643987619924633,
1.0739882269552894,
0.4935103262912805,
-1.1713673989643745,
0.32339055167680386,
-0.8254170399860232,
3.387204293956005,
0.5437302598099015,
-0.6585961721148387,
1.0903163929414719,
-1.5261906384818182,
1.102349928722959,
-0.7373436929054379,
0.1712635196093208,
1.2086697581514199,
-0.4977298668106944,
-0.8828019932256734,

-0.13051782426094366,
0.048744185631251574,
-1.0535030780910155,
0.5224555306604155,
-0.9502111367985572,
-0.2573420205933925,
1.5833039752620888,
1.3806480812363842,
0.3864833677006273,
-0.0006284730712113316,
0.6487740408134013,
1.697829570896293,
-0.20023867528920775,
-0.32438662287273723,
1.3674639807988553,
-1.5592347723565667,
-0.6294582959326964,
1.111157394071659,
1.4895912958038178,
-0.7203901402661329,
-0.027076706863032294,
0.009574786781832467,
1.1437695218822916,
0.11720143179507149,
1.0973549625545624,
-1.0756168303605034,
-1.5785125289756954,
-0.42999007233668873,
0.7592603373612552,
0.7357293999316703,
0.03766014972468404,
1.450035910180884,
-0.08978031589044846,
0.7176267391661779,
-2.3377280976646966,
0.3878354316208249,
-0.8070070584177742,
0.07084117333557359,
-0.008998168433091903,
0.6612796195456235,
0.6287929626789847,
0.8601333603228948,
-0.7786354708155643,
-0.144476679155352,
0.39492950957300044,
-0.20012968159779548,
0.01326107241550192,
-0.5649340596325207,
-0.020398755490040426,
0.6333499066300691,
0.400844220606507,
0.5954749034740784,
0.8700413681534388,
1.315868035063538,
0.9914334712755634,
0.10418309057983809,
1.1095991500751576,
-0.1712743534684691,
-0.2730581386141925,
0.1453142903679217,
-0.3813955795194703,
0.1280814941289577,
-0.5627986213807415,
-1.0576722394029623,
-1.98721560251358,
0.519862592148374,
1.03695228961559,
0.21101375663462807,
-1.750252459751414,
-1.1450993785481438,
-2.2517541986316916,
-1.5456964825715331,
0.6026638989139771,
0.47561448130168205,
0.08201672609434518,

0.47464262797792917,
-0.14585857152212733,
0.08636040211648545,
0.7388192170680543,
1.071366965639021,
2.5115163876343742,
-0.4286872636197954,
-0.2977998181526895,
0.46569070003689333,
0.2115019986130443,
-0.5123638542481742,
0.09024611293676764,
-0.3380729423866059,
1.6314769884628497,
1.3338027138849353,
-0.7442166070564612,
1.2186722356710948,
1.584709484399307,
-0.6554175978902271,
0.36887213174672207,
-1.365021965092438,
-0.4623126041921459,
0.4869420326570461],
[0.5118290521669792,
0.9034573591840084,
1.0514460879099101,
1.1283814725964603,
1.1293980629167455,
1.6874068541059999,
1.8362001710030256,
2.594529848453243,
2.5949023258749775,
4.384420392533986,
4.804388533064279,
4.8044974726668155,
4.925049921883437,
4.925049921883437,
6.055903002618935,
6.0650337145794,
6.06510024924717,
6.06510024924717,
7.719643893437192,
7.719643893437192,
8.110116534451196,
8.374448879129652,
8.374448879129652,
9.29045698473019,
9.2929923765699,
9.977025927736127,
9.977025927736127,
9.977438303479875,
11.277250876863835,
11.588901828478656,
-0.5970483257864091,
1.048395642841534,
0.3563299433581316,
2.04182497417867,
-1.0659189653240388,
-1.5711224255679066,
-0.6854026923334804,
1.3002799990574343,
0.334979527312172,
-0.41771661057992626,
-0.5255960701407614,
0.9200835209290303,
0.5840294212204432,
0.4309612994747514,
0.010788324288593443,
0.8398650983618882,
0.35555521736822654,
0.33383672576306345,
-0.03972798241435421,
-0.5808926346218187,
-0.6217506829607715,
-0.022449055124049946,

-0.4671428723737863,
1.3840583406465645,
3.321823997583478,
1.9516123562000514,
2.109430068020724,
0.17138900458361278,
1.0734765554988877,
1.1591682463434823,
-1.5085197494005436,
0.3709514368460093,
-0.08354812013321816,
-0.6849414455501801,
-1.1807064172516142,
0.1889544406654655,
0.45125852432694835,
0.40946867023093825,
0.9591749175330944,
0.862406962512403,
-0.47565451021668054,
0.14791469386649075,
0.012323642821740255,
-0.3613715829994197,
-0.6520104899254531,
1.1500641632479522,
0.11480411675389499,
0.37929728325133344,
-0.12818017639377846,
1.5548403554359806,
-0.08803696907532751,
0.16058007713541472,
-0.2835300528775952,
1.4317946638041426,
0.04218231832431045,
0.6164913379423986,
-0.5546192578057587,
-0.0016176435191898148,
0.3455446109244303,
-0.7983090776945799,
0.06548494602750325,
0.09601011255154575,
-1.274882751616874,
-0.6639296860891427,
0.08468574728059233,
0.09178155492125434,
-1.9746588496307496,
1.363083624655819,
-0.11754172722968637,
-1.7454296260872237,
0.689437867548643,
0.09536601432398076,
2.1875690275532618,
1.9737367040865754,
0.19623670188051764,
1.3618525198389628,
-1.6551377680800423,
-1.2796187335858575,
-0.02979150349567955,
0.3655877448114123,
-1.339933806945279,
-1.6131280858420662,
1.1268021774842836,
0.7853899025443938,
-1.214954346638235,
1.881374469676006,
-2.2108375055617597,
0.38299461206746305,
-0.062378913993248956,
-1.0812916442950207,
0.19113842494911862,
0.43947065709017896,
1.4340329690981122,
-2.1914706359927933,
1.683520172253517,
-0.4151753501170658,
0.2652280013518245,

```

-0.09829123134965576,
-0.4033858255054015,
-0.09488290407421365,
1.1087820722198931,
1.3758250784849964,
0.3693308876448478,
-1.4774754586810708,
0.45790576106810227,
-1.318961759734687,
1.2898775251023542,
0.5291693857421421,
-0.7225860054677886,
-0.6024865845145562,
1.1258609582335735,
1.4545810101615688,
0.4988359655120556,
0.39968483637691726,
1.4778005154845164,
1.6358502222921723,
0.07081811420225437,
1.5336807499569445,
-0.7595143926860392,
0.07947795439670366]],
'run_time_sec': 68705.11171293259}}],
'zne': [ { 'config': { 'ansatz': { 'gateset': 1,
                                'layer': 30,
                                'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                            'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                            'r': 0},
                                'time': { 'max': 10.0, 'min': 0.0},
                                'type': 'ising'}}},
'init_param': { 'value': [ 0.32980400918867037,
0.7818596814437996,
0.9680306497310072,
1.0586432954274991,
1.7663116348413028,
1.8956491295658664,
2.0436116769319477,
2.0436116769319477,
2.0436116769319477,
2.1846619555479685,
2.1846619555479685,
4.516997836642403,
4.516997836642403,
5.3166564351730035,
5.320130741320412,
5.399447086770547,
5.400046100307029,
5.400046100307029,
5.400046100307029,
5.400046100307029,
7.097750404951498,
7.618222769006738,
7.956666567531361,
7.956666567531361,
8.961109688308255,
8.961109688308255,
11.028500695271937,
11.5809418583784,
11.5809418583784,
11.5809418583784,
0.24114173221382337,
1.3754512844042854,
0.9720111739434927,
-0.29877116871955545,
-0.029737313498699618,
0.3185153459444675,
-0.31840745367437817,
0.6801602139502809,
-0.015407845998522466,
-1.114484654283524,
0.22997676136047449,
1.5201821520192487,
0.3217486942799661,
0.43678760134757944,

```

0.17618749246228715,
0.7010359103792874,
0.3446701255696699,
1.4700166379981474,
0.3471333265905846,
-0.968444017764766,
-1.1485269521507444,
0.3282504398407059,
-0.7898554413193151,
-0.22157985789400653,
0.4570931682042821,
-0.22158635023930806,
-0.36549836384277606,
-0.2366283880333898,
0.4114765778384287,
-0.7694643480660831,
-0.40507000819737765,
-0.6576468499882752,
0.2023600853039859,
-1.2917280200640486,
0.32702771491506877,
-0.07546137553843744,
-0.7713328193108817,
-0.8163562225239717,
-0.07772489804090182,
-0.32841576637855896,
-0.05728139293162625,
0.8564947637293437,
0.15384434981345546,
0.41944080031994563,
0.16884394015055976,
-0.010490661309829376,
-1.3198249528292274,
0.07128101806419536,
0.03515395872296211,
-0.09826644736389979,
-0.06882299183557238,
-0.7062443088386591,
-0.5280915567870916,
0.14860690208212682,
0.338235073517239,
0.012130023587481156,
-0.037483081002571056,
0.8881056862099682,
0.06312673968269537,
-0.44738898868242327,
-0.2463284936792685,
1.1708672891281489,
0.5297819270424092,
-0.5463182826640249,
-1.667403353493104,
1.3713545814180228,
0.9413366263710182,
-0.021951472934173096,
1.0345046397029698,
-0.8002739446090432,
1.076242700462428,
0.04994674089563993,
-1.446235170371471,
-0.3268246709561185,
0.6264234635598642,
-1.2465048991503196,
0.1169092097763034,
1.2453937722962254,
0.7220287063888942,
0.8114212311039525,
0.18333674920471413,
0.7345776889004225,
0.9807678368065545,
0.8703572591074042,
0.28720991162605214,
-0.9085817320602367,
1.1216621938695932,
1.1327477502874412,
-1.8285395156530388,


```

1.4231155243966302,
0.08156271721069372,
0.18027327076782884,
-0.10302960773112253,
-0.5560482514096969,
-0.09484541903718098,
0.5886430341587933,
-1.1674014904638685,
-0.03940950630369794,
0.2792515765657198,
0.08214125135098017,
0.04344924016389119,
-0.7787970529711157,
0.07887586110139429,
0.7532574519080738,
0.17107625040009872,
-0.4128415687827808,
1.3723943620028143,
1.4278581324344195,
0.5879977529754344,
0.28910810409117477,
-1.4412046993697643,
1.624844367134446,
0.19366683662945536,
0.3179841250458877,
0.5357240606734743,
1.5680682524509713,
0.04418323830986127,
0.2311137213204739,
2.8741283660232546,
0.9612408402592636]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric2_sample#5'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.720969040841223],
[12, 6, 0, 0, -5.577003279994283]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 3.504753112792969e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -7.292951921264693,
'others': { 'beta_coefficients': [1.5, -0.5],
'cost_richardson_zne': 2.5,
'sorted_expectation_vals': [ -6.720969040841223,
-5.577003279994283],
'sorted_noise': [6, 18]},
'sampled data': [ [4, 2, 0, 0, -6.720969040841223],
[12, 6, 0, 0, -5.577003279994283]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [ -4.1290471568429383e-17,

```

0.028387941050870547,
0.028387941050870516,
0.028387941050870544,
0.4764187545019148,
0.485154879376271,
0.7094706976105217,
2.998376901501724,
3.486264965503922,
3.487775573446841,
4.058897205630428,
5.020552086463093,
5.02158492362958,
5.02158492362958,
5.0846878698400815,
5.0846878698400815,
6.794704827143902,
6.794704827143902,
6.794704827143902,
8.013356083289752,
8.21076169573117,
8.21076169573117,
8.633075344303478,
8.633075344303478,
8.930018414781724,
8.930018414781724,
8.930018414781724,
9.990329186240457,
9.990329186240457,
12.915390927357043,
0.8289215528981712,
0.6465354442587259,
-0.06922368073773534,
-2.0853926237797484,
0.25605409461249207,
-0.765586831155604,
0.9589723041272441,
0.27455658215646966,
-0.07189982532198122,
-0.3449139525561357,
-0.46800027418386986,
1.016480689947782,
-0.9706976058410632,
-0.9556868223128826,
0.9323574793538898,
-0.4376613900889883,
0.44138701419965265,
1.4418520162063444,
0.8420619129497768,
-0.025530845682239494,
-2.3076344184440156,
0.37458463155346333,
-0.47031338116346383,
0.07338001085955462,
-0.869249007732584,
1.6824786550253577,
1.719346297136726,
-0.9622509832303179,
0.5779076253870294,
0.4760527439731688,
-0.2451715046346434,
1.5906513054768954,
0.35918458386075697,
-0.44817294000256097,
-0.37416377940751877,
0.37048554789855764,
0.693137718643464,
2.288219219235325,
-0.26501198620577204,
0.3689053264668303,
0.10166782579927121,
-0.3356073800920907,
-0.6413954053758932,
0.28636656044125136,
-0.8690978212322298,
1.166030172665772,

-0.46365263204645707,
-0.8505396438729699,
2.4328989003329693,
-1.4554443951257197,
2.09134311034601,
1.7695145471360614,
1.1170184462314887,
0.20325573406500014,
0.34206355374635444,
0.016804831103183593,
-0.19882507083844791,
1.4011478969931443,
-0.0572616468882171,
-1.3773340344964602,
1.1031086912886459,
0.47433623341773395,
-0.037679845343088075,
0.7450954028512253,
1.3114871459373192,
-0.18651966717154955,
1.0750199871192778,
0.48905379729033566,
-0.6281827821606347,
1.322074656325554,
-1.4761484868509396,
0.2859918816514792,
-1.0033541595176405,
0.8264053813863532,
0.622010811103528,
1.1636747207646185,
-0.31883271288389914,
-0.0013981850979601159,
-0.09999405171802342,
1.3085280853238759,
-1.3727969089464465,
-0.08341159264840771,
1.6249369276484606,
-0.28082331570516694,
1.863294400254821,
2.2932652495309953,
1.4849536675442596,
0.6811668802751195,
1.5380110980874968,
-0.6615225065895972,
0.4000828281024264,
0.33932121588009856,
0.48855764825038867,
2.5371111232777053,
-0.7260430040504564,
0.09005940379987407,
-0.8762980817992282,
-0.23800314057977784,
0.9728027685352436,
-0.20857834368009137,
0.6148333457084475,
-1.5171898336040224,
0.8658542069129427,
-1.6478287868490311,
0.5770049367106069,
0.134497700004433,
-0.11612046371881249,
1.5826352840237945,
1.6045820840261675,
-0.08805669980595342,
-1.507745125921201,
-0.11276651576435986,
0.07206005105970639,
-1.7788122679933258,
-0.002988765738482436,
0.7255804452314191,
2.7661804504965604,
-3.1943627011957507,
-0.07088434770936546,
0.10949711075632612]],

'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},

1.3224575871686013,
0.0349265192498484,
0.36138711032415904,
-0.5043488106619078,
0.7634801084306476,
-0.3536814593617201,
0.9606809354246906,
-1.1143894720984433,
-0.45289061390722213,
-0.1605333784989486,
-1.594210268458609,
-0.09465166736510336,
0.11132251138190599,
1.004235666711761,
-0.23115493220428254,
0.9451761882751959,
-0.3707284170975133,
-0.17782948231634454,
-0.18433417990028922,
-0.06421460341189031,
-1.155489001827845,
-0.2074908306187579,
-0.036360325422879626,
0.5283181819327893,
0.3603171208434518,
0.6819349944270952,
-0.2733083993242672,
0.7075879191324372,
-0.042779623335368125,
0.7032435595714176,
-0.7363942983542189,
0.9583235184546098,
-0.5892472427173001,
-0.7139877518112785,
-0.12446625268371646,
-0.439646171766664,
0.7129808957109575,
0.3373075268097138,
0.3345550414369601,
-0.38068761370203624,
0.07864964370676188,
-0.3383553862900091,
0.6633419895127639,
-0.578720703894673,
0.486162302530746,
-0.06112668401522289,
0.34417654454775265,
-0.09182266750822224,
0.012215000731405658,
0.5658291917160285,
-0.16728488191683846,
0.3754616489195623,
-0.30497315054621493,
-0.7559846133933803,
0.9474959005813702,
-0.5904076686561035,
-0.2620011537974264,
0.1328515248519479,
-0.44066408562446224,
-0.7849425610737308,
0.44059474279745625,
-0.04646983134951628,
0.27897770882356554,
0.36838349877106535,
1.1610009997739648,
0.9058411506824863,
0.3119955912029684,
-0.5265683545543336,
1.0661822357546584,
0.8560044406571757,
0.5004409227892788,
0.5192086876555008,
-0.0019665558968806392,
-0.23949800878404184,
0.32563956832297103,

```

0.795323960796817,
1.5213873243487468,
0.8637563634588281,
0.07945031145196896,
1.7583315097859118,
0.5335864884320034,
-0.1366270407456156,
-0.9111102828185913,
-0.41650979281126865,
0.20926691692934202,
0.2776061293997892,
1.4257231544028384,
-0.10754544247245298,
1.0035199546983695,
1.1162390986419686,
1.7211522778091677,
-1.4332245925979028,
1.4417595659646931,
-2.106784537892007,
-0.6878309432904478,
-0.20360639354575213,
0.6373705186717347,
0.051333236332203484,
-0.35868991955772395,
-0.24365907607080922,
1.619736185871098,
0.5895182658245096,
-0.6086170057348131,
0.39774539931448055,
1.409837948786624,
0.17438880158294706,
-0.0415548590378924,
0.1413623732119859,
-0.8192475896231625,
0.5823982141653503,
-0.8462302367353051,
-0.60454163385331,
1.8216629921858374,
1.2634551976945552,
-0.40200420270308274,
-1.5954759495917508,
-1.7732438129518737]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0, 0],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric2_sample#3'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.1514222647937595],
                          [12, 6, 0, 0, -4.906882414400225]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 3.7670135498046875e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -6.773692189990527,
                           'others': { 'beta_coefficients': [1.5, -0.5],
                                         'cost_richardson_zne': 2.5,
                                         'sorted_expectation_vals': [ -6.1514222647937595,
                                                                       -4.906882414400225],
                                         'sorted_noise': [6, 18]},
                           'status': True,
                           'type': 'depolarizing'}},

```

```

'sampled data': [ [4, 2, 0, 0, -6.1514222647937595],
                  [12, 6, 0, 0, -4.906882414400225]],
'sampling': 'default'}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                    'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                    'r': 0},
                        'time': {'max': 10.0, 'min': 0.0},
                        'type': 'ising'}}},
  'init_param': { 'value': [ 0.27654223573266234,
                             0.27654223573266246,
                             0.27654223573266246,
                             0.4643738922966213,
                             0.6582164862517121,
                             0.6582164862517121,
                             0.6582164862517121,
                             1.3424523496852623,
                             1.47635793598756,
                             1.47635793598756,
                             3.385550316280557,
                             3.385550316280557,
                             3.385550316280557,
                             4.494062371253785,
                             4.494062371253785,
                             5.0204758683966135,
                             5.023404188660082,
                             7.05704094573064,
                             7.087391629397132,
                             7.533802135057517,
                             8.245970909352765,
                             8.390887842007308,
                             8.390887842007308,
                             8.703391612007538,
                             8.703391612007538,
                             9.485545381639698,
                             9.485545381639698,
                             9.849437794456351,
                             9.849437794456351,
                             10.799155842585746,
                             -0.12708089143278917,
                             1.4168148977025445,
                             -0.13519536748631492,
                             -0.32308714516775877,
                             0.06981336221044306,
                             -0.034321895006426455,
                             -0.18209443759399158,
                             -0.024858958594746107,
                             0.664488174834645,
                             0.6897317665852637,
                             -0.008774731192519907,
                             0.625642551337688,
                             0.6862410821653743,
                             -0.7670419540542446,
                             -0.48055390132433673,
                             1.573964677292243,
                             -1.0555745677813198,
                             1.3159721165631335,
                             -0.9138502497993347,
                             -0.4641566084568084,
                             -0.06966564000847691,
                             0.039012746519005555,
                             1.0827641697579717,
                             1.2190093772283843,
                             1.5391150382377632,
                             -0.6675581491933664,
                             0.09104662559232415,
                             0.5283818781579022,
                             0.04478118244404365,
                             1.5836075485651113,
                             0.251932560991488,
                             -0.3949560381811925,
                             0.8854639388207172,
                             -0.8611267164776638,
                             1.566928747241295,

```

-0.15445151319573347,
-0.01396050433941564,
0.8172620115356718,
-0.014658132241289147,
-0.6623010385304499,
0.6299284572753584,
-0.15026947869759505,
-1.2293112897827836,
-0.5768656731979434,
1.231557922062332,
-1.447872926116422,
1.4918722706131171,
0.31945486783385635,
0.43870804207253483,
1.3582636256308331,
-0.8478398442626237,
0.799518029891469,
0.5539369896048497,
0.13453197415145846,
0.7167498104862234,
-0.06589109804263343,
0.041038724923082276,
1.151975756926885,
-0.052026838729968425,
0.3952978000808587,
-1.1754748970457833,
-0.01965607042871015,
-0.2456324144680547,
-0.01136184217607491,
0.022503168314325715,
1.1777923723104653,
0.14554117253938412,
-1.2287946167385297,
-0.9683737383161766,
1.073329622962756,
-0.7143583802799215,
-0.3545002682598469,
-0.31906110571926777,
-0.9597094503848127,
0.8825903987559673,
-0.08273415521500459,
0.2895039435279609,
-0.3388594197258312,
0.2406459434933063,
0.3824128671502841,
-0.5390555829405563,
0.3757684495816124,
-0.5015719636333678,
0.3958257688275076,
-0.6428354095046468,
-0.9178624992467067,
0.29327306512581747,
-0.09193285724698198,
0.1770091195095364,
0.6803523512851801,
1.05525770275555,
-0.4646437845139076,
2.2560908642344026,
-0.7500939262299234,
1.92959092052175,
-0.36895777993274226,
-0.04538283416771957,
1.2529586558275392,
-0.57805717638909,
0.3080585810325818,
0.031910210614104206,
0.18519950010488512,
1.3888566879174211,
1.087260914864616,
0.15259355623633358,
-0.08810167996863522,
0.07467874103808375,
1.4444893382870363,
-1.446220680213735,
0.8215180856176293,


```

2.0151723649306295,
0.13454991965456886,
0.21923381008732426,
0.31112145477192793,
-0.8977876743954992,
0.10239876571985529,
-1.0132922278363503,
-0.3794863035801566,
0.8437803973608401,
1.111671691788496]],
'noise_profile': { 'noise_on_init_param': { 'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noise_free_time_evo_ric2_sample#2'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.51493088468928],
[12, 6, 0, 0, -5.310706408751286]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 3.361701965332031e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -7.117043122658277,
'others': { 'beta_coefficients': [1.5, -0.5],
'cost_richardson_zne': 2.5,
'sorted_expectation_vals': [ -6.51493088468928,
-5.310706408751286],
'sorted_noise': [6, 18]},
'sampled data': [ [4, 2, 0, 0, -6.51493088468928],
[12, 6, 0, 0, -5.310706408751286]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [ 0.46756606893882113,
0.46983859896604685,
0.5133985927484581,
0.5133985927484581,
0.5133985927484581,
1.4016794842576101,
1.6914116785996751,
1.691411678599675,
1.6941756924801525,
1.6941756924801528,
3.749864451084885,
3.749864451084885,
3.749864451084885,
3.749864451084885,
4.969636107010246,
4.969636107010246,
4.969636107010246,
4.969636107010246,
5.983107385388112,
5.983107385388112,
8.378949246097575,
8.378949246097575,

```

8.635680633753974,
8.635680633753974,
9.108807770997622,
9.108807770997622,
9.108807770997622,
9.960036411132767,
10.436643041587223,
10.716533289695375,
0.5466515694267952,
3.318518267704871,
0.40678728382369284,
0.6290091030083763,
-0.4770983795300032,
-0.7455483390932742,
0.1630064768968011,
0.4889724943271746,
0.10596004146863507,
-1.4759536729524354,
0.957604974713893,
0.6004514101153323,
0.0725728265044025,
1.4994700617905987,
0.15508641282397265,
-0.15494303336158094,
-0.5663845144966579,
2.6663065398173496,
-0.27086336898318036,
-1.1828784814640632,
-0.295045685212881,
0.6236994649007899,
1.1769360493970402,
1.7159795997719172,
-0.3019600780315931,
0.651545330446025,
-2.0782382222089724,
0.3186344830290561,
-0.2815476257680432,
0.4067635126151388,
-0.8228051832090519,
-0.4154478828430832,
2.3723163321732827,
0.931436930203099,
0.9804374423714106,
0.058895326584719035,
0.3167253661454302,
0.3390503539155689,
-0.5024758650485814,
-0.2425872659725522,
-0.6703364135359525,
-0.147558399690948,
1.1644296968010859,
0.5475507579506065,
0.1736068872874216,
0.30547527216138975,
0.5300773267023443,
0.7722201656261689,
-0.6858549484151734,
-0.16073113081558088,
-1.197458252004488,
-0.507307082981415,
-0.16437738354723153,
0.2647407403580543,
-0.45930133674563567,
-0.22673992276266391,
1.8127715449849986,
-1.4334128748253496,
-0.759404859696163,
1.2431110601081345,
0.8953271645792334,
0.6963251389999259,
-0.23147129445607595,
0.3638861144885483,
2.2394622330281906,
1.543816783446588,
1.7535830084845025,

```

-0.3169637330795974,
0.2453194636587019,
2.0076391670843647,
0.286017506499548,
0.6600449842625801,
0.19782118041633945,
0.05463577684798935,
0.4755005320975566,
0.055207215781380096,
0.06969685609102444,
0.2645328829673777,
-0.01235845210357445,
1.01982876424508,
2.262492417600915,
1.3443784687440867,
0.10698144982242018,
0.428081928456963,
-1.5494921893757727,
1.0701294130786736,
0.39987519173641656,
0.06301738855484718,
-1.6711192593807125,
0.26920159401030147,
-0.08358711295948693,
0.4054786916979446,
-0.5013194648307361,
2.126550690771933,
0.10959959792145137,
0.4205360694942305,
-0.586878230803869,
-0.3381608637889511,
-0.5055401953468239,
-0.45995744817477163,
0.07999668782445096,
-0.4269018063306222,
0.9588276598807326,
1.042187521522261,
0.07821502449201799,
0.6777167152964713,
0.4177387822172794,
-0.18721284346239236,
-0.8064458441104513,
2.8712851588196218,
0.7141862154478698,
-0.9976723444987857,
-0.8529435737233043,
1.9602636639541733,
1.2634521849205176,
-0.7170870121313951,
0.31227949023655405,
-1.0451293221946831,
-0.6787852158143914,
1.8581833428208436]],
'noise_profile': { 'noise_on_init_param': { 'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0, 0],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric2_sample#6'},
'redundant': { 'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.4878948983695235],
                          [12, 6, 0, 0, -5.297876077058435]],
        'degree': 0,

```

```

        'method': 'richardson',
        'sampling': 'default'}}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.9591064453125e-05,
            'zne_values': { 'degree': 0,
                            'extrapolated_value': -7.082904309025067,
                            'others': { 'beta_coeffieicients': [1.5, -0.5],
                                         'cost_richardson_zne': 2.5,
                                         'sorted_expectation_vals': [ -6.4878948983695235,
                                                                       -5.297876077058435],
                                         'sorted_noise': [6, 18]},
                            'sampled data': [ [4, 2, 0, 0, -6.4878948983695235],
                                                [12, 6, 0, 0, -5.297876077058435]],
                            'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                         'r': 0},
                                      'time': {'max': 10.0, 'min': 0.0},
                                      'type': 'ising'}}},
  'init_param': { 'value': [ 0.4319322451408806,
                              0.4441934852523559,
                              0.5878315575931067,
                              0.5878315575931067,
                              0.5912427958750637,
                              0.5947187380620651,
                              0.5947187380620652,
                              1.5628074958209335,
                              1.568718083448373,
                              2.013035808837258,
                              2.013035808837258,
                              2.013035808837258,
                              2.0277481086167364,
                              4.114573484042011,
                              4.1154343976538055,
                              5.358852938788934,
                              5.358852938788934,
                              5.549639110185835,
                              5.667644288942511,
                              5.6706149635077026,
                              7.063797010237488,
                              7.658675772413245,
                              8.123010743902626,
                              8.633070512181217,
                              8.7849794748515,
                              8.7849794748515,
                              10.749769122252356,
                              10.89899813672382,
                              11.413879450074099,
                              11.413879450074099,
                              -0.04151001581527148,
                              -2.751209241920508,
                              -0.5056705094410174,
                              -0.406548832581749,
                              0.8633340895103881,
                              0.17259893989954428,
                              -0.21235924407313156,
                              -0.5485334561985523,
                              0.2012951527025238,
                              -0.23858034563229433,
                              2.1364176057207223,
                              0.7817014982866842,
                              -0.1108899918027214,
                              0.6048531661014985,
                              1.809261328558185,
                              -1.1786415575821785,
                              -1.0827170099436865,
                              0.4109384498547832,
                              0.8090945308822337,
                              -0.9356585252219308,
                              0.28697429006677655,
                              -0.15332459566133277,
                              1.2998079093160566,
                              0.3870967504567973,

```

-0.2413465423953762,
1.001771196320206,
0.1138394509020578,
1.3220786271535379,
-1.1806565063083097,
2.2789659185869464,
0.11875741943281123,
-0.15631084464392242,
-2.14680662310284,
0.18035127914430246,
-0.260880073181003,
0.21031876899961,
0.7641152777599183,
1.203568210035907,
1.2767673297334832,
1.2585129786889195,
0.24752046596725819,
0.7203551788145919,
-0.17420447991641552,
-1.3486030918499698,
0.2145369555808404,
0.8896874485404291,
-1.6470089821954121,
-0.37928143051952584,
-0.29903232984244293,
0.6984471945997809,
0.14906235834494008,
0.054873949093549826,
1.3877615433857144,
0.04193983551007985,
1.2014028569521673,
-0.02338341588216991,
-0.05255422822816251,
-0.1905875093002492,
0.16425167437144692,
0.2542835721118732,
1.1822299841866795,
1.2032222480152739,
0.30142488752060526,
0.09125464470109583,
-0.45638140154025475,
0.050260497156630585,
-0.04000614529085093,
0.9096133936431033,
-0.03555378439095736,
0.4405325841695877,
-0.48129885373423525,
-0.3617221403442356,
-2.5216506086368518,
-0.5081934521569421,
0.041332600726076626,
0.7413771443671474,
0.2204029987501584,
-0.7517920276590782,
-0.21022249714251526,
1.5607830526955413,
0.16615947769508344,
-0.17023919527139658,
-0.4295619963603494,
-1.1838803442651602,
0.6319564057885301,
-0.3935277768061919,
0.005550337258741166,
-0.4226065324972882,
-0.3715846249960284,
-0.48395498614671884,
-0.4790957128536121,
1.4427293239187895,
0.5875094653073935,
1.404722056846091,
0.6973465258635152,
0.8623508528879442,
-0.5460563718875903,
0.8326491581513842,
-2.4015389244466907,

```

-0.04583861440999851,
-0.36265388938085896,
-0.5347868312224912,
0.49674658852292397,
-0.6488379607428911,
1.1568053403721041,
0.30662269781395407,
-2.013151608263867,
-0.17383275003878457,
0.5036662539248705,
1.9638910157847163,
3.118155075975211,
0.8117246453310248,
-1.4517859264576565,
-1.4946803709676548,
-2.7792982516572895,
1.1174424252005506,
2.333430423546859,
0.6903927508165344,
-0.2883899927957993,
-0.8542826896862586]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric2_sample#7'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.662379321078875],
[12, 6, 0, 0, -5.394498644176584]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 3.886222839355469e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -7.296319659530022,
'others': { 'beta_coefficients': [1.5, -0.5],
'cost_richardson_zne': 2.5,
'sorted_expectation_vals': [ -6.662379321078875,
-5.394498644176584],
'sorted_noise': [6, 18]},
'sampled data': [ [4, 2, 0, 0, -6.662379321078875],
[12, 6, 0, 0, -5.394498644176584]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [ 0.23818326834520384,
0.242015507589488,
0.5292100600091615,
0.5598196968290768,
0.5598196968290768,
2.3451854381830244,
2.3451854381830244,
3.8453025861748578,
3.8453025861748578,
4.395137112369032,
4.408833101975425,

```

4.829585475162733,
4.867115107997544,
5.058958687371625,
5.058958687371625,
5.058958687371625,
5.581243202757385,
6.011433944644334,
6.011433944644334,
8.358003932454157,
8.358003932454157,
8.358003932454157,
8.371420896509267,
8.783744346022278,
8.783744346022278,
9.34189587380724,
9.34189587380724,
9.39608576624051,
9.726727464435275,
10.95102910414735,
0.14721560345391302,
1.0205767628001141,
0.40564926087548847,
-1.9238321605823567,
-0.04729505851188324,
0.482541600628026,
0.1152155677775581,
0.8524320040134185,
0.42380767990264745,
-0.6219665889019169,
-0.05759638705357143,
0.8161328242190486,
0.7425402472775383,
0.7535392998290764,
1.8314745216576223,
1.3334023135030668,
0.05552193989453376,
0.9691374289368472,
-0.07338372949421543,
-0.6690664310785921,
1.168533104165511,
-0.01962836639606316,
0.6576122461884292,
-0.037637936325233226,
-0.0009647519828692927,
0.4524839512418326,
-0.013115244288784337,
1.6270372625341938,
1.182284078721198,
0.9533845712158122,
1.0956662941001023,
-0.9647918240366662,
0.1897778948773928,
-0.04047520420354409,
0.5790278460715794,
0.631719469583885,
-0.3335418357294817,
1.0603186443819097,
-0.7449471003423949,
0.1630231005006795,
-0.07563007210422526,
-1.1009042311153838,
-0.34587951137957407,
-0.22408475871037167,
-0.6364303490666092,
-0.5245531829065874,
-0.7807390048523051,
-0.7767575864285188,
-0.9679768325508518,
-1.8691843526791756,
0.4783530568164554,
0.6133546874260105,
-0.20020986269221935,
-0.41532071148562183,
-0.8557118152516873,
-0.16985308585729458,

```

0.8428338023785568,
0.6750667051632169,
1.2795031602880573,
-0.41384224674171627,
0.3213362409234008,
1.2340309605381181,
1.0506369357289735,
-0.6415500741723684,
-0.9628453188140567,
1.935214357876566,
1.4159154638103797,
-2.0864443113151867,
-0.07426020994129089,
-1.0538315394110076,
0.2589745455351295,
-0.2955978932252484,
-0.030644528099012087,
-0.8602288471584197,
-0.09490606960216369,
0.4190622694749842,
-1.5667390101922392,
-0.7804920738383083,
0.1255849265457167,
0.5884338319364165,
-0.46028089859616994,
0.2488277677353687,
-0.7513271615815256,
-1.0561258734551267,
0.3697237180501531,
-0.7108432439071602,
0.5137418203820399,
0.946767478727941,
1.619957961978311,
1.2944079622561346,
0.21625880379379211,
0.007168703139473973,
-0.334764182643075,
-0.9086121674985949,
-0.8247776105092918,
-0.5734460393428641,
-0.4381826298821608,
-0.44306191141826123,
-0.23860269421390362,
-0.03343496826073402,
-0.008972211649351379,
1.2196121588758198,
1.5985131140815898,
0.060325450296581484,
-0.7312434635304047,
0.9413542695101463,
-0.6701271691421761,
-1.0680925542186157,
-0.5192506372311365,
1.5594784335685845,
0.17928043514789443,
0.2455246384495121,
-0.053106843359071614,
0.7488220810664211,
1.3030207698426615,
-1.787387332504871,
-1.1184519607509558,
-0.2511275398302805,
1.445676213762956,
1.5991220174606833]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0, 0],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix':

```



```

'AUTOMATE_ising_noisefree_time_evo_ric2_sample#4'},
  'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
  'run': 'zne',
  'state': 'dmatrix',
  'vqe': { 'iteration': 1,
    'optimization': { 'algorithm': 'SLSQP',
      'constraint': True,
      'status': False}},
  'zne': { 'data_points': [ [4, 2, 0, 0, -6.630764461698827],
    [12, 6, 0, 0, -5.461578440310041]],
    'degree': 0,
    'method': 'richardson',
    'sampling': 'default'}},
  'output': { 'exact_sol': -8.566772233505624,
    'run_time_sec': 4.00543212890625e-05,
    'zne_values': { 'degree': 0,
      'extrapolated_value': -7.215357472393219,
      'others': { 'beta_coefficients': [1.5, -0.5],
        'cost_richardson_zne': 2.5,
        'sorted_expectation_vals': [ -6.630764461698827,
          -5.461578440310041],
        'sorted_noise': [6, 18]},
      'sampled data': [ [4, 2, 0, 0, -6.630764461698827],
        [12, 6, 0, 0, -5.461578440310041]],
      'sampling': 'default'}}},
  { 'config': { 'ansatz': { 'gateset': 1,
    'layer': 30,
    'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
      'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
      'r': 0},
    'time': {'max': 10.0, 'min': 0.0},
    'type': 'ising'}}},
    'init_param': { 'value': [ 0.48023040681030116,
      0.7029544315470944,
      0.722671987246733,
      0.9307929858945541,
      0.9307929858945543,
      2.546598088688794,
      2.546598088688794,
      2.546598088688794,
      2.546598088688794,
      5.184334435330857,
      5.195111458618343,
      5.47436411399969,
      5.47436411399969,
      5.47436411399969,
      5.47436411399969,
      5.47436411399969,
      6.068336244524409,
      6.068336244524409,
      7.838605251840715,
      7.912250316107233,
      7.912250316107233,
      8.562517202814183,
      9.063385768597811,
      9.216218862905647,
      9.216218862905647,
      10.84047371877396,
      10.84047371877396,
      11.044358714867027,
      11.375406566932424,
      11.375406566932424,
      0.18886549713610382,
      0.6040291443391601,
      -0.9727809979878402,
      3.008209947509642,
      0.3239743149147096,
      -0.643987619924633,
      1.0739882269552894,
      0.4935103262912805,
      -1.1713673989643745,
      0.32339055167680386,
      -0.8254170399860232,
      3.387204293956005,
      0.5437302598099015,

```

-0.6585961721148387,
1.0903163929414719,
-1.5261906384818182,
1.102349928722959,
-0.7373436929054379,
0.1712635196093208,
1.2086697581514199,
-0.4977298668106944,
-0.8828019932256734,
-0.13051782426094366,
0.048744185631251574,
-1.0535030780910155,
0.5224555306604155,
-0.9502111367985572,
-0.2573420205933925,
1.5833039752620888,
1.3806480812363842,
0.3864833677006273,
-0.0006284730712113316,
0.6487740408134013,
1.697829570896293,
-0.20023867528920775,
-0.32438662287273723,
1.3674639807988553,
-1.5592347723565667,
-0.6294582959326964,
1.11157394071659,
1.4895912958038178,
-0.7203901402661329,
-0.027076706863032294,
0.009574786781832467,
1.1437695218822916,
0.11720143179507149,
1.0973549625545624,
-1.0756168303605034,
-1.5785125289756954,
-0.42999007233668873,
0.7592603373612552,
0.7357293999316703,
0.03766014972468404,
1.450035910180884,
-0.08978031589044846,
0.7176267391661779,
-2.3377280976646966,
0.3878354316208249,
-0.8070070584177742,
0.07084117333557359,
-0.008998168433091903,
0.6612796195456235,
0.6287929626789847,
0.8601333603228948,
-0.7786354708155643,
-0.144476679155352,
0.39492950957300044,
-0.20012968159779548,
0.01326107241550192,
-0.5649340596325207,
-0.020398755490040426,
0.6333499066300691,
0.400844220606507,
0.5954749034740784,
0.8700413681534388,
1.315868035063538,
0.9914334712755634,
0.10418309057983809,
1.1095991500751576,
-0.1712743534684691,
-0.2730581386141925,
0.1453142903679217,
-0.3813955795194703,
0.1280814941289577,
-0.5627986213807415,
-1.0576722394029623,
-1.98721560251358,
0.519862592148374,


```
'init_param': { 'value': [ 0.5118290521669792,  
0.9034573591840084,  
1.0514460879099101,  
1.1283814725964603,  
1.1293980629167455,  
1.6874068541059999,  
1.8362001710030256,  
2.594529848453243,  
2.5949023258749775,  
4.384420392533986,  
4.804388533064279,  
4.8044974726668155,  
4.925049921883437,  
4.925049921883437,  
6.055903002618935,  
6.0650337145794,  
6.06510024924717,  
6.06510024924717,  
7.719643893437192,  
7.719643893437192,  
8.110116534451196,  
8.374448879129652,  
8.374448879129652,  
9.29045698473019,  
9.2929923765699,  
9.977025927736127,  
9.977025927736127,  
9.977438303479875,  
11.277250876863835,  
11.588901828478656,  
-0.5970483257864091,  
1.048395642841534,  
0.3563299433581316,  
2.04182497417867,  
-1.0659189653240388,  
-1.5711224255679066,  
-0.6854026923334804,  
1.3002799990574343,  
0.334979527312172,  
-0.41771661057992626,  
-0.5255960701407614,  
0.9200835209290303,  
0.5840294212204432,  
0.4309612994747514,  
0.010788324288593443,  
0.8398650983618882,  
0.35555521736822654,  
0.33383672576306345,  
-0.03972798241435421,  
-0.5808926346218187,  
-0.6217506829607715,  
-0.022449055124049946,  
-0.4671428723737863,  
1.3840583406465645,  
3.321823997583478,  
1.9516123562000514,  
2.109430068020724,  
0.17138900458361278,  
1.0734765554988877,  
1.1591682463434823,  
-1.5085197494005436,  
0.3709514368460093,  
-0.08354812013321816,  
-0.6849414455501801,  
-1.1807064172516142,  
0.1889544406654655,  
0.45125852432694835,  
0.40946867023093825,  
0.9591749175330944,  
0.862406962512403,  
-0.47565451021668054,  
0.14791469386649075,  
0.012323642821740255,  
-0.3613715829994197,  
-0.6520104899254531,
```

1.1500641632479522,
0.11480411675389499,
0.37929728325133344,
-0.12818017639377846,
1.5548403554359806,
-0.08803696907532751,
0.16058007713541472,
-0.2835300528775952,
1.4317946638041426,
0.04218231832431045,
0.6164913379423986,
-0.5546192578057587,
-0.0016176435191898148,
0.3455446109244303,
-0.7983090776945799,
0.06548494602750325,
0.09601011255154575,
-1.274882751616874,
-0.6639296860891427,
0.08468574728059233,
0.09178155492125434,
-1.9746588496307496,
1.363083624655819,
-0.11754172722968637,
-1.7454296260872237,
0.689437867548643,
0.09536601432398076,
2.1875690275532618,
1.9737367040865754,
0.19623670188051764,
1.3618525198389628,
-1.6551377680800423,
-1.2796187335858575,
-0.02979150349567955,
0.3655877448114123,
-1.339933806945279,
-1.6131280858420662,
1.1268021774842836,
0.7853899025443938,
-1.214954346638235,
1.881374469676006,
-2.2108375055617597,
0.38299461206746305,
-0.062378913993248956,
-1.0812916442950207,
0.19113842494911862,
0.43947065709017896,
1.4340329690981122,
-2.1914706359927933,
1.683520172253517,
-0.4151753501170658,
0.2652280013518245,
-0.09829123134965576,
-0.4033858255054015,
-0.09488290407421365,
1.1087820722198931,
1.3758250784849964,
0.3693308876448478,
-1.4774754586810708,
0.45790576106810227,
-1.318961759734687,
1.2898775251023542,
0.5291693857421421,
-0.7225860054677886,
-0.6024865845145562,
1.1258609582335735,
1.4545810101615688,
0.4988359655120556,
0.39968483637691726,
1.4778005154845164,
1.6358502222921723,
0.07081811420225437,
1.5336807499569445,
-0.7595143926860392,
0.07947795439670366]]},

0.7852252158845567,
-0.4957764525834907,
-0.3354719566017423,
0.16097458435951068,
-0.21971891747261646,
1.5370438656668401,
0.26531517225969137,
-0.6386119707402834,
0.37748437574064975,
1.2333814560270282,
0.5934062959304571,
1.633059089666515,
0.31170790405459564,
0.3995146629536743,
-0.11155094016301184,
0.8138063699882787,
0.019467271717771087,
-0.38186317682975884,
-2.778803401027002,
0.7887082829537533,
-3.1348050840084074,
-1.5109224736536222,
-0.38589710406642097,
-1.105927053527023,
0.3451707237315739,
0.6290043291617069,
-1.834722402137916,
-1.0551446159380258,
-1.3581260193610523,
0.9666180881078981,
-0.05759676522292653,
1.1866392284178628,
-1.1308471740259582,
0.34327443183487805,
0.17662613760194443,
-0.043135596735193626,
-0.03667546733606037,
-0.4901895513664937,
-1.0399227541005371,
0.4880459765792483,
0.053089003166299456,
-0.6259966396796703,
0.03159579372412852,
-0.09798805529491862,
0.052872660107769434,
1.490194875693949,
0.22167423812356596,
-0.33497064714209324,
1.0144679614160885,
0.01608415027143934,
0.49167899744128424,
0.12491439548600353,
-0.1329609067962543,
0.8639852596635488,
-1.6607871811173882,
0.2801513494447602,
0.3908501333463579,
0.6044891435983025,
0.08637544621327581,
1.2270127042754213,
0.11860545946003472,
-0.13066468162372125,
-0.998222267933649,
-1.2643438207146234,
0.7835210648968657,
0.8370312354478452,
0.06527740704106412,
-1.0235950282739155,
-0.705070781232052,
0.037682760290170314,
1.4778132318704873,
1.7053977225841355,
-0.1291331271719845,
0.1648924115529835,
2.208642048449397,

```

1.1885953055185114,
-0.8138037079028133,
0.6715189891287302,
0.7819572861394978,
-0.4022064636590845,
-0.3927127310055937,
-0.4118784323717791,
1.3463239978578432,
-0.3323606385924751,
0.6664683657180488,
1.58852177854836,
-1.212203169826861,
-0.031436536468697034,
0.5094262403458202,
-0.0616688967239856,
-0.0666169691754888,
-0.7277012125404719,
-0.12684196683036533,
0.5325519345188151,
2.2778420441533447,
0.8735933622003313,
0.9907279467768721,
1.1448787224294625,
0.4291077507842401,
-0.21356701746617238,
1.7174041100732933,
0.1885112766205208,
0.3630143936992631,
0.6395640633025061,
-0.2937369198596852,
-0.18128670495631546,
0.9504587478277511,
-0.6241293783488098,
1.8733903679237267,
0.4875344831143031,
-0.9182747577406853,
0.850791268752179,
-0.46183623376516114,
-0.634145646812118,
-0.9165900492445601,
-0.8973746003077631,
-0.879107286657044,
0.15645801311276636]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric2_sample#8'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.672780190342767],
[12, 6, 0, 0, -5.522238922989663]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 4.00543212890625e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -7.248050824019319,
'others': { 'beta_coefficients': [1.5, -0.5],
'cost_richardson_zne': 2.5,
'sorted_expectation_vals': [ -6.672780190342767,
-5.522238922989663],

```



```

'sorted_noise': [6, 18]],
'sampled_data': [ [4, 2, 0, 0, -6.672780190342767],
                  [12, 6, 0, 0, -5.522238922989663]],
'sampling': 'default']}]},
'ising-ric3-uniform-scaling': { 'noiseoff': [ { 'config': { 'ansatz': { 'gateset': 1,
                                'layer': 30,
                                'ugate': { 'coefficients': { 'bn': [ 0,
                                                                    0,
                                                                    0,
                                                                    0,
                                                                    0,
                                                                    0],
                                                                    'cn': [ 0.5,
                                                                    0.5,
                                                                    0.5,
                                                                    0.5,
                                                                    0.5,
                                                                    0.5],
                                                                    'r': 0},
                                'time': {'max': 10.0, 'min': 0.0},
                                'type': 'ising'}},
                                'init_param': {'value': 'random'},
                                'noise_profile': { 'noise_on_init_param': { 'status':
                                                                    'value': 0},
                                                                    'noise_prob': [ 0.001,
                                                                    0.001,
                                                                    0.001,
                                                                    0.001],
                                                                    'status': False,
                                                                    'type': 'depolarizing'},
                                'nqubits': 7,
                                'observable': { 'coefficients': { 'bn': [ 1.0,
                                                                    1.0,
                                                                    1.0,
                                                                    1.0,
                                                                    1.0,
                                                                    1.0,
                                                                    1.0],
                                                                    'cn': [ 0.5,
                                                                    0.5,
                                                                    0.5,
                                                                    0.5,
                                                                    0.5,
                                                                    0.5],
                                                                    'r': 1},
                                'def': 'ising'},
                                'output': { 'draw': { 'fig_dpi': 100,
                                                        'status': True,
                                                        'type': 'png'},
                                'file_name_prefix': 'ising_noise_off'},
                                'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                                                    [1, 1, 0, 0]]},
                                'run': 'vqe',
                                'state': 'dmatrix',
                                'vqe': { 'iteration': 10,
                                        'optimization': { 'algorithm': 'SLSQP',
                                                            'constraint': True,
                                                            'status': True}},
                                'zne': { 'data_points': None,
                                        'degree': 1,
                                        'method': 'richardson',
                                        'sampling': 'default'}},
                                'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
                                                                    'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
                                                                    '4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
                                                                    'X 5 X 6 + 1 Z 5 + 1 Z 6',
                                        'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 + '
                                                                    '1 Z 0 + 1 X '
                                                                    '1 X 2 + 1 Z '
                                                                    '1 + 1 X 2 X '
                                                                    '3 + 1 Z 2 + '
                                                                    '1 X 3 X 4 + '
                                                                    '1 Z 3 + 1 X '

```

False,

'4 X 5 + 1 Z '
'4 + 1 X 5 X '
'6 + 1 Z 5 + '
'1 Z 6']},

'output': { 'exact_sol': -8.56677223350563,
 'initial_cost_history': [3.7901552515724086,
 3.396779462430669,
 3.6416626483897634,
 5.191980344851134,
 3.300383519902229,
 4.2302694452267025,
 5.5832854157035126,
 3.424340408479732,
 4.575860085354091,
 3.843028199719871],
 'noise_details': { 'gates_num': [4, 1, 1, 0],
 'identity_factors': [0, 0, 0, 0],
 'noise_level': [0, 0, 0, 0],
 'noise_profile': {

'noise_on_init_param': { 'status': False,
 'value': 0},

'noise_prob': [

0.001,
0.001,
0.001,
0.001],

'status': False,
'type':

'depolarizing'},

'odd_wires': 4},
'optimized_minimum_cost': [-7.3566708947935595,
 -7.585318923493526,
 -7.222021821934782,
 -7.637732542535524,
 -6.9717347656162385,
 -7.183899566070664,
 -7.312238847468353,
 -7.55546648538883,
 -6.871387778407254,
 -7.103750879845002],
'optimized_parameters': [[-8.981486547411966e-18,
 0.2355269100006643,
 0.6734351680867104,
 0.7281872519006076,
 0.8287057436356319,
 0.9968962697957876,
 0.9968962697957877,
 0.9968962697957875,
 0.9968962697957875,
 1.6732799868313266,
 1.9939389913716405,
 1.9939389913716405,
 1.9939389913716405,
 2.7786795690350794,
 2.7786795690350794,
 4.834741825646661,
 4.843194997300159,
 5.442664851556481,
 6.006041306704694,
 6.006041306704694,
 6.42159854018659,
 6.42159854018659,
 7.172079495244572,
 7.835006027334925,
 7.835006027334925,
 7.835006027334926,
 9.27390458565848,
 10.142272229921797,
 10.547148172870333,
 11.945646083493756,
 0.30631338507375855,
 0.950240081652421,
 0.3197394260348401,
 -0.8989315334324052,
 -0.19972246213391193,

1.1515465399564957,
0.5794269610995502,
0.029234461946744308,
-0.3033609133811924,
-0.5068802243159589,
0.41891242044729254,
1.2272962791271989,
0.3217023759538914,
-0.0389595358377685,
-0.2818159573379153,
0.40344301100056285,
0.012601476380185236,
-0.7249227034698249,
-0.28859136632269766,
0.9177183017829931,
0.3192674751581855,
-0.6960323093564068,
0.9954617266198262,
-0.0960673339612117,
0.30357952531427623,
0.9148253966896076,
-0.41157308092318357,
-0.2244725515886058,
0.900815469140735,
-0.8704821270800919,
-0.2201886465573435,
-0.058451834208670006,
-0.5564504264806815,
1.0028418241392123,
0.6862825176439941,
-0.07515432929364027,
-0.132716959754057,
0.3156193912732197,
-0.5571752224261073,
-1.1266280252389607,
-0.2496012410462972,
0.4568813812573679,
0.5040642955504137,
-0.27382094178370653,
0.738802930675346,
-2.1017467172452315,
1.7134969759675782,
0.19045289169065632,
0.4959480509883087,
0.2725041426894662,
-0.020113534287112555,
-0.27730132318570966,
0.9121222342974794,
0.7771919661145833,
1.2410984938270941,
0.698373852916785,
0.049079898076347495,
0.4715079446630061,
-0.0012225173391828686,
1.2380103358912495,
0.18126256323161893,
0.02822806167971892,
0.9624172506978782,
-0.1017646649934255,
0.10712069411267353,
0.11059700546555196,
0.1764728469149672,
1.0536625920020957,
-0.43614672067179516,
1.1290800077402758,
-0.7544321351970764,
0.20178203991689547,
0.545960403958859,
-1.271331788368515,
0.3827893670869708,
0.514944994226915,
0.9987612988518954,
0.9130737051848579,
-0.33354580533033523,
0.4110572637290035,

-1.0065548546569214,
0.05504586363332423,
-0.8123464439408392,
0.9622209015425294,
-1.7981345279534349,
0.8489846299822246,
1.9191971402543575,
-2.019266402915899,
-1.6151954323496909,
1.693158829823371,
0.8368996699819713,
-0.7925788637806501,
-1.416264150106769,
-0.7484853165625206,
-1.026066230288144,
0.2591646216069995,
0.9485984853376032,
1.624644962520762,
0.1389264160191751,
1.7901079295930897,
0.28173803525757823,
-0.06727915670909099,
-0.39998196135024877,
-0.911581975644718,
-0.8802717995009537,
-0.6496644186002655,
0.9474433579476823,
0.5810351318438188,
-0.5928504020318089,
0.40429180463166087,
0.19955746243713188,
0.14940534343654552,
1.148493268597809,
-0.6171972430901704,
2.358530954611547,
1.6132804954395525,
1.5230819806307347,
0.07100911239249047,
-0.9821849785093056,
-1.2186503325370892],
[0.43242608326371745,
0.47036005705701023,
0.6035896599876003,
0.6406575109863006,
1.589410377648682,
1.589410377648682,
1.589410377648682,
1.9902163938334159,
1.9902163938334156,
2.6096176446013035,
2.6096176446013035,
3.039961339509649,
3.039961339509649,
3.0408788822093955,
3.0408788822093955,
4.1325863285768,
4.1325863285768,
4.801738885893826,
5.558430388585627,
5.558430388585628,
6.48703427557745,
6.599293033589678,
7.119130597501129,
7.119130597501129,
8.096879981584715,
8.096879981584715,
11.989689272176165,
12.289334556064002,
12.289334556064002,
12.695171119049377,
-2.1320303145314323,
1.164574641976517,
0.2281426394568994,
0.5098216286601104,
-0.15959745874570574,

-0.06246168319786304,
-1.335183444898994,
1.1253611654826523,
-2.453373641849758,
1.542670605477509,
-0.23459513075014657,
0.7629240608803527,
0.7066862387679962,
0.023354672976759586,
-0.26545157517882206,
0.8070755399487665,
0.621829996842247,
-0.8206657495436546,
-0.6456938685697818,
-0.8432536462697657,
1.5481421634749726,
0.030458967225015475,
-0.6362611752642527,
-0.5779049484598997,
0.15136987691256515,
0.37940285865801,
-0.13025289482332936,
-0.6088352960517447,
1.0137338691570092,
0.9312563664781934,
-0.06260332826127091,
0.16422266217724538,
-0.44919022316429513,
0.7120810093352475,
-0.045811140982536094,
1.4693096124875311,
1.3068674578470139,
0.3099696428239748,
0.05726716998373588,
1.4138231816748696,
1.3251150202531472,
2.4661338582136705,
-1.9221360663147242,
1.6309773912120944,
1.012572768899168,
1.1304323717308846,
-0.10014704960087417,
1.2138621646932324,
0.9150507542177275,
-0.1931179153053463,
-0.19983886142442753,
-0.37113288193216537,
1.0783926703777422,
0.7626802083907843,
-0.5396621733108105,
-1.0558614970998554,
-0.5031372556139323,
-0.1733591918410679,
-0.05810492443974475,
0.15252588011535415,
-1.3613790038949474,
0.14268023162896296,
2.1247307661901087,
-0.2996908036461345,
0.2682896527620792,
-0.6487468384363545,
0.0174900962257196,
0.7865320417875226,
-0.4365770411878935,
0.12280152026315115,
0.2668485276831852,
1.0055493052017264,
-0.13783144983938653,
0.16428177903351826,
-0.4107244428852575,
-0.5006989958309392,
0.4728442136126457,
-0.5756228808846314,
-0.2555605201147205,
-0.6655897533222209,

0.27354627255859315,
1.056546145946755,
-1.0108463744632321,
0.8570822622523233,
1.1003650202540056,
-0.13364446072464353,
0.19686136380997882,
-2.015333107317771,
-0.3474605408709381,
0.5127772990877489,
0.8873132714146144,
-0.1538651896386833,
-0.49691154598038334,
2.4337764506921564,
0.11687121421819974,
0.2637773919076603,
1.1119719051108947,
-1.1407171454187541,
0.293218856784229,
0.9089141681578352,
1.0587297353590328,
0.41354403585027527,
0.26331197174432663,
0.04763876798960274,
2.8376965303594455,
0.038400591301352385,
0.732378529151854,
1.259237861002353,
-0.09371603030550141,
-2.6524545055620123,
0.166545686266708,
0.8383948517861464,
0.5625425863873136,
0.6352389159281269,
-0.3793199251622615,
1.4353166704457385,
-0.06088962240819872,
0.5575840323453971,
-0.4370866693930651,
-1.3477496560670819],
[0.5485274765250757,
0.6924497584354614,
0.77926902254116,
1.4968846113254817,
1.4968846113254817,
1.617988193716464,
1.617988193716464,
2.3157329555892914,
2.3157329555892914,
2.3157329555892914,
2.6599448622122046,
2.6599448622122046,
3.198693955827293,
3.198693955827293,
4.93646112072991,
5.0183990860356085,
5.166530398168251,
5.732062846348599,
6.452081956191476,
6.950653712152026,
6.950653712152026,
7.527182443059662,
8.141470761550243,
8.141470761550243,
8.23161566953153,
8.23161566953153,
8.263835107953264,
9.693553204732913,
9.696078071259171,
10.987802073160276,
0.31236348113638823,
-1.370738548013981,
-0.19640846640845944,
-1.409184053738275,
-0.00366718677866519,

0.6361615892126344,
-0.6796219270984741,
-1.7803162192131878,
-1.2406782697025376,
-0.17521451360574944,
-0.2890470362486656,
-0.7870591534070972,
0.6214527622340972,
-0.5055171901576345,
0.3099897895999387,
0.6495941239175653,
0.4137491249725721,
0.292781258130194,
-0.18467758047440086,
-0.2764099099100952,
-0.12795113266111025,
0.14328500307003156,
0.1623766240720685,
0.09884205682735484,
0.07093196196575538,
0.17766773916349385,
0.1810420204025737,
-0.037056767883675096,
-0.38325428434024456,
-1.808138895166064,
-1.2174086587283277,
0.2723541238647832,
0.0911451526301191,
0.829454165789877,
-0.7076888645983359,
-1.1582847892183987,
-0.2789235155040628,
-0.1868914176693055,
-0.03618003850886602,
1.0797789643516258,
0.010710633626313776,
0.3041103423483441,
0.027606917069623423,
0.09715752300035431,
-0.025081467189021912,
0.7307804473426243,
-0.028312324696826655,
-0.5439567135695698,
1.0353579775860826,
0.03274794094556986,
0.338229999884877,
0.003923288912346083,
0.011059712392702849,
-0.24419721784886275,
0.0538500196620971,
-0.29704802197132957,
-0.9714959737951582,
-1.3776571886055275,
-0.4387141762272093,
-0.9013896619503009,
-0.2890811474761574,
-0.5664350037938545,
-1.265940570830118,
-0.36500389888052437,
-0.11634811830012508,
-1.0092353619257808,
-0.6849489189933503,
0.9856942680495377,
0.8277230147066852,
0.2903619348614201,
-0.3836733140597432,
-0.530059482018533,
0.8169168177018395,
0.5323656053240163,
0.649672850550592,
-0.3032256260984974,
-0.8829469814645045,
-0.00635052379210004,
0.35535568318583244,
-0.06185634463022188,

0.10533559478429602,
1.2902435643868964,
0.8825397316379378,
0.3070411861797082,
-0.07934132673084492,
0.09844318572655962,
0.7695889603351249,
0.14399374014816757,
1.8827194490661865,
0.013305728022345127,
1.2201617371808122,
0.13458781619262966,
-0.42760236848753097,
1.3132997483594182,
0.29586095969137777,
0.14782133458354627,
1.1539512755729704,
0.8255769419022573,
2.1284509830181357,
1.5067425074661367,
1.2286171121451426,
0.8625402575232176,
0.5254367589291086,
0.8683165111711829,
0.35804094605055825,
-1.3678290318224675,
-0.3952095755717416,
2.8373763512718595,
-1.2381307915304474,
0.21836664096244152,
-1.0373469190584237,
0.35616351546904684,
-0.48865262743063154,
0.5395297676370602,
-0.0068350558769426075,
-1.6240423184716037,
-1.005087252746991,
-0.324331708942255,
1.56936745287968,
1.3955283874562228],
[0.3026146462879133,
0.3869510743764029,
0.3869510743764029,
0.5382374643481795,
0.7459322218289088,
0.7459322218289088,
0.8083924716224152,
0.8083924716224152,
3.219627607910341,
3.2253302931763046,
3.9244184219369176,
4.114742178023509,
4.115220647252454,
4.115220647252454,
4.115220647252454,
4.65196408394345,
5.196968879903224,
5.736027504566688,
5.736027504566688,
6.321573696596394,
6.7009287553652,
6.7009287553652,
6.7009287553652,
8.383675014305652,
8.383675014305652,
8.383675014305652,
8.383675014305652,
9.235844817182018,
9.235844817182018,
12.415877726713473,
1.4636821166030158,
0.8436356038076634,
-0.6467230542935267,
0.2472607630354758,
0.028721536942984512,

0.5720112007049558,
-0.179388461505878,
0.6605644899444229,
-0.2997484495809327,
-0.5848737649454969,
0.7567373384637915,
-0.3113371232415861,
1.5925608798994126,
0.15551153490893768,
1.144256524486279,
-1.0086304067563407,
0.4692085043130235,
-0.5289416707772752,
0.6895537558918402,
-0.9023920029674053,
0.6687255575647186,
1.3626463195064957,
0.18435277462186048,
0.04101567185284027,
1.2334404589145347,
-0.8099334611041354,
0.47217953762833137,
-0.19021445340255347,
-0.47126223762254077,
0.21535783789089086,
-0.13160146544027274,
-0.14745399432080403,
-0.2131109368071074,
0.6116235033000932,
0.7187755412423116,
-0.7690585507561546,
0.333613273647484,
0.29688326543860977,
0.022443242058892903,
-0.12904022561026712,
1.1137507228270036,
-0.1754869613868649,
0.6791619259450926,
0.2688019369707634,
-0.33561997076117317,
-0.4910450017308718,
-0.2921449581476723,
0.07017675650466139,
-0.5628878746231101,
0.23633646017547374,
-0.4408658192483561,
0.6337746650828207,
-0.5275874765915898,
-0.11397090424872018,
0.7713694979315344,
-0.24077591679338844,
0.7998056369050096,
0.5449521212196076,
0.6440774810386566,
0.0012540722532346088,
0.4406697804838639,
-0.4697708168210915,
-0.38183296272166767,
0.4926783027265995,
0.012102728676941394,
0.9995268367701109,
-0.489600116502774,
-0.9378926015671718,
0.7538025600192173,
-0.9270581698564858,
0.23395173940139535,
-1.5047568644302813,
-0.6335501933328295,
-0.1851947866597326,
-0.18416301345971506,
-0.15010118329037675,
-0.30085163208099847,
-0.030658108221894752,
0.48141025496380413,
1.2807653372993688,

-0.27688172784497056,
-0.9404124429777033,
0.07772158149530461,
1.2206520244647234,
0.07167829739929571,
1.5514296753518346,
-0.22739832257997444,
-0.5457307695824859,
-0.05085143495318925,
1.4086504148457157,
-0.02595768387677813,
1.2057536466860386,
-0.5009834455732999,
1.0234750156328314,
-0.34572346527257924,
1.4975050133157242,
0.6424858734227697,
-0.5412025600486546,
-0.965195784498948,
0.8078930733387817,
-0.8747920106317195,
1.6444412493640082,
-1.3120143788008718,
0.7469227938848356,
0.7428309989793587,
0.25043937013438866,
-0.7176977899841391,
1.6961228018637329,
0.8286720852002186,
-0.08923208660690117,
-0.4564959840968765,
-0.10389957398275913,
-0.04652285532782134,
0.6308845069083873,
-0.025008046054888575,
-1.0109619026173717,
0.31698532529812806,
-0.07816069204265776,
2.517427548138029,
1.300984357739734],
[0.48696254731677435,
0.9433356258260287,
1.8515151481415904,
3.298386332934612,
3.298386332934612,
3.298386332934612,
3.298386332934612,
3.298386332934612,
3.374346159341203,
3.374346159341203,
5.043620376468771,
5.043620376468771,
5.218159502227583,
5.683079438253876,
5.683079438253876,
5.902572186502105,
5.902572186502105,
5.902572186502105,
6.17408053545979,
6.17408053545979,
6.17408053545979,
6.17408053545979,
8.600010985810668,
8.600010985810668,
9.424738936566825,
9.424738936566825,
9.43568443936613,
10.707654072137746,
10.707654072137748,
11.243950101093004,
0.30560526927017156,
-0.6023681078531374,
-0.5052271452160477,
-1.0944465846035194,
0.24881605409968105,

1.5229622975948298,
1.730509331816328,
-1.3834281548913137,
-0.8373914729231547,
-1.186133470726964,
-0.983750614650609,
1.3432121337567746,
1.6569377678026054,
0.14131844528439758,
2.9464329846729154,
1.3448953312948555,
0.7896925151532842,
-0.22765810177877452,
-1.6238177025568765,
1.1679097532601286,
1.5429785141845107,
0.44641536771583223,
-0.4990561654458825,
0.2890079097559007,
0.07664205286689205,
1.0463374337388847,
1.0582154531020835,
-0.7032713068941808,
-0.5914102053027229,
0.4481794192989112,
-0.597054072340342,
0.6503285491619032,
-1.3548677382206493,
0.3073247134537545,
-0.42900928099090013,
0.8290869769130189,
0.25661934620723303,
-0.41883276922189405,
-0.25409113110415216,
-0.2750324278847333,
1.8535327289298686,
0.7375047938415388,
0.21982546706442144,
0.3355829615333414,
-0.18885187042138774,
-0.09263125664317409,
0.8768150768205386,
-0.31105067773622297,
0.11882768143806702,
0.007542537022557154,
0.6689687664434811,
0.7431113887615772,
0.9447406593038509,
1.0052341467989638,
1.4439497926806117,
0.4202121214316228,
0.3604666875525164,
-1.6888556500060994,
0.1396130584965534,
0.10094819946612499,
0.5021137054548022,
0.3961649599525077,
0.4932381662816783,
1.3418440833635417,
0.09854079710503719,
-0.9989986964474209,
1.2672439763832217,
-0.07601477770236317,
0.46767000685442095,
0.3227070865546575,
-0.7175096141500144,
0.2758047915288214,
-0.24095833900456404,
0.058134638501602325,
0.6051082814423787,
0.5442716652871094,
0.3025208071236723,
-0.6679934596266708,
0.8499265434495678,
0.1803663678751026,

-0.3707988225819567,
0.6812577092957876,
1.2284831501706344,
0.20883808756632122,
0.03358446867125293,
0.026609877934757803,
-0.40919242519291843,
-0.07640273708110212,
-1.1643751408115468,
-0.16769301556045021,
-1.2911947285322207,
0.037223344859154565,
-0.05786091979914895,
-1.8047938425602224,
0.014158596153457605,
0.534864509903787,
0.006329420481561668,
1.239578876475735,
0.2639865904342132,
1.2160651224901187,
-0.2035728984920286,
0.6511657116444571,
0.3064364057804899,
2.45196454382623,
0.8340823063941794,
1.113505687918428,
0.049528306503774544,
0.12947314602533921,
-0.5399313265794791,
-1.2647403981540062,
-0.3482737106911781,
0.09697429466359891,
2.20804968215061,
-0.15489516431237316,
0.1250815357650574,
-0.13157603031624435,
-1.2324441762654297,
-0.4179909399666764,
0.3779753490492196,
1.6472121643813018],
[-1.1990978117137884e-16,
0.0939706539344674,
0.583570497583306,
0.583570497583306,
1.7474308983833136,
1.7474308983833136,
2.435631059267761,
2.4356310592677604,
2.435631059267761,
2.4356310592677612,
2.9757043839921162,
2.975704383992116,
4.1141402268542535,
4.1141402268542535,
4.889622687428616,
5.085553065152385,
5.14502423979873,
6.38608549233117,
6.405094454380333,
6.807050165246778,
6.807050165246778,
6.997225211250308,
7.82493752524197,
8.419597935808826,
8.419597935808826,
8.756502393683652,
9.514795580339912,
10.006329694888725,
10.008671195739254,
10.582332645019955,
-0.18325409339400697,
-1.8356335907661927,
-0.6014425812689169,
1.272854521157504,
0.38561099010389943,

-1.3354103698578268,
1.2089446007576305,
-0.16684037696101606,
1.5272415494549387,
-0.21803857489548387,
0.659293576456549,
-0.7629850988015884,
-2.0880787600158244,
-0.2200967935369742,
-0.21257917133409554,
0.05335328813343252,
0.740881120097457,
0.002552459203101907,
2.0381278489789163,
0.07009602349011958,
-0.027486349535234322,
-0.5162309983105858,
-0.018107734929441965,
0.3459554204227654,
1.1337361200247116,
0.6158218713410994,
-0.5251391279578093,
0.2758552924607382,
1.227089028823027,
-0.16405595633516176,
0.7587337472599636,
-1.1046949665522132,
0.7157795020690486,
-0.9666557350608269,
0.5941166908158014,
-0.3035426485157296,
-1.1713314559961463,
-0.3285214016177699,
0.15201600725182188,
1.158582769628082,
-0.2335509332665805,
0.021624313631101504,
-0.7981749327134954,
0.12400359670343174,
-0.00041124409742626503,
0.10665347356443583,
-0.030437347790731137,
-0.22360998336164295,
0.21313944649575595,
0.08788738041696632,
-1.2878790640853577,
-0.028446563531964782,
-0.12898316604526575,
-0.29469285393680966,
-0.07114900510733681,
-0.8104482788171727,
0.6394923422746198,
0.6811407094584414,
0.15127654478155184,
0.5799864823630293,
-0.55116978803857,
-0.5805840448115108,
0.9630930231089359,
0.9222833564508055,
-0.28733988453131243,
2.645322235680672,
0.02138581028992107,
-0.12421624460779133,
-1.4608772192659922,
-1.289711141721096,
-1.433469567857053,
-0.01041716118532159,
0.3377052351749707,
-1.74263853738129,
-0.6445874089376945,
0.6790378089998735,
0.0019732068694448615,
0.6260241949545162,
-1.3812938119718974,
0.06110738334511842,

0.919936405723882,
-2.761538990972181,
0.01773377957551518,
0.6041489324828,
-0.281847147959406,
0.36931104439326196,
0.14008962906529285,
1.0402771661903438,
0.40062198057764514,
-0.03775531840468896,
-0.4974133017905023,
0.5298490799692606,
-1.7184458207495306,
-0.7561470763062167,
-0.5888363413103725,
-0.9669502040570058,
1.3911426640670486,
-1.922807929753264,
-0.7347523142785206,
-0.6976214468215837,
-0.9022103009000106,
-0.17554401086360705,
-0.7004215075729389,
-1.1648825394224225,
-1.1285064327861984,
1.3794410327712907,
0.7524041737227146,
-0.600926776988818,
0.724704963452131,
0.24574404734520428,
1.3856850730269805,
0.505698953761027,
-0.030535393994814766,
-0.4836980977841504,
0.6166564120021005,
2.4235804988343936,
0.6827837636480912,
0.3673788389460804,
0.11748797515103059,
1.4924513209248702],
[1.112287766890805e-16,
0.11579095179987547,
1.7588846075757274,
1.7588846075757274,
1.998030831219005,
3.2454648629288743,
3.248106797240174,
3.248106797240174,
4.259596531772135,
4.958175754196259,
5.7210454174100525,
5.861260496697951,
5.861260496697951,
6.7588095170100075,
6.7588095170100075,
6.758809517010007,
7.044198424034494,
7.044198424034494,
7.044198424034494,
7.044198424034494,
8.232813771743604,
8.791432218349115,
8.791432218349115,
8.791432218349115,
9.616347292478379,
9.616347292478379,
9.869492613451865,
9.869492613451865,
10.182429226961672,
13.25758469112181,
0.5705572906646224,
0.09014038635403764,
0.1263845657016112,
1.0203985977202104,
-1.095649826264738,

0.019398597884167954,
-1.3544747652103108,
0.8866931148929111,
-0.18940184666069662,
1.0130589658671383,
0.15424354259331335,
-0.8117851912876255,
0.28413415623416743,
1.542699745821886,
1.836477396464815,
0.7160099188075301,
-0.9589348876176362,
0.00854407549207699,
0.36750908199249616,
0.7288091138524155,
0.9748756553152056,
0.7257579234395075,
-1.9128288506616224,
0.515489770530508,
-0.8485176532957428,
0.21376115901922352,
1.0834633235702664,
0.404913552038143,
-0.12863840971929708,
0.8333794222571905,
2.496813054711946,
-0.7354302042700512,
0.27199375272502646,
-1.0107781002238316,
-1.201703662816619,
-0.3331724830549245,
1.3030593505928814,
-0.8138384777541797,
1.9586180892259586,
1.1713603302631173,
1.0819642334098143,
-1.1784039537689197,
-0.12146178641002908,
-0.1717950232658355,
0.13833462928205204,
-0.8697724612180706,
0.7515471411202548,
1.166125613326729,
0.37941686042382794,
0.17074885851048915,
-0.33600856179263133,
-0.13669943521352546,
0.548834657399268,
-0.2522233718783481,
0.009665778262181637,
-0.5020120590858954,
0.16617168999240312,
-0.46390045473242114,
-0.05573261329441784,
1.1182956755615927,
0.0937037366596967,
-0.4441962997721035,
-0.21050297984767188,
-1.0381458298882043,
0.830180513233935,
-0.8157309834659043,
-0.5833742339287631,
-1.5354704640903527,
0.44493118131447734,
0.13057382050598434,
-0.7034976447329113,
0.6218444459919024,
-0.2420386731361673,
-0.7867967091164098,
-0.168935029173583,
-1.6928247400206122,
0.22571922404505704,
0.1126763335131705,
-0.22160264498444046,
2.299575126838168,

2.5415010654218975,
-0.5449604787415219,
-1.005432959888493,
-1.722931012111005,
-0.6732620373630841,
-0.0004890899721480893,
1.492586488782192,
-0.9055437852306896,
1.1298576359465664,
-0.5182549439372126,
0.6586606791613716,
0.7929632788713356,
0.08462141435268096,
0.5066414628201175,
0.6503902510816232,
0.06456835392989443,
-1.687283870544211,
-0.2816102525113523,
0.9601910446201565,
-1.0364808528540121,
0.32603884641661884,
-0.37671408634672476,
-0.16068551007585027,
0.6157148912247653,
2.0105220675930333,
1.7580796652062145,
1.7229426093703843,
-0.5199627188331776,
-1.3760925920420213,
0.8290526991464299,
-0.08765267414283408,
-0.456659399307068,
1.4313668535261046,
0.635052489001378,
2.136142869643981,
-1.5755126698917075,
-0.40116032394931217,
-0.004321541181387126,
3.301925406547275,
0.7673504276272846],
[0.49826187507447023,
0.6110785201427625,
0.6558173388435118,
1.2433731466916595,
1.2468055416987345,
1.9464512542038723,
2.109950686395802,
3.7879039161756585,
3.7879039161756585,
3.7879039161756585,
3.7879039161756585,
3.7879039161756585,
3.7879039161756585,
5.054414830549875,
5.106961698737131,
5.488040579082762,
5.488040579082762,
6.832391146999917,
6.967864562238176,
6.967864562238176,
7.490508971727458,
8.20118277218113,
8.612503180124596,
9.093659709060372,
9.807153774862888,
10.372385504551621,
10.372385504551621,
11.272736556325881,
11.767303580595307,
11.767303580595307,
0.035679258194077725,
-0.8723307875716305,
-1.7560663493087938,
1.9629136553479551,
1.9120807210302333,

-0.39357388790453784,
0.036182026687196715,
-0.21816401975370772,
0.8090968256508778,
0.22775227657680963,
-0.9402388060640036,
1.9199330583190146,
-0.6293921442826633,
-0.37521158500092616,
-2.4122776248196542,
0.47614923103710605,
-0.2356719275662886,
0.5930119757607211,
-0.08067436637158562,
-0.6975308853470512,
-0.1533962892581602,
0.6838091266999453,
0.243129377869582,
0.5896632240559074,
0.12048284254834898,
1.186903891075025,
0.22948374273850122,
-0.9592798973456383,
1.431786126792839,
1.099938502501145,
-0.6421065764259075,
-0.4570128400176702,
-0.40361390122591434,
-0.030699975686000303,
0.5709596867047622,
0.8129177753256275,
0.0978347154858261,
-0.1791413540365364,
1.316347643480991,
1.0669242344323897,
0.9542637314281797,
0.2732710750226029,
-0.24925074976969727,
0.14351675204102743,
-2.058043419369877,
-0.5772495977232713,
0.4759355346590471,
-0.3040209890415461,
0.07360159687611288,
0.11769956200453384,
0.01709491725825118,
1.1325236656251834,
0.008658928128351846,
-0.21852721169551936,
-0.1826844911726131,
0.39535377862993026,
0.035729500082236576,
-1.0747644579938396,
-0.15992271631488378,
0.8234408989300128,
2.679077704708449,
0.2559692131167249,
1.5592174030452497,
0.3453465662341248,
-0.24046357809578378,
-0.6876498012434461,
-0.01750342347144341,
0.34704200676536884,
0.2033679229118891,
-1.5920113668255755,
-0.5283783493172445,
0.4351355192977228,
-0.8281455342425702,
0.2974457926897622,
-1.336557258227178,
0.2885982325597283,
-1.2111847441179915,
0.698629224522434,
-1.072563817644129,
-1.2092878769637225,

-0.33153669598628793,
-0.2558676939050336,
-0.039300685015777354,
0.5730206646846686,
0.6772024906875076,
-0.5431857921175166,
-0.32242263836861185,
-1.2435651530921474,
0.6447338068594848,
-1.1796614873773261,
-0.5517178991847818,
-0.9950127711245446,
-0.750107619545704,
1.546476758712725,
0.7798072495873917,
-1.3969546244577882,
-1.3869625997614359,
-0.5499269608373658,
-2.0698482195340895,
-0.31472418351029113,
1.3858906356684872,
-2.3234389152233144,
0.8580050890595776,
0.5574876161662472,
0.12663545081840322,
0.0024286165622580055,
1.4351555562364873,
-0.018100010671135307,
-1.4466701603237728,
1.0091335554802272,
-0.25739419705547106,
0.8584486110461004,
-1.6353613496727522,
1.3481381905015994,
1.664215570319441,
-1.4688501218242656,
0.6630999772073086,
-1.3598408860414706,
-0.9086993193406631,
-0.7720609621308895],
[0.03916643192220903,
0.09260151508853934,
0.5125042140553085,
0.5342494495137248,
1.4070055183169814,
1.4076098722130033,
1.4080234932826805,
1.9886432369288551,
1.988643236970097,
2.4501760969485846,
2.4501760969572235,
2.6141080657383133,
2.614108094671839,
2.6141080978874367,
2.614108199851818,
2.614108200075538,
2.6141082008692407,
4.121207830152765,
4.126135109264967,
4.126137501750553,
4.129417665319773,
4.731890053629371,
4.734221022017094,
4.783748744413943,
6.888546551790504,
7.435120796142169,
9.27338967383107,
9.884598996480708,
9.884598996488108,
10.57466535326793,
0.042050619766987055,
1.2200437115279086,
-0.1897539908832087,
0.3883222993215419,
-0.3953259161886337,

0.9433481261689909,
0.03976732857871137,
0.380475950476648,
-1.6480061036631655,
0.19200671798130073,
1.8964236385178486,
0.5825466621902844,
-0.7017933974726621,
0.46167134168781887,
1.3455780978346443,
1.000341055668334,
0.3220859397336833,
0.4901021103591929,
1.7507755726531005,
0.22815940378303187,
-0.27415827045663116,
-0.06669089502017891,
-1.2681856141960832,
-1.4186851222947996,
-0.7484212831710693,
-0.2624358161393499,
0.22137315279362701,
0.8266616228870926,
-0.28382763940078026,
0.4526868734709094,
-0.03876262869445461,
0.028933904781200354,
0.2124235174903725,
0.5275601102845284,
-0.035224229477880745,
-0.11459385773709553,
0.4537223625267385,
1.3135706165796082,
-0.4103066311241145,
-0.06740865621875465,
0.07865262902016133,
-0.8220672300227196,
0.05131914396597353,
-0.012779296277522511,
1.6650491211728184,
-0.5271258879235212,
-0.2286028431320702,
0.9810985313826684,
0.5796827949136435,
-1.4216581678953852,
0.908254540613963,
0.7809397666182989,
0.23060365236949015,
-1.629683736567043,
0.08687379015275534,
0.5302421766128769,
0.858319872120405,
-1.2816124200556078,
0.08255098737682609,
0.49190330494839835,
1.2241314669476433,
-1.6043734869556,
0.35480117328822147,
-1.3692690448046365,
0.08801115679730828,
-0.3416815622800032,
0.5349891528238044,
0.23599371094677032,
-0.2955967206364773,
0.7351110566185237,
-0.1358629473726594,
-0.2538927978152746,
-0.18822579896800806,
0.023625415865148318,
-0.47104955898400697,
0.3192033261234258,
1.3512870657881308,
-0.7816632422470443,
-1.6469850516046918,
0.27638573599623373,

-0.3090269715855155,
-0.4193062343229584,
-0.1377874030219536,
0.0004776579492072504,
0.08325659247733753,
1.4611032338266408,
0.021431340914286272,
0.20066912699708425,
0.29328305426482204,
2.42997753887945,
-0.5690256391604135,
-1.480840559800266,
-0.1363946771767577,
-1.376439435407693,
0.06387998538825271,
-1.110029094439266,
0.9868538484965683,
0.962243917255456,
-0.41126656322102295,
-0.657103189069447,
1.1434637843768656,
-0.9757973897576084,
1.448315222432508,
1.3110742046722172,
2.3333185683193554,
0.7520157773640208,
-0.20203562372486106,
0.34414613752805073,
0.5926366260275047,
-1.985044802604821,
2.6078832019778186,
-1.6883872397325923,
-1.6784323905215754,
-0.4297482144956373,
0.6572781395405978,
0.02277885553642172,
1.3256214503718338,
0.25673693454286806,
-0.6358308725577859,
-1.6133180674191239],
[0.4366123913610663,
0.4768198412180804,
0.9843286193666714,
1.2674678431689148,
1.5775702096396949,
1.5775702096396946,
2.2591837513222837,
2.2591837513222837,
3.168921311818864,
3.1689213118188637,
3.1689213118188637,
3.2207076420855234,
3.220707642085524,
4.00024951310582,
4.479169433031019,
5.572620903211533,
5.572620903211533,
7.169574827830268,
7.200887753316444,
7.200887753316444,
7.4320670533634186,
7.4320670533634186,
7.668889448658714,
7.668889448658714,
8.088052955990618,
8.127936129548912,
8.127936129548912,
8.129700499301142,
8.129700499301142,
13.234708916500516,
-0.8409114197195183,
0.13809956722013594,
-1.3698910408430074,
-0.1271619514190957,
-1.0028759735951214,

-0.7519751568518348,
-1.7027753599694198,
-0.5003000098057334,
1.7803062126367422,
-0.6956057702973542,
-0.5292841478808504,
1.3371604427134443,
-0.6271315116651272,
-0.9849660401714492,
-1.0245042646602094,
1.7746289077120558,
0.789646250683077,
2.3274439120303834,
1.0250145144124314,
0.236963759763461,
0.6449585291401012,
0.5302895792637682,
-0.17332551626046228,
0.13296008629005646,
1.3001365941190501,
-0.5491163203391332,
0.8939708861040915,
-0.1183728413911535,
-0.27672563627574315,
1.7953720191393387,
-0.00825054985875676,
-0.6170579146731886,
1.801545679753067,
-0.6786704865717594,
-0.5702663839537956,
-1.2235657195316856,
0.059502931568887765,
0.15944146767128192,
-0.5445588526685949,
1.0645821939979259,
-0.3246614360002631,
1.6342475668463907,
-0.6365693512803821,
-0.25644806680599874,
0.9153791587574333,
-1.5555971818784813,
1.7626038696684259,
0.29250830199243494,
0.4597874603079029,
-0.4983420660721544,
0.22206047990995867,
0.11089125969329257,
1.9712857917750568,
-0.8482820514299444,
1.7711968287204813,
-0.3356021753581158,
-2.239225908466879,
-0.021402454584719684,
1.6641712424719513,
0.5059308187330879,
1.9296669762106724,
0.7155610478862724,
1.024294278133781,
0.2881031759975837,
0.06318491357276658,
0.8308741396333285,
0.16800005476886462,
-0.26845892946467115,
-0.378371667085078,
-2.374474400887392,
-0.26943703763693594,
-0.698442208248677,
1.5029972614102114,
0.7682274655447995,
1.7790823677975025,
1.1233449399645796,
-0.1962677980677998,
-0.6013220679983903,
1.243366811759978,
0.23851493156650586,

```

-1.3516830763738037,
0.4546075088056473,
0.45461165179145885,
0.29299766155391044,
0.9768506591689141,
-1.3614782254940123,
-0.1864841683590143,
0.03417328589702677,
-0.12392810093414776,
1.4042342901901068,
0.4181387945528863,
0.6171624447156228,
0.5887687464980093,
0.07995045192475739,
0.12406497411608935,
-0.1304256749231809,
0.5172871404602656,
1.1235454838537668,
1.5976273357881063,
-0.7195312648953682,
-0.7697726746437508,
-0.1718403740134208,
0.48757574872507614,
0.7291603816887677,
1.3754811895843486,
0.06384938170591244,
0.9009847125572167,
1.0744001081352503,
1.7305303874736466,
0.491454536253661,
0.11643540469065636,
0.5925455840333324,
-1.2877724397991297,
0.052924560657905566,
-0.8760423919600488,
0.6313576901954856,
2.736167065037608,
0.029295634649204143,
2.1044233481865446,
0.6660786050345296]],
'run_time_sec': 43335.76831364632}}],
'redundant': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}}},
'init_param': { 'value': [ 0.4319322451408806,
0.4441934852523559,
0.5878315575931067,
0.5878315575931067,
0.5912427958750637,
0.5947187380620651,
0.5947187380620652,
1.5628074958209335,
1.568718083448373,
2.013035808837258,
2.013035808837258,
2.013035808837258,
2.0277481086167364,
4.114573484042011,
4.1154343976538055,
5.358852938788934,

```

5.358852938788934,
5.549639110185835,
5.667644288942511,
5.6706149635077026,
7.063797010237488,
7.658675772413245,
8.123010743902626,
8.633070512181217,
8.7849794748515,
8.7849794748515,
10.749769122252356,
10.89899813672382,
11.413879450074099,
11.413879450074099,
-0.04151001581527148,
-2.751209241920508,
-0.5056705094410174,
-0.406548832581749,
0.8633340895103881,
0.17259893989954428,
-0.21235924407313156,
-0.5485334561985523,
0.2012951527025238,
-0.23858034563229433,
2.1364176057207223,
0.7817014982866842,
-0.1108899918027214,
0.6048531661014985,
1.809261328558185,
-1.1786415575821785,
-1.0827170099436865,
0.4109384498547832,
0.8090945308822337,
-0.9356585252219308,
0.28697429006677655,
-0.15332459566133277,
1.2998079093160566,
0.3870967504567973,
-0.2413465423953762,
1.001771196320206,
0.1138394509020578,
1.3220786271535379,
-1.1806565063083097,
2.2789659185869464,
0.11875741943281123,
-0.15631084464392242,
-2.14680662310284,
0.18035127914430246,
-0.260880073181003,
0.21031876899961,
0.7641152777599183,
1.203568210035907,
1.2767673297334832,
1.2585129786889195,
0.24752046596725819,
0.7203551788145919,
-0.17420447991641552,
-1.3486030918499698,
0.2145369555808404,
0.8896874485404291,
-1.6470089821954121,
-0.37928143051952584,
-0.29903232984244293,
0.6984471945997809,
0.14906235834494008,
0.054873949093549826,
1.3877615433857144,
0.04193983551007985,
1.2014028569521673,
-0.02338341588216991,
-0.05255422822816251,
-0.1905875093002492,
0.16425167437144692,
0.2542835721118732,
1.1822299841866795,

```

1.2032222480152739,
0.30142488752060526,
0.09125464470109583,
-0.45638140154025475,
0.050260497156630585,
-0.04000614529085093,
0.9096133936431033,
-0.03555378439095736,
0.4405325841695877,
-0.48129885373423525,
-0.3617221403442356,
-2.5216506086368518,
-0.5081934521569421,
0.041332600726076626,
0.7413771443671474,
0.2204029987501584,
-0.7517920276590782,
-0.21022249714251526,
1.5607830526955413,
0.16615947769508344,
-0.17023919527139658,
-0.4295619963603494,
-1.1838803442651602,
0.6319564057885301,
-0.3935277768061919,
0.005550337258741166,
-0.4226065324972882,
-0.3715846249960284,
-0.48395498614671884,
-0.4790957128536121,
1.4427293239187895,
0.5875094653073935,
1.404722056846091,
0.6973465258635152,
0.8623508528879442,
-0.5460563718875903,
0.8326491581513842,
-2.4015389244466907,
-0.04583861440999851,
-0.36265388938085896,
-0.5347868312224912,
0.49674658852292397,
-0.6488379607428911,
1.1568053403721041,
0.30662269781395407,
-2.013151608263867,
-0.17383275003878457,
0.5036662539248705,
1.9638910157847163,
3.118155075975211,
0.8117246453310248,
-1.4517859264576565,
-1.4946803709676548,
-2.7792982516572895,
1.1174424252005506,
2.333430423546859,
0.6903927508165344,
-0.2883899927957993,
-0.8542826896862586]],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,

```

False,


```

0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric3_unifrom_scaling_sample#7'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
                                'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
                                '4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
                                'X 5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
                                                         '+ 1 Z 0 + '
                                                         '1 X 1 X 2 '
                                                         '+ 1 Z 1 + '
                                                         '1 X 2 X 3 '
                                                         '+ 1 Z 2 + '
                                                         '1 X 3 X 4 '
                                                         '+ 1 Z 3 + '
                                                         '1 X 4 X 5 '
                                                         '+ 1 Z 4 + '
                                                         '1 X 5 X 6 '
                                                         '+ 1 Z 5 + '
                                                         '1 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -6.662379321078876],
                             [12, 6, 0, 0, -5.394498644176583],
                             [20, 10, 0, 0, -4.374551061230002]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                    'identity_factors': [0, 0, 0, 0],
                    'noise_level': [4, 2, 0, 0],
                    'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [
0.001,
0.001,
0,
0],
'status': True,
'type':
'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [1, 1, 0, 0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [
0.001,
0.001,
0,
0],
'status': True,
'type':
'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],

```

```
'identity_factors': [2, 2, 0, 0],
'noise_level': [20, 10, 0, 0],
'noise_profile': {

'noise_prob': [

0.001,
0.001,
0,
0],

'status': True,
'type':

'depolarizing'},

'odd_wires': 4]],
'run_time_sec': 7.118944406509399}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [ 0,
                                                             0,
                                                             0,
                                                             0,
                                                             0,
                                                             0],
                                                         'cn': [ 0.5,
                                                             0.5,
                                                             0.5,
                                                             0.5,
                                                             0.5,
                                                             0.5],
                                                         'r': 0},
                                'time': {'max': 10.0, 'min': 0.0},
                                'type': 'ising'}}},
  'init_param': { 'value': [-4.1290471568429383e-17,
                             0.028387941050870547,
                             0.028387941050870516,
                             0.028387941050870544,
                             0.4764187545019148,
                             0.485154879376271,
                             0.7094706976105217,
                             2.998376901501724,
                             3.486264965503922,
                             3.487775573446841,
                             4.058897205630428,
                             5.020552086463093,
                             5.02158492362958,
                             5.02158492362958,
                             5.0846878698400815,
                             5.0846878698400815,
                             6.794704827143902,
                             6.794704827143902,
                             6.794704827143902,
                             8.013356083289752,
                             8.21076169573117,
                             8.21076169573117,
                             8.633075344303478,
                             8.633075344303478,
                             8.930018414781724,
                             8.930018414781724,
                             8.930018414781724,
                             9.990329186240457,
                             9.990329186240457,
                             12.915390927357043,
                             0.8289215528981712,
                             0.6465354442587259,
                             -0.06922368073773534,
                             -2.0853926237797484,
                             0.25605409461249207,
                             -0.765586831155604,
                             0.9589723041272441,
                             0.27455658215646966,
                             -0.07189982532198122,
                             -0.3449139525561357,
                             -0.46800027418386986,
                             1.016480689947782,
```

-0.9706976058410632,
-0.9556868223128826,
0.9323574793538898,
-0.4376613900889883,
0.44138701419965265,
1.4418520162063444,
0.8420619129497768,
-0.025530845682239494,
-2.3076344184440156,
0.37458463155346333,
-0.47031338116346383,
0.07338001085955462,
-0.869249007732584,
1.6824786550253577,
1.719346297136726,
-0.9622509832303179,
0.5779076253870294,
0.4760527439731688,
-0.2451715046346434,
1.5906513054768954,
0.35918458386075697,
-0.44817294000256097,
-0.37416377940751877,
0.37048554789855764,
0.693137718643464,
2.288219219235325,
-0.26501198620577204,
0.3689053264668303,
0.10166782579927121,
-0.3356073800920907,
-0.6413954053758932,
0.28636656044125136,
-0.8690978212322298,
1.166030172665772,
-0.46365263204645707,
-0.8505396438729699,
2.4328989003329693,
-1.4554443951257197,
2.09134311034601,
1.7695145471360614,
1.1170184462314887,
0.20325573406500014,
0.34206355374635444,
0.016804831103183593,
-0.19882507083844791,
1.4011478969931443,
-0.0572616468882171,
-1.3773340344964602,
1.1031086912886459,
0.47433623341773395,
-0.037679845343088075,
0.7450954028512253,
1.3114871459373192,
-0.18651966717154955,
1.0750199871192778,
0.48905379729033566,
-0.6281827821606347,
1.322074656325554,
-1.4761484868509396,
0.2859918816514792,
-1.0033541595176405,
0.8264053813863532,
0.622010811103528,
1.1636747207646185,
-0.31883271288389914,
-0.0013981850979601159,
-0.09999405171802342,
1.3085280853238759,
-1.3727969089464465,
-0.08341159264840771,
1.6249369276484606,
-0.28082331570516694,
1.863294400254821,
2.2932652495309953,
1.4849536675442596,

```

0.6811668802751195,
1.5380110980874968,
-0.6615225065895972,
0.4000828281024264,
0.33932121588009856,
0.48855764825038867,
2.5371111232777053,
-0.7260430040504564,
0.09005940379987407,
-0.8762980817992282,
-0.23800314057977784,
0.9728027685352436,
-0.20857834368009137,
0.6148333457084475,
-1.5171898336040224,
0.8658542069129427,
-1.6478287868490311,
0.5770049367106069,
0.134497700004433,
-0.11612046371881249,
1.5826352840237945,
1.6045820840261675,
-0.08805669980595342,
-1.507745125921201,
-0.11276651576435986,
0.07206005105970639,
-1.7788122679933258,
-0.002988765738482436,
0.7255804452314191,
2.7661804504965604,
-3.1943627011957507,
-0.07088434770936546,
0.10949711075632612]],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric3_unifrom_scaling_sample#1'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1
'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X

```

```

'4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
'X 5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
' + 1 Z 0 + '
'1 X 1 X 2 '
' + 1 Z 1 + '
'1 X 2 X 3 '
' + 1 Z 2 + '
'1 X 3 X 4 '
' + 1 Z 3 + '
'1 X 4 X 5 '
' + 1 Z 4 + '
'1 X 5 X 6 '
' + 1 Z 5 + '
'1 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -6.833109640228284],
[12, 6, 0, 0, -5.610198560557999],
[20, 10, 0, 0, -4.609901567200479]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [
0.001,
0.001,
0,
0],
'status': True,
'type':
'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [
0.001,
0.001,
0,
0],
'status': True,
'type':
'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
'identity_factors': [2, 2, 0, 0],
'noise_level': [20, 10, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [
0.001,
0.001,
0,
0],
'status': True,
'type':
'depolarizing'},
'odd_wires': 4}],
'run_time_sec': 6.902796030044556}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
```

```
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [ 0.32980400918867037,
0.7818596814437996,
0.9680306497310072,
1.0586432954274991,
1.7663116348413028,
1.8956491295658664,
2.0436116769319477,
2.0436116769319477,
2.0436116769319477,
2.1846619555479685,
2.1846619555479685,
4.516997836642403,
4.516997836642403,
5.3166564351730035,
5.320130741320412,
5.399447086770547,
5.400046100307029,
5.400046100307029,
5.400046100307029,
5.400046100307029,
7.097750404951498,
7.618222769006738,
7.956666567531361,
7.956666567531361,
8.961109688308255,
8.961109688308255,
11.028500695271937,
11.5809418583784,
11.5809418583784,
11.5809418583784,
0.24114173221382337,
1.3754512844042854,
0.9720111739434927,
-0.29877116871955545,
-0.029737313498699618,
0.3185153459444675,
-0.31840745367437817,
0.6801602139502809,
-0.015407845998522466,
-1.114484654283524,
0.22997676136047449,
1.5201821520192487,
0.3217486942799661,
0.43678760134757944,
0.17618749246228715,
0.7010359103792874,
0.3446701255696699,
1.4700166379981474,
0.3471333265905846,
-0.968444017764766,
-1.1485269521507444,
0.3282504398407059,
-0.7898554413193151,
-0.22157985789400653,
0.4570931682042821,
-0.22158635023930806,
-0.36549836384277606,
-0.2366283880333898,
0.4114765778384287,
-0.7694643480660831,
-0.40507000819737765,
-0.6576468499882752,
0.2023600853039859,
-1.2917280200640486,
0.32702771491506877,
-0.07546137553843744,
-0.7713328193108817,
-0.8163562225239717,
```

-0.07772489804090182,
-0.32841576637855896,
-0.05728139293162625,
0.8564947637293437,
0.15384434981345546,
0.41944080031994563,
0.16884394015055976,
-0.010490661309829376,
-1.3198249528292274,
0.07128101806419536,
0.03515395872296211,
-0.09826644736389979,
-0.06882299183557238,
-0.7062443088386591,
-0.5280915567870916,
0.14860690208212682,
0.338235073517239,
0.012130023587481156,
-0.037483081002571056,
0.8881056862099682,
0.06312673968269537,
-0.44738898868242327,
-0.2463284936792685,
1.1708672891281489,
0.5297819270424092,
-0.5463182826640249,
-1.667403353493104,
1.3713545814180228,
0.9413366263710182,
-0.021951472934173096,
1.0345046397029698,
-0.8002739446090432,
1.076242700462428,
0.04994674089563993,
-1.446235170371471,
-0.3268246709561185,
0.6264234635598642,
-1.2465048991503196,
0.1169092097763034,
1.2453937722962254,
0.7220287063888942,
0.8114212311039525,
0.18333674920471413,
0.7345776889004225,
0.9807678368065545,
0.8703572591074042,
0.28720991162605214,
-0.9085817320602367,
1.1216621938695932,
1.1327477502874412,
-1.8285395156530388,
1.4231155243966302,
0.08156271721069372,
0.18027327076782884,
-0.10302960773112253,
-0.5560482514096969,
-0.09484541903718098,
0.5886430341587933,
-1.1674014904638685,
-0.03940950630369794,
0.2792515765657198,
0.08214125135098017,
0.04344924016389119,
-0.7787970529711157,
0.07887586110139429,
0.7532574519080738,
0.17107625040009872,
-0.4128415687827808,
1.3723943620028143,
1.4278581324344195,
0.5879977529754344,
0.28910810409117477,
-1.4412046993697643,
1.624844367134446,
0.19366683662945536,

```

0.3179841250458877,
0.5357240606734743,
1.5680682524509713,
0.04418323830986127,
0.2311137213204739,
2.8741283660232546,
0.9612408402592636]],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric3_unifrom_scaling_sample#5'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
'4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
'X 5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
'+ 1 Z 0 + '
'1 X 1 X 2 '
'+ 1 Z 1 + '
'1 X 2 X 3 '
'+ 1 Z 2 + '
'1 X 3 X 4 '
'+ 1 Z 3 + '
'1 X 4 X 5 '
'+ 1 Z 4 + '
'1 X 5 X 6 '
'+ 1 Z 5 + '
'1 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -6.720969040841222],
[12, 6, 0, 0, -5.577003279994282],
[20, 10, 0, 0, -4.626982768162376]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [
0.001,

```


7.533802135057517,
8.245970909352765,
8.390887842007308,
8.390887842007308,
8.703391612007538,
8.703391612007538,
9.485545381639698,
9.485545381639698,
9.849437794456351,
9.849437794456351,
10.799155842585746,
-0.12708089143278917,
1.4168148977025445,
-0.13519536748631492,
-0.32308714516775877,
0.06981336221044306,
-0.034321895006426455,
-0.18209443759399158,
-0.024858958594746107,
0.664488174834645,
0.6897317665852637,
-0.008774731192519907,
0.625642551337688,
0.6862410821653743,
-0.7670419540542446,
-0.48055390132433673,
1.573964677292243,
-1.0555745677813198,
1.3159721165631335,
-0.9138502497993347,
-0.4641566084568084,
-0.06966564000847691,
0.039012746519005555,
1.0827641697579717,
1.2190093772283843,
1.5391150382377632,
-0.6675581491933664,
0.09104662559232415,
0.5283818781579022,
0.04478118244404365,
1.5836075485651113,
0.251932560991488,
-0.3949560381811925,
0.8854639388207172,
-0.8611267164776638,
1.566928747241295,
-0.15445151319573347,
-0.01396050433941564,
0.8172620115356718,
-0.014658132241289147,
-0.6623010385304499,
0.6299284572753584,
-0.15026947869759505,
-1.2293112897827836,
-0.5768656731979434,
1.231557922062332,
-1.447872926116422,
1.4918722706131171,
0.31945486783385635,
0.43870804207253483,
1.3582636256308331,
-0.8478398442626237,
0.799518029891469,
0.5539369896048497,
0.13453197415145846,
0.7167498104862234,
-0.06589109804263343,
0.041038724923082276,
1.151975756926885,
-0.052026838729968425,
0.3952978000808587,
-1.1754748970457833,
-0.01965607042871015,
-0.2456324144680547,
-0.01136184217607491,

0.022503168314325715,
1.1777923723104653,
0.14554117253938412,
-1.2287946167385297,
-0.9683737383161766,
1.073329622962756,
-0.7143583802799215,
-0.3545002682598469,
-0.31906110571926777,
-0.9597094503848127,
0.8825903987559673,
-0.08273415521500459,
0.2895039435279609,
-0.3388594197258312,
0.2406459434933063,
0.3824128671502841,
-0.5390555829405563,
0.3757684495816124,
-0.5015719636333678,
0.3958257688275076,
-0.6428354095046468,
-0.9178624992467067,
0.29327306512581747,
-0.09193285724698198,
0.1770091195095364,
0.6803523512851801,
1.05525770275555,
-0.4646437845139076,
2.2560908642344026,
-0.7500939262299234,
1.92959092052175,
-0.36895777993274226,
-0.04538283416771957,
1.2529586558275392,
-0.57805717638909,
0.3080585810325818,
0.031910210614104206,
0.18519950010488512,
1.3888566879174211,
1.087260914864616,
0.15259355623633358,
-0.08810167996863522,
0.07467874103808375,
1.4444893382870363,
-1.446220680213735,
0.8215180856176293,
2.0151723649306295,
0.13454991965456886,
0.21923381008732426,
0.31112145477192793,
-0.8977876743954992,
0.10239876571985529,
-1.0132922278363503,
-0.3794863035801566,
0.8437803973608401,
1.111671691788496]],

```
'noise_profile': { 'noise_on_init_param': { 'status':  
                                                    'value': 0},  
                    'noise_prob': [0.001, 0.001, 0, 0],  
                    'status': True,  
                    'type': 'depolarizing'},  
  'nqubits': 7,  
  'observable': { 'coefficients': { 'bn': [ 1.0,  
                                             1.0,  
                                             1.0,  
                                             1.0,  
                                             1.0,  
                                             1.0],  
                                     'cn': [ 0.5,  
                                             0.5,  
                                             0.5,  
                                             0.5,  
                                             0.5,
```

False,

```

                                0.5],
                                'r': 1},
                                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                        'status': False,
                        'type': 'png'},
            'file_name_prefix':

'AUTOMATE_ising_noisefree_time_evo_ric3_unifrom_scaling_sample#2'},
            'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                                [1, 1, 0, 0],
                                                [2, 2, 0, 0]]},

            'run': 'redundant',
            'state': 'dmatrix',
            'vqe': { 'iteration': 1,
                    'optimization': { 'algorithm': 'SLSQP',
                                      'constraint': True,
                                      'status': False}},

            'zne': { 'data_points': None,
                    'degree': 0,
                    'method': 'richardson',
                    'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
                                'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
                                '4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
                                'X 5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
                                                        '+ 1 Z 0 + '
                                                        '1 X 1 X 2 '
                                                        '+ 1 Z 1 + '
                                                        '1 X 2 X 3 '
                                                        '+ 1 Z 2 + '
                                                        '1 X 3 X 4 '
                                                        '+ 1 Z 3 + '
                                                        '1 X 4 X 5 '
                                                        '+ 1 Z 4 + '
                                                        '1 X 5 X 6 '
                                                        '+ 1 Z 5 + '
                                                        '1 Z 6']],

'output': { 'data_points': [ [4, 2, 0, 0, -6.514930884689279],
                             [12, 6, 0, 0, -5.310706408751285],
                             [20, 10, 0, 0, -4.328532484009807]],

            'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                'identity_factors': [0, 0, 0, 0],
                                'noise_level': [4, 2, 0, 0],
                                'noise_profile': {

'noise_on_init_param': { 'status': False,
                        'value': 0},

                                'noise_prob': [

0.001,
0.001,
0,
0],

                                'status': True,
                                'type':

'depolarizing'},

                                'odd_wires': 4},
                                { 'gates_num': [12, 3, 1, 0],
                                'identity_factors': [1, 1, 0, 0],
                                'noise_level': [12, 6, 0, 0],
                                'noise_profile': {

'noise_on_init_param': { 'status': False,
                        'value': 0},

                                'noise_prob': [

0.001,
0.001,
0,
0],

                                'status': True,
                                'type':

'depolarizing'},

                                'odd_wires': 4},
                                { 'gates_num': [20, 5, 1, 0],
                                'identity_factors': [2, 2, 0, 0],
                                'noise_level': [20, 10, 0, 0],
                                'noise_profile': {

```

```
'noise_on_init_param': { 'status': False,
                          'value': 0},

0.001,
0.001,
0,
0],

'noise_prob': [

'status': True,
'type':

'depolarizing'},

'odd_wires': 4}},
'run_time_sec': 8.669902801513672}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                                'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'ising'}}},
  'init_param': { 'value': [ 0.3463233138801294,
                              0.34632331388012944,
                              0.3463233138801294,
                              0.3463233138801293,
                              1.2403205490491955,
                              1.38033446895038,
                              3.6939218351946623,
                              3.6939218351946623,
                              3.693921835194662,
                              3.693921835194662,
                              3.9412089126587,
                              3.9412089126587,
                              3.9412089126587007,
                              3.9412089126587007,
                              4.8310999601027795,
                              4.870844593548848,
                              4.900228671797898,
                              4.9002286717978984,
                              4.9002286717978984,
                              4.900228671797898,
                              5.70167657474683,
                              6.21188441483336,
                              6.21188441483336,
                              6.686419232103557,
                              6.723504371727371,
                              7.618130890836856,
                              7.618130890836856,
                              9.420364260644014,
                              9.420364260644014,
                              10.775132222422771,
                              -0.3525750090684208,
                              1.2748976558430758,
                              -0.16146998454247521,
                              1.3224575871686013,
                              0.0349265192498484,
                              0.36138711032415904,
                              -0.5043488106619078,
                              0.7634801084306476,
                              -0.3536814593617201,
                              0.9606809354246906,
                              -1.1143894720984433,
                              -0.45289061390722213,
                              -0.1605333784989486,
                              -1.594210268458609,
                              -0.09465166736510336,
```

0.11132251138190599,
1.004235666711761,
-0.23115493220428254,
0.9451761882751959,
-0.3707284170975133,
-0.17782948231634454,
-0.18433417990028922,
-0.06421460341189031,
-1.155489001827845,
-0.2074908306187579,
-0.036360325422879626,
0.5283181819327893,
0.3603171208434518,
0.6819349944270952,
-0.2733083993242672,
0.7075879191324372,
-0.042779623335368125,
0.7032435595714176,
-0.7363942983542189,
0.9583235184546098,
-0.5892472427173001,
-0.7139877518112785,
-0.12446625268371646,
-0.439646171766664,
0.7129808957109575,
0.3373075268097138,
0.3345550414369601,
-0.38068761370203624,
0.07864964370676188,
-0.3383553862900091,
0.6633419895127639,
-0.578720703894673,
0.486162302530746,
-0.06112668401522289,
0.34417654454775265,
-0.09182266750822224,
0.012215000731405658,
0.5658291917160285,
-0.16728488191683846,
0.3754616489195623,
-0.30497315054621493,
-0.7559846133933803,
0.9474959005813702,
-0.5904076686561035,
-0.2620011537974264,
0.1328515248519479,
-0.44066408562446224,
-0.7849425610737308,
0.44059474279745625,
-0.04646983134951628,
0.27897770882356554,
0.36838349877106535,
1.1610009997739648,
0.9058411506824863,
0.3119955912029684,
-0.5265683545543336,
1.0661822357546584,
0.8560044406571757,
0.5004409227892788,
0.5192086876555008,
-0.0019665558968806392,
-0.23949800878404184,
0.32563956832297103,
0.795323960796817,
1.5213873243487468,
0.8637563634588281,
0.07945031145196896,
1.7583315097859118,
0.5335864884320034,
-0.1366270407456156,
-0.9111102828185913,
-0.41650979281126865,
0.20926691692934202,
0.2776061293997892,
1.4257231544028384,

```

-0.10754544247245298,
1.0035199546983695,
1.1162390986419686,
1.7211522778091677,
-1.4332245925979028,
1.4417595659646931,
-2.106784537892007,
-0.6878309432904478,
-0.20360639354575213,
0.6373705186717347,
0.051333236332203484,
-0.35868991955772395,
-0.24365907607080922,
1.619736185871098,
0.5895182658245096,
-0.6086170057348131,
0.39774539931448055,
1.409837948786624,
0.17438880158294706,
-0.0415548590378924,
0.1413623732119859,
-0.8192475896231625,
0.5823982141653503,
-0.8462302367353051,
-0.60454163385331,
1.8216629921858374,
1.2634551976945552,
-0.40200420270308274,
-1.5954759495917508,
-1.7732438129518737]],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric3_unifrom_scaling_sample#3'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
'4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
'X 5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '

```

```
'+ 1 Z 0 + '  
'1 X 1 X 2 '  
'+ 1 Z 1 + '  
'1 X 2 X 3 '  
'+ 1 Z 2 + '  
'1 X 3 X 4 '  
'+ 1 Z 3 + '  
'1 X 4 X 5 '  
'+ 1 Z 4 + '  
'1 X 5 X 6 '  
'+ 1 Z 5 + '  
'1 Z 6']},  
    'output': { 'data_points': [ [4, 2, 0, 0, -6.1514222647937595],  
                                  [12, 6, 0, 0, -4.906882414400225],  
                                  [20, 10, 0, 0, -3.911546483229528]],  
    'noise_details': [ { 'gates_num': [4, 1, 1, 0],  
                          'identity_factors': [0, 0, 0, 0],  
                          'noise_level': [4, 2, 0, 0],  
                          'noise_profile': {  
  
                                'noise_prob': [  
  
0.001,  
0.001,  
0,  
0],  
  
                                'status': True,  
                                'type':  
  
                                'depolarizing'},  
  
                                'odd_wires': 4},  
                        { 'gates_num': [12, 3, 1, 0],  
                          'identity_factors': [1, 1, 0, 0],  
                          'noise_level': [12, 6, 0, 0],  
                          'noise_profile': {  
  
                                'noise_prob': [  
  
0.001,  
0.001,  
0,  
0],  
  
                                'status': True,  
                                'type':  
  
                                'depolarizing'},  
  
                                'odd_wires': 4},  
                        { 'gates_num': [20, 5, 1, 0],  
                          'identity_factors': [2, 2, 0, 0],  
                          'noise_level': [20, 10, 0, 0],  
                          'noise_profile': {  
  
                                'noise_prob': [  
  
0.001,  
0.001,  
0,  
0],  
  
                                'status': True,  
                                'type':  
  
                                'depolarizing'},  
  
                                'odd_wires': 4}],  
    'run_time_sec': 8.115881204605103}},  
    { 'config': { 'ansatz': { 'gateset': 1,  
                              'layer': 30,  
                              'ugate': { 'coefficients': { 'bn': [ 0,  
                                                                0,  
                                                                0,  
                                                                0,  
                                                                0,  
                                                                0],  
                                          'cn': [ 0.5,  
                                                  0.5,  
                                                  0.5,  
                                                  0.5,  
                                                  0.5,
```



```
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [ 0.48023040681030116,
0.7029544315470944,
0.722671987246733,
0.9307929858945541,
0.9307929858945543,
2.546598088688794,
2.546598088688794,
2.546598088688794,
2.546598088688794,
5.184334435330857,
5.195111458618343,
5.47436411399969,
5.47436411399969,
5.47436411399969,
5.47436411399969,
5.47436411399969,
6.068336244524409,
6.068336244524409,
7.838605251840715,
7.912250316107233,
7.912250316107233,
8.562517202814183,
9.063385768597811,
9.216218862905647,
9.216218862905647,
10.84047371877396,
10.84047371877396,
11.044358714867027,
11.375406566932424,
11.375406566932424,
0.18886549713610382,
0.6040291443391601,
-0.9727809979878402,
3.008209947509642,
0.3239743149147096,
-0.643987619924633,
1.0739882269552894,
0.4935103262912805,
-1.1713673989643745,
0.32339055167680386,
-0.8254170399860232,
3.387204293956005,
0.5437302598099015,
-0.6585961721148387,
1.0903163929414719,
-1.5261906384818182,
1.102349928722959,
-0.7373436929054379,
0.1712635196093208,
1.2086697581514199,
-0.4977298668106944,
-0.8828019932256734,
-0.13051782426094366,
0.048744185631251574,
-1.0535030780910155,
0.5224555306604155,
-0.9502111367985572,
-0.2573420205933925,
1.5833039752620888,
1.3806480812363842,
0.3864833677006273,
-0.0006284730712113316,
0.6487740408134013,
1.697829570896293,
-0.20023867528920775,
-0.32438662287273723,
1.3674639807988553,
-1.5592347723565667,
-0.6294582959326964,
1.11157394071659,
1.4895912958038178,
```

-0.7203901402661329,
-0.027076706863032294,
0.009574786781832467,
1.1437695218822916,
0.11720143179507149,
1.0973549625545624,
-1.0756168303605034,
-1.5785125289756954,
-0.42999007233668873,
0.7592603373612552,
0.7357293999316703,
0.03766014972468404,
1.450035910180884,
-0.08978031589044846,
0.7176267391661779,
-2.3377280976646966,
0.3878354316208249,
-0.8070070584177742,
0.07084117333557359,
-0.008998168433091903,
0.6612796195456235,
0.6287929626789847,
0.8601333603228948,
-0.7786354708155643,
-0.144476679155352,
0.39492950957300044,
-0.20012968159779548,
0.01326107241550192,
-0.5649340596325207,
-0.020398755490040426,
0.6333499066300691,
0.400844220606507,
0.5954749034740784,
0.8700413681534388,
1.315868035063538,
0.9914334712755634,
0.10418309057983809,
1.1095991500751576,
-0.1712743534684691,
-0.2730581386141925,
0.1453142903679217,
-0.3813955795194703,
0.1280814941289577,
-0.5627986213807415,
-1.0576722394029623,
-1.98721560251358,
0.519862592148374,
1.03695228961559,
0.21101375663462807,
-1.750252459751414,
-1.1450993785481438,
-2.2517541986316916,
-1.5456964825715331,
0.6026638989139771,
0.47561448130168205,
0.08201672609434518,
0.47464262797792917,
-0.14585857152212733,
0.08636040211648545,
0.7388192170680543,
1.071366965639021,
2.5115163876343742,
-0.4286872636197954,
-0.2977998181526895,
0.46569070003689333,
0.2115019986130443,
-0.5123638542481742,
0.09024611293676764,
-0.3380729423866059,
1.6314769884628497,
1.3338027138849353,
-0.7442166070564612,
1.2186722356710948,
1.584709484399307,
-0.6554175978902271,

```

0.36887213174672207,
-1.365021965092438,
-0.4623126041921459,
0.4869420326570461]],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric3_unifrom_scaling_sample#9'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
'4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
'X 5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
'+ 1 Z 0 + '
'1 X 1 X 2 '
'+ 1 Z 1 + '
'1 X 2 X 3 '
'+ 1 Z 2 + '
'1 X 3 X 4 '
'+ 1 Z 3 + '
'1 X 4 X 5 '
'+ 1 Z 4 + '
'1 X 5 X 6 '
'+ 1 Z 5 + '
'1 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -6.75124151968758],
[12, 6, 0, 0, -5.5757781087309555],
[20, 10, 0, 0, -4.6095181057209125]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [
0.001,
0.001,
0,
0],

```

```

'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,
0,
0],

'depolarizing'},

'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,
0,
0],

'depolarizing'},

'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,
0,
0],

'depolarizing'},

'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [1, 1, 0, 0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {

'noise_prob': [

'status': True,
'type':

'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [2, 2, 0, 0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {

'noise_prob': [

'status': True,
'type':

'odd_wires': 4}},
'run_time_sec': 6.46916651725769}},
{ 'config': { 'ansatz': { 'gateset': 1,
    'layer': 30,
    'ugate': { 'coefficients': { 'bn': [ 0,
    0,
    0,
    0,
    0,
    0,
    0],
    'cn': [ 0.5,
    0.5,
    0.5,
    0.5,
    0.5,
    0.5],
    'r': 0},
    'time': {'max': 10.0, 'min': 0.0},
    'type': 'ising'}},
'init_param': { 'value': [ 0.46756606893882113,
    0.46983859896604685,
    0.5133985927484581,
    0.5133985927484581,
    0.5133985927484581,
    1.4016794842576101,
    1.6914116785996751,
    1.691411678599675,
    1.6941756924801525,
    1.6941756924801528,
    3.749864451084885,
    3.749864451084885,
    3.749864451084885,
    3.749864451084885,
    4.969636107010246,
    4.969636107010246,
    4.969636107010246,
    4.969636107010246,
    5.983107385388112,
    5.983107385388112,
    8.378949246097575,
    8.378949246097575,

```

8.635680633753974,
8.635680633753974,
9.108807770997622,
9.108807770997622,
9.108807770997622,
9.960036411132767,
10.436643041587223,
10.716533289695375,
0.5466515694267952,
3.318518267704871,
0.40678728382369284,
0.6290091030083763,
-0.4770983795300032,
-0.7455483390932742,
0.1630064768968011,
0.4889724943271746,
0.10596004146863507,
-1.4759536729524354,
0.957604974713893,
0.6004514101153323,
0.0725728265044025,
1.4994700617905987,
0.15508641282397265,
-0.15494303336158094,
-0.5663845144966579,
2.6663065398173496,
-0.27086336898318036,
-1.1828784814640632,
-0.295045685212881,
0.6236994649007899,
1.1769360493970402,
1.7159795997719172,
-0.3019600780315931,
0.651545330446025,
-2.0782382222089724,
0.3186344830290561,
-0.2815476257680432,
0.4067635126151388,
-0.8228051832090519,
-0.4154478828430832,
2.3723163321732827,
0.931436930203099,
0.9804374423714106,
0.058895326584719035,
0.3167253661454302,
0.3390503539155689,
-0.5024758650485814,
-0.2425872659725522,
-0.6703364135359525,
-0.147558399690948,
1.1644296968010859,
0.5475507579506065,
0.1736068872874216,
0.30547527216138975,
0.5300773267023443,
0.7722201656261689,
-0.6858549484151734,
-0.16073113081558088,
-1.197458252004488,
-0.507307082981415,
-0.16437738354723153,
0.2647407403580543,
-0.45930133674563567,
-0.22673992276266391,
1.8127715449849986,
-1.4334128748253496,
-0.759404859696163,
1.2431110601081345,
0.8953271645792334,
0.6963251389999259,
-0.23147129445607595,
0.3638861144885483,
2.2394622330281906,
1.543816783446588,
1.7535830084845025,

```

-0.3169637330795974,
0.2453194636587019,
2.0076391670843647,
0.286017506499548,
0.6600449842625801,
0.19782118041633945,
0.05463577684798935,
0.4755005320975566,
0.055207215781380096,
0.06969685609102444,
0.2645328829673777,
-0.01235845210357445,
1.01982876424508,
2.262492417600915,
1.3443784687440867,
0.10698144982242018,
0.428081928456963,
-1.5494921893757727,
1.0701294130786736,
0.39987519173641656,
0.06301738855484718,
-1.6711192593807125,
0.26920159401030147,
-0.08358711295948693,
0.4054786916979446,
-0.5013194648307361,
2.126550690771933,
0.10959959792145137,
0.4205360694942305,
-0.586878230803869,
-0.3381608637889511,
-0.5055401953468239,
-0.45995744817477163,
0.07999668782445096,
-0.4269018063306222,
0.9588276598807326,
1.042187521522261,
0.07821502449201799,
0.6777167152964713,
0.4177387822172794,
-0.18721284346239236,
-0.8064458441104513,
2.8712851588196218,
0.7141862154478698,
-0.9976723444987857,
-0.8529435737233043,
1.9602636639541733,
1.2634521849205176,
-0.7170870121313951,
0.31227949023655405,
-1.0451293221946831,
-0.6787852158143914,
1.8581833428208436]],
'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value': 0},
                    'noise_prob': [0.001, 0.001, 0, 0],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},

```

False,

```

'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':

'AUTOMATE_ising_noisefree_time_evo_ric3_unifrom_scaling_sample#6'},
            'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                                  [1, 1, 0, 0],
                                                  [2, 2, 0, 0]]},

            'run': 'redundant',
            'state': 'dmatrix',
            'vqe': { 'iteration': 1,
                     'optimization': { 'algorithm': 'SLSQP',
                                       'constraint': True,
                                       'status': False}},

            'zne': { 'data_points': None,
                     'degree': 0,
                     'method': 'richardson',
                     'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
                                'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
                                '4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
                                'X 5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
                                                         '+ 1 Z 0 + '
                                                         '1 X 1 X 2 '
                                                         '+ 1 Z 1 + '
                                                         '1 X 2 X 3 '
                                                         '+ 1 Z 2 + '
                                                         '1 X 3 X 4 '
                                                         '+ 1 Z 3 + '
                                                         '1 X 4 X 5 '
                                                         '+ 1 Z 4 + '
                                                         '1 X 5 X 6 '
                                                         '+ 1 Z 5 + '
                                                         '1 Z 6']},

'output': { 'data_points': [ [4, 2, 0, 0, -6.487894898369523],
                              [12, 6, 0, 0, -5.297876077058435],
                              [20, 10, 0, 0, -4.3310003849914125]],
            'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                'identity_factors': [0, 0, 0, 0],
                                'noise_level': [4, 2, 0, 0],
                                'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,
0,
0],

                                'status': True,
                                'type':

'depolarizing'},

                                'odd_wires': 4},
                                { 'gates_num': [12, 3, 1, 0],
                                  'identity_factors': [1, 1, 0, 0],
                                  'noise_level': [12, 6, 0, 0],
                                  'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,
0,
0],

                                'status': True,
                                'type':

'depolarizing'},

                                'odd_wires': 4},
                                { 'gates_num': [20, 5, 1, 0],
                                  'identity_factors': [2, 2, 0, 0],
                                  'noise_level': [20, 10, 0, 0],
                                  'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

```

```
0.001,  
0.001,  
0,  
0],
```

```
'status': True,  
'type':
```

```
'depolarizing'},
```

```
                                'odd_wires': 4}},  
                                'run_time_sec': 7.285674571990967}},  
{ 'config': { 'ansatz': { 'gateset': 1,  
                            'layer': 30,  
                            'ugate': { 'coefficients': { 'bn': [ 0,  
                                                                0,  
                                                                0,  
                                                                0,  
                                                                0,  
                                                                0],  
                                                                'cn': [ 0.5,  
                                                                0.5,  
                                                                0.5,  
                                                                0.5,  
                                                                0.5,  
                                                                0.5],  
                                                                'r': 0},  
                                'time': {'max': 10.0, 'min': 0.0},  
                                'type': 'ising'}}},  
  'init_param': { 'value': [ 0.23818326834520384,  
                              0.242015507589488,  
                              0.5292100600091615,  
                              0.5598196968290768,  
                              0.5598196968290768,  
                              2.3451854381830244,  
                              2.3451854381830244,  
                              3.8453025861748578,  
                              3.8453025861748578,  
                              4.395137112369032,  
                              4.408833101975425,  
                              4.829585475162733,  
                              4.867115107997544,  
                              5.058958687371625,  
                              5.058958687371625,  
                              5.058958687371625,  
                              5.581243202757385,  
                              6.011433944644334,  
                              6.011433944644334,  
                              8.358003932454157,  
                              8.358003932454157,  
                              8.358003932454157,  
                              8.371420896509267,  
                              8.783744346022278,  
                              8.783744346022278,  
                              9.34189587380724,  
                              9.34189587380724,  
                              9.39608576624051,  
                              9.726727464435275,  
                              10.95102910414735,  
                              0.14721560345391302,  
                              1.0205767628001141,  
                              0.40564926087548847,  
                              -1.9238321605823567,  
                              -0.04729505851188324,  
                              0.482541600628026,  
                              0.1152155677775581,  
                              0.8524320040134185,  
                              0.42380767990264745,  
                              -0.6219665889019169,  
                              -0.05759638705357143,  
                              0.8161328242190486,  
                              0.7425402472775383,  
                              0.7535392998290764,  
                              1.8314745216576223,  
                              1.3334023135030668,  
                              0.05552193989453376,  
                              0.9691374289368472,
```


-0.07338372949421543,
-0.6690664310785921,
1.168533104165511,
-0.01962836639606316,
0.6576122461884292,
-0.037637936325233226,
-0.0009647519828692927,
0.4524839512418326,
-0.013115244288784337,
1.6270372625341938,
1.182284078721198,
0.9533845712158122,
1.0956662941001023,
-0.9647918240366662,
0.1897778948773928,
-0.04047520420354409,
0.5790278460715794,
0.631719469583885,
-0.3335418357294817,
1.0603186443819097,
-0.7449471003423949,
0.1630231005006795,
-0.07563007210422526,
-1.1009042311153838,
-0.34587951137957407,
-0.22408475871037167,
-0.6364303490666092,
-0.5245531829065874,
-0.7807390048523051,
-0.7767575864285188,
-0.9679768325508518,
-1.8691843526791756,
0.4783530568164554,
0.6133546874260105,
-0.20020986269221935,
-0.41532071148562183,
-0.8557118152516873,
-0.16985308585729458,
0.8428338023785568,
0.6750667051632169,
1.2795031602880573,
-0.41384224674171627,
0.3213362409234008,
1.2340309605381181,
1.0506369357289735,
-0.6415500741723684,
-0.9628453188140567,
1.935214357876566,
1.4159154638103797,
-2.0864443113151867,
-0.07426020994129089,
-1.0538315394110076,
0.2589745455351295,
-0.2955978932252484,
-0.030644528099012087,
-0.8602288471584197,
-0.09490606960216369,
0.4190622694749842,
-1.5667390101922392,
-0.7804920738383083,
0.1255849265457167,
0.5884338319364165,
-0.46028089859616994,
0.2488277677353687,
-0.7513271615815256,
-1.0561258734551267,
0.3697237180501531,
-0.7108432439071602,
0.5137418203820399,
0.946767478727941,
1.619957961978311,
1.2944079622561346,
0.21625880379379211,
0.007168703139473973,
-0.334764182643075,

```
'AUTOMATE_ising_noisefree_time_evo_ric3_unifrom_scaling_sample#4'},
```

```
'noise_profile': { 'noise_on_init_param': { 'status':
```

```
'value': 0},
```

```
'noise_prob': [0.001, 0.001, 0, 0],
```

```
'status': True,
```

```
'type': 'depolarizing'},
```

```
'nqubits': 7,
```

```
'observable': { 'coefficients': { 'bn': [ 1.0,
```

1.0,

1.0,

1.0,

1.0,

1.0,

1.0]

0.5,

0.5,

0.5,

0.5,

0.5,

0.51

1

```
aw': { 'fig dpi':
```

```

    'status': False

```

```
'type': 'png'},
```

```
e_prefix':
```

```
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
```

 $[1, 1, 0, 0],$ $[2, 2, 0, 0]]\},$

```
'run': 'redundant',
```

```
'state': 'dmatrix',
```

```
'vqe': { 'iteration': 1,
```

```
'optimization': { 'algorithm': 'SLSQP',
```

```
'constraint': True,
```

```
'status': False}},
```

```
'zne': { 'data_points': None,
```

```
'degree': 0,
```

```
'method': 'richardson',
```

```
'sampling': 'default']}]},
```

```
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
```

'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '

'4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '

'X 5 X 6 + 1 Z 5 + 1 Z 6',

```
_hamiltonian_string': [ '1 X 0 X 1 ' ]
```

' + 1 Z 0 + '

'1 X 1 X 2 '

$$+ 1 \quad Z \quad 1 \quad + \quad '$$


```
'type': 'ising'}}},
'init_param': { 'value': [ 0.5118290521669792,
0.9034573591840084,
1.0514460879099101,
1.1283814725964603,
1.1293980629167455,
1.6874068541059999,
1.8362001710030256,
2.594529848453243,
2.5949023258749775,
4.384420392533986,
4.804388533064279,
4.8044974726668155,
4.925049921883437,
4.925049921883437,
6.055903002618935,
6.0650337145794,
6.06510024924717,
6.06510024924717,
7.719643893437192,
7.719643893437192,
8.110116534451196,
8.374448879129652,
8.374448879129652,
9.29045698473019,
9.2929923765699,
9.977025927736127,
9.977025927736127,
9.977438303479875,
11.277250876863835,
11.588901828478656,
-0.5970483257864091,
1.048395642841534,
0.3563299433581316,
2.04182497417867,
-1.0659189653240388,
-1.5711224255679066,
-0.6854026923334804,
1.3002799990574343,
0.334979527312172,
-0.41771661057992626,
-0.5255960701407614,
0.9200835209290303,
0.5840294212204432,
0.4309612994747514,
0.010788324288593443,
0.8398650983618882,
0.35555521736822654,
0.33383672576306345,
-0.03972798241435421,
-0.5808926346218187,
-0.6217506829607715,
-0.022449055124049946,
-0.4671428723737863,
1.3840583406465645,
3.321823997583478,
1.9516123562000514,
2.109430068020724,
0.17138900458361278,
1.0734765554988877,
1.1591682463434823,
-1.5085197494005436,
0.3709514368460093,
-0.08354812013321816,
-0.6849414455501801,
-1.1807064172516142,
0.1889544406654655,
0.45125852432694835,
0.40946867023093825,
0.9591749175330944,
0.862406962512403,
-0.47565451021668054,
0.14791469386649075,
0.012323642821740255,
-0.3613715829994197,
```

-0.6520104899254531,
1.1500641632479522,
0.11480411675389499,
0.37929728325133344,
-0.12818017639377846,
1.5548403554359806,
-0.08803696907532751,
0.16058007713541472,
-0.2835300528775952,
1.4317946638041426,
0.04218231832431045,
0.6164913379423986,
-0.5546192578057587,
-0.0016176435191898148,
0.3455446109244303,
-0.7983090776945799,
0.06548494602750325,
0.09601011255154575,
-1.274882751616874,
-0.6639296860891427,
0.08468574728059233,
0.09178155492125434,
-1.9746588496307496,
1.363083624655819,
-0.11754172722968637,
-1.7454296260872237,
0.689437867548643,
0.09536601432398076,
2.1875690275532618,
1.9737367040865754,
0.19623670188051764,
1.3618525198389628,
-1.6551377680800423,
-1.2796187335858575,
-0.02979150349567955,
0.3655877448114123,
-1.339933806945279,
-1.6131280858420662,
1.1268021774842836,
0.7853899025443938,
-1.214954346638235,
1.881374469676006,
-2.2108375055617597,
0.38299461206746305,
-0.062378913993248956,
-1.0812916442950207,
0.19113842494911862,
0.43947065709017896,
1.4340329690981122,
-2.1914706359927933,
1.683520172253517,
-0.4151753501170658,
0.2652280013518245,
-0.09829123134965576,
-0.4033858255054015,
-0.09488290407421365,
1.1087820722198931,
1.3758250784849964,
0.3693308876448478,
-1.4774754586810708,
0.45790576106810227,
-1.318961759734687,
1.2898775251023542,
0.5291693857421421,
-0.7225860054677886,
-0.6024865845145562,
1.1258609582335735,
1.4545810101615688,
0.4988359655120556,
0.39968483637691726,
1.4778005154845164,
1.6358502222921723,
0.07081811420225437,
1.5336807499569445,
-0.7595143926860392,

False,

```
0.07947795439670366]],
'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value': 0},
                  'noise_prob': [0.001, 0.001, 0, 0],
                  'status': True,
                  'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
               'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric3_unifrom_scaling_sample#10'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
                                'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
                                '4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
                                'X 5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
                                                         '+ 1 Z 0 + '
                                                         '1 X 1 X 2 '
                                                         '+ 1 Z 1 + '
                                                         '1 X 2 X 3 '
                                                         '+ 1 Z 2 + '
                                                         '1 X 3 X 4 '
                                                         '+ 1 Z 3 + '
                                                         '1 X 4 X 5 '
                                                         '+ 1 Z 4 + '
                                                         '1 X 5 X 6 '
                                                         '+ 1 Z 5 + '
                                                         '1 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -6.406033238196167],
                             [12, 6, 0, 0, -5.14768368077025],
                             [20, 10, 0, 0, -4.13060369641699]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                    'identity_factors': [0, 0, 0, 0],
                    'noise_level': [4, 2, 0, 0],
                    'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
                        'noise_prob': [
0.001,
0.001,
0,
0],
                        'status': True,
                        'type':
'depolarizing'},
```

```

'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,
0,
0],

'depolarizing'},

'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,
0,
0],

'depolarizing'},

'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [1, 1, 0, 0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {

'noise_prob': [

'status': True,
'type':

'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [2, 2, 0, 0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {

'noise_prob': [

'status': True,
'type':

'odd_wires': 4}},
'run_time_sec': 7.40142035484314}},
{ 'config': { 'ansatz': { 'gateset': 1,
    'layer': 30,
    'ugate': { 'coefficients': { 'bn': [ 0,
    0,
    0,
    0,
    0,
    0],
    'cn': [ 0.5,
    0.5,
    0.5,
    0.5,
    0.5,
    0.5],
    'r': 0},
    'time': {'max': 10.0, 'min': 0.0},
    'type': 'ising'}},
'init_param': { 'value': [ 0.4349098035102067,
    0.5281397824211302,
    0.6402833431740305,
    0.701719039924861,
    0.8953768954552646,
    1.4118719346144908,
    2.230113621742168,
    2.6548053250922607,
    3.10433397641996,
    3.1043339764199596,
    4.146783795497047,
    4.146783795497047,
    4.149760661232294,
    4.149760661232294,
    4.223539194779984,
    4.223539194779984,
    6.7727886487081435,
    6.7727886487081435,
    7.243551717080481,
    7.282491880183138,
    7.282491880183137,
    7.282491880183138,
    8.806989173945695,
    8.80698917394569,
    9.706285648296737,

```

9.809777023075187,
9.837118725166619,
10.701586770517197,
11.185931122995514,
11.185931122995514,
0.39919688039300266,
1.4533249104921768,
0.7852252158845567,
-0.4957764525834907,
-0.3354719566017423,
0.16097458435951068,
-0.21971891747261646,
1.5370438656668401,
0.26531517225969137,
-0.6386119707402834,
0.37748437574064975,
1.2333814560270282,
0.5934062959304571,
1.633059089666515,
0.31170790405459564,
0.3995146629536743,
-0.11155094016301184,
0.8138063699882787,
0.019467271717771087,
-0.38186317682975884,
-2.778803401027002,
0.7887082829537533,
-3.1348050840084074,
-1.5109224736536222,
-0.38589710406642097,
-1.105927053527023,
0.3451707237315739,
0.6290043291617069,
-1.834722402137916,
-1.0551446159380258,
-1.3581260193610523,
0.9666180881078981,
-0.05759676522292653,
1.1866392284178628,
-1.1308471740259582,
0.34327443183487805,
0.17662613760194443,
-0.043135596735193626,
-0.03667546733606037,
-0.4901895513664937,
-1.0399227541005371,
0.4880459765792483,
0.053089003166299456,
-0.6259966396796703,
0.03159579372412852,
-0.09798805529491862,
0.052872660107769434,
1.490194875693949,
0.22167423812356596,
-0.33497064714209324,
1.0144679614160885,
0.01608415027143934,
0.49167899744128424,
0.12491439548600353,
-0.1329609067962543,
0.8639852596635488,
-1.6607871811173882,
0.2801513494447602,
0.3908501333463579,
0.6044891435983025,
0.08637544621327581,
1.2270127042754213,
0.11860545946003472,
-0.13066468162372125,
-0.998222267933649,
-1.2643438207146234,
0.7835210648968657,
0.8370312354478452,
0.06527740704106412,
-1.0235950282739155,


```

-0.705070781232052,
0.037682760290170314,
1.4778132318704873,
1.7053977225841355,
-0.1291331271719845,
0.1648924115529835,
2.208642048449397,
1.1885953055185114,
-0.8138037079028133,
0.6715189891287302,
0.7819572861394978,
-0.4022064636590845,
-0.3927127310055937,
-0.4118784323717791,
1.3463239978578432,
-0.3323606385924751,
0.6664683657180488,
1.58852177854836,
-1.212203169826861,
-0.031436536468697034,
0.5094262403458202,
-0.0616688967239856,
-0.0666169691754888,
-0.7277012125404719,
-0.12684196683036533,
0.5325519345188151,
2.2778420441533447,
0.8735933622003313,
0.9907279467768721,
1.1448787224294625,
0.4291077507842401,
-0.21356701746617238,
1.7174041100732933,
0.1885112766205208,
0.3630143936992631,
0.6395640633025061,
-0.2937369198596852,
-0.18128670495631546,
0.9504587478277511,
-0.6241293783488098,
1.8733903679237267,
0.4875344831143031,
-0.9182747577406853,
0.850791268752179,
-0.46183623376516114,
-0.634145646812118,
-0.9165900492445601,
-0.8973746003077631,
-0.879107286657044,
0.15645801311276636]],
'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value': 0},
                    'noise_prob': [0.001, 0.001, 0, 0],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [ 0.5,
                                         0.5,
                                         0.5,
                                         0.5,
                                         0.5,
                                         0.5],
                                'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},

```

False,

```
'file_name_prefix':  
'AUTOMATE_ising_noise_free_time_evo_ric3_unifrom_scaling_sample#8'},  
    'redundant': { 'identity_factors': [ [0, 0, 0, 0],  
                                           [1, 1, 0, 0],  
                                           [2, 2, 0, 0]]},  
  
    'run': 'redundant',  
    'state': 'dmatrix',  
    'vqe': { 'iteration': 1,  
              'optimization': { 'algorithm': 'SLSQP',  
                                'constraint': True,  
                                'status': False}},  
  
    'zne': { 'data_points': None,  
             'degree': 0,  
             'method': 'richardson',  
             'sampling': 'default'}},  
  'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '  
                                     'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '  
                                     '4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '  
                                     'X 5 X 6 + 1 Z 5 + 1 Z 6',  
               'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '  
                                                            '+ 1 Z 0 + '  
                                                            '1 X 1 X 2 '  
                                                            '+ 1 Z 1 + '  
                                                            '1 X 2 X 3 '  
                                                            '+ 1 Z 2 + '  
                                                            '1 X 3 X 4 '  
                                                            '+ 1 Z 3 + '  
                                                            '1 X 4 X 5 '  
                                                            '+ 1 Z 4 + '  
                                                            '1 X 5 X 6 '  
                                                            '+ 1 Z 5 + '  
                                                            '1 Z 6']},  
               'output': { 'data_points': [ [4, 2, 0, 0, -6.672780190342767],  
                                             [12, 6, 0, 0, -5.522238922989663],  
                                             [20, 10, 0, 0, -4.5712287338679385]],  
                             'noise_details': [ { 'gates_num': [4, 1, 1, 0],  
                                                    'identity_factors': [0, 0, 0, 0],  
                                                    'noise_level': [4, 2, 0, 0],  
                                                    'noise_profile': {  
  
                                                         'noise_prob': [  
  
0.001,  
0.001,  
0,  
0],  
  
                                                         'status': True,  
                                                         'type':  
  
'depolarizing'},  
  
                                                     'odd_wires': 4},  
                                                  { 'gates_num': [12, 3, 1, 0],  
            'identity_factors': [1, 1, 0, 0],  
            'noise_level': [12, 6, 0, 0],  
            'noise_profile': {  
  
                                                         'noise_prob': [  
  
0.001,  
0.001,  
0,  
0],  
  
                                                         'status': True,  
                                                         'type':  
  
'depolarizing'},  
  
                                                     'odd_wires': 4},  
                                                  { 'gates_num': [20, 5, 1, 0],  
            'identity_factors': [2, 2, 0, 0],  
            'noise_level': [20, 10, 0, 0],  
            'noise_profile': {  
  
                                                         'noise_prob': [  
  
0.001,  
0.001,  
0,
```

0],

'status': True,
'type':

'depolarizing'},

```
                                'odd_wires': 4]],  
                                'run_time_sec': 8.05130386352539}}],  
'vqe': [ { 'config': { 'ansatz': { 'gateset': 1,  
                                    'layer': 30,  
                                    'ugate': { 'coefficients': { 'bn': [ 0,  
                                                                    0,  
                                                                    0,  
                                                                    0,  
                                                                    0,  
                                                                    0],  
                                                                    'cn': [ 0.5,  
                                                                    0.5,  
                                                                    0.5,  
                                                                    0.5,  
                                                                    0.5,  
                                                                    0.5],  
                                                                    'r': 0},  
                                    'time': {'max': 10.0, 'min': 0.0},  
                                    'type': 'ising'}}},  
            'init_param': {'value': 'random'},  
            'noise_profile': { 'noise_on_init_param': { 'status': False,  
                                                         'value': 0},  
                               'noise_prob': [0.001, 0.001, 0, 0],  
                               'status': True,  
                               'type': 'depolarizing'},  
            'nqubits': 7,  
            'observable': { 'coefficients': { 'bn': [ 1.0,  
                                                       1.0,  
                                                       1.0,  
                                                       1.0,  
                                                       1.0,  
                                                       1.0,  
                                                       1.0],  
                                                       'cn': [ 0.5,  
                                                       0.5,  
                                                       0.5,  
                                                       0.5,  
                                                       0.5,  
                                                       0.5],  
                                                       'r': 1},  
                            'def': 'ising'},  
            'output': { 'draw': { 'fig_dpi': 100,  
                                   'status': True,  
                                   'type': 'png'},  
                        'file_name_prefix': 'ising_noisefree_time_evo'},  
            'redundant': { 'identity_factors': [ [0, 0, 0, 0],  
                                                  [1, 1, 0, 0]]},  
            'run': 'vqe',  
            'state': 'dmatrix',  
            'vqe': { 'iteration': 10,  
                    'optimization': { 'algorithm': 'SLSQP',  
                                       'constraint': True,  
                                       'status': True}},  
            'zne': { 'data_points': None,  
                    'degree': 1,  
                    'method': 'richardson',  
                    'sampling': 'default'}},  
            'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + '  
                                             '1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + '  
                                             '1 X 4 X 5 + 1 Z 4 + 1 X 5 X 6 + 1 Z 5 + '  
                                             '1 Z 6',  
                    'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 + 1 Z '  
                                                                  '0 + 1 X 1 X 2 + '  
                                                                  '1 Z 1 + 1 X 2 X '  
                                                                  '3 + 1 Z 2 + 1 X '  
                                                                  '3 X 4 + 1 Z 3 + '  
                                                                  '1 X 4 X 5 + 1 Z '  
                                                                  '4 + 1 X 5 X 6 + '  
                                                                  '1 Z 5 + 1 Z 6']},  
            'output': { 'exact_sol': -8.56677223350563,
```

```
'initial_cost_history': [ 4.2809785500030175,
                          2.611097744705238,
                          3.419414059277099,
                          1.5773527652373136,
                          3.36522638437751,
                          3.1574660994918307,
                          4.256553926466165,
                          4.577016272245339,
                          4.50890344694736,
                          4.056304045999429],
'noise_details': { 'gates_num': [4, 1, 1, 0],
                   'identity_factors': [0, 0, 0, 0],
                   'noise_level': [4, 2, 0, 0],
                   'noise_profile': { 'noise_on_init_param': {
                                     'noise_prob': [ 0.001,
                                                      0.001,
                                                      0,
                                                      0],
                                     'status': True,
                                     'type': 'depolarizing'},
                                   'odd_wires': 4},
'optimized_minimum_cost': [-6.833109640228162,
                           -6.51493088468928,
                           -6.151422264793805,
                           -6.630764461698797,
                           -6.72096904084131,
                           -6.487894898369486,
                           -6.662379321078827,
                           -6.6727801903427855,
                           -6.751241519687556,
                           -6.406033238196173],
'optimized_parameters': [ [-4.1290471568429383e-17,
                           0.028387941050870547,
                           0.028387941050870516,
                           0.028387941050870544,
                           0.4764187545019148,
                           0.485154879376271,
                           0.7094706976105217,
                           2.998376901501724,
                           3.486264965503922,
                           3.487775573446841,
                           4.058897205630428,
                           5.020552086463093,
                           5.02158492362958,
                           5.02158492362958,
                           5.0846878698400815,
                           5.0846878698400815,
                           6.794704827143902,
                           6.794704827143902,
                           6.794704827143902,
                           8.013356083289752,
                           8.21076169573117,
                           8.21076169573117,
                           8.633075344303478,
                           8.633075344303478,
                           8.930018414781724,
                           8.930018414781724,
                           8.930018414781724,
                           9.990329186240457,
                           9.990329186240457,
                           12.915390927357043,
                           0.8289215528981712,
                           0.6465354442587259,
                           -0.06922368073773534,
                           -2.0853926237797484,
                           0.25605409461249207,
                           -0.765586831155604,
                           0.9589723041272441,
                           0.27455658215646966,
                           -0.07189982532198122,
                           -0.3449139525561357,
                           -0.46800027418386986,
                           1.016480689947782,
```

-0.9706976058410632,
-0.9556868223128826,
0.9323574793538898,
-0.4376613900889883,
0.44138701419965265,
1.4418520162063444,
0.8420619129497768,
-0.025530845682239494,
-2.3076344184440156,
0.37458463155346333,
-0.47031338116346383,
0.07338001085955462,
-0.869249007732584,
1.6824786550253577,
1.719346297136726,
-0.9622509832303179,
0.5779076253870294,
0.4760527439731688,
-0.2451715046346434,
1.5906513054768954,
0.35918458386075697,
-0.44817294000256097,
-0.37416377940751877,
0.37048554789855764,
0.693137718643464,
2.288219219235325,
-0.26501198620577204,
0.3689053264668303,
0.10166782579927121,
-0.3356073800920907,
-0.6413954053758932,
0.28636656044125136,
-0.8690978212322298,
1.166030172665772,
-0.46365263204645707,
-0.8505396438729699,
2.4328989003329693,
-1.4554443951257197,
2.09134311034601,
1.7695145471360614,
1.1170184462314887,
0.20325573406500014,
0.34206355374635444,
0.016804831103183593,
-0.19882507083844791,
1.4011478969931443,
-0.0572616468882171,
-1.3773340344964602,
1.1031086912886459,
0.47433623341773395,
-0.037679845343088075,
0.7450954028512253,
1.3114871459373192,
-0.18651966717154955,
1.0750199871192778,
0.48905379729033566,
-0.6281827821606347,
1.322074656325554,
-1.4761484868509396,
0.2859918816514792,
-1.0033541595176405,
0.8264053813863532,
0.622010811103528,
1.1636747207646185,
-0.31883271288389914,
-0.0013981850979601159,
-0.09999405171802342,
1.3085280853238759,
-1.3727969089464465,
-0.08341159264840771,
1.6249369276484606,
-0.28082331570516694,
1.863294400254821,
2.2932652495309953,
1.4849536675442596,

0.6811668802751195,
1.5380110980874968,
-0.6615225065895972,
0.4000828281024264,
0.33932121588009856,
0.48855764825038867,
2.5371111232777053,
-0.7260430040504564,
0.09005940379987407,
-0.8762980817992282,
-0.23800314057977784,
0.9728027685352436,
-0.20857834368009137,
0.6148333457084475,
-1.5171898336040224,
0.8658542069129427,
-1.6478287868490311,
0.5770049367106069,
0.134497700004433,
-0.11612046371881249,
1.5826352840237945,
1.6045820840261675,
-0.08805669980595342,
-1.507745125921201,
-0.11276651576435986,
0.07206005105970639,
-1.7788122679933258,
-0.002988765738482436,
0.7255804452314191,
2.7661804504965604,
-3.1943627011957507,
-0.07088434770936546,
0.10949711075632612],
[0.27654223573266234,
0.27654223573266246,
0.27654223573266246,
0.4643738922966213,
0.6582164862517121,
0.6582164862517121,
0.6582164862517121,
1.3424523496852623,
1.47635793598756,
1.47635793598756,
3.385550316280557,
3.385550316280557,
3.385550316280557,
4.494062371253785,
4.494062371253785,
5.0204758683966135,
5.023404188660082,
7.05704094573064,
7.087391629397132,
7.533802135057517,
8.245970909352765,
8.390887842007308,
8.390887842007308,
8.703391612007538,
8.703391612007538,
9.485545381639698,
9.485545381639698,
9.849437794456351,
9.849437794456351,
10.799155842585746,
-0.12708089143278917,
1.4168148977025445,
-0.13519536748631492,
-0.32308714516775877,
0.06981336221044306,
-0.034321895006426455,
-0.18209443759399158,
-0.024858958594746107,
0.664488174834645,
0.6897317665852637,
-0.008774731192519907,
0.625642551337688,

0.6862410821653743,
-0.7670419540542446,
-0.48055390132433673,
1.573964677292243,
-1.0555745677813198,
1.3159721165631335,
-0.9138502497993347,
-0.4641566084568084,
-0.06966564000847691,
0.039012746519005555,
1.0827641697579717,
1.2190093772283843,
1.5391150382377632,
-0.6675581491933664,
0.09104662559232415,
0.5283818781579022,
0.04478118244404365,
1.5836075485651113,
0.251932560991488,
-0.3949560381811925,
0.8854639388207172,
-0.8611267164776638,
1.566928747241295,
-0.15445151319573347,
-0.01396050433941564,
0.8172620115356718,
-0.014658132241289147,
-0.6623010385304499,
0.6299284572753584,
-0.15026947869759505,
-1.2293112897827836,
-0.5768656731979434,
1.231557922062332,
-1.447872926116422,
1.4918722706131171,
0.31945486783385635,
0.43870804207253483,
1.3582636256308331,
-0.8478398442626237,
0.799518029891469,
0.5539369896048497,
0.13453197415145846,
0.7167498104862234,
-0.06589109804263343,
0.041038724923082276,
1.151975756926885,
-0.052026838729968425,
0.3952978000808587,
-1.1754748970457833,
-0.01965607042871015,
-0.2456324144680547,
-0.01136184217607491,
0.022503168314325715,
1.1777923723104653,
0.14554117253938412,
-1.2287946167385297,
-0.9683737383161766,
1.073329622962756,
-0.7143583802799215,
-0.3545002682598469,
-0.31906110571926777,
-0.9597094503848127,
0.8825903987559673,
-0.08273415521500459,
0.2895039435279609,
-0.3388594197258312,
0.2406459434933063,
0.3824128671502841,
-0.5390555829405563,
0.3757684495816124,
-0.5015719636333678,
0.3958257688275076,
-0.6428354095046468,
-0.9178624992467067,
0.29327306512581747,

-0.09193285724698198,
0.1770091195095364,
0.6803523512851801,
1.05525770275555,
-0.4646437845139076,
2.2560908642344026,
-0.7500939262299234,
1.92959092052175,
-0.36895777993274226,
-0.04538283416771957,
1.2529586558275392,
-0.57805717638909,
0.3080585810325818,
0.031910210614104206,
0.18519950010488512,
1.3888566879174211,
1.087260914864616,
0.15259355623633358,
-0.08810167996863522,
0.07467874103808375,
1.4444893382870363,
-1.446220680213735,
0.8215180856176293,
2.0151723649306295,
0.13454991965456886,
0.21923381008732426,
0.31112145477192793,
-0.8977876743954992,
0.10239876571985529,
-1.0132922278363503,
-0.3794863035801566,
0.8437803973608401,
1.111671691788496],
[0.3463233138801294,
0.34632331388012944,
0.3463233138801294,
0.3463233138801293,
1.2403205490491955,
1.38033446895038,
3.6939218351946623,
3.6939218351946623,
3.693921835194662,
3.693921835194662,
3.9412089126587,
3.9412089126587,
3.9412089126587007,
3.9412089126587007,
4.8310999601027795,
4.870844593548848,
4.900228671797898,
4.9002286717978984,
4.9002286717978984,
4.900228671797898,
5.70167657474683,
6.21188441483336,
6.21188441483336,
6.686419232103557,
6.723504371727371,
7.618130890836856,
7.618130890836856,
9.420364260644014,
9.420364260644014,
10.775132222422771,
-0.3525750090684208,
1.2748976558430758,
-0.16146998454247521,
1.3224575871686013,
0.0349265192498484,
0.36138711032415904,
-0.5043488106619078,
0.7634801084306476,
-0.3536814593617201,
0.9606809354246906,
-1.1143894720984433,
-0.45289061390722213,

-0.1605333784989486,
-1.594210268458609,
-0.09465166736510336,
0.11132251138190599,
1.004235666711761,
-0.23115493220428254,
0.9451761882751959,
-0.3707284170975133,
-0.17782948231634454,
-0.18433417990028922,
-0.06421460341189031,
-1.1554890001827845,
-0.2074908306187579,
-0.036360325422879626,
0.5283181819327893,
0.3603171208434518,
0.6819349944270952,
-0.2733083993242672,
0.7075879191324372,
-0.042779623335368125,
0.7032435595714176,
-0.7363942983542189,
0.9583235184546098,
-0.5892472427173001,
-0.7139877518112785,
-0.12446625268371646,
-0.439646171766664,
0.7129808957109575,
0.3373075268097138,
0.3345550414369601,
-0.38068761370203624,
0.07864964370676188,
-0.3383553862900091,
0.6633419895127639,
-0.578720703894673,
0.486162302530746,
-0.06112668401522289,
0.34417654454775265,
-0.09182266750822224,
0.012215000731405658,
0.5658291917160285,
-0.16728488191683846,
0.3754616489195623,
-0.30497315054621493,
-0.7559846133933803,
0.9474959005813702,
-0.5904076686561035,
-0.2620011537974264,
0.1328515248519479,
-0.44066408562446224,
-0.7849425610737308,
0.44059474279745625,
-0.04646983134951628,
0.27897770882356554,
0.36838349877106535,
1.1610009997739648,
0.9058411506824863,
0.3119955912029684,
-0.5265683545543336,
1.0661822357546584,
0.8560044406571757,
0.5004409227892788,
0.5192086876555008,
-0.0019665558968806392,
-0.23949800878404184,
0.32563956832297103,
0.795323960796817,
1.5213873243487468,
0.8637563634588281,
0.07945031145196896,
1.7583315097859118,
0.5335864884320034,
-0.1366270407456156,
-0.9111102828185913,
-0.41650979281126865,

0.20926691692934202,
0.2776061293997892,
1.4257231544028384,
-0.10754544247245298,
1.0035199546983695,
1.1162390986419686,
1.7211522778091677,
-1.4332245925979028,
1.4417595659646931,
-2.106784537892007,
-0.6878309432904478,
-0.20360639354575213,
0.6373705186717347,
0.051333236332203484,
-0.35868991955772395,
-0.24365907607080922,
1.619736185871098,
0.5895182658245096,
-0.6086170057348131,
0.39774539931448055,
1.409837948786624,
0.17438880158294706,
-0.0415548590378924,
0.1413623732119859,
-0.8192475896231625,
0.5823982141653503,
-0.8462302367353051,
-0.60454163385331,
1.8216629921858374,
1.2634551976945552,
-0.40200420270308274,
-1.5954759495917508,
-1.7732438129518737],
[0.23818326834520384,
0.242015507589488,
0.5292100600091615,
0.5598196968290768,
0.5598196968290768,
2.3451854381830244,
2.3451854381830244,
3.8453025861748578,
3.8453025861748578,
4.395137112369032,
4.408833101975425,
4.829585475162733,
4.867115107997544,
5.058958687371625,
5.058958687371625,
5.058958687371625,
5.581243202757385,
6.011433944644334,
6.011433944644334,
8.358003932454157,
8.358003932454157,
8.358003932454157,
8.371420896509267,
8.783744346022278,
8.783744346022278,
9.34189587380724,
9.34189587380724,
9.39608576624051,
9.726727464435275,
10.95102910414735,
0.14721560345391302,
1.0205767628001141,
0.40564926087548847,
-1.9238321605823567,
-0.04729505851188324,
0.482541600628026,
0.1152155677775581,
0.8524320040134185,
0.42380767990264745,
-0.6219665889019169,
-0.05759638705357143,
0.8161328242190486,

0.7425402472775383,
0.7535392998290764,
1.8314745216576223,
1.3334023135030668,
0.05552193989453376,
0.9691374289368472,
-0.07338372949421543,
-0.6690664310785921,
1.168533104165511,
-0.01962836639606316,
0.6576122461884292,
-0.037637936325233226,
-0.0009647519828692927,
0.4524839512418326,
-0.013115244288784337,
1.6270372625341938,
1.182284078721198,
0.9533845712158122,
1.0956662941001023,
-0.9647918240366662,
0.1897778948773928,
-0.04047520420354409,
0.5790278460715794,
0.631719469583885,
-0.3335418357294817,
1.0603186443819097,
-0.7449471003423949,
0.1630231005006795,
-0.07563007210422526,
-1.1009042311153838,
-0.34587951137957407,
-0.22408475871037167,
-0.6364303490666092,
-0.5245531829065874,
-0.7807390048523051,
-0.7767575864285188,
-0.9679768325508518,
-1.8691843526791756,
0.4783530568164554,
0.6133546874260105,
-0.20020986269221935,
-0.41532071148562183,
-0.8557118152516873,
-0.16985308585729458,
0.8428338023785568,
0.6750667051632169,
1.2795031602880573,
-0.41384224674171627,
0.3213362409234008,
1.2340309605381181,
1.0506369357289735,
-0.6415500741723684,
-0.9628453188140567,
1.935214357876566,
1.4159154638103797,
-2.0864443113151867,
-0.07426020994129089,
-1.0538315394110076,
0.2589745455351295,
-0.2955978932252484,
-0.030644528099012087,
-0.8602288471584197,
-0.09490606960216369,
0.4190622694749842,
-1.5667390101922392,
-0.7804920738383083,
0.1255849265457167,
0.5884338319364165,
-0.46028089859616994,
0.2488277677353687,
-0.7513271615815256,
-1.0561258734551267,
0.3697237180501531,
-0.7108432439071602,
0.5137418203820399,

0.946767478727941,
1.619957961978311,
1.2944079622561346,
0.21625880379379211,
0.007168703139473973,
-0.334764182643075,
-0.9086121674985949,
-0.8247776105092918,
-0.5734460393428641,
-0.4381826298821608,
-0.44306191141826123,
-0.23860269421390362,
-0.03343496826073402,
-0.008972211649351379,
1.2196121588758198,
1.5985131140815898,
0.060325450296581484,
-0.7312434635304047,
0.9413542695101463,
-0.6701271691421761,
-1.0680925542186157,
-0.5192506372311365,
1.5594784335685845,
0.17928043514789443,
0.2455246384495121,
-0.053106843359071614,
0.7488220810664211,
1.3030207698426615,
-1.787387332504871,
-1.1184519607509558,
-0.2511275398302805,
1.445676213762956,
1.5991220174606833],
[0.32980400918867037,
0.7818596814437996,
0.9680306497310072,
1.0586432954274991,
1.7663116348413028,
1.8956491295658664,
2.0436116769319477,
2.0436116769319477,
2.0436116769319477,
2.1846619555479685,
2.1846619555479685,
4.516997836642403,
4.516997836642403,
5.3166564351730035,
5.320130741320412,
5.399447086770547,
5.400046100307029,
5.400046100307029,
5.400046100307029,
5.400046100307029,
7.097750404951498,
7.618222769006738,
7.956666567531361,
7.956666567531361,
8.961109688308255,
8.961109688308255,
11.028500695271937,
11.5809418583784,
11.5809418583784,
11.5809418583784,
0.24114173221382337,
1.3754512844042854,
0.9720111739434927,
-0.29877116871955545,
-0.029737313498699618,
0.3185153459444675,
-0.31840745367437817,
0.6801602139502809,
-0.015407845998522466,
-1.114484654283524,
0.22997676136047449,
1.5201821520192487,

0.3217486942799661,
0.43678760134757944,
0.17618749246228715,
0.7010359103792874,
0.3446701255696699,
1.4700166379981474,
0.3471333265905846,
-0.968444017764766,
-1.1485269521507444,
0.3282504398407059,
-0.7898554413193151,
-0.22157985789400653,
0.4570931682042821,
-0.22158635023930806,
-0.36549836384277606,
-0.2366283880333898,
0.4114765778384287,
-0.7694643480660831,
-0.40507000819737765,
-0.6576468499882752,
0.2023600853039859,
-1.2917280200640486,
0.32702771491506877,
-0.07546137553843744,
-0.7713328193108817,
-0.8163562225239717,
-0.07772489804090182,
-0.32841576637855896,
-0.05728139293162625,
0.8564947637293437,
0.15384434981345546,
0.41944080031994563,
0.16884394015055976,
-0.010490661309829376,
-1.3198249528292274,
0.07128101806419536,
0.03515395872296211,
-0.09826644736389979,
-0.06882299183557238,
-0.7062443088386591,
-0.5280915567870916,
0.14860690208212682,
0.338235073517239,
0.012130023587481156,
-0.037483081002571056,
0.8881056862099682,
0.06312673968269537,
-0.44738898868242327,
-0.2463284936792685,
1.1708672891281489,
0.5297819270424092,
-0.5463182826640249,
-1.667403353493104,
1.3713545814180228,
0.9413366263710182,
-0.021951472934173096,
1.0345046397029698,
-0.8002739446090432,
1.076242700462428,
0.04994674089563993,
-1.446235170371471,
-0.3268246709561185,
0.6264234635598642,
-1.2465048991503196,
0.1169092097763034,
1.2453937722962254,
0.7220287063888942,
0.8114212311039525,
0.18333674920471413,
0.7345776889004225,
0.9807678368065545,
0.8703572591074042,
0.28720991162605214,
-0.9085817320602367,
1.1216621938695932,

1.1327477502874412,
-1.8285395156530388,
1.4231155243966302,
0.08156271721069372,
0.18027327076782884,
-0.10302960773112253,
-0.5560482514096969,
-0.09484541903718098,
0.5886430341587933,
-1.1674014904638685,
-0.03940950630369794,
0.2792515765657198,
0.08214125135098017,
0.04344924016389119,
-0.7787970529711157,
0.07887586110139429,
0.7532574519080738,
0.17107625040009872,
-0.4128415687827808,
1.3723943620028143,
1.4278581324344195,
0.5879977529754344,
0.28910810409117477,
-1.4412046993697643,
1.624844367134446,
0.19366683662945536,
0.3179841250458877,
0.5357240606734743,
1.5680682524509713,
0.04418323830986127,
0.2311137213204739,
2.8741283660232546,
0.9612408402592636],
[0.46756606893882113,
0.46983859896604685,
0.5133985927484581,
0.5133985927484581,
0.5133985927484581,
1.4016794842576101,
1.6914116785996751,
1.691411678599675,
1.6941756924801525,
1.6941756924801528,
3.749864451084885,
3.749864451084885,
3.749864451084885,
3.749864451084885,
4.969636107010246,
4.969636107010246,
4.969636107010246,
4.969636107010246,
5.983107385388112,
5.983107385388112,
8.378949246097575,
8.378949246097575,
8.635680633753974,
8.635680633753974,
9.108807770997622,
9.108807770997622,
9.108807770997622,
9.960036411132767,
10.436643041587223,
10.716533289695375,
0.5466515694267952,
3.318518267704871,
0.40678728382369284,
0.6290091030083763,
-0.4770983795300032,
-0.7455483390932742,
0.1630064768968011,
0.4889724943271746,
0.10596004146863507,
-1.4759536729524354,
0.957604974713893,
0.6004514101153323,

0.0725728265044025,
1.4994700617905987,
0.15508641282397265,
-0.15494303336158094,
-0.5663845144966579,
2.6663065398173496,
-0.27086336898318036,
-1.1828784814640632,
-0.295045685212881,
0.6236994649007899,
1.1769360493970402,
1.7159795997719172,
-0.3019600780315931,
0.651545330446025,
-2.0782382222089724,
0.3186344830290561,
-0.2815476257680432,
0.4067635126151388,
-0.8228051832090519,
-0.4154478828430832,
2.3723163321732827,
0.931436930203099,
0.9804374423714106,
0.058895326584719035,
0.3167253661454302,
0.3390503539155689,
-0.5024758650485814,
-0.2425872659725522,
-0.6703364135359525,
-0.147558399690948,
1.1644296968010859,
0.5475507579506065,
0.1736068872874216,
0.30547527216138975,
0.5300773267023443,
0.7722201656261689,
-0.6858549484151734,
-0.16073113081558088,
-1.197458252004488,
-0.507307082981415,
-0.16437738354723153,
0.2647407403580543,
-0.45930133674563567,
-0.22673992276266391,
1.8127715449849986,
-1.4334128748253496,
-0.759404859696163,
1.2431110601081345,
0.8953271645792334,
0.6963251389999259,
-0.23147129445607595,
0.3638861144885483,
2.2394622330281906,
1.543816783446588,
1.7535830084845025,
-0.3169637330795974,
0.2453194636587019,
2.0076391670843647,
0.286017506499548,
0.6600449842625801,
0.19782118041633945,
0.05463577684798935,
0.4755005320975566,
0.055207215781380096,
0.06969685609102444,
0.2645328829673777,
-0.01235845210357445,
1.01982876424508,
2.262492417600915,
1.3443784687440867,
0.10698144982242018,
0.428081928456963,
-1.5494921893757727,
1.0701294130786736,
0.39987519173641656,

0.06301738855484718,
-1.6711192593807125,
0.26920159401030147,
-0.08358711295948693,
0.4054786916979446,
-0.5013194648307361,
2.126550690771933,
0.10959959792145137,
0.4205360694942305,
-0.586878230803869,
-0.3381608637889511,
-0.5055401953468239,
-0.45995744817477163,
0.07999668782445096,
-0.4269018063306222,
0.9588276598807326,
1.042187521522261,
0.07821502449201799,
0.6777167152964713,
0.4177387822172794,
-0.18721284346239236,
-0.8064458441104513,
2.8712851588196218,
0.7141862154478698,
-0.9976723444987857,
-0.8529435737233043,
1.9602636639541733,
1.2634521849205176,
-0.7170870121313951,
0.31227949023655405,
-1.0451293221946831,
-0.6787852158143914,
1.8581833428208436],
[0.4319322451408806,
0.4441934852523559,
0.5878315575931067,
0.5878315575931067,
0.5912427958750637,
0.5947187380620651,
0.5947187380620652,
1.5628074958209335,
1.568718083448373,
2.013035808837258,
2.013035808837258,
2.013035808837258,
2.0277481086167364,
4.114573484042011,
4.1154343976538055,
5.358852938788934,
5.358852938788934,
5.549639110185835,
5.667644288942511,
5.6706149635077026,
7.063797010237488,
7.658675772413245,
8.123010743902626,
8.633070512181217,
8.7849794748515,
8.7849794748515,
10.749769122252356,
10.89899813672382,
11.413879450074099,
11.413879450074099,
-0.04151001581527148,
-2.751209241920508,
-0.5056705094410174,
-0.406548832581749,
0.8633340895103881,
0.17259893989954428,
-0.21235924407313156,
-0.5485334561985523,
0.2012951527025238,
-0.23858034563229433,
2.1364176057207223,
0.7817014982866842,

-0.1108899918027214,
0.6048531661014985,
1.809261328558185,
-1.1786415575821785,
-1.0827170099436865,
0.4109384498547832,
0.8090945308822337,
-0.9356585252219308,
0.28697429006677655,
-0.15332459566133277,
1.2998079093160566,
0.3870967504567973,
-0.2413465423953762,
1.001771196320206,
0.1138394509020578,
1.3220786271535379,
-1.1806565063083097,
2.2789659185869464,
0.11875741943281123,
-0.15631084464392242,
-2.14680662310284,
0.18035127914430246,
-0.260880073181003,
0.21031876899961,
0.7641152777599183,
1.203568210035907,
1.2767673297334832,
1.2585129786889195,
0.24752046596725819,
0.7203551788145919,
-0.17420447991641552,
-1.3486030918499698,
0.2145369555808404,
0.8896874485404291,
-1.6470089821954121,
-0.37928143051952584,
-0.29903232984244293,
0.6984471945997809,
0.14906235834494008,
0.054873949093549826,
1.3877615433857144,
0.04193983551007985,
1.2014028569521673,
-0.02338341588216991,
-0.05255422822816251,
-0.1905875093002492,
0.16425167437144692,
0.2542835721118732,
1.1822299841866795,
1.2032222480152739,
0.30142488752060526,
0.09125464470109583,
-0.45638140154025475,
0.050260497156630585,
-0.04000614529085093,
0.9096133936431033,
-0.03555378439095736,
0.4405325841695877,
-0.48129885373423525,
-0.3617221403442356,
-2.5216506086368518,
-0.5081934521569421,
0.041332600726076626,
0.7413771443671474,
0.2204029987501584,
-0.7517920276590782,
-0.21022249714251526,
1.5607830526955413,
0.16615947769508344,
-0.17023919527139658,
-0.4295619963603494,
-1.1838803442651602,
0.6319564057885301,
-0.3935277768061919,
0.005550337258741166,

-0.4226065324972882,
-0.3715846249960284,
-0.48395498614671884,
-0.4790957128536121,
1.4427293239187895,
0.5875094653073935,
1.404722056846091,
0.6973465258635152,
0.8623508528879442,
-0.5460563718875903,
0.8326491581513842,
-2.4015389244466907,
-0.04583861440999851,
-0.36265388938085896,
-0.5347868312224912,
0.49674658852292397,
-0.6488379607428911,
1.1568053403721041,
0.30662269781395407,
-2.013151608263867,
-0.17383275003878457,
0.5036662539248705,
1.9638910157847163,
3.118155075975211,
0.8117246453310248,
-1.4517859264576565,
-1.4946803709676548,
-2.7792982516572895,
1.1174424252005506,
2.333430423546859,
0.6903927508165344,
-0.2883899927957993,
-0.8542826896862586],
[0.4349098035102067,
0.5281397824211302,
0.6402833431740305,
0.701719039924861,
0.8953768954552646,
1.4118719346144908,
2.230113621742168,
2.6548053250922607,
3.10433397641996,
3.1043339764199596,
4.146783795497047,
4.146783795497047,
4.149760661232294,
4.149760661232294,
4.223539194779984,
4.223539194779984,
6.7727886487081435,
6.7727886487081435,
7.243551717080481,
7.282491880183138,
7.282491880183137,
7.282491880183138,
8.806989173945695,
8.80698917394569,
9.706285648296737,
9.809777023075187,
9.837118725166619,
10.701586770517197,
11.185931122995514,
11.185931122995514,
0.39919688039300266,
1.4533249104921768,
0.7852252158845567,
-0.4957764525834907,
-0.3354719566017423,
0.16097458435951068,
-0.21971891747261646,
1.5370438656668401,
0.26531517225969137,
-0.6386119707402834,
0.37748437574064975,
1.2333814560270282,

0.5934062959304571,
1.633059089666515,
0.31170790405459564,
0.3995146629536743,
-0.11155094016301184,
0.8138063699882787,
0.019467271717771087,
-0.38186317682975884,
-2.778803401027002,
0.7887082829537533,
-3.1348050840084074,
-1.5109224736536222,
-0.38589710406642097,
-1.105927053527023,
0.3451707237315739,
0.6290043291617069,
-1.834722402137916,
-1.0551446159380258,
-1.3581260193610523,
0.9666180881078981,
-0.05759676522292653,
1.1866392284178628,
-1.1308471740259582,
0.34327443183487805,
0.17662613760194443,
-0.043135596735193626,
-0.03667546733606037,
-0.4901895513664937,
-1.0399227541005371,
0.4880459765792483,
0.053089003166299456,
-0.6259966396796703,
0.03159579372412852,
-0.09798805529491862,
0.052872660107769434,
1.490194875693949,
0.22167423812356596,
-0.33497064714209324,
1.0144679614160885,
0.01608415027143934,
0.49167899744128424,
0.12491439548600353,
-0.1329609067962543,
0.8639852596635488,
-1.6607871811173882,
0.2801513494447602,
0.3908501333463579,
0.6044891435983025,
0.08637544621327581,
1.2270127042754213,
0.11860545946003472,
-0.13066468162372125,
-0.998222267933649,
-1.2643438207146234,
0.7835210648968657,
0.8370312354478452,
0.06527740704106412,
-1.0235950282739155,
-0.705070781232052,
0.037682760290170314,
1.4778132318704873,
1.7053977225841355,
-0.1291331271719845,
0.1648924115529835,
2.208642048449397,
1.1885953055185114,
-0.8138037079028133,
0.6715189891287302,
0.7819572861394978,
-0.4022064636590845,
-0.3927127310055937,
-0.4118784323717791,
1.3463239978578432,
-0.3323606385924751,
0.6664683657180488,

1.58852177854836,
-1.212203169826861,
-0.031436536468697034,
0.5094262403458202,
-0.0616688967239856,
-0.0666169691754888,
-0.7277012125404719,
-0.12684196683036533,
0.5325519345188151,
2.2778420441533447,
0.8735933622003313,
0.9907279467768721,
1.1448787224294625,
0.4291077507842401,
-0.21356701746617238,
1.7174041100732933,
0.1885112766205208,
0.3630143936992631,
0.6395640633025061,
-0.2937369198596852,
-0.18128670495631546,
0.9504587478277511,
-0.6241293783488098,
1.8733903679237267,
0.4875344831143031,
-0.9182747577406853,
0.850791268752179,
-0.46183623376516114,
-0.634145646812118,
-0.9165900492445601,
-0.8973746003077631,
-0.879107286657044,
0.15645801311276636],
[0.48023040681030116,
0.7029544315470944,
0.722671987246733,
0.9307929858945541,
0.9307929858945543,
2.546598088688794,
2.546598088688794,
2.546598088688794,
2.546598088688794,
2.546598088688794,
5.184334435330857,
5.195111458618343,
5.47436411399969,
5.47436411399969,
5.47436411399969,
5.47436411399969,
5.47436411399969,
6.068336244524409,
6.068336244524409,
7.838605251840715,
7.912250316107233,
7.912250316107233,
8.562517202814183,
9.063385768597811,
9.216218862905647,
9.216218862905647,
10.84047371877396,
10.84047371877396,
11.044358714867027,
11.375406566932424,
11.375406566932424,
0.18886549713610382,
0.6040291443391601,
-0.9727809979878402,
3.008209947509642,
0.3239743149147096,
-0.643987619924633,
1.0739882269552894,
0.4935103262912805,
-1.1713673989643745,
0.32339055167680386,
-0.8254170399860232,
3.387204293956005,

0.5437302598099015,
-0.6585961721148387,
1.0903163929414719,
-1.5261906384818182,
1.102349928722959,
-0.7373436929054379,
0.1712635196093208,
1.2086697581514199,
-0.4977298668106944,
-0.8828019932256734,
-0.13051782426094366,
0.048744185631251574,
-1.0535030780910155,
0.5224555306604155,
-0.9502111367985572,
-0.2573420205933925,
1.5833039752620888,
1.3806480812363842,
0.3864833677006273,
-0.0006284730712113316,
0.6487740408134013,
1.697829570896293,
-0.20023867528920775,
-0.32438662287273723,
1.3674639807988553,
-1.5592347723565667,
-0.6294582959326964,
1.111157394071659,
1.4895912958038178,
-0.7203901402661329,
-0.027076706863032294,
0.009574786781832467,
1.1437695218822916,
0.11720143179507149,
1.0973549625545624,
-1.0756168303605034,
-1.5785125289756954,
-0.42999007233668873,
0.7592603373612552,
0.7357293999316703,
0.03766014972468404,
1.450035910180884,
-0.08978031589044846,
0.7176267391661779,
-2.3377280976646966,
0.3878354316208249,
-0.8070070584177742,
0.07084117333557359,
-0.008998168433091903,
0.6612796195456235,
0.6287929626789847,
0.8601333603228948,
-0.7786354708155643,
-0.144476679155352,
0.39492950957300044,
-0.20012968159779548,
0.01326107241550192,
-0.5649340596325207,
-0.020398755490040426,
0.6333499066300691,
0.400844220606507,
0.5954749034740784,
0.8700413681534388,
1.315868035063538,
0.9914334712755634,
0.10418309057983809,
1.1095991500751576,
-0.1712743534684691,
-0.2730581386141925,
0.1453142903679217,
-0.3813955795194703,
0.1280814941289577,
-0.5627986213807415,
-1.0576722394029623,
-1.98721560251358,

0.519862592148374,
1.03695228961559,
0.21101375663462807,
-1.750252459751414,
-1.1450993785481438,
-2.2517541986316916,
-1.5456964825715331,
0.6026638989139771,
0.47561448130168205,
0.08201672609434518,
0.47464262797792917,
-0.14585857152212733,
0.08636040211648545,
0.7388192170680543,
1.071366965639021,
2.5115163876343742,
-0.4286872636197954,
-0.2977998181526895,
0.46569070003689333,
0.2115019986130443,
-0.5123638542481742,
0.09024611293676764,
-0.3380729423866059,
1.6314769884628497,
1.3338027138849353,
-0.7442166070564612,
1.2186722356710948,
1.584709484399307,
-0.6554175978902271,
0.36887213174672207,
-1.365021965092438,
-0.4623126041921459,
0.4869420326570461],
[0.5118290521669792,
0.9034573591840084,
1.0514460879099101,
1.1283814725964603,
1.1293980629167455,
1.6874068541059999,
1.8362001710030256,
2.594529848453243,
2.5949023258749775,
4.384420392533986,
4.804388533064279,
4.8044974726668155,
4.925049921883437,
4.925049921883437,
6.055903002618935,
6.0650337145794,
6.06510024924717,
6.06510024924717,
7.719643893437192,
7.719643893437192,
8.110116534451196,
8.374448879129652,
8.374448879129652,
9.29045698473019,
9.2929923765699,
9.977025927736127,
9.977025927736127,
9.977438303479875,
11.277250876863835,
11.588901828478656,
-0.5970483257864091,
1.048395642841534,
0.3563299433581316,
2.04182497417867,
-1.0659189653240388,
-1.5711224255679066,
-0.6854026923334804,
1.3002799990574343,
0.334979527312172,
-0.41771661057992626,
-0.5255960701407614,
0.9200835209290303,

0.5840294212204432,
0.4309612994747514,
0.010788324288593443,
0.8398650983618882,
0.35555521736822654,
0.33383672576306345,
-0.03972798241435421,
-0.5808926346218187,
-0.6217506829607715,
-0.022449055124049946,
-0.4671428723737863,
1.3840583406465645,
3.321823997583478,
1.9516123562000514,
2.109430068020724,
0.17138900458361278,
1.0734765554988877,
1.1591682463434823,
-1.5085197494005436,
0.3709514368460093,
-0.08354812013321816,
-0.6849414455501801,
-1.1807064172516142,
0.1889544406654655,
0.45125852432694835,
0.40946867023093825,
0.9591749175330944,
0.862406962512403,
-0.47565451021668054,
0.14791469386649075,
0.012323642821740255,
-0.3613715829994197,
-0.6520104899254531,
1.1500641632479522,
0.11480411675389499,
0.37929728325133344,
-0.12818017639377846,
1.5548403554359806,
-0.08803696907532751,
0.16058007713541472,
-0.2835300528775952,
1.4317946638041426,
0.04218231832431045,
0.6164913379423986,
-0.5546192578057587,
-0.0016176435191898148,
0.3455446109244303,
-0.7983090776945799,
0.06548494602750325,
0.09601011255154575,
-1.274882751616874,
-0.6639296860891427,
0.08468574728059233,
0.09178155492125434,
-1.9746588496307496,
1.363083624655819,
-0.11754172722968637,
-1.7454296260872237,
0.689437867548643,
0.09536601432398076,
2.1875690275532618,
1.9737367040865754,
0.19623670188051764,
1.3618525198389628,
-1.6551377680800423,
-1.2796187335858575,
-0.02979150349567955,
0.3655877448114123,
-1.339933806945279,
-1.6131280858420662,
1.1268021774842836,
0.7853899025443938,
-1.214954346638235,
1.881374469676006,
-2.2108375055617597,

```
0.38299461206746305,  
-0.062378913993248956,  
-1.0812916442950207,  
0.19113842494911862,  
0.43947065709017896,  
1.4340329690981122,  
-2.1914706359927933,  
1.683520172253517,  
-0.4151753501170658,  
0.2652280013518245,  
-0.09829123134965576,  
-0.4033858255054015,  
-0.09488290407421365,  
1.1087820722198931,  
1.3758250784849964,  
0.3693308876448478,  
-1.4774754586810708,  
0.45790576106810227,  
-1.318961759734687,  
1.2898775251023542,  
0.5291693857421421,  
-0.7225860054677886,  
-0.6024865845145562,  
1.1258609582335735,  
1.4545810101615688,  
0.4988359655120556,  
0.39968483637691726,  
1.4778005154845164,  
1.6358502222921723,  
0.07081811420225437,  
1.5336807499569445,  
-0.7595143926860392,  
0.07947795439670366]],  
      'run_time_sec': 68705.11171293259}}],  
'zne': [ { 'config': { 'ansatz': { 'gateset': 1,  
                                'layer': 30,  
                                'ugate': { 'coefficients': { 'bn': [ 0,  
                                                                    0,  
                                                                    0,  
                                                                    0,  
                                                                    0,  
                                                                    0,  
                                                                    0],  
                        'cn': [ 0.5,  
                                0.5,  
                                0.5,  
                                0.5,  
                                0.5,  
                                0.5],  
                        'r': 0},  
                    'time': {'max': 10.0, 'min': 0.0},  
                    'type': 'ising'}}},  
    'init_param': { 'value': [ 0.46756606893882113,  
                               0.46983859896604685,  
                               0.5133985927484581,  
                               0.5133985927484581,  
                               0.5133985927484581,  
                               1.4016794842576101,  
                               1.6914116785996751,  
                               1.691411678599675,  
                               1.6941756924801525,  
                               1.6941756924801528,  
                               3.749864451084885,  
                               3.749864451084885,  
                               3.749864451084885,  
                               3.749864451084885,  
                               4.969636107010246,  
                               4.969636107010246,  
                               4.969636107010246,  
                               4.969636107010246,  
                               5.983107385388112,  
                               5.983107385388112,  
                               8.378949246097575,  
                               8.378949246097575,  
                               8.635680633753974,
```


8.635680633753974,
9.108807770997622,
9.108807770997622,
9.108807770997622,
9.960036411132767,
10.436643041587223,
10.716533289695375,
0.5466515694267952,
3.318518267704871,
0.40678728382369284,
0.6290091030083763,
-0.4770983795300032,
-0.7455483390932742,
0.1630064768968011,
0.4889724943271746,
0.10596004146863507,
-1.4759536729524354,
0.957604974713893,
0.6004514101153323,
0.0725728265044025,
1.4994700617905987,
0.15508641282397265,
-0.15494303336158094,
-0.5663845144966579,
2.6663065398173496,
-0.27086336898318036,
-1.1828784814640632,
-0.295045685212881,
0.6236994649007899,
1.1769360493970402,
1.7159795997719172,
-0.3019600780315931,
0.651545330446025,
-2.0782382222089724,
0.3186344830290561,
-0.2815476257680432,
0.4067635126151388,
-0.8228051832090519,
-0.4154478828430832,
2.3723163321732827,
0.931436930203099,
0.9804374423714106,
0.058895326584719035,
0.3167253661454302,
0.3390503539155689,
-0.5024758650485814,
-0.2425872659725522,
-0.6703364135359525,
-0.147558399690948,
1.1644296968010859,
0.5475507579506065,
0.1736068872874216,
0.30547527216138975,
0.5300773267023443,
0.7722201656261689,
-0.6858549484151734,
-0.16073113081558088,
-1.197458252004488,
-0.507307082981415,
-0.16437738354723153,
0.2647407403580543,
-0.45930133674563567,
-0.22673992276266391,
1.8127715449849986,
-1.4334128748253496,
-0.759404859696163,
1.2431110601081345,
0.8953271645792334,
0.6963251389999259,
-0.23147129445607595,
0.3638861144885483,
2.2394622330281906,
1.543816783446588,
1.7535830084845025,
-0.3169637330795974,

```

0.2453194636587019,
2.0076391670843647,
0.286017506499548,
0.6600449842625801,
0.19782118041633945,
0.05463577684798935,
0.4755005320975566,
0.055207215781380096,
0.06969685609102444,
0.2645328829673777,
-0.01235845210357445,
1.01982876424508,
2.262492417600915,
1.3443784687440867,
0.10698144982242018,
0.428081928456963,
-1.5494921893757727,
1.0701294130786736,
0.39987519173641656,
0.06301738855484718,
-1.6711192593807125,
0.26920159401030147,
-0.08358711295948693,
0.4054786916979446,
-0.5013194648307361,
2.126550690771933,
0.10959959792145137,
0.4205360694942305,
-0.586878230803869,
-0.3381608637889511,
-0.5055401953468239,
-0.45995744817477163,
0.07999668782445096,
-0.4269018063306222,
0.9588276598807326,
1.042187521522261,
0.07821502449201799,
0.6777167152964713,
0.4177387822172794,
-0.18721284346239236,
-0.8064458441104513,
2.8712851588196218,
0.7141862154478698,
-0.9976723444987857,
-0.8529435737233043,
1.9602636639541733,
1.2634521849205176,
-0.7170870121313951,
0.31227949023655405,
-1.0451293221946831,
-0.6787852158143914,
1.8581833428208436]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
               'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,

```

```

        'type': 'png'},
        'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric3_unifrom_scaling_sample#6'},
        'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                              [1, 1, 0, 0],
                                              [2, 2, 0, 0]]},
        'run': 'zne',
        'state': 'dmatrix',
        'vqe': { 'iteration': 1,
                  'optimization': { 'algorithm': 'SLSQP',
                                    'constraint': True,
                                    'status': False}},
        'zne': { 'data_points': [ [4, 2, 0, 0, -6.487894898369523],
                                   [12, 6, 0, 0, -5.297876077058435],
                                   [20, 10, 0, 0, -4.3310003849914125]],
                  'degree': 0,
                  'method': 'richardson',
                  'sampling': 'default'}},
        'output': { 'exact_sol': -8.566772233505624,
                     'run_time_sec': 4.029273986816406e-05,
                     'zne_values': { 'degree': 0,
                                     'extrapolated_value': -7.166582982491592,
                                     'others': { 'beta_coeffieicients': [ 1.875,
                                                                           -1.25,
                                                                           0.375]},
                                     'cost_richardson_zne': 5.21875,
                                     'sorted_expectation_vals': [
-6.487894898369523,
-5.297876077058435,
-4.3310003849914125],

```

```

                                     'sorted_noise': [6, 18, 30]],
        'sampled data': [ [ 4,
                             2,
                             0,
                             0,
                             -6.487894898369523],
                           [ 12,
                             6,
                             0,
                             0,
                             -5.297876077058435],
                           [ 20,
                             10,
                             0,
                             0,
                             -4.3310003849914125]],
        'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                            'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                            'r': 0},
                                'time': {'max': 10.0, 'min': 0.0},
                                'type': 'ising'}}},
  'init_param': { 'value': [ 0.27654223573266234,
                             0.27654223573266246,
                             0.27654223573266246,
                             0.4643738922966213,
                             0.6582164862517121,
                             0.6582164862517121,
                             0.6582164862517121,
                             1.3424523496852623,
                             1.47635793598756,
                             1.47635793598756,

```

3.385550316280557,
3.385550316280557,
3.385550316280557,
4.494062371253785,
4.494062371253785,
5.0204758683966135,
5.023404188660082,
7.05704094573064,
7.087391629397132,
7.533802135057517,
8.245970909352765,
8.390887842007308,
8.390887842007308,
8.703391612007538,
8.703391612007538,
9.485545381639698,
9.485545381639698,
9.849437794456351,
9.849437794456351,
10.799155842585746,
-0.12708089143278917,
1.4168148977025445,
-0.13519536748631492,
-0.32308714516775877,
0.06981336221044306,
-0.034321895006426455,
-0.18209443759399158,
-0.024858958594746107,
0.664488174834645,
0.6897317665852637,
-0.008774731192519907,
0.625642551337688,
0.6862410821653743,
-0.7670419540542446,
-0.48055390132433673,
1.573964677292243,
-1.0555745677813198,
1.3159721165631335,
-0.9138502497993347,
-0.4641566084568084,
-0.06966564000847691,
0.039012746519005555,
1.0827641697579717,
1.2190093772283843,
1.5391150382377632,
-0.6675581491933664,
0.09104662559232415,
0.5283818781579022,
0.04478118244404365,
1.5836075485651113,
0.251932560991488,
-0.3949560381811925,
0.8854639388207172,
-0.8611267164776638,
1.566928747241295,
-0.15445151319573347,
-0.01396050433941564,
0.8172620115356718,
-0.014658132241289147,
-0.6623010385304499,
0.6299284572753584,
-0.15026947869759505,
-1.2293112897827836,
-0.5768656731979434,
1.231557922062332,
-1.447872926116422,
1.4918722706131171,
0.31945486783385635,
0.43870804207253483,
1.3582636256308331,
-0.8478398442626237,
0.799518029891469,
0.5539369896048497,
0.13453197415145846,
0.7167498104862234,

```

-0.06589109804263343,
0.041038724923082276,
1.151975756926885,
-0.052026838729968425,
0.3952978000808587,
-1.1754748970457833,
-0.01965607042871015,
-0.2456324144680547,
-0.01136184217607491,
0.022503168314325715,
1.1777923723104653,
0.14554117253938412,
-1.2287946167385297,
-0.9683737383161766,
1.073329622962756,
-0.7143583802799215,
-0.3545002682598469,
-0.31906110571926777,
-0.9597094503848127,
0.8825903987559673,
-0.08273415521500459,
0.2895039435279609,
-0.3388594197258312,
0.2406459434933063,
0.3824128671502841,
-0.5390555829405563,
0.3757684495816124,
-0.5015719636333678,
0.3958257688275076,
-0.6428354095046468,
-0.9178624992467067,
0.29327306512581747,
-0.09193285724698198,
0.1770091195095364,
0.6803523512851801,
1.05525770275555,
-0.4646437845139076,
2.2560908642344026,
-0.7500939262299234,
1.92959092052175,
-0.36895777993274226,
-0.04538283416771957,
1.2529586558275392,
-0.57805717638909,
0.3080585810325818,
0.031910210614104206,
0.18519950010488512,
1.3888566879174211,
1.087260914864616,
0.15259355623633358,
-0.08810167996863522,
0.07467874103808375,
1.4444893382870363,
-1.446220680213735,
0.8215180856176293,
2.0151723649306295,
0.13454991965456886,
0.21923381008732426,
0.31112145477192793,
-0.8977876743954992,
0.10239876571985529,
-1.0132922278363503,
-0.3794863035801566,
0.8437803973608401,
1.111671691788496]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,

```

```

1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric3_unifrom_scaling_sample#2'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.514930884689279],
[12, 6, 0, 0, -5.310706408751285],
[20, 10, 0, 0, -4.328532484009807]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 3.647804260253906e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -7.200312079356969,
'others': { 'beta_coefficients': [ 1.875,
-1.25,
0.375],
'cost_richardson_zne': 5.21875,
'sorted_expectation_vals': [
-6.514930884689279,
-5.310706408751285,
-4.328532484009807],
'sorted_noise': [6, 18, 30]},
'sampled data': [ [ 4,
2,
0,
0,
-6.514930884689279],
[ 12,
6,
0,
0,
-5.310706408751285],
[ 20,
10,
0,
0,
-4.328532484009807]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],

```

```
        'r': 0},
        'time': {'max': 10.0, 'min': 0.0},
        'type': 'ising'}},
'init_param': { 'value': [ 0.48023040681030116,
                           0.7029544315470944,
                           0.722671987246733,
                           0.9307929858945541,
                           0.9307929858945543,
                           2.546598088688794,
                           2.546598088688794,
                           2.546598088688794,
                           2.546598088688794,
                           2.546598088688794,
                           5.184334435330857,
                           5.195111458618343,
                           5.47436411399969,
                           5.47436411399969,
                           5.47436411399969,
                           5.47436411399969,
                           5.47436411399969,
                           6.068336244524409,
                           6.068336244524409,
                           7.838605251840715,
                           7.912250316107233,
                           7.912250316107233,
                           8.562517202814183,
                           9.063385768597811,
                           9.216218862905647,
                           9.216218862905647,
                           10.84047371877396,
                           10.84047371877396,
                           11.044358714867027,
                           11.375406566932424,
                           11.375406566932424,
                           0.18886549713610382,
                           0.6040291443391601,
                           -0.9727809979878402,
                           3.008209947509642,
                           0.3239743149147096,
                           -0.643987619924633,
                           1.0739882269552894,
                           0.4935103262912805,
                           -1.1713673989643745,
                           0.32339055167680386,
                           -0.8254170399860232,
                           3.387204293956005,
                           0.5437302598099015,
                           -0.6585961721148387,
                           1.0903163929414719,
                           -1.5261906384818182,
                           1.102349928722959,
                           -0.7373436929054379,
                           0.1712635196093208,
                           1.2086697581514199,
                           -0.4977298668106944,
                           -0.8828019932256734,
                           -0.13051782426094366,
                           0.048744185631251574,
                           -1.0535030780910155,
                           0.5224555306604155,
                           -0.9502111367985572,
                           -0.2573420205933925,
                           1.5833039752620888,
                           1.3806480812363842,
                           0.3864833677006273,
                           -0.0006284730712113316,
                           0.6487740408134013,
                           1.697829570896293,
                           -0.20023867528920775,
                           -0.32438662287273723,
                           1.3674639807988553,
                           -1.5592347723565667,
                           -0.6294582959326964,
                           1.11157394071659,
                           1.4895912958038178,
                           -0.7203901402661329,
```

-0.027076706863032294,
0.009574786781832467,
1.1437695218822916,
0.11720143179507149,
1.0973549625545624,
-1.0756168303605034,
-1.5785125289756954,
-0.42999007233668873,
0.7592603373612552,
0.7357293999316703,
0.03766014972468404,
1.450035910180884,
-0.08978031589044846,
0.7176267391661779,
-2.3377280976646966,
0.3878354316208249,
-0.8070070584177742,
0.07084117333557359,
-0.008998168433091903,
0.6612796195456235,
0.6287929626789847,
0.8601333603228948,
-0.7786354708155643,
-0.144476679155352,
0.39492950957300044,
-0.20012968159779548,
0.01326107241550192,
-0.5649340596325207,
-0.020398755490040426,
0.6333499066300691,
0.400844220606507,
0.5954749034740784,
0.8700413681534388,
1.315868035063538,
0.9914334712755634,
0.10418309057983809,
1.1095991500751576,
-0.1712743534684691,
-0.2730581386141925,
0.1453142903679217,
-0.3813955795194703,
0.1280814941289577,
-0.5627986213807415,
-1.0576722394029623,
-1.98721560251358,
0.519862592148374,
1.03695228961559,
0.21101375663462807,
-1.750252459751414,
-1.1450993785481438,
-2.2517541986316916,
-1.5456964825715331,
0.6026638989139771,
0.47561448130168205,
0.08201672609434518,
0.47464262797792917,
-0.14585857152212733,
0.08636040211648545,
0.7388192170680543,
1.071366965639021,
2.5115163876343742,
-0.4286872636197954,
-0.2977998181526895,
0.46569070003689333,
0.2115019986130443,
-0.5123638542481742,
0.09024611293676764,
-0.3380729423866059,
1.6314769884628497,
1.3338027138849353,
-0.7442166070564612,
1.2186722356710948,
1.584709484399307,
-0.6554175978902271,
0.36887213174672207,


```

-1.365021965092438,
-0.4623126041921459,
0.4869420326570461]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric3_unifrom_scaling_sample#9'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.75124151968758],
                          [12, 6, 0, 0, -5.5757781087309555],
                          [20, 10, 0, 0, -4.6095181057209125]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 3.9577484130859375e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -7.417424503145861,
                           'others': { 'beta_coeffieicients': [ 1.875,
                                                                -1.25,
                                                                0.375],
                                         'cost_richardson_zne': 5.21875,
                                         'sorted_expectation_vals': [
-6.75124151968758,
-5.5757781087309555,
-4.6095181057209125],

'sorted_noise': [6, 18, 30]},
'sampled data': [ [ 4,
                    2,
                    0,
                    0,
                    -6.75124151968758],
                  [ 12,
                    6,
                    0,
                    0,
                    -5.5757781087309555],
                  [ 20,
                    10,
                    0,
                    0,
                    -4.6095181057209125]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,

```

```
'ugate': { 'coefficients': { 'bn': [ 0,
                                     0,
                                     0,
                                     0,
                                     0,
                                     0,
                                     0],
                               'cn': [ 0.5,
                                     0.5,
                                     0.5,
                                     0.5,
                                     0.5,
                                     0.5],
                               'r': 0},
          'time': {'max': 10.0, 'min': 0.0},
          'type': 'ising'}},
'init_param': { 'value': [ 0.5118290521669792,
                           0.9034573591840084,
                           1.0514460879099101,
                           1.1283814725964603,
                           1.1293980629167455,
                           1.6874068541059999,
                           1.8362001710030256,
                           2.594529848453243,
                           2.5949023258749775,
                           4.384420392533986,
                           4.804388533064279,
                           4.8044974726668155,
                           4.925049921883437,
                           4.925049921883437,
                           6.055903002618935,
                           6.0650337145794,
                           6.06510024924717,
                           6.06510024924717,
                           7.719643893437192,
                           7.719643893437192,
                           8.110116534451196,
                           8.374448879129652,
                           8.374448879129652,
                           9.29045698473019,
                           9.2929923765699,
                           9.977025927736127,
                           9.977025927736127,
                           9.977438303479875,
                           11.277250876863835,
                           11.588901828478656,
                           -0.5970483257864091,
                           1.048395642841534,
                           0.3563299433581316,
                           2.04182497417867,
                           -1.0659189653240388,
                           -1.5711224255679066,
                           -0.6854026923334804,
                           1.3002799990574343,
                           0.334979527312172,
                           -0.41771661057992626,
                           -0.5255960701407614,
                           0.9200835209290303,
                           0.5840294212204432,
                           0.4309612994747514,
                           0.010788324288593443,
                           0.8398650983618882,
                           0.35555521736822654,
                           0.33383672576306345,
                           -0.03972798241435421,
                           -0.5808926346218187,
                           -0.6217506829607715,
                           -0.022449055124049946,
                           -0.4671428723737863,
                           1.3840583406465645,
                           3.321823997583478,
                           1.9516123562000514,
                           2.109430068020724,
                           0.17138900458361278,
                           1.0734765554988877,
```

1.1591682463434823,
-1.5085197494005436,
0.3709514368460093,
-0.08354812013321816,
-0.6849414455501801,
-1.1807064172516142,
0.1889544406654655,
0.45125852432694835,
0.40946867023093825,
0.9591749175330944,
0.862406962512403,
-0.47565451021668054,
0.14791469386649075,
0.012323642821740255,
-0.3613715829994197,
-0.6520104899254531,
1.1500641632479522,
0.11480411675389499,
0.37929728325133344,
-0.12818017639377846,
1.5548403554359806,
-0.08803696907532751,
0.16058007713541472,
-0.2835300528775952,
1.4317946638041426,
0.04218231832431045,
0.6164913379423986,
-0.5546192578057587,
-0.0016176435191898148,
0.3455446109244303,
-0.7983090776945799,
0.06548494602750325,
0.09601011255154575,
-1.274882751616874,
-0.6639296860891427,
0.08468574728059233,
0.09178155492125434,
-1.9746588496307496,
1.363083624655819,
-0.11754172722968637,
-1.7454296260872237,
0.689437867548643,
0.09536601432398076,
2.1875690275532618,
1.9737367040865754,
0.19623670188051764,
1.3618525198389628,
-1.6551377680800423,
-1.2796187335858575,
-0.02979150349567955,
0.3655877448114123,
-1.339933806945279,
-1.6131280858420662,
1.1268021774842836,
0.7853899025443938,
-1.214954346638235,
1.881374469676006,
-2.2108375055617597,
0.38299461206746305,
-0.062378913993248956,
-1.0812916442950207,
0.19113842494911862,
0.43947065709017896,
1.4340329690981122,
-2.1914706359927933,
1.683520172253517,
-0.4151753501170658,
0.2652280013518245,
-0.09829123134965576,
-0.4033858255054015,
-0.09488290407421365,
1.1087820722198931,
1.3758250784849964,
0.3693308876448478,
-1.4774754586810708,

```

0.45790576106810227,
-1.318961759734687,
1.2898775251023542,
0.5291693857421421,
-0.7225860054677886,
-0.6024865845145562,
1.1258609582335735,
1.4545810101615688,
0.4988359655120556,
0.39968483637691726,
1.4778005154845164,
1.6358502222921723,
0.07081811420225437,
1.5336807499569445,
-0.7595143926860392,
0.07947795439670366]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                     'status': False,
                     'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric3_unifrom_scaling_sample#10'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.406033238196167],
                          [12, 6, 0, 0, -5.14768368077025],
                          [20, 10, 0, 0, -4.13060369641699]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 5.459785461425781e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -7.12568410681137,
                           'others': { 'beta_coefficients': [ 1.875,
                                                              -1.25,
                                                              0.375],
                                       'cost_richardson_zne': 5.21875,
                                       'sorted_expectation_vals': [
-6.406033238196167,
-5.14768368077025,
-4.13060369641699],
'sorted_noise': [6, 18, 30]},
'sampled data': [ [ 4,
                    2,
                    0,
                    0,
                    -6.406033238196167],

```

```

[ 12,
  6,
  0,
  0,
  -5.14768368077025],
[ 20,
  10,
  0,
  0,
  -4.13060369641699]],
'sampling': 'default']]],
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                                'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'ising'}},
  'init_param': { 'value': [ -4.1290471568429383e-17,
                              0.028387941050870547,
                              0.028387941050870516,
                              0.028387941050870544,
                              0.4764187545019148,
                              0.485154879376271,
                              0.7094706976105217,
                              2.998376901501724,
                              3.486264965503922,
                              3.487775573446841,
                              4.058897205630428,
                              5.020552086463093,
                              5.02158492362958,
                              5.02158492362958,
                              5.0846878698400815,
                              5.0846878698400815,
                              6.794704827143902,
                              6.794704827143902,
                              6.794704827143902,
                              8.013356083289752,
                              8.21076169573117,
                              8.21076169573117,
                              8.633075344303478,
                              8.633075344303478,
                              8.930018414781724,
                              8.930018414781724,
                              8.930018414781724,
                              9.990329186240457,
                              9.990329186240457,
                              12.915390927357043,
                              0.8289215528981712,
                              0.6465354442587259,
                              -0.06922368073773534,
                              -2.0853926237797484,
                              0.25605409461249207,
                              -0.765586831155604,
                              0.9589723041272441,
                              0.27455658215646966,
                              -0.07189982532198122,
                              -0.3449139525561357,
                              -0.46800027418386986,
                              1.016480689947782,
                              -0.9706976058410632,
                              -0.9556868223128826,
                              0.9323574793538898,
                              -0.4376613900889883,

```

0.44138701419965265,
1.4418520162063444,
0.8420619129497768,
-0.025530845682239494,
-2.3076344184440156,
0.37458463155346333,
-0.47031338116346383,
0.07338001085955462,
-0.869249007732584,
1.6824786550253577,
1.719346297136726,
-0.9622509832303179,
0.5779076253870294,
0.4760527439731688,
-0.2451715046346434,
1.5906513054768954,
0.35918458386075697,
-0.44817294000256097,
-0.37416377940751877,
0.37048554789855764,
0.693137718643464,
2.288219219235325,
-0.26501198620577204,
0.3689053264668303,
0.10166782579927121,
-0.3356073800920907,
-0.6413954053758932,
0.28636656044125136,
-0.8690978212322298,
1.166030172665772,
-0.46365263204645707,
-0.8505396438729699,
2.4328989003329693,
-1.4554443951257197,
2.09134311034601,
1.7695145471360614,
1.1170184462314887,
0.20325573406500014,
0.34206355374635444,
0.016804831103183593,
-0.19882507083844791,
1.4011478969931443,
-0.0572616468882171,
-1.3773340344964602,
1.1031086912886459,
0.47433623341773395,
-0.037679845343088075,
0.7450954028512253,
1.3114871459373192,
-0.18651966717154955,
1.0750199871192778,
0.48905379729033566,
-0.6281827821606347,
1.322074656325554,
-1.4761484868509396,
0.2859918816514792,
-1.0033541595176405,
0.8264053813863532,
0.622010811103528,
1.1636747207646185,
-0.31883271288389914,
-0.0013981850979601159,
-0.09999405171802342,
1.3085280853238759,
-1.3727969089464465,
-0.08341159264840771,
1.6249369276484606,
-0.28082331570516694,
1.863294400254821,
2.2932652495309953,
1.4849536675442596,
0.6811668802751195,
1.5380110980874968,
-0.6615225065895972,
0.4000828281024264,

```

0.33932121588009856,
0.48855764825038867,
2.5371111232777053,
-0.7260430040504564,
0.09005940379987407,
-0.8762980817992282,
-0.23800314057977784,
0.9728027685352436,
-0.20857834368009137,
0.6148333457084475,
-1.5171898336040224,
0.8658542069129427,
-1.6478287868490311,
0.5770049367106069,
0.134497700004433,
-0.11612046371881249,
1.5826352840237945,
1.6045820840261675,
-0.08805669980595342,
-1.507745125921201,
-0.11276651576435986,
0.07206005105970639,
-1.7788122679933258,
-0.002988765738482436,
0.7255804452314191,
2.7661804504965604,
-3.1943627011957507,
-0.07088434770936546,
0.10949711075632612]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric3_unifrom_scaling_sample#1'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.833109640228284],
                           [12, 6, 0, 0, -5.610198560557999],
                           [20, 10, 0, 0, -4.609901567200479]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 3.62396240234375e-05,
            'zne_values': { 'degree': 0,
                            'extrapolated_value': -7.528045462430714,
                            'others': { 'beta_coeficients': [ 1.875,

```

-6.833109640228284,
-5.610198560557999,
-4.609901567200479],

-1.25,
0.375],
'cost_richardson_zne': 5.21875,
'sorted_expectation_vals': [

'sorted_noise': [6, 18, 30]],
'sampled data': [[4,
2,
0,
0,
-6.833109640228284],
[12,
6,
0,
0,
-5.610198560557999],
[20,
10,
0,
0,
-4.609901567200479]],
'sampling': 'default']},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0,
0,
0,
0,
0,
0],
'cn': [0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [0.4319322451408806,
0.4441934852523559,
0.5878315575931067,
0.5878315575931067,
0.5912427958750637,
0.5947187380620651,
0.5947187380620652,
1.5628074958209335,
1.568718083448373,
2.013035808837258,
2.013035808837258,
2.013035808837258,
2.0277481086167364,
4.114573484042011,
4.1154343976538055,
5.358852938788934,
5.358852938788934,
5.549639110185835,
5.667644288942511,
5.6706149635077026,
7.063797010237488,
7.658675772413245,
8.123010743902626,
8.633070512181217,
8.7849794748515,
8.7849794748515,
10.749769122252356,
10.89899813672382,
11.413879450074099,
11.413879450074099,
-0.04151001581527148,
-2.751209241920508,
-0.5056705094410174,

-0.406548832581749,
0.8633340895103881,
0.17259893989954428,
-0.21235924407313156,
-0.5485334561985523,
0.2012951527025238,
-0.23858034563229433,
2.1364176057207223,
0.7817014982866842,
-0.1108899918027214,
0.6048531661014985,
1.809261328558185,
-1.1786415575821785,
-1.0827170099436865,
0.4109384498547832,
0.8090945308822337,
-0.9356585252219308,
0.28697429006677655,
-0.15332459566133277,
1.2998079093160566,
0.3870967504567973,
-0.2413465423953762,
1.001771196320206,
0.1138394509020578,
1.3220786271535379,
-1.1806565063083097,
2.2789659185869464,
0.11875741943281123,
-0.15631084464392242,
-2.14680662310284,
0.18035127914430246,
-0.260880073181003,
0.21031876899961,
0.7641152777599183,
1.203568210035907,
1.2767673297334832,
1.2585129786889195,
0.24752046596725819,
0.7203551788145919,
-0.17420447991641552,
-1.3486030918499698,
0.2145369555808404,
0.8896874485404291,
-1.6470089821954121,
-0.37928143051952584,
-0.29903232984244293,
0.6984471945997809,
0.14906235834494008,
0.054873949093549826,
1.3877615433857144,
0.04193983551007985,
1.2014028569521673,
-0.02338341588216991,
-0.05255422822816251,
-0.1905875093002492,
0.16425167437144692,
0.2542835721118732,
1.1822299841866795,
1.2032222480152739,
0.30142488752060526,
0.09125464470109583,
-0.45638140154025475,
0.050260497156630585,
-0.04000614529085093,
0.9096133936431033,
-0.03555378439095736,
0.4405325841695877,
-0.48129885373423525,
-0.3617221403442356,
-2.5216506086368518,
-0.5081934521569421,
0.041332600726076626,
0.7413771443671474,
0.2204029987501584,
-0.7517920276590782,

```

-0.21022249714251526,
1.5607830526955413,
0.16615947769508344,
-0.17023919527139658,
-0.4295619963603494,
-1.1838803442651602,
0.6319564057885301,
-0.3935277768061919,
0.005550337258741166,
-0.4226065324972882,
-0.3715846249960284,
-0.48395498614671884,
-0.4790957128536121,
1.4427293239187895,
0.5875094653073935,
1.404722056846091,
0.6973465258635152,
0.8623508528879442,
-0.5460563718875903,
0.8326491581513842,
-2.4015389244466907,
-0.04583861440999851,
-0.36265388938085896,
-0.5347868312224912,
0.49674658852292397,
-0.6488379607428911,
1.1568053403721041,
0.30662269781395407,
-2.013151608263867,
-0.17383275003878457,
0.5036662539248705,
1.9638910157847163,
3.118155075975211,
0.8117246453310248,
-1.4517859264576565,
-1.4946803709676548,
-2.7792982516572895,
1.1174424252005506,
2.333430423546859,
0.6903927508165344,
-0.2883899927957993,
-0.8542826896862586]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric3_unifrom_scaling_sample#7'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',

```

```
'sorted_noise': [6, 18, 30]},
'sampled_data': [ [ 4,
                    2,
                    0,
                    0,
                    -6.662379321078876],
                  [ 12,
                    6,
                    0,
                    0,
                    -5.394498644176583],
                  [ 20,
                    10,
                    0,
                    0,
                    -4.374551061230002]],
'sampling': 'default'}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                            'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                            'r': 0},
                                'time': {'max': 10.0, 'min': 0.0},
                                'type': 'ising'},
                          'init_param': { 'value': [ 0.32980400918867037,
                                                       0.7818596814437996,
                                                       0.9680306497310072,
                                                       1.0586432954274991,
                                                       1.7663116348413028,
                                                       1.8956491295658664,
                                                       2.0436116769319477,
                                                       2.0436116769319477,
                                                       2.0436116769319477,
                                                       2.1846619555479685,
                                                       2.1846619555479685,
                                                       4.516997836642403,
                                                       4.516997836642403,
                                                       5.3166564351730035,
                                                       5.320130741320412,
                                                       5.399447086770547,
                                                       5.400046100307029,
                                                       5.400046100307029,
                                                       5.400046100307029,
                                                       5.400046100307029]
            }
        }
    }
}
```

7.097750404951498,
7.618222769006738,
7.956666567531361,
7.956666567531361,
8.961109688308255,
8.961109688308255,
11.028500695271937,
11.5809418583784,
11.5809418583784,
11.5809418583784,
0.24114173221382337,
1.3754512844042854,
0.9720111739434927,
-0.29877116871955545,
-0.029737313498699618,
0.3185153459444675,
-0.31840745367437817,
0.6801602139502809,
-0.015407845998522466,
-1.114484654283524,
0.22997676136047449,
1.5201821520192487,
0.3217486942799661,
0.43678760134757944,
0.17618749246228715,
0.7010359103792874,
0.3446701255696699,
1.4700166379981474,
0.3471333265905846,
-0.968444017764766,
-1.1485269521507444,
0.3282504398407059,
-0.7898554413193151,
-0.22157985789400653,
0.4570931682042821,
-0.22158635023930806,
-0.36549836384277606,
-0.2366283880333898,
0.4114765778384287,
-0.7694643480660831,
-0.40507000819737765,
-0.6576468499882752,
0.2023600853039859,
-1.2917280200640486,
0.32702771491506877,
-0.07546137553843744,
-0.7713328193108817,
-0.8163562225239717,
-0.07772489804090182,
-0.32841576637855896,
-0.05728139293162625,
0.8564947637293437,
0.15384434981345546,
0.41944080031994563,
0.16884394015055976,
-0.010490661309829376,
-1.3198249528292274,
0.07128101806419536,
0.03515395872296211,
-0.09826644736389979,
-0.06882299183557238,
-0.7062443088386591,
-0.5280915567870916,
0.14860690208212682,
0.338235073517239,
0.012130023587481156,
-0.037483081002571056,
0.8881056862099682,
0.06312673968269537,
-0.44738898868242327,
-0.2463284936792685,
1.1708672891281489,
0.5297819270424092,
-0.5463182826640249,
-1.667403353493104,

```

1.3713545814180228,
0.9413366263710182,
-0.021951472934173096,
1.0345046397029698,
-0.8002739446090432,
1.076242700462428,
0.04994674089563993,
-1.446235170371471,
-0.3268246709561185,
0.6264234635598642,
-1.2465048991503196,
0.1169092097763034,
1.2453937722962254,
0.7220287063888942,
0.8114212311039525,
0.18333674920471413,
0.7345776889004225,
0.9807678368065545,
0.8703572591074042,
0.28720991162605214,
-0.9085817320602367,
1.1216621938695932,
1.1327477502874412,
-1.8285395156530388,
1.4231155243966302,
0.08156271721069372,
0.18027327076782884,
-0.10302960773112253,
-0.5560482514096969,
-0.09484541903718098,
0.5886430341587933,
-1.1674014904638685,
-0.03940950630369794,
0.2792515765657198,
0.08214125135098017,
0.04344924016389119,
-0.7787970529711157,
0.07887586110139429,
0.7532574519080738,
0.17107625040009872,
-0.4128415687827808,
1.3723943620028143,
1.4278581324344195,
0.5879977529754344,
0.28910810409117477,
-1.4412046993697643,
1.624844367134446,
0.19366683662945536,
0.3179841250458877,
0.5357240606734743,
1.5680682524509713,
0.04418323830986127,
0.2311137213204739,
2.8741283660232546,
0.9612408402592636]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},

```


3.6939218351946623,
3.693921835194662,
3.693921835194662,
3.9412089126587,
3.9412089126587,
3.9412089126587007,
3.9412089126587007,
4.8310999601027795,
4.870844593548848,
4.900228671797898,
4.9002286717978984,
4.9002286717978984,
4.9002286717978984,
4.900228671797898,
5.70167657474683,
6.21188441483336,
6.21188441483336,
6.686419232103557,
6.723504371727371,
7.618130890836856,
7.618130890836856,
9.420364260644014,
9.420364260644014,
10.775132222422771,
-0.3525750090684208,
1.2748976558430758,
-0.16146998454247521,
1.3224575871686013,
0.0349265192498484,
0.36138711032415904,
-0.5043488106619078,
0.7634801084306476,
-0.3536814593617201,
0.9606809354246906,
-1.1143894720984433,
-0.45289061390722213,
-0.1605333784989486,
-1.594210268458609,
-0.09465166736510336,
0.11132251138190599,
1.004235666711761,
-0.23115493220428254,
0.9451761882751959,
-0.3707284170975133,
-0.17782948231634454,
-0.18433417990028922,
-0.06421460341189031,
-1.155489001827845,
-0.2074908306187579,
-0.036360325422879626,
0.5283181819327893,
0.3603171208434518,
0.6819349944270952,
-0.2733083993242672,
0.7075879191324372,
-0.042779623335368125,
0.7032435595714176,
-0.7363942983542189,
0.9583235184546098,
-0.5892472427173001,
-0.7139877518112785,
-0.12446625268371646,
-0.439646171766664,
0.7129808957109575,
0.3373075268097138,
0.3345550414369601,
-0.38068761370203624,
0.07864964370676188,
-0.3383553862900091,
0.6633419895127639,
-0.578720703894673,
0.486162302530746,
-0.06112668401522289,
0.34417654454775265,
-0.09182266750822224,
0.012215000731405658,

```

0.5658291917160285,
-0.16728488191683846,
0.3754616489195623,
-0.30497315054621493,
-0.7559846133933803,
0.9474959005813702,
-0.5904076686561035,
-0.2620011537974264,
0.1328515248519479,
-0.44066408562446224,
-0.7849425610737308,
0.44059474279745625,
-0.04646983134951628,
0.27897770882356554,
0.36838349877106535,
1.1610009997739648,
0.9058411506824863,
0.3119955912029684,
-0.5265683545543336,
1.0661822357546584,
0.8560044406571757,
0.5004409227892788,
0.5192086876555008,
-0.0019665558968806392,
-0.23949800878404184,
0.32563956832297103,
0.795323960796817,
1.5213873243487468,
0.8637563634588281,
0.07945031145196896,
1.7583315097859118,
0.5335864884320034,
-0.1366270407456156,
-0.9111102828185913,
-0.41650979281126865,
0.20926691692934202,
0.2776061293997892,
1.4257231544028384,
-0.10754544247245298,
1.0035199546983695,
1.1162390986419686,
1.7211522778091677,
-1.4332245925979028,
1.4417595659646931,
-2.106784537892007,
-0.6878309432904478,
-0.20360639354575213,
0.6373705186717347,
0.051333236332203484,
-0.35868991955772395,
-0.24365907607080922,
1.619736185871098,
0.5895182658245096,
-0.6086170057348131,
0.39774539931448055,
1.409837948786624,
0.17438880158294706,
-0.0415548590378924,
0.1413623732119859,
-0.8192475896231625,
0.5823982141653503,
-0.8462302367353051,
-0.60454163385331,
1.8216629921858374,
1.2634551976945552,
-0.40200420270308274,
-1.5954759495917508,
-1.7732438129518737]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,

```



```

1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric3_unifrom_scaling_sample#3'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.1514222647937595],
[12, 6, 0, 0, -4.906882414400225],
[20, 10, 0, 0, -3.911546483229528]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 4.076957702636719e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -6.8671436596990905,
'others': { 'beta_coefficients': [ 1.875,
-1.25,
0.375],
'cost_richardson_zne': 5.21875,
'sorted_expectation_vals': [
-6.1514222647937595,
-4.906882414400225,
-3.911546483229528],
'sorted_noise': [6, 18, 30]},
'sampled data': [ [ 4,
2,
0,
0,
-6.1514222647937595],
[ 12,
6,
0,
0,
-4.906882414400225],
[ 20,
10,
0,
0,
-3.911546483229528]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],

```

```
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [ 0.4349098035102067,
0.5281397824211302,
0.6402833431740305,
0.701719039924861,
0.8953768954552646,
1.4118719346144908,
2.230113621742168,
2.6548053250922607,
3.10433397641996,
3.1043339764199596,
4.146783795497047,
4.146783795497047,
4.149760661232294,
4.149760661232294,
4.223539194779984,
4.223539194779984,
6.7727886487081435,
6.7727886487081435,
7.243551717080481,
7.282491880183138,
7.282491880183137,
7.282491880183138,
8.806989173945695,
8.80698917394569,
9.706285648296737,
9.809777023075187,
9.837118725166619,
10.701586770517197,
11.185931122995514,
11.185931122995514,
0.39919688039300266,
1.4533249104921768,
0.7852252158845567,
-0.4957764525834907,
-0.3354719566017423,
0.16097458435951068,
-0.21971891747261646,
1.5370438656668401,
0.26531517225969137,
-0.6386119707402834,
0.37748437574064975,
1.2333814560270282,
0.5934062959304571,
1.633059089666515,
0.31170790405459564,
0.3995146629536743,
-0.11155094016301184,
0.8138063699882787,
0.019467271717771087,
-0.38186317682975884,
-2.778803401027002,
0.7887082829537533,
-3.1348050840084074,
-1.5109224736536222,
-0.38589710406642097,
-1.105927053527023,
0.3451707237315739,
0.6290043291617069,
-1.834722402137916,
-1.0551446159380258,
-1.3581260193610523,
0.9666180881078981,
-0.05759676522292653,
1.1866392284178628,
-1.1308471740259582,
0.34327443183487805,
0.17662613760194443,
-0.043135596735193626,
-0.03667546733606037,
```

-0.4901895513664937,
-1.0399227541005371,
0.4880459765792483,
0.053089003166299456,
-0.6259966396796703,
0.03159579372412852,
-0.09798805529491862,
0.052872660107769434,
1.490194875693949,
0.22167423812356596,
-0.33497064714209324,
1.0144679614160885,
0.01608415027143934,
0.49167899744128424,
0.12491439548600353,
-0.1329609067962543,
0.8639852596635488,
-1.6607871811173882,
0.2801513494447602,
0.3908501333463579,
0.6044891435983025,
0.08637544621327581,
1.2270127042754213,
0.11860545946003472,
-0.13066468162372125,
-0.998222267933649,
-1.2643438207146234,
0.7835210648968657,
0.8370312354478452,
0.06527740704106412,
-1.0235950282739155,
-0.705070781232052,
0.037682760290170314,
1.4778132318704873,
1.7053977225841355,
-0.1291331271719845,
0.1648924115529835,
2.208642048449397,
1.1885953055185114,
-0.8138037079028133,
0.6715189891287302,
0.7819572861394978,
-0.4022064636590845,
-0.3927127310055937,
-0.4118784323717791,
1.3463239978578432,
-0.3323606385924751,
0.6664683657180488,
1.58852177854836,
-1.212203169826861,
-0.031436536468697034,
0.5094262403458202,
-0.0616688967239856,
-0.0666169691754888,
-0.7277012125404719,
-0.12684196683036533,
0.5325519345188151,
2.2778420441533447,
0.8735933622003313,
0.9907279467768721,
1.1448787224294625,
0.4291077507842401,
-0.21356701746617238,
1.7174041100732933,
0.1885112766205208,
0.3630143936992631,
0.6395640633025061,
-0.2937369198596852,
-0.18128670495631546,
0.9504587478277511,
-0.6241293783488098,
1.8733903679237267,
0.4875344831143031,
-0.9182747577406853,
0.850791268752179,

```

-0.46183623376516114,
-0.634145646812118,
-0.9165900492445601,
-0.8973746003077631,
-0.879107286657044,
0.15645801311276636]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric3_unifrom_scaling_sample#8'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.672780190342767],
                          [12, 6, 0, 0, -5.522238922989663],
                          [20, 10, 0, 0, -4.5712287338679385]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 3.719329833984375e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -7.322874978356086,
                           'others': { 'beta_coeffieicients': [ 1.875,
                                                                -1.25,
                                                                0.375],
                                       'cost_richardson_zne': 5.21875,
                                       'sorted_expectation_vals': [
-6.672780190342767,
-5.522238922989663,
-4.5712287338679385],

'sorted_noise': [6, 18, 30]},
'sampled data': [ [ 4,
                    2,
                    0,
                    0,
                    -6.672780190342767],
                  [ 12,
                    6,
                    0,
                    0,
                    -5.522238922989663],
                  [ 20,
                    10,
                    0,
                    0,
                    -4.5712287338679385]],

```

```

'sampling': 'default']]],
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                                'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'ising'}},
  'init_param': { 'value': [ 0.23818326834520384,
                              0.242015507589488,
                              0.5292100600091615,
                              0.5598196968290768,
                              0.5598196968290768,
                              2.3451854381830244,
                              2.3451854381830244,
                              3.8453025861748578,
                              3.8453025861748578,
                              4.395137112369032,
                              4.408833101975425,
                              4.829585475162733,
                              4.867115107997544,
                              5.058958687371625,
                              5.058958687371625,
                              5.058958687371625,
                              5.581243202757385,
                              6.011433944644334,
                              6.011433944644334,
                              8.358003932454157,
                              8.358003932454157,
                              8.358003932454157,
                              8.371420896509267,
                              8.783744346022278,
                              8.783744346022278,
                              9.34189587380724,
                              9.34189587380724,
                              9.39608576624051,
                              9.726727464435275,
                              10.95102910414735,
                              0.14721560345391302,
                              1.0205767628001141,
                              0.40564926087548847,
                              -1.9238321605823567,
                              -0.04729505851188324,
                              0.482541600628026,
                              0.1152155677775581,
                              0.8524320040134185,
                              0.42380767990264745,
                              -0.6219665889019169,
                              -0.05759638705357143,
                              0.8161328242190486,
                              0.7425402472775383,
                              0.7535392998290764,
                              1.8314745216576223,
                              1.3334023135030668,
                              0.05552193989453376,
                              0.9691374289368472,
                              -0.07338372949421543,
                              -0.6690664310785921,
                              1.168533104165511,
                              -0.01962836639606316,
                              0.6576122461884292,
                              -0.037637936325233226,
                              -0.0009647519828692927,
                              0.4524839512418326,

```

-0.013115244288784337,
1.6270372625341938,
1.182284078721198,
0.9533845712158122,
1.0956662941001023,
-0.9647918240366662,
0.1897778948773928,
-0.04047520420354409,
0.5790278460715794,
0.631719469583885,
-0.3335418357294817,
1.0603186443819097,
-0.7449471003423949,
0.1630231005006795,
-0.07563007210422526,
-1.1009042311153838,
-0.34587951137957407,
-0.22408475871037167,
-0.6364303490666092,
-0.5245531829065874,
-0.7807390048523051,
-0.7767575864285188,
-0.9679768325508518,
-1.8691843526791756,
0.4783530568164554,
0.6133546874260105,
-0.20020986269221935,
-0.41532071148562183,
-0.8557118152516873,
-0.16985308585729458,
0.8428338023785568,
0.6750667051632169,
1.2795031602880573,
-0.41384224674171627,
0.3213362409234008,
1.2340309605381181,
1.0506369357289735,
-0.6415500741723684,
-0.9628453188140567,
1.935214357876566,
1.4159154638103797,
-2.0864443113151867,
-0.07426020994129089,
-1.0538315394110076,
0.2589745455351295,
-0.2955978932252484,
-0.030644528099012087,
-0.8602288471584197,
-0.09490606960216369,
0.4190622694749842,
-1.5667390101922392,
-0.7804920738383083,
0.1255849265457167,
0.5884338319364165,
-0.46028089859616994,
0.2488277677353687,
-0.7513271615815256,
-1.0561258734551267,
0.3697237180501531,
-0.7108432439071602,
0.5137418203820399,
0.946767478727941,
1.619957961978311,
1.2944079622561346,
0.21625880379379211,
0.007168703139473973,
-0.334764182643075,
-0.9086121674985949,
-0.8247776105092918,
-0.5734460393428641,
-0.4381826298821608,
-0.44306191141826123,
-0.23860269421390362,
-0.03343496826073402,
-0.008972211649351379,

```

1.2196121588758198,
1.5985131140815898,
0.060325450296581484,
-0.7312434635304047,
0.9413542695101463,
-0.6701271691421761,
-1.0680925542186157,
-0.5192506372311365,
1.5594784335685845,
0.17928043514789443,
0.2455246384495121,
-0.053106843359071614,
0.7488220810664211,
1.3030207698426615,
-1.787387332504871,
-1.1184519607509558,
-0.2511275398302805,
1.445676213762956,
1.5991220174606833]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric3_unifrom_scaling_sample#4'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.630764461698827],
                          [12, 6, 0, 0, -5.461578440310042],
                          [20, 10, 0, 0, -4.498507346043123]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 7.033348083496094e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -7.292650570063918,
                           'others': { 'beta_coeffieicients': [ 1.875,
                                                                -1.25,
                                                                0.375],
                           'cost_richardson_zne': 5.21875,
                           'sorted_expectation_vals': [
-6.630764461698827,
-5.461578440310042,
-4.498507346043123],

'sorted_noise': [6, 18, 30]],
'sampled data': [ [ 4,
                    2,

```

```

0,
0,
-6.630764461698827],
[ 12,
6,
0,
0,
-5.461578440310042],
[ 20,
10,
0,
0,
-4.498507346043123]],
'sampling': 'default']}]},
'ising-ric4-uniform-scaling': { 'noiseoff': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': {'value': 'random'},
'noise_profile': { 'noise_on_init_param': { 'status':
'value': 0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': False,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': True,
'type': 'png'},
'file_name_prefix': 'ising_noise_off'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0]]},
'run': 'vqe',
'state': 'dmatrix',
'vqe': { 'iteration': 10,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': True}},
'zne': { 'data_points': None,
'degree': 1,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '

```



```
'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
'4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
'X 5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 + '
'1 Z 0 + 1 X '
'1 X 2 + 1 Z '
'1 + 1 X 2 X '
'3 + 1 Z 2 + '
'1 X 3 X 4 + '
'1 Z 3 + 1 X '
'4 X 5 + 1 Z '
'4 + 1 X 5 X '
'6 + 1 Z 5 + '
'1 Z 6']],
'output': { 'exact_sol': -8.56677223350563,
'initial_cost_history': [ 3.7901552515724086,
3.396779462430669,
3.6416626483897634,
5.191980344851134,
3.300383519902229,
4.2302694452267025,
5.5832854157035126,
3.424340408479732,
4.575860085354091,
3.843028199719871],
'noise_details': { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [0, 0, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [
0.001,
0.001,
0.001,
0.001],
'status': False,
'type':
'depolarizing'},
'odd_wires': 4},
'optimized_minimum_cost': [ -7.3566708947935595,
-7.585318923493526,
-7.222021821934782,
-7.637732542535524,
-6.9717347656162385,
-7.183899566070664,
-7.312238847468353,
-7.55546648538883,
-6.871387778407254,
-7.103750879845002],
'optimized_parameters': [ [ -8.981486547411966e-18,
0.2355269100006643,
0.6734351680867104,
0.7281872519006076,
0.8287057436356319,
0.9968962697957876,
0.9968962697957877,
0.9968962697957875,
0.9968962697957875,
1.6732799868313266,
1.9939389913716405,
1.9939389913716405,
1.9939389913716405,
2.7786795690350794,
2.7786795690350794,
4.834741825646661,
4.843194997300159,
5.442664851556481,
6.006041306704694,
6.006041306704694,
6.42159854018659,
6.42159854018659,
7.172079495244572,
7.835006027334925,
7.835006027334925,
```

7.835006027334926,
9.27390458565848,
10.142272229921797,
10.547148172870333,
11.945646083493756,
0.30631338507375855,
0.950240081652421,
0.3197394260348401,
-0.8989315334324052,
-0.19972246213391193,
1.1515465399564957,
0.5794269610995502,
0.029234461946744308,
-0.3033609133811924,
-0.5068802243159589,
0.41891242044729254,
1.2272962791271989,
0.3217023759538914,
-0.0389595358377685,
-0.2818159573379153,
0.40344301100056285,
0.012601476380185236,
-0.7249227034698249,
-0.28859136632269766,
0.9177183017829931,
0.3192674751581855,
-0.6960323093564068,
0.9954617266198262,
-0.0960673339612117,
0.30357952531427623,
0.9148253966896076,
-0.41157308092318357,
-0.2244725515886058,
0.900815469140735,
-0.8704821270800919,
-0.2201886465573435,
-0.058451834208670006,
-0.5564504264806815,
1.0028418241392123,
0.6862825176439941,
-0.07515432929364027,
-0.132716959754057,
0.3156193912732197,
-0.5571752224261073,
-1.1266280252389607,
-0.2496012410462972,
0.4568813812573679,
0.5040642955504137,
-0.27382094178370653,
0.738802930675346,
-2.1017467172452315,
1.7134969759675782,
0.19045289169065632,
0.4959480509883087,
0.2725041426894662,
-0.020113534287112555,
-0.27730132318570966,
0.9121222342974794,
0.7771919661145833,
1.2410984938270941,
0.698373852916785,
0.049079898076347495,
0.4715079446630061,
-0.0012225173391828686,
1.2380103358912495,
0.18126256323161893,
0.02822806167971892,
0.9624172506978782,
-0.1017646649934255,
0.10712069411267353,
0.11059700546555196,
0.1764728469149672,
1.0536625920020957,
-0.43614672067179516,
1.1290800077402758,

-0.7544321351970764,
0.20178203991689547,
0.545960403958859,
-1.271331788368515,
0.3827893670869708,
0.514944994226915,
0.9987612988518954,
0.9130737051848579,
-0.33354580533033523,
0.4110572637290035,
-1.0065548546569214,
0.05504586363332423,
-0.8123464439408392,
0.9622209015425294,
-1.7981345279534349,
0.8489846299822246,
1.9191971402543575,
-2.019266402915899,
-1.6151954323496909,
1.693158829823371,
0.8368996699819713,
-0.7925788637806501,
-1.416264150106769,
-0.7484853165625206,
-1.026066230288144,
0.2591646216069995,
0.9485984853376032,
1.624644962520762,
0.1389264160191751,
1.7901079295930897,
0.28173803525757823,
-0.06727915670909099,
-0.39998196135024877,
-0.911581975644718,
-0.8802717995009537,
-0.6496644186002655,
0.9474433579476823,
0.5810351318438188,
-0.5928504020318089,
0.40429180463166087,
0.19955746243713188,
0.14940534343654552,
1.148493268597809,
-0.6171972430901704,
2.358530954611547,
1.6132804954395525,
1.5230819806307347,
0.07100911239249047,
-0.9821849785093056,
-1.2186503325370892],
[0.43242608326371745,
0.47036005705701023,
0.6035896599876003,
0.6406575109863006,
1.589410377648682,
1.589410377648682,
1.589410377648682,
1.9902163938334159,
1.9902163938334156,
2.6096176446013035,
2.6096176446013035,
3.039961339509649,
3.039961339509649,
3.0408788822093955,
3.0408788822093955,
4.1325863285768,
4.1325863285768,
4.801738885893826,
5.558430388585627,
5.558430388585628,
6.48703427557745,
6.599293033589678,
7.119130597501129,
7.119130597501129,
8.096879981584715,

8.096879981584715,
11.989689272176165,
12.289334556064002,
12.289334556064002,
12.695171119049377,
-2.1320303145314323,
1.164574641976517,
0.2281426394568994,
0.5098216286601104,
-0.15959745874570574,
-0.06246168319786304,
-1.335183444898994,
1.1253611654826523,
-2.453373641849758,
1.542670605477509,
-0.23459513075014657,
0.7629240608803527,
0.7066862387679962,
0.023354672976759586,
-0.26545157517882206,
0.8070755399487665,
0.621829996842247,
-0.8206657495436546,
-0.6456938685697818,
-0.8432536462697657,
1.5481421634749726,
0.030458967225015475,
-0.6362611752642527,
-0.5779049484598997,
0.15136987691256515,
0.37940285865801,
-0.13025289482332936,
-0.6088352960517447,
1.0137338691570092,
0.9312563664781934,
-0.06260332826127091,
0.16422266217724538,
-0.44919022316429513,
0.7120810093352475,
-0.045811140982536094,
1.4693096124875311,
1.3068674578470139,
0.3099696428239748,
0.05726716998373588,
1.4138231816748696,
1.3251150202531472,
2.4661338582136705,
-1.9221360663147242,
1.6309773912120944,
1.012572768899168,
1.1304323717308846,
-0.10014704960087417,
1.2138621646932324,
0.9150507542177275,
-0.1931179153053463,
-0.19983886142442753,
-0.37113288193216537,
1.0783926703777422,
0.7626802083907843,
-0.5396621733108105,
-1.0558614970998554,
-0.5031372556139323,
-0.1733591918410679,
-0.05810492443974475,
0.15252588011535415,
-1.3613790038949474,
0.14268023162896296,
2.1247307661901087,
-0.2996908036461345,
0.2682896527620792,
-0.6487468384363545,
0.0174900962257196,
0.7865320417875226,
-0.4365770411878935,
0.12280152026315115,

0.2668485276831852,
1.0055493052017264,
-0.13783144983938653,
0.16428177903351826,
-0.4107244428852575,
-0.5006989958309392,
0.4728442136126457,
-0.5756228808846314,
-0.2555605201147205,
-0.6655897533222209,
0.27354627255859315,
1.056546145946755,
-1.0108463744632321,
0.8570822622523233,
1.1003650202540056,
-0.13364446072464353,
0.19686136380997882,
-2.015333107317771,
-0.3474605408709381,
0.5127772990877489,
0.8873132714146144,
-0.1538651896386833,
-0.49691154598038334,
2.4337764506921564,
0.11687121421819974,
0.2637773919076603,
1.1119719051108947,
-1.1407171454187541,
0.293218856784229,
0.9089141681578352,
1.0587297353590328,
0.41354403585027527,
0.26331197174432663,
0.04763876798960274,
2.8376965303594455,
0.038400591301352385,
0.732378529151854,
1.259237861002353,
-0.09371603030550141,
-2.6524545055620123,
0.166545686266708,
0.8383948517861464,
0.5625425863873136,
0.6352389159281269,
-0.3793199251622615,
1.4353166704457385,
-0.06088962240819872,
0.5575840323453971,
-0.4370866693930651,
-1.3477496560670819],
[0.5485274765250757,
0.6924497584354614,
0.77926902254116,
1.4968846113254817,
1.4968846113254817,
1.617988193716464,
1.617988193716464,
2.3157329555892914,
2.3157329555892914,
2.3157329555892914,
2.6599448622122046,
2.6599448622122046,
3.198693955827293,
3.198693955827293,
4.93646112072991,
5.0183990860356085,
5.166530398168251,
5.732062846348599,
6.452081956191476,
6.950653712152026,
6.950653712152026,
7.527182443059662,
8.141470761550243,
8.141470761550243,
8.23161566953153,

8.23161566953153,
8.263835107953264,
9.693553204732913,
9.696078071259171,
10.987802073160276,
0.31236348113638823,
-1.370738548013981,
-0.19640846640845944,
-1.409184053738275,
-0.00366718677866519,
0.6361615892126344,
-0.6796219270984741,
-1.7803162192131878,
-1.2406782697025376,
-0.17521451360574944,
-0.2890470362486656,
-0.7870591534070972,
0.6214527622340972,
-0.5055171901576345,
0.3099897895999387,
0.6495941239175653,
0.4137491249725721,
0.292781258130194,
-0.18467758047440086,
-0.2764099099100952,
-0.12795113266111025,
0.14328500307003156,
0.1623766240720685,
0.09884205682735484,
0.07093196196575538,
0.17766773916349385,
0.1810420204025737,
-0.037056767883675096,
-0.38325428434024456,
-1.808138895166064,
-1.2174086587283277,
0.2723541238647832,
0.0911451526301191,
0.829454165789877,
-0.7076888645983359,
-1.1582847892183987,
-0.2789235155040628,
-0.1868914176693055,
-0.03618003850886602,
1.0797789643516258,
0.010710633626313776,
0.3041103423483441,
0.027606917069623423,
0.09715752300035431,
-0.025081467189021912,
0.7307804473426243,
-0.028312324696826655,
-0.5439567135695698,
1.0353579775860826,
0.03274794094556986,
0.338229999884877,
0.003923288912346083,
0.011059712392702849,
-0.24419721784886275,
0.0538500196620971,
-0.29704802197132957,
-0.9714959737951582,
-1.3776571886055275,
-0.4387141762272093,
-0.9013896619503009,
-0.2890811474761574,
-0.5664350037938545,
-1.265940570830118,
-0.36500389888052437,
-0.11634811830012508,
-1.0092353619257808,
-0.6849489189933503,
0.9856942680495377,
0.8277230147066852,
0.2903619348614201,

-0.3836733140597432,
-0.530059482018533,
0.8169168177018395,
0.5323656053240163,
0.649672850550592,
-0.3032256260984974,
-0.8829469814645045,
-0.00635052379210004,
0.35535568318583244,
-0.06185634463022188,
0.10533559478429602,
1.2902435643868964,
0.8825397316379378,
0.3070411861797082,
-0.07934132673084492,
0.09844318572655962,
0.7695889603351249,
0.14399374014816757,
1.8827194490661865,
0.013305728022345127,
1.2201617371808122,
0.13458781619262966,
-0.42760236848753097,
1.3132997483594182,
0.29586095969137777,
0.14782133458354627,
1.1539512755729704,
0.8255769419022573,
2.1284509830181357,
1.5067425074661367,
1.2286171121451426,
0.8625402575232176,
0.5254367589291086,
0.8683165111711829,
0.35804094605055825,
-1.3678290318224675,
-0.3952095755717416,
2.8373763512718595,
-1.2381307915304474,
0.21836664096244152,
-1.0373469190584237,
0.35616351546904684,
-0.48865262743063154,
0.5395297676370602,
-0.0068350558769426075,
-1.6240423184716037,
-1.005087252746991,
-0.324331708942255,
1.56936745287968,
1.3955283874562228],
[0.3026146462879133,
0.3869510743764029,
0.3869510743764029,
0.5382374643481795,
0.7459322218289088,
0.7459322218289088,
0.8083924716224152,
0.8083924716224152,
3.219627607910341,
3.2253302931763046,
3.9244184219369176,
4.114742178023509,
4.115220647252454,
4.115220647252454,
4.115220647252454,
4.65196408394345,
5.196968879903224,
5.736027504566688,
5.736027504566688,
6.321573696596394,
6.7009287553652,
6.7009287553652,
6.7009287553652,
8.383675014305652,
8.383675014305652,

8.383675014305652,
8.383675014305652,
9.235844817182018,
9.235844817182018,
12.415877726713473,
1.4636821166030158,
0.8436356038076634,
-0.6467230542935267,
0.2472607630354758,
0.028721536942984512,
0.5720112007049558,
-0.179388461505878,
0.6605644899444229,
-0.2997484495809327,
-0.5848737649454969,
0.7567373384637915,
-0.3113371232415861,
1.5925608798994126,
0.15551153490893768,
1.144256524486279,
-1.0086304067563407,
0.4692085043130235,
-0.5289416707772752,
0.6895537558918402,
-0.9023920029674053,
0.6687255575647186,
1.3626463195064957,
0.18435277462186048,
0.04101567185284027,
1.2334404589145347,
-0.8099334611041354,
0.47217953762833137,
-0.19021445340255347,
-0.47126223762254077,
0.21535783789089086,
-0.13160146544027274,
-0.14745399432080403,
-0.2131109368071074,
0.6116235033000932,
0.7187755412423116,
-0.7690585507561546,
0.333613273647484,
0.29688326543860977,
0.022443242058892903,
-0.12904022561026712,
1.1137507228270036,
-0.1754869613868649,
0.6791619259450926,
0.2688019369707634,
-0.33561997076117317,
-0.4910450017308718,
-0.2921449581476723,
0.07017675650466139,
-0.5628878746231101,
0.23633646017547374,
-0.4408658192483561,
0.6337746650828207,
-0.5275874765915898,
-0.11397090424872018,
0.7713694979315344,
-0.24077591679338844,
0.7998056369050096,
0.5449521212196076,
0.6440774810386566,
0.0012540722532346088,
0.4406697804838639,
-0.4697708168210915,
-0.38183296272166767,
0.4926783027265995,
0.012102728676941394,
0.9995268367701109,
-0.489600116502774,
-0.9378926015671718,
0.7538025600192173,
-0.9270581698564858,

0.23395173940139535,
-1.5047568644302813,
-0.6335501933328295,
-0.1851947866597326,
-0.18416301345971506,
-0.15010118329037675,
-0.30085163208099847,
-0.030658108221894752,
0.48141025496380413,
1.2807653372993688,
-0.27688172784497056,
-0.9404124429777033,
0.07772158149530461,
1.2206520244647234,
0.07167829739929571,
1.5514296753518346,
-0.22739832257997444,
-0.5457307695824859,
-0.05085143495318925,
1.4086504148457157,
-0.02595768387677813,
1.2057536466860386,
-0.5009834455732999,
1.0234750156328314,
-0.34572346527257924,
1.4975050133157242,
0.6424858734227697,
-0.5412025600486546,
-0.965195784498948,
0.8078930733387817,
-0.8747920106317195,
1.6444412493640082,
-1.3120143788008718,
0.7469227938848356,
0.7428309989793587,
0.25043937013438866,
-0.7176977899841391,
1.6961228018637329,
0.8286720852002186,
-0.08923208660690117,
-0.4564959840968765,
-0.10389957398275913,
-0.04652285532782134,
0.6308845069083873,
-0.025008046054888575,
-1.0109619026173717,
0.31698532529812806,
-0.07816069204265776,
2.517427548138029,
1.300984357739734],
[0.48696254731677435,
0.9433356258260287,
1.8515151481415904,
3.298386332934612,
3.298386332934612,
3.298386332934612,
3.298386332934612,
3.298386332934612,
3.374346159341203,
3.374346159341203,
5.043620376468771,
5.043620376468771,
5.218159502227583,
5.683079438253876,
5.683079438253876,
5.902572186502105,
5.902572186502105,
5.902572186502105,
6.17408053545979,
6.17408053545979,
6.17408053545979,
6.17408053545979,
8.600010985810668,
8.600010985810668,
9.424738936566825,

9.424738936566825,
9.43568443936613,
10.707654072137746,
10.707654072137748,
11.243950101093004,
0.30560526927017156,
-0.6023681078531374,
-0.5052271452160477,
-1.0944465846035194,
0.24881605409968105,
1.5229622975948298,
1.730509331816328,
-1.3834281548913137,
-0.8373914729231547,
-1.186133470726964,
-0.983750614650609,
1.3432121337567746,
1.6569377678026054,
0.14131844528439758,
2.9464329846729154,
1.3448953312948555,
0.7896925151532842,
-0.22765810177877452,
-1.6238177025568765,
1.1679097532601286,
1.5429785141845107,
0.44641536771583223,
-0.4990561654458825,
0.2890079097559007,
0.07664205286689205,
1.0463374337388847,
1.0582154531020835,
-0.7032713068941808,
-0.5914102053027229,
0.4481794192989112,
-0.597054072340342,
0.6503285491619032,
-1.3548677382206493,
0.3073247134537545,
-0.42900928099090013,
0.8290869769130189,
0.25661934620723303,
-0.41883276922189405,
-0.25409113110415216,
-0.2750324278847333,
1.8535327289298686,
0.7375047938415388,
0.21982546706442144,
0.3355829615333414,
-0.18885187042138774,
-0.09263125664317409,
0.8768150768205386,
-0.31105067773622297,
0.11882768143806702,
0.007542537022557154,
0.6689687664434811,
0.7431113887615772,
0.9447406593038509,
1.0052341467989638,
1.4439497926806117,
0.4202121214316228,
0.3604666875525164,
-1.6888556500060994,
0.1396130584965534,
0.10094819946612499,
0.5021137054548022,
0.3961649599525077,
0.4932381662816783,
1.3418440833635417,
0.09854079710503719,
-0.9989986964474209,
1.2672439763832217,
-0.07601477770236317,
0.46767000685442095,
0.3227070865546575,

-0.7175096141500144,
0.2758047915288214,
-0.24095833900456404,
0.058134638501602325,
0.6051082814423787,
0.5442716652871094,
0.3025208071236723,
-0.6679934596266708,
0.8499265434495678,
0.1803663678751026,
-0.3707988225819567,
0.6812577092957876,
1.2284831501706344,
0.20883808756632122,
0.03358446867125293,
0.026609877934757803,
-0.40919242519291843,
-0.07640273708110212,
-1.1643751408115468,
-0.16769301556045021,
-1.2911947285322207,
0.037223344859154565,
-0.05786091979914895,
-1.8047938425602224,
0.014158596153457605,
0.534864509903787,
0.006329420481561668,
1.239578876475735,
0.2639865904342132,
1.2160651224901187,
-0.2035728984920286,
0.6511657116444571,
0.3064364057804899,
2.45196454382623,
0.8340823063941794,
1.113505687918428,
0.049528306503774544,
0.12947314602533921,
-0.5399313265794791,
-1.2647403981540062,
-0.3482737106911781,
0.09697429466359891,
2.20804968215061,
-0.15489516431237316,
0.1250815357650574,
-0.13157603031624435,
-1.2324441762654297,
-0.4179909399666764,
0.3779753490492196,
1.6472121643813018],
[-1.1990978117137884e-16,
0.0939706539344674,
0.583570497583306,
0.583570497583306,
1.7474308983833136,
1.7474308983833136,
2.435631059267761,
2.4356310592677604,
2.435631059267761,
2.4356310592677612,
2.9757043839921162,
2.975704383992116,
4.1141402268542535,
4.1141402268542535,
4.889622687428616,
5.085553065152385,
5.14502423979873,
6.38608549233117,
6.405094454380333,
6.807050165246778,
6.807050165246778,
6.997225211250308,
7.82493752524197,
8.419597935808826,
8.419597935808826,

8.756502393683652,
9.514795580339912,
10.006329694888725,
10.008671195739254,
10.582332645019955,
-0.18325409339400697,
-1.8356335907661927,
-0.6014425812689169,
1.272854521157504,
0.38561099010389943,
-1.3354103698578268,
1.2089446007576305,
-0.16684037696101606,
1.5272415494549387,
-0.21803857489548387,
0.659293576456549,
-0.7629850988015884,
-2.0880787600158244,
-0.2200967935369742,
-0.21257917133409554,
0.05335328813343252,
0.740881120097457,
0.002552459203101907,
2.0381278489789163,
0.07009602349011958,
-0.027486349535234322,
-0.5162309983105858,
-0.018107734929441965,
0.3459554204227654,
1.1337361200247116,
0.6158218713410994,
-0.5251391279578093,
0.2758552924607382,
1.227089028823027,
-0.16405595633516176,
0.7587337472599636,
-1.1046949665522132,
0.7157795020690486,
-0.9666557350608269,
0.5941166908158014,
-0.3035426485157296,
-1.1713314559961463,
-0.3285214016177699,
0.15201600725182188,
1.158582769628082,
-0.2335509332665805,
0.021624313631101504,
-0.7981749327134954,
0.12400359670343174,
-0.00041124409742626503,
0.10665347356443583,
-0.030437347790731137,
-0.22360998336164295,
0.21313944649575595,
0.08788738041696632,
-1.2878790640853577,
-0.028446563531964782,
-0.12898316604526575,
-0.29469285393680966,
-0.07114900510733681,
-0.8104482788171727,
0.6394923422746198,
0.6811407094584414,
0.15127654478155184,
0.5799864823630293,
-0.55116978803857,
-0.5805840448115108,
0.9630930231089359,
0.9222833564508055,
-0.28733988453131243,
2.645322235680672,
0.02138581028992107,
-0.12421624460779133,
-1.4608772192659922,
-1.289711141721096,

-1.433469567857053,
-0.01041716118532159,
0.3377052351749707,
-1.74263853738129,
-0.6445874089376945,
0.6790378089998735,
0.0019732068694448615,
0.6260241949545162,
-1.3812938119718974,
0.06110738334511842,
0.919936405723882,
-2.761538990972181,
0.01773377957551518,
0.6041489324828,
-0.281847147959406,
0.36931104439326196,
0.14008962906529285,
1.0402771661903438,
0.40062198057764514,
-0.03775531840468896,
-0.4974133017905023,
0.5298490799692606,
-1.7184458207495306,
-0.7561470763062167,
-0.5888363413103725,
-0.9669502040570058,
1.3911426640670486,
-1.922807929753264,
-0.7347523142785206,
-0.6976214468215837,
-0.9022103009000106,
-0.17554401086360705,
-0.7004215075729389,
-1.1648825394224225,
-1.1285064327861984,
1.3794410327712907,
0.7524041737227146,
-0.600926776988818,
0.724704963452131,
0.24574404734520428,
1.3856850730269805,
0.505698953761027,
-0.030535393994814766,
-0.4836980977841504,
0.6166564120021005,
2.4235804988343936,
0.6827837636480912,
0.3673788389460804,
0.11748797515103059,
1.4924513209248702],
[1.112287766890805e-16,
0.11579095179987547,
1.7588846075757274,
1.7588846075757274,
1.998030831219005,
3.2454648629288743,
3.248106797240174,
3.248106797240174,
4.259596531772135,
4.958175754196259,
5.7210454174100525,
5.861260496697951,
5.861260496697951,
6.7588095170100075,
6.7588095170100075,
6.758809517010007,
7.044198424034494,
7.044198424034494,
7.044198424034494,
7.044198424034494,
8.232813771743604,
8.791432218349115,
8.791432218349115,
8.791432218349115,
9.616347292478379,

9.616347292478379,
9.869492613451865,
9.869492613451865,
10.182429226961672,
13.25758469112181,
0.5705572906646224,
0.09014038635403764,
0.1263845657016112,
1.0203985977202104,
-1.095649826264738,
0.019398597884167954,
-1.3544747652103108,
0.8866931148929111,
-0.18940184666069662,
1.0130589658671383,
0.15424354259331335,
-0.8117851912876255,
0.28413415623416743,
1.542699745821886,
1.836477396464815,
0.7160099188075301,
-0.9589348876176362,
0.00854407549207699,
0.36750908199249616,
0.7288091138524155,
0.9748756553152056,
0.7257579234395075,
-1.9128288506616224,
0.515489770530508,
-0.8485176532957428,
0.21376115901922352,
1.0834633235702664,
0.404913552038143,
-0.12863840971929708,
0.8333794222571905,
2.496813054711946,
-0.7354302042700512,
0.27199375272502646,
-1.0107781002238316,
-1.201703662816619,
-0.3331724830549245,
1.3030593505928814,
-0.8138384777541797,
1.9586180892259586,
1.1713603302631173,
1.0819642334098143,
-1.1784039537689197,
-0.12146178641002908,
-0.1717950232658355,
0.13833462928205204,
-0.8697724612180706,
0.7515471411202548,
1.166125613326729,
0.37941686042382794,
0.17074885851048915,
-0.33600856179263133,
-0.13669943521352546,
0.548834657399268,
-0.2522233718783481,
0.009665778262181637,
-0.5020120590858954,
0.16617168999240312,
-0.46390045473242114,
-0.05573261329441784,
1.1182956755615927,
0.0937037366596967,
-0.4441962997721035,
-0.21050297984767188,
-1.0381458298882043,
0.830180513233935,
-0.8157309834659043,
-0.5833742339287631,
-1.5354704640903527,
0.44493118131447734,
0.13057382050598434,

-0.7034976447329113,
0.6218444459919024,
-0.2420386731361673,
-0.7867967091164098,
-0.168935029173583,
-1.6928247400206122,
0.22571922404505704,
0.1126763335131705,
-0.22160264498444046,
2.299575126838168,
2.5415010654218975,
-0.5449604787415219,
-1.005432959888493,
-1.722931012111005,
-0.6732620373630841,
-0.0004890899721480893,
1.492586488782192,
-0.9055437852306896,
1.1298576359465664,
-0.5182549439372126,
0.6586606791613716,
0.7929632788713356,
0.08462141435268096,
0.5066414628201175,
0.6503902510816232,
0.06456835392989443,
-1.687283870544211,
-0.2816102525113523,
0.9601910446201565,
-1.0364808528540121,
0.32603884641661884,
-0.37671408634672476,
-0.16068551007585027,
0.6157148912247653,
2.0105220675930333,
1.7580796652062145,
1.7229426093703843,
-0.5199627188331776,
-1.3760925920420213,
0.8290526991464299,
-0.08765267414283408,
-0.456659399307068,
1.4313668535261046,
0.635052489001378,
2.136142869643981,
-1.5755126698917075,
-0.40116032394931217,
-0.004321541181387126,
3.301925406547275,
0.7673504276272846],
[0.49826187507447023,
0.6110785201427625,
0.6558173388435118,
1.2433731466916595,
1.2468055416987345,
1.9464512542038723,
2.109950686395802,
3.7879039161756585,
3.7879039161756585,
3.7879039161756585,
3.7879039161756585,
3.7879039161756585,
3.7879039161756585,
5.054414830549875,
5.106961698737131,
5.488040579082762,
5.488040579082762,
6.832391146999917,
6.967864562238176,
6.967864562238176,
7.490508971727458,
8.20118277218113,
8.612503180124596,
9.093659709060372,
9.807153774862888,

10.372385504551621,
10.372385504551621,
11.272736556325881,
11.767303580595307,
11.767303580595307,
0.035679258194077725,
-0.8723307875716305,
-1.7560663493087938,
1.9629136553479551,
1.9120807210302333,
-0.39357388790453784,
0.036182026687196715,
-0.21816401975370772,
0.8090968256508778,
0.22775227657680963,
-0.9402388060640036,
1.9199330583190146,
-0.6293921442826633,
-0.37521158500092616,
-2.4122776248196542,
0.47614923103710605,
-0.2356719275662886,
0.5930119757607211,
-0.08067436637158562,
-0.6975308853470512,
-0.1533962892581602,
0.6838091266999453,
0.243129377869582,
0.5896632240559074,
0.12048284254834898,
1.186903891075025,
0.22948374273850122,
-0.9592798973456383,
1.431786126792839,
1.099938502501145,
-0.6421065764259075,
-0.4570128400176702,
-0.40361390122591434,
-0.030699975686000303,
0.5709596867047622,
0.8129177753256275,
0.0978347154858261,
-0.1791413540365364,
1.316347643480991,
1.0669242344323897,
0.9542637314281797,
0.2732710750226029,
-0.24925074976969727,
0.14351675204102743,
-2.058043419369877,
-0.5772495977232713,
0.4759355346590471,
-0.3040209890415461,
0.07360159687611288,
0.11769956200453384,
0.01709491725825118,
1.1325236656251834,
0.008658928128351846,
-0.21852721169551936,
-0.1826844911726131,
0.39535377862993026,
0.035729500082236576,
-1.0747644579938396,
-0.15992271631488378,
0.8234408989300128,
2.679077704708449,
0.2559692131167249,
1.5592174030452497,
0.3453465662341248,
-0.24046357809578378,
-0.6876498012434461,
-0.01750342347144341,
0.34704200676536884,
0.2033679229118891,
-1.5920113668255755,

-0.5283783493172445,
0.4351355192977228,
-0.8281455342425702,
0.2974457926897622,
-1.336557258227178,
0.2885982325597283,
-1.2111847441179915,
0.698629224522434,
-1.072563817644129,
-1.2092878769637225,
-0.33153669598628793,
-0.2558676939050336,
-0.039300685015777354,
0.5730206646846686,
0.6772024906875076,
-0.5431857921175166,
-0.32242263836861185,
-1.2435651530921474,
0.6447338068594848,
-1.1796614873773261,
-0.5517178991847818,
-0.9950127711245446,
-0.750107619545704,
1.546476758712725,
0.7798072495873917,
-1.3969546244577882,
-1.3869625997614359,
-0.5499269608373658,
-2.0698482195340895,
-0.31472418351029113,
1.3858906356684872,
-2.3234389152233144,
0.8580050890595776,
0.5574876161662472,
0.12663545081840322,
0.0024286165622580055,
1.4351555562364873,
-0.018100010671135307,
-1.4466701603237728,
1.0091335554802272,
-0.25739419705547106,
0.8584486110461004,
-1.6353613496727522,
1.3481381905015994,
1.664215570319441,
-1.4688501218242656,
0.6630999772073086,
-1.3598408860414706,
-0.9086993193406631,
-0.7720609621308895],
[0.03916643192220903,
0.09260151508853934,
0.5125042140553085,
0.5342494495137248,
1.4070055183169814,
1.4076098722130033,
1.4080234932826805,
1.9886432369288551,
1.988643236970097,
2.4501760969485846,
2.4501760969572235,
2.6141080657383133,
2.614108094671839,
2.6141080978874367,
2.614108199851818,
2.614108200075538,
2.6141082008692407,
4.121207830152765,
4.126135109264967,
4.126137501750553,
4.129417665319773,
4.731890053629371,
4.734221022017094,
4.783748744413943,
6.888546551790504,

7.435120796142169,
9.27338967383107,
9.884598996480708,
9.884598996488108,
10.57466535326793,
0.042050619766987055,
1.2200437115279086,
-0.1897539908832087,
0.3883222993215419,
-0.3953259161886337,
0.9433481261689909,
0.03976732857871137,
0.380475950476648,
-1.6480061036631655,
0.19200671798130073,
1.8964236385178486,
0.5825466621902844,
-0.7017933974726621,
0.46167134168781887,
1.3455780978346443,
1.000341055668334,
0.3220859397336833,
0.4901021103591929,
1.7507755726531005,
0.22815940378303187,
-0.27415827045663116,
-0.06669089502017891,
-1.2681856141960832,
-1.4186851222947996,
-0.7484212831710693,
-0.2624358161393499,
0.22137315279362701,
0.8266616228870926,
-0.28382763940078026,
0.4526868734709094,
-0.03876262869445461,
0.028933904781200354,
0.2124235174903725,
0.5275601102845284,
-0.035224229477880745,
-0.11459385773709553,
0.4537223625267385,
1.3135706165796082,
-0.4103066311241145,
-0.06740865621875465,
0.07865262902016133,
-0.8220672300227196,
0.05131914396597353,
-0.012779296277522511,
1.6650491211728184,
-0.5271258879235212,
-0.2286028431320702,
0.9810985313826684,
0.5796827949136435,
-1.4216581678953852,
0.908254540613963,
0.7809397666182989,
0.23060365236949015,
-1.629683736567043,
0.08687379015275534,
0.5302421766128769,
0.858319872120405,
-1.2816124200556078,
0.08255098737682609,
0.49190330494839835,
1.2241314669476433,
-1.6043734869556,
0.35480117328822147,
-1.3692690448046365,
0.08801115679730828,
-0.3416815622800032,
0.5349891528238044,
0.23599371094677032,
-0.2955967206364773,
0.7351110566185237,

-0.1358629473726594,
-0.2538927978152746,
-0.18822579896800806,
0.023625415865148318,
-0.47104955898400697,
0.3192033261234258,
1.3512870657881308,
-0.7816632422470443,
-1.6469850516046918,
0.27638573599623373,
-0.3090269715855155,
-0.4193062343229584,
-0.1377874030219536,
0.0004776579492072504,
0.08325659247733753,
1.4611032338266408,
0.021431340914286272,
0.20066912699708425,
0.29328305426482204,
2.42997753887945,
-0.5690256391604135,
-1.480840559800266,
-0.1363946771767577,
-1.376439435407693,
0.06387998538825271,
-1.110029094439266,
0.9868538484965683,
0.962243917255456,
-0.41126656322102295,
-0.657103189069447,
1.1434637843768656,
-0.9757973897576084,
1.448315222432508,
1.3110742046722172,
2.3333185683193554,
0.7520157773640208,
-0.20203562372486106,
0.34414613752805073,
0.5926366260275047,
-1.985044802604821,
2.6078832019778186,
-1.6883872397325923,
-1.6784323905215754,
-0.4297482144956373,
0.6572781395405978,
0.02277885553642172,
1.3256214503718338,
0.25673693454286806,
-0.6358308725577859,
-1.6133180674191239],
[0.4366123913610663,
0.4768198412180804,
0.9843286193666714,
1.2674678431689148,
1.5775702096396949,
1.5775702096396946,
2.2591837513222837,
2.2591837513222837,
3.168921311818864,
3.1689213118188637,
3.1689213118188637,
3.2207076420855234,
3.220707642085524,
4.00024951310582,
4.479169433031019,
5.572620903211533,
5.572620903211533,
7.169574827830268,
7.200887753316444,
7.200887753316444,
7.4320670533634186,
7.4320670533634186,
7.668889448658714,
7.668889448658714,
8.088052955990618,

8.127936129548912,
8.127936129548912,
8.129700499301142,
8.129700499301142,
13.234708916500516,
-0.8409114197195183,
0.13809956722013594,
-1.3698910408430074,
-0.1271619514190957,
-1.0028759735951214,
-0.7519751568518348,
-1.7027753599694198,
-0.5003000098057334,
1.7803062126367422,
-0.6956057702973542,
-0.5292841478808504,
1.3371604427134443,
-0.6271315116651272,
-0.9849660401714492,
-1.0245042646602094,
1.7746289077120558,
0.789646250683077,
2.3274439120303834,
1.0250145144124314,
0.236963759763461,
0.6449585291401012,
0.5302895792637682,
-0.17332551626046228,
0.13296008629005646,
1.3001365941190501,
-0.5491163203391332,
0.8939708861040915,
-0.1183728413911535,
-0.27672563627574315,
1.7953720191393387,
-0.00825054985875676,
-0.6170579146731886,
1.801545679753067,
-0.6786704865717594,
-0.5702663839537956,
-1.2235657195316856,
0.059502931568887765,
0.15944146767128192,
-0.5445588526685949,
1.0645821939979259,
-0.3246614360002631,
1.6342475668463907,
-0.6365693512803821,
-0.25644806680599874,
0.9153791587574333,
-1.5555971818784813,
1.7626038696684259,
0.29250830199243494,
0.4597874603079029,
-0.4983420660721544,
0.22206047990995867,
0.11089125969329257,
1.9712857917750568,
-0.8482820514299444,
1.7711968287204813,
-0.3356021753581158,
-2.239225908466879,
-0.021402454584719684,
1.6641712424719513,
0.5059308187330879,
1.9296669762106724,
0.7155610478862724,
1.024294278133781,
0.2881031759975837,
0.06318491357276658,
0.8308741396333285,
0.16800005476886462,
-0.26845892946467115,
-0.378371667085078,
-2.374474400887392,

```

-0.26943703763693594,
-0.698442208248677,
1.5029972614102114,
0.7682274655447995,
1.7790823677975025,
1.1233449399645796,
-0.1962677980677998,
-0.6013220679983903,
1.243366811759978,
0.23851493156650586,
-1.3516830763738037,
0.4546075088056473,
0.45461165179145885,
0.29299766155391044,
0.9768506591689141,
-1.3614782254940123,
-0.1864841683590143,
0.03417328589702677,
-0.12392810093414776,
1.4042342901901068,
0.4181387945528863,
0.6171624447156228,
0.5887687464980093,
0.07995045192475739,
0.12406497411608935,
-0.1304256749231809,
0.5172871404602656,
1.1235454838537668,
1.5976273357881063,
-0.7195312648953682,
-0.7697726746437508,
-0.1718403740134208,
0.48757574872507614,
0.7291603816887677,
1.3754811895843486,
0.06384938170591244,
0.9009847125572167,
1.0744001081352503,
1.7305303874736466,
0.491454536253661,
0.11643540469065636,
0.5925455840333324,
-1.2877724397991297,
0.052924560657905566,
-0.8760423919600488,
0.6313576901954856,
2.736167065037608,
0.029295634649204143,
2.1044233481865446,
0.6660786050345296]],
'run_time_sec': 43335.76831364632}}],
'redundant': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}}},
'init_param': { 'value': [ -4.1290471568429383e-17,
0.028387941050870547,
0.028387941050870516,
0.028387941050870544,
0.4764187545019148,
0.485154879376271,

```

0.7094706976105217,
2.998376901501724,
3.486264965503922,
3.487775573446841,
4.058897205630428,
5.020552086463093,
5.02158492362958,
5.02158492362958,
5.0846878698400815,
5.0846878698400815,
6.794704827143902,
6.794704827143902,
6.794704827143902,
8.013356083289752,
8.21076169573117,
8.21076169573117,
8.633075344303478,
8.633075344303478,
8.930018414781724,
8.930018414781724,
8.930018414781724,
9.990329186240457,
9.990329186240457,
12.915390927357043,
0.8289215528981712,
0.6465354442587259,
-0.06922368073773534,
-2.0853926237797484,
0.25605409461249207,
-0.765586831155604,
0.9589723041272441,
0.27455658215646966,
-0.07189982532198122,
-0.3449139525561357,
-0.46800027418386986,
1.016480689947782,
-0.9706976058410632,
-0.9556868223128826,
0.9323574793538898,
-0.4376613900889883,
0.44138701419965265,
1.4418520162063444,
0.8420619129497768,
-0.025530845682239494,
-2.3076344184440156,
0.37458463155346333,
-0.47031338116346383,
0.07338001085955462,
-0.869249007732584,
1.6824786550253577,
1.719346297136726,
-0.9622509832303179,
0.5779076253870294,
0.4760527439731688,
-0.2451715046346434,
1.5906513054768954,
0.35918458386075697,
-0.44817294000256097,
-0.37416377940751877,
0.37048554789855764,
0.693137718643464,
2.288219219235325,
-0.26501198620577204,
0.3689053264668303,
0.10166782579927121,
-0.3356073800920907,
-0.6413954053758932,
0.28636656044125136,
-0.8690978212322298,
1.166030172665772,
-0.46365263204645707,
-0.8505396438729699,
2.4328989003329693,
-1.4554443951257197,
2.09134311034601,

1.7695145471360614,
1.1170184462314887,
0.20325573406500014,
0.34206355374635444,
0.016804831103183593,
-0.19882507083844791,
1.4011478969931443,
-0.0572616468882171,
-1.3773340344964602,
1.1031086912886459,
0.47433623341773395,
-0.037679845343088075,
0.7450954028512253,
1.3114871459373192,
-0.18651966717154955,
1.0750199871192778,
0.48905379729033566,
-0.6281827821606347,
1.322074656325554,
-1.4761484868509396,
0.2859918816514792,
-1.0033541595176405,
0.8264053813863532,
0.622010811103528,
1.1636747207646185,
-0.31883271288389914,
-0.0013981850979601159,
-0.09999405171802342,
1.3085280853238759,
-1.3727969089464465,
-0.08341159264840771,
1.6249369276484606,
-0.28082331570516694,
1.863294400254821,
2.2932652495309953,
1.4849536675442596,
0.6811668802751195,
1.5380110980874968,
-0.6615225065895972,
0.4000828281024264,
0.33932121588009856,
0.48855764825038867,
2.5371111232777053,
-0.7260430040504564,
0.09005940379987407,
-0.8762980817992282,
-0.23800314057977784,
0.9728027685352436,
-0.20857834368009137,
0.6148333457084475,
-1.5171898336040224,
0.8658542069129427,
-1.6478287868490311,
0.5770049367106069,
0.134497700004433,
-0.11612046371881249,
1.5826352840237945,
1.6045820840261675,
-0.08805669980595342,
-1.507745125921201,
-0.11276651576435986,
0.07206005105970639,
-1.7788122679933258,
-0.002988765738482436,
0.7255804452314191,
2.7661804504965604,
-3.1943627011957507,
-0.07088434770936546,
0.10949711075632612]],

'noise_profile': { 'noise_on_init_param': { 'status':

'value': 0},

'noise_prob': [0.001, 0.001, 0, 0],

'status': True,

'type': 'depolarizing'},

False,

```

'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric4_unifrom_scaling_sample#1'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0],
[3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
'4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
'X 5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
'+ 1 Z 0 + '
'1 X 1 X 2 '
'+ 1 Z 1 + '
'1 X 2 X 3 '
'+ 1 Z 2 + '
'1 X 3 X 4 '
'+ 1 Z 3 + '
'1 X 4 X 5 '
'+ 1 Z 4 + '
'1 X 5 X 6 '
'+ 1 Z 5 + '
'1 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -6.833109640228284],
[12, 6, 0, 0, -5.610198560557998],
[20, 10, 0, 0, -4.60990156720048],
[28, 14, 0, 0, -3.786810221698121]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [
0.001,
0.001,
0,
0],
'status': True,
'type':
'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': {

```



```

'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,
0,
0],

'noise_prob': [

'status': True,
'type':

'depolarizing'},

    'odd_wires': 4},
    { 'gates_num': [20, 5, 1, 0],
      'identity_factors': [2, 2, 0, 0],
      'noise_level': [20, 10, 0, 0],
      'noise_profile': {

'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,
0,
0],

'noise_prob': [

'status': True,
'type':

'depolarizing'},

    'odd_wires': 4},
    { 'gates_num': [28, 7, 1, 0],
      'identity_factors': [3, 3, 0, 0],
      'noise_level': [28, 14, 0, 0],
      'noise_profile': {

'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,
0,
0],

'noise_prob': [

'status': True,
'type':

'depolarizing'},

    'odd_wires': 4}},
    'run_time_sec': 9.735995769500732}},
{ 'config': { 'ansatz': { 'gateset': 1,
    'layer': 30,
    'ugate': { 'coefficients': { 'bn': [ 0,
    0,
    0,
    0,
    0,
    0,
    0],
    'cn': [ 0.5,
    0.5,
    0.5,
    0.5,
    0.5,
    0.5],
    'r': 0},
    'time': {'max': 10.0, 'min': 0.0},
    'type': 'ising'}}},
    'init_param': { 'value': [ 0.46756606893882113,
    0.46983859896604685,
    0.5133985927484581,
    0.5133985927484581,
    0.5133985927484581,
    1.4016794842576101,
    1.6914116785996751,
    1.691411678599675,
    1.6941756924801525,
    1.6941756924801528,
    3.749864451084885,
    3.749864451084885,
    3.749864451084885,
    3.749864451084885,
    4.969636107010246,

```

4.969636107010246,
4.969636107010246,
4.969636107010246,
5.983107385388112,
5.983107385388112,
8.378949246097575,
8.378949246097575,
8.635680633753974,
8.635680633753974,
9.108807770997622,
9.108807770997622,
9.108807770997622,
9.960036411132767,
10.436643041587223,
10.716533289695375,
0.5466515694267952,
3.318518267704871,
0.40678728382369284,
0.6290091030083763,
-0.4770983795300032,
-0.7455483390932742,
0.1630064768968011,
0.4889724943271746,
0.10596004146863507,
-1.4759536729524354,
0.957604974713893,
0.6004514101153323,
0.0725728265044025,
1.4994700617905987,
0.15508641282397265,
-0.15494303336158094,
-0.5663845144966579,
2.6663065398173496,
-0.27086336898318036,
-1.1828784814640632,
-0.295045685212881,
0.6236994649007899,
1.1769360493970402,
1.7159795997719172,
-0.3019600780315931,
0.651545330446025,
-2.0782382222089724,
0.3186344830290561,
-0.2815476257680432,
0.4067635126151388,
-0.8228051832090519,
-0.4154478828430832,
2.3723163321732827,
0.931436930203099,
0.9804374423714106,
0.058895326584719035,
0.3167253661454302,
0.3390503539155689,
-0.5024758650485814,
-0.2425872659725522,
-0.6703364135359525,
-0.147558399690948,
1.1644296968010859,
0.5475507579506065,
0.1736068872874216,
0.30547527216138975,
0.5300773267023443,
0.7722201656261689,
-0.6858549484151734,
-0.16073113081558088,
-1.197458252004488,
-0.507307082981415,
-0.16437738354723153,
0.2647407403580543,
-0.45930133674563567,
-0.22673992276266391,
1.8127715449849986,
-1.4334128748253496,
-0.759404859696163,
1.2431110601081345,

```

0.8953271645792334,
0.6963251389999259,
-0.23147129445607595,
0.3638861144885483,
2.2394622330281906,
1.543816783446588,
1.7535830084845025,
-0.3169637330795974,
0.2453194636587019,
2.0076391670843647,
0.286017506499548,
0.6600449842625801,
0.19782118041633945,
0.05463577684798935,
0.4755005320975566,
0.055207215781380096,
0.06969685609102444,
0.2645328829673777,
-0.01235845210357445,
1.01982876424508,
2.262492417600915,
1.3443784687440867,
0.10698144982242018,
0.428081928456963,
-1.5494921893757727,
1.0701294130786736,
0.39987519173641656,
0.06301738855484718,
-1.6711192593807125,
0.26920159401030147,
-0.08358711295948693,
0.4054786916979446,
-0.5013194648307361,
2.126550690771933,
0.10959959792145137,
0.4205360694942305,
-0.586878230803869,
-0.3381608637889511,
-0.5055401953468239,
-0.45995744817477163,
0.07999668782445096,
-0.4269018063306222,
0.9588276598807326,
1.042187521522261,
0.07821502449201799,
0.6777167152964713,
0.4177387822172794,
-0.18721284346239236,
-0.8064458441104513,
2.8712851588196218,
0.7141862154478698,
-0.9976723444987857,
-0.8529435737233043,
1.9602636639541733,
1.2634521849205176,
-0.7170870121313951,
0.31227949023655405,
-1.0451293221946831,
-0.6787852158143914,
1.8581833428208436]],
'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value': 0},
                    'noise_prob': [0.001, 0.001, 0, 0],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,

```

False,

```

0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':

'AUTOMATE_ising_noisefree_time_evo_ric4_unifrom_scaling_sample#6'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},

'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
                                'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
                                '4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
                                'X 5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
                                                         '+ 1 Z 0 + '
                                                         '1 X 1 X 2 '
                                                         '+ 1 Z 1 + '
                                                         '1 X 2 X 3 '
                                                         '+ 1 Z 2 + '
                                                         '1 X 3 X 4 '
                                                         '+ 1 Z 3 + '
                                                         '1 X 4 X 5 '
                                                         '+ 1 Z 4 + '
                                                         '1 X 5 X 6 '
                                                         '+ 1 Z 5 + '
                                                         '1 Z 6']],
            'output': { 'data_points': [ [4, 2, 0, 0, -6.4878948983695235],
                                         [12, 6, 0, 0, -5.297876077058435],
                                         [20, 10, 0, 0, -4.3310003849914125],
                                         [28, 14, 0, 0, -3.539166940153787]],
                      'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                           'identity_factors': [0, 0, 0, 0],
                                           'noise_level': [4, 2, 0, 0],
                                           'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

'noise_prob': [

0.001,
0.001,
0,
0],

'status': True,
'type':

'depolarizing'},

'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [1, 1, 0, 0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

'noise_prob': [

0.001,
0.001,
0,
0],

'status': True,
'type':

```

```

'depolarizing'},

'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,
0,
0],

'depolarizing'},

'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,
0,
0],

'depolarizing'},

'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
'identity_factors': [2, 2, 0, 0],
'noise_level': [20, 10, 0, 0],
'noise_profile': {

'noise_prob': [

'status': True,
'type':

'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
'identity_factors': [3, 3, 0, 0],
'noise_level': [28, 14, 0, 0],
'noise_profile': {

'noise_prob': [

'status': True,
'type':

'odd_wires': 4}},
'run_time_sec': 10.259121179580688}},
{ 'config': { 'ansatz': { 'gateset': 1,
    'layer': 30,
    'ugate': { 'coefficients': { 'bn': [ 0,
    0,
    0,
    0,
    0,
    0,
    0],
    'cn': [ 0.5,
    0.5,
    0.5,
    0.5,
    0.5,
    0.5],
    'r': 0},
    'time': {'max': 10.0, 'min': 0.0},
    'type': 'ising'}}},
'init_param': { 'value': [ 0.27654223573266234,
    0.27654223573266246,
    0.27654223573266246,
    0.4643738922966213,
    0.6582164862517121,
    0.6582164862517121,
    0.6582164862517121,
    1.3424523496852623,
    1.47635793598756,
    1.47635793598756,
    3.385550316280557,
    3.385550316280557,
    3.385550316280557,
    4.494062371253785,
    4.494062371253785,
    5.0204758683966135,
    5.023404188660082,
    7.05704094573064,
    7.087391629397132,
    7.533802135057517,
    8.245970909352765,
    8.390887842007308,
    8.390887842007308,
    8.703391612007538,

```

8.703391612007538,
9.485545381639698,
9.485545381639698,
9.849437794456351,
9.849437794456351,
10.799155842585746,
-0.12708089143278917,
1.4168148977025445,
-0.13519536748631492,
-0.32308714516775877,
0.06981336221044306,
-0.034321895006426455,
-0.18209443759399158,
-0.024858958594746107,
0.664488174834645,
0.6897317665852637,
-0.008774731192519907,
0.625642551337688,
0.6862410821653743,
-0.7670419540542446,
-0.48055390132433673,
1.573964677292243,
-1.0555745677813198,
1.3159721165631335,
-0.9138502497993347,
-0.4641566084568084,
-0.06966564000847691,
0.039012746519005555,
1.0827641697579717,
1.2190093772283843,
1.5391150382377632,
-0.6675581491933664,
0.09104662559232415,
0.5283818781579022,
0.04478118244404365,
1.5836075485651113,
0.251932560991488,
-0.3949560381811925,
0.8854639388207172,
-0.8611267164776638,
1.566928747241295,
-0.15445151319573347,
-0.01396050433941564,
0.8172620115356718,
-0.014658132241289147,
-0.6623010385304499,
0.6299284572753584,
-0.15026947869759505,
-1.2293112897827836,
-0.5768656731979434,
1.231557922062332,
-1.447872926116422,
1.4918722706131171,
0.31945486783385635,
0.43870804207253483,
1.3582636256308331,
-0.8478398442626237,
0.799518029891469,
0.5539369896048497,
0.13453197415145846,
0.7167498104862234,
-0.06589109804263343,
0.041038724923082276,
1.151975756926885,
-0.052026838729968425,
0.3952978000808587,
-1.1754748970457833,
-0.01965607042871015,
-0.2456324144680547,
-0.01136184217607491,
0.022503168314325715,
1.1777923723104653,
0.14554117253938412,
-1.2287946167385297,
-0.9683737383161766,

```

1.073329622962756,
-0.7143583802799215,
-0.3545002682598469,
-0.31906110571926777,
-0.9597094503848127,
0.8825903987559673,
-0.08273415521500459,
0.2895039435279609,
-0.3388594197258312,
0.2406459434933063,
0.3824128671502841,
-0.5390555829405563,
0.3757684495816124,
-0.5015719636333678,
0.3958257688275076,
-0.6428354095046468,
-0.9178624992467067,
0.29327306512581747,
-0.09193285724698198,
0.1770091195095364,
0.6803523512851801,
1.05525770275555,
-0.4646437845139076,
2.2560908642344026,
-0.7500939262299234,
1.92959092052175,
-0.36895777993274226,
-0.04538283416771957,
1.2529586558275392,
-0.57805717638909,
0.3080585810325818,
0.031910210614104206,
0.18519950010488512,
1.3888566879174211,
1.087260914864616,
0.15259355623633358,
-0.08810167996863522,
0.07467874103808375,
1.4444893382870363,
-1.446220680213735,
0.8215180856176293,
2.0151723649306295,
0.13454991965456886,
0.21923381008732426,
0.31112145477192793,
-0.8977876743954992,
0.10239876571985529,
-1.0132922278363503,
-0.3794863035801566,
0.8437803973608401,
1.111671691788496]],
'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value': 0},
                    'noise_prob': [0.001, 0.001, 0, 0],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [ 0.5,
                                         0.5,
                                         0.5,
                                         0.5,
                                         0.5,
                                         0.5],
                                'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,

```

False,

```

        'type': 'png'},
        'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric4_unifrom_scaling_sample#2'},
        'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                              [1, 1, 0, 0],
                                              [2, 2, 0, 0],
                                              [3, 3, 0, 0]]},
        'run': 'redundant',
        'state': 'dmatrix',
        'vqe': { 'iteration': 1,
                  'optimization': { 'algorithm': 'SLSQP',
                                    'constraint': True,
                                    'status': False}},
        'zne': { 'data_points': None,
                  'degree': 0,
                  'method': 'richardson',
                  'sampling': 'default'}},
        'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
                                          'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
                                          '4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
                                          'X 5 X 6 + 1 Z 5 + 1 Z 6',
                    'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
                                                                '+ 1 Z 0 + '
                                                                '1 X 1 X 2 '
                                                                '+ 1 Z 1 + '
                                                                '1 X 2 X 3 '
                                                                '+ 1 Z 2 + '
                                                                '1 X 3 X 4 '
                                                                '+ 1 Z 3 + '
                                                                '1 X 4 X 5 '
                                                                '+ 1 Z 4 + '
                                                                '1 X 5 X 6 '
                                                                '+ 1 Z 5 + '
                                                                '1 Z 6']},
        'output': { 'data_points': [ [4, 2, 0, 0, -6.514930884689278],
                                       [12, 6, 0, 0, -5.310706408751286],
                                       [20, 10, 0, 0, -4.328532484009807],
                                       [28, 14, 0, 0, -3.520772220894083]],
                    'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                          'identity_factors': [0, 0, 0, 0],
                                          'noise_level': [4, 2, 0, 0],
                                          'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
                                'noise_prob': [
0.001,
0.001,
0,
0],
                                'status': True,
                                'type':
'depolarizing'},
                                'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [1, 1, 0, 0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
                                'noise_prob': [
0.001,
0.001,
0,
0],
                                'status': True,
                                'type':
'depolarizing'},
                                'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [2, 2, 0, 0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
                                'noise_prob': [
0.001,
0.001,
0,
0],
                                'status': True,
                                'type':
'depolarizing'},
                                'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
  'identity_factors': [3, 3, 0, 0],
  'noise_level': [28, 14, 0, 0],
  'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
                                'noise_prob': [
0.001,
0.001,
0,
0],
                                'status': True,
                                'type':
'depolarizing'},
                                'odd_wires': 4}
                    ]}

```



```

0.001,
0.001,
0,
0],

'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,
0,
0],

'depolarizing'},

'noise_prob': [

'status': True,
'type':

'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
'identity_factors': [3, 3, 0, 0],
'noise_level': [28, 14, 0, 0],
'noise_profile': {

'run_time_sec': 9.931480884552002}},
{ 'config': { 'ansatz': { 'gateset': 1,
    'layer': 30,
    'ugate': { 'coefficients': { 'bn': [ 0,
    0,
    0,
    0,
    0,
    0],
    'cn': [ 0.5,
    0.5,
    0.5,
    0.5,
    0.5],
    'r': 0},
    'time': {'max': 10.0, 'min': 0.0},
    'type': 'ising'}}},
'init_param': { 'value': [ 0.5118290521669792,
    0.9034573591840084,
    1.0514460879099101,
    1.1283814725964603,
    1.1293980629167455,
    1.6874068541059999,
    1.8362001710030256,
    2.594529848453243,
    2.5949023258749775,
    4.384420392533986,
    4.804388533064279,
    4.8044974726668155,
    4.925049921883437,
    4.925049921883437,
    6.055903002618935,
    6.0650337145794,
    6.06510024924717,
    6.06510024924717,
    7.719643893437192,
    7.719643893437192,
    8.110116534451196,
    8.374448879129652,
    8.374448879129652,
    9.29045698473019,
    9.2929923765699,
    9.977025927736127,
    9.977025927736127,
    9.977438303479875,
    11.277250876863835,
    11.588901828478656,
    -0.5970483257864091,
    1.048395642841534,
    0.3563299433581316,

```

2.04182497417867,
-1.0659189653240388,
-1.5711224255679066,
-0.6854026923334804,
1.3002799990574343,
0.334979527312172,
-0.41771661057992626,
-0.5255960701407614,
0.9200835209290303,
0.5840294212204432,
0.4309612994747514,
0.010788324288593443,
0.8398650983618882,
0.35555521736822654,
0.33383672576306345,
-0.03972798241435421,
-0.5808926346218187,
-0.6217506829607715,
-0.022449055124049946,
-0.4671428723737863,
1.3840583406465645,
3.321823997583478,
1.9516123562000514,
2.109430068020724,
0.17138900458361278,
1.0734765554988877,
1.1591682463434823,
-1.5085197494005436,
0.3709514368460093,
-0.08354812013321816,
-0.6849414455501801,
-1.1807064172516142,
0.1889544406654655,
0.45125852432694835,
0.40946867023093825,
0.9591749175330944,
0.862406962512403,
-0.47565451021668054,
0.14791469386649075,
0.012323642821740255,
-0.3613715829994197,
-0.6520104899254531,
1.1500641632479522,
0.11480411675389499,
0.37929728325133344,
-0.12818017639377846,
1.5548403554359806,
-0.08803696907532751,
0.16058007713541472,
-0.2835300528775952,
1.4317946638041426,
0.04218231832431045,
0.6164913379423986,
-0.5546192578057587,
-0.0016176435191898148,
0.3455446109244303,
-0.7983090776945799,
0.06548494602750325,
0.09601011255154575,
-1.274882751616874,
-0.6639296860891427,
0.08468574728059233,
0.09178155492125434,
-1.9746588496307496,
1.363083624655819,
-0.11754172722968637,
-1.7454296260872237,
0.689437867548643,
0.09536601432398076,
2.1875690275532618,
1.9737367040865754,
0.19623670188051764,
1.3618525198389628,
-1.6551377680800423,
-1.2796187335858575,

```

-0.02979150349567955,
0.3655877448114123,
-1.339933806945279,
-1.6131280858420662,
1.1268021774842836,
0.7853899025443938,
-1.214954346638235,
1.881374469676006,
-2.2108375055617597,
0.38299461206746305,
-0.062378913993248956,
-1.0812916442950207,
0.19113842494911862,
0.43947065709017896,
1.4340329690981122,
-2.1914706359927933,
1.683520172253517,
-0.4151753501170658,
0.2652280013518245,
-0.09829123134965576,
-0.4033858255054015,
-0.09488290407421365,
1.1087820722198931,
1.3758250784849964,
0.3693308876448478,
-1.4774754586810708,
0.45790576106810227,
-1.318961759734687,
1.2898775251023542,
0.5291693857421421,
-0.7225860054677886,
-0.6024865845145562,
1.1258609582335735,
1.4545810101615688,
0.4988359655120556,
0.39968483637691726,
1.4778005154845164,
1.6358502222921723,
0.07081811420225437,
1.5336807499569445,
-0.7595143926860392,
0.07947795439670366]],
'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value': 0},
                    'noise_prob': [0.001, 0.001, 0, 0],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric4_unifrom_scaling_sample#10'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',

```

False,

```

'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': False}},
'zne': { 'data_points': None,
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
                                'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
                                '4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
                                'X 5 X 6 + 1 Z 5 + 1 Z 6',
        'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
                                                    '+ 1 Z 0 + '
                                                    '1 X 1 X 2 '
                                                    '+ 1 Z 1 + '
                                                    '1 X 2 X 3 '
                                                    '+ 1 Z 2 + '
                                                    '1 X 3 X 4 '
                                                    '+ 1 Z 3 + '
                                                    '1 X 4 X 5 '
                                                    '+ 1 Z 4 + '
                                                    '1 X 5 X 6 '
                                                    '+ 1 Z 5 + '
                                                    '1 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -6.4060332381961675],
                             [12, 6, 0, 0, -5.147683680770249],
                             [20, 10, 0, 0, -4.13060369641699],
                             [28, 14, 0, 0, -3.301960374777364]],
        'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                              'identity_factors': [0, 0, 0, 0],
                              'noise_level': [4, 2, 0, 0],
                              'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},

                                0.001,
                                0.001,
                                0,
                                0],

                                'status': True,
                                'type':

'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [1, 1, 0, 0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},

                                0.001,
                                0.001,
                                0,
                                0],

                                'status': True,
                                'type':

'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [2, 2, 0, 0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},

                                0.001,
                                0.001,
                                0,
                                0],

                                'status': True,
                                'type':

'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],

```


0.5934062959304571,
1.633059089666515,
0.31170790405459564,
0.3995146629536743,
-0.11155094016301184,
0.8138063699882787,
0.019467271717771087,
-0.38186317682975884,
-2.778803401027002,
0.7887082829537533,
-3.1348050840084074,
-1.5109224736536222,
-0.38589710406642097,
-1.105927053527023,
0.3451707237315739,
0.6290043291617069,
-1.834722402137916,
-1.0551446159380258,
-1.3581260193610523,
0.9666180881078981,
-0.05759676522292653,
1.1866392284178628,
-1.1308471740259582,
0.34327443183487805,
0.17662613760194443,
-0.043135596735193626,
-0.03667546733606037,
-0.4901895513664937,
-1.0399227541005371,
0.4880459765792483,
0.053089003166299456,
-0.6259966396796703,
0.03159579372412852,
-0.09798805529491862,
0.052872660107769434,
1.490194875693949,
0.22167423812356596,
-0.33497064714209324,
1.0144679614160885,
0.01608415027143934,
0.49167899744128424,
0.12491439548600353,
-0.1329609067962543,
0.8639852596635488,
-1.6607871811173882,
0.2801513494447602,
0.3908501333463579,
0.6044891435983025,
0.08637544621327581,
1.2270127042754213,
0.11860545946003472,
-0.13066468162372125,
-0.998222267933649,
-1.2643438207146234,
0.7835210648968657,
0.8370312354478452,
0.06527740704106412,
-1.0235950282739155,
-0.705070781232052,
0.037682760290170314,
1.4778132318704873,
1.7053977225841355,
-0.1291331271719845,
0.1648924115529835,
2.208642048449397,
1.1885953055185114,
-0.8138037079028133,
0.6715189891287302,
0.7819572861394978,
-0.4022064636590845,
-0.3927127310055937,
-0.4118784323717791,
1.3463239978578432,
-0.3323606385924751,
0.6664683657180488,

```

1.58852177854836,
-1.212203169826861,
-0.031436536468697034,
0.5094262403458202,
-0.0616688967239856,
-0.0666169691754888,
-0.7277012125404719,
-0.12684196683036533,
0.5325519345188151,
2.2778420441533447,
0.8735933622003313,
0.9907279467768721,
1.1448787224294625,
0.4291077507842401,
-0.21356701746617238,
1.7174041100732933,
0.1885112766205208,
0.3630143936992631,
0.6395640633025061,
-0.2937369198596852,
-0.18128670495631546,
0.9504587478277511,
-0.6241293783488098,
1.8733903679237267,
0.4875344831143031,
-0.9182747577406853,
0.850791268752179,
-0.46183623376516114,
-0.634145646812118,
-0.9165900492445601,
-0.8973746003077631,
-0.879107286657044,
0.15645801311276636]],
'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value': 0},
                    'noise_prob': [0.001, 0.001, 0, 0],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric4_unifrom_scaling_sample#8'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '

```

```

'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
'4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
'X 5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
                                             '+ 1 Z 0 + '
                                             '1 X 1 X 2 '
                                             '+ 1 Z 1 + '
                                             '1 X 2 X 3 '
                                             '+ 1 Z 2 + '
                                             '1 X 3 X 4 '
                                             '+ 1 Z 3 + '
                                             '1 X 4 X 5 '
                                             '+ 1 Z 4 + '
                                             '1 X 5 X 6 '
                                             '+ 1 Z 5 + '
                                             '1 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -6.672780190342767],
                              [12, 6, 0, 0, -5.522238922989663],
                              [20, 10, 0, 0, -4.5712287338679385],
                              [28, 14, 0, 0, -3.7801844878377153]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                    'identity_factors': [0, 0, 0, 0],
                    'noise_level': [4, 2, 0, 0],
                    'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},

'noise_prob': [
0.001,
0.001,
0,
0],

'status': True,
'type':
'depolarizing'},

'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [1, 1, 0, 0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},

'noise_prob': [
0.001,
0.001,
0,
0],

'status': True,
'type':
'depolarizing'},

'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [2, 2, 0, 0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},

'noise_prob': [
0.001,
0.001,
0,
0],

'status': True,
'type':
'depolarizing'},

'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
  'identity_factors': [3, 3, 0, 0],
  'noise_level': [28, 14, 0, 0],
  'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},

'noise_prob': [
0.001,
0.001,
0,
0],

```


0],

'status': True,
'type':

'depolarizing'},

```
        'odd_wires': 4]],  
        'run_time_sec': 9.87885570526123}},  
{ 'config': { 'ansatz': { 'gateset': 1,  
                          'layer': 30,  
                          'ugate': { 'coefficients': { 'bn': [ 0,  
                                                             0,  
                                                             0,  
                                                             0,  
                                                             0,  
                                                             0],  
                                     'cn': [ 0.5,  
                                              0.5,  
                                              0.5,  
                                              0.5,  
                                              0.5],  
                                     'r': 0},  
                          'time': {'max': 10.0, 'min': 0.0},  
                          'type': 'ising'}}},  
  'init_param': { 'value': [ 0.48023040681030116,  
                             0.7029544315470944,  
                             0.722671987246733,  
                             0.9307929858945541,  
                             0.9307929858945543,  
                             2.546598088688794,  
                             2.546598088688794,  
                             2.546598088688794,  
                             2.546598088688794,  
                             5.184334435330857,  
                             5.195111458618343,  
                             5.47436411399969,  
                             5.47436411399969,  
                             5.47436411399969,  
                             5.47436411399969,  
                             5.47436411399969,  
                             6.068336244524409,  
                             6.068336244524409,  
                             7.838605251840715,  
                             7.912250316107233,  
                             7.912250316107233,  
                             8.562517202814183,  
                             9.063385768597811,  
                             9.216218862905647,  
                             9.216218862905647,  
                             10.84047371877396,  
                             10.84047371877396,  
                             11.044358714867027,  
                             11.375406566932424,  
                             11.375406566932424,  
                             0.18886549713610382,  
                             0.6040291443391601,  
                             -0.9727809979878402,  
                             3.008209947509642,  
                             0.3239743149147096,  
                             -0.643987619924633,  
                             1.0739882269552894,  
                             0.4935103262912805,  
                             -1.1713673989643745,  
                             0.32339055167680386,  
                             -0.8254170399860232,  
                             3.387204293956005,  
                             0.5437302598099015,  
                             -0.6585961721148387,  
                             1.0903163929414719,  
                             -1.5261906384818182,  
                             1.102349928722959,  
                             -0.7373436929054379,  
                             0.1712635196093208,  
                             1.2086697581514199,  
                             -0.4977298668106944,
```

-0.8828019932256734,
-0.13051782426094366,
0.048744185631251574,
-1.0535030780910155,
0.5224555306604155,
-0.9502111367985572,
-0.2573420205933925,
1.5833039752620888,
1.3806480812363842,
0.3864833677006273,
-0.0006284730712113316,
0.6487740408134013,
1.697829570896293,
-0.20023867528920775,
-0.32438662287273723,
1.3674639807988553,
-1.5592347723565667,
-0.6294582959326964,
1.111157394071659,
1.4895912958038178,
-0.7203901402661329,
-0.027076706863032294,
0.009574786781832467,
1.1437695218822916,
0.11720143179507149,
1.0973549625545624,
-1.0756168303605034,
-1.5785125289756954,
-0.42999007233668873,
0.7592603373612552,
0.7357293999316703,
0.03766014972468404,
1.450035910180884,
-0.08978031589044846,
0.7176267391661779,
-2.3377280976646966,
0.3878354316208249,
-0.8070070584177742,
0.07084117333557359,
-0.008998168433091903,
0.6612796195456235,
0.6287929626789847,
0.8601333603228948,
-0.7786354708155643,
-0.144476679155352,
0.39492950957300044,
-0.20012968159779548,
0.01326107241550192,
-0.5649340596325207,
-0.020398755490040426,
0.6333499066300691,
0.400844220606507,
0.5954749034740784,
0.8700413681534388,
1.315868035063538,
0.9914334712755634,
0.10418309057983809,
1.1095991500751576,
-0.1712743534684691,
-0.2730581386141925,
0.1453142903679217,
-0.3813955795194703,
0.1280814941289577,
-0.5627986213807415,
-1.0576722394029623,
-1.98721560251358,
0.519862592148374,
1.03695228961559,
0.21101375663462807,
-1.750252459751414,
-1.1450993785481438,
-2.2517541986316916,
-1.5456964825715331,
0.6026638989139771,
0.47561448130168205,

```

0.08201672609434518,
0.47464262797792917,
-0.14585857152212733,
0.08636040211648545,
0.7388192170680543,
1.071366965639021,
2.5115163876343742,
-0.4286872636197954,
-0.2977998181526895,
0.46569070003689333,
0.2115019986130443,
-0.5123638542481742,
0.09024611293676764,
-0.3380729423866059,
1.6314769884628497,
1.3338027138849353,
-0.7442166070564612,
1.2186722356710948,
1.584709484399307,
-0.6554175978902271,
0.36887213174672207,
-1.365021965092438,
-0.4623126041921459,
0.4869420326570461]],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
'value': 0},
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric4_unifrom_scaling_sample#9'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0],
[3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
'4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
'X 5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
'+ 1 Z 0 + '
'1 X 1 X 2 '
'+ 1 Z 1 + '
'1 X 2 X 3 '
'+ 1 Z 2 + '

```

```

'1 X 3 X 4 '
'+ 1 Z 3 + '
'1 X 4 X 5 '
'+ 1 Z 4 + '
'1 X 5 X 6 '
'+ 1 Z 5 + '
'1 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -6.75124151968758],
[12, 6, 0, 0, -5.575778108730956],
[20, 10, 0, 0, -4.6095181057209125],
[28, 14, 0, 0, -3.8092488284193005]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 0, 0],
'noise_profile': {
'noise_prob': [
0.001,
0.001,
0,
0],
'status': True,
'type':
'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': {
'noise_prob': [
0.001,
0.001,
0,
0],
'status': True,
'type':
'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
'identity_factors': [2, 2, 0, 0],
'noise_level': [20, 10, 0, 0],
'noise_profile': {
'noise_prob': [
0.001,
0.001,
0,
0],
'status': True,
'type':
'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
'identity_factors': [3, 3, 0, 0],
'noise_level': [28, 14, 0, 0],
'noise_profile': {
'noise_prob': [
0.001,
0.001,
0,
0],
'status': True,
'type':
'depolarizing'},
'odd_wires': 4}],
'run_time_sec': 10.669064044952393}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,

```

```
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [ 0.3463233138801294,
0.34632331388012944,
0.3463233138801294,
0.3463233138801293,
1.2403205490491955,
1.38033446895038,
3.6939218351946623,
3.6939218351946623,
3.693921835194662,
3.693921835194662,
3.9412089126587,
3.9412089126587,
3.9412089126587007,
3.9412089126587007,
4.8310999601027795,
4.870844593548848,
4.900228671797898,
4.9002286717978984,
4.9002286717978984,
4.900228671797898,
5.70167657474683,
6.21188441483336,
6.21188441483336,
6.686419232103557,
6.723504371727371,
7.618130890836856,
7.618130890836856,
9.420364260644014,
9.420364260644014,
10.775132222422771,
-0.3525750090684208,
1.2748976558430758,
-0.16146998454247521,
1.3224575871686013,
0.0349265192498484,
0.36138711032415904,
-0.5043488106619078,
0.7634801084306476,
-0.3536814593617201,
0.9606809354246906,
-1.1143894720984433,
-0.45289061390722213,
-0.1605333784989486,
-1.594210268458609,
-0.09465166736510336,
0.11132251138190599,
1.004235666711761,
-0.23115493220428254,
0.9451761882751959,
-0.3707284170975133,
-0.17782948231634454,
-0.18433417990028922,
-0.06421460341189031,
-1.155489001827845,
-0.2074908306187579,
-0.036360325422879626,
0.5283181819327893,
0.3603171208434518,
0.6819349944270952,
-0.2733083993242672,
```

0.7075879191324372,
-0.042779623335368125,
0.7032435595714176,
-0.7363942983542189,
0.9583235184546098,
-0.5892472427173001,
-0.7139877518112785,
-0.12446625268371646,
-0.439646171766664,
0.7129808957109575,
0.3373075268097138,
0.3345550414369601,
-0.38068761370203624,
0.07864964370676188,
-0.3383553862900091,
0.6633419895127639,
-0.578720703894673,
0.486162302530746,
-0.06112668401522289,
0.34417654454775265,
-0.09182266750822224,
0.012215000731405658,
0.5658291917160285,
-0.16728488191683846,
0.3754616489195623,
-0.30497315054621493,
-0.7559846133933803,
0.9474959005813702,
-0.5904076686561035,
-0.2620011537974264,
0.1328515248519479,
-0.44066408562446224,
-0.7849425610737308,
0.44059474279745625,
-0.04646983134951628,
0.27897770882356554,
0.36838349877106535,
1.1610009997739648,
0.9058411506824863,
0.3119955912029684,
-0.5265683545543336,
1.0661822357546584,
0.8560044406571757,
0.5004409227892788,
0.5192086876555008,
-0.0019665558968806392,
-0.23949800878404184,
0.32563956832297103,
0.795323960796817,
1.5213873243487468,
0.8637563634588281,
0.07945031145196896,
1.7583315097859118,
0.5335864884320034,
-0.1366270407456156,
-0.9111102828185913,
-0.41650979281126865,
0.20926691692934202,
0.2776061293997892,
1.4257231544028384,
-0.10754544247245298,
1.0035199546983695,
1.1162390986419686,
1.7211522778091677,
-1.4332245925979028,
1.4417595659646931,
-2.106784537892007,
-0.6878309432904478,
-0.20360639354575213,
0.6373705186717347,
0.051333236332203484,
-0.35868991955772395,
-0.24365907607080922,
1.619736185871098,
0.5895182658245096,

False,

```

-0.6086170057348131,
0.39774539931448055,
1.409837948786624,
0.17438880158294706,
-0.0415548590378924,
0.1413623732119859,
-0.8192475896231625,
0.5823982141653503,
-0.8462302367353051,
-0.60454163385331,
1.8216629921858374,
1.2634551976945552,
-0.40200420270308274,
-1.5954759495917508,
-1.7732438129518737]],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
'value': 0},
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric4_unifrom_scaling_sample#3'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0],
[3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
'4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
'X 5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
'+ 1 Z 0 + '
'1 X 1 X 2 '
'+ 1 Z 1 + '
'1 X 2 X 3 '
'+ 1 Z 2 + '
'1 X 3 X 4 '
'+ 1 Z 3 + '
'1 X 4 X 5 '
'+ 1 Z 4 + '
'1 X 5 X 6 '
'+ 1 Z 5 + '
'1 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -6.1514222647937595],
[12, 6, 0, 0, -4.906882414400226],

```

[illegible]


```
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [ 0.23818326834520384,
0.242015507589488,
0.5292100600091615,
0.5598196968290768,
0.5598196968290768,
2.3451854381830244,
2.3451854381830244,
3.8453025861748578,
3.8453025861748578,
4.395137112369032,
4.408833101975425,
4.829585475162733,
4.867115107997544,
5.058958687371625,
5.058958687371625,
5.058958687371625,
5.581243202757385,
6.011433944644334,
6.011433944644334,
8.358003932454157,
8.358003932454157,
8.358003932454157,
8.371420896509267,
8.783744346022278,
8.783744346022278,
9.34189587380724,
9.34189587380724,
9.39608576624051,
9.726727464435275,
10.95102910414735,
0.14721560345391302,
1.0205767628001141,
0.40564926087548847,
-1.9238321605823567,
-0.04729505851188324,
0.482541600628026,
0.1152155677775581,
0.8524320040134185,
0.42380767990264745,
-0.6219665889019169,
-0.05759638705357143,
0.8161328242190486,
0.7425402472775383,
0.7535392998290764,
1.8314745216576223,
1.3334023135030668,
0.05552193989453376,
0.9691374289368472,
-0.07338372949421543,
-0.6690664310785921,
1.168533104165511,
-0.01962836639606316,
0.6576122461884292,
-0.037637936325233226,
-0.0009647519828692927,
0.4524839512418326,
-0.013115244288784337,
1.6270372625341938,
1.182284078721198,
0.9533845712158122,
1.0956662941001023,
-0.9647918240366662,
0.1897778948773928,
-0.04047520420354409,
0.5790278460715794,
0.631719469583885,
-0.3335418357294817,
1.0603186443819097,
-0.7449471003423949,
```

0.1630231005006795,
-0.07563007210422526,
-1.1009042311153838,
-0.34587951137957407,
-0.22408475871037167,
-0.6364303490666092,
-0.5245531829065874,
-0.7807390048523051,
-0.7767575864285188,
-0.9679768325508518,
-1.8691843526791756,
0.4783530568164554,
0.6133546874260105,
-0.20020986269221935,
-0.41532071148562183,
-0.8557118152516873,
-0.16985308585729458,
0.8428338023785568,
0.6750667051632169,
1.2795031602880573,
-0.41384224674171627,
0.3213362409234008,
1.2340309605381181,
1.0506369357289735,
-0.6415500741723684,
-0.9628453188140567,
1.935214357876566,
1.4159154638103797,
-2.0864443113151867,
-0.07426020994129089,
-1.0538315394110076,
0.2589745455351295,
-0.2955978932252484,
-0.030644528099012087,
-0.8602288471584197,
-0.09490606960216369,
0.4190622694749842,
-1.5667390101922392,
-0.7804920738383083,
0.1255849265457167,
0.5884338319364165,
-0.46028089859616994,
0.2488277677353687,
-0.7513271615815256,
-1.0561258734551267,
0.3697237180501531,
-0.7108432439071602,
0.5137418203820399,
0.946767478727941,
1.619957961978311,
1.2944079622561346,
0.21625880379379211,
0.007168703139473973,
-0.334764182643075,
-0.9086121674985949,
-0.8247776105092918,
-0.5734460393428641,
-0.4381826298821608,
-0.44306191141826123,
-0.23860269421390362,
-0.03343496826073402,
-0.008972211649351379,
1.2196121588758198,
1.5985131140815898,
0.060325450296581484,
-0.7312434635304047,
0.9413542695101463,
-0.6701271691421761,
-1.0680925542186157,
-0.5192506372311365,
1.5594784335685845,
0.17928043514789443,
0.2455246384495121,
-0.053106843359071614,
0.7488220810664211,

False,

```
1.3030207698426615,
-1.787387332504871,
-1.1184519607509558,
-0.2511275398302805,
1.445676213762956,
1.5991220174606833]],
'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value': 0},
                  'noise_prob': [0.001, 0.001, 0, 0],
                  'status': True,
                  'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric4_unifrom_scaling_sample#4'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
                                'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
                                '4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
                                'X 5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
                                                         '+ 1 Z 0 + '
                                                         '1 X 1 X 2 '
                                                         '+ 1 Z 1 + '
                                                         '1 X 2 X 3 '
                                                         '+ 1 Z 2 + '
                                                         '1 X 3 X 4 '
                                                         '+ 1 Z 3 + '
                                                         '1 X 4 X 5 '
                                                         '+ 1 Z 4 + '
                                                         '1 X 5 X 6 '
                                                         '+ 1 Z 5 + '
                                                         '1 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -6.630764461698827],
                             [12, 6, 0, 0, -5.461578440310042],
                             [20, 10, 0, 0, -4.498507346043123],
                             [28, 14, 0, 0, -3.699540169020614]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                    'identity_factors': [0, 0, 0, 0],
                    'noise_level': [4, 2, 0, 0],
                    'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
                          'noise_prob': [
```

```

0.001,
0.001,
0,
0],

'status': True,
'type':

'depolarizing'},

'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': {

'noise_prob': [

0.001,
0.001,
0,
0],

'status': True,
'type':

'depolarizing'},

'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
'identity_factors': [2, 2, 0, 0],
'noise_level': [20, 10, 0, 0],
'noise_profile': {

'noise_prob': [

0.001,
0.001,
0,
0],

'status': True,
'type':

'depolarizing'},

'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
'identity_factors': [3, 3, 0, 0],
'noise_level': [28, 14, 0, 0],
'noise_profile': {

'noise_prob': [

0.001,
0.001,
0,
0],

'status': True,
'type':

'depolarizing'},

'odd_wires': 4}],
'run_time_sec': 10.73041558265686}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [ 0.32980400918867037,
0.7818596814437996,
0.9680306497310072,
```

1.0586432954274991,
1.7663116348413028,
1.8956491295658664,
2.0436116769319477,
2.0436116769319477,
2.0436116769319477,
2.1846619555479685,
2.1846619555479685,
4.516997836642403,
4.516997836642403,
5.3166564351730035,
5.320130741320412,
5.399447086770547,
5.400046100307029,
5.400046100307029,
5.400046100307029,
5.400046100307029,
7.097750404951498,
7.618222769006738,
7.956666567531361,
7.956666567531361,
8.961109688308255,
8.961109688308255,
11.028500695271937,
11.5809418583784,
11.5809418583784,
11.5809418583784,
0.24114173221382337,
1.3754512844042854,
0.9720111739434927,
-0.29877116871955545,
-0.029737313498699618,
0.3185153459444675,
-0.31840745367437817,
0.6801602139502809,
-0.015407845998522466,
-1.114484654283524,
0.22997676136047449,
1.5201821520192487,
0.3217486942799661,
0.43678760134757944,
0.17618749246228715,
0.7010359103792874,
0.3446701255696699,
1.4700166379981474,
0.3471333265905846,
-0.968444017764766,
-1.1485269521507444,
0.3282504398407059,
-0.7898554413193151,
-0.22157985789400653,
0.4570931682042821,
-0.22158635023930806,
-0.36549836384277606,
-0.2366283880333898,
0.4114765778384287,
-0.7694643480660831,
-0.40507000819737765,
-0.6576468499882752,
0.2023600853039859,
-1.2917280200640486,
0.32702771491506877,
-0.07546137553843744,
-0.7713328193108817,
-0.8163562225239717,
-0.07772489804090182,
-0.32841576637855896,
-0.05728139293162625,
0.8564947637293437,
0.15384434981345546,
0.41944080031994563,
0.16884394015055976,
-0.010490661309829376,
-1.3198249528292274,
0.07128101806419536,

0.03515395872296211,
-0.09826644736389979,
-0.06882299183557238,
-0.7062443088386591,
-0.5280915567870916,
0.14860690208212682,
0.338235073517239,
0.012130023587481156,
-0.037483081002571056,
0.8881056862099682,
0.06312673968269537,
-0.44738898868242327,
-0.2463284936792685,
1.1708672891281489,
0.5297819270424092,
-0.5463182826640249,
-1.667403353493104,
1.3713545814180228,
0.9413366263710182,
-0.021951472934173096,
1.0345046397029698,
-0.8002739446090432,
1.076242700462428,
0.04994674089563993,
-1.446235170371471,
-0.3268246709561185,
0.6264234635598642,
-1.2465048991503196,
0.1169092097763034,
1.2453937722962254,
0.7220287063888942,
0.8114212311039525,
0.18333674920471413,
0.7345776889004225,
0.9807678368065545,
0.8703572591074042,
0.28720991162605214,
-0.9085817320602367,
1.1216621938695932,
1.1327477502874412,
-1.8285395156530388,
1.4231155243966302,
0.08156271721069372,
0.18027327076782884,
-0.10302960773112253,
-0.5560482514096969,
-0.09484541903718098,
0.5886430341587933,
-1.1674014904638685,
-0.03940950630369794,
0.2792515765657198,
0.08214125135098017,
0.04344924016389119,
-0.7787970529711157,
0.07887586110139429,
0.7532574519080738,
0.17107625040009872,
-0.4128415687827808,
1.3723943620028143,
1.4278581324344195,
0.5879977529754344,
0.28910810409117477,
-1.4412046993697643,
1.624844367134446,
0.1936683662945536,
0.3179841250458877,
0.5357240606734743,
1.5680682524509713,
0.04418323830986127,
0.2311137213204739,
2.8741283660232546,
0.9612408402592636]],

'noise_profile': { 'noise_on_init_param': { 'status':

False,

'value': 0},

```

        'noise_prob': [0.001, 0.001, 0, 0],
        'status': True,
        'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [ 0.5,
                                         0.5,
                                         0.5,
                                         0.5,
                                         0.5,
                                         0.5],
                                'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric4_unifrom_scaling_sample#5'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
                                'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
                                '4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
                                'X 5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
                                                         '+ 1 Z 0 + '
                                                         '1 X 1 X 2 '
                                                         '+ 1 Z 1 + '
                                                         '1 X 2 X 3 '
                                                         '+ 1 Z 2 + '
                                                         '1 X 3 X 4 '
                                                         '+ 1 Z 3 + '
                                                         '1 X 4 X 5 '
                                                         '+ 1 Z 4 + '
                                                         '1 X 5 X 6 '
                                                         '+ 1 Z 5 + '
                                                         '1 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -6.720969040841222],
                             [12, 6, 0, 0, -5.577003279994282],
                             [20, 10, 0, 0, -4.626982768162378],
                             [28, 14, 0, 0, -3.833723359776359]],
            'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                'identity_factors': [0, 0, 0, 0],
                                'noise_level': [4, 2, 0, 0],
                                'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [
0.001,
0.001,
0,
0],
'status': True,
'type':
'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],

```

```

'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,
0,
0],

'depolarizing'},

'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,
0,
0],

'depolarizing'},

'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,
0,
0],

'depolarizing'},

'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': {

'noise_prob': [

'status': True,
'type':

'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
'identity_factors': [2, 2, 0, 0],
'noise_level': [20, 10, 0, 0],
'noise_profile': {

'noise_prob': [

'status': True,
'type':

'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
'identity_factors': [3, 3, 0, 0],
'noise_level': [28, 14, 0, 0],
'noise_profile': {

'noise_prob': [

'status': True,
'type':

'odd_wires': 4}},
'run_time_sec': 9.778919696807861}},
{ 'config': { 'ansatz': { 'gateset': 1,
    'layer': 30,
    'ugate': { 'coefficients': { 'bn': [ 0,
        0,
        0,
        0,
        0,
        0,
        0],
        'cn': [ 0.5,
            0.5,
            0.5,
            0.5,
            0.5,
            0.5],
        'r': 0},
    'time': {'max': 10.0, 'min': 0.0},
    'type': 'ising'}}},
'init_param': { 'value': [ 0.4319322451408806,
    0.4441934852523559,
    0.5878315575931067,
    0.5878315575931067,
    0.5912427958750637,
    0.5947187380620651,
    0.5947187380620652,
    1.5628074958209335,
    1.568718083448373,
    2.013035808837258,
    2.013035808837258,
    2.013035808837258,

```


2.0277481086167364,
4.114573484042011,
4.1154343976538055,
5.358852938788934,
5.358852938788934,
5.549639110185835,
5.667644288942511,
5.6706149635077026,
7.063797010237488,
7.658675772413245,
8.123010743902626,
8.633070512181217,
8.7849794748515,
8.7849794748515,
10.749769122252356,
10.89899813672382,
11.413879450074099,
11.413879450074099,
-0.04151001581527148,
-2.751209241920508,
-0.5056705094410174,
-0.406548832581749,
0.8633340895103881,
0.17259893989954428,
-0.21235924407313156,
-0.5485334561985523,
0.2012951527025238,
-0.23858034563229433,
2.1364176057207223,
0.7817014982866842,
-0.1108899918027214,
0.6048531661014985,
1.809261328558185,
-1.1786415575821785,
-1.0827170099436865,
0.4109384498547832,
0.8090945308822337,
-0.9356585252219308,
0.28697429006677655,
-0.15332459566133277,
1.2998079093160566,
0.3870967504567973,
-0.2413465423953762,
1.001771196320206,
0.1138394509020578,
1.3220786271535379,
-1.1806565063083097,
2.2789659185869464,
0.11875741943281123,
-0.15631084464392242,
-2.14680662310284,
0.18035127914430246,
-0.260880073181003,
0.21031876899961,
0.7641152777599183,
1.203568210035907,
1.2767673297334832,
1.2585129786889195,
0.24752046596725819,
0.7203551788145919,
-0.17420447991641552,
-1.3486030918499698,
0.2145369555808404,
0.8896874485404291,
-1.6470089821954121,
-0.37928143051952584,
-0.29903232984244293,
0.6984471945997809,
0.14906235834494008,
0.054873949093549826,
1.3877615433857144,
0.04193983551007985,
1.2014028569521673,
-0.02338341588216991,
-0.05255422822816251,

-0.1905875093002492,
0.16425167437144692,
0.2542835721118732,
1.1822299841866795,
1.2032222480152739,
0.30142488752060526,
0.09125464470109583,
-0.45638140154025475,
0.050260497156630585,
-0.04000614529085093,
0.9096133936431033,
-0.03555378439095736,
0.4405325841695877,
-0.48129885373423525,
-0.3617221403442356,
-2.5216506086368518,
-0.5081934521569421,
0.041332600726076626,
0.7413771443671474,
0.2204029987501584,
-0.7517920276590782,
-0.21022249714251526,
1.5607830526955413,
0.16615947769508344,
-0.17023919527139658,
-0.4295619963603494,
-1.1838803442651602,
0.6319564057885301,
-0.3935277768061919,
0.005550337258741166,
-0.4226065324972882,
-0.3715846249960284,
-0.48395498614671884,
-0.4790957128536121,
1.4427293239187895,
0.5875094653073935,
1.404722056846091,
0.6973465258635152,
0.8623508528879442,
-0.5460563718875903,
0.8326491581513842,
-2.4015389244466907,
-0.04583861440999851,
-0.36265388938085896,
-0.5347868312224912,
0.49674658852292397,
-0.6488379607428911,
1.1568053403721041,
0.30662269781395407,
-2.013151608263867,
-0.17383275003878457,
0.5036662539248705,
1.9638910157847163,
3.118155075975211,
0.8117246453310248,
-1.4517859264576565,
-1.4946803709676548,
-2.7792982516572895,
1.1174424252005506,
2.333430423546859,
0.6903927508165344,
-0.2883899927957993,
-0.8542826896862586]],

'noise_profile': { 'noise_on_init_param': { 'status':
'value': 0},
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0,
1.0,
1.0,
1.0,
1.0,

False,

```

1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric4_unifrom_scaling_sample#7'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0],
[3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
'4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
'X 5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
'+ 1 Z 0 + '
'1 X 1 X 2 '
'+ 1 Z 1 + '
'1 X 2 X 3 '
'+ 1 Z 2 + '
'1 X 3 X 4 '
'+ 1 Z 3 + '
'1 X 4 X 5 '
'+ 1 Z 4 + '
'1 X 5 X 6 '
'+ 1 Z 5 + '
'1 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -6.662379321078875],
[12, 6, 0, 0, -5.394498644176583],
[20, 10, 0, 0, -4.374551061230002],
[28, 14, 0, 0, -3.546647371824164]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [
0.001,
0.001,
0,
0],
'status': True,
'type':
'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [
0.001,
0.001,
0,
0],

```

[illegible]

```

        'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                    'status': True,
                    'type': 'png'},
            'file_name_prefix': 'ising_noisefree_time_evo'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0]]},
'run': 'vqe',
'state': 'dmatrix',
'vqe': { 'iteration': 10,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': True}},
'zne': { 'data_points': None,
        'degree': 1,
        'method': 'richardson',
        'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + '
                                '1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + '
                                '1 X 4 X 5 + 1 Z 4 + 1 X 5 X 6 + 1 Z 5 + '
                                '1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 + 1 Z '
                                                         '0 + 1 X 1 X 2 + '
                                                         '1 Z 1 + 1 X 2 X '
                                                         '3 + 1 Z 2 + 1 X '
                                                         '3 X 4 + 1 Z 3 + '
                                                         '1 X 4 X 5 + 1 Z '
                                                         '4 + 1 X 5 X 6 + '
                                                         '1 Z 5 + 1 Z 6']},
'output': { 'exact_sol': -8.56677223350563,
            'initial_cost_history': [ 4.2809785500030175,
                                     2.611097744705238,
                                     3.419414059277099,
                                     1.5773527652373136,
                                     3.36522638437751,
                                     3.1574660994918307,
                                     4.256553926466165,
                                     4.577016272245339,
                                     4.50890344694736,
                                     4.056304045999429],
            'noise_details': { 'gates_num': [4, 1, 1, 0],
                              'identity_factors': [0, 0, 0, 0],
                              'noise_level': [4, 2, 0, 0],
                              'noise_profile': { 'noise_on_init_param': {
'noise_prob': [ 0.001,
                0.001,
                0,
                0],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
'optimized_minimum_cost': [ -6.833109640228162,
                           -6.51493088468928,
                           -6.151422264793805,
                           -6.630764461698797,
                           -6.72096904084131,
                           -6.487894898369486,
                           -6.662379321078827,
                           -6.6727801903427855,
                           -6.751241519687556,
                           -6.406033238196173],
'optimized_parameters': [ [ -4.1290471568429383e-17,
                           0.028387941050870547,
                           0.028387941050870516,
                           0.028387941050870544,
                           0.4764187545019148,
                           0.485154879376271,
                           0.7094706976105217,
                           2.998376901501724,
                           3.486264965503922,
                           3.487775573446841,
                           4.058897205630428,
                           5.020552086463093,

```

```

'status': False,
'value': 0},

```

5.02158492362958,
5.02158492362958,
5.0846878698400815,
5.0846878698400815,
6.794704827143902,
6.794704827143902,
6.794704827143902,
8.013356083289752,
8.21076169573117,
8.21076169573117,
8.633075344303478,
8.633075344303478,
8.930018414781724,
8.930018414781724,
8.930018414781724,
9.990329186240457,
9.990329186240457,
12.915390927357043,
0.8289215528981712,
0.6465354442587259,
-0.06922368073773534,
-2.0853926237797484,
0.25605409461249207,
-0.765586831155604,
0.9589723041272441,
0.27455658215646966,
-0.07189982532198122,
-0.3449139525561357,
-0.46800027418386986,
1.016480689947782,
-0.9706976058410632,
-0.9556868223128826,
0.9323574793538898,
-0.4376613900889883,
0.44138701419965265,
1.4418520162063444,
0.8420619129497768,
-0.025530845682239494,
-2.3076344184440156,
0.37458463155346333,
-0.47031338116346383,
0.07338001085955462,
-0.869249007732584,
1.6824786550253577,
1.719346297136726,
-0.9622509832303179,
0.5779076253870294,
0.4760527439731688,
-0.2451715046346434,
1.5906513054768954,
0.35918458386075697,
-0.44817294000256097,
-0.37416377940751877,
0.37048554789855764,
0.693137718643464,
2.288219219235325,
-0.26501198620577204,
0.3689053264668303,
0.10166782579927121,
-0.3356073800920907,
-0.6413954053758932,
0.28636656044125136,
-0.8690978212322298,
1.166030172665772,
-0.46365263204645707,
-0.8505396438729699,
2.4328989003329693,
-1.4554443951257197,
2.09134311034601,
1.7695145471360614,
1.1170184462314887,
0.20325573406500014,
0.34206355374635444,
0.016804831103183593,
-0.19882507083844791,

1.4011478969931443,
-0.0572616468882171,
-1.3773340344964602,
1.1031086912886459,
0.47433623341773395,
-0.037679845343088075,
0.7450954028512253,
1.3114871459373192,
-0.18651966717154955,
1.0750199871192778,
0.48905379729033566,
-0.6281827821606347,
1.322074656325554,
-1.4761484868509396,
0.2859918816514792,
-1.0033541595176405,
0.8264053813863532,
0.622010811103528,
1.1636747207646185,
-0.31883271288389914,
-0.0013981850979601159,
-0.09999405171802342,
1.3085280853238759,
-1.3727969089464465,
-0.08341159264840771,
1.6249369276484606,
-0.28082331570516694,
1.863294400254821,
2.2932652495309953,
1.4849536675442596,
0.6811668802751195,
1.5380110980874968,
-0.6615225065895972,
0.4000828281024264,
0.33932121588009856,
0.48855764825038867,
2.5371111232777053,
-0.7260430040504564,
0.09005940379987407,
-0.8762980817992282,
-0.23800314057977784,
0.9728027685352436,
-0.20857834368009137,
0.6148333457084475,
-1.5171898336040224,
0.8658542069129427,
-1.6478287868490311,
0.5770049367106069,
0.134497700004433,
-0.11612046371881249,
1.5826352840237945,
1.6045820840261675,
-0.08805669980595342,
-1.507745125921201,
-0.11276651576435986,
0.07206005105970639,
-1.7788122679933258,
-0.002988765738482436,
0.7255804452314191,
2.7661804504965604,
-3.1943627011957507,
-0.07088434770936546,
0.10949711075632612],
[0.27654223573266234,
0.27654223573266246,
0.27654223573266246,
0.4643738922966213,
0.6582164862517121,
0.6582164862517121,
0.6582164862517121,
1.3424523496852623,
1.47635793598756,
1.47635793598756,
3.385550316280557,
3.385550316280557,

3.385550316280557,
4.494062371253785,
4.494062371253785,
5.0204758683966135,
5.023404188660082,
7.05704094573064,
7.087391629397132,
7.533802135057517,
8.245970909352765,
8.390887842007308,
8.390887842007308,
8.703391612007538,
8.703391612007538,
9.485545381639698,
9.485545381639698,
9.849437794456351,
9.849437794456351,
10.799155842585746,
-0.12708089143278917,
1.4168148977025445,
-0.13519536748631492,
-0.32308714516775877,
0.06981336221044306,
-0.034321895006426455,
-0.18209443759399158,
-0.024858958594746107,
0.664488174834645,
0.6897317665852637,
-0.008774731192519907,
0.625642551337688,
0.6862410821653743,
-0.7670419540542446,
-0.48055390132433673,
1.573964677292243,
-1.0555745677813198,
1.3159721165631335,
-0.9138502497993347,
-0.4641566084568084,
-0.06966564000847691,
0.039012746519005555,
1.0827641697579717,
1.2190093772283843,
1.5391150382377632,
-0.6675581491933664,
0.09104662559232415,
0.5283818781579022,
0.04478118244404365,
1.5836075485651113,
0.251932560991488,
-0.3949560381811925,
0.8854639388207172,
-0.8611267164776638,
1.566928747241295,
-0.15445151319573347,
-0.01396050433941564,
0.8172620115356718,
-0.014658132241289147,
-0.6623010385304499,
0.6299284572753584,
-0.15026947869759505,
-1.2293112897827836,
-0.5768656731979434,
1.231557922062332,
-1.447872926116422,
1.4918722706131171,
0.31945486783385635,
0.43870804207253483,
1.3582636256308331,
-0.8478398442626237,
0.799518029891469,
0.5539369896048497,
0.13453197415145846,
0.7167498104862234,
-0.06589109804263343,
0.041038724923082276,

1.151975756926885,
-0.052026838729968425,
0.3952978000808587,
-1.1754748970457833,
-0.01965607042871015,
-0.2456324144680547,
-0.01136184217607491,
0.022503168314325715,
1.1777923723104653,
0.14554117253938412,
-1.2287946167385297,
-0.9683737383161766,
1.073329622962756,
-0.7143583802799215,
-0.3545002682598469,
-0.31906110571926777,
-0.9597094503848127,
0.8825903987559673,
-0.08273415521500459,
0.2895039435279609,
-0.3388594197258312,
0.2406459434933063,
0.3824128671502841,
-0.5390555829405563,
0.3757684495816124,
-0.5015719636333678,
0.3958257688275076,
-0.6428354095046468,
-0.9178624992467067,
0.29327306512581747,
-0.09193285724698198,
0.1770091195095364,
0.6803523512851801,
1.05525770275555,
-0.4646437845139076,
2.2560908642344026,
-0.7500939262299234,
1.92959092052175,
-0.36895777993274226,
-0.04538283416771957,
1.2529586558275392,
-0.57805717638909,
0.3080585810325818,
0.031910210614104206,
0.18519950010488512,
1.3888566879174211,
1.087260914864616,
0.15259355623633358,
-0.08810167996863522,
0.07467874103808375,
1.4444893382870363,
-1.446220680213735,
0.8215180856176293,
2.0151723649306295,
0.13454991965456886,
0.21923381008732426,
0.31112145477192793,
-0.8977876743954992,
0.10239876571985529,
-1.0132922278363503,
-0.3794863035801566,
0.8437803973608401,
1.111671691788496],
[0.3463233138801294,
0.34632331388012944,
0.3463233138801294,
0.3463233138801293,
1.2403205490491955,
1.38033446895038,
3.6939218351946623,
3.6939218351946623,
3.693921835194662,
3.693921835194662,
3.9412089126587,
3.9412089126587,

3.9412089126587007,
3.9412089126587007,
4.8310999601027795,
4.870844593548848,
4.900228671797898,
4.9002286717978984,
4.9002286717978984,
4.900228671797898,
5.70167657474683,
6.21188441483336,
6.21188441483336,
6.686419232103557,
6.723504371727371,
7.618130890836856,
7.618130890836856,
9.420364260644014,
9.420364260644014,
10.775132222422771,
-0.3525750090684208,
1.2748976558430758,
-0.16146998454247521,
1.3224575871686013,
0.0349265192498484,
0.36138711032415904,
-0.5043488106619078,
0.7634801084306476,
-0.3536814593617201,
0.9606809354246906,
-1.1143894720984433,
-0.45289061390722213,
-0.1605333784989486,
-1.594210268458609,
-0.09465166736510336,
0.11132251138190599,
1.004235666711761,
-0.23115493220428254,
0.9451761882751959,
-0.3707284170975133,
-0.17782948231634454,
-0.18433417990028922,
-0.06421460341189031,
-1.155489001827845,
-0.2074908306187579,
-0.036360325422879626,
0.5283181819327893,
0.3603171208434518,
0.6819349944270952,
-0.2733083993242672,
0.7075879191324372,
-0.042779623335368125,
0.7032435595714176,
-0.7363942983542189,
0.9583235184546098,
-0.5892472427173001,
-0.7139877518112785,
-0.12446625268371646,
-0.439646171766664,
0.7129808957109575,
0.3373075268097138,
0.3345550414369601,
-0.38068761370203624,
0.07864964370676188,
-0.3383553862900091,
0.6633419895127639,
-0.578720703894673,
0.486162302530746,
-0.06112668401522289,
0.34417654454775265,
-0.09182266750822224,
0.012215000731405658,
0.5658291917160285,
-0.16728488191683846,
0.3754616489195623,
-0.30497315054621493,
-0.7559846133933803,

0.9474959005813702,
-0.5904076686561035,
-0.2620011537974264,
0.1328515248519479,
-0.44066408562446224,
-0.7849425610737308,
0.44059474279745625,
-0.04646983134951628,
0.27897770882356554,
0.36838349877106535,
1.1610009997739648,
0.9058411506824863,
0.3119955912029684,
-0.5265683545543336,
1.0661822357546584,
0.8560044406571757,
0.5004409227892788,
0.5192086876555008,
-0.0019665558968806392,
-0.23949800878404184,
0.32563956832297103,
0.795323960796817,
1.5213873243487468,
0.8637563634588281,
0.07945031145196896,
1.7583315097859118,
0.5335864884320034,
-0.1366270407456156,
-0.9111102828185913,
-0.41650979281126865,
0.20926691692934202,
0.2776061293997892,
1.4257231544028384,
-0.10754544247245298,
1.0035199546983695,
1.1162390986419686,
1.7211522778091677,
-1.4332245925979028,
1.4417595659646931,
-2.106784537892007,
-0.6878309432904478,
-0.20360639354575213,
0.6373705186717347,
0.051333236332203484,
-0.35868991955772395,
-0.24365907607080922,
1.619736185871098,
0.5895182658245096,
-0.6086170057348131,
0.39774539931448055,
1.409837948786624,
0.17438880158294706,
-0.0415548590378924,
0.1413623732119859,
-0.8192475896231625,
0.5823982141653503,
-0.8462302367353051,
-0.60454163385331,
1.8216629921858374,
1.2634551976945552,
-0.40200420270308274,
-1.5954759495917508,
-1.7732438129518737],
[0.23818326834520384,
0.242015507589488,
0.5292100600091615,
0.5598196968290768,
0.5598196968290768,
2.3451854381830244,
2.3451854381830244,
3.8453025861748578,
3.8453025861748578,
4.395137112369032,
4.408833101975425,
4.829585475162733,

4.867115107997544,
5.058958687371625,
5.058958687371625,
5.058958687371625,
5.581243202757385,
6.011433944644334,
6.011433944644334,
8.358003932454157,
8.358003932454157,
8.358003932454157,
8.371420896509267,
8.783744346022278,
8.783744346022278,
9.34189587380724,
9.34189587380724,
9.39608576624051,
9.726727464435275,
10.95102910414735,
0.14721560345391302,
1.0205767628001141,
0.40564926087548847,
-1.9238321605823567,
-0.04729505851188324,
0.482541600628026,
0.1152155677775581,
0.8524320040134185,
0.42380767990264745,
-0.6219665889019169,
-0.05759638705357143,
0.8161328242190486,
0.7425402472775383,
0.7535392998290764,
1.8314745216576223,
1.3334023135030668,
0.05552193989453376,
0.9691374289368472,
-0.07338372949421543,
-0.6690664310785921,
1.168533104165511,
-0.01962836639606316,
0.6576122461884292,
-0.037637936325233226,
-0.0009647519828692927,
0.4524839512418326,
-0.013115244288784337,
1.6270372625341938,
1.182284078721198,
0.9533845712158122,
1.0956662941001023,
-0.9647918240366662,
0.1897778948773928,
-0.04047520420354409,
0.5790278460715794,
0.631719469583885,
-0.3335418357294817,
1.0603186443819097,
-0.7449471003423949,
0.1630231005006795,
-0.07563007210422526,
-1.1009042311153838,
-0.34587951137957407,
-0.22408475871037167,
-0.6364303490666092,
-0.5245531829065874,
-0.7807390048523051,
-0.7767575864285188,
-0.9679768325508518,
-1.8691843526791756,
0.4783530568164554,
0.6133546874260105,
-0.20020986269221935,
-0.41532071148562183,
-0.8557118152516873,
-0.16985308585729458,
0.8428338023785568,

0.6750667051632169,
1.2795031602880573,
-0.41384224674171627,
0.3213362409234008,
1.2340309605381181,
1.0506369357289735,
-0.6415500741723684,
-0.9628453188140567,
1.935214357876566,
1.4159154638103797,
-2.0864443113151867,
-0.07426020994129089,
-1.0538315394110076,
0.2589745455351295,
-0.2955978932252484,
-0.030644528099012087,
-0.8602288471584197,
-0.09490606960216369,
0.4190622694749842,
-1.5667390101922392,
-0.7804920738383083,
0.1255849265457167,
0.5884338319364165,
-0.46028089859616994,
0.2488277677353687,
-0.7513271615815256,
-1.0561258734551267,
0.3697237180501531,
-0.7108432439071602,
0.5137418203820399,
0.946767478727941,
1.619957961978311,
1.2944079622561346,
0.21625880379379211,
0.007168703139473973,
-0.334764182643075,
-0.9086121674985949,
-0.8247776105092918,
-0.5734460393428641,
-0.4381826298821608,
-0.44306191141826123,
-0.23860269421390362,
-0.03343496826073402,
-0.008972211649351379,
1.2196121588758198,
1.5985131140815898,
0.060325450296581484,
-0.7312434635304047,
0.9413542695101463,
-0.6701271691421761,
-1.0680925542186157,
-0.5192506372311365,
1.5594784335685845,
0.17928043514789443,
0.2455246384495121,
-0.053106843359071614,
0.7488220810664211,
1.3030207698426615,
-1.787387332504871,
-1.1184519607509558,
-0.2511275398302805,
1.445676213762956,
1.5991220174606833],
[0.32980400918867037,
0.7818596814437996,
0.9680306497310072,
1.0586432954274991,
1.7663116348413028,
1.8956491295658664,
2.0436116769319477,
2.0436116769319477,
2.0436116769319477,
2.1846619555479685,
2.1846619555479685,
4.516997836642403,

4.516997836642403,
5.3166564351730035,
5.320130741320412,
5.399447086770547,
5.400046100307029,
5.400046100307029,
5.400046100307029,
5.400046100307029,
7.097750404951498,
7.618222769006738,
7.956666567531361,
7.956666567531361,
8.961109688308255,
8.961109688308255,
11.028500695271937,
11.5809418583784,
11.5809418583784,
11.5809418583784,
0.24114173221382337,
1.3754512844042854,
0.9720111739434927,
-0.29877116871955545,
-0.029737313498699618,
0.3185153459444675,
-0.31840745367437817,
0.6801602139502809,
-0.015407845998522466,
-1.114484654283524,
0.22997676136047449,
1.5201821520192487,
0.3217486942799661,
0.43678760134757944,
0.17618749246228715,
0.7010359103792874,
0.3446701255696699,
1.4700166379981474,
0.3471333265905846,
-0.968444017764766,
-1.1485269521507444,
0.3282504398407059,
-0.7898554413193151,
-0.22157985789400653,
0.4570931682042821,
-0.22158635023930806,
-0.36549836384277606,
-0.2366283880333898,
0.4114765778384287,
-0.7694643480660831,
-0.40507000819737765,
-0.6576468499882752,
0.2023600853039859,
-1.2917280200640486,
0.32702771491506877,
-0.07546137553843744,
-0.7713328193108817,
-0.8163562225239717,
-0.07772489804090182,
-0.32841576637855896,
-0.05728139293162625,
0.8564947637293437,
0.15384434981345546,
0.41944080031994563,
0.16884394015055976,
-0.010490661309829376,
-1.3198249528292274,
0.07128101806419536,
0.03515395872296211,
-0.09826644736389979,
-0.06882299183557238,
-0.7062443088386591,
-0.5280915567870916,
0.14860690208212682,
0.338235073517239,
0.012130023587481156,
-0.037483081002571056,

0.8881056862099682,
0.06312673968269537,
-0.44738898868242327,
-0.2463284936792685,
1.1708672891281489,
0.5297819270424092,
-0.5463182826640249,
-1.667403353493104,
1.3713545814180228,
0.9413366263710182,
-0.021951472934173096,
1.0345046397029698,
-0.8002739446090432,
1.076242700462428,
0.04994674089563993,
-1.446235170371471,
-0.3268246709561185,
0.6264234635598642,
-1.2465048991503196,
0.1169092097763034,
1.2453937722962254,
0.7220287063888942,
0.8114212311039525,
0.18333674920471413,
0.7345776889004225,
0.9807678368065545,
0.8703572591074042,
0.28720991162605214,
-0.9085817320602367,
1.1216621938695932,
1.1327477502874412,
-1.8285395156530388,
1.4231155243966302,
0.08156271721069372,
0.18027327076782884,
-0.10302960773112253,
-0.5560482514096969,
-0.09484541903718098,
0.5886430341587933,
-1.1674014904638685,
-0.03940950630369794,
0.2792515765657198,
0.08214125135098017,
0.04344924016389119,
-0.7787970529711157,
0.07887586110139429,
0.7532574519080738,
0.17107625040009872,
-0.4128415687827808,
1.3723943620028143,
1.4278581324344195,
0.5879977529754344,
0.28910810409117477,
-1.4412046993697643,
1.624844367134446,
0.19366683662945536,
0.3179841250458877,
0.5357240606734743,
1.5680682524509713,
0.04418323830986127,
0.2311137213204739,
2.8741283660232546,
0.9612408402592636],
[0.46756606893882113,
0.46983859896604685,
0.5133985927484581,
0.5133985927484581,
0.5133985927484581,
1.4016794842576101,
1.6914116785996751,
1.691411678599675,
1.6941756924801525,
1.6941756924801528,
3.749864451084885,
3.749864451084885,

3.7498644510848855,
3.7498644510848855,
4.969636107010246,
4.969636107010246,
4.969636107010246,
4.969636107010246,
4.969636107010246,
5.983107385388112,
5.983107385388112,
8.378949246097575,
8.378949246097575,
8.635680633753974,
8.635680633753974,
9.108807770997622,
9.108807770997622,
9.108807770997622,
9.960036411132767,
10.436643041587223,
10.716533289695375,
0.5466515694267952,
3.318518267704871,
0.40678728382369284,
0.6290091030083763,
-0.4770983795300032,
-0.7455483390932742,
0.1630064768968011,
0.4889724943271746,
0.10596004146863507,
-1.4759536729524354,
0.957604974713893,
0.6004514101153323,
0.0725728265044025,
1.4994700617905987,
0.15508641282397265,
-0.15494303336158094,
-0.5663845144966579,
2.6663065398173496,
-0.27086336898318036,
-1.1828784814640632,
-0.295045685212881,
0.6236994649007899,
1.1769360493970402,
1.7159795997719172,
-0.3019600780315931,
0.651545330446025,
-2.0782382222089724,
0.3186344830290561,
-0.2815476257680432,
0.4067635126151388,
-0.8228051832090519,
-0.4154478828430832,
2.3723163321732827,
0.931436930203099,
0.9804374423714106,
0.058895326584719035,
0.3167253661454302,
0.3390503539155689,
-0.5024758650485814,
-0.2425872659725522,
-0.6703364135359525,
-0.147558399690948,
1.1644296968010859,
0.5475507579506065,
0.1736068872874216,
0.30547527216138975,
0.5300773267023443,
0.7722201656261689,
-0.6858549484151734,
-0.16073113081558088,
-1.197458252004488,
-0.507307082981415,
-0.16437738354723153,
0.2647407403580543,
-0.45930133674563567,
-0.22673992276266391,
1.8127715449849986,

-1.4334128748253496,
-0.759404859696163,
1.2431110601081345,
0.8953271645792334,
0.6963251389999259,
-0.23147129445607595,
0.3638861144885483,
2.2394622330281906,
1.543816783446588,
1.7535830084845025,
-0.3169637330795974,
0.2453194636587019,
2.0076391670843647,
0.286017506499548,
0.6600449842625801,
0.19782118041633945,
0.05463577684798935,
0.4755005320975566,
0.055207215781380096,
0.06969685609102444,
0.2645328829673777,
-0.01235845210357445,
1.01982876424508,
2.262492417600915,
1.3443784687440867,
0.10698144982242018,
0.428081928456963,
-1.5494921893757727,
1.0701294130786736,
0.39987519173641656,
0.06301738855484718,
-1.6711192593807125,
0.26920159401030147,
-0.08358711295948693,
0.4054786916979446,
-0.5013194648307361,
2.126550690771933,
0.10959959792145137,
0.4205360694942305,
-0.586878230803869,
-0.3381608637889511,
-0.5055401953468239,
-0.45995744817477163,
0.07999668782445096,
-0.4269018063306222,
0.9588276598807326,
1.042187521522261,
0.07821502449201799,
0.6777167152964713,
0.4177387822172794,
-0.18721284346239236,
-0.8064458441104513,
2.8712851588196218,
0.7141862154478698,
-0.9976723444987857,
-0.8529435737233043,
1.9602636639541733,
1.2634521849205176,
-0.7170870121313951,
0.31227949023655405,
-1.0451293221946831,
-0.6787852158143914,
1.8581833428208436],
[0.4319322451408806,
0.4441934852523559,
0.5878315575931067,
0.5878315575931067,
0.5912427958750637,
0.5947187380620651,
0.5947187380620652,
1.5628074958209335,
1.568718083448373,
2.013035808837258,
2.013035808837258,
2.013035808837258,

2.0277481086167364,
4.114573484042011,
4.1154343976538055,
5.358852938788934,
5.358852938788934,
5.549639110185835,
5.667644288942511,
5.6706149635077026,
7.063797010237488,
7.658675772413245,
8.123010743902626,
8.633070512181217,
8.7849794748515,
8.7849794748515,
10.749769122252356,
10.89899813672382,
11.413879450074099,
11.413879450074099,
-0.04151001581527148,
-2.751209241920508,
-0.5056705094410174,
-0.406548832581749,
0.8633340895103881,
0.17259893989954428,
-0.21235924407313156,
-0.5485334561985523,
0.2012951527025238,
-0.23858034563229433,
2.1364176057207223,
0.7817014982866842,
-0.1108899918027214,
0.6048531661014985,
1.809261328558185,
-1.1786415575821785,
-1.0827170099436865,
0.4109384498547832,
0.8090945308822337,
-0.9356585252219308,
0.28697429006677655,
-0.15332459566133277,
1.2998079093160566,
0.3870967504567973,
-0.2413465423953762,
1.001771196320206,
0.1138394509020578,
1.3220786271535379,
-1.1806565063083097,
2.2789659185869464,
0.11875741943281123,
-0.15631084464392242,
-2.14680662310284,
0.18035127914430246,
-0.260880073181003,
0.21031876899961,
0.7641152777599183,
1.203568210035907,
1.2767673297334832,
1.2585129786889195,
0.24752046596725819,
0.7203551788145919,
-0.17420447991641552,
-1.3486030918499698,
0.2145369555808404,
0.8896874485404291,
-1.6470089821954121,
-0.37928143051952584,
-0.29903232984244293,
0.6984471945997809,
0.14906235834494008,
0.054873949093549826,
1.3877615433857144,
0.04193983551007985,
1.2014028569521673,
-0.02338341588216991,
-0.05255422822816251,

-0.1905875093002492,
0.16425167437144692,
0.2542835721118732,
1.1822299841866795,
1.2032222480152739,
0.30142488752060526,
0.09125464470109583,
-0.45638140154025475,
0.050260497156630585,
-0.04000614529085093,
0.9096133936431033,
-0.03555378439095736,
0.4405325841695877,
-0.48129885373423525,
-0.3617221403442356,
-2.5216506086368518,
-0.5081934521569421,
0.041332600726076626,
0.7413771443671474,
0.2204029987501584,
-0.7517920276590782,
-0.21022249714251526,
1.5607830526955413,
0.16615947769508344,
-0.17023919527139658,
-0.4295619963603494,
-1.1838803442651602,
0.6319564057885301,
-0.3935277768061919,
0.005550337258741166,
-0.4226065324972882,
-0.3715846249960284,
-0.48395498614671884,
-0.4790957128536121,
1.4427293239187895,
0.5875094653073935,
1.404722056846091,
0.6973465258635152,
0.8623508528879442,
-0.5460563718875903,
0.8326491581513842,
-2.4015389244466907,
-0.04583861440999851,
-0.36265388938085896,
-0.5347868312224912,
0.49674658852292397,
-0.6488379607428911,
1.1568053403721041,
0.30662269781395407,
-2.013151608263867,
-0.17383275003878457,
0.5036662539248705,
1.9638910157847163,
3.118155075975211,
0.8117246453310248,
-1.4517859264576565,
-1.4946803709676548,
-2.7792982516572895,
1.1174424252005506,
2.333430423546859,
0.6903927508165344,
-0.2883899927957993,
-0.8542826896862586],
[0.4349098035102067,
0.5281397824211302,
0.6402833431740305,
0.701719039924861,
0.8953768954552646,
1.4118719346144908,
2.230113621742168,
2.6548053250922607,
3.10433397641996,
3.1043339764199596,
4.146783795497047,
4.146783795497047,

4.149760661232294,
4.149760661232294,
4.223539194779984,
4.223539194779984,
6.7727886487081435,
6.7727886487081435,
7.243551717080481,
7.282491880183138,
7.282491880183137,
7.282491880183138,
8.806989173945695,
8.80698917394569,
9.706285648296737,
9.809777023075187,
9.837118725166619,
10.701586770517197,
11.185931122995514,
11.185931122995514,
0.39919688039300266,
1.4533249104921768,
0.7852252158845567,
-0.4957764525834907,
-0.3354719566017423,
0.16097458435951068,
-0.21971891747261646,
1.5370438656668401,
0.26531517225969137,
-0.6386119707402834,
0.37748437574064975,
1.2333814560270282,
0.5934062959304571,
1.633059089666515,
0.31170790405459564,
0.3995146629536743,
-0.11155094016301184,
0.8138063699882787,
0.019467271717771087,
-0.38186317682975884,
-2.778803401027002,
0.7887082829537533,
-3.1348050840084074,
-1.5109224736536222,
-0.38589710406642097,
-1.105927053527023,
0.3451707237315739,
0.6290043291617069,
-1.834722402137916,
-1.0551446159380258,
-1.3581260193610523,
0.9666180881078981,
-0.05759676522292653,
1.1866392284178628,
-1.1308471740259582,
0.34327443183487805,
0.17662613760194443,
-0.043135596735193626,
-0.03667546733606037,
-0.4901895513664937,
-1.0399227541005371,
0.4880459765792483,
0.053089003166299456,
-0.6259966396796703,
0.03159579372412852,
-0.09798805529491862,
0.052872660107769434,
1.490194875693949,
0.22167423812356596,
-0.33497064714209324,
1.0144679614160885,
0.01608415027143934,
0.49167899744128424,
0.12491439548600353,
-0.1329609067962543,
0.8639852596635488,
-1.6607871811173882,

0.2801513494447602,
0.3908501333463579,
0.6044891435983025,
0.08637544621327581,
1.2270127042754213,
0.11860545946003472,
-0.13066468162372125,
-0.998222267933649,
-1.2643438207146234,
0.7835210648968657,
0.8370312354478452,
0.06527740704106412,
-1.0235950282739155,
-0.705070781232052,
0.037682760290170314,
1.4778132318704873,
1.7053977225841355,
-0.1291331271719845,
0.1648924115529835,
2.208642048449397,
1.1885953055185114,
-0.8138037079028133,
0.6715189891287302,
0.7819572861394978,
-0.4022064636590845,
-0.3927127310055937,
-0.4118784323717791,
1.3463239978578432,
-0.3323606385924751,
0.6664683657180488,
1.58852177854836,
-1.212203169826861,
-0.031436536468697034,
0.5094262403458202,
-0.0616688967239856,
-0.0666169691754888,
-0.7277012125404719,
-0.12684196683036533,
0.5325519345188151,
2.2778420441533447,
0.8735933622003313,
0.9907279467768721,
1.1448787224294625,
0.4291077507842401,
-0.21356701746617238,
1.7174041100732933,
0.1885112766205208,
0.3630143936992631,
0.6395640633025061,
-0.2937369198596852,
-0.18128670495631546,
0.9504587478277511,
-0.6241293783488098,
1.8733903679237267,
0.4875344831143031,
-0.9182747577406853,
0.850791268752179,
-0.46183623376516114,
-0.634145646812118,
-0.9165900492445601,
-0.8973746003077631,
-0.879107286657044,
0.15645801311276636],
[0.48023040681030116,
0.7029544315470944,
0.722671987246733,
0.9307929858945541,
0.9307929858945543,
2.546598088688794,
2.546598088688794,
2.546598088688794,
2.546598088688794,
5.184334435330857,
5.195111458618343,
5.47436411399969,

5.47436411399969,
5.47436411399969,
5.47436411399969,
5.47436411399969,
6.068336244524409,
6.068336244524409,
7.838605251840715,
7.912250316107233,
7.912250316107233,
8.562517202814183,
9.063385768597811,
9.216218862905647,
9.216218862905647,
10.84047371877396,
10.84047371877396,
11.044358714867027,
11.375406566932424,
11.375406566932424,
0.18886549713610382,
0.6040291443391601,
-0.9727809979878402,
3.008209947509642,
0.3239743149147096,
-0.643987619924633,
1.0739882269552894,
0.4935103262912805,
-1.1713673989643745,
0.32339055167680386,
-0.8254170399860232,
3.387204293956005,
0.5437302598099015,
-0.6585961721148387,
1.0903163929414719,
-1.5261906384818182,
1.102349928722959,
-0.7373436929054379,
0.1712635196093208,
1.2086697581514199,
-0.4977298668106944,
-0.8828019932256734,
-0.13051782426094366,
0.048744185631251574,
-1.0535030780910155,
0.5224555306604155,
-0.9502111367985572,
-0.2573420205933925,
1.5833039752620888,
1.3806480812363842,
0.3864833677006273,
-0.0006284730712113316,
0.6487740408134013,
1.697829570896293,
-0.20023867528920775,
-0.32438662287273723,
1.3674639807988553,
-1.5592347723565667,
-0.6294582959326964,
1.111157394071659,
1.4895912958038178,
-0.7203901402661329,
-0.027076706863032294,
0.009574786781832467,
1.1437695218822916,
0.11720143179507149,
1.0973549625545624,
-1.0756168303605034,
-1.5785125289756954,
-0.42999007233668873,
0.7592603373612552,
0.7357293999316703,
0.03766014972468404,
1.450035910180884,
-0.08978031589044846,
0.7176267391661779,
-2.3377280976646966,

0.3878354316208249,
-0.8070070584177742,
0.07084117333557359,
-0.008998168433091903,
0.6612796195456235,
0.6287929626789847,
0.8601333603228948,
-0.7786354708155643,
-0.144476679155352,
0.39492950957300044,
-0.20012968159779548,
0.01326107241550192,
-0.5649340596325207,
-0.020398755490040426,
0.6333499066300691,
0.400844220606507,
0.5954749034740784,
0.8700413681534388,
1.315868035063538,
0.9914334712755634,
0.10418309057983809,
1.1095991500751576,
-0.1712743534684691,
-0.2730581386141925,
0.1453142903679217,
-0.3813955795194703,
0.1280814941289577,
-0.5627986213807415,
-1.0576722394029623,
-1.98721560251358,
0.519862592148374,
1.03695228961559,
0.21101375663462807,
-1.750252459751414,
-1.1450993785481438,
-2.2517541986316916,
-1.5456964825715331,
0.6026638989139771,
0.47561448130168205,
0.08201672609434518,
0.47464262797792917,
-0.14585857152212733,
0.08636040211648545,
0.7388192170680543,
1.071366965639021,
2.5115163876343742,
-0.4286872636197954,
-0.2977998181526895,
0.46569070003689333,
0.2115019986130443,
-0.5123638542481742,
0.09024611293676764,
-0.3380729423866059,
1.6314769884628497,
1.3338027138849353,
-0.7442166070564612,
1.2186722356710948,
1.584709484399307,
-0.6554175978902271,
0.36887213174672207,
-1.365021965092438,
-0.4623126041921459,
0.4869420326570461],
[0.5118290521669792,
0.9034573591840084,
1.0514460879099101,
1.1283814725964603,
1.1293980629167455,
1.6874068541059999,
1.8362001710030256,
2.594529848453243,
2.5949023258749775,
4.384420392533986,
4.804388533064279,
4.8044974726668155,

4.925049921883437,
4.925049921883437,
6.055903002618935,
6.0650337145794,
6.06510024924717,
6.06510024924717,
7.719643893437192,
7.719643893437192,
8.110116534451196,
8.374448879129652,
8.374448879129652,
9.29045698473019,
9.2929923765699,
9.977025927736127,
9.977025927736127,
9.977438303479875,
11.277250876863835,
11.588901828478656,
-0.5970483257864091,
1.048395642841534,
0.3563299433581316,
2.04182497417867,
-1.0659189653240388,
-1.5711224255679066,
-0.6854026923334804,
1.3002799990574343,
0.334979527312172,
-0.41771661057992626,
-0.5255960701407614,
0.9200835209290303,
0.5840294212204432,
0.4309612994747514,
0.010788324288593443,
0.8398650983618882,
0.35555521736822654,
0.33383672576306345,
-0.03972798241435421,
-0.5808926346218187,
-0.6217506829607715,
-0.022449055124049946,
-0.4671428723737863,
1.3840583406465645,
3.321823997583478,
1.9516123562000514,
2.109430068020724,
0.17138900458361278,
1.0734765554988877,
1.1591682463434823,
-1.5085197494005436,
0.3709514368460093,
-0.08354812013321816,
-0.6849414455501801,
-1.1807064172516142,
0.1889544406654655,
0.45125852432694835,
0.40946867023093825,
0.9591749175330944,
0.862406962512403,
-0.47565451021668054,
0.14791469386649075,
0.012323642821740255,
-0.3613715829994197,
-0.6520104899254531,
1.1500641632479522,
0.11480411675389499,
0.37929728325133344,
-0.12818017639377846,
1.5548403554359806,
-0.08803696907532751,
0.16058007713541472,
-0.2835300528775952,
1.4317946638041426,
0.04218231832431045,
0.6164913379423986,
-0.5546192578057587,

[illegible]

```
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [ 0.4319322451408806,
0.4441934852523559,
0.5878315575931067,
0.5878315575931067,
0.5912427958750637,
0.5947187380620651,
0.5947187380620652,
1.5628074958209335,
1.568718083448373,
2.013035808837258,
2.013035808837258,
2.013035808837258,
2.0277481086167364,
4.114573484042011,
4.1154343976538055,
5.358852938788934,
5.358852938788934,
5.549639110185835,
5.667644288942511,
5.6706149635077026,
7.063797010237488,
7.658675772413245,
8.123010743902626,
8.633070512181217,
8.7849794748515,
8.7849794748515,
10.749769122252356,
10.89899813672382,
11.413879450074099,
11.413879450074099,
-0.04151001581527148,
-2.751209241920508,
-0.5056705094410174,
-0.406548832581749,
0.8633340895103881,
0.17259893989954428,
-0.21235924407313156,
-0.5485334561985523,
0.2012951527025238,
-0.23858034563229433,
2.1364176057207223,
0.7817014982866842,
-0.1108899918027214,
0.6048531661014985,
1.809261328558185,
-1.1786415575821785,
-1.0827170099436865,
0.4109384498547832,
0.8090945308822337,
-0.9356585252219308,
0.28697429006677655,
-0.15332459566133277,
1.2998079093160566,
0.3870967504567973,
-0.2413465423953762,
1.001771196320206,
0.1138394509020578,
1.3220786271535379,
-1.1806565063083097,
2.2789659185869464,
0.11875741943281123,
-0.15631084464392242,
-2.14680662310284,
0.18035127914430246,
-0.260880073181003,
0.21031876899961,
0.7641152777599183,
1.203568210035907,
```

1.2767673297334832,
1.2585129786889195,
0.24752046596725819,
0.7203551788145919,
-0.17420447991641552,
-1.3486030918499698,
0.2145369555808404,
0.8896874485404291,
-1.6470089821954121,
-0.37928143051952584,
-0.29903232984244293,
0.6984471945997809,
0.14906235834494008,
0.054873949093549826,
1.3877615433857144,
0.04193983551007985,
1.2014028569521673,
-0.02338341588216991,
-0.05255422822816251,
-0.1905875093002492,
0.16425167437144692,
0.2542835721118732,
1.1822299841866795,
1.2032222480152739,
0.30142488752060526,
0.09125464470109583,
-0.45638140154025475,
0.050260497156630585,
-0.04000614529085093,
0.9096133936431033,
-0.03555378439095736,
0.4405325841695877,
-0.48129885373423525,
-0.3617221403442356,
-2.5216506086368518,
-0.5081934521569421,
0.041332600726076626,
0.7413771443671474,
0.2204029987501584,
-0.7517920276590782,
-0.21022249714251526,
1.5607830526955413,
0.16615947769508344,
-0.17023919527139658,
-0.4295619963603494,
-1.1838803442651602,
0.6319564057885301,
-0.3935277768061919,
0.005550337258741166,
-0.4226065324972882,
-0.3715846249960284,
-0.48395498614671884,
-0.4790957128536121,
1.4427293239187895,
0.5875094653073935,
1.404722056846091,
0.6973465258635152,
0.8623508528879442,
-0.5460563718875903,
0.8326491581513842,
-2.4015389244466907,
-0.04583861440999851,
-0.36265388938085896,
-0.5347868312224912,
0.49674658852292397,
-0.6488379607428911,
1.1568053403721041,
0.30662269781395407,
-2.013151608263867,
-0.17383275003878457,
0.5036662539248705,
1.9638910157847163,
3.118155075975211,
0.8117246453310248,
-1.4517859264576565,

```

-1.4946803709676548,
-2.7792982516572895,
1.1174424252005506,
2.333430423546859,
0.6903927508165344,
-0.2883899927957993,
-0.8542826896862586]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric4_unifrom_scaling_sample#7'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.662379321078875],
                          [12, 6, 0, 0, -5.394498644176583],
                          [20, 10, 0, 0, -4.374551061230002],
                          [28, 14, 0, 0, -3.546647371824164]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.00543212890625e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -7.40675994489309,
                           'others': { 'beta_coeffieicients': [ 2.1875,
                                                                -2.1875,
                                                                1.3125,
                                                                -0.3125],
                                       'cost_richardson_zne': 11.390625,
                                       'sorted_expectation_vals': [
-6.662379321078875,
-5.394498644176583,
-4.374551061230002,
-3.546647371824164],

'sorted_noise': [6, 18, 30, 42]],
'sampled data': [ [ 4,
                    2,
                    0,
                    0,
                    -6.662379321078875],
                  [ 12,
                    6,
                    0,
                    0,
                    -5.394498644176583],

```

```

[ 20,
 10,
 0,
 0,
-4.374551061230002],
[ 28,
 14,
 0,
 0,
-3.546647371824164]],
'sampling': 'default']}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                                'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'ising'}},
'init_param': { 'value': [ -4.1290471568429383e-17,
                             0.028387941050870547,
                             0.028387941050870516,
                             0.028387941050870544,
                             0.4764187545019148,
                             0.485154879376271,
                             0.7094706976105217,
                             2.998376901501724,
                             3.486264965503922,
                             3.487775573446841,
                             4.058897205630428,
                             5.020552086463093,
                             5.02158492362958,
                             5.02158492362958,
                             5.0846878698400815,
                             5.0846878698400815,
                             6.794704827143902,
                             6.794704827143902,
                             6.794704827143902,
                             8.013356083289752,
                             8.21076169573117,
                             8.21076169573117,
                             8.633075344303478,
                             8.633075344303478,
                             8.930018414781724,
                             8.930018414781724,
                             8.930018414781724,
                             9.990329186240457,
                             9.990329186240457,
                             12.915390927357043,
                             0.8289215528981712,
                             0.6465354442587259,
                             -0.06922368073773534,
                             -2.0853926237797484,
                             0.25605409461249207,
                             -0.765586831155604,
                             0.9589723041272441,
                             0.27455658215646966,
                             -0.07189982532198122,
                             -0.3449139525561357,
                             -0.46800027418386986,
                             1.016480689947782,
                             -0.9706976058410632,
                             -0.9556868223128826,
                             0.9323574793538898,
                             -0.4376613900889883,

```

0.44138701419965265,
1.4418520162063444,
0.8420619129497768,
-0.025530845682239494,
-2.3076344184440156,
0.37458463155346333,
-0.47031338116346383,
0.07338001085955462,
-0.869249007732584,
1.6824786550253577,
1.719346297136726,
-0.9622509832303179,
0.5779076253870294,
0.4760527439731688,
-0.2451715046346434,
1.5906513054768954,
0.35918458386075697,
-0.44817294000256097,
-0.37416377940751877,
0.37048554789855764,
0.693137718643464,
2.288219219235325,
-0.26501198620577204,
0.3689053264668303,
0.10166782579927121,
-0.3356073800920907,
-0.6413954053758932,
0.28636656044125136,
-0.8690978212322298,
1.166030172665772,
-0.46365263204645707,
-0.8505396438729699,
2.4328989003329693,
-1.4554443951257197,
2.09134311034601,
1.7695145471360614,
1.1170184462314887,
0.20325573406500014,
0.34206355374635444,
0.016804831103183593,
-0.19882507083844791,
1.4011478969931443,
-0.0572616468882171,
-1.3773340344964602,
1.1031086912886459,
0.47433623341773395,
-0.037679845343088075,
0.7450954028512253,
1.3114871459373192,
-0.18651966717154955,
1.0750199871192778,
0.48905379729033566,
-0.6281827821606347,
1.322074656325554,
-1.4761484868509396,
0.2859918816514792,
-1.0033541595176405,
0.8264053813863532,
0.622010811103528,
1.1636747207646185,
-0.31883271288389914,
-0.0013981850979601159,
-0.09999405171802342,
1.3085280853238759,
-1.3727969089464465,
-0.08341159264840771,
1.6249369276484606,
-0.28082331570516694,
1.863294400254821,
2.2932652495309953,
1.4849536675442596,
0.6811668802751195,
1.5380110980874968,
-0.6615225065895972,
0.4000828281024264,

```

0.33932121588009856,
0.48855764825038867,
2.5371111232777053,
-0.7260430040504564,
0.09005940379987407,
-0.8762980817992282,
-0.23800314057977784,
0.9728027685352436,
-0.20857834368009137,
0.6148333457084475,
-1.5171898336040224,
0.8658542069129427,
-1.6478287868490311,
0.5770049367106069,
0.134497700004433,
-0.11612046371881249,
1.5826352840237945,
1.6045820840261675,
-0.08805669980595342,
-1.507745125921201,
-0.11276651576435986,
0.07206005105970639,
-1.7788122679933258,
-0.002988765738482436,
0.7255804452314191,
2.7661804504965604,
-3.1943627011957507,
-0.07088434770936546,
0.10949711075632612]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric4_unifrom_scaling_sample#1'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.833109640228284],
                           [12, 6, 0, 0, -5.610198560557998],
                           [20, 10, 0, 0, -4.60990156720048],
                           [28, 14, 0, 0, -3.786810221698121]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.220008850097656e-05,
            'zne_values': { 'degree': 0,

```

```
-6.833109640228284,  
-5.610198560557998,  
-4.60990156720048,  
-3.786810221698121],
```

```
'extrapolated_value': -7.542235599448719,  
'others': { 'beta_coefficients': [ 2.1875,  
                                   -2.1875,  
                                   1.3125,  
                                   -0.3125],  
            'cost_richardson_zne': 11.390625,  
            'sorted_expectation_vals': [  
  
                'sorted_noise': [6, 18, 30, 42]],  
            'sampled_data': [ [ 4,  
                                2,  
                                0,  
                                0,  
                                -6.833109640228284],  
                              [ 12,  
                                6,  
                                0,  
                                0,  
                                -5.610198560557998],  
                              [ 20,  
                                10,  
                                0,  
                                0,  
                                -4.60990156720048],  
                              [ 28,  
                                14,  
                                0,  
                                0,  
                                -3.786810221698121]],  
            'sampling': 'default'}}},  
{ 'config': { 'ansatz': { 'gateset': 1,  
                           'layer': 30,  
                           'ugate': { 'coefficients': { 'bn': [ 0,  
                                                                0,  
                                                                0,  
                                                                0,  
                                                                0,  
                                                                0],  
                                                                'cn': [ 0.5,  
                                                                0.5,  
                                                                0.5,  
                                                                0.5,  
                                                                0.5,  
                                                                0.5],  
                                                                'r': 0},  
                           'time': {'max': 10.0, 'min': 0.0},  
                           'type': 'ising'}}},  
  'init_param': { 'value': [ 0.4349098035102067,  
                              0.5281397824211302,  
                              0.6402833431740305,  
                              0.701719039924861,  
                              0.8953768954552646,  
                              1.4118719346144908,  
                              2.230113621742168,  
                              2.6548053250922607,  
                              3.10433397641996,  
                              3.1043339764199596,  
                              4.146783795497047,  
                              4.146783795497047,  
                              4.149760661232294,  
                              4.149760661232294,  
                              4.223539194779984,  
                              4.223539194779984,  
                              6.7727886487081435,  
                              6.7727886487081435,  
                              7.243551717080481,  
                              7.282491880183138,  
                              7.282491880183137,  
                              7.282491880183138,  
                              8.806989173945695,  
                              8.80698917394569,
```


9.706285648296737,
9.809777023075187,
9.837118725166619,
10.701586770517197,
11.185931122995514,
11.185931122995514,
0.39919688039300266,
1.4533249104921768,
0.7852252158845567,
-0.4957764525834907,
-0.3354719566017423,
0.16097458435951068,
-0.21971891747261646,
1.5370438656668401,
0.26531517225969137,
-0.6386119707402834,
0.37748437574064975,
1.2333814560270282,
0.5934062959304571,
1.633059089666515,
0.31170790405459564,
0.3995146629536743,
-0.11155094016301184,
0.8138063699882787,
0.019467271717771087,
-0.38186317682975884,
-2.778803401027002,
0.7887082829537533,
-3.1348050840084074,
-1.5109224736536222,
-0.38589710406642097,
-1.105927053527023,
0.3451707237315739,
0.6290043291617069,
-1.834722402137916,
-1.0551446159380258,
-1.3581260193610523,
0.9666180881078981,
-0.05759676522292653,
1.1866392284178628,
-1.1308471740259582,
0.34327443183487805,
0.17662613760194443,
-0.043135596735193626,
-0.03667546733606037,
-0.4901895513664937,
-1.0399227541005371,
0.4880459765792483,
0.053089003166299456,
-0.6259966396796703,
0.03159579372412852,
-0.09798805529491862,
0.052872660107769434,
1.490194875693949,
0.22167423812356596,
-0.33497064714209324,
1.0144679614160885,
0.01608415027143934,
0.49167899744128424,
0.12491439548600353,
-0.1329609067962543,
0.8639852596635488,
-1.6607871811173882,
0.2801513494447602,
0.3908501333463579,
0.6044891435983025,
0.08637544621327581,
1.2270127042754213,
0.11860545946003472,
-0.13066468162372125,
-0.998222267933649,
-1.2643438207146234,
0.7835210648968657,
0.8370312354478452,
0.06527740704106412,

```

-1.0235950282739155,
-0.705070781232052,
0.037682760290170314,
1.4778132318704873,
1.7053977225841355,
-0.1291331271719845,
0.1648924115529835,
2.208642048449397,
1.1885953055185114,
-0.8138037079028133,
0.6715189891287302,
0.7819572861394978,
-0.4022064636590845,
-0.3927127310055937,
-0.4118784323717791,
1.3463239978578432,
-0.3323606385924751,
0.6664683657180488,
1.58852177854836,
-1.212203169826861,
-0.031436536468697034,
0.5094262403458202,
-0.0616688967239856,
-0.0666169691754888,
-0.7277012125404719,
-0.12684196683036533,
0.5325519345188151,
2.2778420441533447,
0.8735933622003313,
0.9907279467768721,
1.1448787224294625,
0.4291077507842401,
-0.21356701746617238,
1.7174041100732933,
0.1885112766205208,
0.3630143936992631,
0.6395640633025061,
-0.2937369198596852,
-0.18128670495631546,
0.9504587478277511,
-0.6241293783488098,
1.8733903679237267,
0.4875344831143031,
-0.9182747577406853,
0.850791268752179,
-0.46183623376516114,
-0.634145646812118,
-0.9165900492445601,
-0.8973746003077631,
-0.879107286657044,
0.15645801311276636]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},

```

```

'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric4_unifrom_scaling_sample#8'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0],
[3, 3, 0, 0]]},

'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.672780190342767],
[12, 6, 0, 0, -5.522238922989663],
[20, 10, 0, 0, -4.5712287338679385],
[28, 14, 0, 0, -3.7801844878377153]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 4.124641418457031e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -7.335239083087297,
'others': { 'beta_coeffieicients': [ 2.1875,
-2.1875,
1.3125,
-0.3125],
'cost_richardson_zne': 11.390625,
'sorted_expectation_vals': [
-6.672780190342767,
-5.522238922989663,
-4.5712287338679385,
-3.7801844878377153],

'sorted_noise': [6, 18, 30, 42]],
'sampled data': [ [ 4,
2,
0,
0,
-6.672780190342767],
[ 12,
6,
0,
0,
-5.522238922989663],
[ 20,
10,
0,
0,
-4.5712287338679385],
[ 28,
14,
0,
0,
-3.7801844878377153]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [ 0.32980400918867037,
0.7818596814437996,

```

0.9680306497310072,
1.0586432954274991,
1.7663116348413028,
1.8956491295658664,
2.0436116769319477,
2.0436116769319477,
2.0436116769319477,
2.1846619555479685,
2.1846619555479685,
4.516997836642403,
4.516997836642403,
5.3166564351730035,
5.320130741320412,
5.399447086770547,
5.400046100307029,
5.400046100307029,
5.400046100307029,
5.400046100307029,
5.400046100307029,
7.097750404951498,
7.618222769006738,
7.956666567531361,
7.956666567531361,
8.961109688308255,
8.961109688308255,
11.028500695271937,
11.5809418583784,
11.5809418583784,
11.5809418583784,
0.24114173221382337,
1.3754512844042854,
0.9720111739434927,
-0.29877116871955545,
-0.029737313498699618,
0.3185153459444675,
-0.31840745367437817,
0.6801602139502809,
-0.015407845998522466,
-1.114484654283524,
0.22997676136047449,
1.5201821520192487,
0.3217486942799661,
0.43678760134757944,
0.17618749246228715,
0.7010359103792874,
0.3446701255696699,
1.4700166379981474,
0.3471333265905846,
-0.968444017764766,
-1.1485269521507444,
0.3282504398407059,
-0.7898554413193151,
-0.22157985789400653,
0.4570931682042821,
-0.22158635023930806,
-0.36549836384277606,
-0.2366283880333898,
0.4114765778384287,
-0.7694643480660831,
-0.40507000819737765,
-0.6576468499882752,
0.2023600853039859,
-1.2917280200640486,
0.32702771491506877,
-0.07546137553843744,
-0.7713328193108817,
-0.8163562225239717,
-0.07772489804090182,
-0.32841576637855896,
-0.05728139293162625,
0.8564947637293437,
0.15384434981345546,
0.41944080031994563,
0.16884394015055976,
-0.010490661309829376,
-1.3198249528292274,


```

        'noise_prob': [0.001, 0.001, 0, 0],
        'status': True,
        'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [ 0.5,
                                         0.5,
                                         0.5,
                                         0.5,
                                         0.5,
                                         0.5],
                                'r': 1}},
        'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric4_unifrom_scaling_sample#5'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.720969040841222],
                           [12, 6, 0, 0, -5.577003279994282],
                           [20, 10, 0, 0, -4.626982768162378],
                           [28, 14, 0, 0, -3.833723359776359]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 5.054473876953125e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -7.37730143513569,
                           'others': { 'beta_coeffieicients': [ 2.1875,
                                                                -2.1875,
                                                                1.3125,
                                                                -0.3125],
                                       'cost_richardson_zne': 11.390625,
                                       'sorted_expectation_vals': [
-6.720969040841222,
-5.577003279994282,
-4.626982768162378,
-3.833723359776359],
'sorted_noise': [6, 18, 30, 42]}},
'sampled data': [ [ 4,
                    2,
                    0,
                    0,
                    -6.720969040841222],
                  [ 12,
                    6,
                    0,
                    0,
                    -5.577003279994282],
                  [ 20,
                    10,
                    0,
                    0,
                    -4.626982768162378],
                  [ 28,
                    14,
                    0,
                    0,
                    0],

```

```

-3.833723359776359]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                                'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'ising'}}},
  'init_param': { 'value': [ 0.46756606893882113,
                              0.46983859896604685,
                              0.5133985927484581,
                              0.5133985927484581,
                              0.5133985927484581,
                              1.4016794842576101,
                              1.6914116785996751,
                              1.691411678599675,
                              1.6941756924801525,
                              1.6941756924801528,
                              3.749864451084885,
                              3.749864451084885,
                              3.749864451084885,
                              3.749864451084885,
                              4.969636107010246,
                              4.969636107010246,
                              4.969636107010246,
                              4.969636107010246,
                              5.983107385388112,
                              5.983107385388112,
                              8.378949246097575,
                              8.378949246097575,
                              8.635680633753974,
                              8.635680633753974,
                              9.108807770997622,
                              9.108807770997622,
                              9.108807770997622,
                              9.960036411132767,
                              10.436643041587223,
                              10.716533289695375,
                              0.5466515694267952,
                              3.318518267704871,
                              0.40678728382369284,
                              0.6290091030083763,
                              -0.4770983795300032,
                              -0.7455483390932742,
                              0.1630064768968011,
                              0.4889724943271746,
                              0.10596004146863507,
                              -1.4759536729524354,
                              0.957604974713893,
                              0.6004514101153323,
                              0.0725728265044025,
                              1.4994700617905987,
                              0.15508641282397265,
                              -0.15494303336158094,
                              -0.5663845144966579,
                              2.6663065398173496,
                              -0.27086336898318036,
                              -1.1828784814640632,
                              -0.295045685212881,
                              0.6236994649007899,
                              1.1769360493970402,
                              1.7159795997719172,
                              -0.3019600780315931,

```

0.651545330446025,
-2.0782382222089724,
0.3186344830290561,
-0.2815476257680432,
0.4067635126151388,
-0.8228051832090519,
-0.4154478828430832,
2.3723163321732827,
0.931436930203099,
0.9804374423714106,
0.058895326584719035,
0.3167253661454302,
0.3390503539155689,
-0.5024758650485814,
-0.2425872659725522,
-0.6703364135359525,
-0.147558399690948,
1.1644296968010859,
0.5475507579506065,
0.1736068872874216,
0.30547527216138975,
0.5300773267023443,
0.7722201656261689,
-0.6858549484151734,
-0.16073113081558088,
-1.197458252004488,
-0.507307082981415,
-0.16437738354723153,
0.2647407403580543,
-0.45930133674563567,
-0.22673992276266391,
1.8127715449849986,
-1.4334128748253496,
-0.759404859696163,
1.2431110601081345,
0.8953271645792334,
0.6963251389999259,
-0.23147129445607595,
0.3638861144885483,
2.2394622330281906,
1.543816783446588,
1.7535830084845025,
-0.3169637330795974,
0.2453194636587019,
2.0076391670843647,
0.286017506499548,
0.6600449842625801,
0.19782118041633945,
0.05463577684798935,
0.4755005320975566,
0.055207215781380096,
0.06969685609102444,
0.2645328829673777,
-0.01235845210357445,
1.01982876424508,
2.262492417600915,
1.3443784687440867,
0.10698144982242018,
0.428081928456963,
-1.5494921893757727,
1.0701294130786736,
0.39987519173641656,
0.06301738855484718,
-1.6711192593807125,
0.26920159401030147,
-0.08358711295948693,
0.4054786916979446,
-0.5013194648307361,
2.126550690771933,
0.10959959792145137,
0.4205360694942305,
-0.586878230803869,
-0.3381608637889511,
-0.5055401953468239,
-0.45995744817477163,


```

0.07999668782445096,
-0.4269018063306222,
0.9588276598807326,
1.042187521522261,
0.07821502449201799,
0.6777167152964713,
0.4177387822172794,
-0.18721284346239236,
-0.8064458441104513,
2.8712851588196218,
0.7141862154478698,
-0.9976723444987857,
-0.8529435737233043,
1.9602636639541733,
1.2634521849205176,
-0.7170870121313951,
0.31227949023655405,
-1.0451293221946831,
-0.6787852158143914,
1.8581833428208436]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric4_unifrom_scaling_sample#6'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.4878948983695235],
                          [12, 6, 0, 0, -5.297876077058435],
                          [20, 10, 0, 0, -4.3310003849914125],
                          [28, 14, 0, 0, -3.539166940153787]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 5.91278076171875e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -7.181614508121177,
                           'others': { 'beta_coeffieicients': [ 2.1875,
                                                                -2.1875,
                                                                1.3125,
                                                                -0.3125],
                                         'cost_richardson_zne': 11.390625,
                                         'sorted_expectation_vals': [
-6.4878948983695235,
-5.297876077058435,

```

```
-4.3310003849914125,  
-3.539166940153787],
```

```
        'sorted_noise': [6, 18, 30, 42]],  
    'sampled data': [ [ 4,  
                        2,  
                        0,  
                        0,  
                        -6.4878948983695235],  
                      [ 12,  
                        6,  
                        0,  
                        0,  
                        -5.297876077058435],  
                      [ 20,  
                        10,  
                        0,  
                        0,  
                        -4.3310003849914125],  
                      [ 28,  
                        14,  
                        0,  
                        0,  
                        -3.539166940153787]],  
    'sampling': 'default']},  
{ 'config': { 'ansatz': { 'gateset': 1,  
                          'layer': 30,  
                          'ugate': { 'coefficients': { 'bn': [ 0,  
                                                              0,  
                                                              0,  
                                                              0,  
                                                              0,  
                                                              0],  
                                                              'cn': [ 0.5,  
                                                              0.5,  
                                                              0.5,  
                                                              0.5,  
                                                              0.5],  
                                                              'r': 0},  
                          'time': {'max': 10.0, 'min': 0.0},  
                          'type': 'ising'}},  
    'init_param': { 'value': [ 0.27654223573266234,  
                                0.27654223573266246,  
                                0.27654223573266246,  
                                0.4643738922966213,  
                                0.6582164862517121,  
                                0.6582164862517121,  
                                0.6582164862517121,  
                                1.3424523496852623,  
                                1.47635793598756,  
                                1.47635793598756,  
                                3.385550316280557,  
                                3.385550316280557,  
                                3.385550316280557,  
                                4.494062371253785,  
                                4.494062371253785,  
                                5.0204758683966135,  
                                5.023404188660082,  
                                7.05704094573064,  
                                7.087391629397132,  
                                7.533802135057517,  
                                8.245970909352765,  
                                8.390887842007308,  
                                8.390887842007308,  
                                8.703391612007538,  
                                8.703391612007538,  
                                9.485545381639698,  
                                9.485545381639698,  
                                9.849437794456351,  
                                9.849437794456351,  
                                10.799155842585746,  
                                -0.12708089143278917,  
                                1.4168148977025445,  
                                -0.13519536748631492,
```

-0.32308714516775877,
0.06981336221044306,
-0.034321895006426455,
-0.18209443759399158,
-0.024858958594746107,
0.664488174834645,
0.6897317665852637,
-0.008774731192519907,
0.625642551337688,
0.6862410821653743,
-0.7670419540542446,
-0.48055390132433673,
1.573964677292243,
-1.0555745677813198,
1.3159721165631335,
-0.9138502497993347,
-0.4641566084568084,
-0.06966564000847691,
0.039012746519005555,
1.0827641697579717,
1.2190093772283843,
1.5391150382377632,
-0.6675581491933664,
0.09104662559232415,
0.5283818781579022,
0.04478118244404365,
1.5836075485651113,
0.251932560991488,
-0.3949560381811925,
0.8854639388207172,
-0.8611267164776638,
1.566928747241295,
-0.15445151319573347,
-0.01396050433941564,
0.8172620115356718,
-0.014658132241289147,
-0.6623010385304499,
0.6299284572753584,
-0.15026947869759505,
-1.2293112897827836,
-0.5768656731979434,
1.231557922062332,
-1.447872926116422,
1.4918722706131171,
0.31945486783385635,
0.43870804207253483,
1.3582636256308331,
-0.8478398442626237,
0.799518029891469,
0.5539369896048497,
0.13453197415145846,
0.7167498104862234,
-0.06589109804263343,
0.041038724923082276,
1.151975756926885,
-0.052026838729968425,
0.3952978000808587,
-1.1754748970457833,
-0.01965607042871015,
-0.2456324144680547,
-0.01136184217607491,
0.022503168314325715,
1.1777923723104653,
0.14554117253938412,
-1.2287946167385297,
-0.9683737383161766,
1.073329622962756,
-0.7143583802799215,
-0.3545002682598469,
-0.31906110571926777,
-0.9597094503848127,
0.8825903987559673,
-0.08273415521500459,
0.2895039435279609,
-0.3388594197258312,

```

0.2406459434933063,
0.3824128671502841,
-0.5390555829405563,
0.3757684495816124,
-0.5015719636333678,
0.3958257688275076,
-0.6428354095046468,
-0.9178624992467067,
0.29327306512581747,
-0.09193285724698198,
0.1770091195095364,
0.6803523512851801,
1.05525770275555,
-0.4646437845139076,
2.2560908642344026,
-0.7500939262299234,
1.92959092052175,
-0.36895777993274226,
-0.04538283416771957,
1.2529586558275392,
-0.57805717638909,
0.3080585810325818,
0.031910210614104206,
0.18519950010488512,
1.3888566879174211,
1.087260914864616,
0.1525935562363358,
-0.08810167996863522,
0.07467874103808375,
1.4444893382870363,
-1.446220680213735,
0.8215180856176293,
2.0151723649306295,
0.13454991965456886,
0.21923381008732426,
0.31112145477192793,
-0.8977876743954992,
0.10239876571985529,
-1.0132922278363503,
-0.3794863035801566,
0.8437803973608401,
1.111671691788496]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric4_unifrom_scaling_sample#2'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,

```

```
'sorted_noise': [6, 18, 30, 42]],  
    'sampled_data': [ [ 4,  
                        2,  
                        0,  
                        0,  
                        -6.514930884689278],  
                      [ 12,  
                        6,  
                        0,  
                        0,  
                        -5.310706408751286],  
                      [ 20,  
                        10,  
                        0,  
                        0,  
                        -4.328532484009807],  
                      [ 28,  
                        14,  
                        0,  
                        0,  
                        -3.520772220894083]],  
    'sampling': 'default']},  
{ 'config': { 'ansatz': { 'gateset': 1,  
                          'layer': 30,  
                          'ugate': { 'coefficients': { 'bn': [ 0,  
                                                                0,  
                                                                0,  
                                                                0,  
                                                                0,  
                                                                0,  
                                                                0],  
                                          'cn': [ 0.5,  
                                                    0.5,  
                                                    0.5,  
                                                    0.5,  
                                                    0.5,  
                                                    0.5],  
                                          'r': 0},  
                                'time': {'max': 10.0, 'min': 0.0},  
                                'type': 'ising'}},  
        'init_param': { 'value': [ 0.3463233138801294,  
                                    0.34632331388012944,  
                                    0.3463233138801294,  
                                    0.3463233138801293,  
                                    1.2403205490491955,  
                                    1.38033446895038,  
                                    3.6939218351946623,  
                                    3.6939218351946623,  
                                    3.693921835194662,  
                                    3.693921835194662,  
                                    3.9412089126587,
```

```
'sorted_noise': [6, 18, 30, 42]],  
    'sampled_data': [[ 4,  
                        2,  
                        0,  
                        0,  
                       -6.514930884689278],  
                     [ 12,  
                        6,  
                        0,  
                        0,  
                       -5.310706408751286],  
                     [ 20,  
                        10,  
                        0,  
                        0,  
                       -4.328532484009807],  
                     [ 28,  
                        14,  
                        0,  
                        0,  
                       -3.520772220894083]],  
    'sampling': 'default'}},  
{ 'config': { 'ansatz': { 'gateset': 1,  
                           'layer': 30,  
                           'ugate': { 'coefficients': { 'bn': [ 0,  
                                                                0,  
                                                                0,  
                                                                0,  
                                                                0,  
                                                                0],  
                                                           'cn': [ 0.5,  
                                                                0.5,  
                                                                0.5,  
                                                                0.5,  
                                                                0.5,  
                                                                0.5],  
                                                           'r': 0},  
                           'time': {'max': 10.0, 'min': 0.0},  
                           'type': 'ising'}}},  
  'init_param': { 'value': [ 0.3463233138801294,  
                              0.34632331388012944,  
                              0.3463233138801294,  
                              0.3463233138801293,  
                              1.2403205490491955,  
                              1.38033446895038,  
                              3.6939218351946623,  
                              3.6939218351946623,  
                              3.693921835194662,  
                              3.693921835194662,  
                              3.9412089126587,
```

3.9412089126587,
3.9412089126587007,
3.9412089126587007,
4.8310999601027795,
4.870844593548848,
4.900228671797898,
4.9002286717978984,
4.9002286717978984,
4.900228671797898,
5.70167657474683,
6.21188441483336,
6.21188441483336,
6.686419232103557,
6.723504371727371,
7.618130890836856,
7.618130890836856,
9.420364260644014,
9.420364260644014,
10.775132222422771,
-0.3525750090684208,
1.2748976558430758,
-0.16146998454247521,
1.3224575871686013,
0.0349265192498484,
0.36138711032415904,
-0.5043488106619078,
0.7634801084306476,
-0.3536814593617201,
0.9606809354246906,
-1.1143894720984433,
-0.45289061390722213,
-0.1605333784989486,
-1.594210268458609,
-0.09465166736510336,
0.11132251138190599,
1.004235666711761,
-0.23115493220428254,
0.9451761882751959,
-0.3707284170975133,
-0.17782948231634454,
-0.18433417990028922,
-0.06421460341189031,
-1.155489001827845,
-0.2074908306187579,
-0.036360325422879626,
0.5283181819327893,
0.3603171208434518,
0.6819349944270952,
-0.2733083993242672,
0.7075879191324372,
-0.042779623335368125,
0.7032435595714176,
-0.7363942983542189,
0.9583235184546098,
-0.5892472427173001,
-0.7139877518112785,
-0.12446625268371646,
-0.439646171766664,
0.7129808957109575,
0.3373075268097138,
0.3345550414369601,
-0.38068761370203624,
0.07864964370676188,
-0.3383553862900091,
0.6633419895127639,
-0.578720703894673,
0.486162302530746,
-0.06112668401522289,
0.34417654454775265,
-0.09182266750822224,
0.012215000731405658,
0.5658291917160285,
-0.16728488191683846,
0.3754616489195623,
-0.30497315054621493,

```
-0.7559846133933803,  
0.9474959005813702,  
-0.5904076686561035,  
-0.2620011537974264,  
0.1328515248519479,  
-0.44066408562446224,  
-0.7849425610737308,  
0.44059474279745625,  
-0.04646983134951628,  
0.27897770882356554,  
0.36838349877106535,  
1.1610009997739648,  
0.9058411506824863,  
0.3119955912029684,  
-0.5265683545543336,  
1.0661822357546584,  
0.8560044406571757,  
0.5004409227892788,  
0.5192086876555008,  
-0.0019665558968806392,  
-0.23949800878404184,  
0.32563956832297103,  
0.795323960796817,  
1.5213873243487468,  
0.8637563634588281,  
0.07945031145196896,  
1.7583315097859118,  
0.5335864884320034,  
-0.1366270407456156,  
-0.9111102828185913,  
-0.41650979281126865,  
0.20926691692934202,  
0.2776061293997892,  
1.4257231544028384,  
-0.10754544247245298,  
1.0035199546983695,  
1.1162390986419686,  
1.7211522778091677,  
-1.4332245925979028,  
1.4417595659646931,  
-2.106784537892007,  
-0.6878309432904478,  
-0.20360639354575213,  
0.6373705186717347,  
0.051333236332203484,  
-0.35868991955772395,  
-0.24365907607080922,  
1.619736185871098,  
0.5895182658245096,  
-0.6086170057348131,  
0.39774539931448055,  
1.409837948786624,  
0.17438880158294706,  
-0.0415548590378924,  
0.1413623732119859,  
-0.8192475896231625,  
0.5823982141653503,  
-0.8462302367353051,  
-0.60454163385331,  
1.8216629921858374,  
1.2634551976945552,  
-0.40200420270308274,  
-1.5954759495917508,  
-1.7732438129518737]],  
'noise_profile': { 'noise_on_init_param': { 'status': False,  
                                              'value': 0},  
                    'noise_prob': [0.001, 0.001, 0, 0],  
                    'status': True,  
                    'type': 'depolarizing'},  
'nqubits': 7,  
'observable': { 'coefficients': { 'bn': [ 1.0,  
                                           1.0,  
                                           1.0,  
                                           1.0,  
                                           1.0,
```

[illegible]


```
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [ 0.48023040681030116,
0.7029544315470944,
0.722671987246733,
0.9307929858945541,
0.9307929858945543,
2.546598088688794,
2.546598088688794,
2.546598088688794,
2.546598088688794,
5.184334435330857,
5.195111458618343,
5.47436411399969,
5.47436411399969,
5.47436411399969,
5.47436411399969,
5.47436411399969,
6.068336244524409,
6.068336244524409,
7.838605251840715,
7.912250316107233,
7.912250316107233,
8.562517202814183,
9.063385768597811,
9.216218862905647,
9.216218862905647,
10.84047371877396,
10.84047371877396,
11.044358714867027,
11.375406566932424,
11.375406566932424,
0.18886549713610382,
0.6040291443391601,
-0.9727809979878402,
3.008209947509642,
0.3239743149147096,
-0.643987619924633,
1.0739882269552894,
0.4935103262912805,
-1.1713673989643745,
0.32339055167680386,
-0.8254170399860232,
3.387204293956005,
0.5437302598099015,
-0.6585961721148387,
1.0903163929414719,
-1.5261906384818182,
1.102349928722959,
-0.7373436929054379,
0.1712635196093208,
1.2086697581514199,
-0.4977298668106944,
-0.8828019932256734,
-0.13051782426094366,
0.048744185631251574,
-1.0535030780910155,
0.5224555306604155,
-0.9502111367985572,
-0.2573420205933925,
1.5833039752620888,
1.3806480812363842,
0.3864833677006273,
-0.0006284730712113316,
0.6487740408134013,
1.697829570896293,
```

-0.20023867528920775,
-0.32438662287273723,
1.3674639807988553,
-1.5592347723565667,
-0.6294582959326964,
1.111157394071659,
1.4895912958038178,
-0.7203901402661329,
-0.027076706863032294,
0.009574786781832467,
1.1437695218822916,
0.11720143179507149,
1.0973549625545624,
-1.0756168303605034,
-1.5785125289756954,
-0.42999007233668873,
0.7592603373612552,
0.7357293999316703,
0.03766014972468404,
1.450035910180884,
-0.08978031589044846,
0.7176267391661779,
-2.3377280976646966,
0.3878354316208249,
-0.8070070584177742,
0.07084117333557359,
-0.008998168433091903,
0.6612796195456235,
0.6287929626789847,
0.8601333603228948,
-0.7786354708155643,
-0.144476679155352,
0.39492950957300044,
-0.20012968159779548,
0.01326107241550192,
-0.5649340596325207,
-0.020398755490040426,
0.6333499066300691,
0.400844220606507,
0.5954749034740784,
0.8700413681534388,
1.315868035063538,
0.9914334712755634,
0.10418309057983809,
1.1095991500751576,
-0.1712743534684691,
-0.2730581386141925,
0.1453142903679217,
-0.3813955795194703,
0.1280814941289577,
-0.5627986213807415,
-1.0576722394029623,
-1.98721560251358,
0.519862592148374,
1.03695228961559,
0.21101375663462807,
-1.750252459751414,
-1.1450993785481438,
-2.2517541986316916,
-1.5456964825715331,
0.6026638989139771,
0.47561448130168205,
0.08201672609434518,
0.47464262797792917,
-0.14585857152212733,
0.08636040211648545,
0.7388192170680543,
1.071366965639021,
2.5115163876343742,
-0.4286872636197954,
-0.2977998181526895,
0.46569070003689333,
0.2115019986130443,
-0.5123638542481742,
0.09024611293676764,

```

-0.3380729423866059,
1.6314769884628497,
1.3338027138849353,
-0.7442166070564612,
1.2186722356710948,
1.584709484399307,
-0.6554175978902271,
0.36887213174672207,
-1.365021965092438,
-0.4623126041921459,
0.4869420326570461]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric4_unifrom_scaling_sample#9'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.75124151968758],
                           [12, 6, 0, 0, -5.575778108730956],
                           [20, 10, 0, 0, -4.6095181057209125],
                           [28, 14, 0, 0, -3.8092488284193005]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.363059997558594e-05,
            'zne_values': { 'degree': 0,
                            'extrapolated_value': -7.43092846634528,
                            'others': { 'beta_coeffieicients': [ 2.1875,
                                                                -2.1875,
                                                                1.3125,
                                                                -0.3125],
                            'cost_richardson_zne': 11.390625,
                            'sorted_expectation_vals': [
-6.75124151968758,
-5.575778108730956,
-4.6095181057209125,
-3.8092488284193005],

'sorted_noise': [6, 18, 30, 42]},
'sampled data': [ [ 4,
                    2,
                    0,
                    0,
                    -6.75124151968758],
                  [ 12,
                    6,
                    0,
                    0,
                    -5.575778108730956],
                  [ 20,
                    10,
                    0,
                    0,
                    -4.6095181057209125],
                  [ 28,
                    14,
                    0,
                    0,
                    -3.8092488284193005]]}

```

```

        6,
        0,
        0,
        -5.575778108730956],
    [ 20,
      10,
      0,
      0,
      -4.6095181057209125],
    [ 28,
      14,
      0,
      0,
      -3.8092488284193005]],
    'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                                'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'ising'}}},
  'init_param': { 'value': [ 0.5118290521669792,
                              0.9034573591840084,
                              1.0514460879099101,
                              1.1283814725964603,
                              1.1293980629167455,
                              1.6874068541059999,
                              1.8362001710030256,
                              2.594529848453243,
                              2.5949023258749775,
                              4.384420392533986,
                              4.804388533064279,
                              4.8044974726668155,
                              4.925049921883437,
                              4.925049921883437,
                              6.055903002618935,
                              6.0650337145794,
                              6.06510024924717,
                              6.06510024924717,
                              7.719643893437192,
                              7.719643893437192,
                              8.110116534451196,
                              8.374448879129652,
                              8.374448879129652,
                              9.29045698473019,
                              9.2929923765699,
                              9.977025927736127,
                              9.977025927736127,
                              9.977438303479875,
                              11.277250876863835,
                              11.588901828478656,
                              -0.5970483257864091,
                              1.048395642841534,
                              0.3563299433581316,
                              2.04182497417867,
                              -1.0659189653240388,
                              -1.5711224255679066,
                              -0.6854026923334804,
                              1.3002799990574343,
                              0.334979527312172,
                              -0.41771661057992626,
                              -0.5255960701407614,
                              0.9200835209290303,

```

0.5840294212204432,
0.4309612994747514,
0.010788324288593443,
0.8398650983618882,
0.35555521736822654,
0.33383672576306345,
-0.03972798241435421,
-0.5808926346218187,
-0.6217506829607715,
-0.022449055124049946,
-0.4671428723737863,
1.3840583406465645,
3.321823997583478,
1.9516123562000514,
2.109430068020724,
0.17138900458361278,
1.0734765554988877,
1.1591682463434823,
-1.5085197494005436,
0.3709514368460093,
-0.08354812013321816,
-0.6849414455501801,
-1.1807064172516142,
0.1889544406654655,
0.45125852432694835,
0.40946867023093825,
0.9591749175330944,
0.862406962512403,
-0.47565451021668054,
0.14791469386649075,
0.012323642821740255,
-0.3613715829994197,
-0.6520104899254531,
1.1500641632479522,
0.11480411675389499,
0.37929728325133344,
-0.12818017639377846,
1.5548403554359806,
-0.08803696907532751,
0.16058007713541472,
-0.2835300528775952,
1.4317946638041426,
0.04218231832431045,
0.6164913379423986,
-0.5546192578057587,
-0.0016176435191898148,
0.3455446109244303,
-0.7983090776945799,
0.06548494602750325,
0.09601011255154575,
-1.274882751616874,
-0.6639296860891427,
0.08468574728059233,
0.09178155492125434,
-1.9746588496307496,
1.363083624655819,
-0.11754172722968637,
-1.7454296260872237,
0.689437867548643,
0.09536601432398076,
2.1875690275532618,
1.9737367040865754,
0.19623670188051764,
1.3618525198389628,
-1.6551377680800423,
-1.2796187335858575,
-0.02979150349567955,
0.3655877448114123,
-1.339933806945279,
-1.6131280858420662,
1.1268021774842836,
0.7853899025443938,
-1.214954346638235,
1.881374469676006,
-2.2108375055617597,

```

0.38299461206746305,
-0.062378913993248956,
-1.0812916442950207,
0.19113842494911862,
0.43947065709017896,
1.4340329690981122,
-2.1914706359927933,
1.683520172253517,
-0.4151753501170658,
0.2652280013518245,
-0.09829123134965576,
-0.4033858255054015,
-0.09488290407421365,
1.1087820722198931,
1.3758250784849964,
0.3693308876448478,
-1.4774754586810708,
0.45790576106810227,
-1.318961759734687,
1.2898775251023542,
0.5291693857421421,
-0.7225860054677886,
-0.6024865845145562,
1.1258609582335735,
1.4545810101615688,
0.4988359655120556,
0.39968483637691726,
1.4778005154845164,
1.6358502222921723,
0.07081811420225437,
1.5336807499569445,
-0.7595143926860392,
0.07947795439670366]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric4_unifrom_scaling_sample#10'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.4060332381961675],
                          [12, 6, 0, 0, -5.147683680770249],
                          [20, 10, 0, 0, -4.13060369641699],
                          [28, 14, 0, 0, -3.301960374777364]],
         'degree': 0,
         'method': 'richardson',

```

```

'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 4.267692565917969e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -7.14219439129857,
'others': { 'beta_coeffieicients': [ 2.1875,
-2.1875,
1.3125,
-0.3125],
'cost_richardson_zne': 11.390625,
'sorted_expectation_vals': [
-6.4060332381961675,
-5.147683680770249,
-4.13060369641699,
-3.301960374777364],

'sorted_noise': [6, 18, 30, 42]},
'sampled data': [ [ 4,
2,
0,
0,
-6.4060332381961675],
[ 12,
6,
0,
0,
-5.147683680770249],
[ 20,
10,
0,
0,
-4.13060369641699],
[ 28,
14,
0,
0,
-3.301960374777364]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [ 0.23818326834520384,
0.242015507589488,
0.5292100600091615,
0.5598196968290768,
0.5598196968290768,
2.3451854381830244,
2.3451854381830244,
3.8453025861748578,
3.8453025861748578,
4.395137112369032,
4.408833101975425,
4.829585475162733,
4.867115107997544,
5.058958687371625,
5.058958687371625,
5.058958687371625,
5.581243202757385,
6.011433944644334,
6.011433944644334,
8.358003932454157,

```

8.358003932454157,
8.358003932454157,
8.371420896509267,
8.783744346022278,
8.783744346022278,
9.34189587380724,
9.34189587380724,
9.39608576624051,
9.726727464435275,
10.95102910414735,
0.14721560345391302,
1.0205767628001141,
0.40564926087548847,
-1.9238321605823567,
-0.04729505851188324,
0.482541600628026,
0.1152155677775581,
0.8524320040134185,
0.42380767990264745,
-0.6219665889019169,
-0.05759638705357143,
0.8161328242190486,
0.7425402472775383,
0.7535392998290764,
1.8314745216576223,
1.3334023135030668,
0.05552193989453376,
0.9691374289368472,
-0.07338372949421543,
-0.6690664310785921,
1.168533104165511,
-0.01962836639606316,
0.6576122461884292,
-0.037637936325233226,
-0.0009647519828692927,
0.4524839512418326,
-0.013115244288784337,
1.6270372625341938,
1.182284078721198,
0.9533845712158122,
1.0956662941001023,
-0.9647918240366662,
0.1897778948773928,
-0.04047520420354409,
0.5790278460715794,
0.631719469583885,
-0.3335418357294817,
1.0603186443819097,
-0.7449471003423949,
0.1630231005006795,
-0.07563007210422526,
-1.1009042311153838,
-0.34587951137957407,
-0.22408475871037167,
-0.6364303490666092,
-0.5245531829065874,
-0.7807390048523051,
-0.7767575864285188,
-0.9679768325508518,
-1.8691843526791756,
0.4783530568164554,
0.6133546874260105,
-0.20020986269221935,
-0.41532071148562183,
-0.8557118152516873,
-0.16985308585729458,
0.8428338023785568,
0.6750667051632169,
1.2795031602880573,
-0.41384224674171627,
0.3213362409234008,
1.2340309605381181,
1.0506369357289735,
-0.6415500741723684,
-0.9628453188140567,


```

1.935214357876566,
1.4159154638103797,
-2.0864443113151867,
-0.07426020994129089,
-1.0538315394110076,
0.2589745455351295,
-0.2955978932252484,
-0.030644528099012087,
-0.8602288471584197,
-0.09490606960216369,
0.4190622694749842,
-1.5667390101922392,
-0.7804920738383083,
0.1255849265457167,
0.5884338319364165,
-0.46028089859616994,
0.2488277677353687,
-0.7513271615815256,
-1.0561258734551267,
0.3697237180501531,
-0.7108432439071602,
0.5137418203820399,
0.946767478727941,
1.619957961978311,
1.2944079622561346,
0.21625880379379211,
0.007168703139473973,
-0.334764182643075,
-0.9086121674985949,
-0.8247776105092918,
-0.5734460393428641,
-0.4381826298821608,
-0.44306191141826123,
-0.23860269421390362,
-0.03343496826073402,
-0.008972211649351379,
1.2196121588758198,
1.5985131140815898,
0.060325450296581484,
-0.7312434635304047,
0.9413542695101463,
-0.6701271691421761,
-1.0680925542186157,
-0.5192506372311365,
1.5594784335685845,
0.17928043514789443,
0.2455246384495121,
-0.053106843359071614,
0.7488220810664211,
1.3030207698426615,
-1.787387332504871,
-1.1184519607509558,
-0.2511275398302805,
1.445676213762956,
1.5991220174606833]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},

```

```

        'def': 'ising'},
        'output': { 'draw': { 'fig_dpi': 100,
                                'status': False,
                                'type': 'png'},
                    'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_ric4_unifrom_scaling_sample#4'},
        'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                                [1, 1, 0, 0],
                                                [2, 2, 0, 0],
                                                [3, 3, 0, 0]]},
        'run': 'zne',
        'state': 'dmatrix',
        'vqe': { 'iteration': 1,
                  'optimization': { 'algorithm': 'SLSQP',
                                    'constraint': True,
                                    'status': False}},
        'zne': { 'data_points': [ [4, 2, 0, 0, -6.630764461698827],
                                    [12, 6, 0, 0, -5.461578440310042],
                                    [20, 10, 0, 0, -4.498507346043123],
                                    [28, 14, 0, 0, -3.699540169020614]],
                  'degree': 0,
                  'method': 'richardson',
                  'sampling': 'default'}},
        'output': { 'exact_sol': -8.566772233505624,
                    'run_time_sec': 4.076957702636719e-05,
                    'zne_values': { 'degree': 0,
                                    'extrapolated_value': -7.305779010650624,
                                    'others': { 'beta_coeffieicients': [ 2.1875,
                                                                        -2.1875,
                                                                        1.3125,
                                                                        -0.3125],
                                                'cost_richardson_zne': 11.390625,
                                                'sorted_expectation_vals': [
-6.630764461698827,
-5.461578440310042,
-4.498507346043123,
-3.699540169020614],
'sorted_noise': [6, 18, 30, 42]},
'sampled data': [ [ 4,
                    2,
                    0,
                    0,
                    -6.630764461698827],
                    [ 12,
                    6,
                    0,
                    0,
                    -5.461578440310042],
                    [ 20,
                    10,
                    0,
                    0,
                    -4.498507346043123],
                    [ 28,
                    14,
                    0,
                    0,
                    -3.699540169020614]],
'sampling': 'default'}}}},
        'xy-ric2': { 'noiseoff': [ { 'config': { 'ansatz': { 'gateset': 1,
                                                            'layer': 30,
                                                            'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                                 'cn': [ 0.5,
                                                                 0.5,
                                                                 0.5,
                                                                 0.5,
                                                                 0.5,
                                                                 0.5],
                                                                 'r': 0},
                                                            'time': {'max': 10.0, 'min': 0.0},
                                                            'type': 'xy-iss'}},
'init_param': {'value': 'random'},
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': False,

```


3.800207092295721,
4.798448841518561,
4.812026604031267,
6.02543601067661,
6.02543601067661,
6.5265343452987,
6.5265343452987,
6.526534345298701,
6.528381158760148,
6.528381158760148,
7.113023968622042,
7.113023968622042,
7.593654062389684,
7.5949106716383605,
7.780058793305066,
9.000370275113886,
9.005799799983558,
-0.22105281285875572,
0.07196296170715184,
1.0527781347416576,
-0.4594806493129553,
1.0526039239081737,
-0.01709270509934428,
0.3199675106146978,
-0.0972220164715271,
0.9145097677998977,
0.09808103992344623,
0.5373974901276602,
-0.5786059880820694,
-0.02594763303680847,
-0.22132394348491485,
-0.7274785930689547,
-0.41090279602128,
-0.520884263329148,
-1.0260835302846771,
0.1692597388493397,
1.3250313995601488,
1.510357845739394,
-0.9233604808876308,
-0.2049157632594096,
-0.9894281051832,
0.25772120620526445,
-0.04822142329545429,
-0.3912638297585932,
-0.48486712444934904,
1.4836649713082644,
1.566996182191827,
0.08093959953023218,
-2.3562926055425963,
-0.21876618997457167,
0.10911407834571141,
-1.3776955808248483,
-0.31816531211057325,
0.7083857256813267,
0.5593482503842677,
0.7483381318097894,
0.010042025170353972,
0.05290425882351868,
0.20616876613540233,
-0.8673924682871486,
-0.08017548043713153,
-0.3056222778255953,
2.3981163154468383,
0.37079365521682633,
0.3870558721039623,
0.19341028278759992,
-0.3154248344945196,
-0.12073260037561208,
-1.5858414285227351,
0.59830347262823,
0.30388710178795286,
-1.4558796844932331,
-0.3225618511944297,
0.09243390755822743,
-0.0006426173071943948,

0.7561662109631148,
-1.3902324585417218,
0.1677701056843372,
2.7311884232201735,
-1.1122880815211051,
0.0467986017971284,
-0.020089420529231013,
0.005481851366742631,
1.3205596999819047,
-0.1290403664286252,
0.3870444394307043,
-0.6549197738372419,
-0.17656835190090397,
-0.0008013451895215781,
-0.8769883624496301,
1.6929520910231297,
0.7288671099890552,
-0.6169673343691255,
1.667217439782634,
0.2554635335546482,
-0.029515171705053127,
-0.8290153205719569,
0.042733616172741226,
-0.5148227135977747,
1.2301836750084596,
-0.9175908300980525,
0.5229779878549122,
-1.5901934710811256,
-0.07316585886439231,
1.1195018365185199,
-0.31111306362203306,
0.7254709796877604,
1.1995417479543053,
0.6908225815252583,
-0.2919630814941808,
0.15955998483086176,
1.2871422030548714,
-1.2683991057896014,
-0.3768598572966417,
1.4863954796333156,
-0.4300298743216642,
0.19151460300148976,
0.9665838873771405,
-0.07244306377732504,
-0.04169546816073936,
-0.389029635704794,
1.5218961812832925,
-0.1772066326032777,
-0.70949107362336,
-0.015222466074935298,
0.3587128629768407,
1.5809494716454662,
0.12428164288740734,
1.5083554487395696,
-0.044379618902227067,
-0.03505579308192811,
-0.6786538117837789,
-0.007428157845337949,
1.5281494732580478,
-0.14124856089608578,
2.726718429443767,
-0.003106328084445967],
[0.5270475437101116,
0.7815425625485376,
1.6469610445191378,
1.6712876404917778,
2.122904188877425,
2.131106816606521,
3.7367798442214175,
3.7367798442219424,
4.750276145421292,
4.853973384996016,
4.865098072058574,
4.873980651141647,
4.987599328262889,

5.00246239884106,
5.520902021852277,
5.793284849001337,
5.8087832398624775,
7.432729128414403,
7.432953787975026,
8.144896858715603,
8.172580744083415,
8.788071761144659,
8.792288306719156,
8.828260688191317,
9.011586808740137,
9.011995976067807,
9.538729699920484,
9.576884709106812,
10.486750161390887,
10.48675016212107,
-0.15531374899360717,
-0.0849039080739262,
-0.8434555658556069,
0.9447826609611266,
-0.19482082661427938,
2.4999420394249707,
-2.7164831941210483,
-1.5408169364026967,
-0.8621482593978547,
0.31547458537730877,
1.4220435826823787,
1.348860553316339,
0.40827474166943256,
1.063960383917828,
-0.21906798993221652,
-0.33263312803097533,
0.4889869181158972,
-0.25524084746075687,
-1.3622146811064544,
0.996105149433398,
1.4161210728452394,
-0.6491827941870941,
0.7399329880002848,
-0.11994422281276146,
3.179573825543474,
-0.034943024466848674,
0.6075464711124173,
-0.3019825794696199,
0.029392952712390567,
0.14840732209332994,
0.08592470568620861,
0.016725634393736574,
-1.3697962205589145,
-0.06155375586805023,
0.16508080354919497,
-0.5975014914051775,
0.09635736997937351,
1.254101402191796,
-0.6193953337438176,
0.47602269510847217,
-1.7251540483535392,
0.5378274350019089,
-1.0159384367751108,
1.0968223266795405,
-0.79878134010976,
0.10468891520322528,
-0.2910173593847854,
-0.7078981306941186,
-0.3510837282628223,
0.5371002854277007,
-0.4911274293534503,
1.7273270693440552,
1.3500555714793954,
0.9052059235505011,
0.4771526028272018,
-0.1893747959421651,
0.13989050072973697,
-0.5930955187300607,

0.36780210947560876,
-0.5043370473487999,
-0.4851893292987325,
0.1987440888222681,
-0.5806445532873666,
2.3171646404124093,
0.06032126128170586,
-0.6389920437730593,
0.9985359186149056,
0.04870155999532171,
0.16447399991313041,
0.05315090520136875,
0.15727255736970833,
-1.3722292706433088,
-0.33233649668322635,
0.7758243461945761,
-0.20476613404890107,
0.06972046040604139,
0.3143879960336948,
-0.01452323353492752,
-1.2901366074359686,
-0.31033000747276407,
-0.16264485166390844,
-1.3914094473719103,
-0.5273128620655095,
0.02465459151953875,
0.10942766294132272,
1.0292094835171124,
-1.882689768994833,
-1.553216293090225,
1.3287846024814622,
0.20963913600213377,
0.7195732642990699,
0.0969787732183028,
-0.9744344415633356,
1.5822487402774768,
0.7077471503862973,
-0.766345302327988,
0.3831437091022152,
-0.12028853613988252,
-0.8209919023748664,
-0.21164805512089607,
0.5290684330656528,
-0.1040878870845114,
1.1956945887281514,
0.25703931501419025,
0.6344733295912399,
-0.02497584874729833,
-0.20937697561532842,
-1.4409411147319051,
0.9707320454927133,
1.4039378085761043,
0.9760826987262126,
-0.17314228405542512,
0.16734556762922406,
0.07881248116464283,
-0.2924467918405051,
1.4946430081327182,
0.7516806171134963,
0.799700486923224,
-0.8440600324500036,
-0.01069194371181784],
[0.3388610072211759,
0.3930923343746358,
0.4059472295013711,
0.4597995545870322,
0.4597995545870322,
0.4667083780713653,
0.9924683364283498,
1.054118787032341,
1.3839145947816414,
1.4469616998432806,
3.044048229936016,
3.044048229936016,
4.682975233106521,

4.682975233106521,
5.287925895396742,
5.287925895396742,
5.625677830869467,
5.625677830869467,
6.519005703020383,
6.519005703020383,
6.519005703020383,
6.854665591577039,
6.854665591577039,
8.092962128458302,
8.097830997780322,
8.102159968176109,
8.111954675742565,
8.727038821401255,
8.727038821401255,
8.784424594255395,
-0.8636093730812797,
-0.04742973062196874,
-0.37989856483295564,
0.3750798322441465,
-0.509087961968837,
0.5366875438436245,
0.1232704940065221,
0.5249134835872401,
0.13594138148710633,
0.10802292582448766,
-0.7156772546121367,
0.34756877087651405,
-0.5704812075854907,
0.289395233495819,
0.2674862930849357,
0.8724143866961164,
0.5770602988816299,
0.658777432128999,
-0.47288232304292416,
-0.029663566814993374,
-1.323180236232167,
-0.2890901656039231,
-0.015141732930961212,
1.1846850771343012,
-0.4506688471965015,
0.22241980016839272,
-0.015748484493019736,
1.1522499025192263,
0.2996907655672081,
-1.7370631162182948,
-0.35306232532663057,
0.17322471105923057,
-1.1855180134814352,
-0.31537295018088835,
-1.8120366955290261,
0.6663892425075478,
0.21874911670355193,
1.2729693967005706,
-0.14230256773273223,
0.2012427441067739,
-2.3269317425793643,
-0.03299848401271634,
-0.4074853136565566,
-0.30371905431664253,
-0.12037074206123735,
0.8456676988075262,
-0.21838328071319946,
0.019494764922420988,
0.16383099728733078,
0.056056403895700845,
-3.5632063709291613,
0.6579904510057321,
0.5207861046034454,
-0.017180990359968364,
0.2532807007765998,
-0.03315495616728509,
0.7564905992030696,
-0.11091568858030387,

-0.9281890577202553,
-0.6082181315906088,
-0.4480511658411004,
2.1953283079942025,
-0.30446713497464617,
-0.03359602431311142,
0.28972943746372876,
0.17817064395522006,
0.7965754083210965,
-0.9520780563921613,
0.10150445450145726,
1.3510884636720888,
-0.37849615173878054,
0.17729737647006832,
0.1884683806301832,
-0.03234547530918763,
-1.338763456177623,
-0.6711058962200585,
-0.012976043431998471,
0.027578392175021227,
0.10161286285089129,
-0.00939135247962935,
-1.302164623174096,
1.3715576399721376,
0.737767307868418,
1.5939269674119705,
1.4291758244230544,
0.5601859764306084,
0.3488764286390935,
-1.093762224545935,
-1.1976831420922067,
1.4557381665683378,
0.8924472305311927,
0.5300221300338881,
1.0785280788144638,
-0.6029509417428172,
-0.6790632925762424,
0.4138865114731201,
0.2917474006869537,
1.8050154062536312,
0.05177383423831921,
0.25467777497912664,
1.4671356779039024,
0.4422247342165525,
-1.5431607378377095,
1.8101814873938125,
1.6870467346899987,
0.7517920825884995,
1.3622912362338087,
0.13653050459472865,
1.5210183141257738,
-0.004247168330480351,
0.26137218980413907,
0.5451511562235306,
-0.6705596749039198,
1.2136085126593865,
0.5231529087420831,
-0.03279466120532818,
0.2548954759886831,
1.5797820110558691,
1.075840789720537,
1.5541818084195007],
[3.821020346740443e-17,
1.386496864345847,
1.386496864345847,
1.7951472089989222,
1.8559400139285378,
2.7980544315911504,
2.7980544315911504,
2.82954781812891,
2.82954781812891,
2.82954781812891,
2.956183634338348,
3.840561875918348,
3.840561875918348,

4.79501776182173,
4.79501776182173,
5.670460018404704,
5.670460018404704,
5.670460018404704,
6.865392250052202,
6.865392250052202,
7.506826147384778,
7.506826147384778,
8.737685173635683,
8.737685173635683,
8.737685173635683,
8.737685173635683,
9.601023092540274,
9.601023092540274,
9.853738132880634,
9.853738132880634,
0.5611658219809542,
1.427331325859137,
0.8881601644240952,
-0.8937497791603879,
0.5712044928850715,
1.3195951619089188,
0.5467792160605555,
-0.6816630744322637,
0.06707477836950569,
0.6798692668930644,
0.19576168860111362,
0.5094724850631845,
1.1424414477256812,
0.30471145260454696,
-0.08158911485577636,
0.9585661478313002,
1.0836238583568754,
-0.3718197048926503,
0.7841948506971846,
0.3023515457437066,
0.7662240328883452,
0.9970048707689876,
-0.8946064024093777,
0.27649390200762114,
-0.15342626519190924,
-0.4015720487025046,
1.460926694270634,
-0.055421168101734586,
0.6077050880702092,
-0.7932081335349191,
1.3188332541813446,
0.01929555634963922,
-1.1301845164589392,
-0.1718174564221745,
2.26154163970611,
-0.9882028269623031,
0.8038643847286138,
0.6849304102208338,
0.874798983022982,
-0.016486064274954207,
0.0264221098096002,
-0.3625000384278827,
1.4376331477886695,
0.8499992740594865,
0.10042171561200945,
0.14542648933357244,
1.3513090878714353,
-0.22248922869259902,
-0.4483175327546578,
1.2345758678942387,
-0.18983522552694923,
0.019591210301167533,
-1.3607248526627624,
0.018375533581887468,
-0.3730738776375715,
1.0123537416913542,
0.7412601688930508,
-1.6525919628732422,

-0.8341682190885938,
0.04519729634722039,
-0.06718028327556408,
-1.5649719285752928,
-0.37531851249345394,
-0.11822841604899964,
0.008154106549025474,
-1.4601449692077721,
-0.0030993446167428395,
1.462573870839384,
-0.11887757571245174,
-0.09361960188417112,
-1.2543408854038138,
-1.525280802664792,
-1.241413372948363,
0.008758808605552393,
-1.6676332942965022,
0.13826170532334445,
0.7537363962746654,
0.0833397476047736,
-0.12216949704941273,
0.02084807234445201,
-0.3421949354726438,
0.0017882611822728064,
1.5186851123669147,
-0.4603163345929323,
-0.09956972276089109,
0.7247597830417082,
0.5304935691839561,
0.003193152207841171,
0.5581346990750625,
0.010542051444639388,
0.7553167297004638,
0.5956891015800863,
-0.05566254332595716,
0.5362461335652635,
0.009895282042162012,
-0.3029901133286735,
0.09399206534914867,
0.8818812838713691,
-0.06795832284115338,
-1.580933930101608,
0.3014975711017802,
1.2092022074751856,
-0.019692531397995948,
0.9250981578351692,
0.4673383297638096,
0.03344339272872125,
0.13058472572385768,
0.036262684184263205,
0.458095375934523,
0.6492846129896516,
1.3039221898344189,
0.024764812872192767,
0.7236926243101844,
0.01444840086266692,
-1.0388939706148563,
-0.08359693344439546,
-0.4698004238001188,
-0.48055768545836347,
0.4662000815300196,
-0.041204969991243484],
[0.01798320141016013,
0.3393419127512466,
0.7047697703765823,
0.7064789163091142,
0.7191656785855808,
0.7198672068604162,
0.719867206860416,
1.0840512961729545,
1.0900003904895101,
1.1926153174924548,
1.867054342490295,
1.867054342490295,
2.368094916985588,

2.3683751276938714,
3.6383003552329964,
3.642527762446636,
4.505845683894636,
4.602446412902637,
4.972427659100928,
4.972427659100928,
5.1148164150678035,
5.8066136392845,
5.820681023327439,
6.727088267818048,
8.352653016300794,
8.773334122756978,
9.065902540064004,
9.065902540064004,
9.74880092287043,
11.30145464466351,
-0.13904770480884485,
0.23493722133539818,
0.4497803172772882,
-0.640584995353188,
0.9779304528121034,
0.0011745349607135795,
-0.7175493520168287,
0.373620380836484,
0.08111255661159392,
0.6653458367314794,
0.0037967443269442437,
0.6286122183467432,
-0.03825426016359485,
0.6896824961465278,
-0.23916946602734285,
0.4624850989736848,
0.1419493525208794,
0.7145176662978019,
-0.316909964913161,
0.6605458698076423,
-0.39309909050950365,
0.7623048158369822,
0.7469736169282462,
1.2400892032597353,
0.12266455668869777,
0.19578648318487166,
-0.784043232212839,
-0.10334183265831844,
0.32774480702026265,
-0.011491906972883237,
0.322988869457402,
-0.2027870428092001,
0.1281571522083961,
-0.1947961113336468,
1.7114513053475526,
0.005425684515457419,
0.8136825826883561,
-0.25903007981373005,
-0.5015428579892063,
-0.2600978115589891,
-0.15405473476720474,
-0.4207498663011112,
1.04622682040645,
-0.8642339256273757,
0.6248691295250803,
-0.3539240130838131,
0.011689879721248103,
-0.3068283940685192,
-0.6048660879777619,
-0.11704784670887185,
-1.2745062338511486,
-1.7890038533223112,
-0.7818671277256136,
-0.8877455887286905,
0.3164818198590212,
-0.27477307924208777,
1.8083251930535746,
-0.5031479272825263,

-0.9648993597903888,
-0.41682962131770707,
0.27040705073896654,
1.0892900815418987,
0.17779960873468054,
0.020337033950376556,
1.0551529464315492,
-0.10705558118147128,
-0.8851977322505518,
0.2232945485845247,
-0.5850403569085746,
-0.7690595025978197,
-0.4391673030332381,
-0.9309566421919284,
0.2973030450025941,
0.6780545449896137,
0.14045373030826291,
-0.5109313900859794,
-0.44070010469428217,
1.6940975579604587,
0.17062304524159472,
-0.402681393200546,
-0.7091187255059627,
1.3132357546586793,
0.2729391261599054,
1.0115940175088693,
1.0046673925246778,
0.4982257464819494,
0.04454679168285006,
0.4863023994043421,
-1.1433261311581624,
-1.820308437409531,
0.6035405700431521,
0.04003907881711185,
2.99381748600971,
-1.0437523456620463,
-0.22517480393070408,
0.10746937069246698,
-0.8788513722547997,
1.286531186523234,
-1.3157160458746164,
-0.6979748907629274,
-0.16573593981190854,
0.2529194783090185,
-0.2562517800846549,
-0.7966921479109774,
-0.4446940958312971,
-1.052310031662883,
-0.1506245930250076,
0.1615975776996014,
2.9220364155549214,
-1.7197843955783065,
-0.6592103621593355,
-0.2007498223803255,
-1.6188873615119244,
-0.15193115939751467,
-0.20866967970071015,
1.2910791163230286,
-0.4917152660712325,
-0.42348313202027765,
0.3518303936962707,
-0.10034486495238693],
[0.009039772647041902,
1.1400526086001903,
1.466905028683042,
1.6524670085313564,
1.987797816914966,
1.9900950346150876,
2.1413202332332344,
2.390935514217652,
2.402429399816812,
2.81913686416337,
2.81913686416337,
3.5656810200683298,
3.5656810200683298,

3.5656810200683298,
3.76603447051701,
3.76603447051701,
5.499236323276389,
5.4992363232763894,
5.4992363232763894,
5.532789537471719,
5.532789537471719,
7.371733777832876,
7.371733777832876,
7.680783486275433,
7.706623781446164,
7.706623781446164,
8.457742149729329,
8.457742149729329,
8.95222736351662,
8.95222736351662,
0.18928393791812126,
1.0354889043700823,
0.3703113207409527,
0.30988589523494625,
-0.34669340410550803,
0.09615284519636855,
0.1011898884094612,
0.6160115432573767,
0.6148540393435122,
0.07934943483664206,
0.8557624329627399,
-0.7672633290674925,
-0.14188111026081618,
0.1995320641320393,
0.4594888802421805,
-0.03593187081620776,
1.2926591415659605,
0.31433950962447343,
0.8053272546604554,
0.6634815559763872,
0.311660786273531,
0.0294653444200854,
0.3044394736040779,
0.7599259037372076,
0.22962735656034647,
0.22562971232722423,
0.29511485598499815,
0.02260075636691185,
-0.9497096253675098,
-0.5331371571917215,
0.2192512323868535,
1.104832430155296,
0.8636617903941131,
-0.6756684630428217,
0.4169844309263729,
1.1297527651723116,
-1.160118439748996,
-0.7971836651224,
1.9169656561991728,
-0.007260796290839314,
0.3614132074474742,
0.6243022778229537,
1.8335025630018775,
0.35139722497115833,
-0.7247870129359816,
-0.6744811297738377,
0.49997538785346773,
1.4139521212248116,
-3.0799818401546553,
-0.14768564076593044,
0.06251500852230953,
-0.8547134606957393,
-0.30107856776262504,
-0.7654972812734481,
0.28603020918627825,
0.2129303248959857,
-1.2077141264570097,
-0.8488549318812054,

0.6492574804166855,
0.7751290676155554,
0.40909519807061084,
-1.2750630737105486,
0.4187798718836698,
0.33498424523924863,
-0.19017028209895162,
-0.03232931911977397,
1.0621483440417678,
0.452017889610544,
1.0192767515800205,
0.7290563184502946,
-0.5286462788982386,
-0.05512372609304388,
1.3189151919853837,
0.5865235617306253,
-0.579843833868396,
0.6980158293666034,
0.027004860579276476,
1.2876675233834314,
-0.1718561727774212,
-0.3827017785428823,
-0.7286404996479297,
1.1682112525473984,
0.22765379123755217,
1.4870969768845252,
1.1642898801914527,
-0.03976972562460009,
-0.8163296820868416,
-0.15318057329999143,
0.0482918949035788,
-0.06014345828968787,
1.1478583262090496,
-0.005141116181289305,
-0.5820859555621626,
1.086519910329415,
-1.08558016317555,
1.531511042803357,
0.042734048269580674,
0.7452599788242152,
-1.3094923903018465,
-0.9012205175381317,
-0.0755166851159581,
1.5493661520767013,
-0.2278023712186688,
0.8627212805492942,
1.8523276191392382,
-0.10961709979291216,
0.6746606852631131,
0.22566845572735392,
0.5920154791154506,
1.2709342533151342,
0.26903045177842116,
0.11724485476711638,
0.621942277674204,
0.15491924869980966,
1.472217084758381,
0.8975149794611956,
0.3601026339731485,
1.3234513983254743,
0.0182465793142515,
-0.11914786829986494],
[-5.765348397008027e-18,
0.3428984748224433,
0.3655450682584724,
0.704784031551808,
1.5409081472811441,
1.549300366088437,
2.3787666619954635,
2.3787666619954635,
2.8808650819472406,
2.884987740396357,
3.1308808834009203,
3.1363836212549696,
4.869321728501938,

4.869321728501938,
6.1150174264690085,
6.2582278136815175,
6.271636236200362,
6.551203581243674,
6.59741701640448,
8.230564833747417,
8.294003346338926,
8.31431753208387,
8.369586036193128,
8.415319617883744,
8.70447046242739,
8.709120596281233,
8.918659758853744,
8.99410216914391,
9.002256766523654,
10.218690251738174,
-0.6798627919679444,
-1.1436724543472427,
-0.1278269606092795,
-0.9286552360068459,
0.6524046490818174,
0.06134572949387961,
-0.49761179900021657,
-1.3918471119443894,
0.5858287319319103,
0.18669127045720346,
-0.4586789737568629,
0.2400608953322058,
0.022222124883641912,
0.9282004764805514,
0.6952358394741303,
0.805684979006905,
0.6146729452742722,
0.6777524139082742,
-0.8117842884117104,
0.3426849285038029,
0.7100299235979358,
0.7658186509139059,
-0.05418633990950139,
0.1761917490887145,
0.5744825473612316,
0.31705937472296297,
-0.9188523464356001,
-0.7337446596965294,
0.2061368312243915,
-0.6645694031850036,
-0.009871104765786294,
-0.14388187083546433,
-1.2485707106311008,
2.849540809212292,
-0.5119215846255833,
-0.19381894062193808,
-0.17310717674563775,
0.8465397629359774,
-0.46757240698237135,
-0.05607605828560263,
-0.40986137011816176,
0.07649505002491204,
-2.1522907800348112,
-0.6653346582594515,
0.4084801543759574,
0.4028897820911683,
0.16908117843954765,
-0.07217026591146015,
-0.72804755296126,
-0.13548738477012795,
-2.590679566047418,
-0.40743076961744734,
1.1030856800887578,
0.08388380440455438,
0.25224805905642333,
0.10644300798076774,
1.0474125729404944,
-1.1459016177859003,

-1.6916287310468103,
-0.9886067106540645,
0.24182274960650893,
-0.25703819491337826,
-1.1942818044595898,
-0.7763678643520989,
-0.916718557142268,
0.20086626876005667,
-0.3711023890817823,
-0.006962349053602254,
2.687031223814704,
0.3970791704566904,
-0.6890591654410289,
1.0504536719091986,
-1.2458238157294268,
-1.197330311624414,
-0.12155955689525438,
-0.2969718406476498,
0.15406921803787998,
0.9984735505197111,
-0.2734456365103056,
0.0100494123830882,
0.16172873483083283,
-0.6780409734277772,
0.4539997995464368,
0.7558576390626953,
0.9556605206819043,
-0.10572075101403118,
1.5564628686733721,
0.05397552837650759,
0.7415987024402694,
-0.1648500682327342,
1.052699139325594,
0.6776572023171561,
0.15219630541397192,
0.8843933196559655,
1.3948099389813178,
-0.16710315713046545,
1.2199200746863228,
-0.3825678575328573,
-0.26766985045935626,
2.5777182352113908,
0.23156136798610805,
-0.3377548185037014,
1.0689111835439342,
-0.17216022970038714,
-0.29381807805976945,
-0.5689533077540762,
0.30596827309967156,
0.5260993417881752,
-0.6768480303494689,
-0.1092430795457648,
1.5442082488768039,
-0.10826504240549732,
-0.9555598065181787,
0.5334835330029429,
0.5569227134357198,
1.2665483894271157,
-1.9677332823608626,
1.5559931182978979,
0.7709453943956464,
1.5746506000831744],
[1.7625089228179507e-05,
1.298540182320117,
1.3141870420194457,
1.3305471045820527,
2.0686735179487985,
2.23810597518972,
2.267373737425337,
2.295441377067579,
2.3325388050051976,
2.352484360014207,
2.370891483796666,
3.3117124844820194,
3.363605896627787,

3.858981460194034,
3.8589814603490344,
3.859537867870235,
3.8595381713444072,
4.26458571773699,
4.264585717738593,
6.599549149586233,
6.655963438031886,
6.939612414177192,
6.939612414268938,
7.004302986969205,
7.010544991093791,
7.97314828104894,
7.999280802101656,
8.022628296732579,
8.027896471579398,
10.077254342713585,
-1.0733166032270363,
0.4264387306682,
-0.29490479134596703,
-0.45181113351566254,
-0.9496712390342118,
-0.24412788786515252,
-0.17005038539710493,
2.131747001055919,
-0.8801477965160032,
0.1985900676052623,
-0.17829180427317373,
0.1513896817808816,
-0.2996418627583153,
0.09152392337611555,
1.0155053498240294,
0.6713280773462803,
-0.5915672184524843,
-0.21780814909773427,
-0.23676595566314207,
-0.1539223229682568,
0.9733737893230769,
-0.8325690111887051,
-0.6684349693104433,
0.944923850675747,
-0.8270803600282979,
0.4869668407293288,
0.07342964610078141,
-0.8928903181234079,
-0.04262248204617692,
-0.7602498290394286,
-0.48858825463899247,
1.7340528561515622,
0.7340502315717261,
0.5163401187759543,
-0.5015858040228415,
0.2890335055344555,
-0.46033506221820203,
-0.8185907107929625,
0.4557079379705311,
1.6601520871164626,
0.07125306723268252,
-1.4035119795986863,
-0.6741575848895364,
1.84360490583597,
1.6836338044245691,
-0.33329370458317326,
0.19854317161902654,
2.03917897859608,
-0.3342251938073038,
-1.6015864477635742,
0.35422488479060155,
0.1487121545506934,
-0.2821114888507261,
0.21869934797564045,
0.7523961573071125,
1.3021785537923125,
0.26184859002187644,
1.2441238228144538,

0.039670539596101485,
-0.15687777352236798,
1.1386642271664005,
-0.13559116971858365,
0.39167499360215746,
2.4453261619465274,
-1.0890320328395309,
-0.4289422005081893,
1.425249318116838,
-0.0016733698174337221,
0.023121567661497415,
-0.28325087531692295,
0.940069260590291,
1.163437407349777,
-0.4256527399896147,
-1.4972731615803316,
-0.12624975701746496,
-0.10144651691714318,
1.3307378561466523,
0.041778278155808674,
-1.2746374184036902,
-1.2216848741140482,
1.4048769664929757,
0.8957112974734468,
1.0058068981677006,
-3.0965653057688316,
-0.3344448337959029,
-1.0948102803209163,
-0.038478236121346335,
0.12618687504486675,
-0.0850775611793361,
1.33550885346982,
-0.04334772196309596,
0.2406908654957122,
0.7327249608913058,
0.9966290850109706,
2.2852565843192423,
-0.030570948934034583,
-0.5566407461361556,
0.813246653395058,
0.5946355273642772,
0.07519254658407079,
0.27120989268694123,
-0.15415722381241237,
-1.1239288818585278,
1.893751281002709,
0.7530205722182245,
1.4454588921082565,
0.15037300343080462,
0.1793867334099716,
0.3587259643057371,
-0.006799349268608,
1.0907160355013774,
-0.30186717991666434,
0.8507589785411056,
0.009155709268365956,
-0.8165113788656561,
0.011816515180360782,
1.116430418883973,
1.5343789763879045,
-0.1682835166498154,
-1.5032827361947747],
[0.10956969985098665,
0.4360115975982133,
0.5932904704915389,
0.6050451872391112,
1.3688142364348175,
1.3737597367805747,
1.5226773273508583,
1.5589271816733294,
3.150320547140947,
3.1613053536766147,
4.7635234433865605,
4.823148382609575,
4.823188243925802,

7.171157860626202,
7.171157878485542,
7.171158529865571,
7.175073431930883,
7.183070822592498,
8.102400589897519,
8.10392263292654,
8.103940780725882,
9.021308924987686,
9.02133947231205,
9.082831297490506,
9.094868565409332,
9.236804949782064,
9.237029573386183,
9.26430291432718,
10.00744744746223,
10.032040374677464,
-0.5042382729215367,
1.0180015436638659,
0.10423420212375085,
-0.3355687578372037,
-0.3902319222280216,
-0.09867602213718178,
0.3860296418386605,
0.8522828003851446,
-0.7730596965193267,
0.08803782521712378,
1.175277695996097,
-0.3769225558693855,
-0.4553519820435133,
0.4758962487375975,
-1.0517577693418567,
0.3712029741888282,
0.27334544561774066,
0.03397030216872572,
0.3891396532807883,
0.12152393226873999,
1.5192516885791116,
-0.5124931636551364,
-0.5667003913709334,
-0.08425393582345661,
-0.8152956002657638,
-0.008087141654348068,
-0.20833380477584987,
-0.19186344764752156,
0.309463849209865,
-0.16555810903618162,
-0.28534485935787846,
0.025710068868521854,
-0.9172960495760906,
0.027100436879331523,
-0.9806057417125867,
0.46877133554227174,
0.22597640762440013,
-1.8700902599315918,
-0.6222133609404971,
0.023922176634194332,
-0.6777524378245746,
-0.3883818313368169,
-0.19921738892417568,
-1.3418496115619054,
-0.05365669225714106,
0.38835985529462186,
0.46741217514915123,
-0.1388983196762152,
1.0002905156718758,
1.6633296566254185,
-1.0097136763815113,
1.203957772036831,
0.34033273848251194,
1.1456357751380877,
-0.4637199651146574,
-0.13624703361320328,
0.6785440200843139,
0.21691345067474185,

-0.4166036808107477,
1.6003722740710606,
-0.294881370724797,
0.04198878790454511,
-0.689532047804753,
-1.3911901894570817,
-1.1667039833298627,
0.824824800006074,
0.8047687705752488,
0.5727196814509768,
0.2385364014798047,
-0.512164832149354,
0.17925141295137917,
-0.19704824673887428,
-1.6372757558452697,
0.5441675736491897,
-0.13936070677778437,
1.3935144064781402,
0.10763856884501717,
0.6679788244416961,
-0.33118654939932857,
-0.35796695002153883,
0.2632208027041665,
1.5831503864671255,
-0.25140146740224,
1.135774876905931,
0.5897962040478675,
-0.7664904031063712,
1.092649076983624,
-0.9862462517151122,
0.010216869101717933,
0.5705833807099128,
0.09839127014019662,
-1.034770130526523,
0.20582133073730807,
-0.44428721001967564,
0.5272444819968604,
-0.2627660867388947,
-0.9857421779520247,
-1.3646538516413942,
0.14779052759756187,
1.1207932022043874,
0.30427416833245535,
-1.055109005337564,
0.30548258474769546,
0.8153937738463674,
0.5082537250510629,
-0.2973639305797175,
0.0034742209991892575,
1.5168338311712208,
0.0971904670135119,
0.5305672799342332,
-0.08170159551874663,
0.4909203029282298,
-1.1771354207712392,
0.038725619801271274,
-0.6655680746562402,
1.312626970471223,
-1.0115568894576918,
0.16290676730453105,
0.3074286838975462,
0.013117345918778788],
[0.4866654913190249,
0.4866654913190249,
1.4314116389487666,
1.4314116389487666,
1.548608655381947,
1.5552664901547235,
1.662920497545568,
1.676155881591226,
1.6952532774325564,
1.6952532774325564,
2.46741851740979,
2.50102877426778,
3.636145431120801,

3.6361454311208004,
3.6361454311208004,
3.6361454311208004,
4.0161273218482085,
4.0161273218482085,
4.5411636896776235,
4.5411636896776235,
4.922545565757144,
5.624338470516907,
5.642847901355697,
5.904037934138086,
5.915382265322609,
6.860930229950661,
6.860930229950661,
9.422517992567101,
9.467623480195014,
9.467623480195014,
0.40091998571272736,
-0.3581846438287994,
-0.32033246941904975,
0.24409866553117068,
0.176560178011825,
0.4452285262890565,
-0.37150326481454765,
-1.0185408309777293,
1.6197684873006286,
2.0095194673452026,
-0.20342804182482532,
-0.4471079471088122,
0.37127155242298737,
0.4199607045713712,
0.11145558426636207,
-0.289968410859463,
0.9752004498173846,
-0.11972987215087512,
-0.8215210043660294,
-0.4144105549344152,
0.04966095047933562,
0.6091224412599766,
-0.2481186610805342,
-0.9036735819055022,
0.22922493224752044,
-0.04567884568114995,
-0.40529506738584414,
-0.44609185937563545,
1.1000542207972743,
0.8995834380092431,
-0.4391184290327733,
-1.623814742575614,
0.11121717167805684,
-0.15344289124808663,
0.7072346226342329,
0.013911394934765222,
1.1008239636292285,
1.1342299046285917,
-0.1605718532112884,
-0.8521154510350495,
-0.9726520318623231,
-0.11078467990736285,
1.2893067940177063,
-0.8656847782514395,
-0.4554157535410911,
0.46752563535079134,
1.0115326820592876,
-0.017465678177200683,
0.7555243091795097,
0.2225241128439446,
-0.6129237770264853,
0.36249038141661666,
-0.1470816778834455,
-0.23781939745964606,
0.7410640700988803,
-0.10219672891678527,
-0.9043863030697507,
-0.054446492384892985,

```

-0.23618927113386157,
-0.3971482428910102,
0.42980979959963467,
-0.4511512430809565,
-0.3322123782853245,
0.15031459204805087,
-1.4287283715133035,
-0.026859055017604255,
0.20826999854501216,
-0.016397767746330064,
-0.026711091547853946,
0.15899307013846248,
0.18349959987180348,
-0.019181103141429404,
-1.3777401647916119,
0.05578672583947999,
-0.17191397249005735,
0.08492618339903452,
0.2781511329827625,
1.6853098582374997,
-0.7964285169174421,
0.13628687772581302,
-0.26524580358677924,
1.3615146942088054,
-0.18163849035743365,
-1.5476669195398023,
0.2740670870741208,
0.19740142577560438,
0.18567840551061798,
0.43288784764736754,
-0.8069341399605817,
1.5139336080928714,
-0.7594440385529729,
-0.20443587392639553,
0.0916888700165231,
-0.03922249489745327,
0.4135801157921025,
-0.06731598775715468,
0.5038720973518589,
-0.7705715736697758,
-1.4775324565516457,
0.061950004909204714,
-0.9555167624183505,
3.1277455162888304,
-1.137887484425651,
-0.09852947784569874,
0.042597844143281625,
0.06134264537966588,
0.3384848933516047,
0.012002169875906457,
1.5732417764972468,
-0.3550969359952667,
0.37821230776206083,
1.4914211446267915,
-1.2027836619303884,
0.22421848814312553,
-0.2072711571643093,
0.05122655754886873,
-1.5578500765322636,
1.5918641150141506,
1.2635397829742223,
-1.2452354025631063]],
'run_time_sec': 43720.81102514267}}],
'redundant': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}}},

```

```
'init_param': { 'value': [ 0.5482907189468895,  
0.6413620809492503,  
0.6422482149552934,  
1.5892409709442974,  
1.850708229814058,  
1.850708229814058,  
1.8745552660050262,  
2.400679867939261,  
2.402212278134192,  
2.485190345183722,  
2.485190345183722,  
2.4851903451837223,  
2.488397493491329,  
2.4883974934913287,  
5.077476636438464,  
5.077476636438464,  
5.732990197214016,  
5.806868622878696,  
5.817834245705687,  
6.166757373752113,  
7.3114292874931905,  
7.311858449036949,  
8.346734585466223,  
8.350435170229554,  
8.354614694314481,  
8.35659817980371,  
8.356905245901928,  
9.313534396314816,  
9.313534396314816,  
10.443787047737276,  
-0.5971318801423705,  
-0.6830583120471296,  
0.053446928626708114,  
-1.488251087277441,  
0.7409998345617078,  
-0.020561007687691084,  
-0.47029189134787985,  
-0.9187802118512801,  
-0.07390570205801456,  
0.6091612085770479,  
0.5696050095749661,  
-2.634874245900948,  
0.14733366015521565,  
1.0781100436188797,  
0.3459269460772517,  
1.2820636645787713,  
-1.605623121104658,  
0.856791024529966,  
0.16374864437029604,  
-0.7377840957855112,  
0.7603752030179747,  
0.4958590077418653,  
1.0293890027768824,  
0.4862585296205842,  
-1.0838654128292502,  
1.2632637762941026,  
-1.1043454462318527,  
-0.7867654190654241,  
0.8818217647037212,  
1.0406267750063978,  
0.05770567519039498,  
-0.5922070722918433,  
-0.3271403288282926,  
0.5980786503295974,  
-0.02050375112330311,  
-0.9007241609880184,  
0.5239405469453117,  
0.7601496703420288,  
-0.6449049942439851,  
1.7340858429103823,  
0.10050113333288421,  
0.3143347863780438,  
0.28408386490618526,  
-0.1222389867527315,  
-0.09502282329152913,
```


-1.3004368662949968,
0.16856179320059875,
0.6790103402211991,
0.3355977478652572,
0.0935957266804422,
0.42936820813464754,
0.5148227650107633,
0.10764356588532305,
-0.054251737808208085,
0.06525040933822983,
-0.21143245853075732,
-1.5801988491404158,
0.023513208971767067,
1.0921277689034055,
-0.11501776583679489,
-0.14182911234947893,
1.8716225858725093,
-1.0747761250482988,
-0.01873186129021693,
0.9952348515577564,
1.5589645169221944,
2.4625123021717044,
1.5320934484141422,
0.6264578101560674,
0.015802850908690148,
0.6559295719723718,
0.5030706883609005,
0.010357532831659657,
-0.6971755064049523,
-1.334963536555068,
0.05714132580252865,
0.9478292633237722,
1.541635767019547,
-1.2357775558070643,
1.5753066317572375,
1.2206753513411082,
-0.013882574638756213,
2.102115305768464,
0.6867648509985108,
-1.0810617894892982,
0.608205793929093,
-1.0701207529796843,
-0.008804617073740417,
1.0802724219136723,
1.4472704941581649,
-0.525983160453369,
0.10347757085774918,
0.07451664808136123,
0.8113743025574697,
-0.19830571509994802,
0.20760359505818776,
-0.5709476706639499,
0.1791350390980861,
-0.32061652775092425,
-1.6382844876022806,
2.271046687924862,
0.09972766177435569,
1.671157305178672,
1.0596955078067567,
-0.7395801076185267,
-0.8217251410877643,
1.2660358026872456,
-0.149859182069558,
1.7386861885389688,
-0.003505497297480453,
0.677126408740769,
-1.0541684899696007,
0.9437251450981726,
1.7354017388586316,
-0.8022872949930118,
0.01881044769021433,
-0.013699411839369284,
1.5965466699123847,
0.4486397166383021,
1.5669137279484207]]},

3.1453440534904744,
3.166261678218574,
3.2843704266652405,
4.089250903115526,
4.091989893807717,
4.091989893807717,
5.181052167254873,
5.181052167254873,
5.184972473956635,
5.585122024093685,
5.5990893229720795,
5.630355974178803,
5.704160317443757,
8.658831571647385,
8.66505230970618,
8.712765508205756,
8.712765508205756,
8.712765508205756,
9.168875276976076,
9.176734348833309,
9.262471352200725,
9.262471352200725,
10.578606537988184,
10.578606537988184,
1.5905119084430799,
0.8691136234632737,
0.04907693052707525,
-0.23336553986203207,
0.4284777328779486,
2.2866815267075147,
0.1760352312050406,
0.27820168999860917,
2.067264415281462,
0.5755331494929522,
-0.06858856463703264,
-0.7222667597755238,
0.7750924366303688,
-0.278304785792398,
0.5352624789798245,
-1.1702690163303244,
-0.012717570027659048,
-0.07125250327360101,
1.7927644689789062,
1.3752915621891346,
-1.1981440291126808,
-0.30737164793891514,
0.2657743539912098,
0.04156274567190095,
1.1363043468044227,
0.059217925806269944,
0.7608352104096147,
-0.1513784586666425,
0.025696061144845524,
0.861481321747562,
-0.48544656000194114,
-0.006033515622113467,
0.473431667961849,
1.5469119181224316,
0.1871819193791605,
1.555475823471471,
0.9001349633565042,
-1.5961139272878428,
-0.6749613217725859,
-1.5596468027729833,
0.5296186936496604,
-0.8549575199843406,
1.7649719371938613,
1.5569889217595585,
-0.9244334407213903,
-1.5804326845781602,
0.3840271097002098,
-0.8070678340768227,
-0.48791686278645724,
0.13110829311357092,
-2.023238766590243,

```
-0.8997043584182223,
0.8435119798114825,
0.12108034013815198,
-0.7013572315179941,
0.05813246571029178,
-1.3645229029121433,
-1.704441396215925,
0.6160899239086374,
-1.6580292769388947,
-1.550797918270104,
-0.033040246679508146,
-0.7595754539148374,
0.6024045095197653,
0.446745843265138,
0.9717727329863253,
-1.3190014690683383,
-3.180952616822137,
-0.47685451708111876,
0.03958388950097112,
-0.04894683660460036,
-0.013869967587830365,
0.9715551393376142,
-0.2688733257646762,
-1.7112803723577967,
-0.06346862095088202,
1.2756440828431679,
0.4087049517265435,
-0.4433737535464867,
0.07044159255576982,
0.877714241226598,
-0.33059433004855665,
-0.41683973817974146,
-0.09123983383660174,
1.153451542408442,
1.5381060924707182,
1.4432055127327397,
0.7948381511053778,
0.12126271666707081,
0.3736422837955817,
-0.06595460407520624,
-0.1569219765005705,
1.4053035958769775,
-0.3443177603686477,
0.14604377230684606,
0.733462484703497,
1.2365404639754205,
-0.020105376528163625,
-0.09631958929266826,
-0.16876809298151013,
0.42184799028013376,
0.8803036484986011,
-0.4148292271104988,
0.011464734896518355,
1.0895788780475535,
-0.006017115067308532,
0.6476481463328121,
-0.4792828310008005,
1.7145504838049759,
0.9300252591703124,
1.4147034805691376,
-0.04601267898776636,
-0.34579443398211707,
0.006107561937391137,
2.7608588509789516,
-2.470402787070412,
-0.2193552778493498,
-0.43636419827955597,
0.5036040054264609,
-0.006776779992121664]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                  'noise_prob': [0.001, 0.001, 0, 0],
                  'status': True,
                  'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
```


3.837593499897418,
4.847467397840062,
5.03660309473088,
5.176578602594472,
5.176578602594472,
5.214332760543968,
6.547255537065404,
6.547255537065404,
7.083633091144979,
7.083633091144979,
7.133204014188821,
7.133204014188821,
7.133204014188821,
7.937225733864811,
8.370661235857398,
8.62656679928389,
8.759707620277526,
9.424598849529344,
0.24247733923543782,
0.79275118507479,
-0.012314768690710378,
1.2247395896468714,
-0.6087890007814074,
0.9804698093938512,
0.19554007335952167,
-0.18291383394130525,
-0.08254523307317092,
0.18380200005144362,
-0.06242013495608768,
-0.18902966193120999,
0.4846770511497838,
1.2162916157423216,
0.4476502570273604,
-0.5204828708955642,
1.1525916637240337,
0.9814837501696517,
1.2019836737296188,
0.10039944576408244,
-1.165620769823409,
0.5709783345435084,
0.04690539627488982,
0.3150855883896384,
0.6835833386917337,
0.7350671828581666,
0.15105693185753874,
0.7851129846448224,
-0.290237804763004,
0.02628688844012564,
-0.3427838784588314,
-0.12202424723691215,
0.30368921729231485,
-0.08398845633414893,
-0.34421225485285395,
-1.0992299599969957,
-0.6679166311787512,
-0.4925602784337871,
-0.7607905356843834,
-0.061822787578292515,
-0.33643902903801526,
1.7237644621079549,
1.1087838048318122,
1.4391293107545464,
-0.2688865923867924,
0.07238699078201241,
-0.7262166431073663,
-0.8607492696026928,
0.1828620517189283,
-1.5556021786529124,
-0.5230641559705965,
-0.023465829708141782,
0.5773817105800866,
-0.6189512057198875,
-0.1704298219551506,
0.6935490081871019,
0.43024761601627914,

```

-1.3818053283072975,
0.32069867830077864,
0.11223505487625844,
-0.6642209341325642,
-0.7163563578386999,
0.9533459704680486,
-2.5373567196844773,
-0.3530595330204968,
-0.36744271288254204,
-0.4105700661159024,
-0.529185988706723,
-0.5074466275713505,
0.5969687397167245,
0.1454479675984071,
0.9510340399481164,
-0.16051754924491393,
0.168676911677473,
1.6243103666539225,
-0.3533677275828062,
0.23248347317288842,
1.550100715213732,
0.3540002793004863,
0.11004939052123668,
0.0026309322638186474,
1.0023099891482978,
1.3075603782358884,
0.06614039249633322,
0.12696883216368457,
0.4650745950568721,
0.09288306127511851,
-0.05535633254466518,
1.0893487961438044,
0.1856572010619516,
-0.3369565165204948,
1.1993442715596423,
-0.1637322799431796,
-0.05574034797393315,
-0.6499132821077687,
-1.1854402746414112,
-0.20554021240749126,
-0.697991352679575,
0.7715290435685477,
0.27728540290360154,
-0.7490020286422794,
0.6727458447445039,
-1.3928966895283914,
-1.1123692530480307,
1.2429888713160355,
0.5002310144425463,
1.0425617895549082,
-0.41250228671824735,
-0.20782873770333182,
0.952256354944612,
-0.2180059005687231,
0.8024032545701543,
-1.0975152466418368,
0.6192383355749876,
-0.44420258004771784,
-1.3605235580058401,
1.4196204597477091,
-1.1230853758949062,
0.09475958663430169,
0.9412942473630452]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0, 0],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix':
'AUTOMATE_xy_noiseefree_time_evo_ric2_sample#8'},

```


8.258376681022987,
8.258376681022987,
8.264818959972613,
8.264818959972613,
8.264818959972613,
8.264818959972613,
8.265620267330194,
8.265620267330194,
9.412842541483705,
9.412842541483705,
10.502716174481437,
10.502716174481437,
0.1665558361704551,
0.0766182276072886,
-0.06331706305100791,
0.03225035272845343,
-0.2265177184784691,
0.6859734408919996,
0.358399999202588,
-0.31490649734707366,
0.03084506070978915,
0.8141179515629021,
-0.05751068028622007,
0.9013204998454583,
-0.325467390457223,
-0.16027539072229496,
0.19756436832536928,
0.17875052394899651,
0.07441990946369173,
-1.7121562848350094,
0.5411900571694856,
0.025994665514066265,
-1.0547594680221428,
-0.3558195097906033,
0.2575282902752545,
-0.32652811387218433,
-0.17477771883072452,
0.05603758809983181,
-0.31531793685177084,
0.24953935818155465,
-0.21945102528047247,
0.016201452486715968,
-0.47139285875704856,
0.8354485366702803,
0.28783787212142514,
-1.483190369052958,
-0.05324490057492157,
0.00808064868141892,
-1.3894868155533049,
0.07648399162880297,
0.28128388531065984,
-0.8032486745370345,
0.9961277979922681,
0.7271807873997648,
-0.2598267503182047,
0.058937213125893145,
1.4176939090014022,
1.5623836495300838,
-1.2281014386266311,
1.627596878633186,
0.150437982354294,
0.017798034487339433,
-0.2091243023446671,
0.08776907917710067,
0.637527078885032,
1.2603122563899933,
1.0910453886717337,
0.001707278941889729,
-0.5366140699752073,
-0.04457743756443993,
0.708269991966563,
1.6757260639955311,
0.6610325406900899,
-0.508228690084442,
-0.5178693625521746,

```

-0.2955589105377924,
-0.47790822054112786,
0.31077598178920485,
0.020632196003020267,
1.5538854336639996,
1.2816443511581943,
-0.6926864187150775,
0.73698114320343,
0.3645818015611773,
-0.2097443780477125,
0.1421287040679699,
0.052774836076248374,
-0.7306191772967899,
1.107856618529218,
0.3743379513498811,
1.2959169452837032,
0.8858564921663113,
0.03370518740548656,
-1.0860983069970114,
1.2199055901240408,
-1.0630806559423625,
1.3748402618369016,
0.9943926078324806,
0.7116645059379805,
-0.21098608064682703,
0.04581715611274856,
-0.17117312830066983,
1.2229557830916487,
-1.6273428514067598,
1.8508927721454946,
0.851693435560975,
1.0946646161197402,
0.43088380554694883,
-0.6171985422875839,
1.6866137951313547,
-0.2289359575496427,
-0.09152362655526167,
1.6917359096449291,
0.22722692961233917,
0.49434218233553445,
1.2228162165990006,
-0.4070166799423108,
0.00966082758630671,
-0.3344922737632815,
-1.0755692755599169,
1.1846485271761396,
-0.06070659712292082,
0.6384418838463358,
-0.034216786122488324,
-0.6519049362670448,
-0.0019231105719348242,
0.503764046349436,
0.5779702041064313,
-0.8610978040126155,
0.5832069567020347,
0.07954533618121334,
0.016329041774977193]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0, 0],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix':
'AUTOMATE_xy_noisefree_time_evo_ric2_sample#3'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]],
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,

```


8.950027818739875,
8.950027818739875,
8.950027818739875,
8.950027818739875,
8.950027818739875,
9.275481401407285,
0.034552241786943605,
0.7960682992937013,
-0.16960774395822967,
-0.13488653659207303,
-0.07127010448742209,
0.7213919487775996,
0.07517487951530155,
0.7669403129297154,
-0.0980146039609658,
0.2841753594230342,
0.41897129078802076,
0.7644737129881134,
0.21659836679204617,
0.40978346593820614,
0.23386906952753808,
0.6351364889148511,
0.33021983360135304,
0.3518425622667454,
-0.16759580526603354,
1.1955178055405769,
-0.1119172073880175,
-1.6647459335605923,
-0.09499524124538847,
1.171044098528735,
-0.022137095989983098,
0.46479840085551427,
0.5434533112600137,
0.7953483283824695,
-0.23791626316884204,
-0.34327459028867535,
0.9770824990177593,
0.3972109926583832,
0.031305244787475206,
0.06660894739153407,
-0.4270300066285869,
-1.4580475210852422,
0.18899331747125298,
-0.5789031865333153,
0.2203088633995836,
-0.028805094274594364,
-0.21236041446549625,
0.00835869807102751,
1.8444421334304235,
-0.36021419267454113,
-1.0777998782841605,
0.7201309879688491,
-0.07151832135087417,
0.013523038580686776,
-0.5101061898109359,
1.528477364550823,
-0.24055091899624312,
1.595727742687772,
1.181011187425284,
-0.07205139039345279,
-0.3645912646252544,
0.922259297261975,
0.47504473393792157,
-1.4582846498754758,
-1.1041238978674286,
-0.08619812725702276,
1.5348185477516838,
-0.034799821636765677,
0.34854563969795865,
-0.3057886345189525,
0.3435344949085216,
2.387989506593159,
0.5928805156422549,
-0.02250711897690342,
0.734685803060112,

```

1.1230348216575474,
0.9339212770561773,
0.5429657194529808,
-0.9530889967003539,
1.2607191116533525,
-0.18262479735843284,
0.2292294020716647,
0.484625167272873,
2.2621526738562925,
0.3832060422832924,
0.2504382199240898,
-1.1186566372739195,
1.0354182359738777,
-0.9045076995614423,
1.0282595509077306,
1.805679851173207,
0.9332861023326164,
1.4396403686507162,
1.6178312922032976,
-0.22196274260736226,
0.19030625281227892,
-0.32840557234129963,
1.004592356770436,
1.8720477930131976,
-0.06388076228748135,
0.12235432131563899,
-0.5998878734536057,
-0.5461403115162753,
-0.17116133229966776,
0.6826247841010238,
0.6607748570434985,
0.7854114346995006,
0.5612002866021459,
-0.2559995100181322,
-0.8863827207156498,
0.2553569774973194,
0.6372473265785513,
-0.14017909069433732,
0.016226196464851673,
-0.10437448541592237,
2.5268983957827906,
2.3709647446626767,
-0.8108547070158446,
0.5698393243760288,
-1.2042979310590165,
-0.5004321415825274,
-1.0362881146071494,
-0.18531750144204898,
1.5646886079478448,
0.08968487420602586,
-1.6120268318409705]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0, 0],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix':
'AUTOMATE_xy_noiseefree_time_evo_ric2_sample#10'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]],
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '

```

False,

0},

False,

0},

```
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -7.512595875373562],
[12, 6, 0, 0, -6.004303631668366]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 0, 0],
'noise_profile': { 'noise_on_init_param': { 'status':
'value':
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': { 'noise_on_init_param': { 'status':
'value':
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4}],
'run_time_sec': 4.752339839935303}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.2383708143795501,
0.8137820253985804,
1.0561141201801563,
1.2070683478267785,
1.9518166906091101,
2.08301823027678,
2.6774459967535513,
2.7270096866449576,
3.7369434259122447,
3.7421546540398443,
4.458327272130387,
4.463102384700506,
4.463102384700506,
4.482690585412051,
4.482694652744643,
4.498978826283554,
6.3580266266370415,
7.886395999997723,
7.886532810158634,
8.060529572035655,
8.06234783983503,
8.127555772505229,
8.130166630032164,
8.42408407139234,
8.433074307328475,
9.412401860498063,
9.420584174895616,
9.429609653004679,
9.434851407347677,
11.283524698808058,
```

0.19157417616433814,
0.42257952171340063,
0.3129095694283805,
0.28360255750055535,
-0.7229159635345903,
2.545731433831817,
1.1687892956196542,
0.5222949375721274,
0.9482151911772319,
0.28742832079435787,
1.1627391357840349,
-0.27939056132165146,
-0.07574334812551493,
-0.529119211391362,
-0.08653083237786212,
0.042544229108731915,
-0.09154582517805988,
0.2776265525620834,
0.373637405241428,
-0.666296500917012,
-0.729825790392568,
0.36164641825124294,
1.2066181407380305,
0.18327070025757417,
2.2757552060408917,
-0.004789158452516104,
0.49019439516235463,
2.190622553796523,
0.6131206832540379,
1.0987721265230228,
1.9413219378309259,
0.07975169469268534,
0.17531082109327745,
0.01770145552231998,
0.22429808435825893,
1.1460098869067579,
-0.25766861736092417,
0.4386465248337572,
0.7521614434169624,
0.07039918338308593,
-0.23913727928604284,
0.062497116873310714,
0.27575365987954265,
-0.8268952220573135,
0.8599765782432734,
0.6958325703328606,
-0.16792542378671177,
-0.13186811496768225,
0.17745018848846658,
-0.13764108606901534,
-0.34798347931108986,
0.3901205214779397,
-0.12832331396189006,
0.7623260417931323,
-0.08745728181826747,
0.22448567111633413,
0.5103689813486589,
0.020179740460913417,
-0.4639048953982949,
1.8052783293547816,
-0.06895296635033099,
-0.46137869531525183,
-0.7694587159951923,
-0.1507074275764489,
-3.159084078074864,
1.5556721360426484,
0.9802025173412374,
-1.5980562353917622,
0.9974678614222413,
0.11965749551778265,
-0.5964068365118492,
1.00912773091807,
1.2674306548806673,
1.7418320010188646,
-1.2223756485879664,

```

-0.006270145144671745,
0.7863824204489233,
-0.08410721186616651,
0.13420385894759826,
0.6686584151585806,
-0.7946689395347296,
-1.448768973512125,
-0.4158373205099576,
-0.1003664997933048,
0.5370971852976741,
-0.013369261455621718,
-1.2134070474127567,
2.71822204346588,
-0.8266087910498845,
-0.04274152999618493,
0.627725493238542,
-0.10714625448592663,
1.4055199595750496,
-0.08877591733311402,
-0.6400618055168635,
0.27990653766167567,
0.27230874688701184,
1.8329367216853043,
0.12128377229042309,
0.03719209379786179,
0.7558193864298689,
-0.027085062677120984,
0.7947118159580207,
-0.18810209431195668,
0.008294184744538372,
-0.9110921816475059,
1.3065108206329406,
0.004810940817056788,
1.4030602129547465,
0.03878633149748846,
0.29850504257680605,
-0.9524583070906825,
0.10037229527914636,
0.4277056649647598,
-2.2528322583674503,
0.030724614514628212,
0.8416350822543037,
1.5579013461800135,
0.5696673724928452,
-1.5352194543526818]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_xy_noisefree_time_evo_ric2_sample#4'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '

```



```

'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -7.418732841657522],
                             [12, 6, 0, 0, -5.815573107517698]],
  'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                       'identity_factors': [0, 0, 0, 0],
                       'noise_level': [4, 2, 0, 0],
                       'noise_profile': { 'noise_on_init_param': { 'status':
                                                                    'value':
                                                                    'noise_prob': [0.001, 0.001, 0, 0],
                                                                    'status': True,
                                                                    'type': 'depolarizing'},
                       'odd_wires': 4},
                     { 'gates_num': [12, 3, 1, 0],
                       'identity_factors': [1, 1, 0, 0],
                       'noise_level': [12, 6, 0, 0],
                       'noise_profile': { 'noise_on_init_param': { 'status':
                                                                    'value':
                                                                    'noise_prob': [0.001, 0.001, 0, 0],
                                                                    'status': True,
                                                                    'type': 'depolarizing'},
                       'odd_wires': 4}},
                       'run_time_sec': 4.602649450302124}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                         'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'xy-iss'}},
  'init_param': { 'value': [ 4.1837705414928806e-17,
                             1.1742327239243593,
                             1.1742327239243593,
                             1.174232723924359,
                             1.3021435979041922,
                             1.3883521076730898,
                             1.4901272631800928,
                             1.490127263180093,
                             1.490127263180093,
                             2.542545188872923,
                             2.56739133991554,
                             2.677178516908646,
                             5.624946341909109,
                             5.624946341909109,
                             5.63924283183031,
                             5.672838193807689,
                             5.672838193807689,
                             5.672838193807689,
                             5.672838193807689,
                             5.672838193807689,
                             5.672838193807689,
                             5.805507018952099,
                             5.858678454346443,
                             7.374841183417447,
                             7.374841183417447,
                             9.078346087689514,
                             9.17910617229311,
                             9.185363997877552,
                             9.550522060955204,
                             9.715006087156702,
                             0.5380723684568228,
                             0.20346632529178696,
                             -0.28402043001150323,
                             -0.013694659434132353,
                             -0.8491217472851175,
                             -0.13891219530245408,

```

-0.7902948900320785,
0.02035105936303376,
0.35891510624171696,
0.07011262472067725,
0.2571910630887441,
0.09727807922224965,
-0.4253506785797692,
-0.08201618636277365,
-0.5629942371157008,
0.4984357070329931,
0.22460606922422946,
0.6436861273901548,
0.5581387866543217,
0.6438048424266741,
0.6625810608832491,
-0.2877740698749285,
0.03400434270310955,
0.6598198228171638,
0.15695492486622947,
-0.23277712193728803,
0.3379436543817524,
0.6679629986711755,
0.4913802101172593,
-0.6543304626041586,
1.2398324170787045,
0.10868488610295632,
0.4726806257545104,
-1.0278416218016544,
-0.5660236088762642,
-0.06627186605932975,
0.3527547117645279,
0.3944014355355292,
0.33709190475341266,
0.16707772676088822,
0.29195080738380563,
1.0354529601392413,
0.12500585138196232,
-0.037178956773431956,
0.7823218278561086,
-0.04464215343745706,
-0.36923576305262273,
-0.9863817864437636,
0.039519076991602065,
-0.35646400258617383,
0.04542480304419608,
-0.46125271135496354,
0.40472620693719735,
0.7802480661953589,
-0.6757172179201896,
0.6396142241310521,
0.2327897749192615,
0.8954622299286413,
0.7158405805761733,
0.1382139633537826,
1.0616928944303126,
-0.6122315420380016,
-0.6738539946512315,
1.0433459792659807,
0.494526070112864,
0.20621322780405107,
-0.05318175489586881,
-0.11213659998229918,
0.32941262943060273,
0.13931786347259914,
-0.07640781308377202,
0.31450116184408605,
0.3658109215963646,
0.04460349089074807,
-0.12913310998017427,
0.31890485610090735,
-0.005491508360761248,
1.2692063073436923,
-0.3006174703747285,
-0.41404661321397057,
-0.5199199797590206,

```

0.44225909618051784,
-0.4233007342169607,
0.7456354776212751,
-0.32484988424610756,
0.9268151291603125,
0.8385591284905366,
0.027219364270413358,
-0.5520387125082392,
1.3757901840365965,
0.02933419406725825,
-0.2915193842748469,
0.33917094344303517,
0.10828106345120646,
-0.5650186064180197,
-0.6041651963720519,
-0.19230834783869816,
1.9848826966849449,
0.0632219631903188,
0.04804551511145073,
0.47752350144148015,
1.3406847618100746,
-1.3722262204383469,
-1.0513455672656666,
0.7121308231347245,
0.13278942055995957,
1.9758312031522887,
2.19648222646383,
0.8936788375205645,
0.05510909660880629,
-0.18765678539637448,
0.45031205104496697,
-0.6542716355161553,
0.5783583754336702,
-0.18844184232604352,
-1.5667303012058793,
0.36110899429768645,
-0.275182998952267,
0.30367576290775045,
-0.1532687178202341]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_xy_noisefree_time_evo_ric2_sample#2'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]],
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -7.130214875567141],
[12, 6, 0, 0, -5.622281179564727]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],

```

```

False,
0},

'noise_level': [4, 2, 0, 0],
'noise_profile': { 'noise_on_init_param': { 'status':

'value':

'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},

'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': { 'noise_on_init_param': { 'status':

'value':

'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},

'odd_wires': 4}},
'run_time_sec': 5.198779106140137}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.3442332088020621,
0.7247295953616149,
0.7247295953616149,
0.7610537291022658,
1.7162580174489597,
2.0119282772467635,
2.0119282772467635,
3.369934702704019,
3.369934702704019,
5.00211845115155,
5.212181786011135,
5.346566249705437,
6.103091648910964,
6.103091648910964,
7.050711873868975,
7.050711873868975,
7.492144583370162,
7.492144583370162,
7.606105711064038,
7.606105711064038,
8.717193051662631,
8.753101318062777,
8.877107151002964,
8.889590522678484,
8.982812044989718,
8.982812044989718,
9.499874357008034,
9.499874357008034,
9.659025604148875,
9.671110590806489,
0.02814413440270558,
0.250593445582179,
0.17496868989230024,
-0.19749991731592192,
-0.5878125517240043,
1.1370895496823794,
0.3339578948727643,
0.30833012680275795,
0.21147160367429677,
0.5518340573364652,
0.9816766119937317,
-1.3427949214182668,

```

2.2864376119784486,
0.673664283367827,
0.4767745045632036,
-0.47480445681581174,
0.05912449937800317,
0.9513757199218045,
1.079093875588678,
-0.4313388366547843,
0.43347513808131466,
-0.3381632209207048,
-0.42699598389932686,
-1.1528003206699722,
0.2505681368756031,
1.1900947118566725,
-0.38260053073473543,
-0.6405385159573362,
0.08926968472452403,
-0.04834256297702397,
1.603027736914297,
-0.44672812120859223,
-0.5759172165575565,
1.25237714334681,
0.04254918814060746,
-0.002429292667880943,
0.47732275422632153,
-0.10204026491637587,
0.02357869435971368,
0.06910865492863566,
-0.5942663805386644,
1.1629666490831163,
0.28625726751699415,
0.4183193427753825,
0.33026699893767814,
1.8860632823719743,
-0.37929215782635534,
0.8251027249542411,
-0.13937217921728381,
-0.35829122981546885,
-0.11830319301401378,
-1.1307480811476716,
-0.33746884247229475,
1.7161163093033744,
-0.03517175574182253,
-0.4044561448132672,
-0.7301790593832838,
-0.10496992427986328,
-0.0377875273258637,
0.11576685707785253,
-0.9348023198901909,
2.132577954045408,
-0.47199808297503054,
0.09642080758509694,
-1.5590866370005203,
0.10045413530924742,
-0.3215930726850239,
-0.7020784722315541,
1.2376661808413174,
0.5709879203761018,
-0.820670510834149,
0.1634866307612446,
0.06270250983710676,
-0.03570422816362065,
-0.5699745682010601,
-0.26488761312159137,
1.795803611544861,
-0.4017001375917998,
-1.757541020399501,
-0.05132347348943386,
0.8521664539464072,
0.4812990598336637,
-0.7041980923594973,
0.6941905718406797,
1.5589556954504384,
1.9089912382579748,
0.4849438728523755,

```

0.03939482159422182,
-0.2564597588944596,
-0.16675906759041134,
2.5610800117093855,
0.9927310518963223,
1.3170410278463267,
0.6281459286410694,
-0.07878692694615999,
0.5865838795710373,
0.4894408136056478,
-0.043552207832595204,
0.8988064624972067,
-0.35279792613941824,
1.5088457985023234,
-0.32174328535909336,
0.41978129355629945,
0.014567846908235548,
-0.17446008524494477,
0.35494980160059775,
1.44166715836895,
0.027787407199764528,
-0.2573832607835415,
0.9605798444843694,
-0.08907047543629609,
-0.012116788408097524,
1.2676248549462201,
-0.2512276837340942,
0.7869834357157488,
-2.3607393050467915,
0.1914341630047483,
-1.0542342437530707,
0.8454261921281611,
0.02027118187750096]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_xy_noisefree_time_evo_ric2_sample#5'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -7.407206817517894],
[12, 6, 0, 0, -5.829141582765496]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 0, 0],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
'value':
0},
'noise_prob': [0.001, 0.001, 0, 0],

```

False,
0},

```
        'status': True,
        'type': 'depolarizing'},
    'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [1, 1, 0, 0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value':
                                                    'noise_prob': [0.001, 0.001, 0, 0],
                                                    'status': True,
                                                    'type': 'depolarizing'},
                    'odd_wires': 4}},
  'run_time_sec': 4.433908700942993}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [ 0.5,
                                                                    0.5,
                                                                    0.5,
                                                                    0.5,
                                                                    0.5,
                                                                    0.5],
                                                         'r': 0},
                                      'time': {'max': 10.0, 'min': 0.0},
                                      'type': 'xy-iss'}},
    'init_param': { 'value': [ 0.17961742860022237,
                                0.4029745783400613,
                                0.4271020388774394,
                                0.6998927493507625,
                                1.1444327445959492,
                                1.4463150894355827,
                                1.7859669797147917,
                                1.7859669797147917,
                                1.8910076260348616,
                                2.1568466639380097,
                                2.184715675260106,
                                2.3241261579565067,
                                5.221288254664872,
                                5.360678146313719,
                                5.572749542622228,
                                6.315664711217606,
                                6.332942041057151,
                                7.225695385348529,
                                7.238714694082236,
                                7.767863290353532,
                                7.947954629935657,
                                8.045079231148478,
                                8.080705184182557,
                                8.235317994608641,
                                8.52967785406855,
                                8.920755162601605,
                                8.920755162601605,
                                8.920755162601605,
                                9.75907480462672,
                                11.068402571583377,
                                -0.07026370230905016,
                                1.2758275954101812,
                                0.1960569796315372,
                                -0.7454998985427054,
                                0.5260870873255556,
                                1.3632265149428353,
                                0.06759709485075166,
                                0.7201883678889174,
                                0.7780364770341424,
                                0.7034018811814728,
                                -0.3281225120527201,
                                -0.21523884970562404,
                                0.3525948786581971,
                                0.10058439953616755,
                                0.8489242437750127,
                                0.4102210250930649,
                                -1.1239274808373552,
                                -0.07527028057729333,
```

-0.662823747816413,
-0.09016589559770041,
0.0005078927131911535,
-0.9974030724775158,
1.2973260682530905,
0.6207513445119947,
-0.06959715000945968,
0.01479447508324883,
0.5707366547040263,
0.8024230791007029,
0.11580671533830131,
0.5319238260215059,
2.029703693625361,
0.8476591202353092,
-0.5985490132202251,
-1.0850372047073247,
-0.13885829158476937,
0.6160884575951602,
-0.6119315652443118,
0.25456754270344084,
1.6135141314150674,
1.6101080474118592,
-1.0046888465114787,
-1.2174288348798858,
-0.8801790341822278,
-0.01998793431745051,
-1.007194981866689,
-1.6365882266166358,
3.524880820008825,
1.5399950054830251,
-1.401820445976074,
-0.04945776017633801,
0.9565666812114323,
-0.20490442564626934,
-0.5632271134618189,
1.0033608730635488,
0.12368927199762578,
-0.06153617179775209,
1.0557374829951787,
1.532632985657107,
-0.3069192599686721,
1.8620498010178619,
0.18083984306067874,
-0.9702971127887235,
-0.4866438529829019,
0.2527358034573417,
0.46390718912935613,
0.05253485630073179,
-0.15957905175444287,
0.5064463362180326,
0.6588516997606392,
0.036226022952260704,
-1.5941274572195923,
0.3692191472654626,
0.31840120283705997,
0.8741993879399423,
0.4802653460337955,
0.3012459045041651,
0.5102022084743536,
0.3727970282915065,
0.6665750190815365,
-0.7773615380935764,
0.12852119124454983,
0.6774482070442244,
-0.34235651892527574,
-1.1577831131801597,
-0.2879120353476077,
1.157549789799121,
-0.1818163295634488,
-0.034519908964088415,
-0.6744130553492667,
1.1322140576298956,
0.5044463860518625,
1.5469749343656427,
-0.19451127779487945,


```

0.16868157432748737,
0.3887158384137858,
1.179393002549757,
0.4377598134949622,
0.42273561999328285,
0.17555703442650725,
0.7637154786900613,
0.3204620558964089,
-1.0127445325141415,
-0.3366137672636463,
-0.6128563234467006,
-0.19959799618607235,
-0.22993024268879514,
0.6548635589470768,
0.5449412421525476,
0.11171602982233311,
0.5174688869584806,
-0.0926785574168733,
-0.012796553414024043,
0.872831179387319,
-0.3738274777631416,
0.708424995234135,
1.1207261796968788,
0.39043722964884914,
1.5466373285390471,
-0.0566526099067629,
-1.544789451209757]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_xy_noise_free_time_evo_ric2_sample#9'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 +
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 +
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 +
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 +
'0.5 X 3 X 4 + 0.5 Z 3 Z 4 +
'0.5 X 4 X 5 + 0.5 Z 4 Z 5 +
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -7.395384341095475],
[12, 6, 0, 0, -5.790856073500346]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 0, 0],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
False,
0},

```

False,
0},

```
'noise_profile': { 'noise_on_init_param': { 'status':  
                                              'value':  
                                              'noise_prob': [0.001, 0.001, 0, 0],  
                                              'status': True,  
                                              'type': 'depolarizing'},  
                    'odd_wires': 4}},  
  'run_time_sec': 4.583414316177368}},  
{ 'config': { 'ansatz': { 'gateset': 1,  
                          'layer': 30,  
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],  
                                                         'cn': [ 0.5,  
                                                             0.5,  
                                                             0.5,  
                                                             0.5,  
                                                             0.5],  
                                                         'r': 0},  
                                'time': {'max': 10.0, 'min': 0.0},  
                                'type': 'xy-iss'}},  
    'init_param': { 'value': [ 0.02240293444906766,  
                                0.9863710197203618,  
                                1.3708545038264748,  
                                1.3757718928414557,  
                                1.7844795346244002,  
                                2.029643659915566,  
                                2.029643659915566,  
                                4.135298362317422,  
                                4.181946654536538,  
                                4.392782597821499,  
                                4.473609589126697,  
                                4.474471668054438,  
                                4.475838923576404,  
                                4.4785869343507265,  
                                4.4785869343507265,  
                                5.467009291914798,  
                                5.470281162008999,  
                                5.840541346188122,  
                                5.842392372241278,  
                                6.505508038171383,  
                                6.505508038171383,  
                                7.536348099553183,  
                                7.536348099553183,  
                                9.63019412694799,  
                                9.63019412694799,  
                                9.63019412694799,  
                                9.63019412694799,  
                                9.63019412694799,  
                                9.63019412694799,  
                                1.1150074339792337,  
                                -0.03535947987647509,  
                                1.1918755403807781,  
                                -0.20191103850925227,  
                                -0.268362117189328,  
                                -0.3595192771004685,  
                                1.2614204066704717,  
                                -2.9429228213034166,  
                                -1.054810371869961,  
                                0.9594255841252335,  
                                -0.09538831740389547,  
                                0.3965663197177072,  
                                -0.22182226409641906,  
                                -0.2653718580482209,  
                                0.1686090968331403,  
                                1.6516484362324146,  
                                -0.3049608013883917,  
                                0.49165272629600476,  
                                1.9061569916625292,  
                                1.241422732345657,  
                                -0.11527538328521378,  
                                -0.22895223610637053,  
                                -0.06124477487110871,  
                                0.6193220715745159,
```

1.372932614853267,
-1.1538384701543527,
-0.1139440760455718,
0.16973094341798492,
-0.5943234340378932,
0.21399932498003596,
-0.18156105319883253,
-1.0504702368491705,
0.7232808408606985,
-0.018870518436612777,
-0.29343981279013404,
0.30960592292227684,
0.13511542022922338,
0.5096669954667911,
-1.097576281455009,
0.12047585913747774,
-0.5585503308065869,
1.2279608447421912,
1.3012401380411538,
0.3769592231863052,
-0.5692023175942067,
-0.4567508363134565,
-0.679549924864495,
-0.24647993267323956,
0.6292567546169889,
0.9587256267378846,
1.7268825428143277,
0.1413792259593793,
0.6520451922983643,
0.6152471684019365,
-1.1240317980448378,
-0.9856730455453077,
-1.2455074642265085,
0.2235091354354849,
-0.8870861048674711,
0.5299724059178798,
-0.21820971616414211,
-0.09687180933262383,
-0.9247884680119297,
-0.4566833626656702,
-0.4285291193052864,
-0.6406026728496247,
0.9023578506051637,
0.03748900229192909,
0.7551248935414285,
-0.18980887545414976,
-0.1497604267964229,
-0.4253730408988226,
-0.4157675203773789,
1.677209169409502,
0.2126888490300174,
-0.13481167544410624,
-1.6951458494505263,
-0.006476362637627559,
-0.5490470525961781,
0.3195499617454318,
1.077429849794694,
0.6725724542747005,
-0.10235701747439804,
-0.014129818782246995,
2.382406086835482,
0.008866515589568993,
1.0282503794544984,
-0.6899701634722577,
0.3510134577654823,
0.6918405669356381,
-0.08667968085870373,
-0.0005005962237103964,
1.3805542469189591,
1.011514915181436,
-0.4623191099783579,
1.397645255720587,
-0.7813361449321683,
-0.9940660424170442,
-0.6570734018108508,

```

1.0171740600492052,
0.14480072689925969,
1.157289980717297,
1.019472554492438,
1.0029573776528096,
-0.20406480565208354,
-0.011076077177880707,
-0.30566521016454584,
-0.4775965459909992,
-0.7158801142606765,
1.5610067764386268,
1.5187465408730505,
0.21158961058436962,
-0.38098352608767794,
-0.046999172842648596,
0.5964180792256618,
-0.6260168044949087,
-0.7215592053840503,
-0.5706802740378752,
0.11995346168986934,
0.17053214926260635]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_xy_noisefree_time_evo_ric2_sample#6'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -7.25970305977228],
[12, 6, 0, 0, -5.766278116709839]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 0, 0],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
'value':
0},
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
'value':
0},
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,

```

```

        'type': 'depolarizing'},
        'odd_wires': 4}},
    'run_time_sec': 4.72617769241333}}],
'vqe': [ { 'config': { 'ansatz': { 'gateset': 1,
    'layer': 30,
    'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
        'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
        'r': 0},
        'time': {'max': 10.0, 'min': 0.0},
        'type': 'xy-iss'}},
    'init_param': {'value': 'random'},
    'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
        'noise_prob': [0.001, 0.001, 0, 0],
        'status': True,
        'type': 'depolarizing'},
    'nqubits': 7,
    'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
        'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
        'r': 1},
        'def': 'ising'},
    'output': { 'draw': {'fig_dpi': 100, 'status': True, 'type': 'png'},
        'file_name_prefix': 'xy_noisefree_time_evo'},
    'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
    'run': 'vqe',
    'state': 'dmatrix',
    'vqe': { 'iteration': 10,
        'optimization': { 'algorithm': 'SLSQP',
            'constraint': True,
            'status': True}},
    'zne': { 'data_points': None,
        'degree': 1,
        'method': 'richardson',
        'sampling': 'default'}},
    'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + 1 Z 2 '
        '+ 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X 6 + 1 Z '
        '5 + 1 Z 6',
        'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + 0.5 X 1 '
            'X 2 + 0.5 Z 1 Z 2 + 0.5 X 2 X 3 + '
            '0.5 Z 2 Z 3 + 0.5 X 3 X 4 + 0.5 Z 3 '
            'Z 4 + 0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
            '0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
    'output': { 'exact_sol': -8.56677223350563,
        'initial_cost_history': [ 3.450586259258812,
            2.2577647098356715,
            3.220573399525229,
            2.813569497850412,
            2.356577849154886,
            2.484693017894963,
            2.751192081705436,
            3.0939254363357245,
            3.3455601721392987,
            2.9399815036188865],
        'noise_details': { 'gates_num': [4, 1, 1, 0],
            'identity_factors': [0, 0, 0, 0],
            'noise_level': [4, 2, 0, 0],
            'noise_profile': { 'noise_on_init_param': { 'status': False,
                'value': 0},
                'noise_prob': [0.001, 0.001, 0, 0],
                'status': True,
                'type': 'depolarizing'},
            'odd_wires': 4},
        'optimized_minimum_cost': [ -7.463351332655033,
            -7.130214875567174,
            -7.475700783925261,
            -7.41873284165754,
            -7.407206817517966,
            -7.259703059772282,
            -7.568863804770619,
            -7.15916709982671,
            -7.395384341095537,
            -7.512595875373624],
        'optimized_parameters': [ [ 0.5482907189468895,
            0.6413620809492503,
            0.6422482149552934,
            1.5892409709442974,

```

1.850708229814058,
1.850708229814058,
1.8745552660050262,
2.400679867939261,
2.402212278134192,
2.485190345183722,
2.485190345183722,
2.4851903451837223,
2.488397493491329,
2.4883974934913287,
5.077476636438464,
5.077476636438464,
5.732990197214016,
5.806868622878696,
5.817834245705687,
6.166757373752113,
7.3114292874931905,
7.311858449036949,
8.346734585466223,
8.350435170229554,
8.354614694314481,
8.35659817980371,
8.356905245901928,
9.313534396314816,
9.313534396314816,
10.443787047737276,
-0.5971318801423705,
-0.6830583120471296,
0.053446928626708114,
-1.488251087277441,
0.7409998345617078,
-0.020561007687691084,
-0.47029189134787985,
-0.9187802118512801,
-0.07390570205801456,
0.6091612085770479,
0.5696050095749661,
-2.634874245900948,
0.14733366015521565,
1.0781100436188797,
0.3459269460772517,
1.2820636645787713,
-1.605623121104658,
0.856791024529966,
0.16374864437029604,
-0.7377840957855112,
0.7603752030179747,
0.4958590077418653,
1.0293890027768824,
0.4862585296205842,
-1.0838654128292502,
1.2632637762941026,
-1.1043454462318527,
-0.7867654190654241,
0.8818217647037212,
1.0406267750063978,
0.05770567519039498,
-0.5922070722918433,
-0.3271403288282926,
0.5980786503295974,
-0.02050375112330311,
-0.9007241609880184,
0.5239405469453117,
0.7601496703420288,
-0.6449049942439851,
1.7340858429103823,
0.10050113333288421,
0.3143347863780438,
0.28408386490618526,
-0.1222389867527315,
-0.09502282329152913,
-1.3004368662949968,
0.16856179320059875,
0.6790103402211991,
0.3355977478652572,

0.0935957266804422,
0.42936820813464754,
0.5148227650107633,
0.10764356588532305,
-0.054251737808208085,
0.06525040933822983,
-0.21143245853075732,
-1.5801988491404158,
0.023513208971767067,
1.0921277689034055,
-0.11501776583679489,
-0.14182911234947893,
1.8716225858725093,
-1.0747761250482988,
-0.01873186129021693,
0.9952348515577564,
1.5589645169221944,
2.4625123021717044,
1.5320934484141422,
0.6264578101560674,
0.015802850908690148,
0.6559295719723718,
0.5030706883609005,
0.010357532831659657,
-0.6971755064049523,
-1.334963536555068,
0.05714132580252865,
0.9478292633237722,
1.541635767019547,
-1.2357775558070643,
1.5753066317572375,
1.2206753513411082,
-0.013882574638756213,
2.102115305768464,
0.6867648509985108,
-1.0810617894892982,
0.608205793929093,
-1.0701207529796843,
-0.008804617073740417,
1.0802724219136723,
1.4472704941581649,
-0.525983160453369,
0.10347757085774918,
0.07451664808136123,
0.8113743025574697,
-0.19830571509994802,
0.20760359505818776,
-0.5709476706639499,
0.1791350390980861,
-0.32061652775092425,
-1.6382844876022806,
2.271046687924862,
0.09972766177435569,
1.671157305178672,
1.0596955078067567,
-0.7395801076185267,
-0.8217251410877643,
1.2660358026872456,
-0.149859182069558,
1.7386861885389688,
-0.003505497297480453,
0.677126408740769,
-1.0541684899696007,
0.9437251450981726,
1.7354017388586316,
-0.8022872949930118,
0.01881044769021433,
-0.013699411839369284,
1.5965466699123847,
0.4486397166383021,
1.5669137279484207],
[4.1837705414928806e-17,
1.1742327239243593,
1.1742327239243593,
1.174232723924359,

1.3021435979041922,
1.3883521076730898,
1.4901272631800928,
1.490127263180093,
1.490127263180093,
2.542545188872923,
2.56739133991554,
2.677178516908646,
5.624946341909109,
5.624946341909109,
5.63924283183031,
5.672838193807689,
5.672838193807689,
5.672838193807689,
5.672838193807689,
5.672838193807689,
5.672838193807689,
5.805507018952099,
5.858678454346443,
7.374841183417447,
7.374841183417447,
9.078346087689514,
9.17910617229311,
9.185363997877552,
9.550522060955204,
9.715006087156702,
0.5380723684568228,
0.20346632529178696,
-0.28402043001150323,
-0.013694659434132353,
-0.8491217472851175,
-0.13891219530245408,
-0.7902948900320785,
0.02035105936303376,
0.35891510624171696,
0.07011262472067725,
0.2571910630887441,
0.09727807922224965,
-0.4253506785797692,
-0.08201618636277365,
-0.5629942371157008,
0.4984357070329931,
0.22460606922422946,
0.6436861273901548,
0.5581387866543217,
0.6438048424266741,
0.6625810608832491,
-0.2877740698749285,
0.03400434270310955,
0.6598198228171638,
0.15695492486622947,
-0.23277712193728803,
0.3379436543817524,
0.6679629986711755,
0.4913802101172593,
-0.6543304626041586,
1.2398324170787045,
0.10868488610295632,
0.4726806257545104,
-1.0278416218016544,
-0.5660236088762642,
-0.06627186605932975,
0.3527547117645279,
0.3944014355355292,
0.33709190475341266,
0.16707772676088822,
0.29195080738380563,
1.0354529601392413,
0.12500585138196232,
-0.037178956773431956,
0.7823218278561086,
-0.04464215343745706,
-0.36923576305262273,
-0.9863817864437636,
0.039519076991602065,

-0.35646400258617383,
0.04542480304419608,
-0.46125271135496354,
0.40472620693719735,
0.7802480661953589,
-0.6757172179201896,
0.6396142241310521,
0.2327897749192615,
0.8954622299286413,
0.7158405805761733,
0.1382139633537826,
1.0616928944303126,
-0.6122315420380016,
-0.6738539946512315,
1.0433459792659807,
0.494526070112864,
0.20621322780405107,
-0.05318175489586881,
-0.11213659998229918,
0.32941262943060273,
0.13931786347259914,
-0.07640781308377202,
0.31450116184408605,
0.3658109215963646,
0.04460349089074807,
-0.12913310998017427,
0.31890485610090735,
-0.005491508360761248,
1.2692063073436923,
-0.3006174703747285,
-0.41404661321397057,
-0.5199199797590206,
0.44225909618051784,
-0.4233007342169607,
0.7456354776212751,
-0.32484988424610756,
0.9268151291603125,
0.8385591284905366,
0.027219364270413358,
-0.5520387125082392,
1.3757901840365965,
0.02933419406725825,
-0.2915193842748469,
0.33917094344303517,
0.10828106345120646,
-0.5650186064180197,
-0.6041651963720519,
-0.19230834783869816,
1.9848826966849449,
0.0632219631903188,
0.04804551511145073,
0.47752350144148015,
1.3406847618100746,
-1.3722262204383469,
-1.0513455672656666,
0.7121308231347245,
0.13278942055995957,
1.9758312031522887,
2.19648222646383,
0.8936788375205645,
0.05510909660880629,
-0.18765678539637448,
0.45031205104496697,
-0.6542716355161553,
0.5783583754336702,
-0.18844184232604352,
-1.5667303012058793,
0.36110899429768645,
-0.275182998952267,
0.30367576290775045,
-0.1532687178202341],
[-3.641081222961908e-17,
0.6149788059529836,
0.7190202634465676,
1.227117923902744,

1.282995668258181,
1.320146784613617,
1.320146784613617,
3.1464089486715623,
3.1464089486715623,
5.021180125711677,
5.021180125711677,
5.021180125711677,
6.703692809170789,
6.703692809170789,
7.791601108223033,
7.8369202041803225,
7.951907692650894,
7.98389042115583,
8.258376681022987,
8.258376681022987,
8.264818959972613,
8.264818959972613,
8.264818959972613,
8.264818959972613,
8.265620267330194,
8.265620267330194,
9.412842541483705,
9.412842541483705,
10.502716174481437,
10.502716174481437,
0.1665558361704551,
0.0766182276072886,
-0.06331706305100791,
0.03225035272845343,
-0.2265177184784691,
0.6859734408919996,
0.358399999202588,
-0.31490649734707366,
0.03084506070978915,
0.8141179515629021,
-0.05751068028622007,
0.9013204998454583,
-0.325467390457223,
-0.16027539072229496,
0.19756436832536928,
0.17875052394899651,
0.07441990946369173,
-1.7121562848350094,
0.5411900571694856,
0.025994665514066265,
-1.0547594680221428,
-0.3558195097906033,
0.2575282902752545,
-0.32652811387218433,
-0.17477771883072452,
0.05603758809983181,
-0.31531793685177084,
0.24953935818155465,
-0.21945102528047247,
0.016201452486715968,
-0.47139285875704856,
0.8354485366702803,
0.28783787212142514,
-1.483190369052958,
-0.05324490057492157,
0.00808064868141892,
-1.3894868155533049,
0.07648399162880297,
0.28128388531065984,
-0.8032486745370345,
0.9961277979922681,
0.7271807873997648,
-0.2598267503182047,
0.058937213125893145,
1.4176939090014022,
1.5623836495300838,
-1.2281014386266311,
1.627596878633186,
0.150437982354294,

0.017798034487339433,
-0.2091243023446671,
0.08776907917710067,
0.637527078885032,
1.2603122563899933,
1.0910453886717337,
0.001707278941889729,
-0.5366140699752073,
-0.04457743756443993,
0.708269991966563,
1.6757260639955311,
0.6610325406900899,
-0.508228690084442,
-0.5178693625521746,
-0.2955589105377924,
-0.47790822054112786,
0.31077598178920485,
0.020632196003020267,
1.5538854336639996,
1.2816443511581943,
-0.6926864187150775,
0.73698114320343,
0.3645818015611773,
-0.2097443780477125,
0.1421287040679699,
0.052774836076248374,
-0.7306191772967899,
1.107856618529218,
0.3743379513498811,
1.2959169452837032,
0.8858564921663113,
0.03370518740548656,
-1.0860983069970114,
1.2199055901240408,
-1.0630806559423625,
1.3748402618369016,
0.9943926078324806,
0.7116645059379805,
-0.21098608064682703,
0.04581715611274856,
-0.17117312830066983,
1.2229557830916487,
-1.6273428514067598,
1.8508927721454946,
0.851693435560975,
1.0946646161197402,
0.43088380554694883,
-0.6171985422875839,
1.6866137951313547,
-0.2289359575496427,
-0.09152362655526167,
1.6917359096449291,
0.22722692961233917,
0.49434218233553445,
1.2228162165990006,
-0.4070166799423108,
0.00966082758630671,
-0.3344922737632815,
-1.0755692755599169,
1.1846485271761396,
-0.06070659712292082,
0.6384418838463358,
-0.034216786122488324,
-0.6519049362670448,
-0.0019231105719348242,
0.503764046349436,
0.5779702041064313,
-0.8610978040126155,
0.5832069567020347,
0.07954533618121334,
0.016329041774977193],
[0.2383708143795501,
0.8137820253985804,
1.0561141201801563,
1.2070683478267785,

1.9518166906091101,
2.08301823027678,
2.6774459967535513,
2.7270096866449576,
3.7369434259122447,
3.7421546540398443,
4.458327272130387,
4.463102384700506,
4.463102384700506,
4.482690585412051,
4.482694652744643,
4.498978826283554,
6.3580266266370415,
7.88639599997723,
7.886532810158634,
8.060529572035655,
8.06234783983503,
8.127555772505229,
8.130166630032164,
8.42408407139234,
8.433074307328475,
9.412401860498063,
9.420584174895616,
9.429609653004679,
9.434851407347677,
11.283524698808058,
0.19157417616433814,
0.42257952171340063,
0.3129095694283805,
0.28360255750055535,
-0.7229159635345903,
2.545731433831817,
1.1687892956196542,
0.5222949375721274,
0.9482151911772319,
0.28742832079435787,
1.1627391357840349,
-0.27939056132165146,
-0.07574334812551493,
-0.529119211391362,
-0.08653083237786212,
0.042544229108731915,
-0.09154582517805988,
0.2776265525620834,
0.373637405241428,
-0.666296500917012,
-0.729825790392568,
0.36164641825124294,
1.2066181407380305,
0.18327070025757417,
2.2757552060408917,
-0.004789158452516104,
0.49019439516235463,
2.190622553796523,
0.6131206832540379,
1.0987721265230228,
1.9413219378309259,
0.07975169469268534,
0.17531082109327745,
0.01770145552231998,
0.22429808435825893,
1.1460098869067579,
-0.25766861736092417,
0.4386465248337572,
0.7521614434169624,
0.07039918338308593,
-0.23913727928604284,
0.062497116873310714,
0.27575365987954265,
-0.8268952220573135,
0.8599765782432734,
0.6958325703328606,
-0.16792542378671177,
-0.13186811496768225,
0.17745018848846658,

-0.13764108606901534,
-0.34798347931108986,
0.3901205214779397,
-0.12832331396189006,
0.7623260417931323,
-0.08745728181826747,
0.22448567111633413,
0.5103689813486589,
0.020179740460913417,
-0.4639048953982949,
1.8052783293547816,
-0.06895296635033099,
-0.46137869531525183,
-0.7694587159951923,
-0.1507074275764489,
-3.159084078074864,
1.5556721360426484,
0.9802025173412374,
-1.5980562353917622,
0.9974678614222413,
0.11965749551778265,
-0.5964068365118492,
1.00912773091807,
1.2674306548806673,
1.7418320010188646,
-1.2223756485879664,
-0.006270145144671745,
0.7863824204489233,
-0.08410721186616651,
0.13420385894759826,
0.6686584151585806,
-0.7946689395347296,
-1.448768973512125,
-0.4158373205099576,
-0.1003664997933048,
0.5370971852976741,
-0.013369261455621718,
-1.2134070474127567,
2.71822204346588,
-0.8266087910498845,
-0.04274152999618493,
0.627725493238542,
-0.10714625448592663,
1.4055199595750496,
-0.08877591733311402,
-0.6400618055168635,
0.27990653766167567,
0.27230874688701184,
1.8329367216853043,
0.12128377229042309,
0.03719209379786179,
0.7558193864298689,
-0.027085062677120984,
0.7947118159580207,
-0.18810209431195668,
0.008294184744538372,
-0.9110921816475059,
1.3065108206329406,
0.004810940817056788,
1.4030602129547465,
0.03878633149748846,
0.29850504257680605,
-0.9524583070906825,
0.10037229527914636,
0.4277056649647598,
-2.2528322583674503,
0.030724614514628212,
0.8416350822543037,
1.5579013461800135,
0.5696673724928452,
-1.5352194543526818],
[0.3442332088020621,
0.7247295953616149,
0.7247295953616149,
0.7610537291022658,

1.7162580174489597,
2.0119282772467635,
2.0119282772467635,
3.369934702704019,
3.369934702704019,
5.00211845115155,
5.212181786011135,
5.346566249705437,
6.103091648910964,
6.103091648910964,
7.050711873868975,
7.050711873868975,
7.492144583370162,
7.492144583370162,
7.606105711064038,
7.606105711064038,
8.717193051662631,
8.753101318062777,
8.877107151002964,
8.889590522678484,
8.982812044989718,
8.982812044989718,
9.499874357008034,
9.499874357008034,
9.659025604148875,
9.671110590806489,
0.02814413440270558,
0.250593445582179,
0.17496868989230024,
-0.19749991731592192,
-0.5878125517240043,
1.1370895496823794,
0.3339578948727643,
0.30833012680275795,
0.21147160367429677,
0.5518340573364652,
0.9816766119937317,
-1.3427949214182668,
2.2864376119784486,
0.673664283367827,
0.4767745045632036,
-0.47480445681581174,
0.05912449937800317,
0.9513757199218045,
1.079093875588678,
-0.4313388366547843,
0.43347513808131466,
-0.3381632209207048,
-0.42699598389932686,
-1.1528003206699722,
0.2505681368756031,
1.1900947118566725,
-0.38260053073473543,
-0.6405385159573362,
0.08926968472452403,
-0.04834256297702397,
1.603027736914297,
-0.44672812120859223,
-0.5759172165575565,
1.25237714334681,
0.04254918814060746,
-0.002429292667880943,
0.47732275422632153,
-0.10204026491637587,
0.02357869435971368,
0.06910865492863566,
-0.5942663805386644,
1.1629666490831163,
0.28625726751699415,
0.4183193427753825,
0.33026699893767814,
1.8860632823719743,
-0.37929215782635534,
0.8251027249542411,
-0.13937217921728381,

-0.35829122981546885,
-0.11830319301401378,
-1.1307480811476716,
-0.33746884247229475,
1.7161163093033744,
-0.03517175574182253,
-0.4044561448132672,
-0.7301790593832838,
-0.10496992427986328,
-0.0377875273258637,
0.11576685707785253,
-0.9348023198901909,
2.132577954045408,
-0.47199808297503054,
0.09642080758509694,
-1.5590866370005203,
0.10045413530924742,
-0.3215930726850239,
-0.7020784722315541,
1.2376661808413174,
0.5709879203761018,
-0.820670510834149,
0.1634866307612446,
0.06270250983710676,
-0.03570422816362065,
-0.5699745682010601,
-0.26488761312159137,
1.795803611544861,
-0.4017001375917998,
-1.757541020399501,
-0.05132347348943386,
0.8521664539464072,
0.4812990598336637,
-0.7041980923594973,
0.6941905718406797,
1.5589556954504384,
1.9089912382579748,
0.4849438728523755,
0.03939482159422182,
-0.2564597588944596,
-0.16675906759041134,
2.5610800117093855,
0.9927310518963223,
1.3170410278463267,
0.6281459286410694,
-0.07878692694615999,
0.5865838795710373,
0.4894408136056478,
-0.043552207832595204,
0.8988064624972067,
-0.35279792613941824,
1.5088457985023234,
-0.32174328535909336,
0.41978129355629945,
0.014567846908235548,
-0.17446008524494477,
0.35494980160059775,
1.44166715836895,
0.027787407199764528,
-0.2573832607835415,
0.9605798444843694,
-0.08907047543629609,
-0.012116788408097524,
1.2676248549462201,
-0.2512276837340942,
0.7869834357157488,
-2.3607393050467915,
0.1914341630047483,
-1.0542342437530707,
0.8454261921281611,
0.02027118187750096],
[0.02240293444906766,
0.9863710197203618,
1.3708545038264748,
1.3757718928414557,

1.7844795346244002,
2.029643659915566,
2.029643659915566,
4.135298362317422,
4.181946654536538,
4.392782597821499,
4.473609589126697,
4.474471668054438,
4.475838923576404,
4.4785869343507265,
4.4785869343507265,
5.467009291914798,
5.470281162008999,
5.840541346188122,
5.842392372241278,
6.505508038171383,
6.505508038171383,
7.536348099553183,
7.536348099553183,
9.63019412694799,
9.63019412694799,
9.63019412694799,
9.63019412694799,
9.63019412694799,
9.63019412694799,
9.63019412694799,
1.1150074339792337,
-0.03535947987647509,
1.1918755403807781,
-0.20191103850925227,
-0.268362117189328,
-0.3595192771004685,
1.2614204066704717,
-2.9429228213034166,
-1.054810371869961,
0.9594255841252335,
-0.09538831740389547,
0.3965663197177072,
-0.22182226409641906,
-0.2653718580482209,
0.1686090968331403,
1.6516484362324146,
-0.3049608013883917,
0.49165272629600476,
1.9061569916625292,
1.241422732345657,
-0.11527538328521378,
-0.22895223610637053,
-0.06124477487110871,
0.6193220715745159,
1.372932614853267,
-1.1538384701543527,
-0.1139440760455718,
0.16973094341798492,
-0.5943234340378932,
0.21399932498003596,
-0.18156105319883253,
-1.0504702368491705,
0.7232808408606985,
-0.018870518436612777,
-0.29343981279013404,
0.30960592292227684,
0.13511542022922338,
0.5096669954667911,
-1.097576281455009,
0.12047585913747774,
-0.5585503308065869,
1.2279608447421912,
1.3012401380411538,
0.3769592231863052,
-0.5692023175942067,
-0.4567508363134565,
-0.679549924864495,
-0.24647993267323956,
0.6292567546169889,

0.9587256267378846,
1.7268825428143277,
0.1413792259593793,
0.6520451922983643,
0.6152471684019365,
-1.1240317980448378,
-0.9856730455453077,
-1.2455074642265085,
0.2235091354354849,
-0.8870861048674711,
0.5299724059178798,
-0.21820971616414211,
-0.09687180933262383,
-0.9247884680119297,
-0.4566833626656702,
-0.4285291193052864,
-0.6406026728496247,
0.9023578506051637,
0.03748900229192909,
0.7551248935414285,
-0.18980887545414976,
-0.1497604267964229,
-0.4253730408988226,
-0.4157675203773789,
1.677209169409502,
0.2126888490300174,
-0.13481167544410624,
-1.6951458494505263,
-0.006476362637627559,
-0.5490470525961781,
0.3195499617454318,
1.077429849794694,
0.6725724542747005,
-0.10235701747439804,
-0.014129818782246995,
2.382406086835482,
0.008866515589568993,
1.0282503794544984,
-0.6899701634722577,
0.3510134577654823,
0.6918405669356381,
-0.08667968085870373,
-0.0005005962237103964,
1.3805542469189591,
1.011514915181436,
-0.4623191099783579,
1.397645255720587,
-0.7813361449321683,
-0.9940660424170442,
-0.6570734018108508,
1.0171740600492052,
0.14480072689925969,
1.157289980717297,
1.019472554492438,
1.0029573776528096,
-0.20406480565208354,
-0.011076077177880707,
-0.30566521016454584,
-0.4775965459909992,
-0.7158801142606765,
1.5610067764386268,
1.5187465408730505,
0.21158961058436962,
-0.38098352608767794,
-0.046999172842648596,
0.5964180792256618,
-0.6260168044949087,
-0.7215592053840503,
-0.5706802740378752,
0.11995346168986934,
0.17053214926260635],
[0.003408343124459994,
0.0034083431244599804,
0.006710027629051694,
0.2484817728699434,

1.1672656342280407,
1.167470260322433,
3.1453440534904744,
3.166261678218574,
3.2843704266652405,
4.089250903115526,
4.091989893807717,
4.091989893807717,
5.181052167254873,
5.181052167254873,
5.184972473956635,
5.585122024093685,
5.5990893229720795,
5.630355974178803,
5.704160317443757,
8.658831571647385,
8.66505230970618,
8.712765508205756,
8.712765508205756,
8.712765508205756,
9.168875276976076,
9.176734348833309,
9.262471352200725,
9.262471352200725,
10.578606537988184,
10.578606537988184,
1.5905119084430799,
0.8691136234632737,
0.04907693052707525,
-0.23336553986203207,
0.4284777328779486,
2.2866815267075147,
0.1760352312050406,
0.27820168999860917,
2.067264415281462,
0.5755331494929522,
-0.06858856463703264,
-0.7222667597755238,
0.7750924366303688,
-0.278304785792398,
0.5352624789798245,
-1.1702690163303244,
-0.012717570027659048,
-0.07125250327360101,
1.7927644689789062,
1.3752915621891346,
-1.1981440291126808,
-0.30737164793891514,
0.2657743539912098,
0.04156274567190095,
1.1363043468044227,
0.059217925806269944,
0.7608352104096147,
-0.1513784586666425,
0.025696061144845524,
0.861481321747562,
-0.48544656000194114,
-0.006033515622113467,
0.473431667961849,
1.5469119181224316,
0.1871819193791605,
1.555475823471471,
0.9001349633565042,
-1.5961139272878428,
-0.6749613217725859,
-1.5596468027729833,
0.5296186936496604,
-0.8549575199843406,
1.7649719371938613,
1.5569889217595585,
-0.9244334407213903,
-1.5804326845781602,
0.3840271097002098,
-0.8070678340768227,
-0.48791686278645724,

0.13110829311357092,
-2.023238766590243,
-0.8997043584182223,
0.8435119798114825,
0.12108034013815198,
-0.7013572315179941,
0.05813246571029178,
-1.3645229029121433,
-1.704441396215925,
0.6160899239086374,
-1.6580292769388947,
-1.550797918270104,
-0.033040246679508146,
-0.7595754539148374,
0.6024045095197653,
0.446745843265138,
0.9717727329863253,
-1.3190014690683383,
-3.180952616822137,
-0.47685451708111876,
0.03958388950097112,
-0.04894683660460036,
-0.013869967587830365,
0.9715551393376142,
-0.2688733257646762,
-1.7112803723577967,
-0.06346862095088202,
1.2756440828431679,
0.4087049517265435,
-0.4433737535464867,
0.07044159255576982,
0.877714241226598,
-0.33059433004855665,
-0.41683973817974146,
-0.09123983383660174,
1.153451542408442,
1.5381060924707182,
1.4432055127327397,
0.7948381511053778,
0.12126271666707081,
0.3736422837955817,
-0.06595460407520624,
-0.1569219765005705,
1.4053035958769775,
-0.3443177603686477,
0.14604377230684606,
0.733462484703497,
1.2365404639754205,
-0.020105376528163625,
-0.09631958929266826,
-0.16876809298151013,
0.42184799028013376,
0.8803036484986011,
-0.4148292271104988,
0.011464734896518355,
1.0895788780475535,
-0.006017115067308532,
0.6476481463328121,
-0.4792828310008005,
1.7145504838049759,
0.9300252591703124,
1.4147034805691376,
-0.04601267898776636,
-0.34579443398211707,
0.006107561937391137,
2.7608588509789516,
-2.470402787070412,
-0.2193552778493498,
-0.43636419827955597,
0.5036040054264609,
-0.006776779992121664],
[0.5698358581496117,
1.0111463732621893,
1.116254023022104,
1.116254023022104,

1.4543153683590362,
1.4543153683590362,
1.551144943675129,
1.551144943675129,
2.3321866923810566,
2.4857852178760083,
3.837593499897418,
3.837593499897418,
3.837593499897418,
4.847467397840062,
5.03660309473088,
5.176578602594472,
5.176578602594472,
5.214332760543968,
6.547255537065404,
6.547255537065404,
7.083633091144979,
7.083633091144979,
7.133204014188821,
7.133204014188821,
7.133204014188821,
7.937225733864811,
8.370661235857398,
8.62656679928389,
8.759707620277526,
9.424598849529344,
0.24247733923543782,
0.79275118507479,
-0.012314768690710378,
1.2247395896468714,
-0.6087890007814074,
0.9804698093938512,
0.19554007335952167,
-0.18291383394130525,
-0.08254523307317092,
0.18380200005144362,
-0.06242013495608768,
-0.18902966193120999,
0.4846770511497838,
1.2162916157423216,
0.4476502570273604,
-0.5204828708955642,
1.1525916637240337,
0.9814837501696517,
1.2019836737296188,
0.10039944576408244,
-1.165620769823409,
0.5709783345435084,
0.04690539627488982,
0.3150855883896384,
0.6835833386917337,
0.7350671828581666,
0.15105693185753874,
0.7851129846448224,
-0.290237804763004,
0.02628688844012564,
-0.3427838784588314,
-0.12202424723691215,
0.30368921729231485,
-0.08398845633414893,
-0.34421225485285395,
-1.0992299599969957,
-0.6679166311787512,
-0.4925602784337871,
-0.7607905356843834,
-0.061822787578292515,
-0.33643902903801526,
1.7237644621079549,
1.1087838048318122,
1.4391293107545464,
-0.2688865923867924,
0.07238699078201241,
-0.7262166431073663,
-0.8607492696026928,
0.1828620517189283,

-1.5556021786529124,
-0.5230641559705965,
-0.023465829708141782,
0.5773817105800866,
-0.6189512057198875,
-0.1704298219551506,
0.6935490081871019,
0.43024761601627914,
-1.3818053283072975,
0.32069867830077864,
0.11223505487625844,
-0.6642209341325642,
-0.7163563578386999,
0.9533459704680486,
-2.5373567196844773,
-0.3530595330204968,
-0.36744271288254204,
-0.4105700661159024,
-0.529185988706723,
-0.5074466275713505,
0.5969687397167245,
0.1454479675984071,
0.9510340399481164,
-0.16051754924491393,
0.168676911677473,
1.6243103666539225,
-0.3533677275828062,
0.23248347317288842,
1.550100715213732,
0.3540002793004863,
0.11004939052123668,
0.0026309322638186474,
1.0023099891482978,
1.3075603782358884,
0.06614039249633322,
0.12696883216368457,
0.4650745950568721,
0.09288306127511851,
-0.05535633254466518,
1.0893487961438044,
0.1856572010619516,
-0.3369565165204948,
1.1993442715596423,
-0.1637322799431796,
-0.05574034797393315,
-0.6499132821077687,
-1.1854402746414112,
-0.20554021240749126,
-0.697991352679575,
0.7715290435685477,
0.27728540290360154,
-0.7490020286422794,
0.6727458447445039,
-1.3928966895283914,
-1.1123692530480307,
1.2429888713160355,
0.5002310144425463,
1.0425617895549082,
-0.41250228671824735,
-0.20782873770333182,
0.952256354944612,
-0.2180059005687231,
0.8024032545701543,
-1.0975152466418368,
0.6192383355749876,
-0.44420258004771784,
-1.3605235580058401,
1.4196204597477091,
-1.1230853758949062,
0.09475958663430169,
0.9412942473630452],
[0.17961742860022237,
0.4029745783400613,
0.4271020388774394,
0.6998927493507625,

1.1444327445959492,
1.4463150894355827,
1.7859669797147917,
1.7859669797147917,
1.8910076260348616,
2.1568466639380097,
2.184715675260106,
2.3241261579565067,
5.221288254664872,
5.360678146313719,
5.572749542622228,
6.315664711217606,
6.332942041057151,
7.225695385348529,
7.238714694082236,
7.767863290353532,
7.947954629935657,
8.045079231148478,
8.080705184182557,
8.235317994608641,
8.52967785406855,
8.920755162601605,
8.920755162601605,
8.920755162601605,
9.75907480462672,
11.068402571583377,
-0.07026370230905016,
1.2758275954101812,
0.1960569796315372,
-0.7454998985427054,
0.5260870873255556,
1.3632265149428353,
0.06759709485075166,
0.7201883678889174,
0.7780364770341424,
0.7034018811814728,
-0.3281225120527201,
-0.21523884970562404,
0.3525948786581971,
0.10058439953616755,
0.8489242437750127,
0.4102210250930649,
-1.1239274808373552,
-0.07527028057729333,
-0.662823747816413,
-0.09016589559770041,
0.0005078927131911535,
-0.9974030724775158,
1.2973260682530905,
0.6207513445119947,
-0.06959715000945968,
0.01479447508324883,
0.5707366547040263,
0.8024230791007029,
0.11580671533830131,
0.5319238260215059,
2.029703693625361,
0.8476591202353092,
-0.5985490132202251,
-1.0850372047073247,
-0.13885829158476937,
0.6160884575951602,
-0.6119315652443118,
0.25456754270344084,
1.6135141314150674,
1.6101080474118592,
-1.0046888465114787,
-1.2174288348798858,
-0.8801790341822278,
-0.01998793431745051,
-1.007194981866689,
-1.6365882266166358,
3.524880820008825,
1.5399950054830251,
-1.401820445976074,

-0.04945776017633801,
0.9565666812114323,
-0.20490442564626934,
-0.5632271134618189,
1.0033608730635488,
0.12368927199762578,
-0.06153617179775209,
1.0557374829951787,
1.532632985657107,
-0.3069192599686721,
1.8620498010178619,
0.18083984306067874,
-0.9702971127887235,
-0.4866438529829019,
0.2527358034573417,
0.46390718912935613,
0.05253485630073179,
-0.15957905175444287,
0.5064463362180326,
0.6588516997606392,
0.036226022952260704,
-1.5941274572195923,
0.3692191472654626,
0.31840120283705997,
0.8741993879399423,
0.4802653460337955,
0.3012459045041651,
0.5102022084743536,
0.3727970282915065,
0.6665750190815365,
-0.7773615380935764,
0.12852119124454983,
0.6774482070442244,
-0.34235651892527574,
-1.1577831131801597,
-0.2879120353476077,
1.157549789799121,
-0.1818163295634488,
-0.034519908964088415,
-0.6744130553492667,
1.1322140576298956,
0.5044463860518625,
1.5469749343656427,
-0.19451127779487945,
0.16868157432748737,
0.3887158384137858,
1.179393002549757,
0.4377598134949622,
0.42273561999328285,
0.17555703442650725,
0.7637154786900613,
0.3204620558964089,
-1.0127445325141415,
-0.3366137672636463,
-0.6128563234467006,
-0.19959799618607235,
-0.22993024268879514,
0.6548635589470768,
0.5449412421525476,
0.11171602982233311,
0.5174688869584806,
-0.0926785574168733,
-0.012796553414024043,
0.872831179387319,
-0.3738274777631416,
0.708424995234135,
1.1207261796968788,
0.39043722964884914,
1.5466373285390471,
-0.0566526099067629,
-1.544789451209757],
[0.01970245038838479,
0.18403088824160574,
0.23516669011749736,
0.3619340287619915,

0.7559235356843684,
0.8037553588156275,
0.8037553588156274,
0.8037553588156274,
1.523650120002224,
1.5399034674182392,
3.5267025169505333,
3.526702516950533,
4.657784878324055,
4.735234101549794,
4.768060921347247,
6.910522417024203,
6.910522417024203,
7.937298848041795,
7.937298848041795,
7.937298848041795,
7.937298848041795,
7.937298848041795,
7.937298848041795,
8.950027818739875,
8.950027818739875,
8.950027818739875,
8.950027818739875,
8.950027818739875,
9.275481401407285,
0.034552241786943605,
0.7960682992937013,
-0.16960774395822967,
-0.13488653659207303,
-0.07127010448742209,
0.7213919487775996,
0.07517487951530155,
0.7669403129297154,
-0.0980146039609658,
0.2841753594230342,
0.41897129078802076,
0.7644737129881134,
0.21659836679204617,
0.40978346593820614,
0.23386906952753808,
0.6351364889148511,
0.33021983360135304,
0.3518425622667454,
-0.16759580526603354,
1.1955178055405769,
-0.1119172073880175,
-1.6647459335605923,
-0.09499524124538847,
1.171044098528735,
-0.022137095989983098,
0.46479840085551427,
0.5434533112600137,
0.7953483283824695,
-0.23791626316884204,
-0.34327459028867535,
0.9770824990177593,
0.3972109926583832,
0.031305244787475206,
0.06660894739153407,
-0.4270300066285869,
-1.4580475210852422,
0.18899331747125298,
-0.5789031865333153,
0.2203088633995836,
-0.028805094274594364,
-0.21236041446549625,
0.00835869807102751,
1.8444421334304235,
-0.36021419267454113,
-1.0777998782841605,
0.7201309879688491,
-0.07151832135087417,
0.013523038580686776,
-0.5101061898109359,


```

1.528477364550823,
-0.24055091899624312,
1.595727742687772,
1.181011187425284,
-0.07205139039345279,
-0.3645912646252544,
0.922259297261975,
0.47504473393792157,
-1.4582846498754758,
-1.1041238978674286,
-0.08619812725702276,
1.5348185477516838,
-0.034799821636765677,
0.34854563969795865,
-0.3057886345189525,
0.3435344949085216,
2.387989506593159,
0.5928805156422549,
-0.02250711897690342,
0.734685803060112,
1.1230348216575474,
0.9339212770561773,
0.5429657194529808,
-0.9530889967003539,
1.2607191116533525,
-0.18262479735843284,
0.2292294020716647,
0.484625167272873,
2.2621526738562925,
0.3832060422832924,
0.2504382199240898,
-1.1186566372739195,
1.0354182359738777,
-0.9045076995614423,
1.0282595509077306,
1.805679851173207,
0.9332861023326164,
1.4396403686507162,
1.6178312922032976,
-0.22196274260736226,
0.19030625281227892,
-0.32840557234129963,
1.004592356770436,
1.8720477930131976,
-0.06388076228748135,
0.12235432131563899,
-0.5998878734536057,
-0.5461403115162753,
-0.17116133229966776,
0.6826247841010238,
0.6607748570434985,
0.7854114346995006,
0.5612002866021459,
-0.2559995100181322,
-0.8863827207156498,
0.2553569774973194,
0.6372473265785513,
-0.14017909069433732,
0.016226196464851673,
-0.10437448541592237,
2.5268983957827906,
2.3709647446626767,
-0.8108547070158446,
0.5698393243760288,
-1.2042979310590165,
-0.5004321415825274,
-1.0362881146071494,
-0.18531750144204898,
1.5646886079478448,
0.08968487420602586,
-1.6120268318409705]],
'run_time_sec': 68480.87647795677}}]],
'zne': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],

```

```
        'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
        'r': 0},
    'time': {'max': 10.0, 'min': 0.0},
    'type': 'xy-iss'}},
'init_param': { 'value': [ 0.3442332088020621,
    0.7247295953616149,
    0.7247295953616149,
    0.7610537291022658,
    1.7162580174489597,
    2.0119282772467635,
    2.0119282772467635,
    3.369934702704019,
    3.369934702704019,
    5.00211845115155,
    5.212181786011135,
    5.346566249705437,
    6.103091648910964,
    6.103091648910964,
    7.050711873868975,
    7.050711873868975,
    7.492144583370162,
    7.492144583370162,
    7.606105711064038,
    7.606105711064038,
    8.717193051662631,
    8.753101318062777,
    8.877107151002964,
    8.889590522678484,
    8.982812044989718,
    8.982812044989718,
    9.499874357008034,
    9.499874357008034,
    9.659025604148875,
    9.671110590806489,
    0.02814413440270558,
    0.250593445582179,
    0.17496868989230024,
    -0.19749991731592192,
    -0.5878125517240043,
    1.1370895496823794,
    0.3339578948727643,
    0.30833012680275795,
    0.21147160367429677,
    0.5518340573364652,
    0.9816766119937317,
    -1.3427949214182668,
    2.2864376119784486,
    0.673664283367827,
    0.4767745045632036,
    -0.47480445681581174,
    0.05912449937800317,
    0.9513757199218045,
    1.079093875588678,
    -0.4313388366547843,
    0.43347513808131466,
    -0.3381632209207048,
    -0.42699598389932686,
    -1.1528003206699722,
    0.2505681368756031,
    1.1900947118566725,
    -0.38260053073473543,
    -0.6405385159573362,
    0.08926968472452403,
    -0.04834256297702397,
    1.603027736914297,
    -0.44672812120859223,
    -0.5759172165575565,
    1.25237714334681,
    0.04254918814060746,
    -0.002429292667880943,
    0.47732275422632153,
    -0.10204026491637587,
    0.02357869435971368,
    0.06910865492863566,
    -0.5942663805386644,
```

1.1629666490831163,
0.28625726751699415,
0.4183193427753825,
0.33026699893767814,
1.8860632823719743,
-0.37929215782635534,
0.8251027249542411,
-0.13937217921728381,
-0.35829122981546885,
-0.11830319301401378,
-1.1307480811476716,
-0.33746884247229475,
1.7161163093033744,
-0.03517175574182253,
-0.4044561448132672,
-0.7301790593832838,
-0.10496992427986328,
-0.0377875273258637,
0.11576685707785253,
-0.9348023198901909,
2.132577954045408,
-0.47199808297503054,
0.09642080758509694,
-1.5590866370005203,
0.10045413530924742,
-0.3215930726850239,
-0.7020784722315541,
1.2376661808413174,
0.5709879203761018,
-0.820670510834149,
0.1634866307612446,
0.06270250983710676,
-0.03570422816362065,
-0.5699745682010601,
-0.26488761312159137,
1.795803611544861,
-0.4017001375917998,
-1.757541020399501,
-0.05132347348943386,
0.8521664539464072,
0.4812990598336637,
-0.7041980923594973,
0.6941905718406797,
1.5589556954504384,
1.9089912382579748,
0.4849438728523755,
0.03939482159422182,
-0.2564597588944596,
-0.16675906759041134,
2.5610800117093855,
0.9927310518963223,
1.3170410278463267,
0.6281459286410694,
-0.07878692694615999,
0.5865838795710373,
0.4894408136056478,
-0.043552207832595204,
0.8988064624972067,
-0.35279792613941824,
1.5088457985023234,
-0.32174328535909336,
0.41978129355629945,
0.014567846908235548,
-0.17446008524494477,
0.35494980160059775,
1.44166715836895,
0.027787407199764528,
-0.2573832607835415,
0.9605798444843694,
-0.08907047543629609,
-0.012116788408097524,
1.2676248549462201,
-0.2512276837340942,
0.7869834357157488,
-2.3607393050467915,

```

0.1914341630047483,
-1.0542342437530707,
0.8454261921281611,
0.02027118187750096]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noise_free_time_evo_ric2_sample#5'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.407206817517894],
[12, 6, 0, 0, -5.829141582765496]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 4.076957702636719e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -8.196239434894093,
'others': { 'beta_coefficients': [1.5, -0.5],
'cost_richardson_zne': 2.5,
'sorted_expectation_vals': [ -7.407206817517894,
-5.829141582765496],
'sorted_noise': [6, 18]},
'sampled data': [ [4, 2, 0, 0, -7.407206817517894],
[12, 6, 0, 0, -5.829141582765496]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.003408343124459994,
0.0034083431244599804,
0.006710027629051694,
0.2484817728699434,
1.1672656342280407,
1.167470260322433,
3.1453440534904744,
3.166261678218574,
3.2843704266652405,
4.089250903115526,
4.091989893807717,
4.091989893807717,
5.181052167254873,
5.181052167254873,
5.184972473956635,
5.585122024093685,
5.5990893229720795,
5.630355974178803,
5.704160317443757,
8.658831571647385,
8.66505230970618,
8.712765508205756,
8.712765508205756,
8.712765508205756,
9.168875276976076,
9.176734348833309,
9.262471352200725,
9.262471352200725,
10.578606537988184,

```

10.578606537988184,
1.5905119084430799,
0.8691136234632737,
0.04907693052707525,
-0.23336553986203207,
0.4284777328779486,
2.2866815267075147,
0.1760352312050406,
0.27820168999860917,
2.067264415281462,
0.5755331494929522,
-0.06858856463703264,
-0.7222667597755238,
0.7750924366303688,
-0.278304785792398,
0.5352624789798245,
-1.1702690163303244,
-0.012717570027659048,
-0.07125250327360101,
1.7927644689789062,
1.3752915621891346,
-1.1981440291126808,
-0.30737164793891514,
0.2657743539912098,
0.04156274567190095,
1.1363043468044227,
0.059217925806269944,
0.7608352104096147,
-0.1513784586666425,
0.025696061144845524,
0.861481321747562,
-0.48544656000194114,
-0.006033515622113467,
0.473431667961849,
1.5469119181224316,
0.1871819193791605,
1.555475823471471,
0.9001349633565042,
-1.5961139272878428,
-0.6749613217725859,
-1.5596468027729833,
0.5296186936496604,
-0.8549575199843406,
1.7649719371938613,
1.5569889217595585,
-0.9244334407213903,
-1.5804326845781602,
0.3840271097002098,
-0.8070678340768227,
-0.48791686278645724,
0.13110829311357092,
-2.023238766590243,
-0.8997043584182223,
0.8435119798114825,
0.12108034013815198,
-0.7013572315179941,
0.05813246571029178,
-1.3645229029121433,
-1.704441396215925,
0.6160899239086374,
-1.6580292769388947,
-1.550797918270104,
-0.033040246679508146,
-0.7595754539148374,
0.6024045095197653,
0.446745843265138,
0.9717727329863253,
-1.3190014690683383,
-3.180952616822137,
-0.47685451708111876,
0.03958388950097112,
-0.04894683660460036,
-0.013869967587830365,
0.9715551393376142,
-0.2688733257646762,

```
-1.7112803723577967,
-0.06346862095088202,
1.2756440828431679,
0.4087049517265435,
-0.4433737535464867,
0.07044159255576982,
0.877714241226598,
-0.33059433004855665,
-0.41683973817974146,
-0.09123983383660174,
1.153451542408442,
1.5381060924707182,
1.4432055127327397,
0.7948381511053778,
0.12126271666707081,
0.3736422837955817,
-0.06595460407520624,
-0.1569219765005705,
1.4053035958769775,
-0.3443177603686477,
0.14604377230684606,
0.733462484703497,
1.2365404639754205,
-0.020105376528163625,
-0.09631958929266826,
-0.16876809298151013,
0.42184799028013376,
0.8803036484986011,
-0.4148292271104988,
0.011464734896518355,
1.0895788780475535,
-0.006017115067308532,
0.6476481463328121,
-0.4792828310008005,
1.7145504838049759,
0.9300252591703124,
1.4147034805691376,
-0.04601267898776636,
-0.34579443398211707,
0.006107561937391137,
2.7608588509789516,
-2.470402787070412,
-0.2193552778493498,
-0.43636419827955597,
0.5036040054264609,
-0.006776779992121664]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noise_free_evo_ric2_sample#7'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.568863804770512],
[12, 6, 0, 0, -6.084665473601482]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 3.6716461181640625e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -8.310962970355027,
'others': { 'beta_coefficients': [1.5, -0.5],
'cost_richardson_zne': 2.5,
```

```

'sorted_expectation_vals': [ -7.568863804770512,
                              -6.084665473601482],
'sorted_noise': [6, 18]],
'sampled data': [ [4, 2, 0, 0, -7.568863804770512],
                  [12, 6, 0, 0, -6.084665473601482]],
'sampling': 'default']},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                         'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'xy-iss'}},
  'init_param': { 'value': [ 0.5482907189468895,
                              0.6413620809492503,
                              0.6422482149552934,
                              1.5892409709442974,
                              1.850708229814058,
                              1.850708229814058,
                              1.8745552660050262,
                              2.400679867939261,
                              2.402212278134192,
                              2.485190345183722,
                              2.485190345183722,
                              2.4851903451837223,
                              2.488397493491329,
                              2.4883974934913287,
                              5.077476636438464,
                              5.077476636438464,
                              5.732990197214016,
                              5.806868622878696,
                              5.817834245705687,
                              6.166757373752113,
                              7.3114292874931905,
                              7.311858449036949,
                              8.346734585466223,
                              8.350435170229554,
                              8.354614694314481,
                              8.35659817980371,
                              8.356905245901928,
                              9.313534396314816,
                              9.313534396314816,
                              10.443787047737276,
                              -0.5971318801423705,
                              -0.6830583120471296,
                              0.053446928626708114,
                              -1.488251087277441,
                              0.7409998345617078,
                              -0.020561007687691084,
                              -0.47029189134787985,
                              -0.9187802118512801,
                              -0.07390570205801456,
                              0.6091612085770479,
                              0.5696050095749661,
                              -2.634874245900948,
                              0.14733366015521565,
                              1.0781100436188797,
                              0.3459269460772517,
                              1.2820636645787713,
                              -1.605623121104658,
                              0.856791024529966,
                              0.16374864437029604,
                              -0.7377840957855112,
                              0.7603752030179747,
                              0.4958590077418653,
                              1.0293890027768824,
                              0.4862585296205842,
                              -1.0838654128292502,
                              1.2632637762941026,
                              -1.1043454462318527,
                              -0.7867654190654241,
                              0.8818217647037212,
                              1.0406267750063978,
                              0.05770567519039498,
                              -0.5922070722918433,

```

-0.3271403288282926,
0.5980786503295974,
-0.02050375112330311,
-0.9007241609880184,
0.5239405469453117,
0.7601496703420288,
-0.6449049942439851,
1.7340858429103823,
0.10050113333288421,
0.3143347863780438,
0.28408386490618526,
-0.1222389867527315,
-0.09502282329152913,
-1.3004368662949968,
0.16856179320059875,
0.6790103402211991,
0.3355977478652572,
0.0935957266804422,
0.42936820813464754,
0.5148227650107633,
0.10764356588532305,
-0.054251737808208085,
0.06525040933822983,
-0.21143245853075732,
-1.5801988491404158,
0.023513208971767067,
1.0921277689034055,
-0.11501776583679489,
-0.14182911234947893,
1.8716225858725093,
-1.0747761250482988,
-0.01873186129021693,
0.9952348515577564,
1.5589645169221944,
2.4625123021717044,
1.5320934484141422,
0.6264578101560674,
0.015802850908690148,
0.6559295719723718,
0.5030706883609005,
0.010357532831659657,
-0.6971755064049523,
-1.334963536555068,
0.05714132580252865,
0.9478292633237722,
1.541635767019547,
-1.2357775558070643,
1.5753066317572375,
1.2206753513411082,
-0.013882574638756213,
2.102115305768464,
0.6867648509985108,
-1.0810617894892982,
0.608205793929093,
-1.0701207529796843,
-0.008804617073740417,
1.0802724219136723,
1.4472704941581649,
-0.525983160453369,
0.10347757085774918,
0.07451664808136123,
0.8113743025574697,
-0.19830571509994802,
0.20760359505818776,
-0.5709476706639499,
0.1791350390980861,
-0.32061652775092425,
-1.6382844876022806,
2.271046687924862,
0.09972766177435569,
1.671157305178672,
1.0596955078067567,
-0.7395801076185267,
-0.8217251410877643,
1.2660358026872456,

5.672838193807689,
5.805507018952099,
5.858678454346443,
7.374841183417447,
7.374841183417447,
9.078346087689514,
9.17910617229311,
9.185363997877552,
9.550522060955204,
9.715006087156702,
0.5380723684568228,
0.20346632529178696,
-0.28402043001150323,
-0.013694659434132353,
-0.8491217472851175,
-0.13891219530245408,
-0.7902948900320785,
0.02035105936303376,
0.35891510624171696,
0.07011262472067725,
0.2571910630887441,
0.09727807922224965,
-0.4253506785797692,
-0.08201618636277365,
-0.5629942371157008,
0.4984357070329931,
0.22460606922422946,
0.6436861273901548,
0.5581387866543217,
0.6438048424266741,
0.6625810608832491,
-0.2877740698749285,
0.03400434270310955,
0.6598198228171638,
0.15695492486622947,
-0.23277712193728803,
0.3379436543817524,
0.6679629986711755,
0.4913802101172593,
-0.6543304626041586,
1.2398324170787045,
0.10868488610295632,
0.4726806257545104,
-1.0278416218016544,
-0.5660236088762642,
-0.06627186605932975,
0.3527547117645279,
0.3944014355355292,
0.33709190475341266,
0.16707772676088822,
0.29195080738380563,
1.0354529601392413,
0.12500585138196232,
-0.037178956773431956,
0.7823218278561086,
-0.04464215343745706,
-0.36923576305262273,
-0.9863817864437636,
0.039519076991602065,
-0.35646400258617383,
0.04542480304419608,
-0.46125271135496354,
0.40472620693719735,
0.7802480661953589,
-0.6757172179201896,
0.6396142241310521,
0.2327897749192615,
0.8954622299286413,
0.7158405805761733,
0.1382139633537826,
1.0616928944303126,
-0.6122315420380016,
-0.6738539946512315,
1.0433459792659807,
0.494526070112864,

```

0.20621322780405107,
-0.05318175489586881,
-0.11213659998229918,
0.32941262943060273,
0.13931786347259914,
-0.07640781308377202,
0.31450116184408605,
0.3658109215963646,
0.04460349089074807,
-0.12913310998017427,
0.31890485610090735,
-0.005491508360761248,
1.2692063073436923,
-0.3006174703747285,
-0.41404661321397057,
-0.5199199797590206,
0.44225909618051784,
-0.4233007342169607,
0.7456354776212751,
-0.32484988424610756,
0.9268151291603125,
0.8385591284905366,
0.027219364270413358,
-0.5520387125082392,
1.3757901840365965,
0.02933419406725825,
-0.2915193842748469,
0.33917094344303517,
0.10828106345120646,
-0.5650186064180197,
-0.6041651963720519,
-0.19230834783869816,
1.9848826966849449,
0.0632219631903188,
0.04804551511145073,
0.47752350144148015,
1.3406847618100746,
-1.3722262204383469,
-1.0513455672656666,
0.7121308231347245,
0.13278942055995957,
1.9758312031522887,
2.19648222646383,
0.8936788375205645,
0.05510909660880629,
-0.18765678539637448,
0.45031205104496697,
-0.6542716355161553,
0.5783583754336702,
-0.18844184232604352,
-1.5667303012058793,
0.36110899429768645,
-0.275182998952267,
0.30367576290775045,
-0.1532687178202341]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noise_free_time_evo_ric2_sample#2'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.130214875567141],
[12, 6, 0, 0, -5.622281179564727]],

```

```

        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 5.91278076171875e-05,
            'zne_values': { 'degree': 0,
                            'extrapolated_value': -7.884181723568348,
                            'others': { 'beta_coeffieicients': [1.5, -0.5],
                                         'cost_richardson_zne': 2.5,
                                         'sorted_expectation_vals': [ -7.130214875567141,
                                                                      -5.622281179564727],
                                         'sorted_noise': [6, 18]},
                            'sampled data': [ [4, 2, 0, 0, -7.130214875567141],
                                              [12, 6, 0, 0, -5.622281179564727]],
                            'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                         'r': 0},
                                      'time': {'max': 10.0, 'min': 0.0},
                                      'type': 'xy-iss'}}},
  'init_param': { 'value': [ 0.02240293444906766,
                             0.9863710197203618,
                             1.3708545038264748,
                             1.3757718928414557,
                             1.7844795346244002,
                             2.029643659915566,
                             2.029643659915566,
                             4.135298362317422,
                             4.181946654536538,
                             4.392782597821499,
                             4.473609589126697,
                             4.474471668054438,
                             4.475838923576404,
                             4.4785869343507265,
                             4.4785869343507265,
                             5.467009291914798,
                             5.470281162008999,
                             5.840541346188122,
                             5.842392372241278,
                             6.505508038171383,
                             6.505508038171383,
                             7.536348099553183,
                             7.536348099553183,
                             9.63019412694799,
                             9.63019412694799,
                             9.63019412694799,
                             9.63019412694799,
                             9.63019412694799,
                             9.63019412694799,
                             9.63019412694799,
                             1.1150074339792337,
                             -0.03535947987647509,
                             1.1918755403807781,
                             -0.20191103850925227,
                             -0.268362117189328,
                             -0.3595192771004685,
                             1.2614204066704717,
                             -2.9429228213034166,
                             -1.054810371869961,
                             0.9594255841252335,
                             -0.09538831740389547,
                             0.3965663197177072,
                             -0.22182226409641906,
                             -0.2653718580482209,
                             0.1686090968331403,
                             1.6516484362324146,
                             -0.3049608013883917,
                             0.49165272629600476,
                             1.9061569916625292,
                             1.241422732345657,
                             -0.11527538328521378,
                             -0.22895223610637053,
                             -0.06124477487110871,

```

0.6193220715745159,
1.372932614853267,
-1.1538384701543527,
-0.1139440760455718,
0.16973094341798492,
-0.5943234340378932,
0.21399932498003596,
-0.18156105319883253,
-1.0504702368491705,
0.7232808408606985,
-0.018870518436612777,
-0.29343981279013404,
0.30960592292227684,
0.13511542022922338,
0.5096669954667911,
-1.097576281455009,
0.12047585913747774,
-0.5585503308065869,
1.2279608447421912,
1.3012401380411538,
0.3769592231863052,
-0.5692023175942067,
-0.4567508363134565,
-0.679549924864495,
-0.24647993267323956,
0.6292567546169889,
0.9587256267378846,
1.7268825428143277,
0.1413792259593793,
0.6520451922983643,
0.6152471684019365,
-1.1240317980448378,
-0.9856730455453077,
-1.2455074642265085,
0.2235091354354849,
-0.8870861048674711,
0.5299724059178798,
-0.21820971616414211,
-0.09687180933262383,
-0.9247884680119297,
-0.4566833626656702,
-0.4285291193052864,
-0.6406026728496247,
0.9023578506051637,
0.03748900229192909,
0.7551248935414285,
-0.18980887545414976,
-0.1497604267964229,
-0.4253730408988226,
-0.4157675203773789,
1.677209169409502,
0.2126888490300174,
-0.13481167544410624,
-1.6951458494505263,
-0.006476362637627559,
-0.5490470525961781,
0.3195499617454318,
1.077429849794694,
0.6725724542747005,
-0.10235701747439804,
-0.014129818782246995,
2.382406086835482,
0.008866515589568993,
1.0282503794544984,
-0.6899701634722577,
0.3510134577654823,
0.6918405669356381,
-0.08667968085870373,
-0.0005005962237103964,
1.3805542469189591,
1.011514915181436,
-0.4623191099783579,
1.397645255720587,
-0.7813361449321683,
-0.9940660424170442,

```

-0.6570734018108508,
1.0171740600492052,
0.14480072689925969,
1.157289980717297,
1.019472554492438,
1.0029573776528096,
-0.20406480565208354,
-0.011076077177880707,
-0.30566521016454584,
-0.477596545909992,
-0.7158801142606765,
1.5610067764386268,
1.5187465408730505,
0.21158961058436962,
-0.38098352608767794,
-0.046999172842648596,
0.5964180792256618,
-0.6260168044949087,
-0.7215592053840503,
-0.5706802740378752,
0.11995346168986934,
0.17053214926260635]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0, 0],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisefree_time_evo_ric2_sample#6'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.25970305977228],
                           [12, 6, 0, 0, -5.766278116709839]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 3.504753112792969e-05,
            'zne_values': { 'degree': 0,
                            'extrapolated_value': -8.006415531303501,
                            'others': { 'beta_coeffiecients': [1.5, -0.5],
                                         'cost_richardson_zne': 2.5,
                                         'sorted_expectation_vals': [ -7.25970305977228,
                                                                       -5.766278116709839],
                                         'sorted_noise': [6, 18]},
                            'sampled data': [ [4, 2, 0, 0, -7.25970305977228],
                                                [12, 6, 0, 0, -5.766278116709839]],
                            'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                         'r': 0},
                                      'time': {'max': 10.0, 'min': 0.0},
                                      'type': 'xy-iss'}},
            'init_param': { 'value': [ 0.5698358581496117,
                                         1.0111463732621893,
                                         1.116254023022104,
                                         1.116254023022104,
                                         1.4543153683590362,
                                         1.4543153683590362,
                                         1.551144943675129,
                                         1.551144943675129,
                                         2.3321866923810566,
                                         2.4857852178760083,
                                         3.837593499897418,
                                         0.6570734018108508,
                                         1.0171740600492052,
                                         0.14480072689925969,
                                         1.157289980717297,
                                         1.019472554492438,
                                         1.0029573776528096,
                                         -0.20406480565208354,
                                         -0.011076077177880707,
                                         -0.30566521016454584,
                                         -0.477596545909992,
                                         -0.7158801142606765,
                                         1.5610067764386268,
                                         1.5187465408730505,
                                         0.21158961058436962,
                                         -0.38098352608767794,
                                         -0.046999172842648596,
                                         0.5964180792256618,
                                         -0.6260168044949087,
                                         -0.7215592053840503,
                                         -0.5706802740378752,
                                         0.11995346168986934,
                                         0.17053214926260635],
                               'status': True,
                               'type': 'depolarizing'},
            'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                              'noise_prob': [0.001, 0.001, 0, 0],
                              'status': True,
                              'type': 'depolarizing'},
            'nqubits': 7,
            'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                               'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                               'r': 1},
                            'def': 'ising'},
            'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
                      'file_name_prefix': 'AUTOMATE_xy_noisefree_time_evo_ric2_sample#6'},
            'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
            'run': 'zne',
            'state': 'dmatrix',
            'vqe': { 'iteration': 1,
                     'optimization': { 'algorithm': 'SLSQP',
                                       'constraint': True,
                                       'status': False}},
            'zne': { 'data_points': [ [4, 2, 0, 0, -7.25970305977228],
                                       [12, 6, 0, 0, -5.766278116709839]],
                    'degree': 0,
                    'method': 'richardson',
                    'sampling': 'default'}},
            'output': { 'exact_sol': -8.566772233505624,
                      'run_time_sec': 3.504753112792969e-05,
                      'zne_values': { 'degree': 0,
                                      'extrapolated_value': -8.006415531303501,
                                      'others': { 'beta_coeffiecients': [1.5, -0.5],
                                                  'cost_richardson_zne': 2.5,
                                                  'sorted_expectation_vals': [ -7.25970305977228,
                                                                              -5.766278116709839],
                                                  'sorted_noise': [6, 18]},
                                      'sampled data': [ [4, 2, 0, 0, -7.25970305977228],
                                                         [12, 6, 0, 0, -5.766278116709839]],
                                      'sampling': 'default'}}}},
            'config': { 'ansatz': { 'gateset': 1,
                                    'layer': 30,
                                    'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                                   'r': 0},
                                                'time': {'max': 10.0, 'min': 0.0},
                                                'type': 'xy-iss'}},
                        'init_param': { 'value': [ 0.5698358581496117,
                                                    1.0111463732621893,
                                                    1.116254023022104,
                                                    1.116254023022104,
                                                    1.4543153683590362,
                                                    1.4543153683590362,
                                                    1.551144943675129,
                                                    1.551144943675129,
                                                    2.3321866923810566,
                                                    2.4857852178760083,
                                                    3.837593499897418,
                                                    0.6570734018108508,
                                                    1.0171740600492052,
                                                    0.14480072689925969,
                                                    1.157289980717297,
                                                    1.019472554492438,
                                                    1.0029573776528096,
                                                    -0.20406480565208354,
                                                    -0.011076077177880707,
                                                    -0.30566521016454584,
                                                    -0.477596545909992,
                                                    -0.7158801142606765,
                                                    1.5610067764386268,
                                                    1.5187465408730505,
                                                    0.21158961058436962,
                                                    -0.38098352608767794,
                                                    -0.046999172842648596,
                                                    0.5964180792256618,
                                                    -0.6260168044949087,
                                                    -0.7215592053840503,
                                                    -0.5706802740378752,
                                                    0.11995346168986934,
                                                    0.17053214926260635],
                                              'status': True,
                                              'type': 'depolarizing'},
                        'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                                          'noise_prob': [0.001, 0.001, 0, 0],
                                          'status': True,
                                          'type': 'depolarizing'},
                        'nqubits': 7,
                        'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                                           'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                           'r': 1},
                                          'def': 'ising'},
                        'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
                                  'file_name_prefix': 'AUTOMATE_xy_noisefree_time_evo_ric2_sample#6'},
                        'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
                        'run': 'zne',
                        'state': 'dmatrix',
                        'vqe': { 'iteration': 1,
                               'optimization': { 'algorithm': 'SLSQP',
                                                 'constraint': True,
                                                 'status': False}},
                        'zne': { 'data_points': [ [4, 2, 0, 0, -7.25970305977228],
                                                  [12, 6, 0, 0, -5.766278116709839]],
                                'degree': 0,
                                'method': 'richardson',
                                'sampling': 'default'}},
                        'output': { 'exact_sol
```

3.837593499897418,
3.837593499897418,
4.847467397840062,
5.03660309473088,
5.176578602594472,
5.176578602594472,
5.214332760543968,
6.547255537065404,
6.547255537065404,
7.083633091144979,
7.083633091144979,
7.133204014188821,
7.133204014188821,
7.133204014188821,
7.937225733864811,
8.370661235857398,
8.62656679928389,
8.759707620277526,
9.424598849529344,
0.24247733923543782,
0.79275118507479,
-0.012314768690710378,
1.2247395896468714,
-0.6087890007814074,
0.9804698093938512,
0.19554007335952167,
-0.18291383394130525,
-0.08254523307317092,
0.18380200005144362,
-0.06242013495608768,
-0.18902966193120999,
0.4846770511497838,
1.2162916157423216,
0.4476502570273604,
-0.5204828708955642,
1.1525916637240337,
0.9814837501696517,
1.2019836737296188,
0.10039944576408244,
-1.165620769823409,
0.5709783345435084,
0.04690539627488982,
0.3150855883896384,
0.6835833386917337,
0.7350671828581666,
0.15105693185753874,
0.7851129846448224,
-0.290237804763004,
0.02628688844012564,
-0.3427838784588314,
-0.12202424723691215,
0.30368921729231485,
-0.08398845633414893,
-0.34421225485285395,
-1.0992299599969957,
-0.6679166311787512,
-0.4925602784337871,
-0.7607905356843834,
-0.061822787578292515,
-0.33643902903801526,
1.7237644621079549,
1.1087838048318122,
1.4391293107545464,
-0.2688865923867924,
0.07238699078201241,
-0.7262166431073663,
-0.8607492696026928,
0.1828620517189283,
-1.5556021786529124,
-0.5230641559705965,
-0.023465829708141782,
0.5773817105800866,
-0.6189512057198875,
-0.1704298219551506,
0.6935490081871019,

```

0.43024761601627914,
-1.3818053283072975,
0.32069867830077864,
0.11223505487625844,
-0.6642209341325642,
-0.7163563578386999,
0.9533459704680486,
-2.5373567196844773,
-0.3530595330204968,
-0.36744271288254204,
-0.4105700661159024,
-0.529185988706723,
-0.5074466275713505,
0.5969687397167245,
0.1454479675984071,
0.9510340399481164,
-0.16051754924491393,
0.168676911677473,
1.6243103666539225,
-0.3533677275828062,
0.23248347317288842,
1.550100715213732,
0.3540002793004863,
0.11004939052123668,
0.0026309322638186474,
1.0023099891482978,
1.3075603782358884,
0.06614039249633322,
0.12696883216368457,
0.4650745950568721,
0.09288306127511851,
-0.05535633254466518,
1.0893487961438044,
0.1856572010619516,
-0.3369565165204948,
1.1993442715596423,
-0.1637322799431796,
-0.05574034797393315,
-0.6499132821077687,
-1.1854402746414112,
-0.20554021240749126,
-0.697991352679575,
0.7715290435685477,
0.27728540290360154,
-0.7490020286422794,
0.6727458447445039,
-1.3928966895283914,
-1.1123692530480307,
1.2429888713160355,
0.5002310144425463,
1.0425617895549082,
-0.41250228671824735,
-0.20782873770333182,
0.952256354944612,
-0.2180059005687231,
0.8024032545701543,
-1.0975152466418368,
0.6192383355749876,
-0.44420258004771784,
-1.3605235580058401,
1.4196204597477091,
-1.1230853758949062,
0.09475958663430169,
0.9412942473630452]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0, 0],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisefree_time_evo_ric2_sample#8'},

```


-0.08653083237786212,
0.042544229108731915,
-0.09154582517805988,
0.2776265525620834,
0.373637405241428,
-0.666296500917012,
-0.729825790392568,
0.36164641825124294,
1.2066181407380305,
0.18327070025757417,
2.2757552060408917,
-0.004789158452516104,
0.49019439516235463,
2.190622553796523,
0.6131206832540379,
1.0987721265230228,
1.9413219378309259,
0.07975169469268534,
0.17531082109327745,
0.01770145552231998,
0.22429808435825893,
1.1460098869067579,
-0.25766861736092417,
0.4386465248337572,
0.7521614434169624,
0.07039918338308593,
-0.23913727928604284,
0.062497116873310714,
0.27575365987954265,
-0.8268952220573135,
0.8599765782432734,
0.6958325703328606,
-0.16792542378671177,
-0.13186811496768225,
0.17745018848846658,
-0.13764108606901534,
-0.34798347931108986,
0.3901205214779397,
-0.12832331396189006,
0.7623260417931323,
-0.08745728181826747,
0.22448567111633413,
0.5103689813486589,
0.020179740460913417,
-0.4639048953982949,
1.8052783293547816,
-0.06895296635033099,
-0.46137869531525183,
-0.7694587159951923,
-0.1507074275764489,
-3.159084078074864,
1.5556721360426484,
0.9802025173412374,
-1.5980562353917622,
0.9974678614222413,
0.11965749551778265,
-0.5964068365118492,
1.00912773091807,
1.2674306548806673,
1.7418320010188646,
-1.2223756485879664,
-0.006270145144671745,
0.7863824204489233,
-0.08410721186616651,
0.13420385894759826,
0.6686584151585806,
-0.7946689395347296,
-1.448768973512125,
-0.4158373205099576,
-0.1003664997933048,
0.5370971852976741,
-0.013369261455621718,
-1.2134070474127567,
2.71822204346588,
-0.8266087910498845,

```

-0.04274152999618493,
0.627725493238542,
-0.10714625448592663,
1.4055199595750496,
-0.08877591733311402,
-0.6400618055168635,
0.27990653766167567,
0.27230874688701184,
1.8329367216853043,
0.12128377229042309,
0.03719209379786179,
0.7558193864298689,
-0.027085062677120984,
0.7947118159580207,
-0.18810209431195668,
0.008294184744538372,
-0.9110921816475059,
1.3065108206329406,
0.004810940817056788,
1.4030602129547465,
0.03878633149748846,
0.29850504257680605,
-0.9524583070906825,
0.10037229527914636,
0.4277056649647598,
-2.2528322583674503,
0.030724614514628212,
0.8416350822543037,
1.5579013461800135,
0.5696673724928452,
-1.5352194543526818]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noise_free_time_evo_ric2_sample#4'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.418732841657522],
[12, 6, 0, 0, -5.815573107517698]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 4.172325134277344e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -8.220312708727434,
'others': { 'beta_coefficients': [1.5, -0.5],
'cost_richardson_zne': 2.5,
'sorted_expectation_vals': [ -7.418732841657522,
-5.815573107517698],
'sorted_noise': [6, 18]},
'sampled data': [ [4, 2, 0, 0, -7.418732841657522],
[12, 6, 0, 0, -5.815573107517698]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ -3.641081222961908e-17,
0.6149788059529836,

```

0.7190202634465676,
1.227117923902744,
1.282995668258181,
1.320146784613617,
1.320146784613617,
3.1464089486715623,
3.1464089486715623,
5.021180125711677,
5.021180125711677,
5.021180125711677,
6.703692809170789,
6.703692809170789,
7.791601108223033,
7.8369202041803225,
7.951907692650894,
7.98389042115583,
8.258376681022987,
8.258376681022987,
8.264818959972613,
8.264818959972613,
8.264818959972613,
8.264818959972613,
8.265620267330194,
8.265620267330194,
9.412842541483705,
9.412842541483705,
10.502716174481437,
10.502716174481437,
0.1665558361704551,
0.0766182276072886,
-0.06331706305100791,
0.03225035272845343,
-0.2265177184784691,
0.6859734408919996,
0.358399999202588,
-0.31490649734707366,
0.03084506070978915,
0.8141179515629021,
-0.05751068028622007,
0.9013204998454583,
-0.325467390457223,
-0.16027539072229496,
0.19756436832536928,
0.17875052394899651,
0.07441990946369173,
-1.7121562848350094,
0.5411900571694856,
0.025994665514066265,
-1.0547594680221428,
-0.3558195097906033,
0.2575282902752545,
-0.32652811387218433,
-0.17477771883072452,
0.05603758809983181,
-0.31531793685177084,
0.24953935818155465,
-0.21945102528047247,
0.016201452486715968,
-0.47139285875704856,
0.8354485366702803,
0.28783787212142514,
-1.483190369052958,
-0.05324490057492157,
0.00808064868141892,
-1.3894868155533049,
0.07648399162880297,
0.28128388531065984,
-0.8032486745370345,
0.9961277979922681,
0.7271807873997648,
-0.2598267503182047,
0.058937213125893145,
1.4176939090014022,
1.5623836495300838,
-1.2281014386266311,


```

        'status': True,
        'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noise_free_time_evo_ric2_sample#3'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.47570078392518],
                          [12, 6, 0, 0, -5.92114928278248]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 3.6716461181640625e-05,
            'zne_values': { 'degree': 0,
                            'extrapolated_value': -8.25297653449653,
                            'others': { 'beta_coefficients': [1.5, -0.5],
                                         'cost_richardson_zne': 2.5,
                                         'sorted_expectation_vals': [ -7.47570078392518,
                                                                       -5.92114928278248],
                                         'sorted_noise': [6, 18]},
                            'sampled data': [ [4, 2, 0, 0, -7.47570078392518],
                                              [12, 6, 0, 0, -5.92114928278248]],
                            'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                         'r': 0},
                                      'time': {'max': 10.0, 'min': 0.0},
                                      'type': 'xy-iss'}}},
  'init_param': { 'value': [ 0.01970245038838479,
                              0.18403088824160574,
                              0.23516669011749736,
                              0.3619340287619915,
                              0.7559235356843684,
                              0.8037553588156275,
                              0.8037553588156274,
                              0.8037553588156274,
                              1.523650120002224,
                              1.5399034674182392,
                              3.5267025169505333,
                              3.526702516950533,
                              4.657784878324055,
                              4.735234101549794,
                              4.768060921347247,
                              6.910522417024203,
                              6.910522417024203,
                              7.937298848041795,
                              7.937298848041795,
                              7.937298848041795,
                              7.937298848041795,
                              7.937298848041795,
                              7.937298848041795,
                              7.937298848041795,
                              8.950027818739875,
                              8.950027818739875,
                              8.950027818739875,
                              8.950027818739875,
                              8.950027818739875,
                              9.275481401407285,
                              0.034552241786943605,
                              0.7960682992937013,
                              -0.16960774395822967,
                              -0.13488653659207303,
                              -0.07127010448742209,

```

0.7213919487775996,
0.07517487951530155,
0.7669403129297154,
-0.0980146039609658,
0.2841753594230342,
0.41897129078802076,
0.7644737129881134,
0.21659836679204617,
0.40978346593820614,
0.23386906952753808,
0.6351364889148511,
0.33021983360135304,
0.3518425622667454,
-0.16759580526603354,
1.1955178055405769,
-0.1119172073880175,
-1.6647459335605923,
-0.09499524124538847,
1.171044098528735,
-0.022137095989983098,
0.46479840085551427,
0.5434533112600137,
0.7953483283824695,
-0.23791626316884204,
-0.34327459028867535,
0.9770824990177593,
0.3972109926583832,
0.031305244787475206,
0.06660894739153407,
-0.4270300066285869,
-1.4580475210852422,
0.18899331747125298,
-0.5789031865333153,
0.2203088633995836,
-0.028805094274594364,
-0.21236041446549625,
0.00835869807102751,
1.8444421334304235,
-0.36021419267454113,
-1.0777998782841605,
0.7201309879688491,
-0.07151832135087417,
0.013523038580686776,
-0.5101061898109359,
1.528477364550823,
-0.24055091899624312,
1.595727742687772,
1.181011187425284,
-0.07205139039345279,
-0.3645912646252544,
0.922259297261975,
0.47504473393792157,
-1.4582846498754758,
-1.1041238978674286,
-0.08619812725702276,
1.5348185477516838,
-0.034799821636765677,
0.34854563969795865,
-0.3057886345189525,
0.3435344949085216,
2.387989506593159,
0.5928805156422549,
-0.02250711897690342,
0.734685803060112,
1.1230348216575474,
0.9339212770561773,
0.5429657194529808,
-0.9530889967003539,
1.2607191116533525,
-0.18262479735843284,
0.2292294020716647,
0.484625167272873,
2.2621526738562925,
0.3832060422832924,
0.2504382199240898,

```

-1.1186566372739195,
1.0354182359738777,
-0.9045076995614423,
1.0282595509077306,
1.805679851173207,
0.9332861023326164,
1.4396403686507162,
1.6178312922032976,
-0.22196274260736226,
0.19030625281227892,
-0.32840557234129963,
1.004592356770436,
1.8720477930131976,
-0.06388076228748135,
0.12235432131563899,
-0.5998878734536057,
-0.5461403115162753,
-0.17116133229966776,
0.6826247841010238,
0.6607748570434985,
0.7854114346995006,
0.5612002866021459,
-0.2559995100181322,
-0.8863827207156498,
0.2553569774973194,
0.6372473265785513,
-0.14017909069433732,
0.016226196464851673,
-0.10437448541592237,
2.5268983957827906,
2.3709647446626767,
-0.8108547070158446,
0.5698393243760288,
-1.2042979310590165,
-0.5004321415825274,
-1.0362881146071494,
-0.18531750144204898,
1.5646886079478448,
0.08968487420602586,
-1.6120268318409705]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0, 0],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noise_free_time_evo_ric2_sample#10'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.512595875373562],
                          [12, 6, 0, 0, -6.004303631668366]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.076957702636719e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.266741997226159,
                           'others': { 'beta_coefficients': [1.5, -0.5],
                                         'cost_richardson_zne': 2.5,
                                         'sorted_expectation_vals': [ -7.512595875373562,
                                                                        -6.004303631668366],
                                         'sorted_noise': [6, 18]},
                           'sampled data': [ [4, 2, 0, 0, -7.512595875373562],
                                              [12, 6, 0, 0, -6.004303631668366]],
                           'sampling': 'default'}}},

```



```
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                    'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                    'r': 0},
                        'time': {'max': 10.0, 'min': 0.0},
                        'type': 'xy-iss'}},
  'init_param': { 'value': [ 0.17961742860022237,
                             0.4029745783400613,
                             0.4271020388774394,
                             0.6998927493507625,
                             1.1444327445959492,
                             1.4463150894355827,
                             1.7859669797147917,
                             1.7859669797147917,
                             1.8910076260348616,
                             2.1568466639380097,
                             2.184715675260106,
                             2.3241261579565067,
                             5.221288254664872,
                             5.360678146313719,
                             5.572749542622228,
                             6.315664711217606,
                             6.332942041057151,
                             7.225695385348529,
                             7.238714694082236,
                             7.767863290353532,
                             7.947954629935657,
                             8.045079231148478,
                             8.080705184182557,
                             8.235317994608641,
                             8.52967785406855,
                             8.920755162601605,
                             8.920755162601605,
                             8.920755162601605,
                             9.75907480462672,
                             11.068402571583377,
                             -0.07026370230905016,
                             1.2758275954101812,
                             0.1960569796315372,
                             -0.7454998985427054,
                             0.5260870873255556,
                             1.3632265149428353,
                             0.06759709485075166,
                             0.7201883678889174,
                             0.7780364770341424,
                             0.7034018811814728,
                             -0.3281225120527201,
                             -0.21523884970562404,
                             0.3525948786581971,
                             0.10058439953616755,
                             0.8489242437750127,
                             0.4102210250930649,
                             -1.1239274808373552,
                             -0.07527028057729333,
                             -0.662823747816413,
                             -0.09016589559770041,
                             0.0005078927131911535,
                             -0.9974030724775158,
                             1.2973260682530905,
                             0.6207513445119947,
                             -0.06959715000945968,
                             0.01479447508324883,
                             0.5707366547040263,
                             0.8024230791007029,
                             0.11580671533830131,
                             0.5319238260215059,
                             2.029703693625361,
                             0.8476591202353092,
                             -0.5985490132202251,
                             -1.0850372047073247,
                             -0.13885829158476937,
                             0.6160884575951602,
                             -0.6119315652443118,
                             0.25456754270344084,
```

1.6135141314150674,
1.6101080474118592,
-1.0046888465114787,
-1.2174288348798858,
-0.8801790341822278,
-0.01998793431745051,
-1.007194981866689,
-1.6365882266166358,
3.524880820008825,
1.5399950054830251,
-1.401820445976074,
-0.04945776017633801,
0.9565666812114323,
-0.20490442564626934,
-0.5632271134618189,
1.0033608730635488,
0.12368927199762578,
-0.06153617179775209,
1.0557374829951787,
1.532632985657107,
-0.3069192599686721,
1.8620498010178619,
0.18083984306067874,
-0.9702971127887235,
-0.4866438529829019,
0.2527358034573417,
0.46390718912935613,
0.05253485630073179,
-0.15957905175444287,
0.5064463362180326,
0.6588516997606392,
0.036226022952260704,
-1.5941274572195923,
0.3692191472654626,
0.31840120283705997,
0.8741993879399423,
0.4802653460337955,
0.3012459045041651,
0.5102022084743536,
0.3727970282915065,
0.6665750190815365,
-0.7773615380935764,
0.12852119124454983,
0.6774482070442244,
-0.34235651892527574,
-1.1577831131801597,
-0.2879120353476077,
1.157549789799121,
-0.1818163295634488,
-0.034519908964088415,
-0.6744130553492667,
1.1322140576298956,
0.5044463860518625,
1.5469749343656427,
-0.19451127779487945,
0.16868157432748737,
0.3887158384137858,
1.179393002549757,
0.4377598134949622,
0.42273561999328285,
0.17555703442650725,
0.7637154786900613,
0.3204620558964089,
-1.0127445325141415,
-0.3366137672636463,
-0.6128563234467006,
-0.19959799618607235,
-0.22993024268879514,
0.6548635589470768,
0.5449412421525476,
0.11171602982233311,
0.5174688869584806,
-0.0926785574168733,
-0.012796553414024043,
0.872831179387319,

```

-0.3738274777631416,
0.708424995234135,
1.1207261796968788,
0.39043722964884914,
1.5466373285390471,
-0.0566526099067629,
-1.544789451209757]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisefree_time_evo_ric2_sample#9'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.395384341095475],
[12, 6, 0, 0, -5.790856073500346]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 3.719329833984375e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -8.19764847489304,
'others': { 'beta_coeffieicients': [1.5, -0.5],
'cost_richardson_zne': 2.5,
'sorted_expectation_vals': [ -7.395384341095475,
-5.790856073500346],
'sorted_noise': [6, 18]},
'sampled data': [ [4, 2, 0, 0, -7.395384341095475],
[12, 6, 0, 0, -5.790856073500346]],
'sampling': 'default'}}}],
'xy-ric3-uniform-scaling': { 'noiseoff': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': {'value': 'random'},
'noise_profile': { 'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': False,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisefree_time_evo_ric3_sample#9'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.395384341095475],
[12, 6, 0, 0, -5.790856073500346]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 3.719329833984375e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -8.19764847489304,
'others': { 'beta_coeffieicients': [1.5, -0.5],
'cost_richardson_zne': 2.5,
'sorted_expectation_vals': [ -7.395384341095475,
-5.790856073500346],
'sorted_noise': [6, 18]},
'sampled data': [ [4, 2, 0, 0, -7.395384341095475],
[12, 6, 0, 0, -5.790856073500346]],
'sampling': 'default'}}}],
'xy-ric3-uniform-scaling': { 'noiseoff': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': {'value': 'random'},
'noise_profile': { 'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': False,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisefree_time_evo_ric3_sample#9'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.395384341095475],
[12, 6, 0, 0, -5.790856073500346]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 3.719329833984375e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -8.19764847489304,
'others': { 'beta_coeffieicients': [1.5, -0.5],
'cost_richardson_zne': 2.5,
'sorted_expectation_vals': [ -7.395384341095475,
-5.790856073500346],
'sorted_noise': [6, 18]},
'sampled data': [ [4, 2, 0, 0, -7.395384341095475],
[12, 6, 0, 0, -5.790856073500346]],
'sampling': 'default'}}}],
'xy-ric3-uniform-scaling': { 'noiseoff': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': {'value': 'random'},
'noise_profile': { 'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': False,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisefree_time_evo_ric3_sample#9'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.395384341095475],
[12, 6, 0, 0, -5.790856073500346]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 3.719329833984375e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -8.19764847489304,
'others': { 'beta_coeffieicients': [1.5, -0.5],
'cost_richardson_zne': 2.5,
'sorted_expectation_vals': [ -7.395384341095475,
-5.790856073500346],
'sorted_noise': [6, 18]},
'sampled data': [ [4, 2, 0, 0, -7.395384341095475],
[12, 6, 0, 0, -5.790856073500346]],
'sampling': 'default'}}}],
'xy-ric3-uniform-scaling': { 'noiseoff': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': {'value': 'random'},
'noise_profile': { 'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': False,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisefree_time_evo_ric3_sample#9'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.395384341095475],
[12, 6, 0, 0, -5.790856073500346]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 3.719329833984375e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -8.19764847489304,
'others': { 'beta_coeffieicients': [1.5, -0.5],
'cost_richardson_zne': 2.5,
'sorted_expectation_vals': [ -7.395384341095475,
-5.790856073500346],
'sorted_noise': [6, 18]},
'sampled data': [
```

```

1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': True,
'type': 'png'},
'file_name_prefix': 'xy_noise_off'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0]]},
'run': 'vqe',
'state': 'dmatrix',
'vqe': { 'iteration': 10,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': True}},
'zne': { 'data_points': None,
'degree': 1,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 '
'+ 1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + 1 Z '
'3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X 6 + 1 '
'Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + '
'0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + '
'0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + '
'0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + '
'0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + '
'0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + '
'0.5 Z 5 Z 6']},
'output': { 'exact_sol': -8.56677223350563,
'initial_cost_history': [ 4.118038614266066,
2.900590937440303,
2.874944433071395,
3.9296977116518517,
3.1841729388967344,
3.695928991667305,
3.180227821654077,
3.0475045550195774,
2.4669562192398296,
3.4124916083581294],
'noise_details': { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [0, 0, 0, 0],
'noise_profile': { 'noise_on_init_param':
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': False,
'type':
'depolarizing'},
'odd_wires': 4},
'optimized_minimum_cost': [ -8.433468022982382,
-8.464942606128835,
-8.43123094137328,
-8.435401809869298,
-7.9812014353834515,
-8.371080050195895,
-8.46198057547588,
-8.19415563172473,

```

```
-8.438078664602543,  
-8.481886373274177],  
'optimized_parameters': [ [ 0.4631008843346199,  
0.8090636887146863,  
1.1101507421405439,  
1.3400184709676968,  
1.7452332568602784,  
1.9308700883292553,  
2.1752679291102544,  
2.1784212211010603,  
3.800207092295721,  
3.800207092295721,  
3.800207092295721,  
3.800207092295721,  
3.800207092295721,  
3.800207092295721,  
4.798448841518561,  
4.812026604031267,  
6.02543601067661,  
6.02543601067661,  
6.5265343452987,  
6.5265343452987,  
6.526534345298701,  
6.528381158760148,  
6.528381158760148,  
7.113023968622042,  
7.113023968622042,  
7.593654062389684,  
7.5949106716383605,  
7.780058793305066,  
9.000370275113886,  
9.005799799983558,  
-0.22105281285875572,  
0.07196296170715184,  
1.0527781347416576,  
-0.4594806493129553,  
1.0526039239081737,  
-0.01709270509934428,  
0.3199675106146978,  
-0.0972220164715271,  
0.9145097677998977,  
0.09808103992344623,  
0.5373974901276602,  
-0.5786059880820694,  
-0.02594763303680847,  
-0.22132394348491485,  
-0.7274785930689547,  
-0.41090279602128,  
-0.520884263329148,  
-1.0260835302846771,  
0.1692597388493397,  
1.3250313995601488,  
1.510357845739394,  
-0.9233604808876308,  
-0.2049157632594096,  
-0.9894281051832,  
0.25772120620526445,  
-0.04822142329545429,  
-0.3912638297585932,  
-0.48486712444934904,  
1.4836649713082644,  
1.566996182191827,  
0.08093959953023218,  
-2.3562926055425963,  
-0.21876618997457167,  
0.10911407834571141,  
-1.3776955808248483,  
-0.31816531211057325,  
0.7083857256813267,  
0.5593482503842677,  
0.7483381318097894,  
0.010042025170353972,  
0.05290425882351868,  
0.20616876613540233,  
-0.8673924682871486,
```

-0.08017548043713153,
-0.3056222778255953,
2.3981163154468383,
0.37079365521682633,
0.3870558721039623,
0.19341028278759992,
-0.3154248344945196,
-0.12073260037561208,
-1.5858414285227351,
0.59830347262823,
0.30388710178795286,
-1.4558796844932331,
-0.3225618511944297,
0.09243390755822743,
-0.0006426173071943948,
0.7561662109631148,
-1.3902324585417218,
0.1677701056843372,
2.7311884232201735,
-1.1122880815211051,
0.0467986017971284,
-0.020089420529231013,
0.005481851366742631,
1.3205596999819047,
-0.1290403664286252,
0.3870444394307043,
-0.6549197738372419,
-0.17656835190090397,
-0.0008013451895215781,
-0.8769883624496301,
1.6929520910231297,
0.7288671099890552,
-0.6169673343691255,
1.667217439782634,
0.2554635335546482,
-0.029515171705053127,
-0.8290153205719569,
0.042733616172741226,
-0.5148227135977747,
1.2301836750084596,
-0.9175908300980525,
0.5229779878549122,
-1.5901934710811256,
-0.07316585886439231,
1.1195018365185199,
-0.31111306362203306,
0.7254709796877604,
1.1995417479543053,
0.6908225815252583,
-0.2919630814941808,
0.15955998483086176,
1.2871422030548714,
-1.2683991057896014,
-0.3768598572966417,
1.4863954796333156,
-0.4300298743216642,
0.19151460300148976,
0.9665838873771405,
-0.07244306377732504,
-0.04169546816073936,
-0.389029635704794,
1.5218961812832925,
-0.1772066326032777,
-0.70949107362336,
-0.015222466074935298,
0.3587128629768407,
1.5809494716454662,
0.12428164288740734,
1.5083554487395696,
-0.044379618902227067,
-0.03505579308192811,
-0.6786538117837789,
-0.007428157845337949,
1.5281494732580478,
-0.14124856089608578,

2.726718429443767,
-0.003106328084445967],
[0.5270475437101116,
0.7815425625485376,
1.6469610445191378,
1.6712876404917778,
2.122904188877425,
2.131106816606521,
3.7367798442214175,
3.7367798442219424,
4.750276145421292,
4.853973384996016,
4.865098072058574,
4.873980651141647,
4.987599328262889,
5.00246239884106,
5.520902021852277,
5.793284849001337,
5.8087832398624775,
7.432729128414403,
7.432953787975026,
8.144896858715603,
8.172580744083415,
8.788071761144659,
8.792288306719156,
8.828260688191317,
9.011586808740137,
9.011995976067807,
9.538729699920484,
9.576884709106812,
10.486750161390887,
10.48675016212107,
-0.15531374899360717,
-0.0849039080739262,
-0.8434555658556069,
0.9447826609611266,
-0.19482082661427938,
2.4999420394249707,
-2.7164831941210483,
-1.5408169364026967,
-0.8621482593978547,
0.31547458537730877,
1.4220435826823787,
1.348860553316339,
0.40827474166943256,
1.063960383917828,
-0.21906798993221652,
-0.33263312803097533,
0.4889869181158972,
-0.25524084746075687,
-1.3622146811064544,
0.996105149433398,
1.4161210728452394,
-0.6491827941870941,
0.7399329880002848,
-0.11994422281276146,
3.179573825543474,
-0.034943024466848674,
0.6075464711124173,
-0.3019825794696199,
0.029392952712390567,
0.14840732209332994,
0.08592470568620861,
0.016725634393736574,
-1.3697962205589145,
-0.06155375586805023,
0.16508080354919497,
-0.5975014914051775,
0.09635736997937351,
1.254101402191796,
-0.6193953337438176,
0.47602269510847217,
-1.7251540483535392,
0.5378274350019089,
-1.0159384367751108,

1.0968223266795405,
-0.79878134010976,
0.10468891520322528,
-0.2910173593847854,
-0.7078981306941186,
-0.3510837282628223,
0.5371002854277007,
-0.4911274293534503,
1.7273270693440552,
1.3500555714793954,
0.9052059235505011,
0.4771526028272018,
-0.1893747959421651,
0.13989050072973697,
-0.5930955187300607,
0.36780210947560876,
-0.5043370473487999,
-0.4851893292987325,
0.1987440888222681,
-0.5806445532873666,
2.3171646404124093,
0.06032126128170586,
-0.6389920437730593,
0.9985359186149056,
0.04870155999532171,
0.16447399991313041,
0.05315090520136875,
0.15727255736970833,
-1.3722292706433088,
-0.33233649668322635,
0.7758243461945761,
-0.20476613404890107,
0.06972046040604139,
0.3143879960336948,
-0.01452323353492752,
-1.2901366074359686,
-0.31033000747276407,
-0.16264485166390844,
-1.3914094473719103,
-0.5273128620655095,
0.02465459151953875,
0.10942766294132272,
1.0292094835171124,
-1.882689768994833,
-1.553216293090225,
1.3287846024814622,
0.20963913600213377,
0.7195732642990699,
0.0969787732183028,
-0.9744344415633356,
1.5822487402774768,
0.7077471503862973,
-0.766345302327988,
0.3831437091022152,
-0.12028853613988252,
-0.8209919023748664,
-0.21164805512089607,
0.5290684330656528,
-0.1040878870845114,
1.1956945887281514,
0.25703931501419025,
0.6344733295912399,
-0.02497584874729833,
-0.20937697561532842,
-1.4409411147319051,
0.9707320454927133,
1.4039378085761043,
0.9760826987262126,
-0.17314228405542512,
0.16734556762922406,
0.07881248116464283,
-0.2924467918405051,
1.4946430081327182,
0.7516806171134963,
0.799700486923224,

-0.8440600324500036,
-0.01069194371181784],
[0.3388610072211759,
0.3930923343746358,
0.4059472295013711,
0.4597995545870322,
0.4597995545870322,
0.4667083780713653,
0.9924683364283498,
1.054118787032341,
1.3839145947816414,
1.4469616998432806,
3.044048229936016,
3.044048229936016,
4.682975233106521,
4.682975233106521,
5.287925895396742,
5.287925895396742,
5.625677830869467,
5.625677830869467,
6.519005703020383,
6.519005703020383,
6.519005703020383,
6.854665591577039,
6.854665591577039,
8.092962128458302,
8.097830997780322,
8.102159968176109,
8.111954675742565,
8.727038821401255,
8.727038821401255,
8.784424594255395,
-0.8636093730812797,
-0.04742973062196874,
-0.37989856483295564,
0.3750798322441465,
-0.509087961968837,
0.5366875438436245,
0.1232704940065221,
0.5249134835872401,
0.13594138148710633,
0.10802292582448766,
-0.7156772546121367,
0.34756877087651405,
-0.5704812075854907,
0.289395233495819,
0.2674862930849357,
0.8724143866961164,
0.5770602988816299,
0.658777432128999,
-0.47288232304292416,
-0.029663566814993374,
-1.323180236232167,
-0.2890901656039231,
-0.015141732930961212,
1.1846850771343012,
-0.4506688471965015,
0.22241980016839272,
-0.015748484493019736,
1.1522499025192263,
0.2996907655672081,
-1.7370631162182948,
-0.35306232532663057,
0.17322471105923057,
-1.1855180134814352,
-0.31537295018088835,
-1.8120366955290261,
0.6663892425075478,
0.21874911670355193,
1.2729693967005706,
-0.14230256773273223,
0.2012427441067739,
-2.3269317425793643,
-0.03299848401271634,
-0.4074853136565566,

-0.30371905431664253,
-0.12037074206123735,
0.8456676988075262,
-0.21838328071319946,
0.019494764922420988,
0.16383099728733078,
0.056056403895700845,
-3.5632063709291613,
0.6579904510057321,
0.5207861046034454,
-0.017180990359968364,
0.2532807007765998,
-0.03315495616728509,
0.7564905992030696,
-0.11091568858030387,
-0.9281890577202553,
-0.6082181315906088,
-0.4480511658411004,
2.1953283079942025,
-0.30446713497464617,
-0.03359602431311142,
0.28972943746372876,
0.17817064395522006,
0.7965754083210965,
-0.9520780563921613,
0.10150445450145726,
1.3510884636720888,
-0.37849615173878054,
0.17729737647006832,
0.1884683806301832,
-0.03234547530918763,
-1.338763456177623,
-0.6711058962200585,
-0.012976043431998471,
0.027578392175021227,
0.10161286285089129,
-0.00939135247962935,
-1.302164623174096,
1.3715576399721376,
0.737767307868418,
1.5939269674119705,
1.4291758244230544,
0.5601859764306084,
0.3488764286390935,
-1.093762224545935,
-1.1976831420922067,
1.4557381665683378,
0.8924472305311927,
0.5300221300338881,
1.0785280788144638,
-0.6029509417428172,
-0.6790632925762424,
0.4138865114731201,
0.2917474006869537,
1.8050154062536312,
0.05177383423831921,
0.25467777497912664,
1.4671356779039024,
0.4422247342165525,
-1.5431607378377095,
1.8101814873938125,
1.6870467346899987,
0.7517920825884995,
1.3622912362338087,
0.13653050459472865,
1.5210183141257738,
-0.004247168330480351,
0.26137218980413907,
0.5451511562235306,
-0.6705596749039198,
1.2136085126593865,
0.5231529087420831,
-0.03279466120532818,
0.2548954759886831,
1.5797820110558691,

1.075840789720537,
1.5541818084195007],
[3.821020346740443e-17,
1.386496864345847,
1.386496864345847,
1.7951472089989222,
1.8559400139285378,
2.7980544315911504,
2.7980544315911504,
2.82954781812891,
2.82954781812891,
2.82954781812891,
2.956183634338348,
3.840561875918348,
3.840561875918348,
4.79501776182173,
4.79501776182173,
5.670460018404704,
5.670460018404704,
5.670460018404704,
6.865392250052202,
6.865392250052202,
7.506826147384778,
7.506826147384778,
8.737685173635683,
8.737685173635683,
8.737685173635683,
8.737685173635683,
9.601023092540274,
9.601023092540274,
9.853738132880634,
9.853738132880634,
0.5611658219809542,
1.427331325859137,
0.8881601644240952,
-0.8937497791603879,
0.5712044928850715,
1.3195951619089188,
0.5467792160605555,
-0.6816630744322637,
0.06707477836950569,
0.6798692668930644,
0.19576168860111362,
0.5094724850631845,
1.1424414477256812,
0.30471145260454696,
-0.08158911485577636,
0.9585661478313002,
1.0836238583568754,
-0.3718197048926503,
0.7841948506971846,
0.3023515457437066,
0.7662240328883452,
0.9970048707689876,
-0.8946064024093777,
0.27649390200762114,
-0.15342626519190924,
-0.4015720487025046,
1.460926694270634,
-0.055421168101734586,
0.6077050880702092,
-0.7932081335349191,
1.3188332541813446,
0.01929555634963922,
-1.1301845164589392,
-0.1718174564221745,
2.26154163970611,
-0.9882028269623031,
0.8038643847286138,
0.6849304102208338,
0.874798983022982,
-0.016486064274954207,
0.0264221098096002,
-0.3625000384278827,
1.4376331477886695,

0.8499992740594865,
0.10042171561200945,
0.14542648933357244,
1.3513090878714353,
-0.22248922869259902,
-0.4483175327546578,
1.2345758678942387,
-0.18983522552694923,
0.019591210301167533,
-1.3607248526627624,
0.018375533581887468,
-0.3730738776375715,
1.0123537416913542,
0.7412601688930508,
-1.6525919628732422,
-0.8341682190885938,
0.04519729634722039,
-0.06718028327556408,
-1.5649719285752928,
-0.37531851249345394,
-0.11822841604899964,
0.008154106549025474,
-1.4601449692077721,
-0.0030993446167428395,
1.462573870839384,
-0.11887757571245174,
-0.09361960188417112,
-1.2543408854038138,
-1.525280802664792,
-1.241413372948363,
0.008758808605552393,
-1.6676332942965022,
0.13826170532334445,
0.7537363962746654,
0.0833397476047736,
-0.12216949704941273,
0.02084807234445201,
-0.3421949354726438,
0.0017882611822728064,
1.5186851123669147,
-0.4603163345929323,
-0.09956972276089109,
0.7247597830417082,
0.5304935691839561,
0.003193152207841171,
0.5581346990750625,
0.010542051444639388,
0.7553167297004638,
0.5956891015800863,
-0.05566254332595716,
0.5362461335652635,
0.009895282042162012,
-0.3029901133286735,
0.09399206534914867,
0.8818812838713691,
-0.06795832284115338,
-1.580933930101608,
0.3014975711017802,
1.2092022074751856,
-0.019692531397995948,
0.9250981578351692,
0.4673383297638096,
0.03344339272872125,
0.13058472572385768,
0.036262684184263205,
0.458095375934523,
0.6492846129896516,
1.3039221898344189,
0.024764812872192767,
0.7236926243101844,
0.01444840086266692,
-1.0388939706148563,
-0.08359693344439546,
-0.4698004238001188,
-0.48055768545836347,

0.4662000815300196,
-0.041204969991243484],
[0.01798320141016013,
0.3393419127512466,
0.7047697703765823,
0.7064789163091142,
0.7191656785855808,
0.7198672068604162,
0.719867206860416,
1.0840512961729545,
1.0900003904895101,
1.1926153174924548,
1.867054342490295,
1.867054342490295,
2.368094916985588,
2.3683751276938714,
3.6383003552329964,
3.642527762446636,
4.505845683894636,
4.602446412902637,
4.972427659100928,
4.972427659100928,
5.1148164150678035,
5.8066136392845,
5.820681023327439,
6.727088267818048,
8.352653016300794,
8.773334122756978,
9.065902540064004,
9.065902540064004,
9.74880092287043,
11.30145464466351,
-0.13904770480884485,
0.23493722133539818,
0.4497803172772882,
-0.640584995353188,
0.9779304528121034,
0.0011745349607135795,
-0.7175493520168287,
0.373620380836484,
0.08111255661159392,
0.6653458367314794,
0.0037967443269442437,
0.6286122183467432,
-0.03825426016359485,
0.6896824961465278,
-0.23916946602734285,
0.4624850989736848,
0.1419493525208794,
0.7145176662978019,
-0.316909964913161,
0.6605458698076423,
-0.39309909050950365,
0.7623048158369822,
0.7469736169282462,
1.2400892032597353,
0.12266455668869777,
0.19578648318487166,
-0.784043232212839,
-0.10334183265831844,
0.32774480702026265,
-0.011491906972883237,
0.322988869457402,
-0.2027870428092001,
0.1281571522083961,
-0.1947961113336468,
1.7114513053475526,
0.005425684515457419,
0.8136825826883561,
-0.25903007981373005,
-0.5015428579892063,
-0.2600978115589891,
-0.15405473476720474,
-0.4207498663011112,
1.04622682040645,

-0.8642339256273757,
0.6248691295250803,
-0.3539240130838131,
0.011689879721248103,
-0.3068283940685192,
-0.6048660879777619,
-0.11704784670887185,
-1.2745062338511486,
-1.7890038533223112,
-0.7818671277256136,
-0.8877455887286905,
0.3164818198590212,
-0.27477307924208777,
1.8083251930535746,
-0.5031479272825263,
-0.9648993597903888,
-0.41682962131770707,
0.27040705073896654,
1.0892900815418987,
0.17779960873468054,
0.020337033950376556,
1.0551529464315492,
-0.10705558118147128,
-0.8851977322505518,
0.2232945485845247,
-0.5850403569085746,
-0.7690595025978197,
-0.4391673030332381,
-0.9309566421919284,
0.2973030450025941,
0.6780545449896137,
0.14045373030826291,
-0.5109313900859794,
-0.44070010469428217,
1.6940975579604587,
0.17062304524159472,
-0.402681393200546,
-0.7091187255059627,
1.3132357546586793,
0.2729391261599054,
1.0115940175088693,
1.0046673925246778,
0.4982257464819494,
0.04454679168285006,
0.4863023994043421,
-1.1433261311581624,
-1.820308437409531,
0.6035405700431521,
0.04003907881711185,
2.99381748600971,
-1.0437523456620463,
-0.22517480393070408,
0.10746937069246698,
-0.8788513722547997,
1.286531186523234,
-1.3157160458746164,
-0.6979748907629274,
-0.16573593981190854,
0.2529194783090185,
-0.2562517800846549,
-0.7966921479109774,
-0.4446940958312971,
-1.052310031662883,
-0.1506245930250076,
0.1615975776996014,
2.9220364155549214,
-1.7197843955783065,
-0.6592103621593355,
-0.2007498223803255,
-1.6188873615119244,
-0.15193115939751467,
-0.20866967970071015,
1.2910791163230286,
-0.4917152660712325,
-0.42348313202027765,

0.3518303936962707,
-0.10034486495238693],
[0.009039772647041902,
1.1400526086001903,
1.466905028683042,
1.6524670085313564,
1.987797816914966,
1.9900950346150876,
2.141320233232344,
2.390935514217652,
2.402429399816812,
2.81913686416337,
2.81913686416337,
3.5656810200683298,
3.5656810200683298,
3.5656810200683298,
3.76603447051701,
3.76603447051701,
5.499236323276389,
5.4992363232763894,
5.4992363232763894,
5.532789537471719,
5.532789537471719,
7.371733777832876,
7.371733777832876,
7.680783486275433,
7.706623781446164,
7.706623781446164,
8.457742149729329,
8.457742149729329,
8.95222736351662,
8.95222736351662,
0.18928393791812126,
1.0354889043700823,
0.3703113207409527,
0.30988589523494625,
-0.34669340410550803,
0.09615284519636855,
0.1011898884094612,
0.6160115432573767,
0.6148540393435122,
0.07934943483664206,
0.8557624329627399,
-0.7672633290674925,
-0.14188111026081618,
0.1995320641320393,
0.4594888802421805,
-0.03593187081620776,
1.2926591415659605,
0.31433950962447343,
0.8053272546604554,
0.6634815559763872,
0.311660786273531,
0.0294653444200854,
0.3044394736040779,
0.7599259037372076,
0.22962735656034647,
0.22562971232722423,
0.29511485598499815,
0.02260075636691185,
-0.9497096253675098,
-0.5331371571917215,
0.2192512323868535,
1.104832430155296,
0.8636617903941131,
-0.6756684630428217,
0.4169844309263729,
1.1297527651723116,
-1.160118439748996,
-0.7971836651224,
1.9169656561991728,
-0.007260796290839314,
0.3614132074474742,
0.6243022778229537,
1.8335025630018775,

0.35139722497115833,
-0.7247870129359816,
-0.6744811297738377,
0.49997538785346773,
1.4139521212248116,
-3.0799818401546553,
-0.14768564076593044,
0.06251500852230953,
-0.8547134606957393,
-0.30107856776262504,
-0.7654972812734481,
0.28603020918627825,
0.2129303248959857,
-1.2077141264570097,
-0.8488549318812054,
0.6492574804166855,
0.7751290676155554,
0.40909519807061084,
-1.2750630737105486,
0.4187798718836698,
0.33498424523924863,
-0.19017028209895162,
-0.03232931911977397,
1.0621483440417678,
0.452017889610544,
1.0192767515800205,
0.7290563184502946,
-0.5286462788982386,
-0.05512372609304388,
1.3189151919853837,
0.5865235617306253,
-0.579843833868396,
0.6980158293666034,
0.027004860579276476,
1.2876675233834314,
-0.1718561727774212,
-0.3827017785428823,
-0.7286404996479297,
1.1682112525473984,
0.22765379123755217,
1.4870969768845252,
1.1642898801914527,
-0.03976972562460009,
-0.8163296820868416,
-0.15318057329999143,
0.0482918949035788,
-0.06014345828968787,
1.1478583262090496,
-0.005141116181289305,
-0.5820859555621626,
1.086519910329415,
-1.08558016317555,
1.531511042803357,
0.042734048269580674,
0.7452599788242152,
-1.3094923903018465,
-0.9012205175381317,
-0.0755166851159581,
1.5493661520767013,
-0.2278023712186688,
0.8627212805492942,
1.8523276191392382,
-0.10961709979291216,
0.6746606852631131,
0.22566845572735392,
0.5920154791154506,
1.2709342533151342,
0.26903045177842116,
0.11724485476711638,
0.621942277674204,
0.15491924869980966,
1.472217084758381,
0.8975149794611956,
0.3601026339731485,
1.3234513983254743,

0.0182465793142515,
-0.11914786829986494],
[-5.765348397008027e-18,
0.3428984748224433,
0.3655450682584724,
0.704784031551808,
1.5409081472811441,
1.549300366088437,
2.3787666619954635,
2.3787666619954635,
2.8808650819472406,
2.884987740396357,
3.1308808834009203,
3.1363836212549696,
4.869321728501938,
4.869321728501938,
6.1150174264690085,
6.2582278136815175,
6.271636236200362,
6.551203581243674,
6.59741701640448,
8.230564833747417,
8.294003346338926,
8.31431753208387,
8.369586036193128,
8.415319617883744,
8.70447046242739,
8.709120596281233,
8.918659758853744,
8.99410216914391,
9.002256766523654,
10.218690251738174,
-0.6798627919679444,
-1.1436724543472427,
-0.1278269606092795,
-0.9286552360068459,
0.6524046490818174,
0.06134572949387961,
-0.49761179900021657,
-1.3918471119443894,
0.5858287319319103,
0.18669127045720346,
-0.4586789737568629,
0.2400608953322058,
0.022222124883641912,
0.9282004764805514,
0.6952358394741303,
0.805684979006905,
0.6146729452742722,
0.6777524139082742,
-0.8117842884117104,
0.3426849285038029,
0.7100299235979358,
0.7658186509139059,
-0.05418633990950139,
0.1761917490887145,
0.5744825473612316,
0.31705937472296297,
-0.9188523464356001,
-0.7337446596965294,
0.2061368312243915,
-0.6645694031850036,
-0.009871104765786294,
-0.14388187083546433,
-1.2485707106311008,
2.849540809212292,
-0.5119215846255833,
-0.19381894062193808,
-0.17310717674563775,
0.8465397629359774,
-0.46757240698237135,
-0.05607605828560263,
-0.40986137011816176,
0.07649505002491204,
-2.1522907800348112,

-0.6653346582594515,
0.4084801543759574,
0.4028897820911683,
0.16908117843954765,
-0.07217026591146015,
-0.72804755296126,
-0.13548738477012795,
-2.590679566047418,
-0.40743076961744734,
1.1030856800887578,
0.08388380440455438,
0.25224805905642333,
0.10644300798076774,
1.0474125729404944,
-1.1459016177859003,
-1.6916287310468103,
-0.9886067106540645,
0.24182274960650893,
-0.25703819491337826,
-1.1942818044595898,
-0.7763678643520989,
-0.916718557142268,
0.20086626876005667,
-0.3711023890817823,
-0.006962349053602254,
2.687031223814704,
0.3970791704566904,
-0.6890591654410289,
1.0504536719091986,
-1.2458238157294268,
-1.197330311624414,
-0.12155955689525438,
-0.2969718406476498,
0.15406921803787998,
0.9984735505197111,
-0.2734456365103056,
0.0100494123830882,
0.16172873483083283,
-0.6780409734277772,
0.4539997995464368,
0.7558576390626953,
0.9556605206819043,
-0.10572075101403118,
1.5564628686733721,
0.05397552837650759,
0.7415987024402694,
-0.1648500682327342,
1.052699139325594,
0.6776572023171561,
0.15219630541397192,
0.8843933196559655,
1.3948099389813178,
-0.16710315713046545,
1.2199200746863228,
-0.3825678575328573,
-0.26766985045935626,
2.5777182352113908,
0.23156136798610805,
-0.3377548185037014,
1.0689111835439342,
-0.17216022970038714,
-0.29381807805976945,
-0.5689533077540762,
0.30596827309967156,
0.5260993417881752,
-0.6768480303494689,
-0.1092430795457648,
1.5442082488768039,
-0.10826504240549732,
-0.9555598065181787,
0.5334835330029429,
0.5569227134357198,
1.2665483894271157,
-1.9677332823608626,
1.5559931182978979,

0.7709453943956464,
1.5746506000831744],
[1.7625089228179507e-05,
1.298540182320117,
1.3141870420194457,
1.3305471045820527,
2.0686735179487985,
2.23810597518972,
2.267373737425337,
2.295441377067579,
2.3325388050051976,
2.352484360014207,
2.370891483796666,
3.3117124844820194,
3.363605896627787,
3.858981460194034,
3.8589814603490344,
3.859537867870235,
3.8595381713444072,
4.26458571773699,
4.264585717738593,
6.599549149586233,
6.655963438031886,
6.939612414177192,
6.939612414268938,
7.004302986969205,
7.010544991093791,
7.97314828104894,
7.999280802101656,
8.022628296732579,
8.027896471579398,
10.077254342713585,
-1.0733166032270363,
0.4264387306682,
-0.29490479134596703,
-0.45181113351566254,
-0.9496712390342118,
-0.24412788786515252,
-0.17005038539710493,
2.131747001055919,
-0.8801477965160032,
0.1985900676052623,
-0.17829180427317373,
0.1513896817808816,
-0.2996418627583153,
0.09152392337611555,
1.0155053498240294,
0.6713280773462803,
-0.5915672184524843,
-0.21780814909773427,
-0.23676595566314207,
-0.1539223229682568,
0.9733737893230769,
-0.8325690111887051,
-0.6684349693104433,
0.944923850675747,
-0.8270803600282979,
0.4869668407293288,
0.07342964610078141,
-0.8928903181234079,
-0.04262248204617692,
-0.7602498290394286,
-0.48858825463899247,
1.7340528561515622,
0.7340502315717261,
0.5163401187759543,
-0.5015858040228415,
0.2890335055344555,
-0.46033506221820203,
-0.8185907107929625,
0.4557079379705311,
1.6601520871164626,
0.07125306723268252,
-1.4035119795986863,
-0.6741575848895364,

1.84360490583597,
1.6836338044245691,
-0.33329370458317326,
0.19854317161902654,
2.03917897859608,
-0.3342251938073038,
-1.6015864477635742,
0.35422488479060155,
0.1487121545506934,
-0.2821114888507261,
0.21869934797564045,
0.7523961573071125,
1.3021785537923125,
0.26184859002187644,
1.2441238228144538,
0.039670539596101485,
-0.15687777352236798,
1.1386642271664005,
-0.13559116971858365,
0.39167499360215746,
2.4453261619465274,
-1.0890320328395309,
-0.4289422005081893,
1.425249318116838,
-0.0016733698174337221,
0.023121567661497415,
-0.28325087531692295,
0.940069260590291,
1.163437407349777,
-0.4256527399896147,
-1.4972731615803316,
-0.12624975701746496,
-0.10144651691714318,
1.3307378561466523,
0.041778278155808674,
-1.2746374184036902,
-1.2216848741140482,
1.4048769664929757,
0.8957112974734468,
1.0058068981677006,
-3.0965653057688316,
-0.3344448337959029,
-1.0948102803209163,
-0.038478236121346335,
0.12618687504486675,
-0.0850775611793361,
1.33550885346982,
-0.04334772196309596,
0.2406908654957122,
0.7327249608913058,
0.9966290850109706,
2.2852565843192423,
-0.030570948934034583,
-0.5566407461361556,
0.813246653395058,
0.5946355273642772,
0.07519254658407079,
0.27120989268694123,
-0.15415722381241237,
-1.1239288818585278,
1.893751281002709,
0.7530205722182245,
1.4454588921082565,
0.15037300343080462,
0.1793867334099716,
0.3587259643057371,
-0.006799349268608,
1.0907160355013774,
-0.30186717991666434,
0.8507589785411056,
0.009155709268365956,
-0.8165113788656561,
0.011816515180360782,
1.116430418883973,
1.5343789763879045,

-0.1682835166498154,
-1.5032827361947747],
[0.10956969985098665,
0.4360115975982133,
0.5932904704915389,
0.6050451872391112,
1.3688142364348175,
1.3737597367805747,
1.5226773273508583,
1.5589271816733294,
3.150320547140947,
3.1613053536766147,
4.7635234433865605,
4.823148382609575,
4.823188243925802,
7.171157860626202,
7.171157878485542,
7.171158529865571,
7.175073431930883,
7.183070822592498,
8.102400589897519,
8.10392263292654,
8.103940780725882,
9.021308924987686,
9.02133947231205,
9.082831297490506,
9.094868565409332,
9.236804949782064,
9.237029573386183,
9.26430291432718,
10.00744744746223,
10.032040374677464,
-0.5042382729215367,
1.0180015436638659,
0.10423420212375085,
-0.3355687578372037,
-0.390231922280216,
-0.09867602213718178,
0.3860296418386605,
0.8522828003851446,
-0.7730596965193267,
0.08803782521712378,
1.175277695996097,
-0.3769225558693855,
-0.4553519820435133,
0.4758962487375975,
-1.0517577693418567,
0.3712029741888282,
0.27334544561774066,
0.03397030216872572,
0.3891396532807883,
0.12152393226873999,
1.5192516885791116,
-0.5124931636551364,
-0.5667003913709334,
-0.08425393582345661,
-0.8152956002657638,
-0.008087141654348068,
-0.20833380477584987,
-0.19186344764752156,
0.309463849209865,
-0.16555810903618162,
-0.28534485935787846,
0.025710068868521854,
-0.9172960495760906,
0.027100436879331523,
-0.9806057417125867,
0.46877133554227174,
0.22597640762440013,
-1.8700902599315918,
-0.6222133609404971,
0.023922176634194332,
-0.6777524378245746,
-0.3883818313368169,
-0.19921738892417568,

-1.3418496115619054,
-0.05365669225714106,
0.38835985529462186,
0.46741217514915123,
-0.1388983196762152,
1.0002905156718758,
1.6633296566254185,
-1.0097136763815113,
1.203957772036831,
0.34033273848251194,
1.1456357751380877,
-0.4637199651146574,
-0.13624703361320328,
0.6785440200843139,
0.21691345067474185,
-0.4166036808107477,
1.6003722740710606,
-0.294881370724797,
0.04198878790454511,
-0.689532047804753,
-1.3911901894570817,
-1.1667039833298627,
0.824824800006074,
0.8047687705752488,
0.5727196814509768,
0.2385364014798047,
-0.512164832149354,
0.17925141295137917,
-0.19704824673887428,
-1.6372757558452697,
0.5441675736491897,
-0.13936070677778437,
1.3935144064781402,
0.10763856884501717,
0.6679788244416961,
-0.33118654939932857,
-0.35796695002153883,
0.2632208027041665,
1.5831503864671255,
-0.25140146740224,
1.135774876905931,
0.5897962040478675,
-0.7664904031063712,
1.092649076983624,
-0.9862462517151122,
0.010216869101717933,
0.5705833807099128,
0.09839127014019662,
-1.034770130526523,
0.20582133073730807,
-0.44428721001967564,
0.5272444819968604,
-0.2627660867388947,
-0.9857421779520247,
-1.3646538516413942,
0.14779052759756187,
1.1207932022043874,
0.30427416833245535,
-1.055109005337564,
0.30548258474769546,
0.8153937738463674,
0.5082537250510629,
-0.2973639305797175,
0.0034742209991892575,
1.5168338311712208,
0.0971904670135119,
0.5305672799342332,
-0.08170159551874663,
0.4909203029282298,
-1.1771354207712392,
0.038725619801271274,
-0.6655680746562402,
1.312626970471223,
-1.0115568894576918,
0.16290676730453105,

0.3074286838975462,
0.013117345918778788],
[0.4866654913190249,
0.4866654913190249,
1.4314116389487666,
1.4314116389487666,
1.548608655381947,
1.5552664901547235,
1.662920497545568,
1.676155881591226,
1.6952532774325564,
1.6952532774325564,
2.46741851740979,
2.50102877426778,
3.636145431120801,
3.6361454311208004,
3.6361454311208004,
3.6361454311208004,
4.0161273218482085,
4.0161273218482085,
4.5411636896776235,
4.5411636896776235,
4.922545565757144,
5.624338470516907,
5.642847901355697,
5.904037934138086,
5.915382265322609,
6.860930229950661,
6.860930229950661,
9.422517992567101,
9.467623480195014,
9.467623480195014,
0.40091998571272736,
-0.3581846438287994,
-0.32033246941904975,
0.24409866553117068,
0.176560178011825,
0.4452285262890565,
-0.37150326481454765,
-1.0185408309777293,
1.6197684873006286,
2.0095194673452026,
-0.20342804182482532,
-0.4471079471088122,
0.37127155242298737,
0.4199607045713712,
0.11145558426636207,
-0.289968410859463,
0.9752004498173846,
-0.11972987215087512,
-0.8215210043660294,
-0.4144105549344152,
0.04966095047933562,
0.6091224412599766,
-0.2481186610805342,
-0.9036735819055022,
0.22922493224752044,
-0.04567884568114995,
-0.40529506738584414,
-0.44609185937563545,
1.1000542207972743,
0.8995834380092431,
-0.4391184290327733,
-1.623814742575614,
0.11121717167805684,
-0.15344289124808663,
0.7072346226342329,
0.013911394934765222,
1.1008239636292285,
1.1342299046285917,
-0.1605718532112884,
-0.8521154510350495,
-0.9726520318623231,
-0.11078467990736285,
1.2893067940177063,

-0.8656847782514395,
-0.4554157535410911,
0.46752563535079134,
1.0115326820592876,
-0.017465678177200683,
0.7555243091795097,
0.2225241128439446,
-0.6129237770264853,
0.36249038141661666,
-0.1470816778834455,
-0.23781939745964606,
0.7410640700988803,
-0.10219672891678527,
-0.9043863030697507,
-0.054446492384892985,
-0.23618927113386157,
-0.3971482428910102,
0.42980979959963467,
-0.4511512430809565,
-0.3322123782853245,
0.15031459204805087,
-1.4287283715133035,
-0.026859055017604255,
0.20826999854501216,
-0.016397767746330064,
-0.026711091547853946,
0.15899307013846248,
0.18349959987180348,
-0.019181103141429404,
-1.3777401647916119,
0.05578672583947999,
-0.17191397249005735,
0.08492618339903452,
0.2781511329827625,
1.6853098582374997,
-0.7964285169174421,
0.13628687772581302,
-0.26524580358677924,
1.3615146942088054,
-0.18163849035743365,
-1.5476669195398023,
0.2740670870741208,
0.19740142577560438,
0.18567840551061798,
0.43288784764736754,
-0.8069341399605817,
1.5139336080928714,
-0.7594440385529729,
-0.20443587392639553,
0.0916888700165231,
-0.03922249489745327,
0.4135801157921025,
-0.06731598775715468,
0.5038720973518589,
-0.7705715736697758,
-1.4775324565516457,
0.061950004909204714,
-0.9555167624183505,
3.1277455162888304,
-1.137887484425651,
-0.09852947784569874,
0.042597844143281625,
0.06134264537966588,
0.3384848933516047,
0.012002169875906457,
1.5732417764972468,
-0.3550969359952667,
0.37821230776206083,
1.4914211446267915,
-1.2027836619303884,
0.22421848814312553,
-0.2072711571643093,
0.05122655754886873,
-1.5578500765322636,
1.5918641150141506,


```
1.2635397829742223,  
-1.2452354025631063]],  
    'run_time_sec': 43720.81102514267}}],  
'redundant': [ { 'config': { 'ansatz': { 'gateset': 1,  
    'layer': 30,  
    'ugate': { 'coefficients': { 'bn': [ 0,  
    0,  
    0,  
    0,  
    0,  
    0],  
    'cn': [ 0.5,  
    0.5,  
    0.5,  
    0.5,  
    0.5,  
    0.5],  
    'r': 0},  
    'time': {'max': 10.0, 'min': 0.0},  
    'type': 'xy-iss'}},  
    'init_param': { 'value': [ 0.01970245038838479,  
    0.18403088824160574,  
    0.23516669011749736,  
    0.3619340287619915,  
    0.7559235356843684,  
    0.8037553588156275,  
    0.8037553588156274,  
    0.8037553588156274,  
    1.523650120002224,  
    1.5399034674182392,  
    3.5267025169505333,  
    3.526702516950533,  
    4.657784878324055,  
    4.735234101549794,  
    4.768060921347247,  
    6.910522417024203,  
    6.910522417024203,  
    7.937298848041795,  
    7.937298848041795,  
    7.937298848041795,  
    7.937298848041795,  
    7.937298848041795,  
    7.937298848041795,  
    8.950027818739875,  
    8.950027818739875,  
    8.950027818739875,  
    8.950027818739875,  
    8.950027818739875,  
    9.275481401407285,  
    0.034552241786943605,  
    0.7960682992937013,  
    -0.16960774395822967,  
    -0.13488653659207303,  
    -0.07127010448742209,  
    0.7213919487775996,  
    0.07517487951530155,  
    0.7669403129297154,  
    -0.0980146039609658,  
    0.2841753594230342,  
    0.41897129078802076,  
    0.7644737129881134,  
    0.21659836679204617,  
    0.40978346593820614,  
    0.23386906952753808,  
    0.6351364889148511,  
    0.33021983360135304,  
    0.3518425622667454,  
    -0.16759580526603354,  
    1.1955178055405769,  
    -0.1119172073880175,  
    -1.6647459335605923,  
    -0.09499524124538847,  
    1.171044098528735,
```

-0.022137095989983098,
0.46479840085551427,
0.5434533112600137,
0.7953483283824695,
-0.23791626316884204,
-0.34327459028867535,
0.9770824990177593,
0.3972109926583832,
0.031305244787475206,
0.06660894739153407,
-0.4270300066285869,
-1.4580475210852422,
0.18899331747125298,
-0.5789031865333153,
0.2203088633995836,
-0.028805094274594364,
-0.21236041446549625,
0.00835869807102751,
1.8444421334304235,
-0.36021419267454113,
-1.0777998782841605,
0.7201309879688491,
-0.07151832135087417,
0.013523038580686776,
-0.5101061898109359,
1.528477364550823,
-0.24055091899624312,
1.595727742687772,
1.181011187425284,
-0.07205139039345279,
-0.3645912646252544,
0.922259297261975,
0.47504473393792157,
-1.4582846498754758,
-1.1041238978674286,
-0.08619812725702276,
1.5348185477516838,
-0.034799821636765677,
0.34854563969795865,
-0.3057886345189525,
0.3435344949085216,
2.387989506593159,
0.5928805156422549,
-0.02250711897690342,
0.734685803060112,
1.1230348216575474,
0.9339212770561773,
0.5429657194529808,
-0.9530889967003539,
1.2607191116533525,
-0.18262479735843284,
0.2292294020716647,
0.484625167272873,
2.2621526738562925,
0.3832060422832924,
0.2504382199240898,
-1.1186566372739195,
1.0354182359738777,
-0.9045076995614423,
1.0282595509077306,
1.805679851173207,
0.9332861023326164,
1.4396403686507162,
1.6178312922032976,
-0.22196274260736226,
0.19030625281227892,
-0.32840557234129963,
1.004592356770436,
1.8720477930131976,
-0.06388076228748135,
0.12235432131563899,
-0.5998878734536057,
-0.5461403115162753,
-0.17116133229966776,
0.6826247841010238,

[illegible]

```

'0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -7.512595875373561],
                              [12, 6, 0, 0, -6.004303631668366],
                              [20, 10, 0, 0, -4.83491199768215]],
  'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                      'identity_factors': [0, 0, 0, 0],
                      'noise_level': [4, 2, 0, 0],
                      'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
0.001,
0.001,
'noise_prob': [
0,
0],
'status': True,
'type':
'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [1, 1, 0, 0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
0.001,
0.001,
'noise_prob': [
0,
0],
'status': True,
'type':
'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [2, 2, 0, 0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
0.001,
0.001,
'noise_prob': [
0,
0],
'status': True,
'type':
'depolarizing'},
'odd_wires': 4}],
'run_time_sec': 7.652128219604492}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [ 0,
                                                            0,
                                                            0,
                                                            0,
                                                            0,
                                                            0,
                                                            0],
                                                    'cn': [ 0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5],
                                                    'r': 0},
                        'time': {'max': 10.0, 'min': 0.0},
                        'type': 'xy-iss'}},
  'init_param': { 'value': [ 0.2383708143795501,
                             0.8137820253985804,
                             1.0561141201801563,
                             1.2070683478267785,
                             1.9518166906091101,
                             2.08301823027678,
                             2.6774459967535513,

```

2.7270096866449576,
3.7369434259122447,
3.7421546540398443,
4.458327272130387,
4.463102384700506,
4.463102384700506,
4.482690585412051,
4.482694652744643,
4.498978826283554,
6.3580266266370415,
7.88639599997723,
7.886532810158634,
8.060529572035655,
8.06234783983503,
8.127555772505229,
8.130166630032164,
8.42408407139234,
8.433074307328475,
9.412401860498063,
9.420584174895616,
9.429609653004679,
9.434851407347677,
11.283524698808058,
0.19157417616433814,
0.42257952171340063,
0.3129095694283805,
0.28360255750055535,
-0.7229159635345903,
2.545731433831817,
1.1687892956196542,
0.5222949375721274,
0.9482151911772319,
0.28742832079435787,
1.1627391357840349,
-0.27939056132165146,
-0.07574334812551493,
-0.529119211391362,
-0.08653083237786212,
0.042544229108731915,
-0.09154582517805988,
0.2776265525620834,
0.373637405241428,
-0.666296500917012,
-0.729825790392568,
0.36164641825124294,
1.2066181407380305,
0.18327070025757417,
2.2757552060408917,
-0.004789158452516104,
0.49019439516235463,
2.190622553796523,
0.6131206832540379,
1.0987721265230228,
1.9413219378309259,
0.07975169469268534,
0.17531082109327745,
0.01770145552231998,
0.22429808435825893,
1.1460098869067579,
-0.25766861736092417,
0.4386465248337572,
0.7521614434169624,
0.07039918338308593,
-0.23913727928604284,
0.062497116873310714,
0.27575365987954265,
-0.8268952220573135,
0.8599765782432734,
0.6958325703328606,
-0.16792542378671177,
-0.13186811496768225,
0.17745018848846658,
-0.13764108606901534,
-0.34798347931108986,
0.3901205214779397,

```

-0.12832331396189006,
0.7623260417931323,
-0.08745728181826747,
0.22448567111633413,
0.5103689813486589,
0.020179740460913417,
-0.4639048953982949,
1.8052783293547816,
-0.06895296635033099,
-0.46137869531525183,
-0.7694587159951923,
-0.1507074275764489,
-3.159084078074864,
1.5556721360426484,
0.9802025173412374,
-1.5980562353917622,
0.9974678614222413,
0.11965749551778265,
-0.5964068365118492,
1.00912773091807,
1.2674306548806673,
1.7418320010188646,
-1.2223756485879664,
-0.006270145144671745,
0.7863824204489233,
-0.08410721186616651,
0.13420385894759826,
0.6686584151585806,
-0.7946689395347296,
-1.448768973512125,
-0.4158373205099576,
-0.1003664997933048,
0.5370971852976741,
-0.013369261455621718,
-1.2134070474127567,
2.71822204346588,
-0.8266087910498845,
-0.04274152999618493,
0.627725493238542,
-0.10714625448592663,
1.4055199595750496,
-0.08877591733311402,
-0.6400618055168635,
0.27990653766167567,
0.27230874688701184,
1.8329367216853043,
0.12128377229042309,
0.03719209379786179,
0.7558193864298689,
-0.027085062677120984,
0.7947118159580207,
-0.18810209431195668,
0.008294184744538372,
-0.9110921816475059,
1.3065108206329406,
0.004810940817056788,
1.4030602129547465,
0.03878633149748846,
0.29850504257680605,
-0.9524583070906825,
0.10037229527914636,
0.4277056649647598,
-2.2528322583674503,
0.030724614514628212,
0.8416350822543037,
1.5579013461800135,
0.5696673724928452,
-1.5352194543526818]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,

```



```

0,
0],
'status': True,
'type':
'depolarizing'},

'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
'identity_factors': [2, 2, 0, 0],
'noise_level': [20, 10, 0, 0],
'noise_profile': {

'noise_prob': [

0,
0],
'status': True,
'type':
'depolarizing'},

'odd_wires': 4}},
'run_time_sec': 8.07330322265625}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 4.1837705414928806e-17,
1.1742327239243593,
1.1742327239243593,
1.174232723924359,
1.3021435979041922,
1.3883521076730898,
1.4901272631800928,
1.490127263180093,
1.490127263180093,
2.542545188872923,
2.56739133991554,
2.677178516908646,
5.624946341909109,
5.624946341909109,
5.63924283183031,
5.672838193807689,
5.672838193807689,
5.672838193807689,
5.672838193807689,
5.672838193807689,
5.805507018952099,
5.858678454346443,
7.374841183417447,
7.374841183417447,
9.078346087689514,
9.17910617229311,
9.185363997877552,
9.550522060955204,
9.715006087156702,
0.5380723684568228,
0.20346632529178696,
-0.28402043001150323,
-0.013694659434132353,
-0.8491217472851175,

```


-0.13891219530245408,
-0.7902948900320785,
0.02035105936303376,
0.35891510624171696,
0.07011262472067725,
0.2571910630887441,
0.09727807922224965,
-0.4253506785797692,
-0.08201618636277365,
-0.5629942371157008,
0.4984357070329931,
0.22460606922422946,
0.6436861273901548,
0.5581387866543217,
0.6438048424266741,
0.6625810608832491,
-0.2877740698749285,
0.03400434270310955,
0.6598198228171638,
0.15695492486622947,
-0.23277712193728803,
0.3379436543817524,
0.6679629986711755,
0.4913802101172593,
-0.6543304626041586,
1.2398324170787045,
0.10868488610295632,
0.4726806257545104,
-1.0278416218016544,
-0.5660236088762642,
-0.06627186605932975,
0.3527547117645279,
0.3944014355355292,
0.33709190475341266,
0.16707772676088822,
0.29195080738380563,
1.0354529601392413,
0.12500585138196232,
-0.037178956773431956,
0.7823218278561086,
-0.04464215343745706,
-0.36923576305262273,
-0.9863817864437636,
0.039519076991602065,
-0.35646400258617383,
0.04542480304419608,
-0.46125271135496354,
0.40472620693719735,
0.7802480661953589,
-0.6757172179201896,
0.6396142241310521,
0.2327897749192615,
0.8954622299286413,
0.7158405805761733,
0.1382139633537826,
1.0616928944303126,
-0.6122315420380016,
-0.6738539946512315,
1.0433459792659807,
0.494526070112864,
0.20621322780405107,
-0.05318175489586881,
-0.11213659998229918,
0.32941262943060273,
0.13931786347259914,
-0.07640781308377202,
0.31450116184408605,
0.3658109215963646,
0.04460349089074807,
-0.12913310998017427,
0.31890485610090735,
-0.005491508360761248,
1.2692063073436923,
-0.3006174703747285,
-0.41404661321397057,

```

-0.5199199797590206,
0.44225909618051784,
-0.4233007342169607,
0.7456354776212751,
-0.32484988424610756,
0.9268151291603125,
0.8385591284905366,
0.027219364270413358,
-0.5520387125082392,
1.3757901840365965,
0.02933419406725825,
-0.2915193842748469,
0.33917094344303517,
0.10828106345120646,
-0.5650186064180197,
-0.6041651963720519,
-0.19230834783869816,
1.9848826966849449,
0.0632219631903188,
0.04804551511145073,
0.47752350144148015,
1.3406847618100746,
-1.3722262204383469,
-1.0513455672656666,
0.7121308231347245,
0.13278942055995957,
1.9758312031522887,
2.19648222646383,
0.8936788375205645,
0.05510909660880629,
-0.18765678539637448,
0.45031205104496697,
-0.6542716355161553,
0.5783583754336702,
-0.18844184232604352,
-1.5667303012058793,
0.36110899429768645,
-0.275182998952267,
0.30367576290775045,
-0.1532687178202341]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [ 0.5,
                                         0.5,
                                         0.5,
                                         0.5,
                                         0.5,
                                         0.5],
                                'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_xy_noisefree_time_evo_ric3_unifrom_scaling_sample#2'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},

```

```
'zne': { 'data_points': None,  
        'degree': 0,  
        'method': 'richardson',  
        'sampling': 'default'}},  
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z '  
        '1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + '  
        '1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '  
        '6 + 1 Z 5 + 1 Z 6',  
        'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + '  
        '0.5 Z 0 Z 1 + '  
        '0.5 X 1 X 2 + '  
        '0.5 Z 1 Z 2 + '  
        '0.5 X 2 X 3 + '  
        '0.5 Z 2 Z 3 + '  
        '0.5 X 3 X 4 + '  
        '0.5 Z 3 Z 4 + '  
        '0.5 X 4 X 5 + '  
        '0.5 Z 4 Z 5 + '  
        '0.5 X 5 X 6 + '  
        '0.5 Z 5 Z 6']},  
'output': { 'data_points': [ [4, 2, 0, 0, -7.130214875567141],  
        [12, 6, 0, 0, -5.622281179564727],  
        [20, 10, 0, 0, -4.4547314200293675]],  
        'noise_details': [ { 'gates_num': [4, 1, 1, 0],  
        'identity_factors': [0, 0, 0, 0],  
        'noise_level': [4, 2, 0, 0],  
        'noise_profile': {  
  
            'noise_prob': [  
  
                0,  
                0],  
            'status': True,  
            'type':  
  
                'depolarizing'},  
            'odd_wires': 4},  
        { 'gates_num': [12, 3, 1, 0],  
        'identity_factors': [1, 1, 0, 0],  
        'noise_level': [12, 6, 0, 0],  
        'noise_profile': {  
  
            'noise_prob': [  
  
                0,  
                0],  
            'status': True,  
            'type':  
  
                'depolarizing'},  
            'odd_wires': 4},  
        { 'gates_num': [20, 5, 1, 0],  
        'identity_factors': [2, 2, 0, 0],  
        'noise_level': [20, 10, 0, 0],  
        'noise_profile': {  
  
            'noise_prob': [  
  
                0,  
                0],  
            'status': True,  
            'type':  
  
                'depolarizing'},  
            'odd_wires': 4}]}],  
    'run_time_sec': 7.930702447891235}},  
    { 'config': { 'ansatz': { 'gateset': 1,  
        'layer': 30,  
        'ugate': { 'coefficients': { 'bn': [ 0,  
        0,  
        0,  
        0,
```

```
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.3442332088020621,
0.7247295953616149,
0.7247295953616149,
0.7610537291022658,
1.7162580174489597,
2.0119282772467635,
2.0119282772467635,
3.369934702704019,
3.369934702704019,
5.00211845115155,
5.212181786011135,
5.346566249705437,
6.103091648910964,
6.103091648910964,
7.050711873868975,
7.050711873868975,
7.492144583370162,
7.492144583370162,
7.606105711064038,
7.606105711064038,
8.717193051662631,
8.753101318062777,
8.877107151002964,
8.889590522678484,
8.982812044989718,
8.982812044989718,
9.499874357008034,
9.499874357008034,
9.659025604148875,
9.671110590806489,
0.02814413440270558,
0.250593445582179,
0.17496868989230024,
-0.19749991731592192,
-0.5878125517240043,
1.1370895496823794,
0.3339578948727643,
0.30833012680275795,
0.21147160367429677,
0.5518340573364652,
0.9816766119937317,
-1.3427949214182668,
2.2864376119784486,
0.673664283367827,
0.4767745045632036,
-0.47480445681581174,
0.05912449937800317,
0.9513757199218045,
1.079093875588678,
-0.4313388366547843,
0.43347513808131466,
-0.3381632209207048,
-0.42699598389932686,
-1.1528003206699722,
0.2505681368756031,
1.1900947118566725,
-0.38260053073473543,
-0.6405385159573362,
0.08926968472452403,
-0.04834256297702397,
1.603027736914297,
-0.44672812120859223,
-0.5759172165575565,
```

1.25237714334681,
0.04254918814060746,
-0.002429292667880943,
0.47732275422632153,
-0.10204026491637587,
0.02357869435971368,
0.06910865492863566,
-0.5942663805386644,
1.1629666490831163,
0.28625726751699415,
0.4183193427753825,
0.33026699893767814,
1.8860632823719743,
-0.37929215782635534,
0.8251027249542411,
-0.13937217921728381,
-0.35829122981546885,
-0.11830319301401378,
-1.1307480811476716,
-0.33746884247229475,
1.7161163093033744,
-0.03517175574182253,
-0.4044561448132672,
-0.7301790593832838,
-0.10496992427986328,
-0.0377875273258637,
0.11576685707785253,
-0.9348023198901909,
2.132577954045408,
-0.47199808297503054,
0.09642080758509694,
-1.5590866370005203,
0.10045413530924742,
-0.3215930726850239,
-0.7020784722315541,
1.2376661808413174,
0.5709879203761018,
-0.820670510834149,
0.1634866307612446,
0.06270250983710676,
-0.03570422816362065,
-0.5699745682010601,
-0.26488761312159137,
1.795803611544861,
-0.4017001375917998,
-1.757541020399501,
-0.05132347348943386,
0.8521664539464072,
0.4812990598336637,
-0.7041980923594973,
0.6941905718406797,
1.5589556954504384,
1.9089912382579748,
0.4849438728523755,
0.03939482159422182,
-0.2564597588944596,
-0.16675906759041134,
2.5610800117093855,
0.9927310518963223,
1.3170410278463267,
0.6281459286410694,
-0.07878692694615999,
0.5865838795710373,
0.4894408136056478,
-0.043552207832595204,
0.8988064624972067,
-0.35279792613941824,
1.5088457985023234,
-0.32174328535909336,
0.41978129355629945,
0.014567846908235548,
-0.17446008524494477,
0.35494980160059775,
1.44166715836895,
0.027787407199764528,

```

-0.2573832607835415,
0.9605798444843694,
-0.08907047543629609,
-0.012116788408097524,
1.2676248549462201,
-0.2512276837340942,
0.7869834357157488,
-2.3607393050467915,
0.1914341630047483,
-1.0542342437530707,
0.8454261921281611,
0.02027118187750096]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                  'noise_prob': [0.001, 0.001, 0, 0],
                  'status': True,
                  'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_xy_noisefree_time_evo_ric3_unifrom_scaling_sample#5'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z '
                                '1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + '
                                '1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '
                                '6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + '
                                                         '0.5 Z 0 Z 1 + '
                                                         '0.5 X 1 X 2 + '
                                                         '0.5 Z 1 Z 2 + '
                                                         '0.5 X 2 X 3 + '
                                                         '0.5 Z 2 Z 3 + '
                                                         '0.5 X 3 X 4 + '
                                                         '0.5 Z 3 Z 4 + '
                                                         '0.5 X 4 X 5 + '
                                                         '0.5 Z 4 Z 5 + '
                                                         '0.5 X 5 X 6 + '
                                                         '0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -7.407206817517894],
                             [12, 6, 0, 0, -5.829141582765496],
                             [20, 10, 0, 0, -4.618573032874505]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                     'identity_factors': [0, 0, 0, 0],
                     'noise_level': [4, 2, 0, 0],
                     'noise_profile': {
'noise_on_init_param': { 'status': False,

```

```

'value': 0},

0.001,
0.001,

'noise_prob': [
0,
0],
'status': True,
'type':

'depolarizing'},

'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': {

'noise_prob': [
0,
0],
'status': True,
'type':

'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
'identity_factors': [2, 2, 0, 0],
'noise_level': [20, 10, 0, 0],
'noise_profile': {

'noise_prob': [
0,
0],
'status': True,
'type':

'noise_on_init_param': { 'status': False,
'value': 0},

0.001,
0.001,

'depolarizing'},

'noise_on_init_param': { 'status': False,
'value': 0},

0.001,
0.001,

'depolarizing'},

'odd_wires': 4}},
'run_time_sec': 7.742349624633789}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.5482907189468895,
0.6413620809492503,
0.6422482149552934,
1.5892409709442974,
1.850708229814058,
1.850708229814058,
1.8745552660050262,
2.400679867939261,
2.402212278134192,
2.485190345183722,
2.485190345183722,
2.4851903451837223,
2.488397493491329,
2.4883974934913287,
5.077476636438464,
5.077476636438464,

```

5.732990197214016,
5.806868622878696,
5.817834245705687,
6.166757373752113,
7.3114292874931905,
7.311858449036949,
8.346734585466223,
8.350435170229554,
8.354614694314481,
8.35659817980371,
8.356905245901928,
9.313534396314816,
9.313534396314816,
10.443787047737276,
-0.5971318801423705,
-0.6830583120471296,
0.053446928626708114,
-1.488251087277441,
0.7409998345617078,
-0.020561007687691084,
-0.47029189134787985,
-0.9187802118512801,
-0.07390570205801456,
0.6091612085770479,
0.5696050095749661,
-2.634874245900948,
0.14733366015521565,
1.0781100436188797,
0.3459269460772517,
1.2820636645787713,
-1.605623121104658,
0.856791024529966,
0.16374864437029604,
-0.7377840957855112,
0.7603752030179747,
0.4958590077418653,
1.0293890027768824,
0.4862585296205842,
-1.0838654128292502,
1.2632637762941026,
-1.1043454462318527,
-0.7867654190654241,
0.8818217647037212,
1.0406267750063978,
0.05770567519039498,
-0.5922070722918433,
-0.3271403288282926,
0.5980786503295974,
-0.02050375112330311,
-0.9007241609880184,
0.5239405469453117,
0.7601496703420288,
-0.6449049942439851,
1.7340858429103823,
0.10050113333288421,
0.3143347863780438,
0.28408386490618526,
-0.1222389867527315,
-0.09502282329152913,
-1.3004368662949968,
0.16856179320059875,
0.6790103402211991,
0.3355977478652572,
0.0935957266804422,
0.42936820813464754,
0.5148227650107633,
0.10764356588532305,
-0.054251737808208085,
0.06525040933822983,
-0.21143245853075732,
-1.5801988491404158,
0.023513208971767067,
1.0921277689034055,
-0.11501776583679489,
-0.14182911234947893,


```

1.8716225858725093,
-1.0747761250482988,
-0.01873186129021693,
0.9952348515577564,
1.5589645169221944,
2.4625123021717044,
1.5320934484141422,
0.6264578101560674,
0.015802850908690148,
0.6559295719723718,
0.5030706883609005,
0.010357532831659657,
-0.6971755064049523,
-1.334963536555068,
0.05714132580252865,
0.9478292633237722,
1.541635767019547,
-1.2357775558070643,
1.5753066317572375,
1.2206753513411082,
-0.013882574638756213,
2.102115305768464,
0.6867648509985108,
-1.0810617894892982,
0.608205793929093,
-1.0701207529796843,
-0.008804617073740417,
1.0802724219136723,
1.4472704941581649,
-0.525983160453369,
0.10347757085774918,
0.07451664808136123,
0.8113743025574697,
-0.19830571509994802,
0.20760359505818776,
-0.5709476706639499,
0.1791350390980861,
-0.32061652775092425,
-1.6382844876022806,
2.271046687924862,
0.09972766177435569,
1.671157305178672,
1.0596955078067567,
-0.7395801076185267,
-0.8217251410877643,
1.2660358026872456,
-0.149859182069558,
1.7386861885389688,
-0.003505497297480453,
0.677126408740769,
-1.0541684899696007,
0.9437251450981726,
1.7354017388586316,
-0.8022872949930118,
0.01881044769021433,
-0.013699411839369284,
1.5965466699123847,
0.4486397166383021,
1.5669137279484207]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [ 0.5,
                                         0.5,
                                         0.5,
```

```

0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_xy_noisefree_time_evo_ric3_unifrom_scaling_sample#1'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z '
'1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + '
'1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '
'6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + '
'0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + '
'0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + '
'0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + '
'0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + '
'0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + '
'0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -7.463351332654991],
[12, 6, 0, 0, -5.841676652164559],
[20, 10, 0, 0, -4.604434423168806]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [
0.001,
0.001,
0,
0],
'status': True,
'type':
'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [
0.001,
0.001,
0,
0],
'status': True,
'type':
'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
'identity_factors': [2, 2, 0, 0],
'noise_level': [20, 10, 0, 0],

```

```

'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,

'noise_prob': [
    0,
    0],

'status': True,
'type':

'depolarizing'},

'odd_wires': 4]],
'run_time_sec': 7.26769757270813}},
{ 'config': { 'ansatz': { 'gateset': 1,
    'layer': 30,
    'ugate': { 'coefficients': { 'bn': [ 0,
        0,
        0,
        0,
        0,
        0,
        0],
        'cn': [ 0.5,
            0.5,
            0.5,
            0.5,
            0.5,
            0.5],
        'r': 0},
        'time': {'max': 10.0, 'min': 0.0},
        'type': 'xy-iss'}},
    'init_param': { 'value': [ 0.17961742860022237,
        0.4029745783400613,
        0.4271020388774394,
        0.6998927493507625,
        1.1444327445959492,
        1.4463150894355827,
        1.7859669797147917,
        1.7859669797147917,
        1.8910076260348616,
        2.1568466639380097,
        2.184715675260106,
        2.3241261579565067,
        5.221288254664872,
        5.360678146313719,
        5.572749542622228,
        6.315664711217606,
        6.332942041057151,
        7.225695385348529,
        7.238714694082236,
        7.767863290353532,
        7.947954629935657,
        8.045079231148478,
        8.080705184182557,
        8.235317994608641,
        8.52967785406855,
        8.920755162601605,
        8.920755162601605,
        8.920755162601605,
        9.75907480462672,
        11.068402571583377,
        -0.07026370230905016,
        1.2758275954101812,
        0.1960569796315372,
        -0.7454998985427054,
        0.5260870873255556,
        1.3632265149428353,
        0.06759709485075166,
        0.7201883678889174,
        0.7780364770341424,
        0.7034018811814728,
        -0.3281225120527201,
        -0.21523884970562404,
        0.3525948786581971,
        0.10058439953616755,

```

0.8489242437750127,
0.4102210250930649,
-1.1239274808373552,
-0.07527028057729333,
-0.662823747816413,
-0.09016589559770041,
0.0005078927131911535,
-0.9974030724775158,
1.2973260682530905,
0.6207513445119947,
-0.06959715000945968,
0.01479447508324883,
0.5707366547040263,
0.8024230791007029,
0.11580671533830131,
0.5319238260215059,
2.029703693625361,
0.8476591202353092,
-0.5985490132202251,
-1.0850372047073247,
-0.13885829158476937,
0.6160884575951602,
-0.6119315652443118,
0.25456754270344084,
1.6135141314150674,
1.6101080474118592,
-1.0046888465114787,
-1.2174288348798858,
-0.8801790341822278,
-0.01998793431745051,
-1.007194981866689,
-1.6365882266166358,
3.524880820008825,
1.5399950054830251,
-1.401820445976074,
-0.04945776017633801,
0.9565666812114323,
-0.20490442564626934,
-0.5632271134618189,
1.0033608730635488,
0.12368927199762578,
-0.06153617179775209,
1.0557374829951787,
1.532632985657107,
-0.3069192599686721,
1.8620498010178619,
0.18083984306067874,
-0.9702971127887235,
-0.4866438529829019,
0.2527358034573417,
0.46390718912935613,
0.05253485630073179,
-0.15957905175444287,
0.5064463362180326,
0.6588516997606392,
0.036226022952260704,
-1.5941274572195923,
0.3692191472654626,
0.31840120283705997,
0.8741993879399423,
0.4802653460337955,
0.3012459045041651,
0.5102022084743536,
0.3727970282915065,
0.6665750190815365,
-0.7773615380935764,
0.12852119124454983,
0.6774482070442244,
-0.34235651892527574,
-1.1577831131801597,
-0.2879120353476077,
1.157549789799121,
-0.1818163295634488,
-0.034519908964088415,
-0.6744130553492667,

```

1.1322140576298956,
0.5044463860518625,
1.5469749343656427,
-0.19451127779487945,
0.16868157432748737,
0.3887158384137858,
1.179393002549757,
0.4377598134949622,
0.42273561999328285,
0.17555703442650725,
0.7637154786900613,
0.3204620558964089,
-1.0127445325141415,
-0.3366137672636463,
-0.6128563234467006,
-0.19959799618607235,
-0.22993024268879514,
0.6548635589470768,
0.5449412421525476,
0.11171602982233311,
0.5174688869584806,
-0.0926785574168733,
-0.012796553414024043,
0.872831179387319,
-0.3738274777631416,
0.708424995234135,
1.1207261796968788,
0.39043722964884914,
1.5466373285390471,
-0.0566526099067629,
-1.544789451209757]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_xy_noiseefree_time_evo_ric3_unifrom_scaling_sample#9'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z '
                                '1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + '
                                '1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '
                                '6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + '

```



```
        'r': 0},
        'time': {'max': 10.0, 'min': 0.0},
        'type': 'xy-iss'}},
'init_param': { 'value': [ -3.641081222961908e-17,
0.6149788059529836,
0.7190202634465676,
1.227117923902744,
1.282995668258181,
1.320146784613617,
1.320146784613617,
3.1464089486715623,
3.1464089486715623,
5.021180125711677,
5.021180125711677,
5.021180125711677,
6.703692809170789,
6.703692809170789,
7.791601108223033,
7.8369202041803225,
7.951907692650894,
7.98389042115583,
8.258376681022987,
8.258376681022987,
8.264818959972613,
8.264818959972613,
8.264818959972613,
8.264818959972613,
8.265620267330194,
8.265620267330194,
9.412842541483705,
9.412842541483705,
10.502716174481437,
10.502716174481437,
0.1665558361704551,
0.0766182276072886,
-0.06331706305100791,
0.03225035272845343,
-0.2265177184784691,
0.6859734408919996,
0.358399999202588,
-0.31490649734707366,
0.03084506070978915,
0.8141179515629021,
-0.05751068028622007,
0.9013204998454583,
-0.325467390457223,
-0.16027539072229496,
0.19756436832536928,
0.17875052394899651,
0.07441990946369173,
-1.7121562848350094,
0.5411900571694856,
0.025994665514066265,
-1.0547594680221428,
-0.3558195097906033,
0.2575282902752545,
-0.32652811387218433,
-0.17477771883072452,
0.05603758809983181,
-0.31531793685177084,
0.24953935818155465,
-0.21945102528047247,
0.016201452486715968,
-0.47139285875704856,
0.8354485366702803,
0.28783787212142514,
-1.483190369052958,
-0.05324490057492157,
0.00808064868141892,
-1.3894868155533049,
0.07648399162880297,
0.28128388531065984,
-0.8032486745370345,
0.9961277979922681,
0.7271807873997648,
```

-0.2598267503182047,
0.058937213125893145,
1.4176939090014022,
1.5623836495300838,
-1.2281014386266311,
1.627596878633186,
0.150437982354294,
0.017798034487339433,
-0.2091243023446671,
0.08776907917710067,
0.637527078885032,
1.2603122563899933,
1.0910453886717337,
0.001707278941889729,
-0.5366140699752073,
-0.04457743756443993,
0.708269991966563,
1.6757260639955311,
0.6610325406900899,
-0.508228690084442,
-0.5178693625521746,
-0.2955589105377924,
-0.47790822054112786,
0.31077598178920485,
0.020632196003020267,
1.5538854336639996,
1.2816443511581943,
-0.6926864187150775,
0.73698114320343,
0.3645818015611773,
-0.2097443780477125,
0.1421287040679699,
0.052774836076248374,
-0.7306191772967899,
1.107856618529218,
0.3743379513498811,
1.2959169452837032,
0.8858564921663113,
0.03370518740548656,
-1.0860983069970114,
1.2199055901240408,
-1.0630806559423625,
1.3748402618369016,
0.9943926078324806,
0.7116645059379805,
-0.21098608064682703,
0.04581715611274856,
-0.17117312830066983,
1.2229557830916487,
-1.6273428514067598,
1.8508927721454946,
0.851693435560975,
1.0946646161197402,
0.43088380554694883,
-0.6171985422875839,
1.6866137951313547,
-0.2289359575496427,
-0.09152362655526167,
1.6917359096449291,
0.22722692961233917,
0.49434218233553445,
1.2228162165990006,
-0.4070166799423108,
0.00966082758630671,
-0.3344922737632815,
-1.0755692755599169,
1.1846485271761396,
-0.06070659712292082,
0.6384418838463358,
-0.034216786122488324,
-0.6519049362670448,
-0.0019231105719348242,
0.503764046349436,
0.5779702041064313,
-0.8610978040126155,


```

0.5832069567020347,
0.07954533618121334,
0.016329041774977193]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                  'noise_prob': [0.001, 0.001, 0, 0],
                  'status': True,
                  'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                'r': 1},
               'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                     'status': False,
                     'type': 'png'},
           'file_name_prefix':
'AUTOMATE_xy_noisefree_time_evo_ric3_unifrom_scaling_sample#3'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': False}},
'zne': { 'data_points': None,
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z '
                                '1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + '
                                '1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '
                                '6 + 1 Z 5 + 1 Z 6',
          'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + '
                                                       '0.5 Z 0 Z 1 + '
                                                       '0.5 X 1 X 2 + '
                                                       '0.5 Z 1 Z 2 + '
                                                       '0.5 X 2 X 3 + '
                                                       '0.5 Z 2 Z 3 + '
                                                       '0.5 X 3 X 4 + '
                                                       '0.5 Z 3 Z 4 + '
                                                       '0.5 X 4 X 5 + '
                                                       '0.5 Z 4 Z 5 + '
                                                       '0.5 X 5 X 6 + '
                                                       '0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -7.475700783925181],
                             [12, 6, 0, 0, -5.921149282782479],
                             [20, 10, 0, 0, -4.7240834286119355]],
          'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                              'identity_factors': [0, 0, 0, 0],
                              'noise_level': [4, 2, 0, 0],
                              'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
                          'noise_prob': [
0.001,
0.001,
0,
0],
                          'status': True,
                          'type':
'depolarizing'},

```

```
'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,

'depolarizing'},

'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,

'depolarizing'},

'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [1, 1, 0, 0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {

'noise_prob': [

0,
0],

'status': True,
'type':

'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [2, 2, 0, 0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {

'noise_prob': [

0,
0],

'status': True,
'type':

'odd_wires': 4}},
'run_time_sec': 7.921310901641846}},
{ 'config': { 'ansatz': { 'gateset': 1,
    'layer': 30,
    'ugate': { 'coefficients': { 'bn': [ 0,
    0,
    0,
    0,
    0,
    0,
    0],
    'cn': [ 0.5,
    0.5,
    0.5,
    0.5,
    0.5,
    0.5],
    'r': 0}},
    'time': {'max': 10.0, 'min': 0.0},
    'type': 'xy-iss'}},
'init_param': { 'value': [ 0.003408343124459994,
    0.0034083431244599804,
    0.006710027629051694,
    0.2484817728699434,
    1.1672656342280407,
    1.167470260322433,
    3.1453440534904744,
    3.166261678218574,
    3.2843704266652405,
    4.089250903115526,
    4.091989893807717,
    4.091989893807717,
    5.181052167254873,
    5.181052167254873,
    5.184972473956635,
    5.585122024093685,
    5.5990893229720795,
    5.630355974178803,
    5.704160317443757,
    8.658831571647385,
    8.66505230970618,
    8.712765508205756,
    8.712765508205756,
    8.712765508205756,
    9.168875276976076,
```

9.176734348833309,
9.262471352200725,
9.262471352200725,
10.578606537988184,
10.578606537988184,
1.5905119084430799,
0.8691136234632737,
0.04907693052707525,
-0.23336553986203207,
0.4284777328779486,
2.2866815267075147,
0.1760352312050406,
0.27820168999860917,
2.067264415281462,
0.5755331494929522,
-0.06858856463703264,
-0.7222667597755238,
0.7750924366303688,
-0.278304785792398,
0.5352624789798245,
-1.1702690163303244,
-0.012717570027659048,
-0.07125250327360101,
1.7927644689789062,
1.3752915621891346,
-1.1981440291126808,
-0.30737164793891514,
0.2657743539912098,
0.04156274567190095,
1.1363043468044227,
0.059217925806269944,
0.7608352104096147,
-0.1513784586666425,
0.025696061144845524,
0.861481321747562,
-0.48544656000194114,
-0.006033515622113467,
0.473431667961849,
1.5469119181224316,
0.1871819193791605,
1.555475823471471,
0.9001349633565042,
-1.5961139272878428,
-0.6749613217725859,
-1.5596468027729833,
0.5296186936496604,
-0.8549575199843406,
1.7649719371938613,
1.5569889217595585,
-0.9244334407213903,
-1.5804326845781602,
0.3840271097002098,
-0.8070678340768227,
-0.48791686278645724,
0.13110829311357092,
-2.023238766590243,
-0.8997043584182223,
0.8435119798114825,
0.12108034013815198,
-0.7013572315179941,
0.05813246571029178,
-1.3645229029121433,
-1.704441396215925,
0.6160899239086374,
-1.6580292769388947,
-1.550797918270104,
-0.033040246679508146,
-0.7595754539148374,
0.6024045095197653,
0.446745843265138,
0.9717727329863253,
-1.3190014690683383,
-3.180952616822137,
-0.47685451708111876,
0.03958388950097112,

```

-0.04894683660460036,
-0.013869967587830365,
0.9715551393376142,
-0.2688733257646762,
-1.7112803723577967,
-0.06346862095088202,
1.2756440828431679,
0.4087049517265435,
-0.4433737535464867,
0.07044159255576982,
0.877714241226598,
-0.33059433004855665,
-0.41683973817974146,
-0.09123983383660174,
1.153451542408442,
1.5381060924707182,
1.4432055127327397,
0.7948381511053778,
0.12126271666707081,
0.3736422837955817,
-0.06595460407520624,
-0.1569219765005705,
1.4053035958769775,
-0.3443177603686477,
0.14604377230684606,
0.733462484703497,
1.2365404639754205,
-0.020105376528163625,
-0.09631958929266826,
-0.16876809298151013,
0.42184799028013376,
0.8803036484986011,
-0.4148292271104988,
0.011464734896518355,
1.0895788780475535,
-0.006017115067308532,
0.6476481463328121,
-0.4792828310008005,
1.7145504838049759,
0.9300252591703124,
1.4147034805691376,
-0.04601267898776636,
-0.34579443398211707,
0.006107561937391137,
2.7608588509789516,
-2.470402787070412,
-0.2193552778493498,
-0.43636419827955597,
0.5036040054264609,
-0.006776779992121664]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':

```

```

'AUTOMATE_xy_noisefree_time_evo_ric3_unifrom_scaling_sample#7'},
    'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                          [1, 1, 0, 0],
                                          [2, 2, 0, 0]]},

    'run': 'redundant',
    'state': 'dmatrix',
    'vqe': { 'iteration': 1,
              'optimization': { 'algorithm': 'SLSQP',
                                'constraint': True,
                                'status': False}},

    'zne': { 'data_points': None,
              'degree': 0,
              'method': 'richardson',
              'sampling': 'default'}},
  'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z '
                                    '1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + '
                                    '1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '
                                    '6 + 1 Z 5 + 1 Z 6',
              'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + '
                                                           '0.5 Z 0 Z 1 + '
                                                           '0.5 X 1 X 2 + '
                                                           '0.5 Z 1 Z 2 + '
                                                           '0.5 X 2 X 3 + '
                                                           '0.5 Z 2 Z 3 + '
                                                           '0.5 X 3 X 4 + '
                                                           '0.5 Z 3 Z 4 + '
                                                           '0.5 X 4 X 5 + '
                                                           '0.5 Z 4 Z 5 + '
                                                           '0.5 X 5 X 6 + '
                                                           '0.5 Z 5 Z 6']},

  'output': { 'data_points': [ [4, 2, 0, 0, -7.568863804770513],
                                [12, 6, 0, 0, -6.084665473601483],
                                [20, 10, 0, 0, -4.914126798758423]],
    'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                          'identity_factors': [0, 0, 0, 0],
                          'noise_level': [4, 2, 0, 0],
                          'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0,
                                0],

                                'status': True,
                                'type':

'depolarizing'},

                                'odd_wires': 4},
                                { 'gates_num': [12, 3, 1, 0],
                                  'identity_factors': [1, 1, 0, 0],
                                  'noise_level': [12, 6, 0, 0],
                                  'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0,
                                0],

                                'status': True,
                                'type':

'depolarizing'},

                                'odd_wires': 4},
                                { 'gates_num': [20, 5, 1, 0],
                                  'identity_factors': [2, 2, 0, 0],
                                  'noise_level': [20, 10, 0, 0],
                                  'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0,
                                0],

                                'status': True,

```

```

'depolarizing'},

'odd_wires': 4]],
'run_time_sec': 8.946808815002441}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [ 0,
                                                            0,
                                                            0,
                                                            0,
                                                            0,
                                                            0,
                                                            0],
                                                            'cn': [ 0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5],
                                                            'r': 0},
                        'time': {'max': 10.0, 'min': 0.0},
                        'type': 'xy-iss'}},
  'init_param': { 'value': [ 0.5698358581496117,
                             1.0111463732621893,
                             1.116254023022104,
                             1.116254023022104,
                             1.4543153683590362,
                             1.4543153683590362,
                             1.551144943675129,
                             1.551144943675129,
                             2.3321866923810566,
                             2.4857852178760083,
                             3.837593499897418,
                             3.837593499897418,
                             3.837593499897418,
                             4.847467397840062,
                             5.03660309473088,
                             5.176578602594472,
                             5.176578602594472,
                             5.214332760543968,
                             6.547255537065404,
                             6.547255537065404,
                             7.083633091144979,
                             7.083633091144979,
                             7.133204014188821,
                             7.133204014188821,
                             7.133204014188821,
                             7.937225733864811,
                             8.370661235857398,
                             8.62656679928389,
                             8.759707620277526,
                             9.424598849529344,
                             0.24247733923543782,
                             0.79275118507479,
                             -0.012314768690710378,
                             1.2247395896468714,
                             -0.6087890007814074,
                             0.9804698093938512,
                             0.19554007335952167,
                             -0.18291383394130525,
                             -0.08254523307317092,
                             0.18380200005144362,
                             -0.06242013495608768,
                             -0.18902966193120999,
                             0.4846770511497838,
                             1.2162916157423216,
                             0.4476502570273604,
                             -0.5204828708955642,
                             1.1525916637240337,
                             0.9814837501696517,
                             1.2019836737296188,
                             0.10039944576408244,
                             -1.165620769823409,
                             0.5709783345435084,
                             0.04690539627488982,

```

0.3150855883896384,
0.6835833386917337,
0.7350671828581666,
0.15105693185753874,
0.7851129846448224,
-0.290237804763004,
0.02628688844012564,
-0.3427838784588314,
-0.12202424723691215,
0.30368921729231485,
-0.08398845633414893,
-0.34421225485285395,
-1.0992299599969957,
-0.6679166311787512,
-0.4925602784337871,
-0.7607905356843834,
-0.061822787578292515,
-0.33643902903801526,
1.7237644621079549,
1.1087838048318122,
1.4391293107545464,
-0.2688865923867924,
0.07238699078201241,
-0.7262166431073663,
-0.8607492696026928,
0.1828620517189283,
-1.5556021786529124,
-0.5230641559705965,
-0.023465829708141782,
0.5773817105800866,
-0.6189512057198875,
-0.1704298219551506,
0.6935490081871019,
0.43024761601627914,
-1.3818053283072975,
0.32069867830077864,
0.11223505487625844,
-0.6642209341325642,
-0.7163563578386999,
0.9533459704680486,
-2.5373567196844773,
-0.3530595330204968,
-0.36744271288254204,
-0.4105700661159024,
-0.529185988706723,
-0.5074466275713505,
0.5969687397167245,
0.1454479675984071,
0.9510340399481164,
-0.16051754924491393,
0.168676911677473,
1.6243103666539225,
-0.3533677275828062,
0.23248347317288842,
1.550100715213732,
0.3540002793004863,
0.11004939052123668,
0.0026309322638186474,
1.0023099891482978,
1.3075603782358884,
0.06614039249633322,
0.12696883216368457,
0.4650745950568721,
0.09288306127511851,
-0.05535633254466518,
1.0893487961438044,
0.1856572010619516,
-0.3369565165204948,
1.1993442715596423,
-0.1637322799431796,
-0.05574034797393315,
-0.6499132821077687,
-1.1854402746414112,
-0.20554021240749126,
-0.697991352679575,

[illegible]


```

'0.5 X 5 X 6 + '
'0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -7.159167099826597],
                              [12, 6, 0, 0, -5.573237410199788],
                              [20, 10, 0, 0, -4.355904736591592]],
  'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                      'identity_factors': [0, 0, 0, 0],
                      'noise_level': [4, 2, 0, 0],
                      'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
0.001,
0.001,
'noise_prob': [
0,
0],
'status': True,
'type':
'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [1, 1, 0, 0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
0.001,
0.001,
'noise_prob': [
0,
0],
'status': True,
'type':
'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [2, 2, 0, 0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
0.001,
0.001,
'noise_prob': [
0,
0],
'status': True,
'type':
'depolarizing'},
'odd_wires': 4}],
'run_time_sec': 8.069846868515015}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [ 0,
                                                            0,
                                                            0,
                                                            0,
                                                            0,
                                                            0,
                                                            0],
                                                            'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                            'r': 0},
                        'time': {'max': 10.0, 'min': 0.0},
                        'type': 'xy-iss'}},
'init_param': { 'value': [ 0.02240293444906766,
                           0.9863710197203618,
                           1.3708545038264748,
                           1.3757718928414557,
                           1.7844795346244002,
                           2.029643659915566,

```

2.029643659915566,
4.135298362317422,
4.181946654536538,
4.392782597821499,
4.473609589126697,
4.474471668054438,
4.475838923576404,
4.4785869343507265,
4.4785869343507265,
5.467009291914798,
5.470281162008999,
5.840541346188122,
5.842392372241278,
6.505508038171383,
6.505508038171383,
7.536348099553183,
7.536348099553183,
9.63019412694799,
9.63019412694799,
9.63019412694799,
9.63019412694799,
9.63019412694799,
9.63019412694799,
9.63019412694799,
1.1150074339792337,
-0.03535947987647509,
1.1918755403807781,
-0.20191103850925227,
-0.268362117189328,
-0.3595192771004685,
1.2614204066704717,
-2.9429228213034166,
-1.054810371869961,
0.9594255841252335,
-0.09538831740389547,
0.3965663197177072,
-0.22182226409641906,
-0.2653718580482209,
0.1686090968331403,
1.6516484362324146,
-0.3049608013883917,
0.49165272629600476,
1.9061569916625292,
1.241422732345657,
-0.11527538328521378,
-0.22895223610637053,
-0.06124477487110871,
0.6193220715745159,
1.372932614853267,
-1.1538384701543527,
-0.1139440760455718,
0.16973094341798492,
-0.5943234340378932,
0.21399932498003596,
-0.18156105319883253,
-1.0504702368491705,
0.7232808408606985,
-0.018870518436612777,
-0.29343981279013404,
0.30960592292227684,
0.13511542022922338,
0.5096669954667911,
-1.097576281455009,
0.12047585913747774,
-0.5585503308065869,
1.2279608447421912,
1.3012401380411538,
0.3769592231863052,
-0.5692023175942067,
-0.4567508363134565,
-0.679549924864495,
-0.24647993267323956,
0.6292567546169889,
0.9587256267378846,
1.7268825428143277,

```
0.1413792259593793,  
0.6520451922983643,  
0.6152471684019365,  
-1.1240317980448378,  
-0.9856730455453077,  
-1.2455074642265085,  
0.2235091354354849,  
-0.8870861048674711,  
0.5299724059178798,  
-0.21820971616414211,  
-0.09687180933262383,  
-0.9247884680119297,  
-0.4566833626656702,  
-0.4285291193052864,  
-0.6406026728496247,  
0.9023578506051637,  
0.03748900229192909,  
0.7551248935414285,  
-0.18980887545414976,  
-0.1497604267964229,  
-0.4253730408988226,  
-0.4157675203773789,  
1.677209169409502,  
0.2126888490300174,  
-0.13481167544410624,  
-1.6951458494505263,  
-0.006476362637627559,  
-0.5490470525961781,  
0.3195499617454318,  
1.077429849794694,  
0.6725724542747005,  
-0.10235701747439804,  
-0.014129818782246995,  
2.382406086835482,  
0.008866515589568993,  
1.0282503794544984,  
-0.6899701634722577,  
0.3510134577654823,  
0.6918405669356381,  
-0.08667968085870373,  
-0.0005005962237103964,  
1.3805542469189591,  
1.011514915181436,  
-0.4623191099783579,  
1.397645255720587,  
-0.7813361449321683,  
-0.9940660424170442,  
-0.6570734018108508,  
1.0171740600492052,  
0.14480072689925969,  
1.157289980717297,  
1.019472554492438,  
1.0029573776528096,  
-0.20406480565208354,  
-0.011076077177880707,  
-0.30566521016454584,  
-0.4775965459909992,  
-0.7158801142606765,  
1.5610067764386268,  
1.5187465408730505,  
0.21158961058436962,  
-0.38098352608767794,  
-0.046999172842648596,  
0.5964180792256618,  
-0.6260168044949087,  
-0.7215592053840503,  
-0.5706802740378752,  
0.11995346168986934,  
0.17053214926260635]],  
'noise_profile': { 'noise_on_init_param': { 'status': False,  
                                             'value': 0},  
                  'noise_prob': [0.001, 0.001, 0, 0],  
                  'status': True,  
                  'type': 'depolarizing'},  
'nqubits': 7,
```

```

'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_xy_noisefree_time_evo_ric3_unifrom_scaling_sample#6'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z '
'1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + '
'1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '
'6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + '
'0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + '
'0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + '
'0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + '
'0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + '
'0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + '
'0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -7.259703059772281],
[12, 6, 0, 0, -5.766278116709839],
[20, 10, 0, 0, -4.58319338434511]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 0, 0],
'noise_profile': {
'noise_prob': [
0.001,
0.001,
0,
0],
'status': True,
'type':
'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': {
'noise_prob': [
0.001,
0.001,
0,
0],
'status': True,
'type':
'depolarizing'},
'odd_wires': 4}]}]}

```

```

0.001,
0,
0],
'status': True,
'type':
'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
'identity_factors': [2, 2, 0, 0],
'noise_level': [20, 10, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [
0.001,
0.001,
0,
0],
'status': True,
'type':
'depolarizing'},
'odd_wires': 4}},
'run_time_sec': 7.59516716003418}}],
'vqe': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}}},
'init_param': {'value': 'random'},
'noise_profile': { 'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [0.001, 0.001, 0, 0],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': True,
'type': 'png'},
'file_name_prefix': 'xy_noiseefree_time_evo'},
'redundant': { 'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'vqe',
'state': 'dmatrix',
'vqe': { 'iteration': 10,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': True}},
'zne': { 'data_points': None,

```

```

        'degree': 1,
        'method': 'richardson',
        'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 '
        'X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X '
        '4 X 5 + 1 Z 4 + 1 X 5 X 6 + 1 Z 5 + 1 Z 6',
        'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z '
        '0 Z 1 + 0.5 X 1 X 2 '
        '+ 0.5 Z 1 Z 2 + 0.5 '
        'X 2 X 3 + 0.5 Z 2 Z '
        '3 + 0.5 X 3 X 4 + '
        '0.5 Z 3 Z 4 + 0.5 X '
        '4 X 5 + 0.5 Z 4 Z 5 '
        '+ 0.5 X 5 X 6 + 0.5 '
        'Z 5 Z 6']},

'output': { 'exact_sol': -8.56677223350563,
        'initial_cost_history': [ 3.450586259258812,
        2.2577647098356715,
        3.220573399525229,
        2.813569497850412,
        2.356577849154886,
        2.484693017894963,
        2.751192081705436,
        3.0939254363357245,
        3.3455601721392987,
        2.9399815036188865],
        'noise_details': { 'gates_num': [4, 1, 1, 0],
        'identity_factors': [0, 0, 0, 0],
        'noise_level': [4, 2, 0, 0],
        'noise_profile': { 'noise_on_init_param': {

        'noise_prob': [ 0.001,
        0.001,
        0,
        0],

        'status': True,
        'type': 'depolarizing'}},

        'odd_wires': 4},
        'optimized_minimum_cost': [ -7.463351332655033,
        -7.130214875567174,
        -7.475700783925261,
        -7.41873284165754,
        -7.407206817517966,
        -7.259703059772282,
        -7.568863804770619,
        -7.15916709982671,
        -7.395384341095537,
        -7.512595875373624],
        'optimized_parameters': [ [ 0.5482907189468895,
        0.6413620809492503,
        0.6422482149552934,
        1.5892409709442974,
        1.850708229814058,
        1.850708229814058,
        1.8745552660050262,
        2.400679867939261,
        2.402212278134192,
        2.485190345183722,
        2.485190345183722,
        2.4851903451837223,
        2.488397493491329,
        2.4883974934913287,
        5.077476636438464,
        5.077476636438464,
        5.732990197214016,
        5.806868622878696,
        5.817834245705687,
        6.166757373752113,
        7.3114292874931905,
        7.311858449036949,
        8.346734585466223,
        8.350435170229554,
        8.354614694314481,
        8.35659817980371,

```

8.356905245901928,
9.313534396314816,
9.313534396314816,
10.443787047737276,
-0.5971318801423705,
-0.6830583120471296,
0.053446928626708114,
-1.488251087277441,
0.7409998345617078,
-0.020561007687691084,
-0.47029189134787985,
-0.9187802118512801,
-0.07390570205801456,
0.6091612085770479,
0.5696050095749661,
-2.634874245900948,
0.14733366015521565,
1.0781100436188797,
0.3459269460772517,
1.2820636645787713,
-1.605623121104658,
0.856791024529966,
0.16374864437029604,
-0.7377840957855112,
0.7603752030179747,
0.4958590077418653,
1.0293890027768824,
0.4862585296205842,
-1.0838654128292502,
1.2632637762941026,
-1.1043454462318527,
-0.7867654190654241,
0.8818217647037212,
1.0406267750063978,
0.05770567519039498,
-0.5922070722918433,
-0.3271403288282926,
0.5980786503295974,
-0.02050375112330311,
-0.9007241609880184,
0.5239405469453117,
0.7601496703420288,
-0.6449049942439851,
1.7340858429103823,
0.10050113333288421,
0.3143347863780438,
0.28408386490618526,
-0.1222389867527315,
-0.09502282329152913,
-1.3004368662949968,
0.16856179320059875,
0.6790103402211991,
0.3355977478652572,
0.0935957266804422,
0.42936820813464754,
0.5148227650107633,
0.10764356588532305,
-0.054251737808208085,
0.06525040933822983,
-0.21143245853075732,
-1.5801988491404158,
0.023513208971767067,
1.0921277689034055,
-0.11501776583679489,
-0.14182911234947893,
1.8716225858725093,
-1.0747761250482988,
-0.01873186129021693,
0.9952348515577564,
1.5589645169221944,
2.4625123021717044,
1.5320934484141422,
0.6264578101560674,
0.015802850908690148,
0.6559295719723718,

0.5030706883609005,
0.010357532831659657,
-0.6971755064049523,
-1.334963536555068,
0.05714132580252865,
0.9478292633237722,
1.541635767019547,
-1.2357775558070643,
1.5753066317572375,
1.2206753513411082,
-0.013882574638756213,
2.102115305768464,
0.6867648509985108,
-1.0810617894892982,
0.608205793929093,
-1.0701207529796843,
-0.008804617073740417,
1.0802724219136723,
1.4472704941581649,
-0.525983160453369,
0.10347757085774918,
0.07451664808136123,
0.8113743025574697,
-0.19830571509994802,
0.20760359505818776,
-0.5709476706639499,
0.1791350390980861,
-0.32061652775092425,
-1.6382844876022806,
2.271046687924862,
0.09972766177435569,
1.671157305178672,
1.0596955078067567,
-0.7395801076185267,
-0.8217251410877643,
1.2660358026872456,
-0.149859182069558,
1.7386861885389688,
-0.003505497297480453,
0.677126408740769,
-1.0541684899696007,
0.9437251450981726,
1.7354017388586316,
-0.8022872949930118,
0.01881044769021433,
-0.013699411839369284,
1.5965466699123847,
0.4486397166383021,
1.5669137279484207],
[4.1837705414928806e-17,
1.1742327239243593,
1.1742327239243593,
1.174232723924359,
1.3021435979041922,
1.3883521076730898,
1.4901272631800928,
1.490127263180093,
1.490127263180093,
2.542545188872923,
2.56739133991554,
2.677178516908646,
5.624946341909109,
5.624946341909109,
5.63924283183031,
5.672838193807689,
5.672838193807689,
5.672838193807689,
5.672838193807689,
5.672838193807689,
5.672838193807689,
5.672838193807689,
5.805507018952099,
5.858678454346443,
7.374841183417447,
7.374841183417447,
9.078346087689514,

9.17910617229311,
9.185363997877552,
9.550522060955204,
9.715006087156702,
0.5380723684568228,
0.20346632529178696,
-0.28402043001150323,
-0.013694659434132353,
-0.8491217472851175,
-0.13891219530245408,
-0.7902948900320785,
0.02035105936303376,
0.35891510624171696,
0.07011262472067725,
0.2571910630887441,
0.09727807922224965,
-0.4253506785797692,
-0.08201618636277365,
-0.5629942371157008,
0.4984357070329931,
0.22460606922422946,
0.6436861273901548,
0.5581387866543217,
0.6438048424266741,
0.6625810608832491,
-0.2877740698749285,
0.03400434270310955,
0.6598198228171638,
0.15695492486622947,
-0.23277712193728803,
0.3379436543817524,
0.6679629986711755,
0.4913802101172593,
-0.6543304626041586,
1.2398324170787045,
0.10868488610295632,
0.4726806257545104,
-1.0278416218016544,
-0.5660236088762642,
-0.06627186605932975,
0.3527547117645279,
0.3944014355355292,
0.33709190475341266,
0.16707772676088822,
0.29195080738380563,
1.0354529601392413,
0.12500585138196232,
-0.037178956773431956,
0.7823218278561086,
-0.04464215343745706,
-0.36923576305262273,
-0.9863817864437636,
0.039519076991602065,
-0.35646400258617383,
0.04542480304419608,
-0.46125271135496354,
0.40472620693719735,
0.7802480661953589,
-0.6757172179201896,
0.6396142241310521,
0.2327897749192615,
0.8954622299286413,
0.7158405805761733,
0.1382139633537826,
1.0616928944303126,
-0.6122315420380016,
-0.6738539946512315,
1.0433459792659807,
0.494526070112864,
0.20621322780405107,
-0.05318175489586881,
-0.11213659998229918,
0.32941262943060273,
0.13931786347259914,
-0.07640781308377202,

0.31450116184408605,
0.3658109215963646,
0.04460349089074807,
-0.12913310998017427,
0.31890485610090735,
-0.005491508360761248,
1.2692063073436923,
-0.3006174703747285,
-0.41404661321397057,
-0.5199199797590206,
0.44225909618051784,
-0.4233007342169607,
0.7456354776212751,
-0.32484988424610756,
0.9268151291603125,
0.8385591284905366,
0.027219364270413358,
-0.5520387125082392,
1.3757901840365965,
0.02933419406725825,
-0.2915193842748469,
0.33917094344303517,
0.10828106345120646,
-0.5650186064180197,
-0.6041651963720519,
-0.19230834783869816,
1.9848826966849449,
0.0632219631903188,
0.04804551511145073,
0.47752350144148015,
1.3406847618100746,
-1.3722262204383469,
-1.0513455672656666,
0.7121308231347245,
0.13278942055995957,
1.9758312031522887,
2.19648222646383,
0.8936788375205645,
0.05510909660880629,
-0.18765678539637448,
0.45031205104496697,
-0.6542716355161553,
0.5783583754336702,
-0.18844184232604352,
-1.5667303012058793,
0.36110899429768645,
-0.275182998952267,
0.30367576290775045,
-0.1532687178202341],
[-3.641081222961908e-17,
0.6149788059529836,
0.7190202634465676,
1.227117923902744,
1.282995668258181,
1.320146784613617,
1.320146784613617,
3.1464089486715623,
3.1464089486715623,
5.021180125711677,
5.021180125711677,
5.021180125711677,
6.703692809170789,
6.703692809170789,
7.791601108223033,
7.8369202041803225,
7.951907692650894,
7.98389042115583,
8.258376681022987,
8.258376681022987,
8.264818959972613,
8.264818959972613,
8.264818959972613,
8.264818959972613,
8.265620267330194,
8.265620267330194,

9.412842541483705,
9.412842541483705,
10.502716174481437,
10.502716174481437,
0.1665558361704551,
0.0766182276072886,
-0.06331706305100791,
0.03225035272845343,
-0.2265177184784691,
0.6859734408919996,
0.358399999202588,
-0.31490649734707366,
0.03084506070978915,
0.8141179515629021,
-0.05751068028622007,
0.9013204998454583,
-0.325467390457223,
-0.16027539072229496,
0.19756436832536928,
0.17875052394899651,
0.07441990946369173,
-1.7121562848350094,
0.5411900571694856,
0.025994665514066265,
-1.0547594680221428,
-0.3558195097906033,
0.2575282902752545,
-0.32652811387218433,
-0.17477771883072452,
0.05603758809983181,
-0.31531793685177084,
0.24953935818155465,
-0.21945102528047247,
0.016201452486715968,
-0.47139285875704856,
0.8354485366702803,
0.28783787212142514,
-1.483190369052958,
-0.05324490057492157,
0.00808064868141892,
-1.3894868155533049,
0.07648399162880297,
0.28128388531065984,
-0.8032486745370345,
0.9961277979922681,
0.7271807873997648,
-0.2598267503182047,
0.058937213125893145,
1.4176939090014022,
1.5623836495300838,
-1.2281014386266311,
1.627596878633186,
0.150437982354294,
0.017798034487339433,
-0.2091243023446671,
0.08776907917710067,
0.637527078885032,
1.2603122563899933,
1.0910453886717337,
0.001707278941889729,
-0.5366140699752073,
-0.04457743756443993,
0.708269991966563,
1.6757260639955311,
0.6610325406900899,
-0.508228690084442,
-0.5178693625521746,
-0.2955589105377924,
-0.47790822054112786,
0.31077598178920485,
0.020632196003020267,
1.5538854336639996,
1.2816443511581943,
-0.6926864187150775,
0.73698114320343,

0.3645818015611773,
-0.2097443780477125,
0.1421287040679699,
0.052774836076248374,
-0.7306191772967899,
1.107856618529218,
0.3743379513498811,
1.2959169452837032,
0.8858564921663113,
0.03370518740548656,
-1.0860983069970114,
1.2199055901240408,
-1.0630806559423625,
1.3748402618369016,
0.9943926078324806,
0.7116645059379805,
-0.21098608064682703,
0.04581715611274856,
-0.17117312830066983,
1.2229557830916487,
-1.6273428514067598,
1.8508927721454946,
0.851693435560975,
1.0946646161197402,
0.43088380554694883,
-0.6171985422875839,
1.6866137951313547,
-0.2289359575496427,
-0.09152362655526167,
1.6917359096449291,
0.22722692961233917,
0.49434218233553445,
1.2228162165990006,
-0.4070166799423108,
0.00966082758630671,
-0.3344922737632815,
-1.0755692755599169,
1.1846485271761396,
-0.06070659712292082,
0.6384418838463358,
-0.034216786122488324,
-0.6519049362670448,
-0.0019231105719348242,
0.503764046349436,
0.5779702041064313,
-0.8610978040126155,
0.5832069567020347,
0.07954533618121334,
0.016329041774977193],
[0.2383708143795501,
0.8137820253985804,
1.0561141201801563,
1.2070683478267785,
1.9518166906091101,
2.08301823027678,
2.6774459967535513,
2.7270096866449576,
3.7369434259122447,
3.7421546540398443,
4.458327272130387,
4.463102384700506,
4.463102384700506,
4.482690585412051,
4.482694652744643,
4.498978826283554,
6.3580266266370415,
7.88639599997723,
7.886532810158634,
8.060529572035655,
8.06234783983503,
8.127555772505229,
8.130166630032164,
8.42408407139234,
8.433074307328475,
9.412401860498063,

9.420584174895616,
9.429609653004679,
9.434851407347677,
11.283524698808058,
0.19157417616433814,
0.42257952171340063,
0.3129095694283805,
0.28360255750055535,
-0.7229159635345903,
2.545731433831817,
1.1687892956196542,
0.5222949375721274,
0.9482151911772319,
0.28742832079435787,
1.1627391357840349,
-0.27939056132165146,
-0.07574334812551493,
-0.529119211391362,
-0.08653083237786212,
0.042544229108731915,
-0.09154582517805988,
0.2776265525620834,
0.373637405241428,
-0.666296500917012,
-0.729825790392568,
0.36164641825124294,
1.2066181407380305,
0.18327070025757417,
2.2757552060408917,
-0.004789158452516104,
0.49019439516235463,
2.190622553796523,
0.6131206832540379,
1.0987721265230228,
1.9413219378309259,
0.07975169469268534,
0.17531082109327745,
0.01770145552231998,
0.22429808435825893,
1.1460098869067579,
-0.25766861736092417,
0.4386465248337572,
0.7521614434169624,
0.07039918338308593,
-0.23913727928604284,
0.062497116873310714,
0.27575365987954265,
-0.8268952220573135,
0.8599765782432734,
0.6958325703328606,
-0.16792542378671177,
-0.13186811496768225,
0.17745018848846658,
-0.13764108606901534,
-0.34798347931108986,
0.3901205214779397,
-0.12832331396189006,
0.7623260417931323,
-0.08745728181826747,
0.22448567111633413,
0.5103689813486589,
0.020179740460913417,
-0.4639048953982949,
1.8052783293547816,
-0.06895296635033099,
-0.46137869531525183,
-0.7694587159951923,
-0.1507074275764489,
-3.159084078074864,
1.5556721360426484,
0.9802025173412374,
-1.5980562353917622,
0.9974678614222413,
0.11965749551778265,
-0.5964068365118492,

1.00912773091807,
1.2674306548806673,
1.7418320010188646,
-1.2223756485879664,
-0.006270145144671745,
0.7863824204489233,
-0.08410721186616651,
0.13420385894759826,
0.6686584151585806,
-0.7946689395347296,
-1.448768973512125,
-0.4158373205099576,
-0.1003664997933048,
0.5370971852976741,
-0.013369261455621718,
-1.2134070474127567,
2.71822204346588,
-0.8266087910498845,
-0.04274152999618493,
0.627725493238542,
-0.10714625448592663,
1.4055199595750496,
-0.08877591733311402,
-0.6400618055168635,
0.27990653766167567,
0.27230874688701184,
1.8329367216853043,
0.12128377229042309,
0.03719209379786179,
0.7558193864298689,
-0.027085062677120984,
0.7947118159580207,
-0.18810209431195668,
0.008294184744538372,
-0.9110921816475059,
1.3065108206329406,
0.004810940817056788,
1.4030602129547465,
0.03878633149748846,
0.29850504257680605,
-0.9524583070906825,
0.10037229527914636,
0.4277056649647598,
-2.2528322583674503,
0.030724614514628212,
0.8416350822543037,
1.5579013461800135,
0.5696673724928452,
-1.5352194543526818],
[0.3442332088020621,
0.7247295953616149,
0.7247295953616149,
0.7610537291022658,
1.7162580174489597,
2.0119282772467635,
2.0119282772467635,
3.369934702704019,
3.369934702704019,
5.00211845115155,
5.212181786011135,
5.346566249705437,
6.103091648910964,
6.103091648910964,
7.050711873868975,
7.050711873868975,
7.492144583370162,
7.492144583370162,
7.606105711064038,
7.606105711064038,
8.717193051662631,
8.753101318062777,
8.877107151002964,
8.889590522678484,
8.982812044989718,
8.982812044989718,

9.499874357008034,
9.499874357008034,
9.659025604148875,
9.671110590806489,
0.02814413440270558,
0.250593445582179,
0.17496868989230024,
-0.19749991731592192,
-0.5878125517240043,
1.1370895496823794,
0.3339578948727643,
0.30833012680275795,
0.21147160367429677,
0.5518340573364652,
0.9816766119937317,
-1.3427949214182668,
2.2864376119784486,
0.673664283367827,
0.4767745045632036,
-0.47480445681581174,
0.05912449937800317,
0.9513757199218045,
1.079093875588678,
-0.4313388366547843,
0.43347513808131466,
-0.3381632209207048,
-0.42699598389932686,
-1.1528003206699722,
0.2505681368756031,
1.1900947118566725,
-0.38260053073473543,
-0.6405385159573362,
0.08926968472452403,
-0.04834256297702397,
1.603027736914297,
-0.44672812120859223,
-0.5759172165575565,
1.25237714334681,
0.04254918814060746,
-0.002429292667880943,
0.47732275422632153,
-0.10204026491637587,
0.02357869435971368,
0.06910865492863566,
-0.5942663805386644,
1.1629666490831163,
0.28625726751699415,
0.4183193427753825,
0.33026699893767814,
1.8860632823719743,
-0.37929215782635534,
0.8251027249542411,
-0.13937217921728381,
-0.35829122981546885,
-0.11830319301401378,
-1.1307480811476716,
-0.33746884247229475,
1.7161163093033744,
-0.03517175574182253,
-0.4044561448132672,
-0.7301790593832838,
-0.10496992427986328,
-0.0377875273258637,
0.11576685707785253,
-0.9348023198901909,
2.132577954045408,
-0.47199808297503054,
0.09642080758509694,
-1.5590866370005203,
0.10045413530924742,
-0.3215930726850239,
-0.7020784722315541,
1.2376661808413174,
0.5709879203761018,
-0.820670510834149,

0.1634866307612446,
0.06270250983710676,
-0.03570422816362065,
-0.5699745682010601,
-0.26488761312159137,
1.795803611544861,
-0.4017001375917998,
-1.757541020399501,
-0.05132347348943386,
0.8521664539464072,
0.4812990598336637,
-0.7041980923594973,
0.6941905718406797,
1.5589556954504384,
1.9089912382579748,
0.4849438728523755,
0.03939482159422182,
-0.2564597588944596,
-0.16675906759041134,
2.5610800117093855,
0.9927310518963223,
1.3170410278463267,
0.6281459286410694,
-0.07878692694615999,
0.5865838795710373,
0.4894408136056478,
-0.043552207832595204,
0.8988064624972067,
-0.35279792613941824,
1.5088457985023234,
-0.32174328535909336,
0.41978129355629945,
0.014567846908235548,
-0.17446008524494477,
0.35494980160059775,
1.44166715836895,
0.027787407199764528,
-0.2573832607835415,
0.9605798444843694,
-0.08907047543629609,
-0.012116788408097524,
1.2676248549462201,
-0.2512276837340942,
0.7869834357157488,
-2.3607393050467915,
0.1914341630047483,
-1.0542342437530707,
0.8454261921281611,
0.02027118187750096],
[0.02240293444906766,
0.9863710197203618,
1.3708545038264748,
1.3757718928414557,
1.7844795346244002,
2.029643659915566,
2.029643659915566,
4.135298362317422,
4.181946654536538,
4.392782597821499,
4.473609589126697,
4.474471668054438,
4.475838923576404,
4.4785869343507265,
4.4785869343507265,
5.467009291914798,
5.470281162008999,
5.840541346188122,
5.842392372241278,
6.505508038171383,
6.505508038171383,
7.536348099553183,
7.536348099553183,
9.63019412694799,
9.63019412694799,
9.63019412694799,

9.63019412694799,
9.63019412694799,
9.63019412694799,
9.63019412694799,
1.1150074339792337,
-0.03535947987647509,
1.1918755403807781,
-0.20191103850925227,
-0.268362117189328,
-0.3595192771004685,
1.2614204066704717,
-2.9429228213034166,
-1.054810371869961,
0.9594255841252335,
-0.09538831740389547,
0.3965663197177072,
-0.22182226409641906,
-0.2653718580482209,
0.1686090968331403,
1.6516484362324146,
-0.3049608013883917,
0.49165272629600476,
1.9061569916625292,
1.241422732345657,
-0.11527538328521378,
-0.22895223610637053,
-0.06124477487110871,
0.6193220715745159,
1.372932614853267,
-1.1538384701543527,
-0.1139440760455718,
0.16973094341798492,
-0.5943234340378932,
0.21399932498003596,
-0.18156105319883253,
-1.0504702368491705,
0.7232808408606985,
-0.018870518436612777,
-0.29343981279013404,
0.30960592292227684,
0.13511542022922338,
0.5096669954667911,
-1.097576281455009,
0.12047585913747774,
-0.5585503308065869,
1.2279608447421912,
1.3012401380411538,
0.3769592231863052,
-0.5692023175942067,
-0.4567508363134565,
-0.679549924864495,
-0.24647993267323956,
0.6292567546169889,
0.9587256267378846,
1.7268825428143277,
0.1413792259593793,
0.6520451922983643,
0.6152471684019365,
-1.1240317980448378,
-0.9856730455453077,
-1.2455074642265085,
0.2235091354354849,
-0.8870861048674711,
0.5299724059178798,
-0.21820971616414211,
-0.09687180933262383,
-0.9247884680119297,
-0.4566833626656702,
-0.4285291193052864,
-0.6406026728496247,
0.9023578506051637,
0.03748900229192909,
0.7551248935414285,
-0.18980887545414976,
-0.1497604267964229,

-0.4253730408988226,
-0.4157675203773789,
1.677209169409502,
0.2126888490300174,
-0.13481167544410624,
-1.6951458494505263,
-0.006476362637627559,
-0.5490470525961781,
0.3195499617454318,
1.077429849794694,
0.6725724542747005,
-0.10235701747439804,
-0.014129818782246995,
2.382406086835482,
0.008866515589568993,
1.0282503794544984,
-0.6899701634722577,
0.3510134577654823,
0.6918405669356381,
-0.08667968085870373,
-0.0005005962237103964,
1.3805542469189591,
1.011514915181436,
-0.4623191099783579,
1.397645255720587,
-0.7813361449321683,
-0.9940660424170442,
-0.6570734018108508,
1.0171740600492052,
0.14480072689925969,
1.157289980717297,
1.019472554492438,
1.0029573776528096,
-0.20406480565208354,
-0.011076077177880707,
-0.30566521016454584,
-0.4775965459909992,
-0.7158801142606765,
1.5610067764386268,
1.5187465408730505,
0.21158961058436962,
-0.38098352608767794,
-0.046999172842648596,
0.5964180792256618,
-0.6260168044949087,
-0.7215592053840503,
-0.5706802740378752,
0.11995346168986934,
0.17053214926260635],
[0.003408343124459994,
0.0034083431244599804,
0.006710027629051694,
0.2484817728699434,
1.1672656342280407,
1.167470260322433,
3.1453440534904744,
3.166261678218574,
3.2843704266652405,
4.089250903115526,
4.091989893807717,
4.091989893807717,
5.181052167254873,
5.181052167254873,
5.184972473956635,
5.585122024093685,
5.5990893229720795,
5.630355974178803,
5.704160317443757,
8.658831571647385,
8.66505230970618,
8.712765508205756,
8.712765508205756,
8.712765508205756,
9.168875276976076,
9.176734348833309,

9.262471352200725,
9.262471352200725,
10.578606537988184,
10.578606537988184,
1.5905119084430799,
0.8691136234632737,
0.04907693052707525,
-0.23336553986203207,
0.4284777328779486,
2.2866815267075147,
0.1760352312050406,
0.27820168999860917,
2.067264415281462,
0.5755331494929522,
-0.06858856463703264,
-0.7222667597755238,
0.7750924366303688,
-0.278304785792398,
0.5352624789798245,
-1.1702690163303244,
-0.012717570027659048,
-0.07125250327360101,
1.7927644689789062,
1.3752915621891346,
-1.1981440291126808,
-0.30737164793891514,
0.2657743539912098,
0.04156274567190095,
1.1363043468044227,
0.059217925806269944,
0.7608352104096147,
-0.1513784586666425,
0.025696061144845524,
0.861481321747562,
-0.48544656000194114,
-0.006033515622113467,
0.473431667961849,
1.5469119181224316,
0.1871819193791605,
1.555475823471471,
0.9001349633565042,
-1.5961139272878428,
-0.6749613217725859,
-1.5596468027729833,
0.5296186936496604,
-0.8549575199843406,
1.7649719371938613,
1.5569889217595585,
-0.9244334407213903,
-1.5804326845781602,
0.3840271097002098,
-0.8070678340768227,
-0.48791686278645724,
0.13110829311357092,
-2.023238766590243,
-0.8997043584182223,
0.8435119798114825,
0.12108034013815198,
-0.7013572315179941,
0.05813246571029178,
-1.3645229029121433,
-1.704441396215925,
0.6160899239086374,
-1.6580292769388947,
-1.550797918270104,
-0.033040246679508146,
-0.7595754539148374,
0.6024045095197653,
0.446745843265138,
0.9717727329863253,
-1.3190014690683383,
-3.180952616822137,
-0.47685451708111876,
0.03958388950097112,
-0.04894683660460036,

-0.013869967587830365,
0.9715551393376142,
-0.2688733257646762,
-1.7112803723577967,
-0.06346862095088202,
1.2756440828431679,
0.4087049517265435,
-0.4433737535464867,
0.07044159255576982,
0.877714241226598,
-0.33059433004855665,
-0.41683973817974146,
-0.09123983383660174,
1.153451542408442,
1.5381060924707182,
1.4432055127327397,
0.7948381511053778,
0.12126271666707081,
0.3736422837955817,
-0.06595460407520624,
-0.1569219765005705,
1.4053035958769775,
-0.3443177603686477,
0.14604377230684606,
0.733462484703497,
1.2365404639754205,
-0.020105376528163625,
-0.09631958929266826,
-0.16876809298151013,
0.42184799028013376,
0.8803036484986011,
-0.4148292271104988,
0.011464734896518355,
1.0895788780475535,
-0.006017115067308532,
0.6476481463328121,
-0.4792828310008005,
1.7145504838049759,
0.9300252591703124,
1.4147034805691376,
-0.04601267898776636,
-0.34579443398211707,
0.006107561937391137,
2.7608588509789516,
-2.470402787070412,
-0.2193552778493498,
-0.43636419827955597,
0.5036040054264609,
-0.006776779992121664],
[0.5698358581496117,
1.0111463732621893,
1.116254023022104,
1.116254023022104,
1.4543153683590362,
1.4543153683590362,
1.551144943675129,
1.551144943675129,
2.3321866923810566,
2.4857852178760083,
3.837593499897418,
3.837593499897418,
3.837593499897418,
4.847467397840062,
5.03660309473088,
5.176578602594472,
5.176578602594472,
5.214332760543968,
6.547255537065404,
6.547255537065404,
7.083633091144979,
7.083633091144979,
7.133204014188821,
7.133204014188821,
7.133204014188821,
7.937225733864811,

8.370661235857398,
8.62656679928389,
8.759707620277526,
9.424598849529344,
0.24247733923543782,
0.79275118507479,
-0.012314768690710378,
1.2247395896468714,
-0.6087890007814074,
0.9804698093938512,
0.19554007335952167,
-0.18291383394130525,
-0.08254523307317092,
0.18380200005144362,
-0.06242013495608768,
-0.18902966193120999,
0.4846770511497838,
1.2162916157423216,
0.4476502570273604,
-0.5204828708955642,
1.1525916637240337,
0.9814837501696517,
1.2019836737296188,
0.10039944576408244,
-1.165620769823409,
0.5709783345435084,
0.04690539627488982,
0.3150855883896384,
0.6835833386917337,
0.7350671828581666,
0.15105693185753874,
0.7851129846448224,
-0.290237804763004,
0.02628688844012564,
-0.3427838784588314,
-0.12202424723691215,
0.30368921729231485,
-0.08398845633414893,
-0.34421225485285395,
-1.0992299599969957,
-0.6679166311787512,
-0.4925602784337871,
-0.7607905356843834,
-0.061822787578292515,
-0.33643902903801526,
1.7237644621079549,
1.1087838048318122,
1.4391293107545464,
-0.2688865923867924,
0.07238699078201241,
-0.7262166431073663,
-0.8607492696026928,
0.1828620517189283,
-1.5556021786529124,
-0.5230641559705965,
-0.023465829708141782,
0.5773817105800866,
-0.6189512057198875,
-0.1704298219551506,
0.6935490081871019,
0.43024761601627914,
-1.3818053283072975,
0.32069867830077864,
0.11223505487625844,
-0.6642209341325642,
-0.7163563578386999,
0.9533459704680486,
-2.5373567196844773,
-0.3530595330204968,
-0.36744271288254204,
-0.4105700661159024,
-0.529185988706723,
-0.5074466275713505,
0.5969687397167245,
0.1454479675984071,

0.9510340399481164,
-0.16051754924491393,
0.168676911677473,
1.6243103666539225,
-0.3533677275828062,
0.23248347317288842,
1.550100715213732,
0.3540002793004863,
0.11004939052123668,
0.0026309322638186474,
1.0023099891482978,
1.3075603782358884,
0.06614039249633322,
0.12696883216368457,
0.4650745950568721,
0.09288306127511851,
-0.05535633254466518,
1.0893487961438044,
0.1856572010619516,
-0.3369565165204948,
1.1993442715596423,
-0.1637322799431796,
-0.05574034797393315,
-0.6499132821077687,
-1.1854402746414112,
-0.20554021240749126,
-0.697991352679575,
0.7715290435685477,
0.27728540290360154,
-0.7490020286422794,
0.6727458447445039,
-1.3928966895283914,
-1.1123692530480307,
1.2429888713160355,
0.5002310144425463,
1.0425617895549082,
-0.41250228671824735,
-0.20782873770333182,
0.952256354944612,
-0.2180059005687231,
0.8024032545701543,
-1.0975152466418368,
0.6192383355749876,
-0.44420258004771784,
-1.3605235580058401,
1.4196204597477091,
-1.1230853758949062,
0.09475958663430169,
0.9412942473630452],
[0.17961742860022237,
0.4029745783400613,
0.4271020388774394,
0.6998927493507625,
1.1444327445959492,
1.4463150894355827,
1.7859669797147917,
1.7859669797147917,
1.8910076260348616,
2.1568466639380097,
2.184715675260106,
2.3241261579565067,
5.221288254664872,
5.360678146313719,
5.572749542622228,
6.315664711217606,
6.332942041057151,
7.225695385348529,
7.238714694082236,
7.767863290353532,
7.947954629935657,
8.045079231148478,
8.080705184182557,
8.235317994608641,
8.52967785406855,
8.920755162601605,

8.920755162601605,
8.920755162601605,
9.75907480462672,
11.068402571583377,
-0.07026370230905016,
1.2758275954101812,
0.1960569796315372,
-0.7454998985427054,
0.5260870873255556,
1.3632265149428353,
0.06759709485075166,
0.7201883678889174,
0.7780364770341424,
0.7034018811814728,
-0.3281225120527201,
-0.21523884970562404,
0.3525948786581971,
0.10058439953616755,
0.8489242437750127,
0.4102210250930649,
-1.1239274808373552,
-0.07527028057729333,
-0.662823747816413,
-0.09016589559770041,
0.0005078927131911535,
-0.9974030724775158,
1.2973260682530905,
0.6207513445119947,
-0.06959715000945968,
0.01479447508324883,
0.5707366547040263,
0.8024230791007029,
0.11580671533830131,
0.5319238260215059,
2.029703693625361,
0.8476591202353092,
-0.5985490132202251,
-1.0850372047073247,
-0.13885829158476937,
0.6160884575951602,
-0.6119315652443118,
0.25456754270344084,
1.6135141314150674,
1.6101080474118592,
-1.0046888465114787,
-1.2174288348798858,
-0.8801790341822278,
-0.01998793431745051,
-1.007194981866689,
-1.6365882266166358,
3.524880820008825,
1.5399950054830251,
-1.401820445976074,
-0.04945776017633801,
0.9565666812114323,
-0.20490442564626934,
-0.5632271134618189,
1.0033608730635488,
0.12368927199762578,
-0.06153617179775209,
1.0557374829951787,
1.532632985657107,
-0.3069192599686721,
1.8620498010178619,
0.18083984306067874,
-0.9702971127887235,
-0.4866438529829019,
0.2527358034573417,
0.46390718912935613,
0.05253485630073179,
-0.15957905175444287,
0.5064463362180326,
0.6588516997606392,
0.036226022952260704,
-1.5941274572195923,

0.3692191472654626,
0.31840120283705997,
0.8741993879399423,
0.4802653460337955,
0.3012459045041651,
0.5102022084743536,
0.3727970282915065,
0.6665750190815365,
-0.7773615380935764,
0.12852119124454983,
0.6774482070442244,
-0.34235651892527574,
-1.1577831131801597,
-0.2879120353476077,
1.157549789799121,
-0.1818163295634488,
-0.034519908964088415,
-0.6744130553492667,
1.1322140576298956,
0.5044463860518625,
1.5469749343656427,
-0.19451127779487945,
0.16868157432748737,
0.3887158384137858,
1.179393002549757,
0.4377598134949622,
0.42273561999328285,
0.17555703442650725,
0.7637154786900613,
0.3204620558964089,
-1.0127445325141415,
-0.3366137672636463,
-0.6128563234467006,
-0.19959799618607235,
-0.22993024268879514,
0.6548635589470768,
0.5449412421525476,
0.11171602982233311,
0.5174688869584806,
-0.0926785574168733,
-0.012796553414024043,
0.872831179387319,
-0.3738274777631416,
0.708424995234135,
1.1207261796968788,
0.39043722964884914,
1.5466373285390471,
-0.0566526099067629,
-1.544789451209757],
[0.01970245038838479,
0.18403088824160574,
0.23516669011749736,
0.3619340287619915,
0.7559235356843684,
0.8037553588156275,
0.8037553588156274,
0.8037553588156274,
1.523650120002224,
1.5399034674182392,
3.5267025169505333,
3.526702516950533,
4.657784878324055,
4.735234101549794,
4.768060921347247,
6.910522417024203,
6.910522417024203,
7.937298848041795,
7.937298848041795,
7.937298848041795,
7.937298848041795,
7.937298848041795,
7.937298848041795,
8.950027818739875,
8.950027818739875,

8.950027818739875,
8.950027818739875,
8.950027818739875,
9.275481401407285,
0.034552241786943605,
0.7960682992937013,
-0.16960774395822967,
-0.13488653659207303,
-0.07127010448742209,
0.7213919487775996,
0.07517487951530155,
0.7669403129297154,
-0.0980146039609658,
0.2841753594230342,
0.41897129078802076,
0.7644737129881134,
0.21659836679204617,
0.40978346593820614,
0.23386906952753808,
0.6351364889148511,
0.33021983360135304,
0.3518425622667454,
-0.16759580526603354,
1.1955178055405769,
-0.1119172073880175,
-1.6647459335605923,
-0.09499524124538847,
1.171044098528735,
-0.022137095989983098,
0.46479840085551427,
0.5434533112600137,
0.7953483283824695,
-0.23791626316884204,
-0.34327459028867535,
0.9770824990177593,
0.3972109926583832,
0.031305244787475206,
0.06660894739153407,
-0.4270300066285869,
-1.4580475210852422,
0.18899331747125298,
-0.5789031865333153,
0.2203088633995836,
-0.028805094274594364,
-0.21236041446549625,
0.00835869807102751,
1.8444421334304235,
-0.36021419267454113,
-1.0777998782841605,
0.7201309879688491,
-0.07151832135087417,
0.013523038580686776,
-0.5101061898109359,
1.528477364550823,
-0.24055091899624312,
1.595727742687772,
1.181011187425284,
-0.07205139039345279,
-0.3645912646252544,
0.922259297261975,
0.47504473393792157,
-1.4582846498754758,
-1.1041238978674286,
-0.08619812725702276,
1.5348185477516838,
-0.034799821636765677,
0.34854563969795865,
-0.3057886345189525,
0.3435344949085216,
2.387989506593159,
0.5928805156422549,
-0.02250711897690342,
0.734685803060112,
1.1230348216575474,
0.9339212770561773,

3.369934702704019,
3.369934702704019,
5.00211845115155,
5.212181786011135,
5.346566249705437,
6.103091648910964,
6.103091648910964,
7.050711873868975,
7.050711873868975,
7.492144583370162,
7.492144583370162,
7.606105711064038,
7.606105711064038,
8.717193051662631,
8.753101318062777,
8.877107151002964,
8.889590522678484,
8.982812044989718,
8.982812044989718,
9.499874357008034,
9.499874357008034,
9.659025604148875,
9.671110590806489,
0.02814413440270558,
0.250593445582179,
0.17496868989230024,
-0.19749991731592192,
-0.5878125517240043,
1.1370895496823794,
0.3339578948727643,
0.30833012680275795,
0.21147160367429677,
0.5518340573364652,
0.9816766119937317,
-1.3427949214182668,
2.2864376119784486,
0.673664283367827,
0.4767745045632036,
-0.47480445681581174,
0.05912449937800317,
0.9513757199218045,
1.079093875588678,
-0.4313388366547843,
0.43347513808131466,
-0.3381632209207048,
-0.42699598389932686,
-1.1528003206699722,
0.2505681368756031,
1.1900947118566725,
-0.38260053073473543,
-0.6405385159573362,
0.08926968472452403,
-0.04834256297702397,
1.603027736914297,
-0.44672812120859223,
-0.5759172165575565,
1.25237714334681,
0.04254918814060746,
-0.002429292667880943,
0.47732275422632153,
-0.10204026491637587,
0.02357869435971368,
0.06910865492863566,
-0.5942663805386644,
1.1629666490831163,
0.28625726751699415,
0.4183193427753825,
0.33026699893767814,
1.8860632823719743,
-0.37929215782635534,
0.8251027249542411,
-0.13937217921728381,
-0.35829122981546885,
-0.11830319301401378,
-1.1307480811476716,

```

-0.33746884247229475,
1.7161163093033744,
-0.03517175574182253,
-0.4044561448132672,
-0.7301790593832838,
-0.10496992427986328,
-0.0377875273258637,
0.11576685707785253,
-0.9348023198901909,
2.132577954045408,
-0.47199808297503054,
0.09642080758509694,
-1.5590866370005203,
0.10045413530924742,
-0.3215930726850239,
-0.7020784722315541,
1.2376661808413174,
0.5709879203761018,
-0.820670510834149,
0.1634866307612446,
0.06270250983710676,
-0.03570422816362065,
-0.5699745682010601,
-0.26488761312159137,
1.795803611544861,
-0.4017001375917998,
-1.757541020399501,
-0.05132347348943386,
0.8521664539464072,
0.4812990598336637,
-0.7041980923594973,
0.6941905718406797,
1.5589556954504384,
1.9089912382579748,
0.4849438728523755,
0.03939482159422182,
-0.2564597588944596,
-0.16675906759041134,
2.5610800117093855,
0.9927310518963223,
1.3170410278463267,
0.6281459286410694,
-0.07878692694615999,
0.5865838795710373,
0.4894408136056478,
-0.043552207832595204,
0.8988064624972067,
-0.35279792613941824,
1.5088457985023234,
-0.32174328535909336,
0.41978129355629945,
0.014567846908235548,
-0.17446008524494477,
0.35494980160059775,
1.44166715836895,
0.027787407199764528,
-0.2573832607835415,
0.9605798444843694,
-0.08907047543629609,
-0.012116788408097524,
1.2676248549462201,
-0.2512276837340942,
0.7869834357157488,
-2.3607393050467915,
0.1914341630047483,
-1.0542342437530707,
0.8454261921281611,
0.02027118187750096]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,

```

```

1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_xy_noisefree_time_evo_ric3_unifrom_scaling_sample#5'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.407206817517894],
[12, 6, 0, 0, -5.829141582765496],
[20, 10, 0, 0, -4.618573032874505]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 9.202957153320312e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -8.33405069171712,
'others': { 'beta_coeffieicients': [ 1.875,
-1.25,
0.375],
'cost_richardson_zne': 5.21875,
'sorted_expectation_vals': [
-7.407206817517894,
-5.829141582765496,
-4.618573032874505],
'sorted_noise': [6, 18, 30]},
'sampled data': [ [ 4,
2,
0,
0,
-7.407206817517894],
[ 12,
6,
0,
0,
-5.829141582765496],
[ 20,
10,
0,
0,
-4.618573032874505]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],

```

```
0.5,  
0.5,  
0.5],  
      'r': 0},  
    'time': {'max': 10.0, 'min': 0.0},  
    'type': 'xy-iss'}},  
  'init_param': { 'value': [ 0.02240293444906766,  
    0.9863710197203618,  
    1.3708545038264748,  
    1.3757718928414557,  
    1.7844795346244002,  
    2.029643659915566,  
    2.029643659915566,  
    4.135298362317422,  
    4.181946654536538,  
    4.392782597821499,  
    4.473609589126697,  
    4.474471668054438,  
    4.475838923576404,  
    4.4785869343507265,  
    4.4785869343507265,  
    5.467009291914798,  
    5.470281162008999,  
    5.840541346188122,  
    5.842392372241278,  
    6.505508038171383,  
    6.505508038171383,  
    7.536348099553183,  
    7.536348099553183,  
    9.63019412694799,  
    9.63019412694799,  
    9.63019412694799,  
    9.63019412694799,  
    9.63019412694799,  
    9.63019412694799,  
    9.63019412694799,  
    1.1150074339792337,  
    -0.03535947987647509,  
    1.1918755403807781,  
    -0.20191103850925227,  
    -0.268362117189328,  
    -0.3595192771004685,  
    1.2614204066704717,  
    -2.9429228213034166,  
    -1.054810371869961,  
    0.9594255841252335,  
    -0.09538831740389547,  
    0.3965663197177072,  
    -0.22182226409641906,  
    -0.2653718580482209,  
    0.1686090968331403,  
    1.6516484362324146,  
    -0.3049608013883917,  
    0.49165272629600476,  
    1.9061569916625292,  
    1.241422732345657,  
    -0.11527538328521378,  
    -0.22895223610637053,  
    -0.06124477487110871,  
    0.6193220715745159,  
    1.372932614853267,  
    -1.1538384701543527,  
    -0.1139440760455718,  
    0.16973094341798492,  
    -0.5943234340378932,  
    0.21399932498003596,  
    -0.18156105319883253,  
    -1.0504702368491705,  
    0.7232808408606985,  
    -0.018870518436612777,  
    -0.29343981279013404,  
    0.30960592292227684,  
    0.13511542022922338,  
    0.5096669954667911,  
    -1.097576281455009,
```

0.12047585913747774,
-0.5585503308065869,
1.2279608447421912,
1.3012401380411538,
0.3769592231863052,
-0.5692023175942067,
-0.4567508363134565,
-0.679549924864495,
-0.24647993267323956,
0.6292567546169889,
0.9587256267378846,
1.7268825428143277,
0.1413792259593793,
0.6520451922983643,
0.6152471684019365,
-1.1240317980448378,
-0.9856730455453077,
-1.2455074642265085,
0.2235091354354849,
-0.8870861048674711,
0.5299724059178798,
-0.21820971616414211,
-0.09687180933262383,
-0.9247884680119297,
-0.4566833626656702,
-0.4285291193052864,
-0.6406026728496247,
0.9023578506051637,
0.03748900229192909,
0.7551248935414285,
-0.18980887545414976,
-0.1497604267964229,
-0.4253730408988226,
-0.4157675203773789,
1.677209169409502,
0.2126888490300174,
-0.13481167544410624,
-1.6951458494505263,
-0.006476362637627559,
-0.5490470525961781,
0.3195499617454318,
1.077429849794694,
0.6725724542747005,
-0.10235701747439804,
-0.014129818782246995,
2.382406086835482,
0.008866515589568993,
1.0282503794544984,
-0.6899701634722577,
0.3510134577654823,
0.6918405669356381,
-0.08667968085870373,
-0.0005005962237103964,
1.3805542469189591,
1.011514915181436,
-0.4623191099783579,
1.397645255720587,
-0.7813361449321683,
-0.9940660424170442,
-0.6570734018108508,
1.0171740600492052,
0.14480072689925969,
1.157289980717297,
1.019472554492438,
1.0029573776528096,
-0.20406480565208354,
-0.011076077177880707,
-0.30566521016454584,
-0.4775965459909992,
-0.7158801142606765,
1.5610067764386268,
1.5187465408730505,
0.21158961058436962,
-0.38098352608767794,
-0.046999172842648596,

```

0.5964180792256618,
-0.6260168044949087,
-0.7215592053840503,
-0.5706802740378752,
0.11995346168986934,
0.17053214926260635]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_xy_noisefree_time_evo_ric3_unifrom_scaling_sample#6'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.259703059772281],
                          [12, 6, 0, 0, -5.766278116709839],
                          [20, 10, 0, 0, -4.58319338434511]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.1961669921875e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.122793110315145,
                           'others': { 'beta_coeffieicients': [ 1.875,
                                                                -1.25,
                                                                0.375],
                                       'cost_richardson_zne': 5.21875,
                                       'sorted_expectation_vals': [
-7.259703059772281,
-5.766278116709839,
-4.58319338434511],

'sorted_noise': [6, 18, 30]],
'sampled data': [ [ 4,
                    2,
                    0,
                    0,
                    -7.259703059772281],
                  [ 12,
                    6,
                    0,
                    0,
                    -5.766278116709839],
                  [ 20,
                    10,
                    0,
                    0,
                    -4.58319338434511]],

```



```

'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                                'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'xy-iss'}}},
  'init_param': { 'value': [ 0.2383708143795501,
                              0.8137820253985804,
                              1.0561141201801563,
                              1.2070683478267785,
                              1.9518166906091101,
                              2.08301823027678,
                              2.6774459967535513,
                              2.7270096866449576,
                              3.7369434259122447,
                              3.7421546540398443,
                              4.458327272130387,
                              4.463102384700506,
                              4.463102384700506,
                              4.482690585412051,
                              4.482694652744643,
                              4.498978826283554,
                              6.3580266266370415,
                              7.88639599997723,
                              7.886532810158634,
                              8.060529572035655,
                              8.06234783983503,
                              8.127555772505229,
                              8.130166630032164,
                              8.42408407139234,
                              8.433074307328475,
                              9.412401860498063,
                              9.420584174895616,
                              9.429609653004679,
                              9.434851407347677,
                              11.283524698808058,
                              0.19157417616433814,
                              0.42257952171340063,
                              0.3129095694283805,
                              0.28360255750055535,
                              -0.7229159635345903,
                              2.545731433831817,
                              1.1687892956196542,
                              0.5222949375721274,
                              0.9482151911772319,
                              0.28742832079435787,
                              1.1627391357840349,
                              -0.27939056132165146,
                              -0.07574334812551493,
                              -0.529119211391362,
                              -0.08653083237786212,
                              0.042544229108731915,
                              -0.09154582517805988,
                              0.2776265525620834,
                              0.373637405241428,
                              -0.666296500917012,
                              -0.729825790392568,
                              0.36164641825124294,
                              1.2066181407380305,
                              0.18327070025757417,
                              2.2757552060408917,
                              -0.004789158452516104,

```

0.49019439516235463,
2.190622553796523,
0.6131206832540379,
1.0987721265230228,
1.9413219378309259,
0.07975169469268534,
0.17531082109327745,
0.01770145552231998,
0.22429808435825893,
1.1460098869067579,
-0.25766861736092417,
0.4386465248337572,
0.7521614434169624,
0.07039918338308593,
-0.23913727928604284,
0.062497116873310714,
0.27575365987954265,
-0.8268952220573135,
0.8599765782432734,
0.6958325703328606,
-0.16792542378671177,
-0.13186811496768225,
0.17745018848846658,
-0.13764108606901534,
-0.34798347931108986,
0.3901205214779397,
-0.12832331396189006,
0.7623260417931323,
-0.08745728181826747,
0.22448567111633413,
0.5103689813486589,
0.020179740460913417,
-0.4639048953982949,
1.8052783293547816,
-0.06895296635033099,
-0.46137869531525183,
-0.7694587159951923,
-0.1507074275764489,
-3.159084078074864,
1.5556721360426484,
0.9802025173412374,
-1.5980562353917622,
0.9974678614222413,
0.11965749551778265,
-0.5964068365118492,
1.00912773091807,
1.2674306548806673,
1.7418320010188646,
-1.2223756485879664,
-0.006270145144671745,
0.7863824204489233,
-0.08410721186616651,
0.13420385894759826,
0.6686584151585806,
-0.7946689395347296,
-1.448768973512125,
-0.4158373205099576,
-0.1003664997933048,
0.5370971852976741,
-0.013369261455621718,
-1.2134070474127567,
2.71822204346588,
-0.8266087910498845,
-0.04274152999618493,
0.627725493238542,
-0.10714625448592663,
1.4055199595750496,
-0.08877591733311402,
-0.6400618055168635,
0.27990653766167567,
0.27230874688701184,
1.8329367216853043,
0.12128377229042309,
0.03719209379786179,
0.7558193864298689,

```

-0.027085062677120984,
0.7947118159580207,
-0.18810209431195668,
0.008294184744538372,
-0.9110921816475059,
1.3065108206329406,
0.004810940817056788,
1.4030602129547465,
0.03878633149748846,
0.29850504257680605,
-0.9524583070906825,
0.10037229527914636,
0.4277056649647598,
-2.2528322583674503,
0.030724614514628212,
0.8416350822543037,
1.5579013461800135,
0.5696673724928452,
-1.5352194543526818}},
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_xy_noisefree_time_evo_ric3_unifrom_scaling_sample#4'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.418732841657521],
                          [12, 6, 0, 0, -5.815573107517698],
                          [20, 10, 0, 0, -4.600391600363878]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 3.838539123535156e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.365804543847183,
                           'others': { 'beta_coeffieicients': [ 1.875,
                                                                -1.25,
                                                                0.375],
                                       'cost_richardson_zne': 5.21875,
                                       'sorted_expectation_vals': [
-7.418732841657521,
-5.815573107517698,
-4.600391600363878],
'sorted_noise': [6, 18, 30]},
'sampled data': [ [ 4,
                    2,

```

```

0,
0,
-7.418732841657521],
[ 12,
6,
0,
0,
-5.815573107517698],
[ 20,
10,
0,
0,
-4.600391600363878]],
'sampling': 'default']},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.003408343124459994,
0.0034083431244599804,
0.006710027629051694,
0.2484817728699434,
1.1672656342280407,
1.167470260322433,
3.1453440534904744,
3.166261678218574,
3.2843704266652405,
4.089250903115526,
4.091989893807717,
4.091989893807717,
5.181052167254873,
5.181052167254873,
5.184972473956635,
5.585122024093685,
5.5990893229720795,
5.630355974178803,
5.704160317443757,
8.658831571647385,
8.66505230970618,
8.712765508205756,
8.712765508205756,
8.712765508205756,
9.168875276976076,
9.176734348833309,
9.262471352200725,
9.262471352200725,
10.578606537988184,
10.578606537988184,
1.5905119084430799,
0.8691136234632737,
0.04907693052707525,
-0.23336553986203207,
0.4284777328779486,
2.2866815267075147,
0.1760352312050406,
0.27820168999860917,
2.067264415281462,
0.5755331494929522,
-0.06858856463703264,
-0.7222667597755238,
0.7750924366303688,

```

-0.278304785792398,
0.5352624789798245,
-1.1702690163303244,
-0.012717570027659048,
-0.07125250327360101,
1.7927644689789062,
1.3752915621891346,
-1.1981440291126808,
-0.30737164793891514,
0.2657743539912098,
0.04156274567190095,
1.1363043468044227,
0.059217925806269944,
0.7608352104096147,
-0.1513784586666425,
0.025696061144845524,
0.861481321747562,
-0.48544656000194114,
-0.006033515622113467,
0.473431667961849,
1.5469119181224316,
0.1871819193791605,
1.555475823471471,
0.9001349633565042,
-1.5961139272878428,
-0.6749613217725859,
-1.5596468027729833,
0.5296186936496604,
-0.8549575199843406,
1.7649719371938613,
1.5569889217595585,
-0.9244334407213903,
-1.5804326845781602,
0.3840271097002098,
-0.8070678340768227,
-0.48791686278645724,
0.13110829311357092,
-2.023238766590243,
-0.8997043584182223,
0.8435119798114825,
0.12108034013815198,
-0.7013572315179941,
0.05813246571029178,
-1.3645229029121433,
-1.704441396215925,
0.6160899239086374,
-1.6580292769388947,
-1.550797918270104,
-0.033040246679508146,
-0.7595754539148374,
0.6024045095197653,
0.446745843265138,
0.9717727329863253,
-1.3190014690683383,
-3.180952616822137,
-0.47685451708111876,
0.03958388950097112,
-0.04894683660460036,
-0.013869967587830365,
0.9715551393376142,
-0.2688733257646762,
-1.7112803723577967,
-0.06346862095088202,
1.2756440828431679,
0.4087049517265435,
-0.4433737535464867,
0.07044159255576982,
0.877714241226598,
-0.33059433004855665,
-0.41683973817974146,
-0.09123983383660174,
1.153451542408442,
1.5381060924707182,
1.4432055127327397,
0.7948381511053778,

```

0.12126271666707081,
0.3736422837955817,
-0.06595460407520624,
-0.1569219765005705,
1.4053035958769775,
-0.3443177603686477,
0.14604377230684606,
0.733462484703497,
1.2365404639754205,
-0.020105376528163625,
-0.09631958929266826,
-0.16876809298151013,
0.42184799028013376,
0.8803036484986011,
-0.4148292271104988,
0.011464734896518355,
1.0895788780475535,
-0.006017115067308532,
0.6476481463328121,
-0.4792828310008005,
1.7145504838049759,
0.9300252591703124,
1.4147034805691376,
-0.04601267898776636,
-0.34579443398211707,
0.006107561937391137,
2.7608588509789516,
-2.470402787070412,
-0.2193552778493498,
-0.43636419827955597,
0.5036040054264609,
-0.006776779992121664]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_xy_noisefree_time_evo_ric3_unifrom_scaling_sample#7'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.568863804770513],
                          [12, 6, 0, 0, -6.084665473601483],
                          [20, 10, 0, 0, -4.914126798758423]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 3.981590270996094e-05,

```

-7.568863804770513,
-6.084665473601483,
-4.914126798758423],

```
'zne_values': { 'degree': 0,  
                'extrapolated_value': -8.428585341477266,  
                'others': { 'beta_coefficients': [ 1.875,  
                                                    -1.25,  
                                                    0.375],  
                'cost_richardson_zne': 5.21875,  
                'sorted_expectation_vals': [  
  
                        'sorted_noise': [6, 18, 30]],  
                'sampled data': [ [ 4,  
                                    2,  
                                    0,  
                                    0,  
                                    -7.568863804770513],  
                                  [ 12,  
                                    6,  
                                    0,  
                                    0,  
                                    -6.084665473601483],  
                                  [ 20,  
                                    10,  
                                    0,  
                                    0,  
                                    -4.914126798758423]],  
                'sampling': 'default'}}},  
{ 'config': { 'ansatz': { 'gateset': 1,  
                          'layer': 30,  
                          'ugate': { 'coefficients': { 'bn': [ 0,  
                                                                0,  
                                                                0,  
                                                                0,  
                                                                0,  
                                                                0],  
                                                                'cn': [ 0.5,  
                                                                0.5,  
                                                                0.5,  
                                                                0.5,  
                                                                0.5,  
                                                                0.5],  
                                                                'r': 0},  
                          'time': {'max': 10.0, 'min': 0.0},  
                          'type': 'xy-iss'}},  
  'init_param': { 'value': [ -3.641081222961908e-17,  
                              0.6149788059529836,  
                              0.7190202634465676,  
                              1.227117923902744,  
                              1.282995668258181,  
                              1.320146784613617,  
                              1.320146784613617,  
                              3.1464089486715623,  
                              3.1464089486715623,  
                              5.021180125711677,  
                              5.021180125711677,  
                              5.021180125711677,  
                              6.703692809170789,  
                              6.703692809170789,  
                              7.791601108223033,  
                              7.8369202041803225,  
                              7.951907692650894,  
                              7.98389042115583,  
                              8.258376681022987,  
                              8.258376681022987,  
                              8.264818959972613,  
                              8.264818959972613,  
                              8.264818959972613,  
                              8.264818959972613,  
                              8.265620267330194,  
                              8.265620267330194,  
                              9.412842541483705,  
                              9.412842541483705,  
                              10.502716174481437,  
                              10.502716174481437,
```

0.1665558361704551,
0.0766182276072886,
-0.06331706305100791,
0.03225035272845343,
-0.2265177184784691,
0.6859734408919996,
0.358399999202588,
-0.31490649734707366,
0.03084506070978915,
0.8141179515629021,
-0.05751068028622007,
0.9013204998454583,
-0.325467390457223,
-0.16027539072229496,
0.19756436832536928,
0.17875052394899651,
0.07441990946369173,
-1.7121562848350094,
0.5411900571694856,
0.025994665514066265,
-1.0547594680221428,
-0.3558195097906033,
0.2575282902752545,
-0.32652811387218433,
-0.17477771883072452,
0.05603758809983181,
-0.31531793685177084,
0.24953935818155465,
-0.21945102528047247,
0.016201452486715968,
-0.47139285875704856,
0.8354485366702803,
0.28783787212142514,
-1.483190369052958,
-0.05324490057492157,
0.00808064868141892,
-1.3894868155533049,
0.07648399162880297,
0.28128388531065984,
-0.8032486745370345,
0.9961277979922681,
0.7271807873997648,
-0.2598267503182047,
0.058937213125893145,
1.4176939090014022,
1.5623836495300838,
-1.2281014386266311,
1.627596878633186,
0.150437982354294,
0.017798034487339433,
-0.2091243023446671,
0.08776907917710067,
0.637527078885032,
1.2603122563899933,
1.0910453886717337,
0.001707278941889729,
-0.5366140699752073,
-0.04457743756443993,
0.708269991966563,
1.6757260639955311,
0.6610325406900899,
-0.508228690084442,
-0.5178693625521746,
-0.2955589105377924,
-0.47790822054112786,
0.31077598178920485,
0.020632196003020267,
1.5538854336639996,
1.2816443511581943,
-0.6926864187150775,
0.73698114320343,
0.3645818015611773,
-0.2097443780477125,
0.1421287040679699,
0.052774836076248374,


```
-0.7306191772967899,  
1.107856618529218,  
0.3743379513498811,  
1.2959169452837032,  
0.8858564921663113,  
0.03370518740548656,  
-1.0860983069970114,  
1.2199055901240408,  
-1.0630806559423625,  
1.3748402618369016,  
0.9943926078324806,  
0.7116645059379805,  
-0.21098608064682703,  
0.04581715611274856,  
-0.17117312830066983,  
1.2229557830916487,  
-1.6273428514067598,  
1.8508927721454946,  
0.851693435560975,  
1.0946646161197402,  
0.43088380554694883,  
-0.6171985422875839,  
1.6866137951313547,  
-0.2289359575496427,  
-0.09152362655526167,  
1.6917359096449291,  
0.22722692961233917,  
0.49434218233553445,  
1.2228162165990006,  
-0.4070166799423108,  
0.00966082758630671,  
-0.3344922737632815,  
-1.0755692755599169,  
1.1846485271761396,  
-0.06070659712292082,  
0.6384418838463358,  
-0.034216786122488324,  
-0.6519049362670448,  
-0.0019231105719348242,  
0.503764046349436,  
0.5779702041064313,  
-0.8610978040126155,  
0.5832069567020347,  
0.07954533618121334,  
0.016329041774977193]],  
'noise_profile': { 'noise_on_init_param': { 'status': False,  
                                             'value': 0},  
                    'noise_prob': [0.001, 0.001, 0, 0],  
                    'status': True,  
                    'type': 'depolarizing'},  
'nqubits': 7,  
'observable': { 'coefficients': { 'bn': [ 1.0,  
                                           1.0,  
                                           1.0,  
                                           1.0,  
                                           1.0,  
                                           1.0,  
                                           1.0],  
                                     'cn': [ 0.5,  
                                              0.5,  
                                              0.5,  
                                              0.5,  
                                              0.5,  
                                              0.5],  
                                     'r': 1}},  
                  'def': 'ising'},  
'output': { 'draw': { 'fig_dpi': 100,  
                       'status': False,  
                       'type': 'png'},  
             'file_name_prefix':  
'AUTOMATE_xy_noise_free_time_evo_ric3_unifrom_scaling_sample#3'},  
            'redundant': { 'identity_factors': [ [0, 0, 0, 0],  
                                                  [1, 1, 0, 0],  
                                                  [2, 2, 0, 0]]},  
            'run': 'zne',
```


7.937298848041795,
7.937298848041795,
7.937298848041795,
7.937298848041795,
7.937298848041795,
7.937298848041795,
7.937298848041795,
8.950027818739875,
8.950027818739875,
8.950027818739875,
8.950027818739875,
8.950027818739875,
9.275481401407285,
0.034552241786943605,
0.7960682992937013,
-0.16960774395822967,
-0.13488653659207303,
-0.07127010448742209,
0.7213919487775996,
0.07517487951530155,
0.7669403129297154,
-0.0980146039609658,
0.2841753594230342,
0.41897129078802076,
0.7644737129881134,
0.21659836679204617,
0.40978346593820614,
0.23386906952753808,
0.6351364889148511,
0.33021983360135304,
0.3518425622667454,
-0.16759580526603354,
1.1955178055405769,
-0.1119172073880175,
-1.6647459335605923,
-0.09499524124538847,
1.171044098528735,
-0.022137095989983098,
0.46479840085551427,
0.5434533112600137,
0.7953483283824695,
-0.23791626316884204,
-0.34327459028867535,
0.9770824990177593,
0.3972109926583832,
0.031305244787475206,
0.06660894739153407,
-0.4270300066285869,
-1.4580475210852422,
0.18899331747125298,
-0.5789031865333153,
0.2203088633995836,
-0.028805094274594364,
-0.21236041446549625,
0.00835869807102751,
1.8444421334304235,
-0.36021419267454113,
-1.0777998782841605,
0.7201309879688491,
-0.07151832135087417,
0.013523038580686776,
-0.5101061898109359,
1.528477364550823,
-0.24055091899624312,
1.595727742687772,
1.181011187425284,
-0.07205139039345279,
-0.3645912646252544,
0.922259297261975,
0.47504473393792157,
-1.4582846498754758,
-1.1041238978674286,
-0.08619812725702276,
1.5348185477516838,
-0.034799821636765677,

```
0.34854563969795865,  
-0.3057886345189525,  
0.3435344949085216,  
2.387989506593159,  
0.5928805156422549,  
-0.02250711897690342,  
0.734685803060112,  
1.1230348216575474,  
0.9339212770561773,  
0.5429657194529808,  
-0.9530889967003539,  
1.2607191116533525,  
-0.18262479735843284,  
0.2292294020716647,  
0.484625167272873,  
2.2621526738562925,  
0.3832060422832924,  
0.2504382199240898,  
-1.1186566372739195,  
1.0354182359738777,  
-0.9045076995614423,  
1.0282595509077306,  
1.805679851173207,  
0.9332861023326164,  
1.4396403686507162,  
1.6178312922032976,  
-0.22196274260736226,  
0.19030625281227892,  
-0.32840557234129963,  
1.004592356770436,  
1.8720477930131976,  
-0.06388076228748135,  
0.12235432131563899,  
-0.5998878734536057,  
-0.5461403115162753,  
-0.17116133229966776,  
0.6826247841010238,  
0.6607748570434985,  
0.7854114346995006,  
0.5612002866021459,  
-0.2559995100181322,  
-0.8863827207156498,  
0.2553569774973194,  
0.6372473265785513,  
-0.14017909069433732,  
0.016226196464851673,  
-0.10437448541592237,  
2.5268983957827906,  
2.3709647446626767,  
-0.8108547070158446,  
0.5698393243760288,  
-1.2042979310590165,  
-0.5004321415825274,  
-1.0362881146071494,  
-0.18531750144204898,  
1.5646886079478448,  
0.08968487420602586,  
-1.6120268318409705]],  
'noise_profile': { 'noise_on_init_param': { 'status': False,  
                                              'value': 0},  
                    'noise_prob': [0.001, 0.001, 0, 0],  
                    'status': True,  
                    'type': 'depolarizing'},  
'nqubits': 7,  
'observable': { 'coefficients': { 'bn': [ 1.0,  
                                           1.0,  
                                           1.0,  
                                           1.0,  
                                           1.0,  
                                           1.0,  
                                           1.0],  
                                'cn': [ 0.5,  
                                         0.5,  
                                         0.5,  
                                         0.5,
```

```

0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_xy_noisefree_time_evo_ric3_unifrom_scaling_sample#10'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.512595875373561],
                          [12, 6, 0, 0, -6.004303631668366],
                          [20, 10, 0, 0, -4.83491199768215]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.38690185546875e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.393829725870777,
                           'others': { 'beta_coeffieicients': [ 1.875,
                                                                -1.25,
                                                                0.375],
                           'cost_richardson_zne': 5.21875,
                           'sorted_expectation_vals': [
-7.512595875373561,
-6.004303631668366,
-4.83491199768215],
'sorted_noise': [6, 18, 30]],
'sampled data': [ [ 4,
                    2,
                    0,
                    0,
                    -7.512595875373561],
                  [ 12,
                    6,
                    0,
                    0,
                    -6.004303631668366],
                  [ 20,
                    10,
                    0,
                    0,
                    -4.83491199768215]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                          'cn': [ 0.5,
                                  0.5,
                                  0.5,
                                  0.5,
                                  0.5,
                                  0.5],
                          'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'xy-iss'}},
'init_param': { 'value': [ 0.5482907189468895,
                           0.6413620809492503,
                           0.6422482149552934,
                           1.5892409709442974,

```

1.850708229814058,
1.850708229814058,
1.8745552660050262,
2.400679867939261,
2.402212278134192,
2.485190345183722,
2.485190345183722,
2.4851903451837223,
2.488397493491329,
2.4883974934913287,
5.077476636438464,
5.077476636438464,
5.732990197214016,
5.806868622878696,
5.817834245705687,
6.166757373752113,
7.3114292874931905,
7.311858449036949,
8.346734585466223,
8.350435170229554,
8.354614694314481,
8.35659817980371,
8.356905245901928,
9.313534396314816,
9.313534396314816,
10.443787047737276,
-0.5971318801423705,
-0.6830583120471296,
0.053446928626708114,
-1.488251087277441,
0.7409998345617078,
-0.020561007687691084,
-0.47029189134787985,
-0.9187802118512801,
-0.07390570205801456,
0.6091612085770479,
0.5696050095749661,
-2.634874245900948,
0.14733366015521565,
1.0781100436188797,
0.3459269460772517,
1.2820636645787713,
-1.605623121104658,
0.856791024529966,
0.16374864437029604,
-0.7377840957855112,
0.7603752030179747,
0.4958590077418653,
1.0293890027768824,
0.4862585296205842,
-1.0838654128292502,
1.2632637762941026,
-1.1043454462318527,
-0.7867654190654241,
0.8818217647037212,
1.0406267750063978,
0.05770567519039498,
-0.5922070722918433,
-0.3271403288282926,
0.5980786503295974,
-0.02050375112330311,
-0.9007241609880184,
0.5239405469453117,
0.7601496703420288,
-0.6449049942439851,
1.7340858429103823,
0.10050113333288421,
0.3143347863780438,
0.28408386490618526,
-0.1222389867527315,
-0.09502282329152913,
-1.3004368662949968,
0.16856179320059875,
0.6790103402211991,
0.3355977478652572,

```
0.0935957266804422,
0.42936820813464754,
0.5148227650107633,
0.10764356588532305,
-0.054251737808208085,
0.06525040933822983,
-0.21143245853075732,
-1.5801988491404158,
0.023513208971767067,
1.0921277689034055,
-0.11501776583679489,
-0.14182911234947893,
1.8716225858725093,
-1.0747761250482988,
-0.01873186129021693,
0.9952348515577564,
1.5589645169221944,
2.4625123021717044,
1.5320934484141422,
0.6264578101560674,
0.015802850908690148,
0.6559295719723718,
0.5030706883609005,
0.010357532831659657,
-0.6971755064049523,
-1.334963536555068,
0.05714132580252865,
0.9478292633237722,
1.541635767019547,
-1.2357775558070643,
1.5753066317572375,
1.2206753513411082,
-0.013882574638756213,
2.102115305768464,
0.6867648509985108,
-1.0810617894892982,
0.608205793929093,
-1.0701207529796843,
-0.008804617073740417,
1.0802724219136723,
1.4472704941581649,
-0.525983160453369,
0.10347757085774918,
0.07451664808136123,
0.8113743025574697,
-0.19830571509994802,
0.20760359505818776,
-0.5709476706639499,
0.1791350390980861,
-0.32061652775092425,
-1.6382844876022806,
2.271046687924862,
0.09972766177435569,
1.671157305178672,
1.0596955078067567,
-0.7395801076185267,
-0.8217251410877643,
1.2660358026872456,
-0.149859182069558,
1.7386861885389688,
-0.003505497297480453,
0.677126408740769,
-1.0541684899696007,
0.9437251450981726,
1.7354017388586316,
-0.8022872949930118,
0.01881044769021433,
-0.013699411839369284,
1.5965466699123847,
0.4486397166383021,
1.5669137279484207]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
```

[illegible]


```
                                'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                'r': 0},
    'time': {'max': 10.0, 'min': 0.0},
    'type': 'xy-iss'}},
'init_param': { 'value': [ 0.17961742860022237,
                             0.4029745783400613,
                             0.4271020388774394,
                             0.6998927493507625,
                             1.1444327445959492,
                             1.4463150894355827,
                             1.7859669797147917,
                             1.7859669797147917,
                             1.8910076260348616,
                             2.1568466639380097,
                             2.184715675260106,
                             2.3241261579565067,
                             5.221288254664872,
                             5.360678146313719,
                             5.572749542622228,
                             6.315664711217606,
                             6.332942041057151,
                             7.225695385348529,
                             7.238714694082236,
                             7.767863290353532,
                             7.947954629935657,
                             8.045079231148478,
                             8.080705184182557,
                             8.235317994608641,
                             8.52967785406855,
                             8.920755162601605,
                             8.920755162601605,
                             8.920755162601605,
                             9.75907480462672,
                             11.068402571583377,
                             -0.07026370230905016,
                             1.2758275954101812,
                             0.1960569796315372,
                             -0.7454998985427054,
                             0.5260870873255556,
                             1.3632265149428353,
                             0.06759709485075166,
                             0.7201883678889174,
                             0.7780364770341424,
                             0.7034018811814728,
                             -0.3281225120527201,
                             -0.21523884970562404,
                             0.3525948786581971,
                             0.10058439953616755,
                             0.8489242437750127,
                             0.4102210250930649,
                             -1.1239274808373552,
                             -0.07527028057729333,
                             -0.662823747816413,
                             -0.09016589559770041,
                             0.0005078927131911535,
                             -0.9974030724775158,
                             1.2973260682530905,
                             0.6207513445119947,
                             -0.06959715000945968,
                             0.01479447508324883,
                             0.5707366547040263,
                             0.8024230791007029,
                             0.11580671533830131,
                             0.5319238260215059,
                             2.029703693625361,
                             0.8476591202353092,
                             -0.5985490132202251,
                             -1.0850372047073247,
                             -0.13885829158476937,
                             0.6160884575951602,
```

-0.6119315652443118,
0.25456754270344084,
1.6135141314150674,
1.6101080474118592,
-1.0046888465114787,
-1.2174288348798858,
-0.8801790341822278,
-0.01998793431745051,
-1.007194981866689,
-1.6365882266166358,
3.524880820008825,
1.5399950054830251,
-1.401820445976074,
-0.04945776017633801,
0.9565666812114323,
-0.20490442564626934,
-0.5632271134618189,
1.0033608730635488,
0.12368927199762578,
-0.06153617179775209,
1.0557374829951787,
1.532632985657107,
-0.3069192599686721,
1.8620498010178619,
0.18083984306067874,
-0.9702971127887235,
-0.4866438529829019,
0.2527358034573417,
0.46390718912935613,
0.05253485630073179,
-0.15957905175444287,
0.5064463362180326,
0.6588516997606392,
0.036226022952260704,
-1.5941274572195923,
0.3692191472654626,
0.31840120283705997,
0.8741993879399423,
0.4802653460337955,
0.3012459045041651,
0.5102022084743536,
0.3727970282915065,
0.6665750190815365,
-0.7773615380935764,
0.12852119124454983,
0.6774482070442244,
-0.34235651892527574,
-1.1577831131801597,
-0.2879120353476077,
1.157549789799121,
-0.1818163295634488,
-0.034519908964088415,
-0.6744130553492667,
1.1322140576298956,
0.5044463860518625,
1.5469749343656427,
-0.19451127779487945,
0.16868157432748737,
0.3887158384137858,
1.179393002549757,
0.4377598134949622,
0.42273561999328285,
0.17555703442650725,
0.7637154786900613,
0.3204620558964089,
-1.0127445325141415,
-0.3366137672636463,
-0.6128563234467006,
-0.19959799618607235,
-0.22993024268879514,
0.6548635589470768,
0.5449412421525476,
0.11171602982233311,
0.5174688869584806,
-0.0926785574168733,

```

-0.012796553414024043,
0.872831179387319,
-0.3738274777631416,
0.708424995234135,
1.1207261796968788,
0.39043722964884914,
1.5466373285390471,
-0.0566526099067629,
-1.544789451209757]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_xy_noisefree_time_evo_ric3_unifrom_scaling_sample#9'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.395384341095475],
                          [12, 6, 0, 0, -5.790856073500347],
                          [20, 10, 0, 0, -4.565524427603155]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 3.933906555175781e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.339847208029765,
                           'others': { 'beta_coeffieicients': [ 1.875,
                                                                -1.25,
                                                                0.375],
                                         'cost_richardson_zne': 5.21875,
                                         'sorted_expectation_vals': [
-7.395384341095475,
-5.790856073500347,
-4.565524427603155],

'sorted_noise': [6, 18, 30]],
'sampled data': [ [ 4,
                    2,
                    0,
                    0,
                    -7.395384341095475],
                  [ 12,
                    6,
                    0,
                    0,
                    -5.790856073500347],
                  [ 20,
                    10,

```

```
0,
0,
-4.565524427603155]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.5698358581496117,
1.0111463732621893,
1.116254023022104,
1.116254023022104,
1.4543153683590362,
1.4543153683590362,
1.551144943675129,
1.551144943675129,
2.3321866923810566,
2.4857852178760083,
3.837593499897418,
3.837593499897418,
3.837593499897418,
3.837593499897418,
4.847467397840062,
5.03660309473088,
5.176578602594472,
5.176578602594472,
5.214332760543968,
6.547255537065404,
6.547255537065404,
7.083633091144979,
7.083633091144979,
7.133204014188821,
7.133204014188821,
7.133204014188821,
7.937225733864811,
8.370661235857398,
8.62656679928389,
8.759707620277526,
9.424598849529344,
0.24247733923543782,
0.79275118507479,
-0.012314768690710378,
1.2247395896468714,
-0.6087890007814074,
0.9804698093938512,
0.19554007335952167,
-0.18291383394130525,
-0.08254523307317092,
0.18380200005144362,
-0.06242013495608768,
-0.18902966193120999,
0.4846770511497838,
1.2162916157423216,
0.4476502570273604,
-0.5204828708955642,
1.1525916637240337,
0.9814837501696517,
1.2019836737296188,
0.10039944576408244,
-1.165620769823409,
0.5709783345435084,
0.04690539627488982,
```

0.3150855883896384,
0.6835833386917337,
0.7350671828581666,
0.15105693185753874,
0.7851129846448224,
-0.290237804763004,
0.02628688844012564,
-0.3427838784588314,
-0.12202424723691215,
0.30368921729231485,
-0.08398845633414893,
-0.34421225485285395,
-1.0992299599969957,
-0.6679166311787512,
-0.4925602784337871,
-0.7607905356843834,
-0.061822787578292515,
-0.33643902903801526,
1.7237644621079549,
1.1087838048318122,
1.4391293107545464,
-0.2688865923867924,
0.07238699078201241,
-0.7262166431073663,
-0.8607492696026928,
0.1828620517189283,
-1.5556021786529124,
-0.5230641559705965,
-0.023465829708141782,
0.5773817105800866,
-0.6189512057198875,
-0.1704298219551506,
0.6935490081871019,
0.43024761601627914,
-1.3818053283072975,
0.32069867830077864,
0.11223505487625844,
-0.6642209341325642,
-0.7163563578386999,
0.9533459704680486,
-2.5373567196844773,
-0.3530595330204968,
-0.36744271288254204,
-0.4105700661159024,
-0.529185988706723,
-0.5074466275713505,
0.5969687397167245,
0.1454479675984071,
0.9510340399481164,
-0.16051754924491393,
0.168676911677473,
1.6243103666539225,
-0.3533677275828062,
0.23248347317288842,
1.550100715213732,
0.3540002793004863,
0.11004939052123668,
0.0026309322638186474,
1.0023099891482978,
1.3075603782358884,
0.06614039249633322,
0.12696883216368457,
0.4650745950568721,
0.09288306127511851,
-0.05535633254466518,
1.0893487961438044,
0.1856572010619516,
-0.3369565165204948,
1.1993442715596423,
-0.1637322799431796,
-0.05574034797393315,
-0.6499132821077687,
-1.1854402746414112,
-0.20554021240749126,
-0.697991352679575,

```

0.7715290435685477,
0.27728540290360154,
-0.7490020286422794,
0.6727458447445039,
-1.3928966895283914,
-1.1123692530480307,
1.2429888713160355,
0.5002310144425463,
1.0425617895549082,
-0.41250228671824735,
-0.20782873770333182,
0.952256354944612,
-0.2180059005687231,
0.8024032545701543,
-1.0975152466418368,
0.6192383355749876,
-0.44420258004771784,
-1.3605235580058401,
1.4196204597477091,
-1.1230853758949062,
0.09475958663430169,
0.9412942473630452]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
               'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_xy_noisefree_time_evo_ric3_unifrom_scaling_sample#8'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.159167099826597],
                          [12, 6, 0, 0, -5.573237410199788],
                          [20, 10, 0, 0, -4.355904736591592]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 3.695487976074219e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.090355825646983,
                           'others': { 'beta_coeffieicients': [ 1.875,
                                                                -1.25,
                                                                0.375],
                                       'cost_richardson_zne': 5.21875,
                                       'sorted_expectation_vals': [
-7.159167099826597,
-5.573237410199788,
-4.355904736591592],

```

```

'sorted_noise': [6, 18, 30]],
'sampled data': [ [ 4,
                    2,
                    0,
                    0,
                    -7.159167099826597],
                  [ 12,
                    6,
                    0,
                    0,
                    -5.573237410199788],
                  [ 20,
                    10,
                    0,
                    0,
                    -4.355904736591592]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [ 0,
                                                            0,
                                                            0,
                                                            0,
                                                            0,
                                                            0],
                                                            'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                            'r': 0},
                        'time': {'max': 10.0, 'min': 0.0},
                        'type': 'xy-iss'}}},
  'init_param': { 'value': [ 4.1837705414928806e-17,
                             1.1742327239243593,
                             1.1742327239243593,
                             1.174232723924359,
                             1.3021435979041922,
                             1.3883521076730898,
                             1.4901272631800928,
                             1.490127263180093,
                             1.490127263180093,
                             2.542545188872923,
                             2.56739133991554,
                             2.677178516908646,
                             5.624946341909109,
                             5.624946341909109,
                             5.63924283183031,
                             5.672838193807689,
                             5.672838193807689,
                             5.672838193807689,
                             5.672838193807689,
                             5.672838193807689,
                             5.672838193807689,
                             5.805507018952099,
                             5.858678454346443,
                             7.374841183417447,
                             7.374841183417447,
                             9.078346087689514,
                             9.17910617229311,
                             9.185363997877552,
                             9.550522060955204,
                             9.715006087156702,
                             0.5380723684568228,
                             0.20346632529178696,
                             -0.28402043001150323,
                             -0.013694659434132353,
                             -0.8491217472851175,
                             -0.13891219530245408,
                             -0.7902948900320785,
                             0.02035105936303376,
                             0.35891510624171696,
                             0.07011262472067725,

```

0.2571910630887441,
0.09727807922224965,
-0.4253506785797692,
-0.08201618636277365,
-0.5629942371157008,
0.4984357070329931,
0.22460606922422946,
0.6436861273901548,
0.5581387866543217,
0.6438048424266741,
0.6625810608832491,
-0.2877740698749285,
0.03400434270310955,
0.6598198228171638,
0.15695492486622947,
-0.23277712193728803,
0.3379436543817524,
0.6679629986711755,
0.4913802101172593,
-0.6543304626041586,
1.2398324170787045,
0.10868488610295632,
0.4726806257545104,
-1.0278416218016544,
-0.5660236088762642,
-0.06627186605932975,
0.3527547117645279,
0.3944014355355292,
0.33709190475341266,
0.16707772676088822,
0.29195080738380563,
1.0354529601392413,
0.12500585138196232,
-0.037178956773431956,
0.7823218278561086,
-0.04464215343745706,
-0.36923576305262273,
-0.9863817864437636,
0.039519076991602065,
-0.35646400258617383,
0.04542480304419608,
-0.46125271135496354,
0.40472620693719735,
0.7802480661953589,
-0.6757172179201896,
0.6396142241310521,
0.2327897749192615,
0.8954622299286413,
0.7158405805761733,
0.1382139633537826,
1.0616928944303126,
-0.6122315420380016,
-0.6738539946512315,
1.0433459792659807,
0.494526070112864,
0.20621322780405107,
-0.05318175489586881,
-0.11213659998229918,
0.32941262943060273,
0.13931786347259914,
-0.07640781308377202,
0.31450116184408605,
0.3658109215963646,
0.04460349089074807,
-0.12913310998017427,
0.31890485610090735,
-0.005491508360761248,
1.2692063073436923,
-0.3006174703747285,
-0.41404661321397057,
-0.5199199797590206,
0.44225909618051784,
-0.4233007342169607,
0.7456354776212751,
-0.32484988424610756,


```

0.9268151291603125,
0.8385591284905366,
0.027219364270413358,
-0.5520387125082392,
1.3757901840365965,
0.02933419406725825,
-0.2915193842748469,
0.33917094344303517,
0.10828106345120646,
-0.5650186064180197,
-0.6041651963720519,
-0.19230834783869816,
1.9848826966849449,
0.0632219631903188,
0.04804551511145073,
0.47752350144148015,
1.3406847618100746,
-1.3722262204383469,
-1.0513455672656666,
0.7121308231347245,
0.13278942055995957,
1.9758312031522887,
2.19648222646383,
0.8936788375205645,
0.05510909660880629,
-0.18765678539637448,
0.45031205104496697,
-0.6542716355161553,
0.5783583754336702,
-0.18844184232604352,
-1.5667303012058793,
0.36110899429768645,
-0.275182998952267,
0.30367576290775045,
-0.1532687178202341]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_xy_noisefree_time_evo_ric3_unifrom_scaling_sample#2'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.130214875567141],
                          [12, 6, 0, 0, -5.622281179564727],
                          [20, 10, 0, 0, -4.4547314200293675]],
        'degree': 0,
        'method': 'richardson',

```

```

'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 3.7670135498046875e-05,
            'zne_values': { 'degree': 0,
                            'extrapolated_value': -8.011825699743495,
                            'others': { 'beta_coeffieicients': [ 1.875,
                                                                -1.25,
                                                                0.375],
                            'cost_richardson_zne': 5.21875,
                            'sorted_expectation_vals': [
-7.130214875567141,
-5.622281179564727,
-4.4547314200293675],

                            'sorted_noise': [6, 18, 30]},
            'sampled data': [ [ 4,
                                2,
                                0,
                                0,
                                -7.130214875567141],
                              [ 12,
                                6,
                                0,
                                0,
                                -5.622281179564727],
                              [ 20,
                                10,
                                0,
                                0,
                                -4.4547314200293675]],
            'sampling': 'default'}}}],
'xy-ric4-uniform-scaling': { 'noiseoff': [ { 'config': { 'ansatz': { 'gateset': 1,
                            'layer': 30,
                            'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                                'r': 0},
                            'time': {'max': 10.0, 'min': 0.0},
                            'type': 'xy-iss'},
                            'init_param': {'value': 'random'},
                            'noise_profile': { 'noise_on_init_param': { 'status': False,
                                'value': 0},
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': False,
                                'type': 'depolarizing'},
                            'nqubits': 7,
                            'observable': { 'coefficients': { 'bn': [ 1.0,
                                                                1.0,
                                                                1.0,
                                                                1.0,
                                                                1.0,
                                                                1.0],
                                                                'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                                'r': 1},
                            'def': 'ising'},
                            'output': { 'draw': { 'fig_dpi': 100,
                                'status': True,

```


3.800207092295721,
3.800207092295721,
3.800207092295721,
3.800207092295721,
4.798448841518561,
4.812026604031267,
6.02543601067661,
6.02543601067661,
6.5265343452987,
6.5265343452987,
6.526534345298701,
6.528381158760148,
6.528381158760148,
7.113023968622042,
7.113023968622042,
7.593654062389684,
7.5949106716383605,
7.780058793305066,
9.000370275113886,
9.005799799983558,
-0.22105281285875572,
0.07196296170715184,
1.0527781347416576,
-0.4594806493129553,
1.0526039239081737,
-0.01709270509934428,
0.3199675106146978,
-0.0972220164715271,
0.9145097677998977,
0.09808103992344623,
0.5373974901276602,
-0.5786059880820694,
-0.02594763303680847,
-0.22132394348491485,
-0.7274785930689547,
-0.41090279602128,
-0.520884263329148,
-1.0260835302846771,
0.1692597388493397,
1.3250313995601488,
1.510357845739394,
-0.9233604808876308,
-0.2049157632594096,
-0.9894281051832,
0.25772120620526445,
-0.04822142329545429,
-0.3912638297585932,
-0.48486712444934904,
1.4836649713082644,
1.566996182191827,
0.08093959953023218,
-2.3562926055425963,
-0.21876618997457167,
0.10911407834571141,
-1.3776955808248483,
-0.31816531211057325,
0.7083857256813267,
0.5593482503842677,
0.7483381318097894,
0.010042025170353972,
0.05290425882351868,
0.20616876613540233,
-0.8673924682871486,
-0.08017548043713153,
-0.3056222778255953,
2.3981163154468383,
0.37079365521682633,
0.3870558721039623,
0.19341028278759992,
-0.3154248344945196,
-0.12073260037561208,
-1.5858414285227351,
0.59830347262823,
0.30388710178795286,
-1.4558796844932331,

-0.3225618511944297,
0.09243390755822743,
-0.0006426173071943948,
0.7561662109631148,
-1.3902324585417218,
0.1677701056843372,
2.7311884232201735,
-1.1122880815211051,
0.0467986017971284,
-0.020089420529231013,
0.005481851366742631,
1.3205596999819047,
-0.1290403664286252,
0.3870444394307043,
-0.6549197738372419,
-0.17656835190090397,
-0.0008013451895215781,
-0.8769883624496301,
1.6929520910231297,
0.7288671099890552,
-0.6169673343691255,
1.667217439782634,
0.2554635335546482,
-0.029515171705053127,
-0.8290153205719569,
0.042733616172741226,
-0.5148227135977747,
1.2301836750084596,
-0.9175908300980525,
0.5229779878549122,
-1.5901934710811256,
-0.07316585886439231,
1.1195018365185199,
-0.31111306362203306,
0.7254709796877604,
1.1995417479543053,
0.6908225815252583,
-0.2919630814941808,
0.15955998483086176,
1.2871422030548714,
-1.2683991057896014,
-0.3768598572966417,
1.4863954796333156,
-0.4300298743216642,
0.19151460300148976,
0.9665838873771405,
-0.07244306377732504,
-0.04169546816073936,
-0.389029635704794,
1.5218961812832925,
-0.1772066326032777,
-0.70949107362336,
-0.015222466074935298,
0.3587128629768407,
1.5809494716454662,
0.12428164288740734,
1.5083554487395696,
-0.044379618902227067,
-0.03505579308192811,
-0.6786538117837789,
-0.007428157845337949,
1.5281494732580478,
-0.14124856089608578,
2.726718429443767,
-0.003106328084445967],
[0.5270475437101116,
0.7815425625485376,
1.6469610445191378,
1.6712876404917778,
2.122904188877425,
2.131106816606521,
3.7367798442214175,
3.7367798442219424,
4.750276145421292,
4.853973384996016,

4.865098072058574,
4.873980651141647,
4.987599328262889,
5.00246239884106,
5.520902021852277,
5.793284849001337,
5.8087832398624775,
7.432729128414403,
7.432953787975026,
8.144896858715603,
8.172580744083415,
8.788071761144659,
8.792288306719156,
8.828260688191317,
9.011586808740137,
9.011995976067807,
9.538729699920484,
9.576884709106812,
10.486750161390887,
10.48675016212107,
-0.15531374899360717,
-0.0849039080739262,
-0.8434555658556069,
0.9447826609611266,
-0.19482082661427938,
2.4999420394249707,
-2.7164831941210483,
-1.5408169364026967,
-0.8621482593978547,
0.31547458537730877,
1.4220435826823787,
1.348860553316339,
0.40827474166943256,
1.063960383917828,
-0.21906798993221652,
-0.33263312803097533,
0.4889869181158972,
-0.25524084746075687,
-1.3622146811064544,
0.996105149433398,
1.4161210728452394,
-0.6491827941870941,
0.7399329880002848,
-0.11994422281276146,
3.179573825543474,
-0.034943024466848674,
0.6075464711124173,
-0.3019825794696199,
0.029392952712390567,
0.14840732209332994,
0.08592470568620861,
0.016725634393736574,
-1.3697962205589145,
-0.06155375586805023,
0.16508080354919497,
-0.5975014914051775,
0.09635736997937351,
1.254101402191796,
-0.6193953337438176,
0.47602269510847217,
-1.7251540483535392,
0.5378274350019089,
-1.0159384367751108,
1.0968223266795405,
-0.79878134010976,
0.10468891520322528,
-0.2910173593847854,
-0.7078981306941186,
-0.3510837282628223,
0.5371002854277007,
-0.4911274293534503,
1.7273270693440552,
1.3500555714793954,
0.9052059235505011,
0.4771526028272018,

-0.1893747959421651,
0.13989050072973697,
-0.5930955187300607,
0.36780210947560876,
-0.5043370473487999,
-0.4851893292987325,
0.1987440888222681,
-0.5806445532873666,
2.3171646404124093,
0.06032126128170586,
-0.6389920437730593,
0.9985359186149056,
0.04870155999532171,
0.16447399991313041,
0.05315090520136875,
0.15727255736970833,
-1.3722292706433088,
-0.33233649668322635,
0.7758243461945761,
-0.20476613404890107,
0.06972046040604139,
0.3143879960336948,
-0.01452323353492752,
-1.2901366074359686,
-0.31033000747276407,
-0.16264485166390844,
-1.3914094473719103,
-0.5273128620655095,
0.02465459151953875,
0.10942766294132272,
1.0292094835171124,
-1.882689768994833,
-1.553216293090225,
1.3287846024814622,
0.20963913600213377,
0.7195732642990699,
0.0969787732183028,
-0.9744344415633356,
1.5822487402774768,
0.7077471503862973,
-0.766345302327988,
0.3831437091022152,
-0.12028853613988252,
-0.8209919023748664,
-0.21164805512089607,
0.5290684330656528,
-0.1040878870845114,
1.1956945887281514,
0.25703931501419025,
0.6344733295912399,
-0.02497584874729833,
-0.20937697561532842,
-1.4409411147319051,
0.9707320454927133,
1.4039378085761043,
0.9760826987262126,
-0.17314228405542512,
0.16734556762922406,
0.07881248116464283,
-0.2924467918405051,
1.4946430081327182,
0.7516806171134963,
0.799700486923224,
-0.8440600324500036,
-0.01069194371181784],
[0.3388610072211759,
0.3930923343746358,
0.4059472295013711,
0.4597995545870322,
0.4597995545870322,
0.4667083780713653,
0.9924683364283498,
1.054118787032341,
1.3839145947816414,
1.4469616998432806,

3.044048229936016,
3.044048229936016,
4.682975233106521,
4.682975233106521,
5.287925895396742,
5.287925895396742,
5.625677830869467,
5.625677830869467,
6.519005703020383,
6.519005703020383,
6.519005703020383,
6.854665591577039,
6.854665591577039,
8.092962128458302,
8.097830997780322,
8.102159968176109,
8.111954675742565,
8.727038821401255,
8.727038821401255,
8.784424594255395,
-0.8636093730812797,
-0.04742973062196874,
-0.37989856483295564,
0.3750798322441465,
-0.509087961968837,
0.5366875438436245,
0.1232704940065221,
0.5249134835872401,
0.13594138148710633,
0.10802292582448766,
-0.7156772546121367,
0.34756877087651405,
-0.5704812075854907,
0.289395233495819,
0.2674862930849357,
0.8724143866961164,
0.5770602988816299,
0.658777432128999,
-0.47288232304292416,
-0.029663566814993374,
-1.323180236232167,
-0.2890901656039231,
-0.015141732930961212,
1.1846850771343012,
-0.4506688471965015,
0.22241980016839272,
-0.015748484493019736,
1.1522499025192263,
0.2996907655672081,
-1.7370631162182948,
-0.35306232532663057,
0.17322471105923057,
-1.1855180134814352,
-0.31537295018088835,
-1.8120366955290261,
0.6663892425075478,
0.21874911670355193,
1.2729693967005706,
-0.14230256773273223,
0.2012427441067739,
-2.3269317425793643,
-0.03299848401271634,
-0.4074853136565566,
-0.30371905431664253,
-0.12037074206123735,
0.8456676988075262,
-0.21838328071319946,
0.019494764922420988,
0.16383099728733078,
0.056056403895700845,
-3.5632063709291613,
0.6579904510057321,
0.5207861046034454,
-0.017180990359968364,
0.2532807007765998,

-0.03315495616728509,
0.7564905992030696,
-0.11091568858030387,
-0.9281890577202553,
-0.6082181315906088,
-0.4480511658411004,
2.1953283079942025,
-0.30446713497464617,
-0.03359602431311142,
0.28972943746372876,
0.17817064395522006,
0.7965754083210965,
-0.9520780563921613,
0.10150445450145726,
1.3510884636720888,
-0.37849615173878054,
0.17729737647006832,
0.1884683806301832,
-0.03234547530918763,
-1.338763456177623,
-0.6711058962200585,
-0.012976043431998471,
0.027578392175021227,
0.10161286285089129,
-0.00939135247962935,
-1.302164623174096,
1.3715576399721376,
0.737767307868418,
1.5939269674119705,
1.4291758244230544,
0.5601859764306084,
0.3488764286390935,
-1.093762224545935,
-1.1976831420922067,
1.4557381665683378,
0.8924472305311927,
0.5300221300338881,
1.0785280788144638,
-0.6029509417428172,
-0.6790632925762424,
0.4138865114731201,
0.2917474006869537,
1.8050154062536312,
0.05177383423831921,
0.25467777497912664,
1.4671356779039024,
0.4422247342165525,
-1.5431607378377095,
1.8101814873938125,
1.6870467346899987,
0.7517920825884995,
1.3622912362338087,
0.13653050459472865,
1.5210183141257738,
-0.004247168330480351,
0.26137218980413907,
0.5451511562235306,
-0.6705596749039198,
1.2136085126593865,
0.5231529087420831,
-0.03279466120532818,
0.2548954759886831,
1.5797820110558691,
1.075840789720537,
1.5541818084195007],
[3.821020346740443e-17,
1.386496864345847,
1.386496864345847,
1.7951472089989222,
1.8559400139285378,
2.7980544315911504,
2.7980544315911504,
2.82954781812891,
2.82954781812891,
2.82954781812891,

2.956183634338348,
3.840561875918348,
3.840561875918348,
4.79501776182173,
4.79501776182173,
5.670460018404704,
5.670460018404704,
5.670460018404704,
6.865392250052202,
6.865392250052202,
7.506826147384778,
7.506826147384778,
8.737685173635683,
8.737685173635683,
8.737685173635683,
8.737685173635683,
9.601023092540274,
9.601023092540274,
9.853738132880634,
9.853738132880634,
0.5611658219809542,
1.427331325859137,
0.8881601644240952,
-0.8937497791603879,
0.5712044928850715,
1.3195951619089188,
0.5467792160605555,
-0.6816630744322637,
0.06707477836950569,
0.6798692668930644,
0.19576168860111362,
0.5094724850631845,
1.1424414477256812,
0.30471145260454696,
-0.08158911485577636,
0.9585661478313002,
1.0836238583568754,
-0.3718197048926503,
0.7841948506971846,
0.3023515457437066,
0.7662240328883452,
0.9970048707689876,
-0.8946064024093777,
0.27649390200762114,
-0.15342626519190924,
-0.4015720487025046,
1.460926694270634,
-0.055421168101734586,
0.6077050880702092,
-0.7932081335349191,
1.3188332541813446,
0.01929555634963922,
-1.1301845164589392,
-0.1718174564221745,
2.26154163970611,
-0.9882028269623031,
0.8038643847286138,
0.6849304102208338,
0.874798983022982,
-0.016486064274954207,
0.0264221098096002,
-0.3625000384278827,
1.4376331477886695,
0.8499992740594865,
0.10042171561200945,
0.14542648933357244,
1.3513090878714353,
-0.22248922869259902,
-0.4483175327546578,
1.2345758678942387,
-0.18983522552694923,
0.019591210301167533,
-1.3607248526627624,
0.018375533581887468,
-0.3730738776375715,

1.0123537416913542,
0.7412601688930508,
-1.6525919628732422,
-0.8341682190885938,
0.04519729634722039,
-0.06718028327556408,
-1.5649719285752928,
-0.37531851249345394,
-0.11822841604899964,
0.008154106549025474,
-1.4601449692077721,
-0.0030993446167428395,
1.462573870839384,
-0.11887757571245174,
-0.09361960188417112,
-1.2543408854038138,
-1.525280802664792,
-1.241413372948363,
0.008758808605552393,
-1.6676332942965022,
0.13826170532334445,
0.7537363962746654,
0.0833397476047736,
-0.12216949704941273,
0.02084807234445201,
-0.3421949354726438,
0.0017882611822728064,
1.5186851123669147,
-0.4603163345929323,
-0.09956972276089109,
0.7247597830417082,
0.5304935691839561,
0.003193152207841171,
0.5581346990750625,
0.010542051444639388,
0.7553167297004638,
0.5956891015800863,
-0.05566254332595716,
0.5362461335652635,
0.009895282042162012,
-0.3029901133286735,
0.09399206534914867,
0.8818812838713691,
-0.06795832284115338,
-1.580933930101608,
0.3014975711017802,
1.2092022074751856,
-0.019692531397995948,
0.9250981578351692,
0.4673383297638096,
0.03344339272872125,
0.13058472572385768,
0.036262684184263205,
0.458095375934523,
0.6492846129896516,
1.3039221898344189,
0.024764812872192767,
0.7236926243101844,
0.01444840086266692,
-1.0388939706148563,
-0.08359693344439546,
-0.4698004238001188,
-0.48055768545836347,
0.4662000815300196,
-0.041204969991243484],
[0.01798320141016013,
0.3393419127512466,
0.7047697703765823,
0.7064789163091142,
0.7191656785855808,
0.7198672068604162,
0.719867206860416,
1.0840512961729545,
1.0900003904895101,
1.1926153174924548,

1.867054342490295,
1.867054342490295,
2.368094916985588,
2.3683751276938714,
3.6383003552329964,
3.642527762446636,
4.505845683894636,
4.602446412902637,
4.972427659100928,
4.972427659100928,
5.1148164150678035,
5.8066136392845,
5.820681023327439,
6.727088267818048,
8.352653016300794,
8.773334122756978,
9.065902540064004,
9.065902540064004,
9.74880092287043,
11.30145464466351,
-0.13904770480884485,
0.23493722133539818,
0.4497803172772882,
-0.640584995353188,
0.9779304528121034,
0.0011745349607135795,
-0.7175493520168287,
0.373620380836484,
0.08111255661159392,
0.6653458367314794,
0.0037967443269442437,
0.6286122183467432,
-0.03825426016359485,
0.6896824961465278,
-0.23916946602734285,
0.4624850989736848,
0.1419493525208794,
0.7145176662978019,
-0.316909964913161,
0.6605458698076423,
-0.39309909050950365,
0.7623048158369822,
0.7469736169282462,
1.2400892032597353,
0.12266455668869777,
0.19578648318487166,
-0.784043232212839,
-0.10334183265831844,
0.32774480702026265,
-0.011491906972883237,
0.322988869457402,
-0.2027870428092001,
0.1281571522083961,
-0.1947961113336468,
1.7114513053475526,
0.005425684515457419,
0.8136825826883561,
-0.25903007981373005,
-0.5015428579892063,
-0.2600978115589891,
-0.15405473476720474,
-0.4207498663011112,
1.04622682040645,
-0.8642339256273757,
0.6248691295250803,
-0.3539240130838131,
0.011689879721248103,
-0.3068283940685192,
-0.6048660879777619,
-0.11704784670887185,
-1.2745062338511486,
-1.7890038533223112,
-0.7818671277256136,
-0.8877455887286905,
0.3164818198590212,

-0.27477307924208777,
1.8083251930535746,
-0.5031479272825263,
-0.9648993597903888,
-0.41682962131770707,
0.27040705073896654,
1.0892900815418987,
0.17779960873468054,
0.020337033950376556,
1.0551529464315492,
-0.10705558118147128,
-0.8851977322505518,
0.2232945485845247,
-0.5850403569085746,
-0.7690595025978197,
-0.4391673030332381,
-0.9309566421919284,
0.2973030450025941,
0.6780545449896137,
0.14045373030826291,
-0.5109313900859794,
-0.44070010469428217,
1.6940975579604587,
0.17062304524159472,
-0.402681393200546,
-0.7091187255059627,
1.3132357546586793,
0.2729391261599054,
1.0115940175088693,
1.0046673925246778,
0.4982257464819494,
0.04454679168285006,
0.4863023994043421,
-1.1433261311581624,
-1.820308437409531,
0.6035405700431521,
0.04003907881711185,
2.99381748600971,
-1.0437523456620463,
-0.22517480393070408,
0.10746937069246698,
-0.8788513722547997,
1.286531186523234,
-1.3157160458746164,
-0.6979748907629274,
-0.16573593981190854,
0.2529194783090185,
-0.2562517800846549,
-0.7966921479109774,
-0.4446940958312971,
-1.052310031662883,
-0.1506245930250076,
0.1615975776996014,
2.9220364155549214,
-1.7197843955783065,
-0.6592103621593355,
-0.2007498223803255,
-1.6188873615119244,
-0.15193115939751467,
-0.20866967970071015,
1.2910791163230286,
-0.4917152660712325,
-0.42348313202027765,
0.3518303936962707,
-0.10034486495238693],
[0.009039772647041902,
1.1400526086001903,
1.466905028683042,
1.6524670085313564,
1.987797816914966,
1.9900950346150876,
2.1413202332332344,
2.390935514217652,
2.402429399816812,
2.81913686416337,

2.81913686416337,
3.5656810200683298,
3.5656810200683298,
3.5656810200683298,
3.76603447051701,
3.76603447051701,
5.499236323276389,
5.4992363232763894,
5.4992363232763894,
5.532789537471719,
5.532789537471719,
7.371733777832876,
7.371733777832876,
7.680783486275433,
7.706623781446164,
7.706623781446164,
8.457742149729329,
8.457742149729329,
8.95222736351662,
8.95222736351662,
0.18928393791812126,
1.0354889043700823,
0.3703113207409527,
0.30988589523494625,
-0.34669340410550803,
0.09615284519636855,
0.1011898884094612,
0.6160115432573767,
0.6148540393435122,
0.07934943483664206,
0.8557624329627399,
-0.7672633290674925,
-0.14188111026081618,
0.1995320641320393,
0.4594888802421805,
-0.03593187081620776,
1.2926591415659605,
0.31433950962447343,
0.8053272546604554,
0.6634815559763872,
0.311660786273531,
0.0294653444200854,
0.3044394736040779,
0.7599259037372076,
0.22962735656034647,
0.22562971232722423,
0.29511485598499815,
0.02260075636691185,
-0.9497096253675098,
-0.5331371571917215,
0.2192512323868535,
1.104832430155296,
0.8636617903941131,
-0.6756684630428217,
0.4169844309263729,
1.1297527651723116,
-1.160118439748996,
-0.7971836651224,
1.9169656561991728,
-0.007260796290839314,
0.3614132074474742,
0.6243022778229537,
1.8335025630018775,
0.35139722497115833,
-0.7247870129359816,
-0.6744811297738377,
0.49997538785346773,
1.4139521212248116,
-3.0799818401546553,
-0.14768564076593044,
0.06251500852230953,
-0.8547134606957393,
-0.30107856776262504,
-0.7654972812734481,
0.28603020918627825,

0.2129303248959857,
-1.2077141264570097,
-0.8488549318812054,
0.6492574804166855,
0.7751290676155554,
0.40909519807061084,
-1.2750630737105486,
0.4187798718836698,
0.33498424523924863,
-0.19017028209895162,
-0.03232931911977397,
1.0621483440417678,
0.452017889610544,
1.0192767515800205,
0.7290563184502946,
-0.5286462788982386,
-0.05512372609304388,
1.3189151919853837,
0.5865235617306253,
-0.579843833868396,
0.6980158293666034,
0.027004860579276476,
1.2876675233834314,
-0.1718561727774212,
-0.3827017785428823,
-0.7286404996479297,
1.1682112525473984,
0.22765379123755217,
1.4870969768845252,
1.1642898801914527,
-0.03976972562460009,
-0.8163296820868416,
-0.15318057329999143,
0.0482918949035788,
-0.06014345828968787,
1.1478583262090496,
-0.005141116181289305,
-0.5820859555621626,
1.086519910329415,
-1.08558016317555,
1.531511042803357,
0.042734048269580674,
0.7452599788242152,
-1.3094923903018465,
-0.9012205175381317,
-0.0755166851159581,
1.5493661520767013,
-0.2278023712186688,
0.8627212805492942,
1.8523276191392382,
-0.10961709979291216,
0.6746606852631131,
0.22566845572735392,
0.5920154791154506,
1.2709342533151342,
0.26903045177842116,
0.11724485476711638,
0.621942277674204,
0.15491924869980966,
1.472217084758381,
0.8975149794611956,
0.3601026339731485,
1.3234513983254743,
0.0182465793142515,
-0.11914786829986494],
[-5.765348397008027e-18,
0.3428984748224433,
0.3655450682584724,
0.704784031551808,
1.5409081472811441,
1.549300366088437,
2.3787666619954635,
2.3787666619954635,
2.8808650819472406,
2.884987740396357,

3.1308808834009203,
3.1363836212549696,
4.869321728501938,
4.869321728501938,
6.1150174264690085,
6.2582278136815175,
6.271636236200362,
6.551203581243674,
6.59741701640448,
8.230564833747417,
8.294003346338926,
8.31431753208387,
8.369586036193128,
8.415319617883744,
8.70447046242739,
8.709120596281233,
8.918659758853744,
8.99410216914391,
9.002256766523654,
10.218690251738174,
-0.6798627919679444,
-1.1436724543472427,
-0.1278269606092795,
-0.9286552360068459,
0.6524046490818174,
0.06134572949387961,
-0.49761179900021657,
-1.3918471119443894,
0.5858287319319103,
0.18669127045720346,
-0.4586789737568629,
0.2400608953322058,
0.022222124883641912,
0.9282004764805514,
0.6952358394741303,
0.805684979006905,
0.6146729452742722,
0.6777524139082742,
-0.8117842884117104,
0.3426849285038029,
0.7100299235979358,
0.7658186509139059,
-0.05418633990950139,
0.1761917490887145,
0.5744825473612316,
0.31705937472296297,
-0.9188523464356001,
-0.7337446596965294,
0.2061368312243915,
-0.6645694031850036,
-0.009871104765786294,
-0.14388187083546433,
-1.2485707106311008,
2.849540809212292,
-0.5119215846255833,
-0.19381894062193808,
-0.17310717674563775,
0.8465397629359774,
-0.46757240698237135,
-0.05607605828560263,
-0.40986137011816176,
0.07649505002491204,
-2.1522907800348112,
-0.6653346582594515,
0.4084801543759574,
0.4028897820911683,
0.16908117843954765,
-0.07217026591146015,
-0.72804755296126,
-0.13548738477012795,
-2.590679566047418,
-0.40743076961744734,
1.1030856800887578,
0.08388380440455438,
0.25224805905642333,

0.10644300798076774,
1.0474125729404944,
-1.1459016177859003,
-1.6916287310468103,
-0.9886067106540645,
0.24182274960650893,
-0.25703819491337826,
-1.1942818044595898,
-0.7763678643520989,
-0.916718557142268,
0.20086626876005667,
-0.3711023890817823,
-0.006962349053602254,
2.687031223814704,
0.3970791704566904,
-0.6890591654410289,
1.0504536719091986,
-1.2458238157294268,
-1.197330311624414,
-0.12155955689525438,
-0.2969718406476498,
0.15406921803787998,
0.9984735505197111,
-0.2734456365103056,
0.0100494123830882,
0.16172873483083283,
-0.6780409734277772,
0.4539997995464368,
0.7558576390626953,
0.9556605206819043,
-0.10572075101403118,
1.5564628686733721,
0.05397552837650759,
0.7415987024402694,
-0.1648500682327342,
1.052699139325594,
0.6776572023171561,
0.15219630541397192,
0.8843933196559655,
1.3948099389813178,
-0.16710315713046545,
1.2199200746863228,
-0.3825678575328573,
-0.26766985045935626,
2.5777182352113908,
0.23156136798610805,
-0.3377548185037014,
1.0689111835439342,
-0.17216022970038714,
-0.29381807805976945,
-0.5689533077540762,
0.30596827309967156,
0.5260993417881752,
-0.6768480303494689,
-0.1092430795457648,
1.5442082488768039,
-0.10826504240549732,
-0.9555598065181787,
0.5334835330029429,
0.5569227134357198,
1.2665483894271157,
-1.9677332823608626,
1.5559931182978979,
0.7709453943956464,
1.5746506000831744],
[1.7625089228179507e-05,
1.298540182320117,
1.3141870420194457,
1.3305471045820527,
2.0686735179487985,
2.23810597518972,
2.267373737425337,
2.295441377067579,
2.3325388050051976,
2.352484360014207,

2.370891483796666,
3.3117124844820194,
3.363605896627787,
3.858981460194034,
3.8589814603490344,
3.859537867870235,
3.8595381713444072,
4.26458571773699,
4.264585717738593,
6.599549149586233,
6.655963438031886,
6.939612414177192,
6.939612414268938,
7.004302986969205,
7.010544991093791,
7.97314828104894,
7.999280802101656,
8.022628296732579,
8.027896471579398,
10.077254342713585,
-1.0733166032270363,
0.4264387306682,
-0.29490479134596703,
-0.45181113351566254,
-0.9496712390342118,
-0.24412788786515252,
-0.17005038539710493,
2.131747001055919,
-0.8801477965160032,
0.1985900676052623,
-0.17829180427317373,
0.1513896817808816,
-0.2996418627583153,
0.09152392337611555,
1.0155053498240294,
0.6713280773462803,
-0.5915672184524843,
-0.21780814909773427,
-0.23676595566314207,
-0.1539223229682568,
0.9733737893230769,
-0.8325690111887051,
-0.6684349693104433,
0.944923850675747,
-0.8270803600282979,
0.4869668407293288,
0.07342964610078141,
-0.8928903181234079,
-0.04262248204617692,
-0.7602498290394286,
-0.48858825463899247,
1.7340528561515622,
0.7340502315717261,
0.5163401187759543,
-0.5015858040228415,
0.2890335055344555,
-0.46033506221820203,
-0.8185907107929625,
0.4557079379705311,
1.6601520871164626,
0.07125306723268252,
-1.4035119795986863,
-0.6741575848895364,
1.84360490583597,
1.6836338044245691,
-0.33329370458317326,
0.19854317161902654,
2.03917897859608,
-0.3342251938073038,
-1.6015864477635742,
0.35422488479060155,
0.1487121545506934,
-0.2821114888507261,
0.21869934797564045,
0.7523961573071125,

1.3021785537923125,
0.26184859002187644,
1.2441238228144538,
0.039670539596101485,
-0.15687777352236798,
1.1386642271664005,
-0.13559116971858365,
0.39167499360215746,
2.4453261619465274,
-1.0890320328395309,
-0.4289422005081893,
1.425249318116838,
-0.0016733698174337221,
0.023121567661497415,
-0.28325087531692295,
0.940069260590291,
1.163437407349777,
-0.4256527399896147,
-1.4972731615803316,
-0.12624975701746496,
-0.10144651691714318,
1.3307378561466523,
0.041778278155808674,
-1.2746374184036902,
-1.2216848741140482,
1.4048769664929757,
0.8957112974734468,
1.0058068981677006,
-3.0965653057688316,
-0.3344448337959029,
-1.0948102803209163,
-0.038478236121346335,
0.12618687504486675,
-0.0850775611793361,
1.33550885346982,
-0.04334772196309596,
0.2406908654957122,
0.7327249608913058,
0.9966290850109706,
2.2852565843192423,
-0.030570948934034583,
-0.5566407461361556,
0.813246653395058,
0.5946355273642772,
0.07519254658407079,
0.27120989268694123,
-0.15415722381241237,
-1.1239288818585278,
1.893751281002709,
0.7530205722182245,
1.4454588921082565,
0.15037300343080462,
0.1793867334099716,
0.3587259643057371,
-0.006799349268608,
1.0907160355013774,
-0.30186717991666434,
0.8507589785411056,
0.009155709268365956,
-0.8165113788656561,
0.011816515180360782,
1.116430418883973,
1.5343789763879045,
-0.1682835166498154,
-1.5032827361947747],
[0.10956969985098665,
0.4360115975982133,
0.5932904704915389,
0.6050451872391112,
1.3688142364348175,
1.3737597367805747,
1.5226773273508583,
1.5589271816733294,
3.150320547140947,
3.1613053536766147,

4.7635234433865605,
4.823148382609575,
4.823188243925802,
7.171157860626202,
7.171157878485542,
7.171158529865571,
7.175073431930883,
7.183070822592498,
8.102400589897519,
8.10392263292654,
8.103940780725882,
9.021308924987686,
9.02133947231205,
9.082831297490506,
9.094868565409332,
9.236804949782064,
9.237029573386183,
9.26430291432718,
10.00744744746223,
10.032040374677464,
-0.5042382729215367,
1.0180015436638659,
0.10423420212375085,
-0.3355687578372037,
-0.3902319222280216,
-0.09867602213718178,
0.3860296418386605,
0.8522828003851446,
-0.7730596965193267,
0.08803782521712378,
1.175277695996097,
-0.3769225558693855,
-0.4553519820435133,
0.4758962487375975,
-1.0517577693418567,
0.3712029741888282,
0.27334544561774066,
0.03397030216872572,
0.3891396532807883,
0.12152393226873999,
1.5192516885791116,
-0.5124931636551364,
-0.5667003913709334,
-0.08425393582345661,
-0.8152956002657638,
-0.008087141654348068,
-0.20833380477584987,
-0.19186344764752156,
0.309463849209865,
-0.16555810903618162,
-0.28534485935787846,
0.025710068868521854,
-0.9172960495760906,
0.027100436879331523,
-0.9806057417125867,
0.46877133554227174,
0.22597640762440013,
-1.8700902599315918,
-0.6222133609404971,
0.023922176634194332,
-0.6777524378245746,
-0.3883818313368169,
-0.19921738892417568,
-1.3418496115619054,
-0.05365669225714106,
0.38835985529462186,
0.46741217514915123,
-0.1388983196762152,
1.0002905156718758,
1.6633296566254185,
-1.0097136763815113,
1.203957772036831,
0.34033273848251194,
1.1456357751380877,
-0.4637199651146574,

-0.13624703361320328,
0.6785440200843139,
0.21691345067474185,
-0.4166036808107477,
1.6003722740710606,
-0.294881370724797,
0.04198878790454511,
-0.689532047804753,
-1.3911901894570817,
-1.1667039833298627,
0.824824800006074,
0.8047687705752488,
0.5727196814509768,
0.2385364014798047,
-0.512164832149354,
0.17925141295137917,
-0.19704824673887428,
-1.6372757558452697,
0.5441675736491897,
-0.13936070677778437,
1.3935144064781402,
0.10763856884501717,
0.6679788244416961,
-0.33118654939932857,
-0.35796695002153883,
0.2632208027041665,
1.5831503864671255,
-0.25140146740224,
1.135774876905931,
0.5897962040478675,
-0.7664904031063712,
1.092649076983624,
-0.9862462517151122,
0.010216869101717933,
0.5705833807099128,
0.09839127014019662,
-1.034770130526523,
0.20582133073730807,
-0.44428721001967564,
0.5272444819968604,
-0.2627660867388947,
-0.9857421779520247,
-1.3646538516413942,
0.14779052759756187,
1.1207932022043874,
0.30427416833245535,
-1.055109005337564,
0.30548258474769546,
0.8153937738463674,
0.5082537250510629,
-0.2973639305797175,
0.0034742209991892575,
1.5168338311712208,
0.0971904670135119,
0.5305672799342332,
-0.08170159551874663,
0.4909203029282298,
-1.1771354207712392,
0.038725619801271274,
-0.6655680746562402,
1.312626970471223,
-1.0115568894576918,
0.16290676730453105,
0.3074286838975462,
0.013117345918778788],
[0.4866654913190249,
0.4866654913190249,
1.4314116389487666,
1.4314116389487666,
1.548608655381947,
1.5552664901547235,
1.662920497545568,
1.676155881591226,
1.6952532774325564,
1.6952532774325564,

2.46741851740979,
2.50102877426778,
3.636145431120801,
3.6361454311208004,
3.6361454311208004,
3.6361454311208004,
3.6361454311208004,
4.0161273218482085,
4.0161273218482085,
4.5411636896776235,
4.5411636896776235,
4.922545565757144,
5.624338470516907,
5.642847901355697,
5.904037934138086,
5.915382265322609,
6.860930229950661,
6.860930229950661,
9.422517992567101,
9.467623480195014,
9.467623480195014,
0.40091998571272736,
-0.3581846438287994,
-0.32033246941904975,
0.24409866553117068,
0.176560178011825,
0.4452285262890565,
-0.37150326481454765,
-1.0185408309777293,
1.6197684873006286,
2.0095194673452026,
-0.20342804182482532,
-0.4471079471088122,
0.37127155242298737,
0.4199607045713712,
0.11145558426636207,
-0.289968410859463,
0.9752004498173846,
-0.11972987215087512,
-0.8215210043660294,
-0.4144105549344152,
0.04966095047933562,
0.6091224412599766,
-0.2481186610805342,
-0.9036735819055022,
0.22922493224752044,
-0.04567884568114995,
-0.40529506738584414,
-0.44609185937563545,
1.1000542207972743,
0.8995834380092431,
-0.4391184290327733,
-1.623814742575614,
0.11121717167805684,
-0.15344289124808663,
0.7072346226342329,
0.013911394934765222,
1.1008239636292285,
1.1342299046285917,
-0.1605718532112884,
-0.8521154510350495,
-0.9726520318623231,
-0.11078467990736285,
1.2893067940177063,
-0.8656847782514395,
-0.4554157535410911,
0.46752563535079134,
1.0115326820592876,
-0.017465678177200683,
0.7555243091795097,
0.2225241128439446,
-0.6129237770264853,
0.36249038141661666,
-0.1470816778834455,
-0.23781939745964606,
0.7410640700988803,

```

-0.10219672891678527,
-0.9043863030697507,
-0.054446492384892985,
-0.23618927113386157,
-0.3971482428910102,
0.42980979959963467,
-0.4511512430809565,
-0.3322123782853245,
0.15031459204805087,
-1.4287283715133035,
-0.026859055017604255,
0.20826999854501216,
-0.016397767746330064,
-0.026711091547853946,
0.15899307013846248,
0.18349959987180348,
-0.019181103141429404,
-1.3777401647916119,
0.05578672583947999,
-0.17191397249005735,
0.08492618339903452,
0.2781511329827625,
1.6853098582374997,
-0.7964285169174421,
0.13628687772581302,
-0.26524580358677924,
1.3615146942088054,
-0.18163849035743365,
-1.5476669195398023,
0.2740670870741208,
0.19740142577560438,
0.18567840551061798,
0.43288784764736754,
-0.8069341399605817,
1.5139336080928714,
-0.7594440385529729,
-0.20443587392639553,
0.0916888700165231,
-0.03922249489745327,
0.4135801157921025,
-0.06731598775715468,
0.5038720973518589,
-0.7705715736697758,
-1.4775324565516457,
0.061950004909204714,
-0.9555167624183505,
3.1277455162888304,
-1.137887484425651,
-0.09852947784569874,
0.042597844143281625,
0.06134264537966588,
0.3384848933516047,
0.012002169875906457,
1.5732417764972468,
-0.3550969359952667,
0.37821230776206083,
1.4914211446267915,
-1.2027836619303884,
0.22421848814312553,
-0.2072711571643093,
0.05122655754886873,
-1.5578500765322636,
1.5918641150141506,
1.2635397829742223,
-1.2452354025631063]],
'run_time_sec': 43720.81102514267}}],
'redundant': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],

```

```
                                'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                'r': 0},
                                'time': {'max': 10.0, 'min': 0.0},
                                'type': 'xy-iss'}},
'init_param': { 'value': [ 0.2383708143795501,
                             0.8137820253985804,
                             1.0561141201801563,
                             1.2070683478267785,
                             1.9518166906091101,
                             2.08301823027678,
                             2.6774459967535513,
                             2.7270096866449576,
                             3.7369434259122447,
                             3.7421546540398443,
                             4.458327272130387,
                             4.463102384700506,
                             4.463102384700506,
                             4.482690585412051,
                             4.482694652744643,
                             4.498978826283554,
                             6.3580266266370415,
                             7.886395999997723,
                             7.886532810158634,
                             8.060529572035655,
                             8.06234783983503,
                             8.127555772505229,
                             8.130166630032164,
                             8.42408407139234,
                             8.433074307328475,
                             9.412401860498063,
                             9.420584174895616,
                             9.429609653004679,
                             9.434851407347677,
                             11.283524698808058,
                             0.19157417616433814,
                             0.42257952171340063,
                             0.3129095694283805,
                             0.28360255750055535,
                             -0.7229159635345903,
                             2.545731433831817,
                             1.1687892956196542,
                             0.5222949375721274,
                             0.9482151911772319,
                             0.28742832079435787,
                             1.1627391357840349,
                             -0.27939056132165146,
                             -0.07574334812551493,
                             -0.529119211391362,
                             -0.08653083237786212,
                             0.042544229108731915,
                             -0.09154582517805988,
                             0.2776265525620834,
                             0.373637405241428,
                             -0.666296500917012,
                             -0.729825790392568,
                             0.36164641825124294,
                             1.2066181407380305,
                             0.18327070025757417,
                             2.2757552060408917,
                             -0.004789158452516104,
                             0.49019439516235463,
                             2.190622553796523,
                             0.6131206832540379,
                             1.0987721265230228,
                             1.9413219378309259,
                             0.07975169469268534,
                             0.17531082109327745,
                             0.01770145552231998,
                             0.22429808435825893,
                             1.1460098869067579,
```


-0.25766861736092417,
0.4386465248337572,
0.7521614434169624,
0.07039918338308593,
-0.23913727928604284,
0.062497116873310714,
0.27575365987954265,
-0.8268952220573135,
0.8599765782432734,
0.6958325703328606,
-0.16792542378671177,
-0.13186811496768225,
0.17745018848846658,
-0.13764108606901534,
-0.34798347931108986,
0.3901205214779397,
-0.12832331396189006,
0.7623260417931323,
-0.08745728181826747,
0.22448567111633413,
0.5103689813486589,
0.020179740460913417,
-0.4639048953982949,
1.8052783293547816,
-0.06895296635033099,
-0.46137869531525183,
-0.7694587159951923,
-0.1507074275764489,
-3.159084078074864,
1.5556721360426484,
0.9802025173412374,
-1.5980562353917622,
0.9974678614222413,
0.11965749551778265,
-0.5964068365118492,
1.00912773091807,
1.2674306548806673,
1.7418320010188646,
-1.2223756485879664,
-0.006270145144671745,
0.7863824204489233,
-0.08410721186616651,
0.13420385894759826,
0.6686584151585806,
-0.7946689395347296,
-1.448768973512125,
-0.4158373205099576,
-0.1003664997933048,
0.5370971852976741,
-0.013369261455621718,
-1.2134070474127567,
2.71822204346588,
-0.8266087910498845,
-0.04274152999618493,
0.627725493238542,
-0.10714625448592663,
1.4055199595750496,
-0.08877591733311402,
-0.6400618055168635,
0.27990653766167567,
0.27230874688701184,
1.8329367216853043,
0.12128377229042309,
0.03719209379786179,
0.7558193864298689,
-0.027085062677120984,
0.7947118159580207,
-0.18810209431195668,
0.008294184744538372,
-0.9110921816475059,
1.3065108206329406,
0.004810940817056788,
1.4030602129547465,
0.03878633149748846,
0.29850504257680605,

```

-0.9524583070906825,
0.10037229527914636,
0.4277056649647598,
-2.2528322583674503,
0.030724614514628212,
0.8416350822543037,
1.5579013461800135,
0.5696673724928452,
-1.5352194543526818]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_xy_noisefree_time_evo_ric4_unifrom_scaling_sample#4'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z '
                                '1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + '
                                '1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '
                                '6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + '
                                                         '0.5 Z 0 Z 1 + '
                                                         '0.5 X 1 X 2 + '
                                                         '0.5 Z 1 Z 2 + '
                                                         '0.5 X 2 X 3 + '
                                                         '0.5 Z 2 Z 3 + '
                                                         '0.5 X 3 X 4 + '
                                                         '0.5 Z 3 Z 4 + '
                                                         '0.5 X 4 X 5 + '
                                                         '0.5 Z 4 Z 5 + '
                                                         '0.5 X 5 X 6 + '
                                                         '0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -7.418732841657523],
                              [12, 6, 0, 0, -5.8155731075176975],
                              [20, 10, 0, 0, -4.600391600363878],
                              [28, 14, 0, 0, -3.6713467271615334]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                     'identity_factors': [0, 0, 0, 0],
                     'noise_level': [4, 2, 0, 0],
                     'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},

```

```

0.001,
0.001,

'noise_prob': [
0,
0],

'status': True,
'type':

'depolarizing'},

'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': {

'noise_prob': [
0,
0],

'status': True,
'type':

'depolarizing'},

'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
'identity_factors': [2, 2, 0, 0],
'noise_level': [20, 10, 0, 0],
'noise_profile': {

'noise_prob': [
0,
0],

'status': True,
'type':

'depolarizing'},

'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
'identity_factors': [3, 3, 0, 0],
'noise_level': [28, 14, 0, 0],
'noise_profile': {

'noise_prob': [
0,
0],

'status': True,
'type':

'depolarizing'},

'odd_wires': 4}],
'run_time_sec': 10.84847092628479}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.3442332088020621,
0.7247295953616149,

```

0.7247295953616149,
0.7610537291022658,
1.7162580174489597,
2.0119282772467635,
2.0119282772467635,
3.369934702704019,
3.369934702704019,
5.00211845115155,
5.212181786011135,
5.346566249705437,
6.103091648910964,
6.103091648910964,
7.050711873868975,
7.050711873868975,
7.492144583370162,
7.492144583370162,
7.606105711064038,
7.606105711064038,
8.717193051662631,
8.753101318062777,
8.877107151002964,
8.889590522678484,
8.982812044989718,
8.982812044989718,
9.499874357008034,
9.499874357008034,
9.659025604148875,
9.671110590806489,
0.02814413440270558,
0.250593445582179,
0.17496868989230024,
-0.19749991731592192,
-0.5878125517240043,
1.1370895496823794,
0.3339578948727643,
0.30833012680275795,
0.21147160367429677,
0.5518340573364652,
0.9816766119937317,
-1.3427949214182668,
2.2864376119784486,
0.673664283367827,
0.4767745045632036,
-0.47480445681581174,
0.05912449937800317,
0.9513757199218045,
1.079093875588678,
-0.4313388366547843,
0.43347513808131466,
-0.3381632209207048,
-0.42699598389932686,
-1.1528003206699722,
0.2505681368756031,
1.1900947118566725,
-0.38260053073473543,
-0.6405385159573362,
0.08926968472452403,
-0.04834256297702397,
1.603027736914297,
-0.44672812120859223,
-0.5759172165575565,
1.25237714334681,
0.04254918814060746,
-0.002429292667880943,
0.47732275422632153,
-0.10204026491637587,
0.02357869435971368,
0.06910865492863566,
-0.5942663805386644,
1.1629666490831163,
0.28625726751699415,
0.4183193427753825,
0.33026699893767814,
1.8860632823719743,
-0.37929215782635534,


```

        'noise_prob': [0.001, 0.001, 0, 0],
        'status': True,
        'type': 'depolarizing'},
    'nqubits': 7,
    'observable': { 'coefficients': { 'bn': [ 1.0,
        1.0,
        1.0,
        1.0,
        1.0,
        1.0,
        1.0],
        'cn': [ 0.5,
        0.5,
        0.5,
        0.5,
        0.5,
        0.5],
        'r': 1},
        'def': 'ising'},
    'output': { 'draw': { 'fig_dpi': 100,
        'status': False,
        'type': 'png'},
        'file_name_prefix':
'AUTOMATE_xy_noisefree_time_evo_ric4_unifrom_scaling_sample#5'},
    'redundant': { 'identity_factors': [ [0, 0, 0, 0],
        [1, 1, 0, 0],
        [2, 2, 0, 0],
        [3, 3, 0, 0]]},
    'run': 'redundant',
    'state': 'dmatrix',
    'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
        'constraint': True,
        'status': False}},
    'zne': { 'data_points': None,
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
    'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z '
        '1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + '
        '1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '
        '6 + 1 Z 5 + 1 Z 6',
        'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + '
        '0.5 Z 0 Z 1 + '
        '0.5 X 1 X 2 + '
        '0.5 Z 1 Z 2 + '
        '0.5 X 2 X 3 + '
        '0.5 Z 2 Z 3 + '
        '0.5 X 3 X 4 + '
        '0.5 Z 3 Z 4 + '
        '0.5 X 4 X 5 + '
        '0.5 Z 4 Z 5 + '
        '0.5 X 5 X 6 + '
        '0.5 Z 5 Z 6']},
    'output': { 'data_points': [ [4, 2, 0, 0, -7.407206817517893],
        [12, 6, 0, 0, -5.829141582765497],
        [20, 10, 0, 0, -4.618573032874505],
        [28, 14, 0, 0, -3.681441154365551]],
        'noise_details': [ { 'gates_num': [4, 1, 1, 0],
        'identity_factors': [0, 0, 0, 0],
        'noise_level': [4, 2, 0, 0],
        'noise_profile': {
'noise_on_init_param': { 'status': False,
        'value': 0},
        'noise_prob': [
0.001,
0.001,
0,
0],
        'status': True,
        'type':
'depolarizing'},
        'odd_wires': 4},
        { 'gates_num': [12, 3, 1, 0],
        'identity_factors': [1, 1, 0, 0],

```

```

'noise_level': [12, 6, 0, 0],
'noise_profile': {

'noise_prob': [

0,
0],

'status': True,
'type':

'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
'identity_factors': [2, 2, 0, 0],
'noise_level': [20, 10, 0, 0],
'noise_profile': {

'noise_prob': [

0,
0],

'status': True,
'type':

'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
'identity_factors': [3, 3, 0, 0],
'noise_level': [28, 14, 0, 0],
'noise_profile': {

'noise_prob': [

0,
0],

'status': True,
'type':

'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
'identity_factors': [3, 3, 0, 0],
'noise_level': [28, 14, 0, 0],
'noise_profile': {

'noise_prob': [

0,
0],

'status': True,
'type':

'odd_wires': 4}],
'run_time_sec': 11.273851871490479}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.17961742860022237,
0.4029745783400613,
0.4271020388774394,
0.6998927493507625,
1.1444327445959492,
1.4463150894355827,
1.7859669797147917,
1.7859669797147917,
1.8910076260348616,
2.1568466639380097,
2.184715675260106,
2.3241261579565067,
5.221288254664872,

```

5.360678146313719,
5.572749542622228,
6.315664711217606,
6.332942041057151,
7.225695385348529,
7.238714694082236,
7.767863290353532,
7.947954629935657,
8.045079231148478,
8.080705184182557,
8.235317994608641,
8.52967785406855,
8.920755162601605,
8.920755162601605,
8.920755162601605,
9.75907480462672,
11.068402571583377,
-0.07026370230905016,
1.2758275954101812,
0.1960569796315372,
-0.7454998985427054,
0.5260870873255556,
1.3632265149428353,
0.06759709485075166,
0.7201883678889174,
0.7780364770341424,
0.7034018811814728,
-0.3281225120527201,
-0.21523884970562404,
0.3525948786581971,
0.10058439953616755,
0.8489242437750127,
0.4102210250930649,
-1.1239274808373552,
-0.07527028057729333,
-0.662823747816413,
-0.09016589559770041,
0.0005078927131911535,
-0.9974030724775158,
1.2973260682530905,
0.6207513445119947,
-0.06959715000945968,
0.01479447508324883,
0.5707366547040263,
0.8024230791007029,
0.11580671533830131,
0.5319238260215059,
2.029703693625361,
0.8476591202353092,
-0.5985490132202251,
-1.0850372047073247,
-0.13885829158476937,
0.6160884575951602,
-0.6119315652443118,
0.25456754270344084,
1.6135141314150674,
1.6101080474118592,
-1.0046888465114787,
-1.2174288348798858,
-0.8801790341822278,
-0.01998793431745051,
-1.007194981866689,
-1.6365882266166358,
3.524880820008825,
1.5399950054830251,
-1.401820445976074,
-0.04945776017633801,
0.9565666812114323,
-0.20490442564626934,
-0.5632271134618189,
1.0033608730635488,
0.12368927199762578,
-0.06153617179775209,
1.0557374829951787,
1.532632985657107,

[illegible]

```

'cn': [ 0.5,
        0.5,
        0.5,
        0.5,
        0.5,
        0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_xy_noisefree_time_evo_ric4_unifrom_scaling_sample#9'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z '
                                '1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + '
                                '1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '
                                '6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + '
                                                         '0.5 Z 0 Z 1 + '
                                                         '0.5 X 1 X 2 + '
                                                         '0.5 Z 1 Z 2 + '
                                                         '0.5 X 2 X 3 + '
                                                         '0.5 Z 2 Z 3 + '
                                                         '0.5 X 3 X 4 + '
                                                         '0.5 Z 3 Z 4 + '
                                                         '0.5 X 4 X 5 + '
                                                         '0.5 Z 4 Z 5 + '
                                                         '0.5 X 5 X 6 + '
                                                         '0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -7.395384341095474],
                             [12, 6, 0, 0, -5.790856073500346],
                             [20, 10, 0, 0, -4.565524427603155],
                             [28, 14, 0, 0, -3.622528542539368]],
            'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                'identity_factors': [0, 0, 0, 0],
                                'noise_level': [4, 2, 0, 0],
                                'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},

'noise_prob': [
0.001,
0.001,

0,
0],

'status': True,
'type':

'depolarizing'},

'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [1, 1, 0, 0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},

'noise_prob': [
0.001,
0.001,

0,
0],

'status': True,
'type':

```

```
'depolarizing'},
```

```
'init_param': { 'value': [ 0.5482907189468895,  
0.6413620809492503,  
0.6422482149552934,  
1.5892409709442974,  
1.850708229814058,  
1.850708229814058,  
1.8745552660050262,  
2.400679867939261,  
2.402212278134192,  
2.485190345183722,  
2.485190345183722,  
2.4851903451837223,  
2.488397493491329,  
2.4883974934913287,  
5.077476636438464,  
5.077476636438464,  
5.732990197214016,  
5.806868622878696,  
5.817834245705687,  
6.166757373752113,  
7.3114292874931905,  
7.311858449036949,  
8.346734585466223,  
8.350435170229554,
```

8.354614694314481,
8.35659817980371,
8.356905245901928,
9.313534396314816,
9.313534396314816,
10.443787047737276,
-0.5971318801423705,
-0.6830583120471296,
0.053446928626708114,
-1.488251087277441,
0.7409998345617078,
-0.020561007687691084,
-0.47029189134787985,
-0.9187802118512801,
-0.07390570205801456,
0.6091612085770479,
0.5696050095749661,
-2.634874245900948,
0.14733366015521565,
1.0781100436188797,
0.3459269460772517,
1.2820636645787713,
-1.605623121104658,
0.856791024529966,
0.16374864437029604,
-0.7377840957855112,
0.7603752030179747,
0.4958590077418653,
1.0293890027768824,
0.4862585296205842,
-1.0838654128292502,
1.2632637762941026,
-1.1043454462318527,
-0.7867654190654241,
0.8818217647037212,
1.0406267750063978,
0.05770567519039498,
-0.5922070722918433,
-0.3271403288282926,
0.5980786503295974,
-0.02050375112330311,
-0.9007241609880184,
0.5239405469453117,
0.7601496703420288,
-0.6449049942439851,
1.7340858429103823,
0.10050113333288421,
0.3143347863780438,
0.28408386490618526,
-0.1222389867527315,
-0.09502282329152913,
-1.3004368662949968,
0.16856179320059875,
0.6790103402211991,
0.3355977478652572,
0.0935957266804422,
0.42936820813464754,
0.5148227650107633,
0.10764356588532305,
-0.054251737808208085,
0.06525040933822983,
-0.21143245853075732,
-1.5801988491404158,
0.023513208971767067,
1.0921277689034055,
-0.11501776583679489,
-0.14182911234947893,
1.8716225858725093,
-1.0747761250482988,
-0.01873186129021693,
0.9952348515577564,
1.5589645169221944,
2.4625123021717044,
1.5320934484141422,
0.6264578101560674,

```

0.015802850908690148,
0.6559295719723718,
0.5030706883609005,
0.010357532831659657,
-0.6971755064049523,
-1.334963536555068,
0.05714132580252865,
0.9478292633237722,
1.541635767019547,
-1.2357775558070643,
1.5753066317572375,
1.2206753513411082,
-0.013882574638756213,
2.102115305768464,
0.6867648509985108,
-1.0810617894892982,
0.608205793929093,
-1.0701207529796843,
-0.008804617073740417,
1.0802724219136723,
1.4472704941581649,
-0.525983160453369,
0.10347757085774918,
0.07451664808136123,
0.8113743025574697,
-0.19830571509994802,
0.20760359505818776,
-0.5709476706639499,
0.1791350390980861,
-0.32061652775092425,
-1.6382844876022806,
2.271046687924862,
0.09972766177435569,
1.671157305178672,
1.0596955078067567,
-0.7395801076185267,
-0.8217251410877643,
1.2660358026872456,
-0.149859182069558,
1.7386861885389688,
-0.003505497297480453,
0.677126408740769,
-1.0541684899696007,
0.9437251450981726,
1.7354017388586316,
-0.8022872949930118,
0.01881044769021433,
-0.013699411839369284,
1.5965466699123847,
0.4486397166383021,
1.5669137279484207]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},

```

```

'file_name_prefix':
'AUTOMATE_xy_noisefree_time_evo_ric4_unifrom_scaling_sample#1'},
    'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                           [1, 1, 0, 0],
                                           [2, 2, 0, 0],
                                           [3, 3, 0, 0]]},

    'run': 'redundant',
    'state': 'dmatrix',
    'vqe': { 'iteration': 1,
             'optimization': { 'algorithm': 'SLSQP',
                               'constraint': True,
                               'status': False}},

    'zne': { 'data_points': None,
             'degree': 0,
             'method': 'richardson',
             'sampling': 'default'}},
    'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z '
                                     '1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + '
                                     '1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '
                                     '6 + 1 Z 5 + 1 Z 6',
               'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + '
                                                           '0.5 Z 0 Z 1 + '
                                                           '0.5 X 1 X 2 + '
                                                           '0.5 Z 1 Z 2 + '
                                                           '0.5 X 2 X 3 + '
                                                           '0.5 Z 2 Z 3 + '
                                                           '0.5 X 3 X 4 + '
                                                           '0.5 Z 3 Z 4 + '
                                                           '0.5 X 4 X 5 + '
                                                           '0.5 Z 4 Z 5 + '
                                                           '0.5 X 5 X 6 + '
                                                           '0.5 Z 5 Z 6']},

    'output': { 'data_points': [ [4, 2, 0, 0, -7.463351332654992],
                                  [12, 6, 0, 0, -5.841676652164559],
                                  [20, 10, 0, 0, -4.604434423168805],
                                  [28, 14, 0, 0, -3.651723685315627]],
              'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                   'identity_factors': [0, 0, 0, 0],
                                   'noise_level': [4, 2, 0, 0],
                                   'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0,
                                0],

                                'status': True,
                                'type':

'depolarizing'},

                                'odd_wires': 4},
                              { 'gates_num': [12, 3, 1, 0],
                                'identity_factors': [1, 1, 0, 0],
                                'noise_level': [12, 6, 0, 0],
                                'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0,
                                0],

                                'status': True,
                                'type':

'depolarizing'},

                                'odd_wires': 4},
                              { 'gates_num': [20, 5, 1, 0],
                                'identity_factors': [2, 2, 0, 0],
                                'noise_level': [20, 10, 0, 0],
                                'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0,
                                0],

                                'status': True,
                                'type':

'depolarizing'},

                                'odd_wires': 4},
                              { 'gates_num': [28, 7, 1, 0],
                                'identity_factors': [3, 3, 0, 0],
                                'noise_level': [28, 14, 0, 0],
                                'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0,
                                0],

                                'status': True,
                                'type':

'depolarizing'},

                                'odd_wires': 4},
                              { 'gates_num': [36, 9, 1, 0],
                                'identity_factors': [4, 4, 0, 0],
                                'noise_level': [36, 18, 0, 0],
                                'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0,
                                0],

                                'status': True,
                                'type':

'depolarizing'},

                                'odd_wires': 4},
                              { 'gates_num': [44, 11, 1, 0],
                                'identity_factors': [5, 5, 0, 0],
                                'noise_level': [44, 22, 0, 0],
                                'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0,
                                0],

                                'status': True,
                                'type':

'depolarizing'},

                                'odd_wires': 4},
                              { 'gates_num': [52, 13, 1, 0],
                                'identity_factors': [6, 6, 0, 0],
                                'noise_level': [52, 26, 0, 0],
                                'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0,
                                0],

                                'status': True,
                                'type':

'depolarizing'},

                                'odd_wires': 4},
                              { 'gates_num': [60, 15, 1, 0],
                                'identity_factors': [7, 7, 0, 0],
                                'noise_level': [60, 30, 0, 0],
                                'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0,
                                0],

                                'status': True,
                                'type':

'depolarizing'},

                                'odd_wires': 4},
                              { 'gates_num': [68, 17, 1, 0],
                                'identity_factors': [8, 8, 0, 0],
                                'noise_level': [68, 34, 0, 0],
                                'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0,
                                0],

                                'status': True,
                                'type':

'depolarizing'},

                                'odd_wires': 4},
                              { 'gates_num': [76, 19, 1, 0],
                                'identity_factors': [9, 9, 0, 0],
                                'noise_level': [76, 38, 0, 0],
                                'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0,
                                0],

                                'status': True,
                                'type':

'depolarizing'},

                                'odd_wires': 4},
                              { 'gates_num': [84, 21, 1, 0],
                                'identity_factors': [10, 10, 0, 0],
                                'noise_level': [84, 42, 0, 0],
                                'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0,
                                0],

                                'status': True,
                                'type':

'depolarizing'},

                                'odd_wires': 4},
                              { 'gates_num': [92, 23, 1, 0],
                                'identity_factors': [11, 11, 0, 0],
                                'noise_level': [92, 46, 0, 0],
                                'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0,
                                0],

                                'status': True,
                                'type':

'depolarizing'},

                                'odd_wires': 4},
                              { 'gates_num': [100, 25, 1, 0],
                                'identity_factors': [12, 12, 0, 0],
                                'noise_level': [100, 50, 0, 0],
                                'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0,
                                0],

                                'status': True,
                                'type':

'depolarizing'},

                                'odd_wires': 4},
                              { 'gates_num': [108, 27, 1, 0],
                                'identity_factors': [13, 13, 0, 0],
                                'noise_level': [108, 54, 0, 0],
                                'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0,
                                0],

                                'status': True,
                                'type':

'depolarizing'},

                                'odd_wires': 4},
                              { 'gates_num': [116, 29, 1, 0],
                                'identity_factors': [14, 14, 0, 0],
                                'noise_level': [116, 58, 0, 0],
                                'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0,
                                0],

                                'status': True,
                                'type':

'depolarizing'},

                                'odd_wires': 4},
                              { 'gates_num': [124, 31, 1, 0],
                                'identity_factors': [15, 15, 0, 0],
                                'noise_level': [124, 62, 0, 0],
                                'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0,
                                0],

                                'status': True,
                                'type':

'depolarizing'},

                                'odd_wires': 4},
                              { 'gates_num': [132, 33, 1, 0],
                                'identity_factors': [16, 16, 0, 0],
                                'noise_level': [132, 66, 0, 0],
                                'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0,
                                0],

                                'status': True,
                                'type':

'depolarizing'},

                                'odd_wires': 4},
                              { 'gates_num': [140, 35, 1, 0],
                                'identity_factors': [17, 17, 0, 0],
                                'noise_level': [140, 70, 0, 0],
                                'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0,
                                0],

                                'status': True,
                                'type':

'depolarizing'},

                                'odd_wires': 4},
                              { 'gates_num': [148, 37, 1, 0],
                                'identity_factors': [18, 18, 0, 0],
                                'noise_level': [148, 74, 0, 0],
                                'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0,
                                0],

                                'status': True,
                                'type':

'depolarizing'},

                                'odd_wires': 4},
                              { 'gates_num': [156, 39, 1, 0],
                                'identity_factors': [19, 19, 0, 0],
                                'noise_level': [156, 78, 0, 0],
                                'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0,
                                0],

                                'status': True,
                                'type':

'depolarizing'},

                                'odd_wires': 4},
                              { 'gates_num': [164, 41, 1, 0],
                                'identity_factors': [20, 20, 0, 0],
                                'noise_level': [164, 82, 0, 0],
                                'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0,
                                0],

                                'status': True,
                                'type':

'depolarizing'},

                                'odd_wires': 4},
                              { 'gates_num': [172, 43, 1, 0],
                                'identity_factors': [21, 21, 0, 0],
                                'noise_level': [172, 86, 0, 0],
                                'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0,
                                0],

                                'status': True,
                                'type':

'depolarizing'},

                                'odd_wires': 4},
                              { 'gates_num': [180, 45, 1, 0],
                                'identity_factors': [22, 22, 0, 0],
                                'noise_level': [180, 90, 0, 0],
                                'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0,
                                0],

                                'status': True,
                                'type':

'depolarizing'},

                                'odd_wires': 4},
                              { 'gates_num': [188, 47, 1, 0],
                                'identity_factors': [
```

```

0,
0],
'status': True,
'type':
'depolarizing'},

'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
'identity_factors': [3, 3, 0, 0],
'noise_level': [28, 14, 0, 0],
'noise_profile': {

'noise_prob': [

0,
0],
'status': True,
'type':
'depolarizing'},

'odd_wires': 4}},
'run_time_sec': 9.285445213317871}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.5698358581496117,
1.0111463732621893,
1.116254023022104,
1.116254023022104,
1.4543153683590362,
1.4543153683590362,
1.551144943675129,
1.551144943675129,
2.3321866923810566,
2.4857852178760083,
3.837593499897418,
3.837593499897418,
3.837593499897418,
3.837593499897418,
4.847467397840062,
5.03660309473088,
5.176578602594472,
5.176578602594472,
5.214332760543968,
6.547255537065404,
6.547255537065404,
7.083633091144979,
7.083633091144979,
7.133204014188821,
7.133204014188821,
7.133204014188821,
7.937225733864811,
8.370661235857398,
8.62656679928389,
8.759707620277526,
9.424598849529344,
0.24247733923543782,
0.79275118507479,
-0.012314768690710378,
1.2247395896468714,
-0.6087890007814074,
```

0.9804698093938512,
0.19554007335952167,
-0.18291383394130525,
-0.08254523307317092,
0.18380200005144362,
-0.06242013495608768,
-0.18902966193120999,
0.4846770511497838,
1.2162916157423216,
0.4476502570273604,
-0.5204828708955642,
1.1525916637240337,
0.9814837501696517,
1.2019836737296188,
0.10039944576408244,
-1.165620769823409,
0.5709783345435084,
0.04690539627488982,
0.3150855883896384,
0.6835833386917337,
0.7350671828581666,
0.15105693185753874,
0.7851129846448224,
-0.290237804763004,
0.02628688844012564,
-0.3427838784588314,
-0.12202424723691215,
0.30368921729231485,
-0.08398845633414893,
-0.34421225485285395,
-1.0992299599969957,
-0.6679166311787512,
-0.4925602784337871,
-0.7607905356843834,
-0.061822787578292515,
-0.33643902903801526,
1.7237644621079549,
1.1087838048318122,
1.4391293107545464,
-0.2688865923867924,
0.07238699078201241,
-0.7262166431073663,
-0.8607492696026928,
0.1828620517189283,
-1.5556021786529124,
-0.5230641559705965,
-0.023465829708141782,
0.5773817105800866,
-0.6189512057198875,
-0.1704298219551506,
0.6935490081871019,
0.43024761601627914,
-1.3818053283072975,
0.32069867830077864,
0.11223505487625844,
-0.6642209341325642,
-0.7163563578386999,
0.9533459704680486,
-2.5373567196844773,
-0.3530595330204968,
-0.36744271288254204,
-0.4105700661159024,
-0.529185988706723,
-0.5074466275713505,
0.5969687397167245,
0.1454479675984071,
0.9510340399481164,
-0.16051754924491393,
0.168676911677473,
1.6243103666539225,
-0.3533677275828062,
0.23248347317288842,
1.550100715213732,
0.3540002793004863,
0.11004939052123668,


```

0.0026309322638186474,
1.0023099891482978,
1.3075603782358884,
0.06614039249633322,
0.12696883216368457,
0.4650745950568721,
0.09288306127511851,
-0.05535633254466518,
1.0893487961438044,
0.1856572010619516,
-0.3369565165204948,
1.1993442715596423,
-0.1637322799431796,
-0.05574034797393315,
-0.6499132821077687,
-1.1854402746414112,
-0.20554021240749126,
-0.697991352679575,
0.7715290435685477,
0.27728540290360154,
-0.7490020286422794,
0.6727458447445039,
-1.3928966895283914,
-1.1123692530480307,
1.2429888713160355,
0.5002310144425463,
1.0425617895549082,
-0.41250228671824735,
-0.20782873770333182,
0.952256354944612,
-0.2180059005687231,
0.8024032545701543,
-1.0975152466418368,
0.6192383355749876,
-0.44420258004771784,
-1.3605235580058401,
1.4196204597477091,
-1.1230853758949062,
0.09475958663430169,
0.9412942473630452]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_xy_noisefree_time_evo_ric4_unifrom_scaling_sample#8'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,

```

```
'status': False}},
    'zne': { 'data_points': None,
             'degree': 0,
             'method': 'richardson',
             'sampling': 'default'}},
    'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z '
                                     '1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + '
                                     '1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '
                                     '6 + 1 Z 5 + 1 Z 6',
                'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + '
                                                            '0.5 Z 0 Z 1 + '
                                                            '0.5 X 1 X 2 + '
                                                            '0.5 Z 1 Z 2 + '
                                                            '0.5 X 2 X 3 + '
                                                            '0.5 Z 2 Z 3 + '
                                                            '0.5 X 3 X 4 + '
                                                            '0.5 Z 3 Z 4 + '
                                                            '0.5 X 4 X 5 + '
                                                            '0.5 Z 4 Z 5 + '
                                                            '0.5 X 5 X 6 + '
                                                            '0.5 Z 5 Z 6']},
    'output': { 'data_points': [ [4, 2, 0, 0, -7.159167099826595],
                                  [12, 6, 0, 0, -5.573237410199789],
                                  [20, 10, 0, 0, -4.355904736591592],
                                  [28, 14, 0, 0, -3.4138455900908444]],
               'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                    'identity_factors': [0, 0, 0, 0],
                                    'noise_level': [4, 2, 0, 0],
                                    'noise_profile': {
'noise_prob': [
0.001,
0.001,
0,
0],
'status': True,
'type':
'depolarizing'},
                                { 'odd_wires': 4},
                                { 'gates_num': [12, 3, 1, 0],
                                  'identity_factors': [1, 1, 0, 0],
                                  'noise_level': [12, 6, 0, 0],
                                  'noise_profile': {
'noise_prob': [
0.001,
0.001,
0,
0],
'status': True,
'type':
'depolarizing'},
                                { 'odd_wires': 4},
                                { 'gates_num': [20, 5, 1, 0],
                                  'identity_factors': [2, 2, 0, 0],
                                  'noise_level': [20, 10, 0, 0],
                                  'noise_profile': {
'noise_prob': [
0.001,
0.001,
0,
0],
'status': True,
'type':
'depolarizing'},
                                { 'odd_wires': 4},
                                { 'gates_num': [28, 7, 1, 0],
                                  'identity_factors': [3, 3, 0, 0],
                                  'noise_level': [28, 14, 0, 0],
                                  'noise_profile': {
'noise_on_init_param': { 'status': False,
                           'value': 0},
                           'noise_prob': [
0.001,
0.001,
0,
0],
                           'status': True,
                           'type':
'depolarizing'},
                           'odd_wires': 4},
                           { 'gates_num': [12, 3, 1, 0],
                             'identity_factors': [1, 1, 0, 0],
                             'noise_level': [12, 6, 0, 0],
                             'noise_profile': {
'noise_prob': [
0.001,
0.001,
0,
0],
'status': True,
'type':
'depolarizing'},
                           { 'odd_wires': 4},
                           { 'gates_num': [20, 5, 1, 0],
                             'identity_factors': [2, 2, 0, 0],
                             'noise_level': [20, 10, 0, 0],
                             'noise_profile': {
'noise_on_init_param': { 'status': False,
                           'value': 0},
                           'noise_prob': [
0.001,
0.001,
0,
0],
                           'status': True,
                           'type':
'depolarizing'},
                           { 'odd_wires': 4},
                           { 'gates_num': [28, 7, 1, 0],
                             'identity_factors': [3, 3, 0, 0],
                             'noise_level': [28, 14, 0, 0],
                             'noise_profile': {
```

```

'value': 0},

0.001,
0.001,

'noise_prob': [

0,
0],

'status': True,
'type':

'depolarizing'},

'odd_wires': 4]],
'run_time_sec': 11.1983642578125}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0}},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 4.1837705414928806e-17,
1.1742327239243593,
1.1742327239243593,
1.174232723924359,
1.3021435979041922,
1.3883521076730898,
1.4901272631800928,
1.490127263180093,
1.490127263180093,
2.542545188872923,
2.56739133991554,
2.677178516908646,
5.624946341909109,
5.624946341909109,
5.63924283183031,
5.672838193807689,
5.672838193807689,
5.672838193807689,
5.672838193807689,
5.672838193807689,
5.672838193807689,
5.805507018952099,
5.858678454346443,
7.374841183417447,
7.374841183417447,
9.078346087689514,
9.17910617229311,
9.185363997877552,
9.550522060955204,
9.715006087156702,
0.5380723684568228,
0.20346632529178696,
-0.28402043001150323,
-0.013694659434132353,
-0.8491217472851175,
-0.13891219530245408,
-0.7902948900320785,
0.02035105936303376,
0.35891510624171696,
0.07011262472067725,
0.2571910630887441,
0.09727807922224965,
-0.4253506785797692,
-0.08201618636277365,
-0.5629942371157008,
0.4984357070329931,

```

0.22460606922422946,
0.6436861273901548,
0.5581387866543217,
0.6438048424266741,
0.6625810608832491,
-0.2877740698749285,
0.03400434270310955,
0.6598198228171638,
0.15695492486622947,
-0.23277712193728803,
0.3379436543817524,
0.6679629986711755,
0.4913802101172593,
-0.6543304626041586,
1.2398324170787045,
0.10868488610295632,
0.4726806257545104,
-1.0278416218016544,
-0.5660236088762642,
-0.06627186605932975,
0.3527547117645279,
0.3944014355355292,
0.33709190475341266,
0.16707772676088822,
0.29195080738380563,
1.0354529601392413,
0.12500585138196232,
-0.037178956773431956,
0.7823218278561086,
-0.04464215343745706,
-0.36923576305262273,
-0.9863817864437636,
0.039519076991602065,
-0.35646400258617383,
0.04542480304419608,
-0.46125271135496354,
0.40472620693719735,
0.7802480661953589,
-0.6757172179201896,
0.6396142241310521,
0.2327897749192615,
0.8954622299286413,
0.7158405805761733,
0.1382139633537826,
1.0616928944303126,
-0.6122315420380016,
-0.6738539946512315,
1.0433459792659807,
0.494526070112864,
0.20621322780405107,
-0.05318175489586881,
-0.11213659998229918,
0.32941262943060273,
0.13931786347259914,
-0.07640781308377202,
0.31450116184408605,
0.3658109215963646,
0.04460349089074807,
-0.12913310998017427,
0.31890485610090735,
-0.005491508360761248,
1.2692063073436923,
-0.3006174703747285,
-0.41404661321397057,
-0.5199199797590206,
0.44225909618051784,
-0.4233007342169607,
0.7456354776212751,
-0.32484988424610756,
0.9268151291603125,
0.8385591284905366,
0.027219364270413358,
-0.5520387125082392,
1.3757901840365965,
0.02933419406725825,

[illegible]

```

'0.5 X 1 X 2 + '
'0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + '
'0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + '
'0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + '
'0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + '
'0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -7.130214875567141],
                              [12, 6, 0, 0, -5.622281179564727],
                              [20, 10, 0, 0, -4.4547314200293675],
                              [28, 14, 0, 0, -3.544056752302471]],
  'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                      'identity_factors': [0, 0, 0, 0],
                      'noise_level': [4, 2, 0, 0],
                      'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},

0.001,
0.001,

'noise_prob': [

0,
0],

'status': True,
'type':

'depolarizing'},

'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [1, 1, 0, 0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},

0.001,
0.001,

'noise_prob': [

0,
0],

'status': True,
'type':

'depolarizing'},

'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [2, 2, 0, 0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},

0.001,
0.001,

'noise_prob': [

0,
0],

'status': True,
'type':

'depolarizing'},

'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
  'identity_factors': [3, 3, 0, 0],
  'noise_level': [28, 14, 0, 0],
  'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},

0.001,
0.001,

'noise_prob': [

0,
0],

'status': True,
'type':

'depolarizing'},

'odd_wires': 4}],
'run_time_sec': 9.331718683242798}},

```

```
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [ 0,
                                                            0,
                                                            0,
                                                            0,
                                                            0,
                                                            0],
                                                            'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                            'r': 0},
                        'time': {'max': 10.0, 'min': 0.0},
                        'type': 'xy-iss'}}},
  'init_param': { 'value': [ 0.003408343124459994,
                              0.0034083431244599804,
                              0.006710027629051694,
                              0.2484817728699434,
                              1.1672656342280407,
                              1.167470260322433,
                              3.1453440534904744,
                              3.166261678218574,
                              3.2843704266652405,
                              4.089250903115526,
                              4.091989893807717,
                              4.091989893807717,
                              5.181052167254873,
                              5.181052167254873,
                              5.184972473956635,
                              5.585122024093685,
                              5.5990893229720795,
                              5.630355974178803,
                              5.704160317443757,
                              8.658831571647385,
                              8.66505230970618,
                              8.712765508205756,
                              8.712765508205756,
                              8.712765508205756,
                              9.168875276976076,
                              9.176734348833309,
                              9.262471352200725,
                              9.262471352200725,
                              10.578606537988184,
                              10.578606537988184,
                              1.5905119084430799,
                              0.8691136234632737,
                              0.04907693052707525,
                              -0.23336553986203207,
                              0.4284777328779486,
                              2.2866815267075147,
                              0.1760352312050406,
                              0.27820168999860917,
                              2.067264415281462,
                              0.5755331494929522,
                              -0.06858856463703264,
                              -0.7222667597755238,
                              0.7750924366303688,
                              -0.278304785792398,
                              0.5352624789798245,
                              -1.1702690163303244,
                              -0.012717570027659048,
                              -0.07125250327360101,
                              1.7927644689789062,
                              1.3752915621891346,
                              -1.1981440291126808,
                              -0.30737164793891514,
                              0.2657743539912098,
                              0.04156274567190095,
                              1.1363043468044227,
                              0.059217925806269944,
                              0.7608352104096147,
```

-0.1513784586666425,
0.025696061144845524,
0.861481321747562,
-0.48544656000194114,
-0.006033515622113467,
0.473431667961849,
1.5469119181224316,
0.1871819193791605,
1.555475823471471,
0.9001349633565042,
-1.5961139272878428,
-0.6749613217725859,
-1.5596468027729833,
0.5296186936496604,
-0.8549575199843406,
1.7649719371938613,
1.5569889217595585,
-0.9244334407213903,
-1.5804326845781602,
0.3840271097002098,
-0.8070678340768227,
-0.48791686278645724,
0.13110829311357092,
-2.023238766590243,
-0.8997043584182223,
0.8435119798114825,
0.12108034013815198,
-0.7013572315179941,
0.05813246571029178,
-1.3645229029121433,
-1.704441396215925,
0.6160899239086374,
-1.6580292769388947,
-1.550797918270104,
-0.033040246679508146,
-0.7595754539148374,
0.6024045095197653,
0.446745843265138,
0.9717727329863253,
-1.3190014690683383,
-3.180952616822137,
-0.47685451708111876,
0.03958388950097112,
-0.04894683660460036,
-0.013869967587830365,
0.9715551393376142,
-0.2688733257646762,
-1.7112803723577967,
-0.06346862095088202,
1.2756440828431679,
0.4087049517265435,
-0.4433737535464867,
0.07044159255576982,
0.877714241226598,
-0.33059433004855665,
-0.41683973817974146,
-0.09123983383660174,
1.153451542408442,
1.5381060924707182,
1.4432055127327397,
0.7948381511053778,
0.12126271666707081,
0.3736422837955817,
-0.06595460407520624,
-0.1569219765005705,
1.4053035958769775,
-0.3443177603686477,
0.14604377230684606,
0.733462484703497,
1.2365404639754205,
-0.020105376528163625,
-0.09631958929266826,
-0.16876809298151013,
0.42184799028013376,
0.8803036484986011,


```

-0.4148292271104988,
0.011464734896518355,
1.0895788780475535,
-0.006017115067308532,
0.6476481463328121,
-0.4792828310008005,
1.7145504838049759,
0.9300252591703124,
1.4147034805691376,
-0.04601267898776636,
-0.34579443398211707,
0.006107561937391137,
2.7608588509789516,
-2.470402787070412,
-0.2193552778493498,
-0.43636419827955597,
0.5036040054264609,
-0.006776779992121664]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_xy_noisefree_time_evo_ric4_unifrom_scaling_sample#7'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z '
                                '1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + '
                                '1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '
                                '6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + '
                                                         '0.5 Z 0 Z 1 + '
                                                         '0.5 X 1 X 2 + '
                                                         '0.5 Z 1 Z 2 + '
                                                         '0.5 X 2 X 3 + '
                                                         '0.5 Z 2 Z 3 + '
                                                         '0.5 X 3 X 4 + '
                                                         '0.5 Z 3 Z 4 + '
                                                         '0.5 X 4 X 5 + '
                                                         '0.5 Z 4 Z 5 + '
                                                         '0.5 X 5 X 6 + '
                                                         '0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -7.568863804770513],

```



```
0.5,  
0.5,  
0.5,  
0.5],  
  
        'r': 0},  
    'time': {'max': 10.0, 'min': 0.0},  
    'type': 'xy-iss'}},  
'init_param': { 'value': [ 0.01970245038838479,  
                             0.18403088824160574,  
                             0.23516669011749736,  
                             0.3619340287619915,  
                             0.7559235356843684,  
                             0.8037553588156275,  
                             0.8037553588156274,  
                             0.8037553588156274,  
                             1.523650120002224,  
                             1.5399034674182392,  
                             3.5267025169505333,  
                             3.526702516950533,  
                             4.657784878324055,  
                             4.735234101549794,  
                             4.768060921347247,  
                             6.910522417024203,  
                             6.910522417024203,  
                             7.937298848041795,  
                             7.937298848041795,  
                             7.937298848041795,  
                             7.937298848041795,  
                             7.937298848041795,  
                             7.937298848041795,  
                             7.937298848041795,  
                             8.950027818739875,  
                             8.950027818739875,  
                             8.950027818739875,  
                             8.950027818739875,  
                             8.950027818739875,  
                             9.275481401407285,  
                             0.034552241786943605,  
                             0.7960682992937013,  
                             -0.16960774395822967,  
                             -0.13488653659207303,  
                             -0.07127010448742209,  
                             0.7213919487775996,  
                             0.07517487951530155,  
                             0.7669403129297154,  
                             -0.0980146039609658,  
                             0.2841753594230342,  
                             0.41897129078802076,  
                             0.7644737129881134,  
                             0.21659836679204617,  
                             0.40978346593820614,  
                             0.23386906952753808,  
                             0.6351364889148511,  
                             0.33021983360135304,  
                             0.3518425622667454,  
                             -0.16759580526603354,  
                             1.1955178055405769,  
                             -0.1119172073880175,  
                             -1.6647459335605923,  
                             -0.09499524124538847,  
                             1.171044098528735,  
                             -0.022137095989983098,  
                             0.46479840085551427,  
                             0.5434533112600137,  
                             0.7953483283824695,  
                             -0.23791626316884204,  
                             -0.34327459028867535,  
                             0.9770824990177593,  
                             0.3972109926583832,  
                             0.031305244787475206,  
                             0.06660894739153407,  
                             -0.4270300066285869,  
                             -1.4580475210852422,  
                             0.18899331747125298,  
                             -0.5789031865333153,
```

0.2203088633995836,
-0.028805094274594364,
-0.21236041446549625,
0.00835869807102751,
1.8444421334304235,
-0.36021419267454113,
-1.0777998782841605,
0.7201309879688491,
-0.07151832135087417,
0.013523038580686776,
-0.5101061898109359,
1.528477364550823,
-0.24055091899624312,
1.595727742687772,
1.181011187425284,
-0.07205139039345279,
-0.3645912646252544,
0.922259297261975,
0.47504473393792157,
-1.4582846498754758,
-1.1041238978674286,
-0.08619812725702276,
1.5348185477516838,
-0.034799821636765677,
0.34854563969795865,
-0.3057886345189525,
0.3435344949085216,
2.387989506593159,
0.5928805156422549,
-0.02250711897690342,
0.734685803060112,
1.1230348216575474,
0.9339212770561773,
0.5429657194529808,
-0.9530889967003539,
1.2607191116533525,
-0.18262479735843284,
0.2292294020716647,
0.484625167272873,
2.2621526738562925,
0.3832060422832924,
0.2504382199240898,
-1.1186566372739195,
1.0354182359738777,
-0.9045076995614423,
1.0282595509077306,
1.805679851173207,
0.9332861023326164,
1.4396403686507162,
1.6178312922032976,
-0.22196274260736226,
0.19030625281227892,
-0.32840557234129963,
1.004592356770436,
1.8720477930131976,
-0.06388076228748135,
0.12235432131563899,
-0.5998878734536057,
-0.5461403115162753,
-0.17116133229966776,
0.6826247841010238,
0.6607748570434985,
0.7854114346995006,
0.5612002866021459,
-0.2559995100181322,
-0.8863827207156498,
0.2553569774973194,
0.6372473265785513,
-0.14017909069433732,
0.016226196464851673,
-0.10437448541592237,
2.5268983957827906,
2.3709647446626767,
-0.8108547070158446,
0.5698393243760288,

```

-1.2042979310590165,
-0.5004321415825274,
-1.0362881146071494,
-0.18531750144204898,
1.5646886079478448,
0.08968487420602586,
-1.6120268318409705]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_xy_noisefree_time_evo_ric4_unifrom_scaling_sample#10'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z '
                                '1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + '
                                '1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '
                                '6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + '
                                                         '0.5 Z 0 Z 1 + '
                                                         '0.5 X 1 X 2 + '
                                                         '0.5 Z 1 Z 2 + '
                                                         '0.5 X 2 X 3 + '
                                                         '0.5 Z 2 Z 3 + '
                                                         '0.5 X 3 X 4 + '
                                                         '0.5 Z 3 Z 4 + '
                                                         '0.5 X 4 X 5 + '
                                                         '0.5 Z 4 Z 5 + '
                                                         '0.5 X 5 X 6 + '
                                                         '0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -7.512595875373562],
                              [12, 6, 0, 0, -6.004303631668366],
                              [20, 10, 0, 0, -4.83491199768215],
                              [28, 14, 0, 0, -3.919939099798885]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                     'identity_factors': [0, 0, 0, 0],
                     'noise_level': [4, 2, 0, 0],
                     'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
                          'noise_prob': [
0.001,

```

```

0.001,
0,
0],
'status': True,
'type':
'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': {
'noise_prob': [
0,
0],
'status': True,
'type':
'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
'identity_factors': [2, 2, 0, 0],
'noise_level': [20, 10, 0, 0],
'noise_profile': {
'noise_prob': [
0,
0],
'status': True,
'type':
'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
'identity_factors': [3, 3, 0, 0],
'noise_level': [28, 14, 0, 0],
'noise_profile': {
'noise_prob': [
0,
0],
'status': True,
'type':
'depolarizing'},
'odd_wires': 4},
'run_time_sec': 10.707031011581421}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.02240293444906766,
0.9863710197203618,
1.3708545038264748,
1.3757718928414557,

```

1.7844795346244002,
2.029643659915566,
2.029643659915566,
4.135298362317422,
4.181946654536538,
4.392782597821499,
4.473609589126697,
4.474471668054438,
4.475838923576404,
4.4785869343507265,
4.4785869343507265,
5.467009291914798,
5.470281162008999,
5.840541346188122,
5.842392372241278,
6.505508038171383,
6.505508038171383,
7.536348099553183,
7.536348099553183,
9.63019412694799,
9.63019412694799,
9.63019412694799,
9.63019412694799,
9.63019412694799,
9.63019412694799,
9.63019412694799,
1.1150074339792337,
-0.03535947987647509,
1.1918755403807781,
-0.20191103850925227,
-0.268362117189328,
-0.3595192771004685,
1.2614204066704717,
-2.9429228213034166,
-1.054810371869961,
0.9594255841252335,
-0.09538831740389547,
0.3965663197177072,
-0.22182226409641906,
-0.2653718580482209,
0.1686090968331403,
1.6516484362324146,
-0.3049608013883917,
0.49165272629600476,
1.9061569916625292,
1.241422732345657,
-0.11527538328521378,
-0.22895223610637053,
-0.06124477487110871,
0.6193220715745159,
1.372932614853267,
-1.1538384701543527,
-0.1139440760455718,
0.16973094341798492,
-0.5943234340378932,
0.21399932498003596,
-0.18156105319883253,
-1.0504702368491705,
0.7232808408606985,
-0.018870518436612777,
-0.29343981279013404,
0.30960592292227684,
0.13511542022922338,
0.5096669954667911,
-1.097576281455009,
0.12047585913747774,
-0.5585503308065869,
1.2279608447421912,
1.3012401380411538,
0.3769592231863052,
-0.5692023175942067,
-0.4567508363134565,
-0.679549924864495,
-0.24647993267323956,
0.6292567546169889,

```
0.9587256267378846,  
1.7268825428143277,  
0.1413792259593793,  
0.6520451922983643,  
0.6152471684019365,  
-1.1240317980448378,  
-0.9856730455453077,  
-1.2455074642265085,  
0.2235091354354849,  
-0.8870861048674711,  
0.5299724059178798,  
-0.21820971616414211,  
-0.09687180933262383,  
-0.9247884680119297,  
-0.4566833626656702,  
-0.4285291193052864,  
-0.6406026728496247,  
0.9023578506051637,  
0.03748900229192909,  
0.7551248935414285,  
-0.18980887545414976,  
-0.1497604267964229,  
-0.4253730408988226,  
-0.4157675203773789,  
1.677209169409502,  
0.2126888490300174,  
-0.13481167544410624,  
-1.6951458494505263,  
-0.006476362637627559,  
-0.5490470525961781,  
0.3195499617454318,  
1.077429849794694,  
0.6725724542747005,  
-0.10235701747439804,  
-0.014129818782246995,  
2.382406086835482,  
0.008866515589568993,  
1.0282503794544984,  
-0.6899701634722577,  
0.3510134577654823,  
0.6918405669356381,  
-0.08667968085870373,  
-0.0005005962237103964,  
1.3805542469189591,  
1.011514915181436,  
-0.4623191099783579,  
1.397645255720587,  
-0.7813361449321683,  
-0.9940660424170442,  
-0.6570734018108508,  
1.0171740600492052,  
0.14480072689925969,  
1.157289980717297,  
1.019472554492438,  
1.0029573776528096,  
-0.20406480565208354,  
-0.011076077177880707,  
-0.30566521016454584,  
-0.4775965459909992,  
-0.7158801142606765,  
1.5610067764386268,  
1.5187465408730505,  
0.21158961058436962,  
-0.38098352608767794,  
-0.046999172842648596,  
0.5964180792256618,  
-0.6260168044949087,  
-0.7215592053840503,  
-0.5706802740378752,  
0.11995346168986934,  
0.17053214926260635]],  
'noise_profile': { 'noise_on_init_param': { 'status': False,  
                                              'value': 0},  
                  'noise_prob': [0.001, 0.001, 0, 0],  
                  'status': True,
```



```

        'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [ 0.5,
                                         0.5,
                                         0.5,
                                         0.5,
                                         0.5,
                                         0.5],
                                'r': 1},
               'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_xy_noisefree_time_evo_ric4_unifrom_scaling_sample#6'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z '
                                '1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + '
                                '1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '
                                '6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + '
                                                         '0.5 Z 0 Z 1 + '
                                                         '0.5 X 1 X 2 + '
                                                         '0.5 Z 1 Z 2 + '
                                                         '0.5 X 2 X 3 + '
                                                         '0.5 Z 2 Z 3 + '
                                                         '0.5 X 3 X 4 + '
                                                         '0.5 Z 3 Z 4 + '
                                                         '0.5 X 4 X 5 + '
                                                         '0.5 Z 4 Z 5 + '
                                                         '0.5 X 5 X 6 + '
                                                         '0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -7.259703059772281],
                             [12, 6, 0, 0, -5.766278116709839],
                             [20, 10, 0, 0, -4.583193384345111],
                             [28, 14, 0, 0, -3.640157125717274]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                    'identity_factors': [0, 0, 0, 0],
                    'noise_level': [4, 2, 0, 0],
                    'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [
0.001,
0.001,

                                0,
                                0],

                                'status': True,
                                'type':
'depolarizing'},

                                'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [1, 1, 0, 0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {

```

```

'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,

'noise_prob': [
    0,
    0],
    'status': True,
    'type':

'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [2, 2, 0, 0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {

'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,

'noise_prob': [
    0,
    0],
    'status': True,
    'type':

'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
  'identity_factors': [3, 3, 0, 0],
  'noise_level': [28, 14, 0, 0],
  'noise_profile': {

'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,

'noise_prob': [
    0,
    0],
    'status': True,
    'type':

'odd_wires': 4}},
    'run_time_sec': 11.005364656448364}},
{ 'config': { 'ansatz': { 'gateset': 1,
    'layer': 30,
    'ugate': { 'coefficients': { 'bn': [ 0,
        0,
        0,
        0,
        0,
        0],
        'cn': [ 0.5,
            0.5,
            0.5,
            0.5,
            0.5,
            0.5],
        'r': 0},
        'time': {'max': 10.0, 'min': 0.0},
        'type': 'xy-iss'}},
    'init_param': { 'value': [ -3.641081222961908e-17,
        0.6149788059529836,
        0.7190202634465676,
        1.227117923902744,
        1.282995668258181,
        1.320146784613617,
        1.320146784613617,
        3.1464089486715623,
        3.1464089486715623,
        5.021180125711677,
        5.021180125711677,
        5.021180125711677,
        6.703692809170789,
        6.703692809170789,
        7.791601108223033,

```

7.8369202041803225,
7.951907692650894,
7.98389042115583,
8.258376681022987,
8.258376681022987,
8.264818959972613,
8.264818959972613,
8.264818959972613,
8.264818959972613,
8.265620267330194,
8.265620267330194,
9.412842541483705,
9.412842541483705,
10.502716174481437,
10.502716174481437,
0.1665558361704551,
0.0766182276072886,
-0.06331706305100791,
0.03225035272845343,
-0.2265177184784691,
0.6859734408919996,
0.358399999202588,
-0.31490649734707366,
0.03084506070978915,
0.8141179515629021,
-0.05751068028622007,
0.9013204998454583,
-0.325467390457223,
-0.16027539072229496,
0.19756436832536928,
0.17875052394899651,
0.07441990946369173,
-1.7121562848350094,
0.5411900571694856,
0.025994665514066265,
-1.0547594680221428,
-0.3558195097906033,
0.2575282902752545,
-0.32652811387218433,
-0.17477771883072452,
0.05603758809983181,
-0.31531793685177084,
0.24953935818155465,
-0.21945102528047247,
0.016201452486715968,
-0.47139285875704856,
0.8354485366702803,
0.28783787212142514,
-1.483190369052958,
-0.05324490057492157,
0.00808064868141892,
-1.3894868155533049,
0.07648399162880297,
0.28128388531065984,
-0.8032486745370345,
0.9961277979922681,
0.7271807873997648,
-0.2598267503182047,
0.058937213125893145,
1.4176939090014022,
1.5623836495300838,
-1.2281014386266311,
1.627596878633186,
0.150437982354294,
0.017798034487339433,
-0.2091243023446671,
0.08776907917710067,
0.637527078885032,
1.2603122563899933,
1.0910453886717337,
0.001707278941889729,
-0.5366140699752073,
-0.04457743756443993,
0.708269991966563,
1.6757260639955311,

```

0.6610325406900899,
-0.508228690084442,
-0.5178693625521746,
-0.2955589105377924,
-0.47790822054112786,
0.31077598178920485,
0.020632196003020267,
1.5538854336639996,
1.2816443511581943,
-0.6926864187150775,
0.73698114320343,
0.3645818015611773,
-0.2097443780477125,
0.1421287040679699,
0.052774836076248374,
-0.7306191772967899,
1.107856618529218,
0.3743379513498811,
1.2959169452837032,
0.8858564921663113,
0.03370518740548656,
-1.0860983069970114,
1.2199055901240408,
-1.0630806559423625,
1.3748402618369016,
0.9943926078324806,
0.7116645059379805,
-0.21098608064682703,
0.04581715611274856,
-0.17117312830066983,
1.2229557830916487,
-1.6273428514067598,
1.8508927721454946,
0.851693435560975,
1.0946646161197402,
0.43088380554694883,
-0.6171985422875839,
1.6866137951313547,
-0.2289359575496427,
-0.09152362655526167,
1.6917359096449291,
0.22722692961233917,
0.49434218233553445,
1.2228162165990006,
-0.4070166799423108,
0.00966082758630671,
-0.3344922737632815,
-1.0755692755599169,
1.1846485271761396,
-0.06070659712292082,
0.6384418838463358,
-0.034216786122488324,
-0.6519049362670448,
-0.0019231105719348242,
0.503764046349436,
0.5779702041064313,
-0.8610978040126155,
0.5832069567020347,
0.07954533618121334,
0.016329041774977193]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,

```

```

0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_xy_noisefree_time_evo_ric4_unifrom_scaling_sample#3'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z '
                                '1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + '
                                '1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '
                                '6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + '
                                                         '0.5 Z 0 Z 1 + '
                                                         '0.5 X 1 X 2 + '
                                                         '0.5 Z 1 Z 2 + '
                                                         '0.5 X 2 X 3 + '
                                                         '0.5 Z 2 Z 3 + '
                                                         '0.5 X 3 X 4 + '
                                                         '0.5 Z 3 Z 4 + '
                                                         '0.5 X 4 X 5 + '
                                                         '0.5 Z 4 Z 5 + '
                                                         '0.5 X 5 X 6 + '
                                                         '0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -7.47570078392518],
                             [12, 6, 0, 0, -5.921149282782478],
                             [20, 10, 0, 0, -4.7240834286119355],
                             [28, 14, 0, 0, -3.7949056030968538]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                     'identity_factors': [0, 0, 0, 0],
                     'noise_level': [4, 2, 0, 0],
                     'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},

0.001,
0.001,

'noise_prob': [

0,
0],

'status': True,
'type':

'depolarizing'},

'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [1, 1, 0, 0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},

0.001,
0.001,

'noise_prob': [

0,
0],

'status': True,
'type':

'depolarizing'},

'odd_wires': 4},

```

```

        { 'gates_num': [20, 5, 1, 0],
          'identity_factors': [2, 2, 0, 0],
          'noise_level': [20, 10, 0, 0],
          'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
0.001,
0.001,
'noise_prob': [
0,
0],
'status': True,
'type':
'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
  'identity_factors': [3, 3, 0, 0],
  'noise_level': [28, 14, 0, 0],
  'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
0.001,
0.001,
'noise_prob': [
0,
0],
'status': True,
'type':
'depolarizing'},
'odd_wires': 4}},
'run_time_sec': 10.085844993591309]],
'vqe': [ { 'config': { 'ansatz': { 'gateset': 1,
                                  'layer': 30,
                                  'ugate': { 'coefficients': { 'bn': [ 0,
                                                                        0,
                                                                        0,
                                                                        0,
                                                                        0,
                                                                        0,
                                                                        0],
                                                                    'cn': [ 0.5,
                                                                        0.5,
                                                                        0.5,
                                                                        0.5,
                                                                        0.5,
                                                                        0.5],
                                                                    'r': 0},
                                                                    'time': {'max': 10.0, 'min': 0.0},
                                                                    'type': 'xy-iss'}},
                                  'init_param': {'value': 'random'},
                                  'noise_profile': { 'noise_on_init_param': { 'status': False,
                                                                              'value': 0},
                                                    'noise_prob': [0.001, 0.001, 0, 0],
                                                    'status': True,
                                                    'type': 'depolarizing'},
                                  'nqubits': 7,
                                  'observable': { 'coefficients': { 'bn': [ 1.0,
                                                                              1.0,
                                                                              1.0,
                                                                              1.0,
                                                                              1.0,
                                                                              1.0],
                                                                    'cn': [ 0.5,
                                                                        0.5,
                                                                        0.5,
                                                                        0.5,
                                                                        0.5],
                                                                    'r': 1},
                                                                    'def': 'ising'},
                                  'output': { 'draw': { 'fig_dpi': 100,
                                                         'status': True,
                                                         'type': 'png'},
                                                         'file_name_prefix': 'xy_noisefree_time_evo'},
        { 'gates_num': [20, 5, 1, 0],
          'identity_factors': [2, 2, 0, 0],
          'noise_level': [20, 10, 0, 0],
          'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
0.001,
0.001,
'noise_prob': [
0,
0],
'status': True,
'type':
'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
  'identity_factors': [3, 3, 0, 0],
  'noise_level': [28, 14, 0, 0],
  'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
0.001,
0.001,
'noise_prob': [
0,
0],
'status': True,
'type':
'depolarizing'},
'odd_wires': 4}},
'run_time_sec': 10.085844993591309]],
'vqe': [ { 'config': { 'ansatz': { 'gateset': 1,
                                  'layer': 30,
                                  'ugate': { 'coefficients': { 'bn': [ 0,
                                                                        0,
                                                                        0,
                                                                        0,
                                                                        0,
                                                                        0,
                                                                        0],
                                                                    'cn': [ 0.5,
                                                                        0.5,
                                                                        0.5,
                                                                        0.5,
                                                                        0.5,
                                                                        0.5],
                                                                    'r': 0},
                                                                    'time': {'max': 10.0, 'min': 0.0},
                                                                    'type': 'xy-iss'}},
                                  'init_param': {'value': 'random'},
                                  'noise_profile': { 'noise_on_init_param': { 'status': False,
                                                                              'value': 0},
                                                    'noise_prob': [0.001, 0.001, 0, 0],
                                                    'status': True,
                                                    'type': 'depolarizing'},
                                  'nqubits': 7,
                                  'observable': { 'coefficients': { 'bn': [ 1.0,
                                                                              1.0,
                                                                              1.0,
                                                                              1.0,
                                                                              1.0,
                                                                              1.0],
                                                                    'cn': [ 0.5,
                                                                        0.5,
                                                                        0.5,
                                                                        0.5,
                                                                        0.5],
                                                                    'r': 1},
                                                                    'def': 'ising'},
                                  'output': { 'draw': { 'fig_dpi': 100,
                                                         'status': True,
                                                         'type': 'png'},
                                                         'file_name_prefix': 'xy_noisefree_time_evo'},

```


5.817834245705687,
6.166757373752113,
7.3114292874931905,
7.311858449036949,
8.346734585466223,
8.350435170229554,
8.354614694314481,
8.35659817980371,
8.356905245901928,
9.313534396314816,
9.313534396314816,
10.443787047737276,
-0.5971318801423705,
-0.6830583120471296,
0.053446928626708114,
-1.488251087277441,
0.7409998345617078,
-0.020561007687691084,
-0.47029189134787985,
-0.9187802118512801,
-0.07390570205801456,
0.6091612085770479,
0.5696050095749661,
-2.634874245900948,
0.14733366015521565,
1.0781100436188797,
0.3459269460772517,
1.2820636645787713,
-1.605623121104658,
0.856791024529966,
0.16374864437029604,
-0.7377840957855112,
0.7603752030179747,
0.4958590077418653,
1.0293890027768824,
0.4862585296205842,
-1.0838654128292502,
1.2632637762941026,
-1.1043454462318527,
-0.7867654190654241,
0.8818217647037212,
1.0406267750063978,
0.05770567519039498,
-0.5922070722918433,
-0.3271403288282926,
0.5980786503295974,
-0.02050375112330311,
-0.9007241609880184,
0.5239405469453117,
0.7601496703420288,
-0.6449049942439851,
1.7340858429103823,
0.10050113333288421,
0.3143347863780438,
0.28408386490618526,
-0.1222389867527315,
-0.09502282329152913,
-1.3004368662949968,
0.16856179320059875,
0.6790103402211991,
0.3355977478652572,
0.0935957266804422,
0.42936820813464754,
0.5148227650107633,
0.10764356588532305,
-0.054251737808208085,
0.06525040933822983,
-0.21143245853075732,
-1.5801988491404158,
0.023513208971767067,
1.0921277689034055,
-0.11501776583679489,
-0.14182911234947893,
1.8716225858725093,
-1.0747761250482988,

-0.01873186129021693,
0.9952348515577564,
1.5589645169221944,
2.4625123021717044,
1.5320934484141422,
0.6264578101560674,
0.015802850908690148,
0.6559295719723718,
0.5030706883609005,
0.010357532831659657,
-0.6971755064049523,
-1.334963536555068,
0.05714132580252865,
0.9478292633237722,
1.541635767019547,
-1.2357775558070643,
1.5753066317572375,
1.2206753513411082,
-0.013882574638756213,
2.102115305768464,
0.6867648509985108,
-1.0810617894892982,
0.608205793929093,
-1.0701207529796843,
-0.008804617073740417,
1.0802724219136723,
1.4472704941581649,
-0.525983160453369,
0.10347757085774918,
0.07451664808136123,
0.8113743025574697,
-0.19830571509994802,
0.20760359505818776,
-0.5709476706639499,
0.1791350390980861,
-0.32061652775092425,
-1.6382844876022806,
2.271046687924862,
0.09972766177435569,
1.671157305178672,
1.0596955078067567,
-0.7395801076185267,
-0.8217251410877643,
1.2660358026872456,
-0.149859182069558,
1.7386861885389688,
-0.003505497297480453,
0.677126408740769,
-1.0541684899696007,
0.9437251450981726,
1.7354017388586316,
-0.8022872949930118,
0.01881044769021433,
-0.013699411839369284,
1.5965466699123847,
0.4486397166383021,
1.5669137279484207],
[4.1837705414928806e-17,
1.1742327239243593,
1.1742327239243593,
1.174232723924359,
1.3021435979041922,
1.3883521076730898,
1.4901272631800928,
1.490127263180093,
1.490127263180093,
2.542545188872923,
2.56739133991554,
2.677178516908646,
5.624946341909109,
5.624946341909109,
5.63924283183031,
5.672838193807689,
5.672838193807689,
5.672838193807689,

5.672838193807689,
5.672838193807689,
5.672838193807689,
5.805507018952099,
5.858678454346443,
7.374841183417447,
7.374841183417447,
9.078346087689514,
9.17910617229311,
9.185363997877552,
9.550522060955204,
9.715006087156702,
0.5380723684568228,
0.20346632529178696,
-0.28402043001150323,
-0.013694659434132353,
-0.8491217472851175,
-0.13891219530245408,
-0.7902948900320785,
0.02035105936303376,
0.35891510624171696,
0.07011262472067725,
0.2571910630887441,
0.09727807922224965,
-0.4253506785797692,
-0.08201618636277365,
-0.5629942371157008,
0.4984357070329931,
0.22460606922422946,
0.6436861273901548,
0.5581387866543217,
0.6438048424266741,
0.6625810608832491,
-0.2877740698749285,
0.03400434270310955,
0.6598198228171638,
0.15695492486622947,
-0.23277712193728803,
0.3379436543817524,
0.6679629986711755,
0.4913802101172593,
-0.6543304626041586,
1.2398324170787045,
0.10868488610295632,
0.4726806257545104,
-1.0278416218016544,
-0.5660236088762642,
-0.06627186605932975,
0.3527547117645279,
0.3944014355355292,
0.33709190475341266,
0.16707772676088822,
0.29195080738380563,
1.0354529601392413,
0.12500585138196232,
-0.037178956773431956,
0.7823218278561086,
-0.04464215343745706,
-0.36923576305262273,
-0.9863817864437636,
0.039519076991602065,
-0.35646400258617383,
0.04542480304419608,
-0.46125271135496354,
0.40472620693719735,
0.7802480661953589,
-0.6757172179201896,
0.6396142241310521,
0.2327897749192615,
0.8954622299286413,
0.7158405805761733,
0.1382139633537826,
1.0616928944303126,
-0.6122315420380016,
-0.6738539946512315,

1.0433459792659807,
0.494526070112864,
0.20621322780405107,
-0.05318175489586881,
-0.11213659998229918,
0.32941262943060273,
0.13931786347259914,
-0.07640781308377202,
0.31450116184408605,
0.3658109215963646,
0.04460349089074807,
-0.12913310998017427,
0.31890485610090735,
-0.005491508360761248,
1.2692063073436923,
-0.3006174703747285,
-0.41404661321397057,
-0.5199199797590206,
0.44225909618051784,
-0.4233007342169607,
0.7456354776212751,
-0.32484988424610756,
0.9268151291603125,
0.8385591284905366,
0.027219364270413358,
-0.5520387125082392,
1.3757901840365965,
0.02933419406725825,
-0.2915193842748469,
0.33917094344303517,
0.10828106345120646,
-0.5650186064180197,
-0.6041651963720519,
-0.19230834783869816,
1.9848826966849449,
0.0632219631903188,
0.04804551511145073,
0.47752350144148015,
1.3406847618100746,
-1.3722262204383469,
-1.0513455672656666,
0.7121308231347245,
0.13278942055995957,
1.9758312031522887,
2.19648222646383,
0.8936788375205645,
0.05510909660880629,
-0.18765678539637448,
0.45031205104496697,
-0.6542716355161553,
0.5783583754336702,
-0.18844184232604352,
-1.5667303012058793,
0.36110899429768645,
-0.275182998952267,
0.30367576290775045,
-0.1532687178202341],
[-3.641081222961908e-17,
0.6149788059529836,
0.7190202634465676,
1.227117923902744,
1.282995668258181,
1.320146784613617,
1.320146784613617,
3.1464089486715623,
3.1464089486715623,
5.021180125711677,
5.021180125711677,
5.021180125711677,
6.703692809170789,
6.703692809170789,
7.791601108223033,
7.8369202041803225,
7.951907692650894,
7.98389042115583,

8.258376681022987,
8.258376681022987,
8.264818959972613,
8.264818959972613,
8.264818959972613,
8.264818959972613,
8.265620267330194,
8.265620267330194,
9.412842541483705,
9.412842541483705,
10.502716174481437,
10.502716174481437,
0.1665558361704551,
0.0766182276072886,
-0.06331706305100791,
0.03225035272845343,
-0.2265177184784691,
0.6859734408919996,
0.358399999202588,
-0.31490649734707366,
0.03084506070978915,
0.8141179515629021,
-0.05751068028622007,
0.9013204998454583,
-0.325467390457223,
-0.16027539072229496,
0.19756436832536928,
0.17875052394899651,
0.07441990946369173,
-1.7121562848350094,
0.5411900571694856,
0.025994665514066265,
-1.0547594680221428,
-0.3558195097906033,
0.2575282902752545,
-0.32652811387218433,
-0.17477771883072452,
0.05603758809983181,
-0.31531793685177084,
0.24953935818155465,
-0.21945102528047247,
0.016201452486715968,
-0.47139285875704856,
0.8354485366702803,
0.28783787212142514,
-1.483190369052958,
-0.05324490057492157,
0.00808064868141892,
-1.3894868155533049,
0.07648399162880297,
0.28128388531065984,
-0.8032486745370345,
0.9961277979922681,
0.7271807873997648,
-0.2598267503182047,
0.058937213125893145,
1.4176939090014022,
1.5623836495300838,
-1.2281014386266311,
1.627596878633186,
0.150437982354294,
0.017798034487339433,
-0.2091243023446671,
0.08776907917710067,
0.637527078885032,
1.2603122563899933,
1.0910453886717337,
0.001707278941889729,
-0.5366140699752073,
-0.04457743756443993,
0.708269991966563,
1.6757260639955311,
0.6610325406900899,
-0.508228690084442,
-0.5178693625521746,

-0.2955589105377924,
-0.47790822054112786,
0.31077598178920485,
0.020632196003020267,
1.5538854336639996,
1.2816443511581943,
-0.6926864187150775,
0.73698114320343,
0.3645818015611773,
-0.2097443780477125,
0.1421287040679699,
0.052774836076248374,
-0.7306191772967899,
1.107856618529218,
0.3743379513498811,
1.2959169452837032,
0.8858564921663113,
0.03370518740548656,
-1.0860983069970114,
1.2199055901240408,
-1.0630806559423625,
1.3748402618369016,
0.9943926078324806,
0.7116645059379805,
-0.21098608064682703,
0.04581715611274856,
-0.17117312830066983,
1.2229557830916487,
-1.6273428514067598,
1.8508927721454946,
0.851693435560975,
1.0946646161197402,
0.43088380554694883,
-0.6171985422875839,
1.6866137951313547,
-0.2289359575496427,
-0.09152362655526167,
1.6917359096449291,
0.22722692961233917,
0.49434218233553445,
1.2228162165990006,
-0.4070166799423108,
0.00966082758630671,
-0.3344922737632815,
-1.0755692755599169,
1.1846485271761396,
-0.06070659712292082,
0.6384418838463358,
-0.034216786122488324,
-0.6519049362670448,
-0.0019231105719348242,
0.503764046349436,
0.5779702041064313,
-0.8610978040126155,
0.5832069567020347,
0.07954533618121334,
0.016329041774977193],
[0.2383708143795501,
0.8137820253985804,
1.0561141201801563,
1.2070683478267785,
1.9518166906091101,
2.08301823027678,
2.6774459967535513,
2.7270096866449576,
3.7369434259122447,
3.7421546540398443,
4.458327272130387,
4.463102384700506,
4.463102384700506,
4.482690585412051,
4.482694652744643,
4.498978826283554,
6.3580266266370415,
7.886395999997723,

7.886532810158634,
8.060529572035655,
8.06234783983503,
8.127555772505229,
8.130166630032164,
8.42408407139234,
8.433074307328475,
9.412401860498063,
9.420584174895616,
9.429609653004679,
9.434851407347677,
11.283524698808058,
0.19157417616433814,
0.42257952171340063,
0.3129095694283805,
0.28360255750055535,
-0.7229159635345903,
2.545731433831817,
1.1687892956196542,
0.5222949375721274,
0.9482151911772319,
0.28742832079435787,
1.1627391357840349,
-0.27939056132165146,
-0.07574334812551493,
-0.529119211391362,
-0.08653083237786212,
0.042544229108731915,
-0.09154582517805988,
0.2776265525620834,
0.373637405241428,
-0.666296500917012,
-0.729825790392568,
0.36164641825124294,
1.2066181407380305,
0.18327070025757417,
2.2757552060408917,
-0.004789158452516104,
0.49019439516235463,
2.190622553796523,
0.6131206832540379,
1.0987721265230228,
1.9413219378309259,
0.07975169469268534,
0.17531082109327745,
0.01770145552231998,
0.22429808435825893,
1.1460098869067579,
-0.25766861736092417,
0.4386465248337572,
0.7521614434169624,
0.07039918338308593,
-0.23913727928604284,
0.062497116873310714,
0.27575365987954265,
-0.8268952220573135,
0.8599765782432734,
0.6958325703328606,
-0.16792542378671177,
-0.13186811496768225,
0.17745018848846658,
-0.13764108606901534,
-0.34798347931108986,
0.3901205214779397,
-0.12832331396189006,
0.7623260417931323,
-0.08745728181826747,
0.22448567111633413,
0.5103689813486589,
0.020179740460913417,
-0.4639048953982949,
1.8052783293547816,
-0.06895296635033099,
-0.46137869531525183,
-0.7694587159951923,

-0.1507074275764489,
-3.159084078074864,
1.5556721360426484,
0.9802025173412374,
-1.5980562353917622,
0.9974678614222413,
0.11965749551778265,
-0.5964068365118492,
1.00912773091807,
1.2674306548806673,
1.7418320010188646,
-1.2223756485879664,
-0.006270145144671745,
0.7863824204489233,
-0.08410721186616651,
0.13420385894759826,
0.6686584151585806,
-0.7946689395347296,
-1.448768973512125,
-0.4158373205099576,
-0.1003664997933048,
0.5370971852976741,
-0.013369261455621718,
-1.2134070474127567,
2.71822204346588,
-0.8266087910498845,
-0.04274152999618493,
0.627725493238542,
-0.10714625448592663,
1.4055199595750496,
-0.08877591733311402,
-0.6400618055168635,
0.27990653766167567,
0.27230874688701184,
1.8329367216853043,
0.12128377229042309,
0.03719209379786179,
0.7558193864298689,
-0.027085062677120984,
0.7947118159580207,
-0.18810209431195668,
0.008294184744538372,
-0.9110921816475059,
1.3065108206329406,
0.004810940817056788,
1.4030602129547465,
0.03878633149748846,
0.29850504257680605,
-0.9524583070906825,
0.10037229527914636,
0.4277056649647598,
-2.2528322583674503,
0.030724614514628212,
0.8416350822543037,
1.5579013461800135,
0.5696673724928452,
-1.5352194543526818],
[0.3442332088020621,
0.7247295953616149,
0.7247295953616149,
0.7610537291022658,
1.7162580174489597,
2.0119282772467635,
2.0119282772467635,
3.369934702704019,
3.369934702704019,
5.00211845115155,
5.212181786011135,
5.346566249705437,
6.103091648910964,
6.103091648910964,
7.050711873868975,
7.050711873868975,
7.492144583370162,
7.492144583370162,

7.606105711064038,
7.606105711064038,
8.717193051662631,
8.753101318062777,
8.877107151002964,
8.889590522678484,
8.982812044989718,
8.982812044989718,
9.499874357008034,
9.499874357008034,
9.659025604148875,
9.671110590806489,
0.02814413440270558,
0.250593445582179,
0.17496868989230024,
-0.19749991731592192,
-0.5878125517240043,
1.1370895496823794,
0.3339578948727643,
0.30833012680275795,
0.21147160367429677,
0.5518340573364652,
0.9816766119937317,
-1.3427949214182668,
2.2864376119784486,
0.673664283367827,
0.4767745045632036,
-0.47480445681581174,
0.05912449937800317,
0.9513757199218045,
1.079093875588678,
-0.4313388366547843,
0.43347513808131466,
-0.3381632209207048,
-0.42699598389932686,
-1.1528003206699722,
0.2505681368756031,
1.1900947118566725,
-0.38260053073473543,
-0.6405385159573362,
0.08926968472452403,
-0.04834256297702397,
1.603027736914297,
-0.44672812120859223,
-0.5759172165575565,
1.25237714334681,
0.04254918814060746,
-0.002429292667880943,
0.47732275422632153,
-0.10204026491637587,
0.02357869435971368,
0.06910865492863566,
-0.5942663805386644,
1.1629666490831163,
0.28625726751699415,
0.4183193427753825,
0.33026699893767814,
1.8860632823719743,
-0.37929215782635534,
0.8251027249542411,
-0.13937217921728381,
-0.35829122981546885,
-0.11830319301401378,
-1.1307480811476716,
-0.33746884247229475,
1.7161163093033744,
-0.03517175574182253,
-0.4044561448132672,
-0.7301790593832838,
-0.10496992427986328,
-0.0377875273258637,
0.11576685707785253,
-0.9348023198901909,
2.132577954045408,
-0.47199808297503054,

0.09642080758509694,
-1.5590866370005203,
0.10045413530924742,
-0.3215930726850239,
-0.7020784722315541,
1.2376661808413174,
0.5709879203761018,
-0.820670510834149,
0.1634866307612446,
0.06270250983710676,
-0.03570422816362065,
-0.5699745682010601,
-0.26488761312159137,
1.795803611544861,
-0.4017001375917998,
-1.757541020399501,
-0.05132347348943386,
0.8521664539464072,
0.4812990598336637,
-0.7041980923594973,
0.6941905718406797,
1.5589556954504384,
1.9089912382579748,
0.4849438728523755,
0.03939482159422182,
-0.2564597588944596,
-0.16675906759041134,
2.5610800117093855,
0.9927310518963223,
1.3170410278463267,
0.6281459286410694,
-0.07878692694615999,
0.5865838795710373,
0.4894408136056478,
-0.043552207832595204,
0.8988064624972067,
-0.35279792613941824,
1.5088457985023234,
-0.32174328535909336,
0.41978129355629945,
0.014567846908235548,
-0.17446008524494477,
0.35494980160059775,
1.44166715836895,
0.027787407199764528,
-0.2573832607835415,
0.9605798444843694,
-0.08907047543629609,
-0.012116788408097524,
1.2676248549462201,
-0.2512276837340942,
0.7869834357157488,
-2.3607393050467915,
0.1914341630047483,
-1.0542342437530707,
0.8454261921281611,
0.02027118187750096],
[0.02240293444906766,
0.9863710197203618,
1.3708545038264748,
1.3757718928414557,
1.7844795346244002,
2.029643659915566,
2.029643659915566,
4.135298362317422,
4.181946654536538,
4.392782597821499,
4.473609589126697,
4.474471668054438,
4.475838923576404,
4.4785869343507265,
4.4785869343507265,
5.467009291914798,
5.470281162008999,
5.840541346188122,

5.842392372241278,
6.505508038171383,
6.505508038171383,
7.536348099553183,
7.536348099553183,
9.63019412694799,
9.63019412694799,
9.63019412694799,
9.63019412694799,
9.63019412694799,
9.63019412694799,
9.63019412694799,
1.1150074339792337,
-0.03535947987647509,
1.1918755403807781,
-0.20191103850925227,
-0.268362117189328,
-0.3595192771004685,
1.2614204066704717,
-2.9429228213034166,
-1.054810371869961,
0.9594255841252335,
-0.09538831740389547,
0.3965663197177072,
-0.22182226409641906,
-0.2653718580482209,
0.1686090968331403,
1.6516484362324146,
-0.3049608013883917,
0.49165272629600476,
1.9061569916625292,
1.241422732345657,
-0.11527538328521378,
-0.22895223610637053,
-0.06124477487110871,
0.6193220715745159,
1.372932614853267,
-1.1538384701543527,
-0.1139440760455718,
0.16973094341798492,
-0.5943234340378932,
0.21399932498003596,
-0.18156105319883253,
-1.0504702368491705,
0.7232808408606985,
-0.018870518436612777,
-0.29343981279013404,
0.30960592292227684,
0.13511542022922338,
0.5096669954667911,
-1.097576281455009,
0.12047585913747774,
-0.5585503308065869,
1.2279608447421912,
1.3012401380411538,
0.3769592231863052,
-0.5692023175942067,
-0.4567508363134565,
-0.679549924864495,
-0.24647993267323956,
0.6292567546169889,
0.9587256267378846,
1.7268825428143277,
0.1413792259593793,
0.6520451922983643,
0.6152471684019365,
-1.1240317980448378,
-0.9856730455453077,
-1.2455074642265085,
0.2235091354354849,
-0.8870861048674711,
0.5299724059178798,
-0.21820971616414211,
-0.09687180933262383,
-0.9247884680119297,

-0.4566833626656702,
-0.4285291193052864,
-0.6406026728496247,
0.9023578506051637,
0.03748900229192909,
0.7551248935414285,
-0.18980887545414976,
-0.1497604267964229,
-0.4253730408988226,
-0.4157675203773789,
1.677209169409502,
0.2126888490300174,
-0.13481167544410624,
-1.6951458494505263,
-0.006476362637627559,
-0.5490470525961781,
0.3195499617454318,
1.077429849794694,
0.6725724542747005,
-0.10235701747439804,
-0.014129818782246995,
2.382406086835482,
0.008866515589568993,
1.0282503794544984,
-0.6899701634722577,
0.3510134577654823,
0.6918405669356381,
-0.08667968085870373,
-0.0005005962237103964,
1.3805542469189591,
1.011514915181436,
-0.4623191099783579,
1.397645255720587,
-0.7813361449321683,
-0.9940660424170442,
-0.6570734018108508,
1.0171740600492052,
0.14480072689925969,
1.157289980717297,
1.019472554492438,
1.0029573776528096,
-0.20406480565208354,
-0.011076077177880707,
-0.30566521016454584,
-0.4775965459909992,
-0.7158801142606765,
1.5610067764386268,
1.5187465408730505,
0.21158961058436962,
-0.38098352608767794,
-0.046999172842648596,
0.5964180792256618,
-0.6260168044949087,
-0.7215592053840503,
-0.5706802740378752,
0.11995346168986934,
0.17053214926260635],
[0.003408343124459994,
0.0034083431244599804,
0.006710027629051694,
0.2484817728699434,
1.1672656342280407,
1.167470260322433,
3.1453440534904744,
3.166261678218574,
3.2843704266652405,
4.089250903115526,
4.091989893807717,
4.091989893807717,
5.181052167254873,
5.181052167254873,
5.184972473956635,
5.585122024093685,
5.5990893229720795,
5.630355974178803,

5.704160317443757,
8.658831571647385,
8.66505230970618,
8.712765508205756,
8.712765508205756,
8.712765508205756,
9.168875276976076,
9.176734348833309,
9.262471352200725,
9.262471352200725,
10.578606537988184,
10.578606537988184,
1.5905119084430799,
0.8691136234632737,
0.04907693052707525,
-0.23336553986203207,
0.4284777328779486,
2.2866815267075147,
0.1760352312050406,
0.27820168999860917,
2.067264415281462,
0.5755331494929522,
-0.06858856463703264,
-0.7222667597755238,
0.7750924366303688,
-0.278304785792398,
0.5352624789798245,
-1.1702690163303244,
-0.012717570027659048,
-0.07125250327360101,
1.7927644689789062,
1.3752915621891346,
-1.1981440291126808,
-0.30737164793891514,
0.2657743539912098,
0.04156274567190095,
1.1363043468044227,
0.059217925806269944,
0.7608352104096147,
-0.1513784586666425,
0.025696061144845524,
0.861481321747562,
-0.48544656000194114,
-0.006033515622113467,
0.473431667961849,
1.5469119181224316,
0.1871819193791605,
1.555475823471471,
0.9001349633565042,
-1.5961139272878428,
-0.6749613217725859,
-1.5596468027729833,
0.5296186936496604,
-0.8549575199843406,
1.7649719371938613,
1.5569889217595585,
-0.9244334407213903,
-1.5804326845781602,
0.3840271097002098,
-0.8070678340768227,
-0.48791686278645724,
0.13110829311357092,
-2.023238766590243,
-0.8997043584182223,
0.8435119798114825,
0.12108034013815198,
-0.7013572315179941,
0.05813246571029178,
-1.3645229029121433,
-1.704441396215925,
0.6160899239086374,
-1.6580292769388947,
-1.550797918270104,
-0.033040246679508146,
-0.7595754539148374,

0.6024045095197653,
0.446745843265138,
0.9717727329863253,
-1.3190014690683383,
-3.180952616822137,
-0.47685451708111876,
0.03958388950097112,
-0.04894683660460036,
-0.013869967587830365,
0.9715551393376142,
-0.2688733257646762,
-1.7112803723577967,
-0.06346862095088202,
1.2756440828431679,
0.4087049517265435,
-0.4433737535464867,
0.07044159255576982,
0.877714241226598,
-0.33059433004855665,
-0.41683973817974146,
-0.09123983383660174,
1.153451542408442,
1.5381060924707182,
1.4432055127327397,
0.7948381511053778,
0.12126271666707081,
0.3736422837955817,
-0.06595460407520624,
-0.1569219765005705,
1.4053035958769775,
-0.3443177603686477,
0.14604377230684606,
0.733462484703497,
1.2365404639754205,
-0.020105376528163625,
-0.09631958929266826,
-0.16876809298151013,
0.42184799028013376,
0.8803036484986011,
-0.4148292271104988,
0.011464734896518355,
1.0895788780475535,
-0.006017115067308532,
0.6476481463328121,
-0.4792828310008005,
1.7145504838049759,
0.9300252591703124,
1.4147034805691376,
-0.04601267898776636,
-0.34579443398211707,
0.006107561937391137,
2.7608588509789516,
-2.470402787070412,
-0.2193552778493498,
-0.43636419827955597,
0.5036040054264609,
-0.006776779992121664],
[0.5698358581496117,
1.0111463732621893,
1.116254023022104,
1.116254023022104,
1.4543153683590362,
1.4543153683590362,
1.551144943675129,
1.551144943675129,
2.3321866923810566,
2.4857852178760083,
3.837593499897418,
3.837593499897418,
3.837593499897418,
4.847467397840062,
5.03660309473088,
5.176578602594472,
5.176578602594472,
5.214332760543968,

6.547255537065404,
6.547255537065404,
7.083633091144979,
7.083633091144979,
7.133204014188821,
7.133204014188821,
7.133204014188821,
7.937225733864811,
8.370661235857398,
8.62656679928389,
8.759707620277526,
9.424598849529344,
0.24247733923543782,
0.79275118507479,
-0.012314768690710378,
1.2247395896468714,
-0.6087890007814074,
0.9804698093938512,
0.19554007335952167,
-0.18291383394130525,
-0.08254523307317092,
0.18380200005144362,
-0.06242013495608768,
-0.18902966193120999,
0.4846770511497838,
1.2162916157423216,
0.4476502570273604,
-0.5204828708955642,
1.1525916637240337,
0.9814837501696517,
1.2019836737296188,
0.10039944576408244,
-1.165620769823409,
0.5709783345435084,
0.04690539627488982,
0.3150855883896384,
0.6835833386917337,
0.7350671828581666,
0.15105693185753874,
0.7851129846448224,
-0.290237804763004,
0.02628688844012564,
-0.3427838784588314,
-0.12202424723691215,
0.30368921729231485,
-0.08398845633414893,
-0.34421225485285395,
-1.0992299599969957,
-0.6679166311787512,
-0.4925602784337871,
-0.7607905356843834,
-0.061822787578292515,
-0.33643902903801526,
1.7237644621079549,
1.1087838048318122,
1.4391293107545464,
-0.2688865923867924,
0.07238699078201241,
-0.7262166431073663,
-0.8607492696026928,
0.1828620517189283,
-1.5556021786529124,
-0.5230641559705965,
-0.023465829708141782,
0.5773817105800866,
-0.6189512057198875,
-0.1704298219551506,
0.6935490081871019,
0.43024761601627914,
-1.3818053283072975,
0.32069867830077864,
0.11223505487625844,
-0.6642209341325642,
-0.7163563578386999,
0.9533459704680486,

-2.5373567196844773,
-0.3530595330204968,
-0.36744271288254204,
-0.4105700661159024,
-0.529185988706723,
-0.5074466275713505,
0.5969687397167245,
0.1454479675984071,
0.9510340399481164,
-0.16051754924491393,
0.168676911677473,
1.6243103666539225,
-0.3533677275828062,
0.23248347317288842,
1.550100715213732,
0.3540002793004863,
0.11004939052123668,
0.0026309322638186474,
1.0023099891482978,
1.3075603782358884,
0.06614039249633322,
0.12696883216368457,
0.4650745950568721,
0.09288306127511851,
-0.05535633254466518,
1.0893487961438044,
0.1856572010619516,
-0.3369565165204948,
1.1993442715596423,
-0.1637322799431796,
-0.05574034797393315,
-0.6499132821077687,
-1.1854402746414112,
-0.20554021240749126,
-0.697991352679575,
0.7715290435685477,
0.27728540290360154,
-0.7490020286422794,
0.6727458447445039,
-1.3928966895283914,
-1.1123692530480307,
1.2429888713160355,
0.5002310144425463,
1.0425617895549082,
-0.41250228671824735,
-0.20782873770333182,
0.952256354944612,
-0.2180059005687231,
0.8024032545701543,
-1.0975152466418368,
0.6192383355749876,
-0.44420258004771784,
-1.3605235580058401,
1.4196204597477091,
-1.1230853758949062,
0.09475958663430169,
0.9412942473630452],
[0.17961742860022237,
0.4029745783400613,
0.4271020388774394,
0.6998927493507625,
1.1444327445959492,
1.4463150894355827,
1.7859669797147917,
1.7859669797147917,
1.8910076260348616,
2.1568466639380097,
2.184715675260106,
2.3241261579565067,
5.221288254664872,
5.360678146313719,
5.572749542622228,
6.315664711217606,
6.332942041057151,
7.225695385348529,

7.238714694082236,
7.767863290353532,
7.947954629935657,
8.045079231148478,
8.080705184182557,
8.235317994608641,
8.52967785406855,
8.920755162601605,
8.920755162601605,
8.920755162601605,
9.75907480462672,
11.068402571583377,
-0.07026370230905016,
1.2758275954101812,
0.1960569796315372,
-0.7454998985427054,
0.5260870873255556,
1.3632265149428353,
0.06759709485075166,
0.7201883678889174,
0.7780364770341424,
0.7034018811814728,
-0.3281225120527201,
-0.21523884970562404,
0.3525948786581971,
0.10058439953616755,
0.8489242437750127,
0.4102210250930649,
-1.1239274808373552,
-0.07527028057729333,
-0.662823747816413,
-0.09016589559770041,
0.0005078927131911535,
-0.9974030724775158,
1.2973260682530905,
0.6207513445119947,
-0.06959715000945968,
0.01479447508324883,
0.5707366547040263,
0.8024230791007029,
0.11580671533830131,
0.5319238260215059,
2.029703693625361,
0.8476591202353092,
-0.5985490132202251,
-1.0850372047073247,
-0.13885829158476937,
0.6160884575951602,
-0.6119315652443118,
0.25456754270344084,
1.6135141314150674,
1.6101080474118592,
-1.0046888465114787,
-1.2174288348798858,
-0.8801790341822278,
-0.01998793431745051,
-1.007194981866689,
-1.6365882266166358,
3.524880820008825,
1.5399950054830251,
-1.401820445976074,
-0.04945776017633801,
0.9565666812114323,
-0.20490442564626934,
-0.5632271134618189,
1.0033608730635488,
0.12368927199762578,
-0.06153617179775209,
1.0557374829951787,
1.532632985657107,
-0.3069192599686721,
1.8620498010178619,
0.18083984306067874,
-0.9702971127887235,
-0.4866438529829019,

0.2527358034573417,
0.46390718912935613,
0.05253485630073179,
-0.15957905175444287,
0.5064463362180326,
0.6588516997606392,
0.036226022952260704,
-1.5941274572195923,
0.3692191472654626,
0.31840120283705997,
0.8741993879399423,
0.4802653460337955,
0.3012459045041651,
0.5102022084743536,
0.3727970282915065,
0.6665750190815365,
-0.7773615380935764,
0.12852119124454983,
0.6774482070442244,
-0.34235651892527574,
-1.1577831131801597,
-0.2879120353476077,
1.157549789799121,
-0.1818163295634488,
-0.034519908964088415,
-0.6744130553492667,
1.1322140576298956,
0.5044463860518625,
1.5469749343656427,
-0.19451127779487945,
0.16868157432748737,
0.3887158384137858,
1.179393002549757,
0.4377598134949622,
0.42273561999328285,
0.17555703442650725,
0.7637154786900613,
0.3204620558964089,
-1.0127445325141415,
-0.3366137672636463,
-0.6128563234467006,
-0.19959799618607235,
-0.22993024268879514,
0.6548635589470768,
0.5449412421525476,
0.11171602982233311,
0.5174688869584806,
-0.0926785574168733,
-0.012796553414024043,
0.872831179387319,
-0.3738274777631416,
0.708424995234135,
1.1207261796968788,
0.39043722964884914,
1.5466373285390471,
-0.0566526099067629,
-1.544789451209757],
[0.01970245038838479,
0.18403088824160574,
0.23516669011749736,
0.3619340287619915,
0.7559235356843684,
0.8037553588156275,
0.8037553588156274,
0.8037553588156274,
1.523650120002224,
1.5399034674182392,
3.5267025169505333,
3.526702516950533,
4.657784878324055,
4.735234101549794,
4.768060921347247,
6.910522417024203,
6.910522417024203,
7.937298848041795,

7.937298848041795,
7.937298848041795,
7.937298848041795,
7.937298848041795,
7.937298848041795,
7.937298848041795,
8.950027818739875,
8.950027818739875,
8.950027818739875,
8.950027818739875,
8.950027818739875,
9.275481401407285,
0.034552241786943605,
0.7960682992937013,
-0.16960774395822967,
-0.13488653659207303,
-0.07127010448742209,
0.7213919487775996,
0.07517487951530155,
0.7669403129297154,
-0.0980146039609658,
0.2841753594230342,
0.41897129078802076,
0.7644737129881134,
0.21659836679204617,
0.40978346593820614,
0.23386906952753808,
0.6351364889148511,
0.33021983360135304,
0.3518425622667454,
-0.16759580526603354,
1.1955178055405769,
-0.1119172073880175,
-1.6647459335605923,
-0.09499524124538847,
1.171044098528735,
-0.022137095989983098,
0.46479840085551427,
0.5434533112600137,
0.7953483283824695,
-0.23791626316884204,
-0.34327459028867535,
0.9770824990177593,
0.3972109926583832,
0.031305244787475206,
0.06660894739153407,
-0.4270300066285869,
-1.4580475210852422,
0.18899331747125298,
-0.5789031865333153,
0.2203088633995836,
-0.028805094274594364,
-0.21236041446549625,
0.00835869807102751,
1.8444421334304235,
-0.36021419267454113,
-1.0777998782841605,
0.7201309879688491,
-0.07151832135087417,
0.013523038580686776,
-0.5101061898109359,
1.528477364550823,
-0.24055091899624312,
1.595727742687772,
1.181011187425284,
-0.07205139039345279,
-0.3645912646252544,
0.922259297261975,
0.47504473393792157,
-1.4582846498754758,
-1.1041238978674286,
-0.08619812725702276,
1.5348185477516838,
-0.034799821636765677,
0.34854563969795865,

[illegible]

```
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.17961742860022237,
0.4029745783400613,
0.4271020388774394,
0.6998927493507625,
1.1444327445959492,
1.4463150894355827,
1.7859669797147917,
1.7859669797147917,
1.8910076260348616,
2.1568466639380097,
2.184715675260106,
2.3241261579565067,
5.221288254664872,
5.360678146313719,
5.572749542622228,
6.315664711217606,
6.332942041057151,
7.225695385348529,
7.238714694082236,
7.767863290353532,
7.947954629935657,
8.045079231148478,
8.080705184182557,
8.235317994608641,
8.52967785406855,
8.920755162601605,
8.920755162601605,
8.920755162601605,
9.75907480462672,
11.068402571583377,
-0.07026370230905016,
1.2758275954101812,
0.1960569796315372,
-0.7454998985427054,
0.5260870873255556,
1.3632265149428353,
0.06759709485075166,
0.7201883678889174,
0.7780364770341424,
0.7034018811814728,
-0.3281225120527201,
-0.21523884970562404,
0.3525948786581971,
0.10058439953616755,
0.8489242437750127,
0.4102210250930649,
-1.1239274808373552,
-0.07527028057729333,
-0.662823747816413,
-0.09016589559770041,
0.0005078927131911535,
-0.9974030724775158,
1.2973260682530905,
0.6207513445119947,
-0.06959715000945968,
0.01479447508324883,
0.5707366547040263,
0.8024230791007029,
0.11580671533830131,
0.5319238260215059,
2.029703693625361,
0.8476591202353092,
-0.5985490132202251,
-1.0850372047073247,
-0.13885829158476937,
0.6160884575951602,
-0.6119315652443118,
0.25456754270344084,
1.6135141314150674,
1.6101080474118592,
-1.0046888465114787,
-1.2174288348798858,
-0.8801790341822278,
-0.01998793431745051,
```

-1.007194981866689,
-1.6365882266166358,
3.524880820008825,
1.5399950054830251,
-1.401820445976074,
-0.04945776017633801,
0.9565666812114323,
-0.20490442564626934,
-0.5632271134618189,
1.0033608730635488,
0.12368927199762578,
-0.06153617179775209,
1.0557374829951787,
1.532632985657107,
-0.3069192599686721,
1.8620498010178619,
0.18083984306067874,
-0.9702971127887235,
-0.4866438529829019,
0.2527358034573417,
0.46390718912935613,
0.05253485630073179,
-0.15957905175444287,
0.5064463362180326,
0.6588516997606392,
0.036226022952260704,
-1.5941274572195923,
0.3692191472654626,
0.31840120283705997,
0.8741993879399423,
0.4802653460337955,
0.3012459045041651,
0.5102022084743536,
0.3727970282915065,
0.6665750190815365,
-0.7773615380935764,
0.12852119124454983,
0.6774482070442244,
-0.34235651892527574,
-1.1577831131801597,
-0.2879120353476077,
1.157549789799121,
-0.1818163295634488,
-0.034519908964088415,
-0.6744130553492667,
1.1322140576298956,
0.5044463860518625,
1.5469749343656427,
-0.19451127779487945,
0.16868157432748737,
0.3887158384137858,
1.179393002549757,
0.4377598134949622,
0.42273561999328285,
0.17555703442650725,
0.7637154786900613,
0.3204620558964089,
-1.0127445325141415,
-0.3366137672636463,
-0.6128563234467006,
-0.19959799618607235,
-0.22993024268879514,
0.6548635589470768,
0.5449412421525476,
0.11171602982233311,
0.5174688869584806,
-0.0926785574168733,
-0.012796553414024043,
0.872831179387319,
-0.3738274777631416,
0.708424995234135,
1.1207261796968788,
0.39043722964884914,
1.5466373285390471,
-0.0566526099067629,

```

-1.544789451209757]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                  'noise_prob': [0.001, 0.001, 0, 0],
                  'status': True,
                  'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
               'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_xy_noisefree_time_evo_ric4_unifrom_scaling_sample#9'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.395384341095474],
                          [12, 6, 0, 0, -5.790856073500346],
                          [20, 10, 0, 0, -4.565524427603155],
                          [28, 14, 0, 0, -3.622528542539368]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.172325134277344e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.370116227049929,
                           'others': { 'beta_coeffiecients': [ 2.1875,
                                                                -2.1875,
                                                                1.3125,
                                                                -0.3125],
                                       'cost_richardson_zne': 11.390625,
                                       'sorted_expectation_vals': [
-7.395384341095474,
-5.790856073500346,
-4.565524427603155,
-3.622528542539368],

'sorted_noise': [6, 18, 30, 42]}},
'sampled data': [ [ 4,
                    2,
                    0,
                    0,
                    -7.395384341095474],
                  [ 12,
                    6,
                    0,
                    0,
                    -5.790856073500346],
                  [ 20,
                    10,
                    0,
                    0,
                    -4.565524427603155],
                  [ 28,

```

```

14,
0,
0,
-3.622528542539368]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.02240293444906766,
0.9863710197203618,
1.3708545038264748,
1.3757718928414557,
1.7844795346244002,
2.029643659915566,
2.029643659915566,
4.135298362317422,
4.181946654536538,
4.392782597821499,
4.473609589126697,
4.474471668054438,
4.475838923576404,
4.4785869343507265,
4.4785869343507265,
5.467009291914798,
5.470281162008999,
5.840541346188122,
5.842392372241278,
6.505508038171383,
6.505508038171383,
7.536348099553183,
7.536348099553183,
9.63019412694799,
9.63019412694799,
9.63019412694799,
9.63019412694799,
9.63019412694799,
9.63019412694799,
9.63019412694799,
1.1150074339792337,
-0.03535947987647509,
1.1918755403807781,
-0.20191103850925227,
-0.268362117189328,
-0.3595192771004685,
1.2614204066704717,
-2.9429228213034166,
-1.054810371869961,
0.9594255841252335,
-0.09538831740389547,
0.3965663197177072,
-0.22182226409641906,
-0.2653718580482209,
0.1686090968331403,
1.6516484362324146,
-0.3049608013883917,
0.49165272629600476,
1.9061569916625292,
1.241422732345657,
-0.11527538328521378,
-0.22895223610637053,

```

-0.06124477487110871,
0.6193220715745159,
1.372932614853267,
-1.1538384701543527,
-0.1139440760455718,
0.16973094341798492,
-0.5943234340378932,
0.21399932498003596,
-0.18156105319883253,
-1.0504702368491705,
0.7232808408606985,
-0.018870518436612777,
-0.29343981279013404,
0.30960592292227684,
0.13511542022922338,
0.5096669954667911,
-1.097576281455009,
0.12047585913747774,
-0.5585503308065869,
1.2279608447421912,
1.3012401380411538,
0.3769592231863052,
-0.5692023175942067,
-0.4567508363134565,
-0.679549924864495,
-0.24647993267323956,
0.6292567546169889,
0.9587256267378846,
1.7268825428143277,
0.1413792259593793,
0.6520451922983643,
0.6152471684019365,
-1.1240317980448378,
-0.9856730455453077,
-1.2455074642265085,
0.2235091354354849,
-0.8870861048674711,
0.5299724059178798,
-0.21820971616414211,
-0.09687180933262383,
-0.9247884680119297,
-0.4566833626656702,
-0.4285291193052864,
-0.6406026728496247,
0.9023578506051637,
0.03748900229192909,
0.7551248935414285,
-0.18980887545414976,
-0.1497604267964229,
-0.4253730408988226,
-0.4157675203773789,
1.677209169409502,
0.2126888490300174,
-0.13481167544410624,
-1.6951458494505263,
-0.006476362637627559,
-0.5490470525961781,
0.3195499617454318,
1.077429849794694,
0.6725724542747005,
-0.10235701747439804,
-0.014129818782246995,
2.382406086835482,
0.008866515589568993,
1.0282503794544984,
-0.6899701634722577,
0.3510134577654823,
0.6918405669356381,
-0.08667968085870373,
-0.0005005962237103964,
1.3805542469189591,
1.011514915181436,
-0.4623191099783579,
1.397645255720587,
-0.7813361449321683,

[illegible]

```
-7.259703059772281,  
-5.766278116709839,  
-4.583193384345111,  
-3.640157125717274],
```

```
'sorted_expectation_vals': [  
  
    'sorted_noise': [6, 18, 30, 42]],  
    'sampled data': [ [ 4,  
                        2,  
                        0,  
                        0,  
                        -7.259703059772281],  
                      [ 12,  
                        6,  
                        0,  
                        0,  
                        -5.766278116709839],  
                      [ 20,  
                        10,  
                        0,  
                        0,  
                        -4.583193384345111],  
                      [ 28,  
                        14,  
                        0,  
                        0,  
                        -3.640157125717274]],  
    'sampling': 'default']},  
{ 'config': { 'ansatz': { 'gateset': 1,  
                          'layer': 30,  
                          'ugate': { 'coefficients': { 'bn': [ 0,  
                                                             0,  
                                                             0,  
                                                             0,  
                                                             0,  
                                                             0],  
                                                             'cn': [ 0.5,  
                                                             0.5,  
                                                             0.5,  
                                                             0.5,  
                                                             0.5],  
                                                             'r': 0},  
                          'time': {'max': 10.0, 'min': 0.0},  
                          'type': 'xy-iss'}},  
  'init_param': { 'value': [ 0.3442332088020621,  
                             0.7247295953616149,  
                             0.7247295953616149,  
                             0.7610537291022658,  
                             1.7162580174489597,  
                             2.0119282772467635,  
                             2.0119282772467635,  
                             3.369934702704019,  
                             3.369934702704019,  
                             5.00211845115155,  
                             5.212181786011135,  
                             5.346566249705437,  
                             6.103091648910964,  
                             6.103091648910964,  
                             7.050711873868975,  
                             7.050711873868975,  
                             7.492144583370162,  
                             7.492144583370162,  
                             7.606105711064038,  
                             7.606105711064038,  
                             8.717193051662631,  
                             8.753101318062777,  
                             8.877107151002964,  
                             8.889590522678484,  
                             8.982812044989718,  
                             8.982812044989718,  
                             9.499874357008034,  
                             9.499874357008034,  
                             9.659025604148875,  
                             9.671110590806489,
```

0.02814413440270558,
0.250593445582179,
0.17496868989230024,
-0.19749991731592192,
-0.5878125517240043,
1.1370895496823794,
0.3339578948727643,
0.30833012680275795,
0.21147160367429677,
0.5518340573364652,
0.9816766119937317,
-1.3427949214182668,
2.2864376119784486,
0.673664283367827,
0.4767745045632036,
-0.47480445681581174,
0.05912449937800317,
0.9513757199218045,
1.079093875588678,
-0.4313388366547843,
0.43347513808131466,
-0.3381632209207048,
-0.42699598389932686,
-1.1528003206699722,
0.2505681368756031,
1.1900947118566725,
-0.38260053073473543,
-0.6405385159573362,
0.08926968472452403,
-0.04834256297702397,
1.603027736914297,
-0.44672812120859223,
-0.5759172165575565,
1.25237714334681,
0.04254918814060746,
-0.002429292667880943,
0.47732275422632153,
-0.10204026491637587,
0.02357869435971368,
0.06910865492863566,
-0.5942663805386644,
1.1629666490831163,
0.28625726751699415,
0.4183193427753825,
0.33026699893767814,
1.8860632823719743,
-0.37929215782635534,
0.8251027249542411,
-0.13937217921728381,
-0.35829122981546885,
-0.11830319301401378,
-1.1307480811476716,
-0.33746884247229475,
1.7161163093033744,
-0.03517175574182253,
-0.4044561448132672,
-0.7301790593832838,
-0.10496992427986328,
-0.0377875273258637,
0.11576685707785253,
-0.9348023198901909,
2.132577954045408,
-0.47199808297503054,
0.09642080758509694,
-1.5590866370005203,
0.10045413530924742,
-0.3215930726850239,
-0.7020784722315541,
1.2376661808413174,
0.5709879203761018,
-0.820670510834149,
0.1634866307612446,
0.06270250983710676,
-0.03570422816362065,
-0.5699745682010601,

```

-0.26488761312159137,
1.795803611544861,
-0.4017001375917998,
-1.757541020399501,
-0.05132347348943386,
0.8521664539464072,
0.4812990598336637,
-0.7041980923594973,
0.6941905718406797,
1.5589556954504384,
1.9089912382579748,
0.4849438728523755,
0.03939482159422182,
-0.2564597588944596,
-0.16675906759041134,
2.5610800117093855,
0.9927310518963223,
1.3170410278463267,
0.6281459286410694,
-0.07878692694615999,
0.5865838795710373,
0.4894408136056478,
-0.043552207832595204,
0.8988064624972067,
-0.35279792613941824,
1.5088457985023234,
-0.32174328535909336,
0.41978129355629945,
0.014567846908235548,
-0.17446008524494477,
0.35494980160059775,
1.44166715836895,
0.027787407199764528,
-0.2573832607835415,
0.9605798444843694,
-0.08907047543629609,
-0.012116788408097524,
1.2676248549462201,
-0.2512276837340942,
0.7869834357157488,
-2.3607393050467915,
0.1914341630047483,
-1.0542342437530707,
0.8454261921281611,
0.02027118187750096]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_xy_noisefree_time_evo_ric4_unifrom_scaling_sample#5'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},

```

```

'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.407206817517893],
                          [12, 6, 0, 0, -5.829141582765497],
                          [20, 10, 0, 0, -4.618573032874505],
                          [28, 14, 0, 0, -3.681441154365551]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.2438507080078125e-05,
            'zne_values': { 'degree': 0,
                          'extrapolated_value': -8.363444445929417,
                          'others': { 'beta_coeffieicients': [ 2.1875,
                                                                -2.1875,
                                                                1.3125,
                                                                -0.3125],
                                      'cost_richardson_zne': 11.390625,
                                      'sorted_expectation_vals': [
-7.407206817517893,
-5.829141582765497,
-4.618573032874505,
-3.681441154365551],
                                                                'sorted_noise': [6, 18, 30, 42]},
                          'sampled data': [ [ 4,
                                                2,
                                                0,
                                                0,
                                                -7.407206817517893],
                                              [ 12,
                                                6,
                                                0,
                                                0,
                                                -5.829141582765497],
                                              [ 20,
                                                10,
                                                0,
                                                0,
                                                -4.618573032874505],
                                              [ 28,
                                                14,
                                                0,
                                                0,
                                                -3.681441154365551]],
                          'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [ 0,
                                                              0,
                                                              0,
                                                              0,
                                                              0,
                                                              0,
                                                              0],
                                                            'cn': [ 0.5,
                                                                  0.5,
                                                                  0.5,
                                                                  0.5,
                                                                  0.5,
                                                                  0.5],
                                                            'r': 0},
                        'time': {'max': 10.0, 'min': 0.0},
                        'type': 'xy-iss'}},
  'init_param': { 'value': [ 0.5482907189468895,
                             0.6413620809492503,
                             0.6422482149552934,
                             1.5892409709442974,
                             1.850708229814058,
                             1.850708229814058,
                             1.8745552660050262,
                             2.400679867939261,

```

2.402212278134192,
2.485190345183722,
2.485190345183722,
2.4851903451837223,
2.488397493491329,
2.4883974934913287,
5.077476636438464,
5.077476636438464,
5.732990197214016,
5.806868622878696,
5.817834245705687,
6.166757373752113,
7.3114292874931905,
7.311858449036949,
8.346734585466223,
8.350435170229554,
8.354614694314481,
8.35659817980371,
8.356905245901928,
9.313534396314816,
9.313534396314816,
10.443787047737276,
-0.5971318801423705,
-0.6830583120471296,
0.053446928626708114,
-1.488251087277441,
0.7409998345617078,
-0.020561007687691084,
-0.47029189134787985,
-0.9187802118512801,
-0.07390570205801456,
0.6091612085770479,
0.5696050095749661,
-2.634874245900948,
0.14733366015521565,
1.0781100436188797,
0.3459269460772517,
1.2820636645787713,
-1.605623121104658,
0.856791024529966,
0.16374864437029604,
-0.7377840957855112,
0.7603752030179747,
0.4958590077418653,
1.0293890027768824,
0.4862585296205842,
-1.0838654128292502,
1.2632637762941026,
-1.1043454462318527,
-0.7867654190654241,
0.8818217647037212,
1.0406267750063978,
0.05770567519039498,
-0.5922070722918433,
-0.3271403288282926,
0.5980786503295974,
-0.02050375112330311,
-0.9007241609880184,
0.5239405469453117,
0.7601496703420288,
-0.6449049942439851,
1.7340858429103823,
0.10050113333288421,
0.3143347863780438,
0.28408386490618526,
-0.1222389867527315,
-0.09502282329152913,
-1.3004368662949968,
0.16856179320059875,
0.6790103402211991,
0.3355977478652572,
0.0935957266804422,
0.42936820813464754,
0.5148227650107633,
0.10764356588532305,

```

-0.054251737808208085,
0.06525040933822983,
-0.21143245853075732,
-1.5801988491404158,
0.023513208971767067,
1.0921277689034055,
-0.11501776583679489,
-0.14182911234947893,
1.8716225858725093,
-1.0747761250482988,
-0.01873186129021693,
0.9952348515577564,
1.5589645169221944,
2.4625123021717044,
1.5320934484141422,
0.6264578101560674,
0.015802850908690148,
0.6559295719723718,
0.5030706883609005,
0.010357532831659657,
-0.6971755064049523,
-1.334963536555068,
0.05714132580252865,
0.9478292633237722,
1.541635767019547,
-1.2357775558070643,
1.5753066317572375,
1.2206753513411082,
-0.013882574638756213,
2.102115305768464,
0.6867648509985108,
-1.0810617894892982,
0.608205793929093,
-1.0701207529796843,
-0.008804617073740417,
1.0802724219136723,
1.4472704941581649,
-0.525983160453369,
0.10347757085774918,
0.07451664808136123,
0.8113743025574697,
-0.19830571509994802,
0.20760359505818776,
-0.5709476706639499,
0.1791350390980861,
-0.32061652775092425,
-1.6382844876022806,
2.271046687924862,
0.09972766177435569,
1.671157305178672,
1.0596955078067567,
-0.7395801076185267,
-0.8217251410877643,
1.2660358026872456,
-0.149859182069558,
1.7386861885389688,
-0.003505497297480453,
0.677126408740769,
-1.0541684899696007,
0.9437251450981726,
1.7354017388586316,
-0.8022872949930118,
0.01881044769021433,
-0.013699411839369284,
1.5965466699123847,
0.4486397166383021,
1.5669137279484207]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,

```

```

1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_xy_noisefree_time_evo_ric4_unifrom_scaling_sample#1'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0],
[3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.463351332654992],
[12, 6, 0, 0, -5.841676652164559],
[20, 10, 0, 0, -4.604434423168805],
[28, 14, 0, 0, -3.651723685315627]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 4.410743713378906e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -8.449569892320747,
'others': { 'beta_coeffieicients': [ 2.1875,
-2.1875,
1.3125,
-0.3125],
'cost_richardson_zne': 11.390625,
'sorted_expectation_vals': [
-7.463351332654992,
-5.841676652164559,
-4.604434423168805,
-3.651723685315627],
'sorted_noise': [6, 18, 30, 42]}},
'sampled data': [ [ 4,
2,
0,
0,
-7.463351332654992],
[ 12,
6,
0,
0,
-5.841676652164559],
[ 20,
10,
0,
0,
-4.604434423168805],
[ 28,
14,
0,
0,
-3.651723685315627]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,

```



```
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.2383708143795501,
0.8137820253985804,
1.0561141201801563,
1.2070683478267785,
1.9518166906091101,
2.08301823027678,
2.6774459967535513,
2.7270096866449576,
3.7369434259122447,
3.7421546540398443,
4.458327272130387,
4.463102384700506,
4.463102384700506,
4.482690585412051,
4.482694652744643,
4.498978826283554,
6.3580266266370415,
7.88639599997723,
7.886532810158634,
8.060529572035655,
8.06234783983503,
8.127555772505229,
8.130166630032164,
8.42408407139234,
8.433074307328475,
9.412401860498063,
9.420584174895616,
9.429609653004679,
9.434851407347677,
11.283524698808058,
0.19157417616433814,
0.42257952171340063,
0.3129095694283805,
0.28360255750055535,
-0.7229159635345903,
2.545731433831817,
1.1687892956196542,
0.5222949375721274,
0.9482151911772319,
0.28742832079435787,
1.1627391357840349,
-0.27939056132165146,
-0.07574334812551493,
-0.529119211391362,
-0.08653083237786212,
0.042544229108731915,
-0.09154582517805988,
0.2776265525620834,
0.373637405241428,
-0.666296500917012,
-0.729825790392568,
0.36164641825124294,
1.2066181407380305,
0.18327070025757417,
2.2757552060408917,
-0.004789158452516104,
0.49019439516235463,
2.190622553796523,
0.6131206832540379,
1.0987721265230228,
1.9413219378309259,
```

0.07975169469268534,
0.17531082109327745,
0.01770145552231998,
0.22429808435825893,
1.1460098869067579,
-0.25766861736092417,
0.4386465248337572,
0.7521614434169624,
0.07039918338308593,
-0.23913727928604284,
0.062497116873310714,
0.27575365987954265,
-0.8268952220573135,
0.8599765782432734,
0.6958325703328606,
-0.16792542378671177,
-0.13186811496768225,
0.17745018848846658,
-0.13764108606901534,
-0.34798347931108986,
0.3901205214779397,
-0.12832331396189006,
0.7623260417931323,
-0.08745728181826747,
0.22448567111633413,
0.5103689813486589,
0.020179740460913417,
-0.4639048953982949,
1.8052783293547816,
-0.06895296635033099,
-0.46137869531525183,
-0.7694587159951923,
-0.1507074275764489,
-3.159084078074864,
1.5556721360426484,
0.9802025173412374,
-1.5980562353917622,
0.9974678614222413,
0.11965749551778265,
-0.5964068365118492,
1.00912773091807,
1.2674306548806673,
1.7418320010188646,
-1.2223756485879664,
-0.006270145144671745,
0.7863824204489233,
-0.08410721186616651,
0.13420385894759826,
0.6686584151585806,
-0.7946689395347296,
-1.448768973512125,
-0.4158373205099576,
-0.1003664997933048,
0.5370971852976741,
-0.013369261455621718,
-1.2134070474127567,
2.71822204346588,
-0.8266087910498845,
-0.04274152999618493,
0.627725493238542,
-0.10714625448592663,
1.4055199595750496,
-0.08877591733311402,
-0.6400618055168635,
0.27990653766167567,
0.27230874688701184,
1.8329367216853043,
0.12128377229042309,
0.03719209379786179,
0.7558193864298689,
-0.027085062677120984,
0.7947118159580207,
-0.18810209431195668,
0.008294184744538372,
-0.9110921816475059,

```

1.3065108206329406,
0.004810940817056788,
1.4030602129547465,
0.03878633149748846,
0.29850504257680605,
-0.9524583070906825,
0.10037229527914636,
0.4277056649647598,
-2.2528322583674503,
0.030724614514628212,
0.8416350822543037,
1.5579013461800135,
0.5696673724928452,
-1.5352194543526818]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_xy_noisefree_time_evo_ric4_unifrom_scaling_sample#4'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.418732841657523],
                          [12, 6, 0, 0, -5.8155731075176975],
                          [20, 10, 0, 0, -4.600391600363878],
                          [28, 14, 0, 0, -3.6713467271615334]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.315376281738281e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.397630041670476,
                           'others': { 'beta_coeffieicients': [ 2.1875,
                                                                -2.1875,
                                                                1.3125,
                                                                -0.3125],
                                         'cost_richardson_zne': 11.390625,
                                         'sorted_expectation_vals': [
-7.418732841657523,
-5.8155731075176975,
-4.600391600363878,
-3.6713467271615334],

'sorted_noise': [6, 18, 30, 42]],
'sampled data': [ [ 4,
                    2,
                    0,

```

```

0,
-7.418732841657523],
[ 12,
6,
0,
0,
-5.8155731075176975],
[ 20,
10,
0,
0,
-4.600391600363878],
[ 28,
14,
0,
0,
-3.6713467271615334]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ -3.641081222961908e-17,
0.6149788059529836,
0.7190202634465676,
1.227117923902744,
1.282995668258181,
1.320146784613617,
1.320146784613617,
3.1464089486715623,
3.1464089486715623,
5.021180125711677,
5.021180125711677,
5.021180125711677,
6.703692809170789,
6.703692809170789,
7.791601108223033,
7.8369202041803225,
7.951907692650894,
7.98389042115583,
8.258376681022987,
8.258376681022987,
8.264818959972613,
8.264818959972613,
8.264818959972613,
8.264818959972613,
8.265620267330194,
8.265620267330194,
9.412842541483705,
9.412842541483705,
10.502716174481437,
10.502716174481437,
0.1665558361704551,
0.0766182276072886,
-0.06331706305100791,
0.03225035272845343,
-0.2265177184784691,
0.6859734408919996,
0.358399999202588,
-0.31490649734707366,
0.03084506070978915,

```

0.8141179515629021,
-0.05751068028622007,
0.9013204998454583,
-0.325467390457223,
-0.16027539072229496,
0.19756436832536928,
0.17875052394899651,
0.07441990946369173,
-1.7121562848350094,
0.5411900571694856,
0.025994665514066265,
-1.0547594680221428,
-0.3558195097906033,
0.2575282902752545,
-0.32652811387218433,
-0.17477771883072452,
0.05603758809983181,
-0.31531793685177084,
0.24953935818155465,
-0.21945102528047247,
0.016201452486715968,
-0.47139285875704856,
0.8354485366702803,
0.28783787212142514,
-1.483190369052958,
-0.05324490057492157,
0.00808064868141892,
-1.3894868155533049,
0.07648399162880297,
0.28128388531065984,
-0.8032486745370345,
0.9961277979922681,
0.7271807873997648,
-0.2598267503182047,
0.058937213125893145,
1.4176939090014022,
1.5623836495300838,
-1.2281014386266311,
1.627596878633186,
0.150437982354294,
0.017798034487339433,
-0.2091243023446671,
0.08776907917710067,
0.637527078885032,
1.2603122563899933,
1.0910453886717337,
0.001707278941889729,
-0.5366140699752073,
-0.04457743756443993,
0.708269991966563,
1.6757260639955311,
0.6610325406900899,
-0.508228690084442,
-0.5178693625521746,
-0.2955589105377924,
-0.47790822054112786,
0.31077598178920485,
0.020632196003020267,
1.5538854336639996,
1.2816443511581943,
-0.6926864187150775,
0.73698114320343,
0.3645818015611773,
-0.2097443780477125,
0.1421287040679699,
0.052774836076248374,
-0.7306191772967899,
1.107856618529218,
0.3743379513498811,
1.2959169452837032,
0.8858564921663113,
0.03370518740548656,
-1.0860983069970114,
1.2199055901240408,
-1.0630806559423625,

[illegible]

-7.47570078392518,
-5.921149282782478,
-4.7240834286119355,
-3.7949056030968538],

```
        [28, 14, 0, 0, -3.7949056030968538]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 3.886222839355469e-05,
            'zne_values': { 'degree': 0,
                            'extrapolated_value': -8.415032907835059,
                            'others': { 'beta_coefficients': [ 2.1875,
                                                                -2.1875,
                                                                1.3125,
                                                                -0.3125],
                            'cost_richardson_zne': 11.390625,
                            'sorted_expectation_vals': [
-7.47570078392518,
-5.921149282782478,
-4.7240834286119355,
-3.7949056030968538],
                            'sorted_noise': [6, 18, 30, 42]},
            'sampled_data': [ [4, 2, 0, 0, -7.47570078392518],
                              [ 12,
                                6,
                                0,
                                0,
                                -5.921149282782478],
                              [ 20,
                                10,
                                0,
                                0,
                                -4.7240834286119355],
                              [ 28,
                                14,
                                0,
                                0,
                                -3.7949056030968538]],
            'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                                'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'xy-iss'}},
  'init_param': { 'value': [ 0.5698358581496117,
                              1.0111463732621893,
                              1.116254023022104,
                              1.116254023022104,
                              1.4543153683590362,
                              1.4543153683590362,
                              1.551144943675129,
                              1.551144943675129,
                              2.3321866923810566,
                              2.4857852178760083,
                              3.837593499897418,
                              3.837593499897418,
                              3.837593499897418,
                              4.847467397840062,
                              5.03660309473088,
                              5.176578602594472,
                              5.176578602594472,
                              5.214332760543968,
                              6.547255537065404,
                              6.547255537065404,
                              7.083633091144979,
```

7.083633091144979,
7.133204014188821,
7.133204014188821,
7.133204014188821,
7.937225733864811,
8.370661235857398,
8.62656679928389,
8.759707620277526,
9.424598849529344,
0.24247733923543782,
0.79275118507479,
-0.012314768690710378,
1.2247395896468714,
-0.6087890007814074,
0.9804698093938512,
0.19554007335952167,
-0.18291383394130525,
-0.08254523307317092,
0.18380200005144362,
-0.06242013495608768,
-0.18902966193120999,
0.4846770511497838,
1.2162916157423216,
0.4476502570273604,
-0.5204828708955642,
1.1525916637240337,
0.9814837501696517,
1.2019836737296188,
0.10039944576408244,
-1.165620769823409,
0.5709783345435084,
0.04690539627488982,
0.3150855883896384,
0.6835833386917337,
0.7350671828581666,
0.15105693185753874,
0.7851129846448224,
-0.290237804763004,
0.02628688844012564,
-0.3427838784588314,
-0.12202424723691215,
0.30368921729231485,
-0.08398845633414893,
-0.34421225485285395,
-1.0992299599969957,
-0.6679166311787512,
-0.4925602784337871,
-0.7607905356843834,
-0.061822787578292515,
-0.33643902903801526,
1.7237644621079549,
1.1087838048318122,
1.4391293107545464,
-0.2688865923867924,
0.07238699078201241,
-0.7262166431073663,
-0.8607492696026928,
0.1828620517189283,
-1.5556021786529124,
-0.5230641559705965,
-0.023465829708141782,
0.5773817105800866,
-0.6189512057198875,
-0.1704298219551506,
0.6935490081871019,
0.43024761601627914,
-1.3818053283072975,
0.32069867830077864,
0.11223505487625844,
-0.6642209341325642,
-0.7163563578386999,
0.9533459704680486,
-2.5373567196844773,
-0.3530595330204968,
-0.36744271288254204,


```

-0.4105700661159024,
-0.529185988706723,
-0.5074466275713505,
0.5969687397167245,
0.1454479675984071,
0.9510340399481164,
-0.16051754924491393,
0.168676911677473,
1.6243103666539225,
-0.3533677275828062,
0.23248347317288842,
1.550100715213732,
0.3540002793004863,
0.11004939052123668,
0.0026309322638186474,
1.0023099891482978,
1.3075603782358884,
0.06614039249633322,
0.12696883216368457,
0.4650745950568721,
0.09288306127511851,
-0.05535633254466518,
1.0893487961438044,
0.1856572010619516,
-0.3369565165204948,
1.1993442715596423,
-0.1637322799431796,
-0.05574034797393315,
-0.6499132821077687,
-1.1854402746414112,
-0.20554021240749126,
-0.697991352679575,
0.7715290435685477,
0.27728540290360154,
-0.7490020286422794,
0.6727458447445039,
-1.3928966895283914,
-1.1123692530480307,
1.2429888713160355,
0.5002310144425463,
1.0425617895549082,
-0.41250228671824735,
-0.20782873770333182,
0.952256354944612,
-0.2180059005687231,
0.8024032545701543,
-1.0975152466418368,
0.6192383355749876,
-0.44420258004771784,
-1.3605235580058401,
1.4196204597477091,
-1.1230853758949062,
0.09475958663430169,
0.9412942473630452]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},

```

[illegible]

```
'type': 'xy-iss'}},
'init_param': { 'value': [ 4.1837705414928806e-17,
1.1742327239243593,
1.1742327239243593,
1.174232723924359,
1.3021435979041922,
1.3883521076730898,
1.4901272631800928,
1.490127263180093,
1.490127263180093,
2.542545188872923,
2.56739133991554,
2.677178516908646,
5.624946341909109,
5.624946341909109,
5.63924283183031,
5.672838193807689,
5.672838193807689,
5.672838193807689,
5.672838193807689,
5.672838193807689,
5.672838193807689,
5.805507018952099,
5.858678454346443,
7.374841183417447,
7.374841183417447,
9.078346087689514,
9.17910617229311,
9.185363997877552,
9.550522060955204,
9.715006087156702,
0.5380723684568228,
0.20346632529178696,
-0.28402043001150323,
-0.013694659434132353,
-0.8491217472851175,
-0.13891219530245408,
-0.7902948900320785,
0.02035105936303376,
0.35891510624171696,
0.07011262472067725,
0.2571910630887441,
0.0972780792224965,
-0.4253506785797692,
-0.08201618636277365,
-0.5629942371157008,
0.4984357070329931,
0.22460606922422946,
0.6436861273901548,
0.5581387866543217,
0.6438048424266741,
0.6625810608832491,
-0.2877740698749285,
0.03400434270310955,
0.6598198228171638,
0.15695492486622947,
-0.23277712193728803,
0.3379436543817524,
0.6679629986711755,
0.4913802101172593,
-0.6543304626041586,
1.2398324170787045,
0.10868488610295632,
0.4726806257545104,
-1.0278416218016544,
-0.5660236088762642,
-0.06627186605932975,
0.3527547117645279,
0.3944014355355292,
0.33709190475341266,
0.16707772676088822,
0.29195080738380563,
1.0354529601392413,
0.12500585138196232,
-0.037178956773431956,
```

0.7823218278561086,
-0.04464215343745706,
-0.36923576305262273,
-0.9863817864437636,
0.039519076991602065,
-0.35646400258617383,
0.04542480304419608,
-0.46125271135496354,
0.40472620693719735,
0.7802480661953589,
-0.6757172179201896,
0.6396142241310521,
0.2327897749192615,
0.8954622299286413,
0.7158405805761733,
0.1382139633537826,
1.0616928944303126,
-0.6122315420380016,
-0.6738539946512315,
1.0433459792659807,
0.494526070112864,
0.20621322780405107,
-0.05318175489586881,
-0.11213659998229918,
0.32941262943060273,
0.13931786347259914,
-0.07640781308377202,
0.31450116184408605,
0.3658109215963646,
0.04460349089074807,
-0.12913310998017427,
0.31890485610090735,
-0.005491508360761248,
1.2692063073436923,
-0.3006174703747285,
-0.41404661321397057,
-0.5199199797590206,
0.44225909618051784,
-0.4233007342169607,
0.7456354776212751,
-0.32484988424610756,
0.9268151291603125,
0.8385591284905366,
0.027219364270413358,
-0.5520387125082392,
1.3757901840365965,
0.02933419406725825,
-0.2915193842748469,
0.33917094344303517,
0.10828106345120646,
-0.5650186064180197,
-0.6041651963720519,
-0.19230834783869816,
1.9848826966849449,
0.0632219631903188,
0.04804551511145073,
0.47752350144148015,
1.3406847618100746,
-1.3722262204383469,
-1.0513455672656666,
0.7121308231347245,
0.13278942055995957,
1.9758312031522887,
2.19648222646383,
0.8936788375205645,
0.05510909660880629,
-0.18765678539637448,
0.45031205104496697,
-0.6542716355161553,
0.5783583754336702,
-0.18844184232604352,
-1.5667303012058793,
0.36110899429768645,
-0.275182998952267,
0.30367576290775045,

```

-0.1532687178202341]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                  'noise_prob': [0.001, 0.001, 0, 0],
                  'status': True,
                  'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
               'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_xy_noisefree_time_evo_ric4_unifrom_scaling_sample#2'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.130214875567141],
                          [12, 6, 0, 0, -5.622281179564727],
                          [20, 10, 0, 0, -4.4547314200293675],
                          [28, 14, 0, 0, -3.544056752302471]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.00543212890625e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.037922213699304,
                           'others': { 'beta_coeffiecients': [ 2.1875,
                                                                -2.1875,
                                                                1.3125,
                                                                -0.3125],
                                       'cost_richardson_zne': 11.390625,
                                       'sorted_expectation_vals': [
-7.130214875567141,
-5.622281179564727,
-4.4547314200293675,
-3.544056752302471],

'sorted_noise': [6, 18, 30, 42]],
'sampled data': [ [ 4,
                    2,
                    0,
                    0,
                    -7.130214875567141],
                  [ 12,
                    6,
                    0,
                    0,
                    -5.622281179564727],
                  [ 20,
                    10,
                    0,
                    0,
                    -4.4547314200293675],
                  [ 28,

```

```

14,
0,
0,
-3.544056752302471]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}}},
'init_param': { 'value': [ 0.01970245038838479,
0.18403088824160574,
0.23516669011749736,
0.3619340287619915,
0.7559235356843684,
0.8037553588156275,
0.8037553588156274,
0.8037553588156274,
1.523650120002224,
1.5399034674182392,
3.5267025169505333,
3.526702516950533,
4.657784878324055,
4.735234101549794,
4.768060921347247,
6.910522417024203,
6.910522417024203,
7.937298848041795,
7.937298848041795,
7.937298848041795,
7.937298848041795,
7.937298848041795,
7.937298848041795,
7.937298848041795,
7.937298848041795,
8.950027818739875,
8.950027818739875,
8.950027818739875,
8.950027818739875,
8.950027818739875,
9.275481401407285,
0.034552241786943605,
0.7960682992937013,
-0.16960774395822967,
-0.13488653659207303,
-0.07127010448742209,
0.7213919487775996,
0.07517487951530155,
0.7669403129297154,
-0.0980146039609658,
0.2841753594230342,
0.41897129078802076,
0.7644737129881134,
0.21659836679204617,
0.40978346593820614,
0.23386906952753808,
0.6351364889148511,
0.33021983360135304,
0.3518425622667454,
-0.16759580526603354,
1.1955178055405769,
-0.1119172073880175,
-1.6647459335605923,

```

-0.09499524124538847,
1.171044098528735,
-0.022137095989983098,
0.46479840085551427,
0.5434533112600137,
0.7953483283824695,
-0.23791626316884204,
-0.34327459028867535,
0.9770824990177593,
0.3972109926583832,
0.031305244787475206,
0.06660894739153407,
-0.4270300066285869,
-1.4580475210852422,
0.18899331747125298,
-0.5789031865333153,
0.2203088633995836,
-0.028805094274594364,
-0.21236041446549625,
0.00835869807102751,
1.8444421334304235,
-0.36021419267454113,
-1.0777998782841605,
0.7201309879688491,
-0.07151832135087417,
0.013523038580686776,
-0.5101061898109359,
1.528477364550823,
-0.24055091899624312,
1.595727742687772,
1.181011187425284,
-0.07205139039345279,
-0.3645912646252544,
0.922259297261975,
0.47504473393792157,
-1.4582846498754758,
-1.1041238978674286,
-0.08619812725702276,
1.5348185477516838,
-0.034799821636765677,
0.34854563969795865,
-0.3057886345189525,
0.3435344949085216,
2.387989506593159,
0.5928805156422549,
-0.02250711897690342,
0.734685803060112,
1.1230348216575474,
0.9339212770561773,
0.5429657194529808,
-0.9530889967003539,
1.2607191116533525,
-0.18262479735843284,
0.2292294020716647,
0.484625167272873,
2.2621526738562925,
0.3832060422832924,
0.2504382199240898,
-1.1186566372739195,
1.0354182359738777,
-0.9045076995614423,
1.0282595509077306,
1.805679851173207,
0.9332861023326164,
1.4396403686507162,
1.6178312922032976,
-0.22196274260736226,
0.19030625281227892,
-0.32840557234129963,
1.004592356770436,
1.8720477930131976,
-0.06388076228748135,
0.12235432131563899,
-0.5998878734536057,
-0.5461403115162753,

[illegible]

-7.512595875373562,
-6.004303631668366,
-4.83491199768215,
-3.919939099798885],

```
'sorted_expectation_vals': [  
  
    'sorted_noise': [6, 18, 30, 42]],  
    'sampled_data': [ [ 4,  
                        2,  
                        0,  
                        0,  
                        -7.512595875373562],  
                      [ 12,  
                        6,  
                        0,  
                        0,  
                        -6.004303631668366],  
                      [ 20,  
                        10,  
                        0,  
                        0,  
                        -4.83491199768215],  
                      [ 28,  
                        14,  
                        0,  
                        0,  
                        -3.919939099798885]],  
    'sampling': 'default']},  
{ 'config': { 'ansatz': { 'gateset': 1,  
                          'layer': 30,  
                          'ugate': { 'coefficients': { 'bn': [ 0,  
                                                            0,  
                                                            0,  
                                                            0,  
                                                            0,  
                                                            0],  
                                                            'cn': [ 0.5,  
                                                            0.5,  
                                                            0.5,  
                                                            0.5,  
                                                            0.5,  
                                                            0.5],  
                                                            'r': 0},  
                          'time': {'max': 10.0, 'min': 0.0},  
                          'type': 'xy-iss'}},  
  'init_param': { 'value': [ 0.003408343124459994,  
                             0.0034083431244599804,  
                             0.006710027629051694,  
                             0.2484817728699434,  
                             1.1672656342280407,  
                             1.167470260322433,  
                             3.1453440534904744,  
                             3.166261678218574,  
                             3.2843704266652405,  
                             4.089250903115526,  
                             4.091989893807717,  
                             4.091989893807717,  
                             5.181052167254873,  
                             5.181052167254873,  
                             5.184972473956635,  
                             5.585122024093685,  
                             5.5990893229720795,  
                             5.630355974178803,  
                             5.704160317443757,  
                             8.658831571647385,  
                             8.66505230970618,  
                             8.712765508205756,  
                             8.712765508205756,  
                             8.712765508205756,  
                             9.168875276976076,  
                             9.176734348833309,  
                             9.262471352200725,  
                             9.262471352200725,  
                             10.578606537988184,  
                             10.578606537988184,
```

1.5905119084430799,
0.8691136234632737,
0.04907693052707525,
-0.23336553986203207,
0.4284777328779486,
2.2866815267075147,
0.1760352312050406,
0.27820168999860917,
2.067264415281462,
0.5755331494929522,
-0.06858856463703264,
-0.7222667597755238,
0.7750924366303688,
-0.278304785792398,
0.5352624789798245,
-1.1702690163303244,
-0.012717570027659048,
-0.07125250327360101,
1.7927644689789062,
1.3752915621891346,
-1.1981440291126808,
-0.30737164793891514,
0.2657743539912098,
0.04156274567190095,
1.1363043468044227,
0.059217925806269944,
0.7608352104096147,
-0.1513784586666425,
0.025696061144845524,
0.861481321747562,
-0.48544656000194114,
-0.006033515622113467,
0.473431667961849,
1.5469119181224316,
0.1871819193791605,
1.555475823471471,
0.9001349633565042,
-1.5961139272878428,
-0.6749613217725859,
-1.5596468027729833,
0.5296186936496604,
-0.8549575199843406,
1.7649719371938613,
1.5569889217595585,
-0.9244334407213903,
-1.5804326845781602,
0.3840271097002098,
-0.8070678340768227,
-0.48791686278645724,
0.13110829311357092,
-2.023238766590243,
-0.8997043584182223,
0.8435119798114825,
0.12108034013815198,
-0.7013572315179941,
0.05813246571029178,
-1.3645229029121433,
-1.704441396215925,
0.6160899239086374,
-1.6580292769388947,
-1.550797918270104,
-0.033040246679508146,
-0.7595754539148374,
0.6024045095197653,
0.446745843265138,
0.9717727329863253,
-1.3190014690683383,
-3.180952616822137,
-0.47685451708111876,
0.03958388950097112,
-0.04894683660460036,
-0.013869967587830365,
0.9715551393376142,
-0.2688733257646762,
-1.7112803723577967,

```

-0.06346862095088202,
1.2756440828431679,
0.4087049517265435,
-0.4433737535464867,
0.07044159255576982,
0.877714241226598,
-0.33059433004855665,
-0.41683973817974146,
-0.09123983383660174,
1.153451542408442,
1.5381060924707182,
1.4432055127327397,
0.7948381511053778,
0.12126271666707081,
0.3736422837955817,
-0.06595460407520624,
-0.1569219765005705,
1.4053035958769775,
-0.3443177603686477,
0.14604377230684606,
0.733462484703497,
1.2365404639754205,
-0.020105376528163625,
-0.09631958929266826,
-0.16876809298151013,
0.42184799028013376,
0.8803036484986011,
-0.4148292271104988,
0.011464734896518355,
1.0895788780475535,
-0.006017115067308532,
0.6476481463328121,
-0.4792828310008005,
1.7145504838049759,
0.9300252591703124,
1.4147034805691376,
-0.04601267898776636,
-0.34579443398211707,
0.006107561937391137,
2.7608588509789516,
-2.470402787070412,
-0.2193552778493498,
-0.43636419827955597,
0.5036040054264609,
-0.006776779992121664]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0, 0],
                   'status': True,
                   'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_xy_noisefree_time_evo_ric4_unifrom_scaling_sample#7'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},

```

```

'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.568863804770513],
                          [12, 6, 0, 0, -6.084665473601484],
                          [20, 10, 0, 0, -4.914126798758423],
                          [28, 14, 0, 0, -3.985131466280292]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.744529724121094e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.451121689590092,
                           'others': { 'beta_coefficients': [ 2.1875,
                                                              -2.1875,
                                                              1.3125,
                                                              -0.3125],
                                       'cost_richardson_zne': 11.390625,
                                       'sorted_expectation_vals': [

```

```

-7.568863804770513,
-6.084665473601484,
-4.914126798758423,
-3.985131466280292],

```

```

'sorted_noise': [6, 18, 30, 42]],
'sampled data': [ [ 4,
                    2,
                    0,
                    0,
                    -7.568863804770513],
                  [ 12,
                    6,
                    0,
                    0,
                    -6.084665473601484],
                  [ 20,
                    10,
                    0,
                    0,
                    -4.914126798758423],
                  [ 28,
                    14,
                    0,
                    0,
                    -3.985131466280292]],
'sampling': 'default'}}]]}}

```